

Д. Д. Дзудцова, Л. Б. Бестаева

ОКИСЛИТЕЛЬНО- ВОССТАНОВИТЕЛЬНЫЕ реакции



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Д. Д. Дзудцова, Л. Б. Бестаева

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ДРОФА

Москва • 2005

УДК 373.167.1:54
ББК 24.1я72
Д43

Серия основана в 2001 году

Дзудцова, Д. Д.
Д43 Окислительно-восстановительные реакции / Д. Д. Дзудцова, Л. Б. Бестаева. — М. : Дрофа, 2005. — 318, [2] с. — (Темы школьного курса).

ISBN 5-7107-8550-4

Данное издание посвящено одной из тем школьного курса химии — «Окислительно-восстановительные реакции». Уникальность пособия в том, что в нем рассмотрены теоретические вопросы данной темы, а также приведены более 1000 примеров уравнений окислительно-восстановительных реакций с участием неорганических и органических веществ и способы подбора коэффициентов в этих реакциях методами электронного и электронно-ионного баланса.

Пособие адресовано учащимся средних школ, абитуриентам, студентам и учителям химии.

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ISBN 5-7107-8550-4

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Данное издание, предназначенное учащимся, абитуриентам, студентам и учителям, является пособием по составлению уравнений окислительно-восстановительных реакций (ОВР).

Подобрать коэффициенты, особенно в сложных уравнениях реакций, часто затруднительно, поэтому основное назначение пособия — помочь разобраться в способах подбора коэффициентов в уравнениях ОВР. Данное издание можно использовать и как справочное по окислительно-восстановительным свойствам неорганических или органических веществ.

Уникальность данного пособия в том, что в нем приведено более 1000 примеров уравнений ОВР с участием неорганических и органических веществ.

В части I пособия рассмотрены теоретические вопросы, которые необходимо изучить для составления уравнений ОВР. В части II приведены уравнения ОВР, в которых коэффициенты подбираются методом электронного баланса, в части III — методом электронно-ионного баланса (методом полуреакций).

Части II и III имеют следующую структуру: правила оформления уравнений реакций, схемы ОВР (условия задач), которые характерны для определенного элемента или класса веществ, решения задач. Схемы расположены в порядке усложнения уравнений реакций. В каждой схеме на первом месте записана формула основного реагента (согласно заглавию). Вторые реагенты также по возможности сгруппированы, что облегчает нахождение необходимых формул.

В пособии имеется формульный указатель неорганических, органических и комплексных соединений. Цифры, стоящие рядом с формулой, указывают номера уравнений, в которых они встречаются. Это поможет быстро собрать информацию об окислительно-восстановительных свойствах данного вещества.

Авторы выражают признательность доктору технических наук, профессору кафедры общей химии МГУ им. М. В. Ломоносова Л. С. Гузею, кандидату химических наук, доценту, декану химико-технологического факультета СОГУ С. В. Кабанову, кандидату химических наук, доценту, зав. кафедрой общей химии СОГУ А. П. Алихановой, доктору химических наук, профессору Р. А. Лидину за ценные указания по содержанию рукописи.

Все пожелания в отношении содержания и материала будут приняты с благодарностью и учтены в дальнейшей работе.

Рекомендации по использованию пособия

1. Ознакомьтесь с теорией и правилами по составлению уравнений ОВР.

2. Самостоятельно подберите коэффициенты в схемах уравнений ОВР. Обращайтесь к решениям задач только в случае затруднений.

3. Для проверки умения составлять уравнения ОВР можно выполнять следующие виды заданий:

1) согласно схеме уравнения под номером ... составьте уравнение ОВР методом электронного баланса;

2) согласно схеме уравнения под номером ... составьте уравнение ОВР методом электронно-ионного баланса;

3) закройте правую часть схемы, оставив один продукт, и составьте уравнение ОВР;

4) закройте правую часть схемы и составьте уравнение ОВР;

5) восстановите левую часть уравнения под номером ..., если даны формулы продуктов реакции с коэффициентами;

6) восстановите левую часть уравнения под номером ..., если даны формулы продуктов реакции без коэффициентов.

Принятые в пособии сокращения

ам. — аморфный;
акт. мет. — активный металл;
атомарн. — атомарный;
бел. — белый;
возд. — воздух;
вар. — вариант;
вос. — восстановитель;
Г — галоген;
г — газ;
гор. — горячий;
дым. — дымящий;
ē — электрон;
ж — жидкий;
изб. — избыток;
кат. — катализатор;
к. с. — кислотная среда;
красн. — красный;
конц. — концентрированный;
коэф. — коэффициент;
недост. — недостаток;
насыщ. — насыщенный;

н. с. — нейтральная среда;
ОВР — окислительно-восстановительная реакция;
ок. — окислитель;
оч. разб. — очень разбавленный раствор;
разб. — разбавленный раствор;
р-р — раствор;
с. о. — степень окисления;
ср. конц. — средняя концентрация;
сплав. — сплавление;
т — твердый;
хол. — холодный;
щ. с. — щелочная среда;
ЭБ — электронный баланс;
ЭИБ — электронно-ионный баланс;
эо — электроотрицательность.

Окислительно-восстановительные реакции: теоретические вопросы

Степень окисления

Степень окисления (с. о.) — это условный, формальный заряд, который приписывают атомам в электронейтральной молекуле или многоатомном ионе.

Степень окисления может быть положительной, отрицательной или равняться нулю, что зависит от природы соответствующих соединений. Положительную или отрицательную степень окисления элемент получает в том случае, если электронные плотности химических связей полностью смещены к атомам элемента с более высоким значением электроотрицательности¹ (эо). Это относится к веществам с ионной связью. В соединениях с сильно- или слабополярными ковалентными связями связи условно рассматриваются как ионные, как если бы молекулы состояли из положительно или отрицательно заряженных ионов.

Относительные электроотрицательности элементов приведены в приложении 1. Расположив элементы в порядке увеличения значений эо, получают шкалу эо (по Полингу):

Si, Sb, B, As, H, P, Se, I, C, S, Br, Cl, N, O, F,

где F — самый электроотрицательный элемент.

Одни элементы имеют постоянные степени окисления, другие — переменные. Например, к элементам с постоянной положительной с. о. относят щелочные металлы: Li^{+1} , Na^{+1} , K^{+1} , Rb^{+1} , Cs^{+1} , Fr^{+1} , следующие элементы II группы периодической системы: Be^{+2} , Mg^{+2} , Ca^{+2} , Sr^{+2} , Ba^{+2} , Ra^{+2} , Zn^{+2} , а также — Al^{+3} и т. д.

¹ Электроотрицательность — это способность атомов притягивать и удерживать около себя электроны других атомов.

Металлы в соединениях всегда имеют положительную степень окисления.

Из неметаллов постоянную отрицательную с. о. (-1) имеет F.

В простых веществах, образованных атомами металлов или неметаллов, с. о. элементов равны нулю, например: Na^0 , Al^0 , Fe^0 , H_2^0 , O_2^0 , O_3^0 , F_2^0 , Cl_2^0 , Br_2^0 , N_2^0 , C^0 , S^0 , Si^0 .

Для водорода характерны с. о.: +1 (H_2O), -1 (NaH).

Для кислорода характерны с. о.: -2 (H_2O), -1 (H_2O_2), 0 (O_2 , FOCl), +1 (F_2O_2), +2 (OF_2).

Следует помнить, что в целом молекула электронейтральна, поэтому в любой молекуле алгебраическая сумма степеней окисления равна нулю.

Сумма степеней окисления атомов многоатомного иона по знаку и величине равна заряду этого иона.

Пользуясь приведенными выше сведениями, можно рассчитать с. о. элементов в соединении.

ПРИМЕР 1. Найдите с. о. марганца в соединении, формула которого MnO_2 .

Способ 1. Обозначим степень окисления марганца через x : $\text{Mn}^x\text{O}_2^{-2}$, откуда $x = -\frac{(\text{число атомов кислорода}) \cdot (-2)}{(\text{число атомов марганца})} = -\frac{2 \cdot (-2)}{1} = +4$; с. о. Mn = +4; $\text{Mn}^{+4}\text{O}_2^{-2}$.

Способ 2. Искомую степень окисления элемента можно найти и из уравнения, которое составляют с учетом того, что алгебраическая сумма степеней окисления всех элементов в соединении равна нулю, а в сложном ионе — заряду иона. Обозначив с. о. марганца через x и умножив известную степень окисления кислорода (-2) на число его атомов (2), получим уравнение: $1 \cdot x + 2 \cdot (-2) = 0$, откуда $x = +4$.

ПРИМЕР 2. Найдите с. о. хрома в соединении, формула которого Cr_2O_3 .

Способ 1. $\text{Cr}_2^x\text{O}_3^{-2}$, $x = -\frac{3 \cdot (-2)}{2} = +3$; $\text{Cr}_2^{+3}\text{O}_3^{-2}$.

Способ 2. $\text{Cr}_2^x\text{O}_3^{-2}$; составляем уравнение: $2x + 3 \cdot (-2) = 0$, откуда $x = +3$; $\text{Cr}_2^{+3}\text{O}_3^{-2}$.

Если в соединении три элемента и с. о. двух из них известны, то легко можно вычислить с. о. третьего элемента.

ПРИМЕР 3. Найдите с. о. хрома в дихромате калия $K_2Cr_2O_7$.

Способ 1. Обозначим с. о. хрома через x : $K_2^{+1}Cr_x^xO_7^{-2}$. Составим уравнение: $2 \cdot (+1) + 2x + 7 \cdot (-2) = 0$; $2 + 2x - 14 = 0$; $2x = +14 - 2$; $2x = +12$; $x = +6$, т. е. с. о. хрома в $K_2Cr_2O_7 = +6$; $K_2^{+1}Cr_2^{+6}O_7^{-2}$.

Способ 2. 1) Подсчитаем число отрицательных зарядов: $7 \cdot (-2) = -14$.

2) Подсчитаем число положительных зарядов:

$$-14 - 2 \cdot (+1) = -12,$$

далее изменим знак разности $(-)$ на противоположный знак $(+)$: $+12$.

3) Так как в формуле два атома хрома, 12 делим на два: $12 : 2 = +6$ — это с. о. хрома.

Проверка: $K_2^{+1}Cr_2^{+6}O_7^{-2}$

$2 \cdot (+1) + 2 \cdot (+6) = +14$ | $7 \cdot (-2) = -14$ — алгебраическая сумма положительных и отрицательных степеней окисления элементов равна нулю, «молекула» электро-нейтральна.

ПРИМЕР 4. Найдите с. о. серы в сульфате алюминия.

Способ 1. Для нахождения с. о. серы в $Al_2(SO_4)_3$, где с. о. Al $(+3)$, O (-2) , S (x) , производят следующие расчеты.

Составляем уравнение: $2 \cdot (+3) + 3x + 3 \cdot 4 \cdot (-2) = 0$, откуда $x = +6$, т. е. с. о. серы в $Al_2(SO_4)_3$ равна $+6$; $Al_2^{+3}(S^{+6}O_4^{-2})_3$.

Способ 2. Находим:

1) число отрицательных зарядов: $3 \cdot 4 \cdot (-2) = -24$;

2) число положительных зарядов: $-24 - 2 \cdot (+3) = -18$;
меняем знак разности на противоположный: $+18$;

3) так как в формуле три атома серы, 18 делим на три: $18 : 3 = +6$ — это с. о. серы в $Al_2(SO_4)_3$.

Проверка: $Al_2^{+3}(S^{+6}O_4^{-2})_3$

$$2 \cdot (+3) + 3 \cdot (+6) = +24$$
 | $3 \cdot 4 \cdot (-2) = -24.$

ПРИМЕР 5. Определите с. о. фосфора в веществах, формулы которых: а) H_3PO_4 ; б) $K_4P_2O_7$.

а) Способ 1. $\text{H}_3^{+1}\text{P}^x\text{O}_4^{-2}$; $3 \cdot (+1) + x + 4 \cdot (-2) = 0$; $x = +5$.

Способ 2. $4 \cdot (-2) = -8$; $8 - 3 \cdot (+1) = +5$.

Ответ: $\text{H}_3^{+1}\text{P}^{+5}\text{O}_4^{-2}$.

б) Способ 1. $\text{K}_4^{+1}\text{P}_2^x\text{O}_7^{-2}$.

$4 \cdot (+1) + 2x + 7 \cdot (-2) = 0$; $x = +5$.

Способ 2. $7 \cdot (-2) = -14$; $14 - 4 \cdot (+1) = 10$; $10 : 2 = +5$.

Ответ: $\text{K}_4^{+1}\text{P}_2^{+5}\text{O}_7^{-2}$.

ПРИМЕР 6. Найдите с. о. фосфора и хлора в сложных ионах: а) $(\text{HPO}_4)^{2-}$; б) $(\text{ClO}_3)^-$.

а) $(\text{H}^{+1}\text{P}^x\text{O}_4^{-2})^{2-}$; $1 + x + 4 \cdot (-2) = -2$, $x = +5$.

Ответ: $(\text{H}^{+1}\text{P}^{+5}\text{O}_4^{-2})^{2-}$.

б) $(\text{Cl}^x\text{O}_3^{-2})^-$; $x + 3 \cdot (-2) = -1$, $x = +5$.

Ответ: $(\text{Cl}^{+5}\text{O}_3^{-2})^-$.

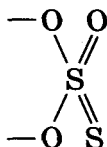
ПРИМЕР 7. Определите с. о. серы в тиосульфате калия $\text{K}_2\text{S}_2\text{O}_3$, или $(\text{K}_2\text{SO}_3)\text{S}$.

$\text{K}_2^{+1}\text{S}_2^x\text{O}_3^{-2}$.

Составляем уравнение: $2 \cdot (+1) + 2x + 3 \cdot (-2) = 0$; $x = +2$ (формально).

Ответ: $\text{K}_2^{+1}\text{S}_2^{+2}\text{O}_3^{-2}$.

Для тиосульфат-иона можно предложить структурную формулу:



Исходя из электронейтральности «молекулы» $\text{K}_2\text{SO}_3\text{S}$ степени окисления двух атомов серы могут быть: S^{+6} , S^{-2} или S^{+4} , S^0 . Для нахождения коэффициентов в уравнении можно использовать любой вариант с. о. серы.

ПРИМЕР 8. Определите с. о. серы в пероксодисульфате калия $\text{K}_2\text{S}_2\text{O}_8$.

$\text{K}_2^{+1}\text{S}_2^x\text{O}_8^{-2}$; $2 \cdot (+1) + 2x + 8 \cdot (-2) = 0$; $x = +7$, формально получается, что степень окисления серы больше, чем номер группы (например, см. примеры 188—192).

ПРИМЕР 9. Найдите с. о. углерода в веществах, формулы которых: 1) CH_4 ; 2) C_2H_6 ; 3) C_2H_4 ; 4) C_6H_6 ; 5) C_7H_8 ; 6) $\text{C}_3\text{H}_8\text{O}_3$; 7) HCOH ; 8) HCOOH ; 9) CO_2 .

1) $\text{C}^x\text{H}_4^{+1}$; $x + 4 \cdot (+1) = 0$; $x = -4$; $\text{C}^{-4}\text{H}_4^{+1}$.

2) $\text{C}_2^x\text{H}_6^{+1}$; $2x + 6 \cdot (+1) = 0$; $x = -3$; $\text{C}_2^{-3}\text{H}_6^{+1}$.

3) $\text{C}_2^x\text{H}_4^{+1}$; $2x + 4 \cdot (+1) = 0$; $x = -2$; $\text{C}_2^{-2}\text{H}_4^{+1}$.

4) $\text{C}_6^x\text{H}_6^{+1}$; $6x + 6 \cdot (+1) = 0$; $x = -1$; $\text{C}_6^{-1}\text{H}_6^{+1}$.

5) $\text{C}_7^x\text{H}_8^{+1}$; $7x + 8 \cdot (+1) = 0$; $x = -8/7$; $\text{C}_7^{-8/7}\text{H}_8^{+1}$.

6) $\text{C}_3^x\text{H}_8^{+1}\text{O}_3^{-2}$; $3x + 8 \cdot (+1) + 3 \cdot (-2) = 0$; $x = -2/3$;
 $\text{C}_3^{-2/3}\text{H}_8^{+1}\text{O}_3^{-2}$.

7) $\text{H}^{+1}\text{C}^x\text{O}^{-2}\text{H}^{+1}$; $2 \cdot (+1) + x + 1 \cdot (-2) = 0$; $x = 0$;
 $\text{H}^{+1}\text{C}^0\text{O}^{-2}\text{H}^{+1}$.

8) $\text{H}^{+1}\text{C}^x\text{O}_2^{-2}\text{H}^{+1}$; $2 \cdot (+1) + x + 2 \cdot (-2) = 0$; $x = +2$;
 $\text{H}^{+1}\text{C}^{+2}\text{O}_2^{-2}\text{H}^{+1}$.

9) $\text{C}^x\text{O}_2^{-2}$; $x + 2 \cdot (-2) = 0$; $x = +4$; $\text{C}^{+4}\text{O}_2^{-2}$.

Следует обратить внимание на то, что с. о. углерода во всех соединениях разная, а валентность его неизменно равна четырем.

ПРИМЕР 10. Найдите с. о. железа в комплексном соединении $\text{K}_4[\text{Fe}(\text{CN})_6]$; $(\text{CN})^-$ — это анион циановодородной кислоты HCN . (Ее структурная формула: $\text{H}-\text{C}\equiv\text{N}$, где с. о. элементов: $\text{H}^{+1}\text{C}^{+2}\text{N}^{-3}$.)

Способ 1. $\text{K}_4^{+1}[\text{Fe}^x(\text{CN})_6]^-$; $4 \cdot (+1) + x + 6 \cdot (-1) = 0$; $x = +2$.

Способ 2. С учетом известных степеней окисления элементов: $\text{K}_4^{+1}[\text{Fe}^x(\text{C}^{+2}\text{N}^{-3})_6]^-$; $4 \cdot (+1) + x + 6 \cdot (+2) + 6 \cdot (-3) = 0$; $x = +2$.

Способ 3. С учетом заряда комплексного аниона: $[\text{Fe}^x(\text{CN})_6]^{-1}{}^{4-}$; $x + 6 \cdot (-1) = -4$; $x = +2$. Степень окисления железа +2.

Название комплексной соли — гексацианоферрат (II) калия.

Как видно из примеров, понятие «степень окисления» формальное, условное, не отражает действительные, эффективные заряды атомов, например в HClO_4 с. о. хлора +7, а кислорода — (−2). Реально же их эффективные заряды равны +1,0 и −0,5 соответственно.

Степень окисления и валентность часто не совпадают, потому что с. о. не отражает распределение валентных электронов, образующих химические связи в молекуле, как, например, в ионе аммония NH_4^+ степень окисления азота (-3), а валентность IV. Однако знание степеней окисления элементов помогает находить стехиометрические коэффициенты в уравнениях окислительно-восстановительных реакций.

Электронные уравнения

Для нахождения коэффициентов в уравнениях ОВР важно правильно уравнивать заряды левой и правой частей схемы электронного баланса. Разберем электронные уравнения на примере азота (табл. 1).

Т а б л и ц а 1

Примеры составления электронных уравнений
(схем ЭБ) на примере азота*

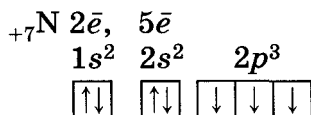
Схема	Расчет числа принятых или отданных электронов	Оформление электронного уравнения (схемы ЭБ)
1) $\text{N}^0 \longrightarrow \text{N}^{+5}$	$0 \longrightarrow +5; 0 - (-5) = +5$	$\text{N}^0 - 5\bar{e} \longrightarrow \text{N}^{+5}$
2) $\text{N}^{+5} \longrightarrow \text{N}^0$	$+5 \longrightarrow 0; +5 + (-5) = 0$	$\text{N}^{+5} + 5\bar{e} \longrightarrow \text{N}^0$
3) $\text{N}^0 \longrightarrow \text{N}^{-3}$	$0 \longrightarrow -3; 0 + (-3) = -3$	$\text{N}^0 + 3\bar{e} \longrightarrow \text{N}^{-3}$
4) $\text{N}^{-3} \longrightarrow \text{N}^0$	$-3 \longrightarrow 0; -3 - (-3) = 0$	$\text{N}^{-3} - 3\bar{e} \longrightarrow \text{N}^0$
5) $\text{N}^{-3} \longrightarrow \text{N}^{+5}$	$-3 \longrightarrow +5; -3 - (-8) = +5$	$\text{N}^{-3} - 8\bar{e} \longrightarrow \text{N}^{+5}$
6) $\text{N}^{+5} \longrightarrow \text{N}^{-3}$	$+5 \longrightarrow -3; +5 + (-8) = -3$	$\text{N}^{+5} + 8\bar{e} \longrightarrow \text{N}^{-3}$
7) $\text{N}^{+5} \longrightarrow \text{N}^{+2}$	$+5 \longrightarrow +2; +5 + (-3) = +2$	$\text{N}^{+5} + 3\bar{e} \longrightarrow \text{N}^{+2}$
8) $\text{N}^{+5} \longrightarrow \text{N}^{+4}$	$+5 \longrightarrow +4; +5 + (-1) = +4$	$\text{N}^{+5} + 1\bar{e} \longrightarrow \text{N}^{+4}$
9) $\text{N}^{+2} \longrightarrow \text{N}^{+3}$	$+2 \longrightarrow +3; +2 - (-1) = +3$	$\text{N}^{+2} - 1\bar{e} \longrightarrow \text{N}^{+3}$
10) $\text{N}^{+3} \longrightarrow \text{N}^0$	$+3 \longrightarrow 0; +3 + (-3) = 0$	$\text{N}^{+3} + 3\bar{e} \longrightarrow \text{N}^0$
11) $\text{N}^{+5} \longrightarrow \text{N}^{+3}$	$+5 \longrightarrow +3; +5 + (-2) = +3$	$\text{N}^{+5} + 2\bar{e} \longrightarrow \text{N}^{+3}$

Схема	Расчет числа принятых или отданных электронов	Оформление электронного уравнения (схемы ЭВ)
12) $N^0 \longrightarrow N^{+1}$	$0 \longrightarrow +1; 0 - (-1) = +1$	$N^0 - 1\bar{e} \longrightarrow N^{+1}$
13) $N^{+5} \longrightarrow N^{+1}$	$+5 \longrightarrow +1; +5 + (-4) = +1$	$N^{+5} + 4\bar{e} \longrightarrow N^{+1}$
14) $N^{+4} \longrightarrow N^{+3}$	$+4 \longrightarrow +3; +4 + (-1) = +3$	$N^{+4} + 1\bar{e} \longrightarrow N^{+3}$
15) $N^{+4} \longrightarrow N^{+2}$	$+4 \longrightarrow +2; +4 + (-2) = +2$	$N^{+4} + 2\bar{e} \longrightarrow N^{+2}$
16) $N^{+1} \longrightarrow N^{+5}$	$+1 \longrightarrow +5; +1 - (-4) = +5$	$N^{+1} - 4\bar{e} \longrightarrow N^{+5}$
17) $N^{+1} \longrightarrow N^{+4}$	$+1 \longrightarrow +4; +1 - (-3) = +4$	$N^{+1} - 3\bar{e} \longrightarrow N^{+4}$
18) $N^{+1} \longrightarrow N^{+3}$	$+1 \longrightarrow +3; +1 - (-2) = +3$	$N^{+1} - 2\bar{e} \longrightarrow N^{+3}$
19) $N^{+1} \longrightarrow N^{+2}$	$+1 \longrightarrow +2; +1 - (-1) = +2$	$N^{+1} - 1\bar{e} \longrightarrow N^{+2}$
20) $N^{+3} \longrightarrow N^{+1}$	$+3 \longrightarrow +1; +3 + (-2) = +1$	$N^{+3} + 2\bar{e} \longrightarrow N^{+1}$
21) $N^{+2} \longrightarrow N^{+4}$	$+2 \longrightarrow +4; +2 - (-2) = +4$	$N^{+2} - 2\bar{e} \longrightarrow N^{+4}$
22) $N^{+2} \longrightarrow N^{-3}$	$+2 \longrightarrow -3; +2 + (-5) = -3$	$N^{+2} + 5\bar{e} \longrightarrow N^{-3}$
23) $N^{+2} \longrightarrow N^{-1}$	$+2 \longrightarrow -1; +2 + (-3) = -1$	$N^{+2} + 3\bar{e} \longrightarrow N^{-1}$
24) $N^{+1} \longrightarrow N^{-2}$	$+1 \longrightarrow -2; +1 + (-3) = -2$	$N^{+1} + 3\bar{e} \longrightarrow N^{-2}$
25) $N^{+3} \longrightarrow N^{-3}$	$+3 \longrightarrow -3; +3 + (-6) = -3$	$N^{+3} + 6\bar{e} \longrightarrow N^{-3}$

* Обратите внимание на то, что во второй графе таблицы перед цифрой, показывающей число принятых или отданных электронов, стоит знак «минус», так как электроны — отрицательно заряженные частицы.

Азот — элемент главной подгруппы V группы, 2-го периода периодической системы.

Схема строения электронной оболочки атома азота:



У азота на внешнем энергетическом уровне пять валентных электронов. Он может отдавать от одного до пяти электронов, получая положительные степени окисления: N^{+1} , N^{+2} , N^{+3} , N^{+4} , N^{+5} , а может и принимать, но не более трех электронов, получая при этом отрицательные степени окисления: N^{-1} , N^{-2} , N^{-3} .

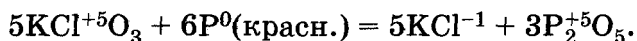
Приведенные примеры — это только часть из 72 возможных электронных переходов для азота.

Количество отданных или принятых электронов можно рассчитать и по-другому. Выпишем степени окисления азота из 2-й графы 20-го примера вышеприведенной таблицы: $+3 \rightarrow +1$. Перенесем $+1$ с противоположным знаком в левую часть и произведем расчет: $+3 - 1 = +2$. «Плюс два» означает, что в электронном уравнении к N^{+3} следует добавить $2\bar{e}$: $N^{+3} + 2\bar{e} \rightarrow N^{+1}$.

Окислительно-восстановительные свойства некоторых веществ.

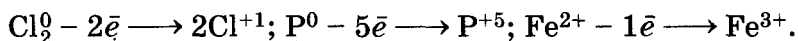
Типичные окислители и восстановители

Окислительно-восстановительные реакции — это такие реакции, в которых одновременно протекают процессы окисления и восстановления и, как правило, изменяются степени окисления элементов. Например:

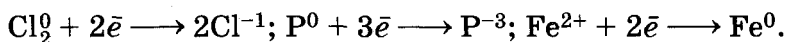


В приведенном примере степени окисления меняют хлор и фосфор.

Процесс отдачи электронов молекулой, атомом, ионом называется **окислением**, например:



Процесс присоединения электронов молекулой, атомом, ионом называется **восстановлением**, например:



Восстановитель, отдавая электроны, окисляется, а окислитель, принимая их, восстанавливается. В вышеприведенных примерах:

$P^0 - 5\bar{e} \rightarrow P^{+5}$, P^0 — восстановитель, окисляется до P^{+5} ;

$\text{Cl}^{+5} + 6\bar{e} \longrightarrow \text{Cl}^{-1}$, Cl^{+5} — окислитель, восстанавливается до Cl^{-1} .

Степень окисления хлора понижается от +5 до -1, степень окисления фосфора повышается от 0 до +5.

Число отдаваемых электронов восстановителем равняется количеству электронов, принимаемых окислителем.

Если элемент имеет свою наивысшую степень окисления, то он может быть только окислителем, например в HN^{+5}O_3 азот в состоянии +5 может быть только окислителем и принимать электроны.

Только восстановителем может быть элемент, находящийся в низшей степени окисления, например в N^{-3}H_3 азот в состоянии -3 может отдавать электроны, т. е. является восстановителем. Соответствующие примеры — 224, 225 и т. д.

Однако аммиак NH_3 формально может быть и окислителем (примеры 227, 228 и т. д.) за счет атомов водорода.

По окислительно-восстановительным свойствам вещества можно разделить на три группы: 1) окислители; 2) восстановители; 3) окислители-восстановители.

Окислители — акцепторы электронов

1. Окислители, проявляющие окислительные свойства при повышенной или очень высокой температуре:

Cl_2 , F_2 , KClO_3 , K_2FeO_4 , KMnO_4 , KNO_3 , $\text{K}_2\text{S}_2\text{O}_8$, MnO_2 , NaBiO_3 , Na_2O_2 , O_2 , PbO_2 , $(\text{Pb}_2^{\text{II}}\text{Pb}^{\text{IV}})\text{O}_4$.

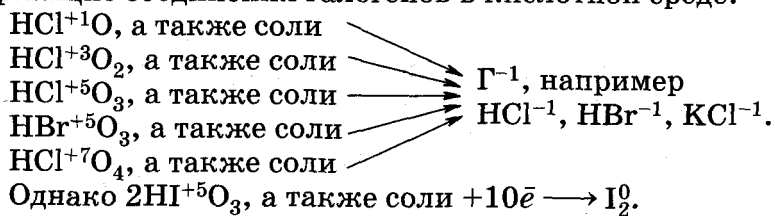
2. Окислители, проявляющие окислительные свойства в кислотной среде (вещества расположены по уменьшению окислительной способности):

F_2 , Na_2O_2 , $\text{NiO}(\text{OH})$, $(\text{Pb}_2^{\text{II}}\text{Pb}^{\text{IV}})\text{O}_4$, O_3 , $\text{K}_2\text{S}_2\text{O}_8$, K_2FeO_4 , NaBiO_3 , $\text{CoO}(\text{OH})$, H_2O_2 , KMnO_4 , KBrO_3 , PbO_2 , Cl_2 , $\text{K}_2\text{Cr}_2\text{O}_7$, MnO_2 , O_2 , KNO_2 , KIO_3 , Br_2 , HNO_3 (конц.), I_2 , H_2SO_4 (конц.), H^+ (разб.).

3. Окислители, проявляющие окислительные свойства в щелочной среде:

F_2 , $\text{K}_2\text{S}_2\text{O}_8$, Cl_2O_3 , Na_2O_2 , Br_2 , H_2O_2 , NaClO , NaBrO , KMnO_4 , I_2 , O_2 , PbO_2 , $(\text{Pb}_2^{\text{II}}\text{Pb}^{\text{IV}})\text{O}_4$, K_2CrO_4 , H_2O .

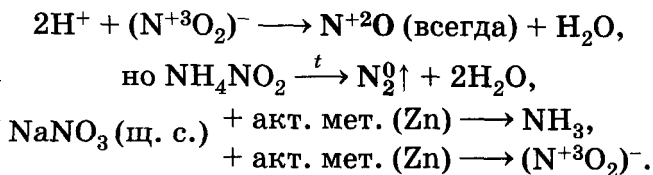
4. Окислительные свойства проявляют кислородсодержащие соединения галогенов в кислотной среде:



5. Окислительные свойства проявляет азотная кислота.

Разбавленная HNO_3	Продукты	HNO_3 концентрированная
Очень разб. (2—3%) $\text{Na, Al, Ca, Mg, Zn} \longrightarrow$	$\text{NH}_3, \text{NH}_4^+$	
Сред. разб., акт. мет. $\text{K, Ca, Mg, Zn, Al} \longrightarrow$	$\text{N}_2\text{O}, \text{N}_2$	
Сред. акт. мет. $\text{Fe, Cr, Ni, Bi} \longrightarrow$	$\text{NO}_2, \text{NO}, \text{N}_2\text{O}$	
$\text{Pb, Cu, Hg, Ag} \longrightarrow$	NO	
	NO	$\longleftarrow \text{Ca, Mg, Zn, (HNO}_3 - 30\%)$
	NO_2	$\longleftarrow \text{S, C, P, I}_2, \text{B, Sb, Sn, Pb, Cu, Hg, Ag (Al, t)}$
Не реагируют на холоде: Al, Cr, Ni, Fe . Пассивируются: Au, Pt, Os, Ir .		

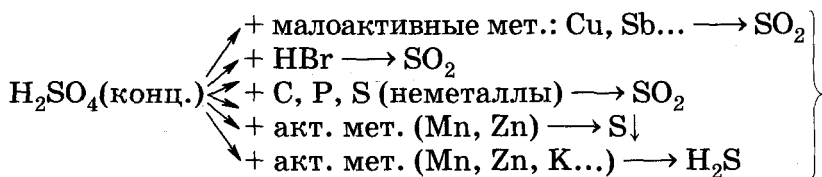
6. Окислительные свойства проявляет азотистая кислота, а также нитриты и нитраты:



7. Окислительные свойства проявляет оксид серы (IV) SO_2 :

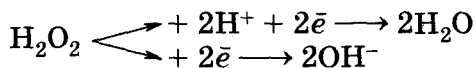


8. Окислительные свойства проявляет концентрированная серная кислота:



могут образоваться одновременно в разных соотношениях.

9. Окислительные свойства проявляет пероксид водорода:



Восстановители — доноры электронов

1. Вещества, проявляющие восстановительные свойства при повышенной или очень высокой температуре, например при сплавлении твердых веществ:

Al, C(кокс), CO, Ca, H₂, K, Mg, Na, S, Si, Se.

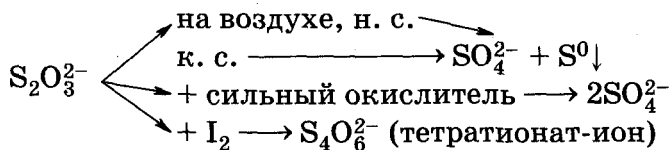
2. Восстановители в кислотной среде (вещества расположены в порядке уменьшения восстановительной способности):

Ca, Na, Mg, CaH₂, H(атомарный), (NH₃OH)Cl, Al, Zn, H₃PO₂, H₂C₂O₄, H₃PO₃, (N₂H₅)Cl, Na₂SO₃, H₂, TiCl₃, H[SnCl₃], H₂S, SO₂, C₂H₅OH, H₂O₂, KI, FeSO₄, KNO₂, HCl(конц.).

3. Восстановители в щелочной среде:

NH₂OH(конц.), Ca, H(атомарный), Mg, Li[AlH₄], Al, CaH₂, Na(PH₂O₂), SO₂, Na₂(PHO₃), Zn, N₂H₄(конц.), Na[Sn(OH)₃], Na₂SO₃, H₂, Na₂S.

4. Восстановительные свойства тиосульфат-иона:

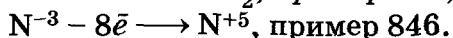
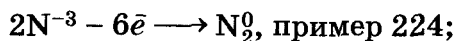


5. Восстановительные свойства проявляют соединения неметаллов.

Бинарные соединения	Увеличивается восстановительная активность →
Водородные соединения неметаллов	$\text{H}_2\text{S}, \text{H}_2\text{Se}, \text{H}_2\text{Te}$
Галогеноводородные кислоты	$\text{HCl}, \text{HBr}, \text{HI}$
Гидриды	$\text{LiH}, \text{NaH}, \text{KH}, \text{RbH}$
Сульфиды	$\text{Li}_2\text{S}, \text{Na}_2\text{S}, \text{K}_2\text{S}, \text{Rb}_2\text{S}$
Нитриды	$\text{Li}_3\text{N}, \text{Na}_3\text{N}, \text{K}_3\text{N}, \text{Rb}_3\text{N}$
Фосфиды	$\text{Li}_3\text{P}, \text{Na}_3\text{P}, \text{K}_3\text{P}, \text{Rb}_3\text{P}$
Галогениды	$\text{NaF}, \text{NaCl}, \text{NaBr}, \text{NaI}$

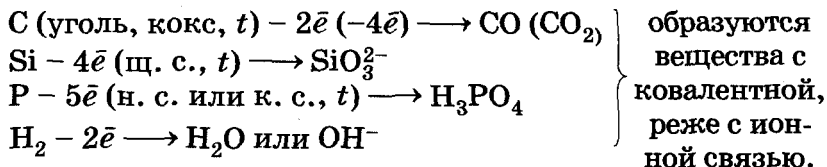
Аналогично — бориды, селениды, теллуриды, арсениды, силициды.

Отрицательно заряженные ионы могут окисляться до нуля или положительной высшей степени окисления:



6. Восстановительные свойства некоторых неметаллов.

Неметаллы проявляют восстановительные свойства при нагревании или при очень высоких температурах:



7. Восстановительные свойства некоторых металлов:

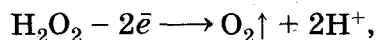
$\text{Zn} - 2\bar{e} \xrightarrow{\text{щ. с.}} \text{ZnO}_2^{2-}, (\text{Na}_2\text{ZnO}_2 \text{ или } \text{Na}_2[\text{Zn}(\text{OH})_4]); \text{при-}$
мер 1018;

$\text{Al} - 3\bar{e} \xrightarrow{\text{ш. с.}} [\text{Al}(\text{OH})_4]^-$, $\text{Na}[\text{Al}(\text{OH})_4]$; пример 1024;

$\text{Pb} - 2\bar{e} \xrightarrow{\text{ш. с.}} [\text{Pb}(\text{OH})_3]^{4-}$, $\text{K}[\text{Pb}(\text{OH})_3]$; пример 1016;

$\text{K} - 1\bar{e} \longrightarrow \text{K}^+$; пример 216 и др.

8. Восстановительные свойства пероксида водорода.



$\text{H}_2\text{O}_2 + 2\text{OH}^- - 2\bar{e} \longrightarrow \text{O}_2\uparrow + 2\text{H}_2\text{O}$ (восстановительный распад).

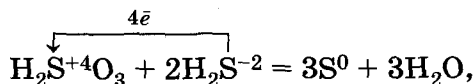
9. В органических соединениях в ряду:

алкан < алкен < алкин < спирт < альдегид <
< карбоновая кислота < оксид углерода (II)

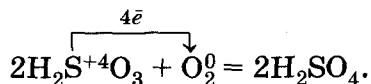
повышается с. о. углерода, уменьшается способность к окислению.

Окислители-восстановители

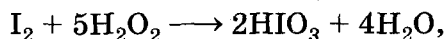
Некоторые вещества, которые в составе имеют элементы с промежуточной степенью окисления, могут проявлять как окислительные, так и восстановительные свойства. Например, в реакции, уравнение которой



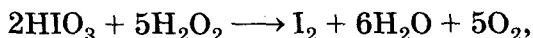
H_2SO_3 — окислитель, так как сероводород, где с. о. серы -2 , может быть только восстановителем. При взаимодействии сернистой кислоты с кислородом H_2SO_3 — восстановитель, так как кислород сильный окислитель:



Среда, в которой идет химическая реакция, также влияет на окислительно-восстановительные свойства веществ. Например, в реакции, уравнение которой

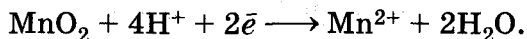
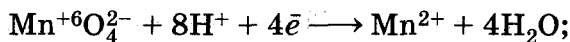


при $\text{pH} = 1,0$ пероксид водорода — окислитель, а в реакции, уравнение которой:

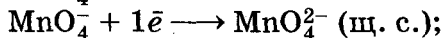
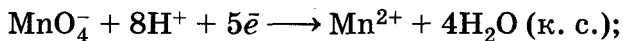


при $\text{pH} = 2,0$ пероксид водорода — восстановитель.

Манганат калия K_2MnO_4 и оксид марганца (IV) MnO_2 проявляют окислительные свойства только в кислотной среде:



Перманганат калия $KMnO_4$ проявляет окислительные свойства в любой среде:



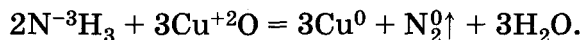
$MnO_4^- + 2H_2O + 3e^- \longrightarrow MnO_2 + 4OH^-$ (н. с., слабокислая и слабощелочная среда).

Сила окислителей увеличивается в кислотной среде, а восстановителей — в щелочной среде.

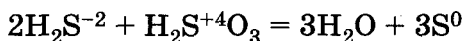
Классификация окислительно-восстановительных реакции

Различают три типа окислительно-восстановительных реакций: межмолекулярные, внутримолекулярные и реакции диспропорционирования (внутримолекулярное самоокисление — самовосстановление), или дисмутации.

К межмолекулярным ОВР относятся те, в которых окислитель и восстановитель находятся в разных веществах:

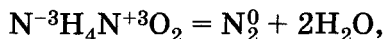


К этому типу относят и реакции конмутации:



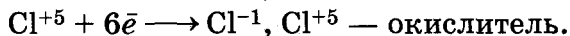
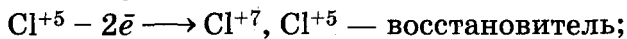
(у образовавшейся серы степень окисления стала промежуточной).

К внутримолекулярным ОВР относятся те, в которых окислитель и восстановитель находятся в одном и том же веществе:



К реакциям диспропорционирования (дисмутации) относят те ОВР, где в одном и том же веществе содержит-

ся элемент с промежуточной степенью окисления, исполняющий роль и окислителя, и восстановителя одновременно: $4\text{KCl}^{+5}\text{O}_3 = 3\text{KCl}^{+7}\text{O}_4 + \text{KCl}^{-1}$



Часть хлора со степенью окисления +5 окисляется, другая восстанавливается.

Сильные и слабые электролиты

Известно, что сильные электролиты — кислоты, щелочи, соли (см. ниже) — в водных растворах полностью диссоциируют на ионы, поэтому в ионных уравнениях, а также в полуреакциях их записывают в виде ионов.

Примеры сильных электролитов:

- кислоты: HClO_4 , HClO_3 , HNO_3 , H_2SO_4 , H_2SeO_4 , HCl , HBr , HI , HBrO_3 , HBrO_4 , HIO_3 , HIO_4 , HNCS , HMnO_4 , $\text{H}_2\text{Cr}_2\text{O}_7$, H_3PO_2 ;

- щелочи: LiOH , NaOH , KOH , RbOH , $\text{Sr}(\text{OH})_2$, $\text{Ba}(\text{OH})_2$, $\text{Ca}(\text{OH})_2$;

- соли — почти все.

Слабые электролиты в водных растворах не полностью диссоциируют на ионы. В ионных уравнениях, а также в полуреакциях формулы слабых электролитов, оксидов, веществ в твердом и газообразном состоянии записываются в молекулярном виде.

Примеры слабых электролитов:

- кислоты: HClO_2 , HClO , HNO_2 , H_2SO_3 , H_2CO_3 , H_2SiO_3 , H_3PO_4 , H_3PO_3 , HPO_2 , HBO_3 , HIO_6 , HIO , H_2S , HCN , HF , HBrO , CH_3COOH и почти все органические кислоты;

- основания: $\text{Fe}(\text{OH})_2$, $\text{Fe}(\text{OH})_3$, $\text{Cu}(\text{OH})_2$, NH_4OH , $\text{Al}(\text{OH})_3$, $\text{Zn}(\text{OH})_2$, $\text{Cr}(\text{OH})_3$, все амфотерные гидроксиды;

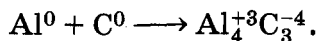
- вода H_2O .

Метод электронного баланса

Правила оформления уравнений ОВР методом электронного баланса

ПРИМЕР 1. Расставьте коэффициенты в реакции, схема которой $\text{Al} + \text{C} \longrightarrow \text{Al}_4\text{C}_3$, методом электронного баланса.

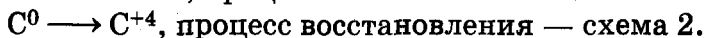
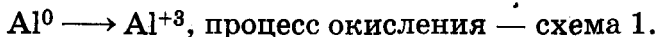
1. Определяют элементы, изменяющие степень окисления:



1-й вариант оформления ЭБ.

Расчет ведут на один атом алюминия и один атом углерода.

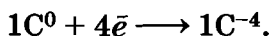
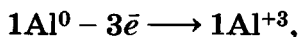
2. Составляют схемы окисления и восстановления.



3. Уравнивают заряды.

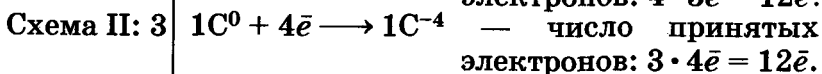
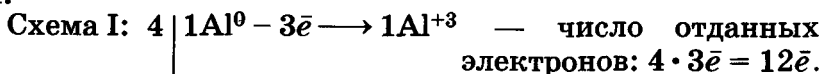
В левой части схемы I заряд равен 0, в правой — +3, следовательно, из левой части отнимают $3\bar{e}$ ($0 - (-3) = +3$).

В схеме II в левой части заряд равен 0, в правой — (-4), поэтому в левую часть схемы добавляют $4\bar{e}$ ($0 + (-4) = -4$).



(Коэффициент 1 перед Al и C поставлен для наглядности, обычно его не записывают.)

4. Количество отданных и принятых электронов всегда одинаково, поэтому, чтобы сбалансировать, уравнять их, числа 3 и 4, называемые множителями, выносят за вертикальную черту, предварительно поменяв их местами:



5. Определяют стехиометрические коэффициенты, которые будут стоять перед формулами веществ в молекулярном уравнении.

Подсчитывают число атомов Al^0 , для этого множитель 4 умножают на коэффициент, который стоит перед Al^0 , т. е. на единицу: $4 \cdot 1Al^0 = 4Al^0$. Эта запись показывает, что в реакцию вступают 4 атома алюминия.

Теперь множитель 4 умножают на коэффициент, стоящий перед Al^{+3} , т. е. на единицу: $4 \cdot 1Al^{+3} = 4Al^{+3}$. Эта запись показывает, что в результате реакции образуется $4Al^{+3}$ в составе $Al_4^{+3}C_3^{-4}$.

Такие же действия производятся и в схеме II: $3 \cdot 1C^0 = 3C^0$; $3 \cdot 1C^{-4} = 3C^{-4}$ в составе $Al_4^{+3}C_3^{-4}$.

Итоговое молекулярное уравнение: $4Al^0 + 3C^0 = Al_4^{+3}C_3^{-4}$.

2-й вариант оформления ЭБ.

Здесь расчет ведут с учетом того, что в карбиде алюминия Al_4C_3 4 атома Al и 3 атома углерода: $Al^0 \longrightarrow 4Al^{+3}$; $C^0 \longrightarrow 3C^{-4}$.

В этом случае следует уравнивать число атомов алюминия и углерода в схемах ЭБ, а потом и заряды:

1	12	$4Al^0 - 12\bar{e} \longrightarrow 4Al^{+3}$	— число отданных электронов $1 \cdot 12\bar{e} = 12\bar{e}$.
1	12	$3C^0 + 12\bar{e} \longrightarrow 3C^{-4}$	— число принятых электронов $1 \cdot 12\bar{e} = 12\bar{e}$.

Множители сократились на 12. Чтобы найти коэффициент, который будет стоять в молекулярном уравнении перед Al^0 , умножают множитель 1 на коэффициент, стоящий перед Al^0 в схеме I, т. е. на четыре: $1 \cdot 4Al^0 = 4Al^0$, после стрелки: $1 \cdot 4Al^{+3} = 4Al^{+3}$. Такие же действия производятся и в схеме II: $1 \cdot 3C^0 = 3C^0$; $1 \cdot 3C^{-4} = 3C^{-4}$; $4Al^0 + 3C^0 = Al_4^{+3}C_3^{-4}$.

ПРИМЕР 2. Расставьте коэффициенты в уравнении реакции, схема которой $HNO_3 + I_2 \longrightarrow HIO_3 + NO + H_2O$, методом электронного баланса.

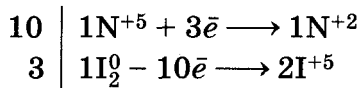
1. Определяют элементы, меняющие степень окисления:



2. Составляют схемы ЭБ: $N^{+5} \longrightarrow N^{+2}$; $I_2^0 \longrightarrow 2I^{+5}$.

Если молекула двухатомна, как I_2 , то в схеме ее записывают в молекулярном виде — I_2 .

3. Уравнивают заряды. В схеме I, в левой части заряд равен +5, в правой — +2, поэтому в левую часть добавляют $3\bar{e}$. В схеме II в левой части заряд 0, в правой — +10, поэтому из левой части отнимается $10\bar{e}$.



ЭБ показывает, что до реакции было $10 \cdot 1N^{+5} = 10N^{+5}$ в составе $10HNO_3$ и образовалось $10 \cdot 1N^{+2} = 10N^{+2}$ ($10NO$); вступили в реакцию $3 \cdot 2$ (два атома иода) = 6 атомов, или $3 \cdot 1I_2 = 3I_2$ (три молекулы иода). После реакции образовалось $3 \cdot 2I^{+5} = 6I^{+5}$ ($6HIO_3$).

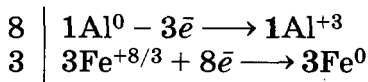
4. Расставляют найденные коэффициенты в уравнении:



ПРИМЕР 3. Расставьте коэффициенты в уравнении реакции, схема которой $Fe_3^{+8/3}O_4 + Al^0 \longrightarrow Al_2^{+3}O_3 + Fe^0$, методом электронного баланса.

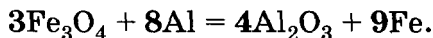
Как видно из примера, степень окисления у железа дробная.

1-й вариант оформления ЭБ.

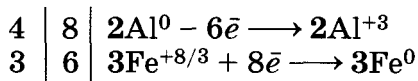


В левой части схемы II ЭБ заряд +8 ($3 \cdot (+8/3)$), справа 0, поэтому в левую часть добавляют $8\bar{e}$ ($+8 + (-8) = 0$). Теперь заряды справа и слева одинаковы. Коэффициенты: $8 \cdot 1Al^0 = 8Al^0$, после стрелки — $8 \cdot 1Al^{+3} = 8Al^{+3}$ в составе $4Al_2O_3$. В схеме II — $3 \cdot 3Fe^{+8/3} = 9Fe^{+8/3}$ в составе $3Fe_3O_4$; после стрелки — $3 \cdot 3Fe^0 = 9Fe^0$.

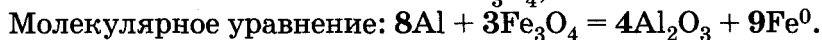
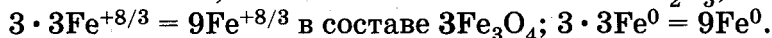
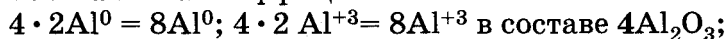
Коэффициенты в молекулярном уравнении:



2-й вариант оформления ЭБ.

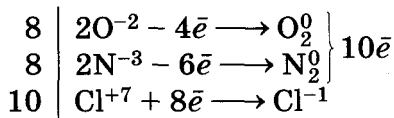


Расстановка коэффициентов:



ПРИМЕР 4. Расставьте коэффициенты в уравнении реакции, схема которой $\text{N}^{-3}\text{H}_4\text{Cl}^{+7}\text{O}_4 \xrightarrow{t} \text{N}_2^0 + \text{O}_2^0 + \text{HCl}^{-1} + \text{H}_2\text{O}$, методом электронного баланса.

При расстановке коэффициентов в уравнении необходимо учесть тот факт, что число атомов азота и хлора в NH_4ClO_4 одинаково. Это означает, что коэффициенты перед NH_4ClO_4 и перед HCl должны быть одинаковыми.

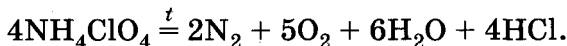


$8 \cdot 2\text{N}^{-3} = 16\text{N}^{-3}$ — в левой части уравнения, тогда в правой части перед HCl следует ставить коэффициент 16.

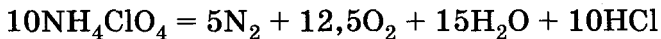
1-й вариант.



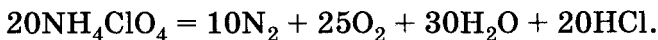
После сокращения коэффициентов:



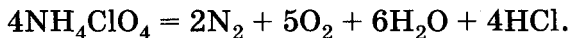
2-й вариант. В качестве коэффициента можно воспользоваться множителем 10:



и далее:

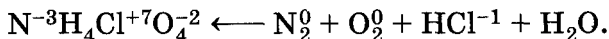


После сокращения коэффициентов:

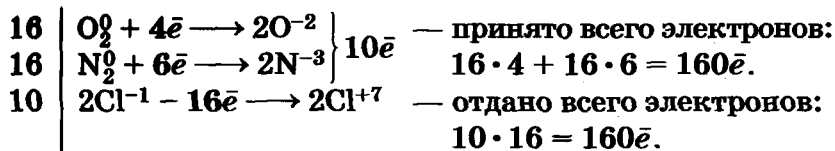


Кислород уравнивают в последнюю очередь, так как только часть его из NH_4ClO_4 переходит в O_2 , а другая — без изменения с. о. переходит в H_2O .

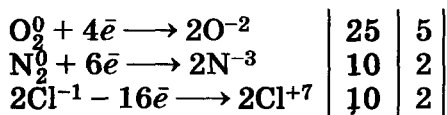
3-й вариант. Так как кислород из NH_4ClO_4 окисляется не полностью, коэффициенты можно подобрать методом ЭБ, считая, что реакция протекает справа налево:



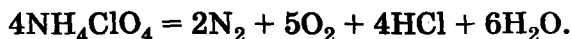
Схемы ЭВ:



С учетом равенства числа атомов азота и хлора подбираются коэффициенты:

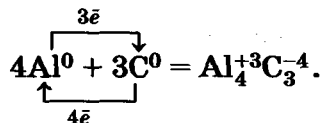


Множитель **25** подбирают таким образом, чтобы общая сумма принятых электронов равнялась $160 - (10 \cdot 6\bar{e} = 60\bar{e}; 160\bar{e} - 60\bar{e} = 100\bar{e}; 100\bar{e} : 4\bar{e} = 25)$. После сокращения всех множителей, коэффициенты переносятся соответственно в молекулярное уравнение:

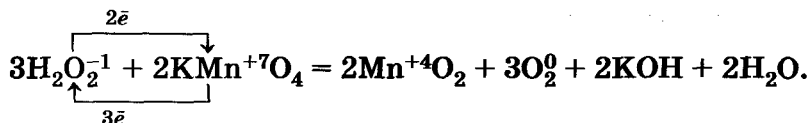


С приобретением навыков уравнивания уравнений методом ЭВ можно упростить оформление ЭВ и соответственно процесс расстановки коэффициентов. Помним, что в ЭВ числа, показывающие количества отданных и принятых электронов, становятся множителями, а потом коэффициентами. Множитель восстановителя становится коэффициентом для окислителя и в уравнении ставится перед ним, а множитель окислителя становится коэффициентом для восстановителя.

Например, чтобы уравнивать оперативно уравнение $\text{Al}^0 + \text{C}^0 \longrightarrow \text{Al}_4^{+3}\text{C}_3^{-4}$, в уме определяем количества отдаваемых и приобретаемых электронов, они равны соответственно 3 и 4, что можно оформить следующим образом:



Другой пример:



Схемы ОВР (условия задач)¹

Галогены и их соединения

- 1*. $\text{Cl}_2 + \text{O}_3 \longrightarrow \text{ClO}_3$
2. $\text{Cl}_2 + \text{H}_2 \longrightarrow \text{HCl}$
3. $\text{Cl}_2 + \text{F}_2 \longrightarrow \text{ClF}$
4. $\text{Cl}_2 + \text{KBr} \longrightarrow \text{Br}_2 + \text{KCl}$
5. $\text{Cl}_2 + \text{KI} + \text{H}_2\text{O} \longrightarrow \text{HCl} + \text{KIO}_3$
6. $\text{Cl}_2 + \text{AgClO}_3 \longrightarrow \text{AgCl} + \text{ClO}_2\uparrow + \text{O}_2\uparrow$
- 7*. $\text{Cl}_2 + \text{S} \longrightarrow \text{S}_2\text{Cl}_2$
8. $\text{Cl}_2 + \text{H}_2\text{S} + \text{H}_2\text{O} \longrightarrow \text{H}_2\text{SO}_4 + \text{HCl}$
- 9*. $\text{Cl}_2 + \text{SO}_2 \longrightarrow \text{SCl}_2\text{O}_2$
10. $\text{Cl}_2 + \text{SO}_2 + \text{H}_2\text{O} \longrightarrow \text{HCl} + \text{H}_2\text{SO}_4$
- 11*. $\text{Cl}_2 + \text{NO} \longrightarrow \text{NOCl}$
- 12*. $\text{Cl}_2 + \text{NH}_4\text{Cl} \longrightarrow \text{Cl}_3\text{N} + \text{HCl}$
- 13*. $\text{Cl}_2 + \text{NH}_3(\text{ж}) \longrightarrow \text{NH}_4\text{Cl} + \text{Cl}_3\text{N}$
- 14*. $\text{Cl}_2 + \text{NH}_3 \longrightarrow \text{N}_2\uparrow + \text{NH}_4\text{Cl}$
15. $\text{Cl}_2 + \text{AgNO}_3 \longrightarrow \text{AgCl} + \text{N}_2\text{O}_5\uparrow + \text{O}_2\uparrow$
16. $\text{Cl}_2 + \text{PH}_3 \longrightarrow \text{PCl}_3 + \text{HCl}$
17. $\text{Cl}_2 + \text{As} + \text{H}_2\text{O} \longrightarrow \text{H}_3\text{AsO}_4 + \text{HCl}$
- 18*. $\text{Cl}_2 + \text{CO} \longrightarrow \text{CCl}_2\text{O}$
19. $\text{Cl}_2 + \text{CaC}_2 \longrightarrow \text{CaCl}_2 + \text{CCl}_4$
20. $\text{Cl}_2 + \text{Al}_4\text{C}_3 \longrightarrow \text{AlCl}_3 + \text{CCl}_4$
21. $\text{Cl}_2 + \text{SiC} \longrightarrow \text{SiCl}_4 + \text{C}$
22. $\text{Cl}_2 + \text{KOH}(\text{хол.}) \longrightarrow \text{KClO} + \text{KCl} + \text{H}_2\text{O}$
23. $\text{Cl}_2 + \text{KOH}(\text{гор.}) \longrightarrow \text{KClO}_3 + \text{KCl} + \text{H}_2\text{O}$
24. $\text{Cl}_2 + \text{Ca}(\text{OH})_2 \longrightarrow \text{CaCl}_2 + \text{Ca}(\text{ClO})_2 + \text{H}_2\text{O}$
25. $\text{Cl}_2 + \text{H}_2\text{O}_2 \longrightarrow \text{HCl} + \text{O}_2\uparrow + \text{H}_2\text{O}$
26. $\text{Cl}_2 + \text{H}_2\text{O}(\text{пар}) + \text{C}(\text{кокс}) \longrightarrow \text{CO}_2 + \text{HCl}$
27. $\text{Cl}_2 + \text{H}_2\text{O} \longrightarrow \text{HCl} + \text{HClO}$
28. $\text{Cl}_2 + \text{H}_2\text{O} \longrightarrow \text{HCl} + \text{O}_2\uparrow(\text{на свету})$
29. $\text{Cl}_2 + \text{K}_2\text{MnO}_4 \longrightarrow \text{KMnO}_4 + \text{KCl}$
30. $\text{Cl}_2 + \text{K}_4[\text{Fe}(\text{CN})_6] \longrightarrow \text{K}_3[\text{Fe}(\text{CN})_6] + \text{KCl}$

¹ Решения задач, отмеченных звездочкой, содержат дополнительную информацию.

- 31*. $\text{Cl}_2 + \text{Fe} \longrightarrow \text{FeCl}_3$
32. $\text{Cl}_2 + \text{Br}_2 + \text{H}_2\text{O} \longrightarrow \text{HBrO}_3 + \text{HCl}$
33. $\text{Br}_2 + \text{HI} \longrightarrow \text{I}_2 + \text{HBr}$
34. $\text{Br}_2 + \text{KClO}_3 \longrightarrow \text{KBrO}_3 + \text{Cl}_2 \uparrow$
35. $\text{Br}_2 + \text{P} \longrightarrow \text{PBr}_5$
36. $\text{Br}_2 + \text{HgO} \longrightarrow \text{HgBr}_2 + \text{Br}_2\text{O}$
37. $\text{Br}_2 + \text{NaOH}(\text{разб.}) \longrightarrow \text{NaBrO} + \text{NaBr} + \text{H}_2\text{O}$
38. $\text{I}_2 + \text{Cl}_2 + \text{H}_2\text{O} \longrightarrow \text{HCl} + \text{HIO}_3$
39. $\text{I}_2 + \text{HClO}_3 \longrightarrow \text{HIO}_3 + \text{Cl}_2 \uparrow$
40. $\text{I}_2 + \text{HClO}_4 \longrightarrow \text{HIO}_4 + \text{Cl}_2 \uparrow$
41. $\text{I}_2 + \text{KBrO}_3 \longrightarrow \text{KIO}_3 + \text{Br}_2$
42. $\text{I}_2 + \text{H}_2\text{S} \longrightarrow \text{S} \downarrow + \text{HI}$
43. $\text{I}_2 + \text{SO}_2 + \text{H}_2\text{O} \longrightarrow \text{HI} + \text{H}_2\text{SO}_4$
44. $\text{I}_2 + \text{K}_2\text{SO}_3 + \text{KOH} \longrightarrow \text{K}_2\text{SO}_4 + \text{KI} + \text{H}_2\text{O}$
45. $\text{I}_2 + \text{HNO}_3(\text{разб.}) \longrightarrow \text{HIO}_3 + \text{NO} \uparrow + \text{H}_2\text{O}$
46. $\text{I}_2 + \text{P} \longrightarrow \text{PI}_3$
47. $\text{I}_2 + \text{P} + \text{H}_2\text{O} \longrightarrow \text{H}_3\text{PO}_3 + \text{HI}$
48. $\text{I}_2(\text{иодная вода}) + \text{H}_2\text{O} + \text{PH}_3 \longrightarrow \text{HI} + \text{H}(\text{PH}_2\text{O}_2)$
49. $\text{I}_2 + \text{NaAsO}_2 + \text{Na}_2\text{CO}_3 + \text{H}_2\text{O} \longrightarrow \text{NaH}_2\text{AsO}_4 + \text{NaI} + \text{CO}_2 \uparrow$
50. $\text{I}_2(\text{т}) + \text{H}_2\text{O}_2(30\%) \longrightarrow \text{HIO}_3 + \text{H}_2\text{O}$
51. $\text{I}_2 + \text{NaOH}(\text{гор.}) \longrightarrow \text{NaI} + \text{NaIO}_3 + \text{H}_2\text{O}$
52. $\text{I}_2 + \text{Al} \longrightarrow \text{AlI}_3$
53. $\text{F}_2 + \text{S} \longrightarrow \text{SF}_6$
- 54*. $\text{F}_2 + \text{N}_2 \longrightarrow \text{NF}_3$
55. $\text{F}_2 + \text{Si} \longrightarrow \text{SiF}_4$
56. $\text{F}_2 + \text{SiO}_2 \longrightarrow \text{SiF}_4 + \text{O}_2 \uparrow$
57. $\text{F}_2 + \text{NaOH} \longrightarrow \text{NaF} + \text{OF}_2 \uparrow + \text{H}_2\text{O}$
58. $\text{F}_2 + \text{H}_2\text{O} \longrightarrow \text{HF} + \text{O}_2 \uparrow$
59. $\text{HCl} + \text{O}_2 \longrightarrow \text{Cl}_2 \uparrow + \text{H}_2\text{O}$
- 60*. $\text{HCl} + \text{KMnO}_4 \longrightarrow \text{Cl}_2 \uparrow + \text{KCl} + \text{MnCl}_2 + \text{H}_2\text{O}$
61. $\text{HCl} + \text{Cr} + \text{O}_2(\text{на воздухе}) \longrightarrow \text{CrCl}_3 + \text{H}_2\text{O}$
62. $\text{HCl}(\text{конц.}) + \text{K}_2\text{MnO}_4 \longrightarrow \text{Cl}_2 + \text{KCl} + \text{MnCl}_2 + \text{H}_2\text{O}$
63. $\text{HCl}(\text{разб.}) + \text{Ca} \longrightarrow \text{CaCl}_2 + \text{H}_2 \uparrow$
64. $\text{HCl}(\text{разб.}) + \text{CaH}_2 \longrightarrow \text{CaCl}_2 + \text{H}_2 \uparrow$
65. $\text{FeCl}_2 + \text{H}_2 \longrightarrow \text{HCl} + \text{Fe}(\text{особо чистое Fe})$

- 66*. $\text{FeCl}_2 + \text{Cl}_2 + \text{H}_2\text{O} \longrightarrow \text{FeO}(\text{OH})\downarrow + \text{HCl}$
- 67*. $\text{FeCl}_3 + \text{O}_2(\text{возд.}) \longrightarrow \text{Fe}_2\text{O}_3 + \text{Cl}_2\uparrow$
68. $\text{FeCl}_3 + \text{H}_2 \longrightarrow \text{FeCl}_2 + \text{HCl}$
69. $\text{CaCl}_2 + \text{H}_2 \longrightarrow \text{CaH}_2 + \text{HCl}$
70. $\text{CuCl}_2 + \text{Al} \longrightarrow \text{AlCl}_3 + \text{Cu}$
71. $\text{NH}_4\text{Cl} + \text{KNO}_3 \longrightarrow \text{N}_2\text{O}\uparrow + \text{H}_2\text{O} + \text{KCl}$
72. $\text{NH}_4\text{Cl}(\text{конц.}) + \text{Mg} \longrightarrow \text{H}_2\uparrow + \text{MgCl}_2 + \text{NH}_3\uparrow$
73. $\text{NaClO} \xrightarrow{t} \text{NaClO}_3 + \text{NaCl}$
- 74*. $\text{KClO}_3 \longrightarrow \text{KCl} + \text{O}_2\uparrow$
75. $\text{KClO}_3 + \text{Al} \longrightarrow \text{Al}_2\text{O}_3 + \text{KCl}$
76. $\text{KClO}_3(\text{т}) + \text{HCl}(\text{конц.}) \longrightarrow \text{KCl} + \text{Cl}_2\uparrow + \text{H}_2\text{O}$
77. $\text{KClO}_3(\text{т}) + \text{H}_2\text{SO}_4(\text{конц.}) \longrightarrow \text{ClO}_2\uparrow + \text{KClO}_4 + \text{KHSO}_4 + \text{H}_2\text{O}$
78. $\text{KClO}_3 + \text{S} \longrightarrow \text{KCl} + \text{SO}_2\uparrow$
79. $\text{KClO}_3 + \text{P}(\text{красн.}) \longrightarrow \text{KCl} + \text{P}_2\text{O}_5$ (аналогично с HClO_3)
80. $\text{KClO}_3 + \text{CaH}_2 \longrightarrow \text{CaO} + \text{H}_2\text{O} + \text{KCl}$
81. $\text{HClO}_4 \longrightarrow \text{ClO}_2\uparrow + \text{O}_2\uparrow + \text{H}_2\text{O}$
82. $\text{KClO}_4 \longrightarrow \text{KCl} + \text{O}_2\uparrow$
83. $\text{NaClO}_2 \longrightarrow \text{NaCl} + \text{NaClO}_3$
84. $\text{Ca}(\text{ClO})_2 + \text{H}_2\text{O}_2 \longrightarrow \text{CaCl}_2 + \text{H}_2\text{O} + \text{O}_2\uparrow$
85. $\text{HClO} + \text{HI} \longrightarrow \text{I}_2\downarrow + \text{HCl} + \text{H}_2\text{O}$
86. $\text{Cl}_2\text{O} \longrightarrow \text{ClO}_2 + \text{Cl}_2\uparrow$
87. $\text{ClO}_2 \xrightarrow{t} \text{Cl}_2\uparrow + \text{O}_2\uparrow$ (взрыв)
88. $\text{ClO}_2 + \text{KOH} \longrightarrow \text{KClO}_2 + \text{KClO}_3 + \text{H}_2\text{O}$
- 89*. $\text{ClO}_2 + \text{O}_3 \longrightarrow \text{Cl}_2\text{O}_6 + \text{O}_2\uparrow$
90. $\text{ClO}_2 + \text{H}_2\text{O}_2 \longrightarrow \text{HClO}_2 + \text{O}_2\uparrow$
91. $\text{Cl}_2\text{O}_6 + \text{KOH} \longrightarrow \text{KClO}_3 + \text{KClO}_4 + \text{H}_2\text{O}$
92. $\text{Cl}_2\text{O}_7 \longrightarrow \text{Cl}_2 + \text{O}_2$
93. $\text{HI} + \text{O}_2 \longrightarrow \text{I}_2\downarrow + \text{H}_2\text{O}$
94. $\text{HI} + \text{H}_2\text{SO}_4 \longrightarrow \text{S}\downarrow + \text{I}_2\downarrow + \text{H}_2\text{O}$
95. $\text{HI} + \text{NO}_2 \longrightarrow \text{NO}\uparrow + \text{I}_2\downarrow + \text{H}_2\text{O}$
96. $\text{NaI} + \text{NaIO}_3 + \text{H}_2\text{SO}_4 \longrightarrow \text{Na}_2\text{SO}_4 + \text{I}_2\downarrow + \text{H}_2\text{O}$
97. $\text{HI} + \text{HNO}_3(60\%) \longrightarrow \text{HIO}_3 + \text{NO}_2\uparrow + \text{H}_2\text{O}$
98. $\text{HI} + \text{FeCl}_3 \longrightarrow \text{FeCl}_2 + \text{I}_2\downarrow + \text{HCl}$
99. $\text{KI}(\text{т}) + \text{H}_2\text{SO}_4(\text{конц.}) \longrightarrow \text{I}_2\downarrow + \text{H}_2\text{S}\uparrow + \text{K}_2\text{SO}_4 + \text{H}_2\text{O}$

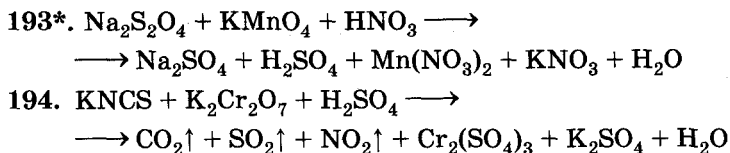
100. $\text{KI} + \text{K}_3[\text{Fe}(\text{CN})_6] \longrightarrow \text{I}_2\downarrow + \text{K}_4[\text{Fe}(\text{CN})_6]$
 101. $\text{KI} + \text{O}_3 + \text{H}_2\text{O} \longrightarrow \text{I}_2\downarrow + \text{O}_2\uparrow + \text{KOH}$
 102. $\text{I}_2\text{O}_5 \longrightarrow \text{I}_2 + \text{O}_2$
 103. $\text{I}_2\text{O}_5 + \text{CO} \longrightarrow \text{CO}_2\uparrow + \text{I}_2\downarrow$
 104. $\text{NaBrO}_3 + \text{F}_2 + \text{NaOH} \longrightarrow \text{NaF} + \text{NaBrO}_4 + \text{H}_2\text{O}$

Сера и ее соединения

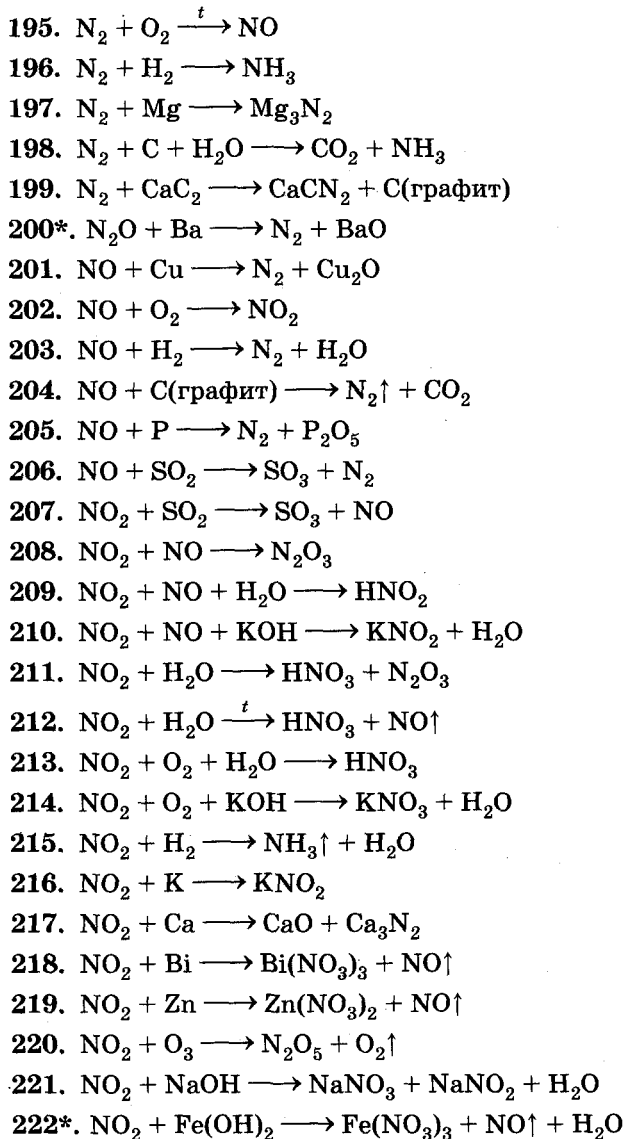
105. $\text{S} + \text{O}_2 \longrightarrow \text{SO}_2$
 106. $\text{S} + \text{O}_2 + \text{H}_2\text{O} \longrightarrow \text{H}_2\text{SO}_4$
 107. $\text{S} + \text{H}_2 \longrightarrow \text{H}_2\text{S}$
 108. $\text{S} + \text{Al} \longrightarrow \text{Al}_2\text{S}_3$
 109. $\text{S} + \text{Na} \longrightarrow \text{Na}_2\text{S}$
 110. $\text{S} + \text{Fe} \longrightarrow \text{FeS}$
 111. $\text{S} + \text{KClO}_3 \longrightarrow \text{KCl} + \text{SO}_2\uparrow$
 112*. $\text{S} + \text{H}_2\text{SO}_4(\text{конц.}) \xrightarrow{t} \text{SO}_2\uparrow + \text{H}_2\text{O}$
 113. $\text{S} + \text{NO}_2 \longrightarrow \text{N}_2\uparrow + \text{SO}_2\uparrow$
 114. $\text{S} + \text{HNO}_3(\text{конц.}) \xrightarrow{t} \text{H}_2\text{SO}_4 + \text{NO}_2\uparrow + \text{H}_2\text{O}$
 115. $\text{S} + \text{KNO}_3 \longrightarrow \text{KNO}_2 + \text{SO}_2\uparrow$
 116*. $\text{S} + \text{Cr}_2\text{O}_3 \longrightarrow \text{Cr}_2\text{S}_3 + \text{SO}_2\uparrow$
 117. $\text{S} + \text{K}_2\text{Cr}_2\text{O}_7 \longrightarrow \text{Cr}_2\text{O}_3 + \text{K}_2\text{SO}_4$
 118. $\text{S} + \text{KMnO}_4 \longrightarrow \text{K}_2\text{SO}_4 + \text{MnO}_2$
 119*. $\text{S} + \text{KO}_2 \longrightarrow \text{K}_2\text{SO}_4$
 120. $\text{S} + \text{Na}_2\text{O}_2 \longrightarrow \text{Na}_2\text{SO}_3 + \text{Na}_2\text{O}$
 121*. $\text{S} + \text{NaOH}(\text{конц.}) \xrightarrow{t} \text{Na}_2\text{SO}_3\text{S} + \text{Na}_2\text{S} + \text{H}_2\text{O}$
 122. $\text{SO}_2 + \text{HClO}_4 + \text{H}_2\text{O} \longrightarrow \text{H}_2\text{SO}_4 + \text{HCl}$
 123. $\text{SO}_2 + \text{KMnO}_4 + \text{KOH} \longrightarrow \text{K}_2\text{MnO}_4 + \text{K}_2\text{SO}_4 + \text{H}_2\text{O}$
 124. $\text{SO}_2 + \text{KMnO}_4 + \text{H}_2\text{O} \longrightarrow \text{MnSO}_4 + \text{K}_2\text{SO}_4 + \text{H}_2\text{SO}_4$
 125. $\text{SO}_2 + \text{SeO}_2 \longrightarrow \text{Se} + \text{SO}_3$
 126. $\text{SCl}_2\text{O} + \text{Br}_2 + \text{H}_2\text{O} \longrightarrow \text{H}_2\text{SO}_4 + \text{HCl} + \text{HBr}$
 127. а) $\text{H}_2\text{S} + \text{O}_2(\text{недост.}) \longrightarrow \text{S}\downarrow + \text{H}_2\text{O};$
 б) $\text{H}_2\text{S} + \text{O}_2(\text{изб.}) \longrightarrow \text{SO}_2\uparrow + \text{H}_2\text{O}$
 128. $\text{H}_2\text{S} + \text{O}_2 + \text{Fe} \longrightarrow \text{FeS} + \text{H}_2\text{O}$
 129. $\text{H}_2\text{S} + \text{O}_2 + \text{Ag} \longrightarrow \text{Ag}_2\text{S} + \text{H}_2\text{O}$
 130. $\text{H}_2\text{S} + \text{Zn} \longrightarrow \text{ZnS} + \text{H}_2\uparrow$

131. $\text{H}_2\text{S} + \text{SO}_2 \longrightarrow \text{S} + \text{H}_2\text{O}$
132. $\text{H}_2\text{S} + \text{H}_2\text{SO}_4(\text{конц.}) \longrightarrow \text{S}\downarrow + \text{SO}_2\uparrow + \text{H}_2\text{O}$
133. $\text{H}_2\text{S} + \text{H}_2\text{SO}_4(\text{конц.}) \longrightarrow \text{SO}_2\uparrow + \text{H}_2\text{O}$
- 134*. $\text{H}_2\text{S} + \text{Fe}_2\text{O}_3 \longrightarrow \text{FeS} + \text{H}_2\text{O} + \text{S}\downarrow$
135. $\text{H}_2\text{S} + \text{K}_2\text{Cr}_2\text{O}_7 + \text{H}_2\text{SO}_4 \longrightarrow \text{S}\downarrow + \text{Cr}_2(\text{SO}_4)_3 + \text{K}_2\text{SO}_4 + \text{H}_2\text{O}$
136. $\text{H}_2\text{S} + \text{KMnO}_4 \longrightarrow \text{MnO}_2 + \text{S}\downarrow + \text{KOH} + \text{H}_2\text{O}$
137. $\text{H}_2\text{S} + \text{FeCl}_3 \longrightarrow \text{FeCl}_2 + \text{FeS}_2 + \text{HCl}$
138. $\text{ZnS} + \text{O}_2 \longrightarrow \text{ZnSO}_4$
139. $\text{Ca}(\text{HS})_2 + \text{HNO}_3(\text{конц.}) \xrightarrow{t} \text{CaSO}_4\downarrow + \text{H}_2\text{SO}_4 + \text{NO}_2\uparrow + \text{H}_2\text{O}$
140. $\text{PbS} + \text{H}_2\text{O}_2 \longrightarrow \text{PbSO}_4 + \text{H}_2\text{O}$
141. $\text{MoS}_2 + \text{O}_2 \longrightarrow \text{MoO}_3 + \text{SO}_2\uparrow$
- 142*. $\text{Cu}_2\text{S} + \text{O}_2 + \text{CaCO}_3 \longrightarrow \text{CuO} + \text{CaSO}_4 + \text{CO}_2$
143. $\text{Cu}_2\text{S} + \text{O}_2 \xrightarrow{t} \text{CuO} + \text{SO}_2\uparrow$
144. $\text{Na}_2\text{S} + \text{H}_2\text{SO}_4(\text{конц.}) \longrightarrow \text{SO}_2\uparrow + \text{S}\downarrow + \text{H}_2\text{O} + \text{NaHSO}_4$
145. $\text{Na}_2\text{S} + \text{FeCl}_3 \longrightarrow \text{FeS}\downarrow + \text{S}\downarrow + \text{NaCl}$
146. $\text{Na}_2\text{S} + \text{H}_2\text{O}_2(\text{конц.}) \longrightarrow \text{Na}_2\text{SO}_4 + \text{H}_2\text{O}$
147. $\text{Na}_2\text{S} + \text{HNO}_3(\text{конц.}) \longrightarrow \text{NO}_2\uparrow + \text{S}\downarrow + \text{H}_2\text{O} + \text{NaNO}_3$
148. $\text{Al}_2\text{S}_3 + \text{HNO}_3(\text{конц.}) \longrightarrow \text{Al}_2(\text{SO}_4)_3 + \text{NO}_2\uparrow + \text{H}_2\text{O}$
149. $\text{Al}_2\text{S}_3 + \text{O}_2 \longrightarrow \text{Al}_2\text{O}_3 + \text{SO}_2\uparrow$
150. $\text{FeS} + \text{H}_2\text{SO}_4(\text{конц.}) + \text{HNO}_3(\text{конц.}) \longrightarrow$
 $\longrightarrow \text{Fe}_2(\text{SO}_4)_3 + \text{NO}_2\uparrow + \text{H}_2\text{O}$
- 151*. $\text{FeS}_2 + \text{H}_2\text{SO}_4(\text{конц.}) \xrightarrow{t} \text{Fe}_2(\text{SO}_4)_3 + \text{SO}_2\uparrow + \text{H}_2\text{O}$
152. $\text{FeS}_2 + \text{HNO}_3(\text{конц.}) \longrightarrow \text{Fe}(\text{NO}_3)_3 + \text{H}_2\text{SO}_4 + \text{NO}_2\uparrow + \text{H}_2\text{O}$
- 153*. $\text{FeS}_2 + \text{O}_2 \longrightarrow \text{SO}_2\uparrow + \text{Fe}_2\text{O}_3$ (обжиг)
154. $\text{FeS} + \text{O}_2 \xrightarrow{t} \text{Fe}_2\text{O}_3 + \text{SO}_2\uparrow$
155. $\text{H}_2\text{SO}_4(\text{конц.}) + \text{Cu} \longrightarrow \text{CuSO}_4 + \text{SO}_2\uparrow + \text{H}_2\text{O}$
156. $\text{H}_2\text{SO}_4(\text{конц.}) + \text{Zn} \longrightarrow \text{ZnSO}_4 + \text{H}_2\text{S}\uparrow + \text{H}_2\text{O}$
157. $\text{H}_2\text{SO}_4(\text{конц.}) + \text{Al} \longrightarrow \text{Al}_2(\text{SO}_4)_3 + \text{H}_2\text{S}\uparrow + \text{H}_2\text{O}$
158. $\text{H}_2\text{SO}_4(\text{разб.}) + \text{K} \longrightarrow \text{K}_2\text{SO}_4 + \text{S}\downarrow + \text{SO}_2\uparrow + \text{H}_2\text{O}$
159. $\text{H}_2\text{SO}_4 + \text{K}_2\text{FeO}_4 \longrightarrow \text{Fe}_2(\text{SO}_4)_3 + \text{O}_2\uparrow + \text{H}_2\text{O} + \text{K}_2\text{SO}_4$
160. а) $\text{H}_2\text{SO}_4(\text{конц.}) + \text{Fe} \xrightarrow{t} \text{Fe}_2(\text{SO}_4)_3 + \text{SO}_2\uparrow + \text{H}_2\text{O};$
 б) $\text{H}_2\text{SO}_4(\text{разб.}) + \text{Fe} \longrightarrow \text{FeSO}_4 + \text{H}_2\uparrow$
161. $\text{H}_2\text{SO}_4(\text{конц.}) + \text{C} \longrightarrow \text{H}_2\text{O} + \text{CO}_2\uparrow + \text{SO}_2\uparrow$

162. $\text{Na}_2\text{SO}_4 + \text{C(кокс)} \longrightarrow \text{Na}_2\text{S} + \text{CO}\uparrow$
163. $\text{Na}_2\text{SO}_4 + \text{C(кокс)} + \text{CaCO}_3 \longrightarrow \text{Na}_2\text{CO}_3 + \text{CaS} + \text{CO}_2\uparrow$
164. $\text{Na}_2\text{SO}_4 + \text{H}_2 \longrightarrow \text{Na}_2\text{S} + \text{H}_2\text{O}$
165. $\text{CaSO}_4 + \text{C(кокс)} \longrightarrow \text{CaS} + \text{CO}\uparrow + \text{CO}_2\uparrow$
166. $\text{Na}_2\text{SO}_4 + \text{Na}_2\text{S} + \text{HCl} \longrightarrow \text{NaCl} + \text{S}\downarrow + \text{H}_2\text{O}$
167. $\text{FeSO}_4 + \text{H}_2\text{O} + \text{O}_2(\text{возд.}) \longrightarrow \text{FeSO}_4(\text{OH})\downarrow$
168. $\text{FeSO}_4(\text{конц.}) + \text{CuSO}_4 \longrightarrow \text{Cu}\downarrow + \text{Fe}_2(\text{SO}_4)_3$
169. $\text{FeSO}_4 + \text{KClO}_3 + \text{KOH} \longrightarrow \text{K}_2\text{FeO}_4 + \text{K}_2\text{SO}_4 + \text{KCl} + \text{H}_2\text{O}$
170. $\text{FeSO}_4 + \text{KMnO}_4 + \text{H}_2\text{SO}_4 \longrightarrow \text{Fe}_2(\text{SO}_4)_3 + \text{MnSO}_4 + \text{H}_2\text{O} + \text{K}_2\text{SO}_4$
171. $\text{FeSO}_4 + \text{KClO}_3 + \text{H}_2\text{SO}_4 \longrightarrow \text{Fe}_2(\text{SO}_4)_3 + \text{KCl} + \text{H}_2\text{O}$
172. $\text{FeSO}_4 + \text{K}_2\text{Cr}_2\text{O}_7 + \text{H}_2\text{SO}_4 \longrightarrow$
 $\longrightarrow \text{Cr}_2(\text{SO}_4)_3 + \text{Fe}_2(\text{SO}_4)_3 + \text{K}_2\text{SO}_4 + \text{H}_2\text{O}$
173. $\text{FeSO}_4 + \text{O}_2 + \text{H}_2\text{SO}_4 \longrightarrow \text{Fe}_2(\text{SO}_4)_3 + \text{H}_2\text{O}$
174. $\text{CuSO}_4 \longrightarrow \text{CuO} + \text{SO}_2\uparrow + \text{O}_2\uparrow$
175. $\text{Al}_2(\text{SO}_4)_3 \longrightarrow \text{Al}_2\text{O}_3 + \text{SO}_2\uparrow + \text{O}_2\uparrow$
176. $\text{MnSO}_4 + \text{NaBiO}_3 + \text{HNO}_3 \longrightarrow$
 $\longrightarrow \text{HMnO}_4 + \text{Bi}(\text{NO}_3)_3 + \text{Na}_2\text{SO}_4 + \text{NaNO}_3 + \text{H}_2\text{O}$
177. $\text{Cr}_2(\text{SO}_4)_3 + \text{KMnO}_4 + \text{KOH} \longrightarrow \text{K}_2\text{CrO}_4 + \text{MnO}_2\downarrow + \text{K}_2\text{SO}_4 + \text{H}_2\text{O}$
178. $\text{Ti}_2(\text{SO}_4)_3 + \text{KClO}_3 + \text{H}_2\text{O} \longrightarrow \text{TiSO}_4\text{O} + \text{KCl} + \text{H}_2\text{SO}_4$
179. $\text{Na}_2\text{SO}_3 + \text{KIO}_3 + \text{H}_2\text{SO}_4 \longrightarrow \text{I}_2\downarrow + \text{Na}_2\text{SO}_4 + \text{K}_2\text{SO}_4 + \text{H}_2\text{O}$
180. $\text{Na}_2\text{SO}_3 + \text{K}_2\text{Cr}_2\text{O}_7 + \text{H}_2\text{O} \longrightarrow \text{Na}_2\text{SO}_4 + \text{KOH} + \text{Cr}(\text{OH})_3\downarrow$
181. $\text{Na}_2\text{SO}_3 \xrightarrow{t} \text{Na}_2\text{S} + \text{Na}_2\text{SO}_4$
182. $\text{K}_2\text{SO}_3 + \text{KMnO}_4 + \text{KOH(конц.)} \longrightarrow \text{K}_2\text{MnO}_4 + \text{K}_2\text{SO}_4 + \text{H}_2\text{O}$
183. $\text{K}_2\text{SO}_3 + \text{Ba}(\text{ClO})_2 \longrightarrow \text{BaSO}_4 + \text{K}_2\text{SO}_4 + \text{KCl}$
- 184*. $\text{Na}_2\text{SO}_3\text{S} + \text{O}_2(\text{возд.}) \longrightarrow \text{Na}_2\text{SO}_4 + \text{S}\downarrow$
185. $\text{Na}_2\text{SO}_3\text{S} + \text{H}_2\text{O}_2 + \text{NaOH} \longrightarrow \text{Na}_2\text{SO}_4 + \text{H}_2\text{O}$
186. $\text{K}_2\text{SO}_3\text{S} + \text{HCl} \longrightarrow \text{SO}_2\uparrow + \text{S}\downarrow + \text{KCl} + \text{H}_2\text{O}$
187. $\text{K}_2\text{SO}_3\text{S} + \text{KMnO}_4 + \text{KOH(конц.)} \longrightarrow \text{K}_2\text{MnO}_4 + \text{K}_2\text{SO}_4 + \text{H}_2\text{O}$
- 188*. $\text{K}_2\text{S}_2\text{O}_6(\text{O}_2) + \text{Fe}(\text{OH})_2 + \text{KOH} \longrightarrow \text{K}_2\text{SO}_4 + \text{FeO}(\text{OH})\downarrow + \text{H}_2\text{O}$
189. $\text{K}_2\text{S}_2\text{O}_6(\text{O}_2) + \text{H}_2\text{S} + \text{H}_2\text{O} \longrightarrow \text{H}_2\text{SO}_4 + \text{K}_2\text{SO}_4$
- 190*. $\text{K}_2\text{S}_2\text{O}_6(\text{O}_2) + \text{I}_2 + \text{KOH} \longrightarrow \text{K}_2\text{SO}_4 + \text{KIO}_3 + \text{H}_2\text{O}$
191. $\text{K}_2\text{S}_2\text{O}_6(\text{O}_2) + \text{K}_2\text{SO}_3\text{S} + \text{H}_2\text{O} \longrightarrow \text{K}_2\text{SO}_4 + \text{H}_2\text{SO}_4$
192. $(\text{NH}_4)_2\text{S}_2\text{O}_6(\text{O}_2) + \text{Ag} \longrightarrow \text{Ag}_2\text{SO}_4\downarrow + (\text{NH}_4)_2\text{SO}_4$



Азот и его соединения



- 223*. $\text{NO}_2 + \text{FeI}_2 \xrightarrow{t} \text{Fe}(\text{NO}_3)_3 + \text{I}_2 + \text{NO}\uparrow$
224. $\text{NH}_3 + \text{O}_2 \xrightarrow{t} \text{N}_2\uparrow + \text{H}_2\text{O}$ (без кат.)
- $\text{NH}_3 + \text{O}_2 \xrightarrow{t} \text{NO} + \text{H}_2\text{O}$ (кат. Pt)
225. $\text{NH}_3 + \text{F}_2 \longrightarrow \text{NH}_4\text{F} + \text{NF}_3$
226. $\text{NH}_3 + \text{Na} \longrightarrow \text{NaNH}_2 + \text{H}_2\uparrow$
- 227*. $\text{NH}_3 + \text{Ca} \longrightarrow \text{CaH}_2 + \text{Ca}_3\text{N}_2$
228. $\text{NH}_3 + \text{Al} \longrightarrow \text{AlN} + \text{H}_2\uparrow$
229. $\text{NH}_3 + \text{KO}_2 \longrightarrow \text{KOH} + \text{N}_2\uparrow + \text{H}_2\text{O}$
230. $\text{NH}_3 + \text{N}_2\text{O} \longrightarrow \text{N}_2\uparrow + \text{H}_2\text{O}$
231. $\text{NH}_3 + \text{SeO}_2 \longrightarrow \text{Se} + \text{N}_2\uparrow + \text{H}_2\text{O}$
232. $\text{NH}_3 + \text{CuO} \xrightarrow{t} \text{Cu} + \text{N}_2\uparrow + \text{H}_2\text{O}$
233. $\text{NH}_3 + \text{H}_2\text{O}_2 \longrightarrow \text{N}_2\uparrow + \text{H}_2\text{O}$
234. $\text{NH}_3 + \text{NaClO} \longrightarrow \text{N}_2\text{H}_4 + \text{NaCl} + \text{H}_2\text{O}$
235. $\text{NH}_3 + \text{KMnO}_4 \longrightarrow \text{N}_2\uparrow + \text{MnO}_2\downarrow + \text{KOH} + \text{H}_2\text{O}$
236. $\text{NH}_3 + \text{KMnO}_4 + \text{KOH} \longrightarrow \text{N}_2\uparrow + \text{K}_2\text{MnO}_4 + \text{H}_2\text{O}$
237. $\text{NH}_3 + \text{KMnO}_4 + \text{H}_2\text{SO}_4 \longrightarrow \text{N}_2\uparrow + \text{MnSO}_4 + \text{K}_2\text{SO}_4 + \text{H}_2\text{O}$
- 238*. $\text{HN}_3 + \text{I}_2 \longrightarrow \text{N}_2 + \text{HI}$
- 239*. $\text{NaN}_3 + \text{NaNO}_3 \longrightarrow \text{N}_2 + \text{Na}_2\text{O}$
240. $\text{NF}_3 + \text{H}_2 \longrightarrow \text{N}_2 + \text{HF}$
241. $\text{NH}_4\text{Cl} + \text{CuO} \longrightarrow \text{Cu} + \text{CuCl}_2 + \text{N}_2\uparrow + \text{H}_2\text{O}$
242. $\text{N}_2\text{H}_4 + \text{H}_2\text{O}_2 \longrightarrow \text{N}_2\uparrow + \text{H}_2\text{O}$
- 243*. $\text{NH}_2\text{OH} + \text{HNO}_3 \longrightarrow \text{N}_2\text{O}\uparrow + \text{NO}\uparrow + \text{H}_2\text{O}$
244. $\text{NH}_3 \cdot \text{H}_2\text{O} + \text{K}_2\text{FeO}_4 \longrightarrow \text{FeO}(\text{OH})\downarrow + \text{N}_2\uparrow + \text{KOH} + \text{H}_2\text{O}$
245. $\text{NH}_4\text{HS} + \text{HNO}_3(\text{разб.}) \longrightarrow \text{S}\downarrow + \text{NO}\uparrow + \text{NH}_4\text{NO}_3 + \text{H}_2\text{O}$
246. $\text{NH}_4\text{HS} + \text{HNO}_3(\text{конц.}) \longrightarrow \text{S}\downarrow + \text{NO}_2\uparrow + \text{NH}_4\text{NO}_3 + \text{H}_2\text{O}$
247. $\text{HNO}_3(\text{конц.}) + \text{B} \xrightarrow{t} \text{B}(\text{OH})_3 + \text{NO}_2\uparrow$
248. $\text{HNO}_3(\text{конц.}) + \text{C} \xrightarrow{t} \text{CO}_2\uparrow + \text{H}_2\text{O} + \text{NO}_2\uparrow$
249. $\text{HNO}_3 + \text{Se} + \text{H}_2\text{O} \longrightarrow \text{H}_2\text{SeO}_3 + \text{NO}\uparrow$
250. $\text{HNO}_3(\text{конц.}) + \text{P}(\text{красн.}) \longrightarrow \text{H}_3\text{PO}_4 + \text{H}_2\text{O} + \text{NO}_2\uparrow$
251. $\text{HNO}_3(\text{конц.}) + \text{I}_2 \longrightarrow \text{HIO}_3 + \text{NO}_2\uparrow + \text{H}_2\text{O}$
252. $\text{HNO}_3(\text{конц.}) + \text{Cu} \longrightarrow \text{Cu}(\text{NO}_3)_2 + \text{NO}_2\uparrow + \text{H}_2\text{O}$
253. $\text{HNO}_3(\text{разб.}) + \text{Cu} \longrightarrow \text{Cu}(\text{NO}_3)_2 + \text{NO}\uparrow + \text{H}_2\text{O}$
254. $\text{HNO}_3(\text{оч. разб.}) + \text{Al} \longrightarrow \text{NH}_4\text{NO}_3 + \text{Al}(\text{NO}_3)_3 + \text{H}_2\text{O}$

255. $\text{HNO}_3(\text{разб.}) + \text{Al} \longrightarrow \text{Al}(\text{NO}_3)_3 + \text{N}_2\text{O}\uparrow + \text{H}_2\text{O}$
256. $\text{HNO}_3(\text{конц.}) + \text{Al} \xrightarrow{t} \text{Al}(\text{NO}_3)_3 + \text{NO}_2\uparrow + \text{H}_2\text{O}$
257. $\text{HNO}_3(\text{разб.}) + \text{Fe} \longrightarrow \text{Fe}(\text{NO}_3)_3 + \text{NO}\uparrow + \text{H}_2\text{O}$
258. $\text{HNO}_3(\text{разб.}) + \text{Ca} \longrightarrow \text{N}_2\text{O}\uparrow + \text{Ca}(\text{NO}_3)_2 + \text{H}_2\text{O}$
259. $\text{HNO}_3(\text{оч. разб.}) + \text{Ca} \longrightarrow \text{NH}_4\text{NO}_3 + \text{Ca}(\text{NO}_3)_2 + \text{H}_2\text{O}$
260. $\text{HNO}_3 + \text{HCl}(\text{«царская водка»}) + \text{Pt} \longrightarrow \text{NO}\uparrow + \text{H}_2[\text{PtCl}_6] + \text{H}_2\text{O}$
261. $\text{HNO}_3(\text{конц.}) + \text{Te} \longrightarrow \text{TeO}_2 + \text{NO}_2\uparrow + \text{H}_2\text{O}$
262. $\text{HNO}_3(\text{конц.}) + \text{As} \longrightarrow \text{H}_3\text{AsO}_4 + \text{NO}_2\uparrow + \text{H}_2\text{O}$
263. $\text{HNO}_3(\text{конц.}) + \text{FeO} \longrightarrow \text{Fe}(\text{NO}_3)_3 + \text{NO}_2\uparrow + \text{H}_2\text{O}$
264. $\text{HNO}_3(\text{конц.}) + (\text{FeFe}_2)\text{O}_4 \longrightarrow \text{Fe}(\text{NO}_3)_3 + \text{NO}_2\uparrow + \text{H}_2\text{O}$
265. $\text{HNO}_3(\text{конц.}) + \text{Fe}_3\text{C} \longrightarrow \text{Fe}(\text{NO}_3)_3 + \text{CO}_2\uparrow + \text{NO}_2\uparrow + \text{H}_2\text{O}$
266. $\text{HNO}_3(\text{конц.}) + \text{Cu}_2\text{S} \longrightarrow \text{H}_2\text{SO}_4 + \text{Cu}(\text{NO}_3)_2 + \text{NO}_2\uparrow + \text{H}_2\text{O}$
267. $\text{HNO}_3(\text{конц.}) + \text{As}_2\text{S}_5 \longrightarrow \text{H}_3\text{AsO}_4 + \text{H}_2\text{SO}_4 + \text{NO}_2\uparrow + \text{H}_2\text{O}$
268. $\text{HNO}_3(\text{конц.}) + \text{H}_2\text{S} \longrightarrow \text{H}_2\text{SO}_4 + \text{NO}_2\uparrow + \text{H}_2\text{O}$
- 269*. $\text{HNO}_3(\text{дым.}) + \text{H}_2(\text{HPO}_3) \longrightarrow \text{H}_3\text{PO}_4 + \text{NO}\uparrow + \text{H}_2\text{O}$
270. $\text{HNO}_3(\text{на свету}) \longrightarrow \text{NO}_2\uparrow + \text{O}_2\uparrow + \text{H}_2\text{O}$
271. $\text{NaNO}_3 + \text{Cu} + \text{H}_2\text{SO}_4 \longrightarrow \text{CuSO}_4 + \text{Na}_2\text{SO}_4 + \text{NO}_2\uparrow + \text{H}_2\text{O}$
272. $\text{KNO}_3 + \text{Mg} + \text{H}_2\text{O} \longrightarrow \text{NH}_3\uparrow + \text{Mg}(\text{OH})_2\downarrow + \text{KOH}$
273. $\text{KNO}_3 + \text{Mg} + \text{H}_2\text{O} \longrightarrow \text{N}_2\text{H}_4 + \text{Mg}(\text{OH})_2\downarrow + \text{KOH}$
274. $\text{KNO}_3 + \text{Mg} + \text{H}_2\text{O} \longrightarrow \text{NH}_2\text{OH} + \text{Mg}(\text{OH})_2\downarrow + \text{KOH}$
275. $\text{KNO}_3 + \text{NH}_4\text{Cl} \xrightarrow{t} \text{H}_2\text{O} + \text{KCl} + \text{N}_2\text{O}\uparrow$
276. $\text{KNO}_3 + \text{Al} + \text{KOH} + \text{H}_2\text{O} \longrightarrow \text{K}[\text{Al}(\text{OH})_4] + \text{NH}_3\uparrow$
277. $\text{KNO}_3 + \text{Zn} + \text{KOH} + \text{H}_2\text{O} \longrightarrow \text{K}_2[\text{Zn}(\text{OH})_4] + \text{NH}_3\uparrow$
278. $\text{KNO}_3 \xrightarrow{t} \text{O}_2\uparrow + \text{KNO}_2$
279. $\text{Pb}(\text{NO}_3)_2 + \text{H}_2\text{O}_2 + \text{NaOH} \longrightarrow \text{PbO}_2\downarrow + \text{NaNO}_3 + \text{H}_2\text{O}$
280. $\text{Pb}(\text{NO}_3)_2 + \text{KClO} + \text{KOH} \xrightarrow{\text{сплав.}} \text{K}_2\text{PbO}_3 + \text{KNO}_3 + \text{H}_2\text{O} + \text{KCl}$
281. $\text{Mn}(\text{NO}_3)_2 + \text{Pb} + \text{HNO}_3(\text{конц.}) \longrightarrow$
 $\longrightarrow \text{HMnO}_4 + \text{Pb}(\text{NO}_3)_2 + \text{H}_2\text{O} + \text{NO}_2\uparrow$
282. $\text{Hg}(\text{NO}_3)_2 + \text{H}_2\text{O}_2 + \text{NaOH} \longrightarrow \text{NaNO}_3 + \text{O}_2\uparrow + \text{H}_2\text{O} + \text{Hg}\downarrow$
283. $\text{Fe}(\text{NO}_3)_2 + \text{Al} \xrightarrow{t} \text{Fe}_2\text{O}_3 + \text{Al}_2\text{O}_3 + \text{NO}_2\uparrow$
284. $\text{Fe}(\text{NO}_3)_3 \xrightarrow{t} \text{NO}_2\uparrow + \text{O}_2\uparrow + \text{Fe}_2\text{O}_3$
285. $\text{AgNO}_3 \xrightarrow{t} \text{NO}_2\uparrow + \text{O}_2\uparrow + \text{Ag}$
286. $\text{NH}_4\text{NO}_3 + \text{Ba} \xrightarrow{t} \text{BaO} + \text{Ba}_3\text{N}_2 + \text{BaH}_2$

287. $\text{NH}_4\text{NO}_3 \xrightarrow{t} \text{N}_2\text{O}\uparrow + \text{H}_2\text{O}$
 288. $\text{NH}_4\text{NO}_3 \xrightarrow{t} \text{N}_2\uparrow + \text{O}_2\uparrow + \text{H}_2\text{O}$
 289. $\text{Pb}(\text{NO}_3)_2 \xrightarrow{t} \text{PbO} + \text{NO}_2\uparrow + \text{O}_2\uparrow$
 290. $\text{HNO}_2 + \text{KMnO}_4 + \text{H}_2\text{SO}_4 \longrightarrow \text{HNO}_3 + \text{MnSO}_4 + \text{K}_2\text{SO}_4 + \text{H}_2\text{O}$
 291. $\text{HNO}_2 + \text{HMnO}_4 \longrightarrow \text{Mn}(\text{NO}_3)_2 + \text{HNO}_3 + \text{H}_2\text{O}$
 292. $\text{HNO}_2 \longrightarrow \text{NO}\uparrow + \text{NO}_2\uparrow + \text{H}_2\text{O}$
 293. $\text{HNO}_2 + \text{O}_2(\text{возд.}) \longrightarrow \text{HNO}_3$
 294. $\text{HNO}_2 \longrightarrow \text{HNO}_3 + \text{NO}\uparrow + \text{H}_2\text{O}$
 295. $\text{HNO}_2 + \text{Cl}_2 + \text{H}_2\text{O} \longrightarrow \text{HNO}_3 + \text{HCl}$
 296. $\text{HNO}_2 + \text{HI} \longrightarrow \text{I}_2\downarrow + \text{NO}\uparrow + \text{H}_2\text{O}$
 297. $\text{NaNO}_2 + \text{K}_2\text{Cr}_2\text{O}_7 + \text{H}_2\text{SO}_4 \longrightarrow$
 $\longrightarrow \text{K}_2\text{SO}_4 + \text{NaNO}_3 + \text{Cr}_2(\text{SO}_4)_3 + \text{H}_2\text{O}$
 298. $\text{NaNO}_2 + \text{FeCl}_2 + \text{HCl} \longrightarrow \text{FeCl}_3 + \text{NO}\uparrow + \text{NaCl} + \text{H}_2\text{O}$
 299. $\text{NaNO}_2 \longrightarrow \text{Na}_2\text{O} + \text{N}_2\uparrow + \text{O}_2\uparrow$
 300. $\text{NaNO}_2 + \text{NH}_4\text{Cl} \longrightarrow \text{N}_2\uparrow + \text{NaCl} + \text{H}_2\text{O}$

Фосфор и его соединения

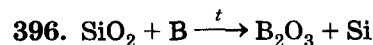
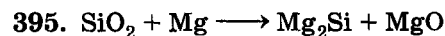
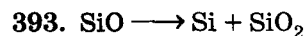
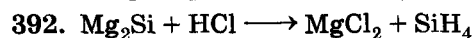
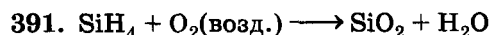
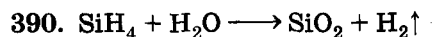
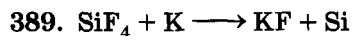
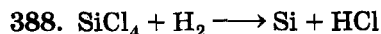
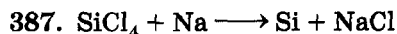
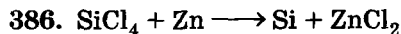
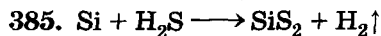
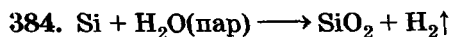
301. $\text{P} + \text{O}_2 \longrightarrow \text{P}_2\text{O}_5$
 302. $\text{P} + \text{O}_2 \longrightarrow \text{P}_2\text{O}_3$ (недостаток O_2)
 303. $\text{P} + \text{H}_2 \longrightarrow \text{PH}_3\uparrow$
 304. $\text{P} + \text{S} \longrightarrow \text{P}_2\text{S}_5$
 305*. $\text{P}(\text{бел.}) + \text{P}_2\text{O}_5 \longrightarrow \text{P}_2\text{O}_3$
 306. $\text{P}_4 + \text{H}_2\text{O} \longrightarrow \text{H}_3\text{PO}_4 + \text{H}_2\uparrow$
 307. $\text{P}(\text{бел.}) + \text{KOH} + \text{H}_2\text{O} \xrightarrow{t} \text{PH}_3\uparrow + \text{K}(\text{PH}_2\text{O}_2)$
 308. $\text{P} + \text{Cl}_2 \longrightarrow \text{PCl}_5$
 309. $\text{P} + \text{S} \longrightarrow \text{P}_2\text{S}_3$
 310. $\text{P} + \text{Ca} \longrightarrow \text{Ca}_3\text{P}_2$
 311. $\text{P}(\text{бел.}) + \text{H}_2\text{SO}_4(\text{конц.}) \longrightarrow \text{H}_2(\text{PHO}_3) + \text{SO}_2\uparrow$
 312. $\text{P}(\text{бел.}) + \text{Ba}(\text{OH})_2 + \text{H}_2\text{O} \longrightarrow \text{PH}_3\uparrow + \text{Ba}(\text{PH}_2\text{O}_2)_2$
 313. $\text{P} + \text{KMnO}_4 + \text{H}_2\text{O}(\text{гор.}) \longrightarrow \text{MnO}_2 + \text{K}_2\text{HPO}_4 + \text{KH}_2\text{PO}_4 + \text{KOH}$
 314. $\text{PCl}_3 + \text{Br}_2 + \text{H}_2\text{O} \longrightarrow \text{H}_3\text{PO}_4 + \text{HBr} + \text{HCl}$
 315. $\text{PCl}_3 + \text{O}_2 \longrightarrow \text{PCl}_3\text{O}$
 316. $\text{PCl}_5 + \text{P}(\text{бел.}) \longrightarrow \text{PCl}_3$
 317. $\text{PH}_3 + \text{O}_2 \longrightarrow \text{H}_3\text{PO}_4$

318. $P_2O_3 + O_2 \longrightarrow P_2O_5$
 319. $P_2O_3 + AgNO_3 + H_2O \longrightarrow Ag\downarrow + H_3PO_4 + HNO_3$
 320. $P_2O_3 + Cu(NO_3)_2 \xrightarrow{t} CuO + P_2O_5 + NO\uparrow$ (может образов. NO_2)
 321. $Na_3P + HNO_3(\text{конц.}) \longrightarrow NaH_2PO_4 + NO_2\uparrow + H_2O + NaNO_3$
 322. $Ca_3P_2 + Ca_3(PO_4)_2 \longrightarrow CaO + P$
 323. $Ca_3P_2 + KMnO_4 + H_2SO_4 \longrightarrow Ca_3(PO_4)_2 + MnSO_4 + K_2SO_4 + H_2O$
 324. $Ca_3P_2 + KMnO_4 + H_2O \longrightarrow Ca(OH)_2 + KOH + MnO_2\downarrow + P$
 325. $Ca_3P_2 + Cl_2 + H_2O \longrightarrow Ca_3(PO_4)_2 + HCl$
 326*. $H_3PO_4(\text{разб.}) + Fe \longrightarrow FeHPO_4\downarrow + Fe_3(PO_4)_2\downarrow + H_2\uparrow$
 327. $H_3PO_4(\text{разб.}) + Mg \longrightarrow Mg_3(PO_4)_2 + H_2\uparrow$
 328. $H_2(PHO_3) + AgNO_3 + H_2O \longrightarrow H_3PO_4 + Ag\downarrow + HNO_3$
 329*. $H(PH_2O_2) + HgCl_2 + H_2O \longrightarrow H_3PO_3 + Hg\downarrow + HCl$

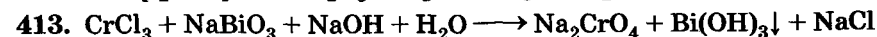
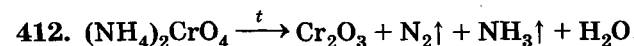
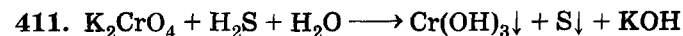
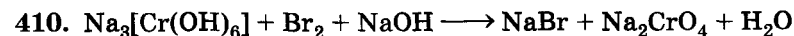
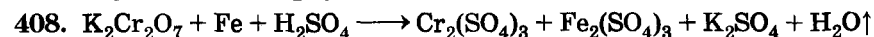
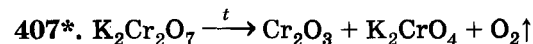
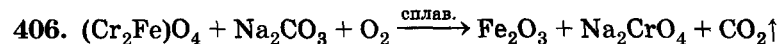
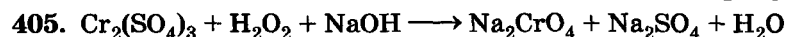
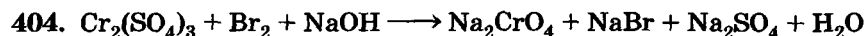
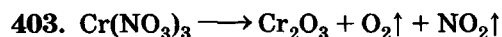
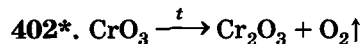
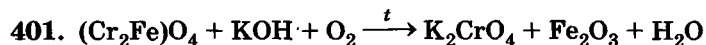
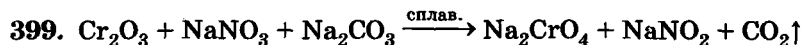
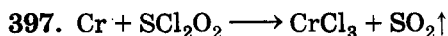
Углерод, кремний и их соединения

330. $C + F_2 \longrightarrow CF_4$
 331. $C + Al \longrightarrow Al_4C_3$
 332. $C + Ca \longrightarrow CaC_2$
 333. $C + Al_2O_3 \longrightarrow Al_4C_3 + CO\uparrow$
 334. $C + H_2O \longrightarrow CO\uparrow + H_2\uparrow$
 335. $C + SO_2 \xrightarrow{t} S + CO_2\uparrow$
 336. $C + NO \longrightarrow CO\uparrow + N_2\uparrow$
 337. $C + NO_2 \longrightarrow CO_2\uparrow + N_2\uparrow$
 338. $C + CaO + H_2O \longrightarrow CaCO_3 + H_2\uparrow$
 339. $C + CaO \xrightarrow{t} CaC_2 + CO\uparrow$
 340. $C + CuO \longrightarrow Cu + CO\uparrow$
 341. $C + P_2O_5 \longrightarrow P + CO\uparrow$
 342. $C + PbO \longrightarrow Pb + CO_2\uparrow$
 343. $C + SiO_2 \longrightarrow Si + CO\uparrow$
 344. $C + SiO_2 \longrightarrow SiC + CO\uparrow$
 345. $C + BaSO_4 \xrightarrow{t} BaS + CO\uparrow$
 346. $C + BaCO_3 \longrightarrow BaO + CO\uparrow$
 347. $C + (Cr_2Fe)O_4 \longrightarrow Fe + Cr + CO\uparrow$
 348. $C + K_2Cr_2O_7 \longrightarrow Cr_2O_3 + K_2CO_3 + CO_2\uparrow$
 349. $C + CaCO_3 + Na_2SO_4 \longrightarrow CO_2\uparrow + Na_2CO_3 + CaS$

350. $C + KNO_3 \longrightarrow K_2CO_3 + CO_2\uparrow + N_2\uparrow$
351. $C + Ca_3(PO_4)_2 + SiO_2 \longrightarrow CaSiO_3 + P + CO\uparrow$
352. $C + Ca_3(PO_4)_2 \longrightarrow CO\uparrow + Ca_3P_2$
353. $C + CaSO_4 \longrightarrow SO_2 + CaO + CO_2\uparrow$
354. $C + Na_2SO_4 \xrightarrow{t} Na_2S + CO_2\uparrow$
355. $C + S \longrightarrow CS_2$
356. $C + S + KNO_3 \longrightarrow N_2 + CO_2\uparrow + K_2S$
357. $C + Cl_2 + Cr_2O_3 \longrightarrow CrCl_3 + CO\uparrow$
358. $C + N_2 + BaCO_3 \longrightarrow Ba(CN)_2 + CO\uparrow$
359. $C + K_2CO_3 + NH_3 \longrightarrow KCN + H_2O$
360. $CO + (FeFe_2)O_4 \longrightarrow Fe + CO_2\uparrow$
361. $CO + O_2 \longrightarrow CO_2\uparrow$
362. $CO + I_2O_5 \longrightarrow CO_2\uparrow + I_2$
363. $CO + CaO + H_2O \xrightarrow{t} CaCO_3 + H_2\uparrow$
364. $CO + PdCl_2 + H_2O \longrightarrow CO_2\uparrow + HCl + Pd\downarrow$
365. $CO + H_2 \longrightarrow CH_3OH$
366. $CO + H_2 \longrightarrow CH_4\uparrow + H_2O$
367. $CO + Na_2O_2 \longrightarrow Na_2CO_3$
368. $CO_2 + H_2 \longrightarrow CH_4\uparrow + H_2O$
369. $CO_2 + C(\text{кокс}) \xrightarrow{t} CO$
370. $CO_2 + Mg \longrightarrow C + MgO$
371. $CO_2 + Ca \longrightarrow CaC_2 + CaO$
372. $HCN + O_2 \longrightarrow CO_2 + H_2O + N_2\uparrow$
373. $BaCO_3 + Al \longrightarrow BaO + C + Al_2O_3$
374. $BaCO_3 + Al \longrightarrow BaO + Al_4C_3 + Al_2O_3$
375. $Ag_2CO_3 \longrightarrow Ag + CO_2\uparrow + O_2\uparrow$
376. $Si + N_2 \longrightarrow Si_3N_4$
377. $Si + O_2 \longrightarrow SiO_2$
378. $Si + MoO_3 \longrightarrow SiO_2 + Mo$
379. $Si + Mg \longrightarrow Mg_2Si$
380. $Si + NH_3 \longrightarrow Si_3N_4 + H_2\uparrow$
381. $Si + NaOH(\text{конц.}) \longrightarrow Na_4SiO_4 + H_2\uparrow$
382. $Si + NaOH + H_2O \longrightarrow Na_2SiO_3 + H_2\uparrow$



Хром. Соединения хрома и марганца



414. $\text{MnO}_2 + \text{H}_2\text{SO}_4 \longrightarrow \text{MnSO}_4 + \text{O}_2\uparrow + \text{H}_2\text{O}$
 415. $(\text{Mn}^{\text{II}}\text{Mn}_2^{\text{III}})\text{O}_4 + \text{Al} \longrightarrow \text{Al}_2\text{O}_3 + \text{Mn}$
 416. $\text{Mn}(\text{OH})_2 + \text{O}_2 \xrightarrow{t} \text{MnO}_2 + \text{H}_2\text{O}$
 417. $\text{Mn}(\text{OH})_2 + \text{H}_2\text{O}_2(\text{конц.}) \longrightarrow \text{MnO}_2 + \text{H}_2\text{O}$
 418. $\text{MnSO}_4 + \text{PbO}_2 + \text{HNO}_3 \longrightarrow$
 $\longrightarrow \text{HMnO}_4 + \text{Pb}(\text{HSO}_4)_2 + \text{Pb}(\text{NO}_3)_2 + \text{H}_2\text{O}$
 419. $\text{MnSO}_4 + \text{Br}_2 + \text{NaOH} \longrightarrow \text{MnO}_2\downarrow + \text{NaBr} + \text{Na}_2\text{SO}_4 + \text{H}_2\text{O}$
 420*. $\text{K}_2\text{MnO}_4 + \text{H}_2\text{O} \longrightarrow \text{MnO}_2\downarrow + \text{KMnO}_4 + \text{KOH}$
 421. $\text{HMnO}_4 \longrightarrow \text{MnO}_2\downarrow + \text{O}_2\uparrow + \text{H}_2\text{O}$
 422. $\text{KMnO}_4 + \text{MnSO}_4 + \text{H}_2\text{O} \longrightarrow \text{MnO}_2\downarrow + \text{H}_2\text{SO}_4 + \text{K}_2\text{SO}_4$
 423. $\text{KMnO}_4 + \text{KOH} \longrightarrow \text{K}_2\text{MnO}_4 + \text{O}_2\uparrow + \text{H}_2\text{O}$
 424. $\text{KMnO}_4 + \text{HCl} \longrightarrow \text{Cl}_2\uparrow + \text{KCl} + \text{MnCl}_2 + \text{H}_2\text{O}$
 425. $\text{KMnO}_4 + \text{K}_2\text{SO}_3 + \text{H}_2\text{SO}_4 \longrightarrow \text{MnSO}_4 + \text{K}_2\text{SO}_4 + \text{H}_2\text{O}$
 426. $\text{KMnO}_4 + \text{K}_2\text{SO}_3 + \text{H}_2\text{O} \longrightarrow \text{MnO}_2\downarrow + \text{K}_2\text{SO}_4 + \text{KOH}$
 427. $\text{KMnO}_4 + \text{K}_2\text{SO}_3 + \text{KOH} \longrightarrow \text{K}_2\text{MnO}_4 + \text{K}_2\text{SO}_4 + \text{H}_2\text{O}$
 428. $\text{KMnO}_4 + \text{H}_2\text{O}_2 + \text{H}_2\text{SO}_4 \longrightarrow \text{MnSO}_4 + \text{O}_2\uparrow + \text{K}_2\text{SO}_4 + \text{H}_2\text{O}$
 429. $\text{KMnO}_4 + \text{PBr}_3 \longrightarrow \text{PBr}_3\text{O} + \text{K}_2\text{MnO}_4 + \text{MnO}_2$
 430*. $\text{KMnO}_4 + \text{PBr}_3 + \text{H}_2\text{O} \longrightarrow \text{Mn}(\text{H}_2\text{PO}_4)_2 + \text{Br}_2 + \text{MnBr}_2\downarrow + \text{KBr}$
 431. $\text{KMnO}_4 \xrightarrow{t} \text{K}_2\text{MnO}_4 + \text{MnO}_2 + \text{O}_2\uparrow$

Пероксиды, надпероксиды и озониды

432. $\text{H}_2\text{O}_2 \longrightarrow \text{H}_2\text{O} + \text{O}_2\uparrow$
 433. $\text{H}_2\text{O}_2 + \text{Fe}(\text{OH})_2 \longrightarrow \text{FeO}(\text{OH})\downarrow + \text{H}_2\text{O}$
 434. $\text{H}_2\text{O}_2 + \text{KClO}_3 \longrightarrow \text{KCl} + \text{H}_2\text{O} + \text{O}_2\uparrow$
 435. $\text{H}_2\text{O}_2 + \text{KMnO}_4 \longrightarrow \text{MnO}_2\downarrow + \text{O}_2\uparrow + \text{KOH} + \text{H}_2\text{O}$
 436. $\text{H}_2\text{O}_2 + \text{KMnO}_4 + \text{H}_2\text{SO}_4 \longrightarrow \text{K}_2\text{SO}_4 + \text{MnSO}_4 + \text{O}_2\uparrow + \text{H}_2\text{O}$
 437. $\text{H}_2\text{O}_2(\text{конц.}) + \text{Cl}_2(\text{насыщ.}) \longrightarrow \text{HCl} + \text{O}_2\uparrow$
 438. $\text{H}_2\text{O}_2 + \text{KNO}_2 \longrightarrow \text{KNO}_3 + \text{H}_2\text{O}$
 439. $\text{H}_2\text{O}_2 + \text{KI} \longrightarrow \text{I}_2 + \text{KOH}$
 440. $\text{H}_2\text{O}_2 + \text{Ag}_2\text{O} \longrightarrow \text{Ag}\downarrow + \text{O}_2\uparrow + \text{H}_2\text{O}$
 441. $\text{H}_2\text{O}_2 + \text{AuCl}_3 \longrightarrow \text{Au}\downarrow + \text{O}_2\uparrow + \text{HCl}$
 442. $\text{Na}_2\text{O}_2 \xrightarrow{t} \text{Na}_2\text{O} + \text{O}_2\uparrow$
 443. $\text{Na}_2\text{O}_2 + \text{H}_2\text{O} \xrightarrow{t} \text{NaOH} + \text{O}_2\uparrow$
 444. $\text{Na}_2\text{O}_2 + \text{HCl}(\text{разб.}) \xrightarrow{t} \text{NaCl} + \text{H}_2\text{O} + \text{O}_2\uparrow$

445. $\text{Na}_2\text{O}_2 + \text{Na}^0 \longrightarrow \text{Na}_2\text{O}$
446. $\text{Na}_2\text{O}_2 + \text{Al}(\text{порошок}) \xrightarrow{t} \text{NaAlO}_2 + \text{Na}_2\text{O}$
447. $\text{Na}_2\text{O}_2 + \text{S} \longrightarrow \text{Na}_2\text{SO}_3 + \text{Na}_2\text{O}$
448. $\text{Na}_2\text{O}_2 + \text{C} \longrightarrow \text{Na}_2\text{CO}_3 + \text{Na}_2\text{O}$
449. $\text{Na}_2\text{O}_2 + \text{CO} \longrightarrow \text{Na}_2\text{CO}_3$
450. $\text{Na}_2\text{O}_2 + \text{CO}_2 \longrightarrow \text{O}_2\uparrow + \text{Na}_2\text{CO}_3$
451. $\text{Na}_2\text{O}_2 + \text{KO}_2 + \text{CO}_2 \longrightarrow \text{Na}_2\text{CO}_3 + \text{K}_2\text{CO}_3 + \text{O}_2\uparrow$
452. $\text{Na}_2\text{O}_2 + \text{FeS}_2 \longrightarrow \text{Na}_2\text{SO}_4 + \text{Na}_2\text{O} + \text{Fe}_2\text{O}_3$
453. $\text{K}_2\text{O}_2 + \text{H}_2\text{SO}_4 + \text{KI} \longrightarrow \text{I}_2\downarrow + \text{K}_2\text{SO}_4 + \text{H}_2\text{O}$
454. $\text{KO}_2 + \text{HCl} \longrightarrow \text{H}_2\text{O}_2 + \text{O}_2\uparrow + \text{KCl}$
455. $\text{KO}_2 + \text{NO}_2 \longrightarrow \text{KNO}_3 + \text{O}_2\uparrow$
456. $\text{KO}_2 + \text{K} \longrightarrow \text{K}_2\text{O}$
457. $\text{KO}_2 + \text{CO}_2 \longrightarrow \text{K}_2\text{CO}_3 + \text{O}_2\uparrow$
458. $\text{KO}_2 + \text{H}_2\text{O} \longrightarrow \text{KOH} + \text{H}_2\text{O}_2 + \text{O}_2\uparrow$
459. $\text{KO}_3 + \text{H}_2\text{O} \longrightarrow \text{KOH} + \text{O}_2\uparrow$
460. $\text{CsO}_3 + \text{HCl} \longrightarrow \text{CsCl} + \text{Cl}_2\uparrow + \text{O}_2\uparrow + \text{H}_2\text{O}$

Гидриды и оксиды металлов

461. $\text{CaH}_2 + \text{O}_2 \longrightarrow \text{CaO} + \text{H}_2\text{O}$
462. $\text{CaH}_2 + \text{H}_2\text{O} \longrightarrow \text{Ca}(\text{OH})_2 + \text{H}_2\uparrow$
463. $\text{CaH}_2 + \text{N}_2 \longrightarrow \text{Ca}_3\text{N}_2 + \text{H}_2\uparrow$
464. $\text{CaH}_2 + \text{Al}_2\text{O}_3 \longrightarrow \text{CaO} + \text{H}_2\uparrow + \text{Al}$
465. $\text{KH}(\text{т}) + \text{Cl}_2 \longrightarrow \text{KCl} + \text{HCl}$
466. $\text{PbO}_2 + \text{H}_2\text{O}_2 \longrightarrow \text{Pb}(\text{OH})_2\downarrow + \text{O}_2\uparrow$
467. $(\text{Fe}^{\text{II}}\text{Fe}^{\text{III}})_2\text{O}_4 + \text{Al} \longrightarrow \text{Al}_2\text{O}_3 + \text{Fe}$
468. $\text{Fe}_2\text{O}_3 + \text{H}_2 \longrightarrow \text{Fe} + \text{H}_2\text{O}$
469. $(\text{Fe}^{\text{II}}\text{Fe}^{\text{III}})_2\text{O}_4 + \text{HNO}_3(\text{конц.}) \longrightarrow \text{Fe}(\text{NO}_3)_3 + \text{NO}_2\uparrow + \text{H}_2\text{O}$
470. $\text{Cu}_2\text{O} + \text{O}_2 \longrightarrow \text{CuO}$
471. $\text{Cu}_2\text{O} + \text{C} \longrightarrow \text{Cu} + \text{CO}_2\uparrow$
472. $\text{Cu}_2\text{O} + \text{CO} \longrightarrow \text{Cu} + \text{CO}_2\uparrow$
473. $\text{Cu}_2\text{O} + \text{HNO}_3(\text{конц.}) \longrightarrow \text{Cu}(\text{NO}_3)_2 + \text{NO}_2\uparrow + \text{H}_2\text{O}$
474. $\text{Cu}_2\text{O} + \text{H}_2\text{SO}_4(\text{конц.}) \longrightarrow \text{CuSO}_4 + \text{SO}_2\uparrow + \text{H}_2\text{O}$
475. $\text{Na}_2\text{O} \longrightarrow \text{Na}_2\text{O}_2 + \text{Na}$
476. $\text{Na}_2\text{O} + \text{O}_2 \longrightarrow \text{Na}_2\text{O}_2$

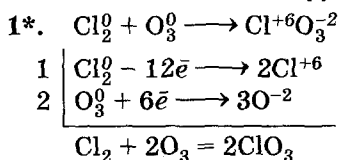
Органические соединения

477. $\text{C}_3\text{H}_8 + \text{O}_2 \longrightarrow \text{CO}_2\uparrow + \text{H}_2\text{O}\uparrow$
478. $\text{C}_7\text{H}_8 + \text{O}_2 \longrightarrow \text{CO}_2\uparrow + \text{H}_2\text{O}\uparrow$
479. $\text{C}_2\text{H}_6\text{O} + \text{O}_2 \longrightarrow \text{CO}_2\uparrow + \text{H}_2\text{O}\uparrow$
480. $\text{C}_2\text{H}_6 + \text{O}_2 \longrightarrow \text{CO}_2\uparrow + \text{H}_2\text{O}\uparrow$
481. $\text{C}_6\text{H}_6 + \text{O}_2 \longrightarrow \text{CO}_2\uparrow + \text{H}_2\text{O}\uparrow$
482. $\text{C}_4\text{H}_{10} + \text{O}_2 \xrightarrow{\text{кат.}} \text{CH}_3\text{COOH} + \text{H}_2\text{O}$
483. $\text{C}_{12}\text{H}_{22}\text{O}_{11} + \text{O}_2 \longrightarrow \text{CO}_2\uparrow + \text{H}_2\text{O}\uparrow$
484. $\text{C}_2\text{H}_2 + \text{O}_2 \longrightarrow \text{CO}_2\uparrow + \text{H}_2\text{O}\uparrow$
485. $\text{C}_3\text{H}_7\text{OH} + \text{O}_2 \longrightarrow \text{CO}_2\uparrow + \text{H}_2\text{O}\uparrow$
486. $\text{C}_3\text{H}_7\text{OH} + \text{CuO} \xrightarrow{t} \text{C}_2\text{H}_5\text{C(H)O} + \text{Cu} + \text{H}_2\text{O}$
487. $\text{C}_2\text{H}_2 + \text{KMnO}_4 \longrightarrow \text{MnO}_2\downarrow + \text{K}_2\text{C}_2\text{O}_4 + \text{KOH} + \text{H}_2\text{O}$
- 488*. $\text{CH}_3\text{OH} + \text{K}_2\text{Cr}_2\text{O}_7 + \text{HCl} \longrightarrow \text{HCOOH} + \text{CrCl}_3 + \text{KCl} + \text{H}_2\text{O}$
489. $\text{C}_2\text{H}_5\text{OH} + \text{KMnO}_4 \longrightarrow \text{CH}_3\text{COOK} + \text{MnO}_2\downarrow + \text{KOH} + \text{H}_2\text{O}$
490. $\text{C}_6\text{H}_{12}\text{O}_6 + \text{KMnO}_4 + \text{H}_2\text{SO}_4 \longrightarrow \text{K}_2\text{SO}_4 + \text{MnSO}_4 + \text{CO}_2\uparrow + \text{H}_2\text{O}$
491. $\text{CH}_3\text{OH} + \text{KMnO}_4 + \text{H}_2\text{SO}_4 \longrightarrow \text{HCOOH} + \text{MnSO}_4 + \text{K}_2\text{SO}_4 + \text{H}_2\text{O}$
492. $\text{C}_5\text{H}_{10} + \text{K}_2\text{Cr}_2\text{O}_7 + \text{H}_2\text{SO}_4 \longrightarrow$
 $\longrightarrow \text{C}_2\text{H}_5\text{COOH} + \text{CH}_3\text{COOH} + \text{Cr}_2(\text{SO}_4)_3 + \text{K}_2\text{SO}_4 + \text{H}_2\text{O}$
493. $\text{C}_6\text{H}_{12} + \text{KMnO}_4 + \text{H}_2\text{SO}_4 \longrightarrow$
 $\longrightarrow \text{CO}_2\uparrow + \text{C}_5\text{H}_{10}\text{O} + \text{MnSO}_4 + \text{K}_2\text{SO}_4 + \text{H}_2\text{O}$
494. $\text{CH}_3-\text{CH}=\text{C}(\text{CH}_3)-\text{CH}_2-\text{CH}_3 + \text{KMnO}_4 + \text{H}_2\text{SO}_4 \longrightarrow$
 $\longrightarrow \text{CH}_3\text{COOH} + \text{CH}_3\text{C(O)C}_2\text{H}_5 + \text{K}_2\text{SO}_4 + \text{MnSO}_4 + \text{H}_2\text{O}$
495. $\text{KOOC}-\text{CH}_2-\text{COOK} + \text{O}_2 \longrightarrow \text{CO}_2\uparrow + \text{H}_2\text{O} + \text{K}_2\text{CO}_3$
496. $\text{CH}_3-\text{CH}=\text{CH}_2 + \text{KMnO}_4 + \text{KOH} \longrightarrow$
 $\longrightarrow \text{CH}_3\text{CH(OH)CH}_2(\text{OH})_2 + \text{K}_2\text{MnO}_4$
497. $\text{CH}_3-\underset{\text{O}}{\underset{\parallel}{\text{C}}}-\text{H} + [\text{Ag}(\text{NH}_3)_2]\text{OH} + \text{NH}_4\text{OH} \longrightarrow$
 $\longrightarrow \text{CH}_3\text{COONH}_4 + \text{Ag}\downarrow + \text{NH}_3\uparrow + \text{H}_2\text{O}$
498. $\text{C}_6\text{H}_5-\text{CH}_2-\text{CH}_3 + \text{K}_2\text{Cr}_2\text{O}_7 + \text{H}_2\text{SO}_4 \longrightarrow$
 $\longrightarrow \text{C}_6\text{H}_5\text{COOH} + \text{CO}_2\uparrow + \text{K}_2\text{SO}_4 + \text{Cr}_2(\text{SO}_4)_3 + \text{H}_2\text{O}$
499. $\text{NH}_4\text{HS} + \text{C}_6\text{H}_5\text{NO}_2 \longrightarrow \text{C}_6\text{H}_5\text{NH}_2 + \text{S}\downarrow + \text{NH}_3\uparrow + \text{H}_2\text{O}$
500. $\text{H}_2\text{NCH}_2\text{COOH} + \text{O}_2 \longrightarrow \text{CO}_2\uparrow + \text{H}_2\text{O} + \text{N}_2\uparrow$
501. $\text{CH}_3\text{COOC}_2\text{H}_5 + \text{O}_2 \longrightarrow \text{CO}_2\uparrow + \text{H}_2\text{O}$

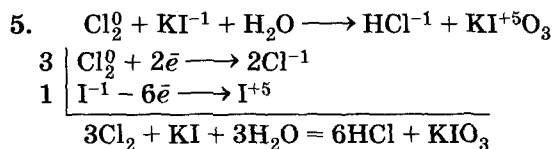
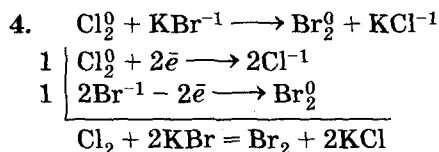
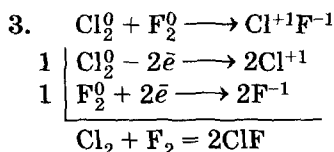
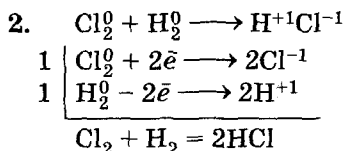
502. $(\text{CH}_3)_2\text{NH} + \text{Cu}_2\text{O} \xrightarrow{t} \text{Cu} + \text{CO}_2\uparrow + \text{H}_2\text{O} + \text{N}_2\uparrow$
503. $(\text{CH}_3)_2\text{NH} + \text{Cu}_2\text{O}(\text{недост.}) \xrightarrow{t} \text{Cu} + \text{CO}\uparrow + \text{H}_2\text{O} + \text{N}_2\uparrow$
504. $\text{C}_3\text{H}_5\text{Cl} + \text{O}_2 \longrightarrow \text{CO}_2\uparrow + \text{H}_2\text{O} + \text{HCl}$
505. $\text{C}_5\text{H}_3\text{NCl}_2 + \text{NO}_2 \longrightarrow \text{CO}_2\uparrow + \text{HCl} + \text{H}_2\text{O} + \text{NO}\uparrow + \text{N}_2\uparrow$
506. $\text{C}_5\text{H}_{10}\text{O}_5 + \text{P} + \text{HI} + \text{H}_2\text{O} \longrightarrow \text{C}_5\text{H}_{11}\text{I} + \text{H}_3\text{PO}_3$
507. $\text{C}_2\text{H}_5\text{NH}_2 + \text{O}_2 \longrightarrow \text{CO}_2\uparrow + \text{H}_2\text{O} + \text{N}_2\uparrow$

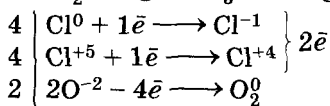
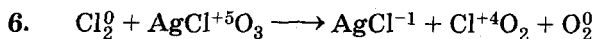
Решения задач

Галогены и их соединения

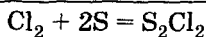
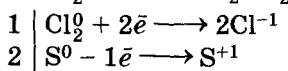
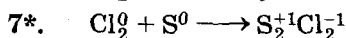
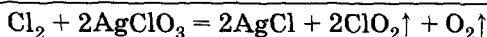
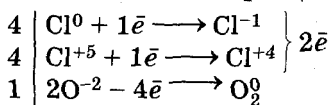


Пояснение. Если в ОВР участвуют простые вещества, молекулы которых образованы несколькими атомами (Cl_2 , O_2 , H_2 , O_3 ...), то в ЭБ число приобретенных и отданных электронов рассчитывают с учетом числа атомов в молекуле, например: $\text{Cl}_2^0 + 2\bar{e} \longrightarrow 2\text{Cl}^{-1}$.

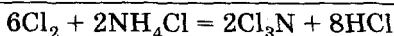
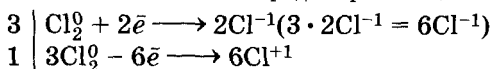
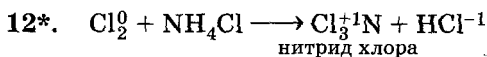
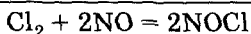
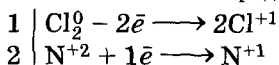
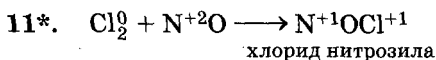
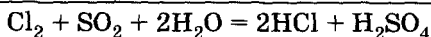
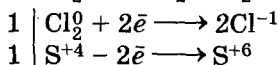
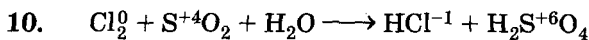
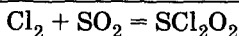
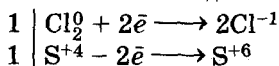
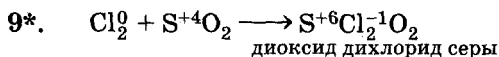
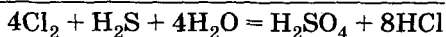
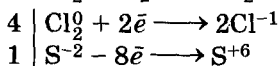
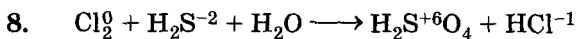




Множители сокращаются:

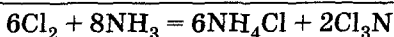
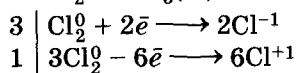
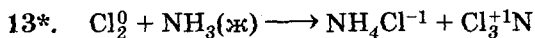
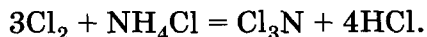


П о я с н е н и е. При других условиях в реакции могут образовываться SCl_2 или SCl_4 .



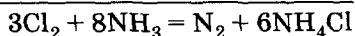
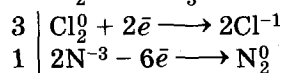
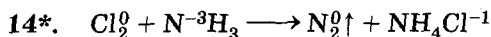
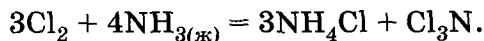
П о я с н е н и е. Коэффициент 8 перед HCl получается, если суммировать 6Cl^{-1} и 2Cl^{-1} из $2\text{NH}_4\text{Cl}$.

После сокращения коэффициентов:

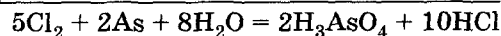
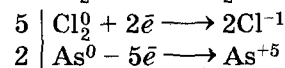
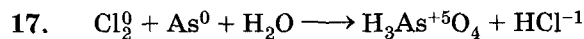
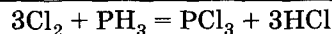
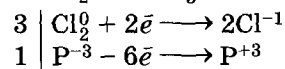
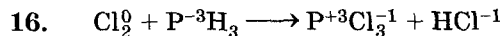
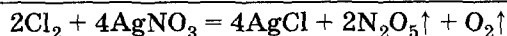
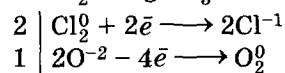
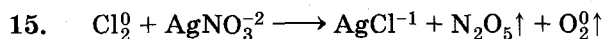


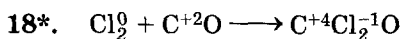
П о я с н е н и е. 6Cl_2 получается, если суммировать $3\text{Cl}_2(3 \cdot \text{Cl}_2^0)$ из 1-й схемы ЭБ и 3Cl_2 из 2-й схемы ЭБ.

После сокращения коэффициентов:

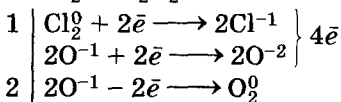
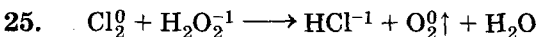
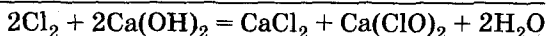
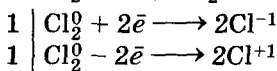
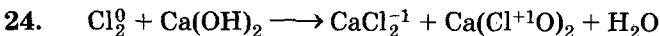
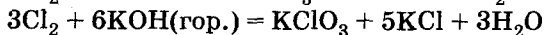
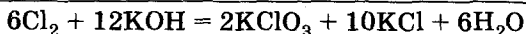
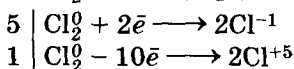
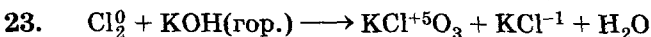
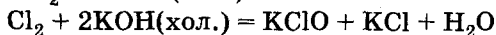
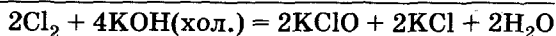
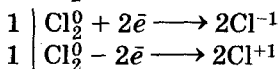
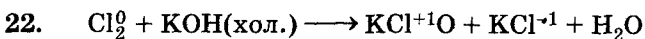
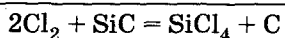
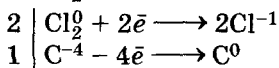
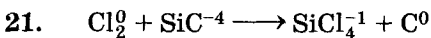
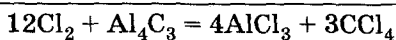
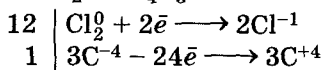
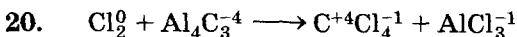
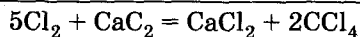
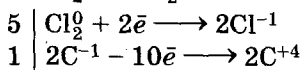
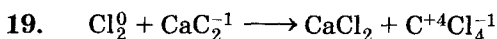
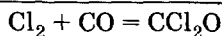
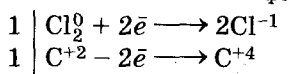


П о я с н е н и е. В этом случае из аммиака NH_3 окисляется только часть азота, другая переходит без изменения степени окисления в состав NH_4Cl , поэтому коэффициент перед NH_3 ставят в последнюю очередь, считав число атомов азота в правой части уравнения.





фосген



26. $\text{Cl}_2^0 + \text{H}_2\text{O}(\text{пар}) + \text{C}^0(\text{кокс}) \longrightarrow \text{C}^{+4}\text{O}_2 + \text{HCl}^{-1}$

$$\begin{array}{l} 2 \left| \text{Cl}_2^0 + 2\bar{e} \longrightarrow 2\text{Cl}^{-1} \right. \\ 1 \left| \text{C}^0 - 4\bar{e} \longrightarrow \text{C}^{+4} \right. \end{array}$$

$$2\text{Cl}_2 + 2\text{H}_2\text{O}(\text{пар}) + \text{C}(\text{кокс}) = \text{CO}_2 + 4\text{HCl}$$

27. $\text{Cl}_2^0 + \text{H}_2\text{O} \longrightarrow \text{HCl}^{-1} + \text{HCl}^{+1}\text{O}$

$$\begin{array}{l} 1 \left| \text{Cl}_2^0 - 2\bar{e} \longrightarrow 2\text{Cl}^{+1} \right. \\ 1 \left| \text{Cl}_2^0 + 2\bar{e} \longrightarrow 2\text{Cl}^{-1} \right. \end{array}$$

$$2\text{Cl}_2 + 2\text{H}_2\text{O} = 2\text{HCl} + 2\text{HClO}$$

$$\text{Cl}_2 + \text{H}_2\text{O} = \text{HCl} + \text{HClO}$$

28. $\text{Cl}_2^0 + \text{H}_2\text{O}^{-2} \longrightarrow \text{HCl}^{-1} + \text{O}_2^0 (\text{на свету})$

$$\begin{array}{l} 2 \left| \text{Cl}_2^0 + 2\bar{e} \longrightarrow 2\text{Cl}^{-1} \right. \\ 1 \left| 2\text{O}^{-2} - 4\bar{e} \longrightarrow \text{O}_2^0 \right. \end{array}$$

$$2\text{Cl}_2 + 2\text{H}_2\text{O} = 4\text{HCl} + \text{O}_2\uparrow$$

29. $\text{Cl}_2^0 + \text{K}_2\text{Mn}^{+6}\text{O}_4 \longrightarrow \text{KMn}^{+7}\text{O}_4 + \text{KCl}^{-1}$

$$\begin{array}{l} 1 \left| \text{Cl}_2^0 + 2\bar{e} \longrightarrow 2\text{Cl}^{-1} \right. \\ 2 \left| \text{Mn}^{+6} - 1\bar{e} \longrightarrow \text{Mn}^{+7} \right. \end{array}$$

$$\text{Cl}_2 + 2\text{K}_2\text{MnO}_4 = 2\text{KMnO}_4 + 2\text{KCl}$$

30. $\text{Cl}_2^0 + \text{K}_4[\text{Fe}^{+2}(\text{CN})_6] \longrightarrow \text{K}_3[\text{Fe}^{+3}(\text{CN})_6] + \text{KCl}^{-1}$

$$\begin{array}{l} 1 \left| \text{Cl}_2^0 + 2\bar{e} \longrightarrow 2\text{Cl}^{-1} \right. \\ 2 \left| \text{Fe}^{+2} - 1\bar{e} \longrightarrow \text{Fe}^{+3} \right. \end{array}$$

$$\text{Cl}_2 + 2\text{K}_4[\text{Fe}(\text{CN})_6] = 2\text{K}_3[\text{Fe}(\text{CN})_6] + 2\text{KCl}$$

31*. $\text{Cl}_2^0 + \text{Fe}^0 \longrightarrow \text{Fe}^{+3}\text{Cl}_3^{-1}$

$$\begin{array}{l} 3 \left| \text{Cl}_2^0 + 2\bar{e} \longrightarrow 2\text{Cl}^{-1} \right. \\ 2 \left| \text{Fe}^0 - 3\bar{e} \longrightarrow \text{Fe}^{+3} \right. \end{array}$$

$$3\text{Cl}_2 + 2\text{Fe} = 2\text{FeCl}_3$$

Пояснение. Если слева и справа соответственно четное и нечетное число атомов элемента, то удобнее сначала уравнивать их в ЭБ. Например, в схеме $\text{Cl}_2 + \text{Fe} \longrightarrow \text{FeCl}_3$ слева два, справа три атома хлора. Уравниваем хлор: $3\text{Cl}_2^0 \longrightarrow 6\text{Cl}^{-1}$. Дальше уравниваем заряды: $3\text{Cl}_2^0 + 6\bar{e} \longrightarrow 6\text{Cl}^{-1}$. Эта запись означает, что в левой части уравнения должны быть 3 молекулы хлора, а справа — 6 хлорид-ионов.

32. $\text{Cl}_2^0 + \text{Br}_2^0 + \text{H}_2\text{O} \longrightarrow \text{HBr}^{+5}\text{O}_3 + \text{HCl}^{-1}$

$$\begin{array}{l} 5 \left| \text{Cl}_2^0 + 2\bar{e} \longrightarrow 2\text{Cl}^{-1} \right. \\ 1 \left| \text{Br}_2^0 - 10\bar{e} \longrightarrow 2\text{Br}^{+5} \right. \end{array}$$

$$5\text{Cl}_2 + \text{Br}_2 + 6\text{H}_2\text{O} = 2\text{HBrO}_3 + 10\text{HCl}$$

33. $\text{Br}_2^0 + \text{HI}^{-1} \longrightarrow \text{I}_2^0 + \text{HBr}^{-1}$

$$\begin{array}{l} 1 \quad \text{Br}_2^0 + 2\bar{e} \longrightarrow 2\text{Br}^{-1} \\ 1 \quad 2\text{I}^{-1} - 2\bar{e} \longrightarrow \text{I}_2^0 \\ \hline \text{Br}_2 + 2\text{HI} = \text{I}_2 + 2\text{HBr} \end{array}$$
34. $\text{Br}_2^0 + \text{KCl}^{+5}\text{O}_3 \longrightarrow \text{KBr}^{+5}\text{O}_3 + \text{Cl}_2^0 \uparrow$

$$\begin{array}{l} 1 \quad \text{Br}_2^0 - 10\bar{e} \longrightarrow 2\text{Br}^{+5} \\ 1 \quad 2\text{Cl}^{+5} + 10\bar{e} \longrightarrow \text{Cl}_2^0 \\ \hline \text{Br}_2 + 2\text{KClO}_3 = 2\text{KBrO}_3 + \text{Cl}_2 \uparrow \end{array}$$
35. $\text{Br}_2^0 + \text{P}^0 \longrightarrow \text{P}^{+5}\text{Br}_5^{-1}$

$$\begin{array}{l} 5 \quad \text{Br}_2^0 + 2\bar{e} \longrightarrow 2\text{Br}^{-1} \\ 2 \quad \text{P}^0 - 5\bar{e} \longrightarrow \text{P}^{+5} \\ \hline 5\text{Br}_2 + 2\text{P} = 2\text{PBr}_5 \end{array}$$
36. $\text{Br}_2^0 + \text{HgO} \longrightarrow \text{HgBr}_2^{-1} + \text{Br}_2^{+1}\text{O}$

$$\begin{array}{l} 1 \quad \text{Br}_2^0 + 2\bar{e} \longrightarrow 2\text{Br}^{-1} \\ 1 \quad \text{Br}_2^0 - 2\bar{e} \longrightarrow 2\text{Br}^{+1} \\ \hline 2\text{Br}_2 + \text{HgO} = \text{HgBr}_2 + \text{Br}_2\text{O} \end{array}$$
37. $\text{Br}_2^0 + \text{NaOH}(\text{разб.}) \longrightarrow \text{NaBr}^{+1}\text{O} + \text{NaBr}^{-1} + \text{H}_2\text{O}$

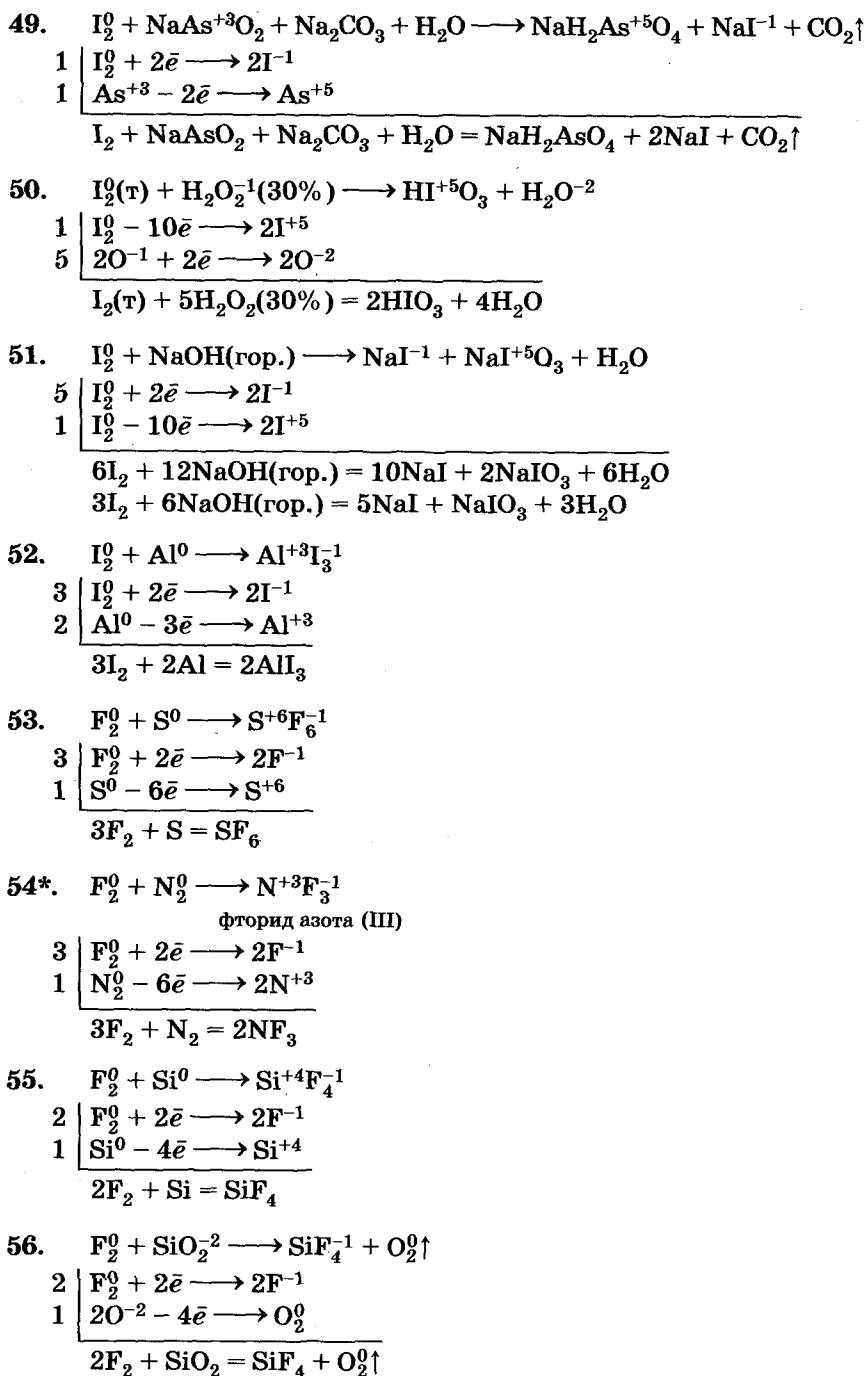
$$\begin{array}{l} 1 \quad \text{Br}_2^0 + 2\bar{e} \longrightarrow 2\text{Br}^{-1} \\ 1 \quad \text{Br}_2^0 - 2\bar{e} \longrightarrow 2\text{Br}^{+1} \\ \hline \text{Br}_2 + 2\text{NaOH}(\text{разб.}) = \text{NaBrO} + \text{NaBr} + \text{H}_2\text{O} \end{array}$$
38. $\text{I}_2^0 + \text{Cl}_2^0 + \text{H}_2\text{O} \longrightarrow \text{HCl}^{-1} + \text{HI}^{+5}\text{O}_3$

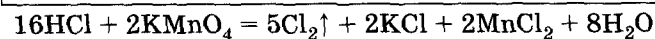
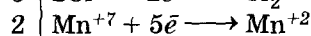
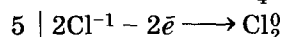
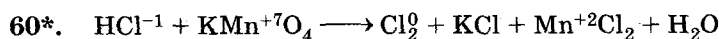
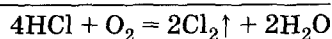
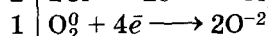
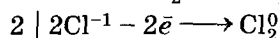
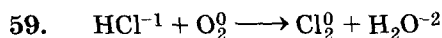
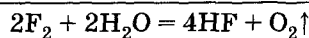
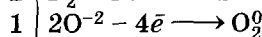
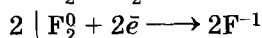
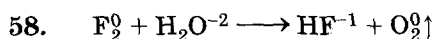
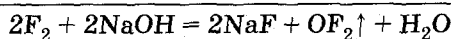
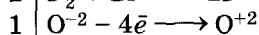
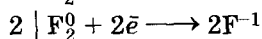
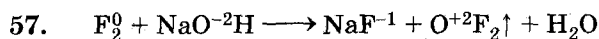
$$\begin{array}{l} 1 \quad \text{I}_2^0 - 10\bar{e} \longrightarrow 2\text{I}^{+5} \\ 5 \quad \text{Cl}_2^0 + 2\bar{e} \longrightarrow 2\text{Cl}^{-1} \\ \hline \text{I}_2 + 5\text{Cl}_2 + 6\text{H}_2\text{O} = 10\text{HCl} + 2\text{HIO}_3 \end{array}$$
39. $\text{I}_2^0 + \text{HCl}^{+5}\text{O}_3 \longrightarrow \text{HI}^{+5}\text{O}_3 + \text{Cl}_2^0 \uparrow$

$$\begin{array}{l} 1 \quad \text{I}_2^0 - 10\bar{e} \longrightarrow 2\text{I}^{+5} \\ 1 \quad 2\text{Cl}^{+5} + 10\bar{e} \longrightarrow \text{Cl}_2^0 \\ \hline \text{I}_2 + 2\text{HClO}_3 = 2\text{HIO}_3 + \text{Cl}_2 \uparrow \end{array}$$
40. $\text{I}_2^0 + \text{HCl}^{+7}\text{O}_4 \longrightarrow \text{HI}^{+7}\text{O}_4 + \text{Cl}_2^0 \uparrow$

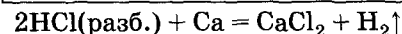
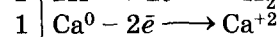
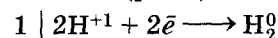
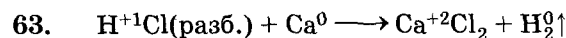
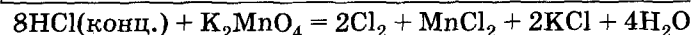
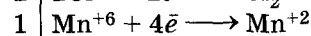
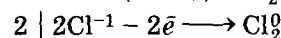
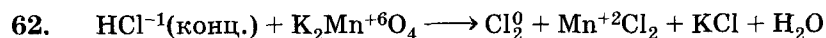
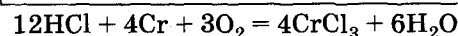
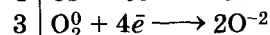
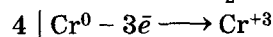
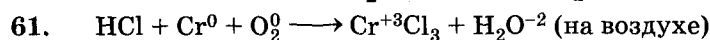
$$\begin{array}{l} 1 \quad \text{I}_2^0 - 14\bar{e} \longrightarrow 2\text{I}^{+7} \\ 1 \quad 2\text{Cl}^{+7} + 14\bar{e} \longrightarrow \text{Cl}_2^0 \\ \hline \text{I}_2 + 2\text{HClO}_4 = 2\text{HIO}_4 + \text{Cl}_2 \uparrow \end{array}$$

41. $I_2^0 + KBr^{+5}O_3 \longrightarrow KI^{+5}O_3 + Br_2^0$
- $$\begin{array}{l} 1 \quad I_2^0 - 10\bar{e} \longrightarrow 2I^{+5} \\ 1 \quad 2Br^{+5} + 10\bar{e} \longrightarrow Br_2^0 \\ \hline I_2 + 2KBrO_3 = 2KIO_3 + Br_2 \end{array}$$
42. $I_2^0 + H_2S^{-2} \longrightarrow S^0 \downarrow + HI^{-1}$
- $$\begin{array}{l} 1 \quad I_2^0 + 2\bar{e} \longrightarrow 2I^{-1} \\ 1 \quad S^{-2} - 2\bar{e} \longrightarrow S^0 \\ \hline I_2 + H_2S = S + 2HI \end{array}$$
43. $I_2^0 + S^{+4}O_2 + H_2O \longrightarrow HI^{-1} + H_2S^{+6}O_4$
- $$\begin{array}{l} 1 \quad I_2^0 + 2\bar{e} \longrightarrow 2I^{-1} \\ 1 \quad S^{+4} - 2\bar{e} \longrightarrow S^{+6} \\ \hline I_2 + SO_2 + 2H_2O = 2HI + H_2SO_4 \end{array}$$
44. $I_2^0 + K_2S^{+4}O_3 + KOH \longrightarrow K_2S^{+6}O_4 + KI^{-1} + H_2O$
- $$\begin{array}{l} 1 \quad I_2^0 + 2\bar{e} \longrightarrow 2I^{-1} \\ 1 \quad S^{+4} - 2\bar{e} \longrightarrow S^{+6} \\ \hline I_2 + K_2SO_3 + 2KOH = K_2SO_4 + 2KI + H_2O \end{array}$$
45. $I_2^0 + HN^{+5}O_3(\text{разб.}) \longrightarrow HI^{+5}O_3 + N^{+2}O \uparrow + H_2O$
- $$\begin{array}{l} 3 \quad I_2^0 - 10\bar{e} \longrightarrow 2I^{+5} \\ 10 \quad N^{+5} + 3\bar{e} \longrightarrow N^{+2} \\ \hline 3I_2 + 10HNO_3(\text{разб.}) = 6HIO_3 + 10NO \uparrow + 2H_2O \end{array}$$
46. $I_2^0 + P^0 \longrightarrow P^{+3}I_3^{-1}$
- $$\begin{array}{l} 1 \quad 3I_2^0 + 6\bar{e} \longrightarrow 6I^{-1} \\ 2 \quad P^0 - 3\bar{e} \longrightarrow P^{+3} \\ \hline 3I_2 + 2P = 2PI_3 \end{array}$$
47. $I_2^0 + P^0 + H_2O \longrightarrow H_3P^{+3}O_3 + HI^{-1}$
- $$\begin{array}{l} 3 \quad I_2^0 + 2\bar{e} \longrightarrow 2I^{-1} \\ 2 \quad P^0 - 3\bar{e} \longrightarrow P^{+3} \\ \hline 3I_2 + 2P + 6H_2O = 2H_3PO_3 + 6HI \end{array}$$
48. $I_2^0 + H_2O + P^{-3}H_3 \longrightarrow HI^{-1} + H(P^{+1}H_2O_2)$
иодная вода
- $$\begin{array}{l} 2 \quad I_2^0 + 2\bar{e} \longrightarrow 2I^{-1} \\ 1 \quad P^{-3} - 4\bar{e} \longrightarrow P^{+1} \\ \hline 2I_2 + 2H_2O + PH_3 = 4HI + H_3PO_2 \end{array}$$

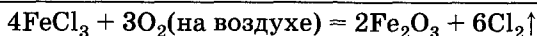
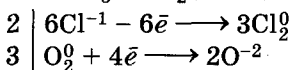
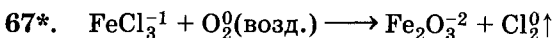
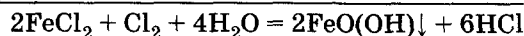
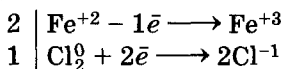
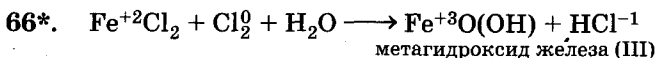
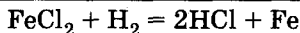
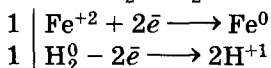
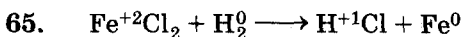
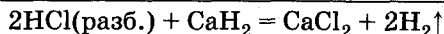
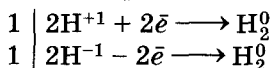
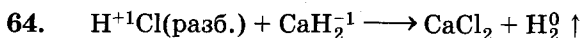




Пояснение. Часть хлорид-ионов из HCl окисляется до Cl₂, часть — переходит без изменения с. о. в состав KCl и MnCl₂, поэтому коэффициент перед HCl ставят в последнюю очередь, после того как будут расставлены коэффициенты перед Cl₂, KCl и MnCl₂.

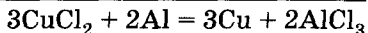
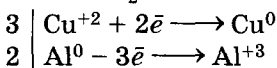
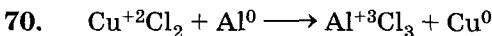
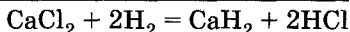
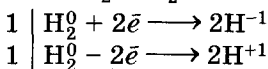
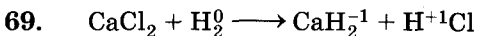
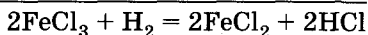
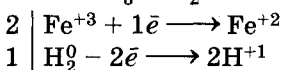
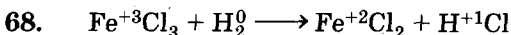
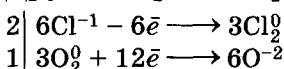


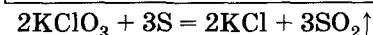
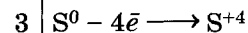
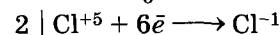
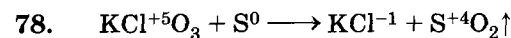
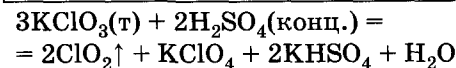
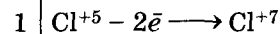
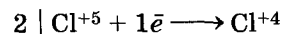
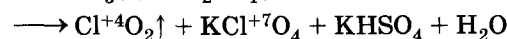
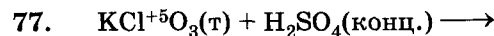
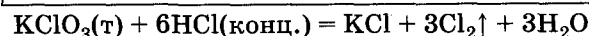
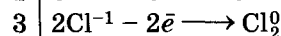
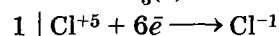
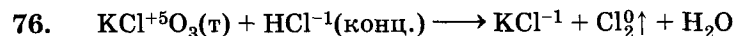
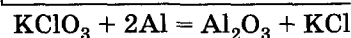
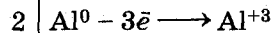
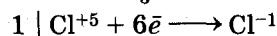
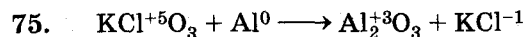
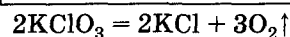
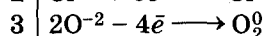
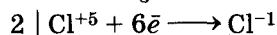
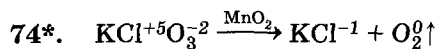
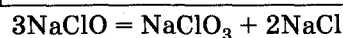
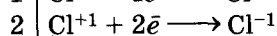
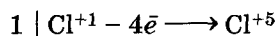
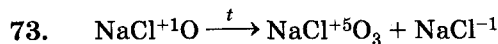
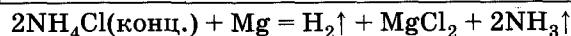
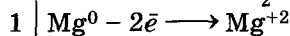
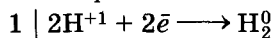
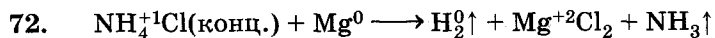
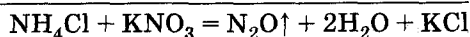
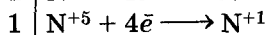
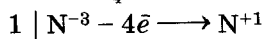
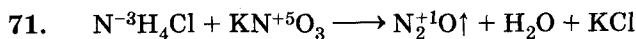
¹ Цифры, записанные под уравнением реакции, показывают порядок расстановки коэффициентов.

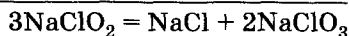
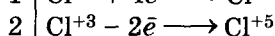
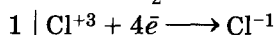
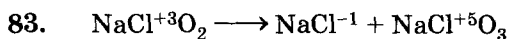
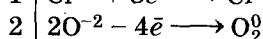
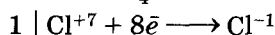
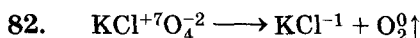
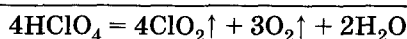
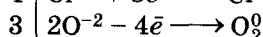
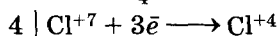
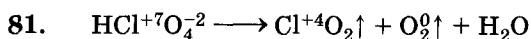
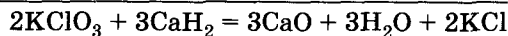
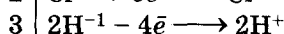
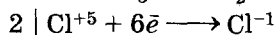
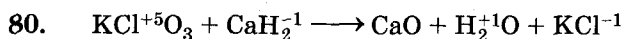
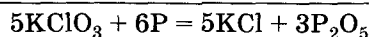
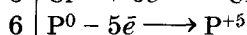
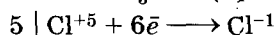
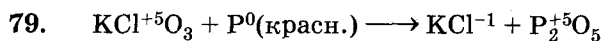


Пояснение. ЭБ показывает, что в левой части уравнения должно быть 12Cl^{-1} ($2 \cdot 6\text{Cl}^{-1}$), в правой — 6Cl_2 ($2 \cdot 3\text{Cl}_2$). В составе FeCl_3 есть 3Cl^{-1} , поэтому перед FeCl_3 ставят коэффициент 4.

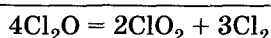
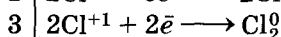
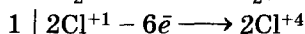
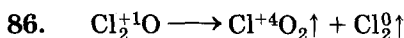
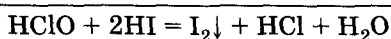
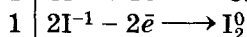
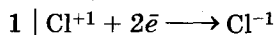
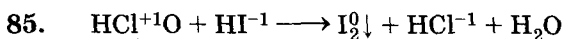
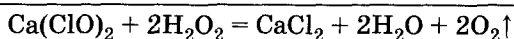
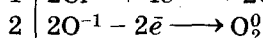
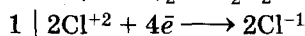
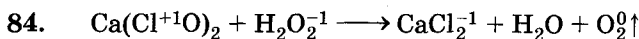
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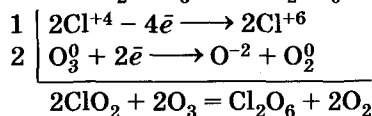
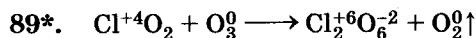
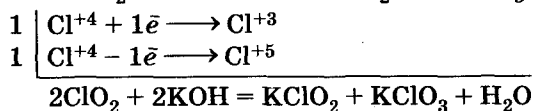
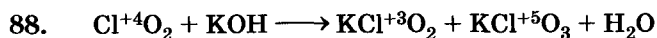
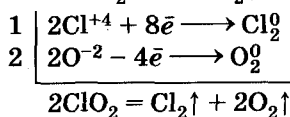
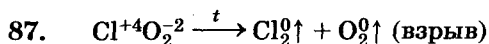




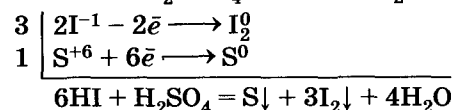
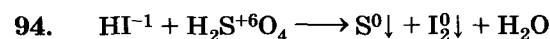
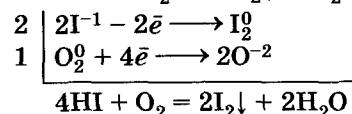
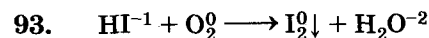
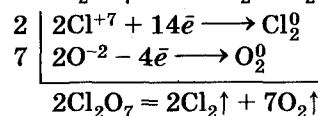
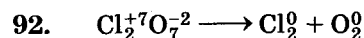
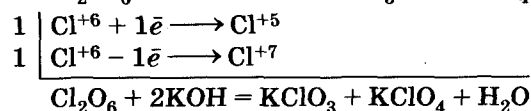
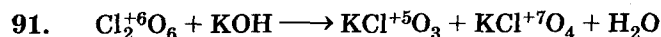
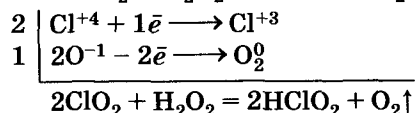
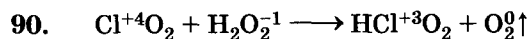


хлорит натрия





П о я с н е н и е. Часть атомов кислорода из O_3 переходит в состав Cl_2O_6 , из другой части образуются молекулы O_2 .



95. $\text{HI}^{-1} + \text{N}^{+4}\text{O}_2 \longrightarrow \text{N}^{+2}\text{O}\uparrow + \text{I}_2^0\downarrow + \text{H}_2\text{O}$

$$\begin{array}{l} 1 \quad 2\text{I}^{-1} - 2\bar{e} \longrightarrow \text{I}_2^0 \\ 1 \quad \text{N}^{+4} + 2\bar{e} \longrightarrow \text{N}^{+2} \end{array}$$

$$2\text{HI} + \text{NO}_2 = \text{NO}\uparrow + \text{I}_2\downarrow + \text{H}_2\text{O}$$
96. $\text{NaI}^{-1} + \text{NaI}^{+5}\text{O}_3 + \text{H}_2\text{SO}_4 \longrightarrow \text{Na}_2\text{SO}_4 + \text{I}_2^0\downarrow + \text{H}_2\text{O}$

$$\begin{array}{l} 5 \quad 2\text{I}^{-1} - 2\bar{e} \longrightarrow \text{I}_2^0 \\ 1 \quad 2\text{I}^{+5} + 10\bar{e} \longrightarrow \text{I}_2^0 \end{array}$$

$$10\text{NaI} + 2\text{NaIO}_3 + 6\text{H}_2\text{SO}_4 = 6\text{Na}_2\text{SO}_4 + 6\text{I}_2 + 6\text{H}_2\text{O}$$

$$5\text{NaI} + \text{NaIO}_3 + 3\text{H}_2\text{SO}_4 = 3\text{Na}_2\text{SO}_4 + 3\text{I}_2 + 3\text{H}_2\text{O}$$
97. $\text{HI}^{-1} + \text{H}^{+5}\text{NO}_3(60\%) \longrightarrow \text{H}^{+5}\text{IO}_3 + \text{N}^{+4}\text{O}_2\uparrow + \text{H}_2\text{O}$

$$\begin{array}{l} 1 \quad \text{I}^{-1} - 6\bar{e} \longrightarrow \text{I}^{+5} \\ 6 \quad \text{N}^{+5} + 1\bar{e} \longrightarrow \text{N}^{+4} \end{array}$$

$$\text{HI} + 6\text{HNO}_3 = \text{HIO}_3 + 6\text{NO}_2\uparrow + 3\text{H}_2\text{O}$$
98. $\text{HI}^{-1} + \text{Fe}^{+3}\text{Cl}_3 \longrightarrow \text{Fe}^{+2}\text{Cl}_2 + \text{I}_2^0\downarrow + \text{HCl}$

$$\begin{array}{l} 1 \quad 2\text{I}^{-1} - 2\bar{e} \longrightarrow \text{I}_2^0 \\ 2 \quad \text{Fe}^{+3} + 1\bar{e} \longrightarrow \text{Fe}^{+2} \end{array}$$

$$2\text{HI} + 2\text{FeCl}_3 = 2\text{FeCl}_2 + \text{I}_2\downarrow + 2\text{HCl}$$
99. $\text{KI}^{-1}(\text{T}) + \text{H}_2\text{S}^{+6}\text{O}_4(\text{конц.}) \longrightarrow \text{I}_2^0\downarrow + \text{H}_2\text{S}^{-2}\uparrow + \text{K}_2\text{SO}_4 + \text{H}_2\text{O}$

$$\begin{array}{l} 4 \quad 2\text{I}^{-1} - 2\bar{e} \longrightarrow \text{I}_2^0\downarrow \\ 1 \quad \text{S}^{+6} + 8\bar{e} \longrightarrow \text{S}^{-2} \end{array}$$

$$\underset{1}{8\text{KI}(\text{T})} + \underset{5}{5\text{H}_2\text{SO}_4(\text{конц.})} = \underset{2}{4\text{I}_2^0\downarrow} + \underset{3}{\text{H}_2\text{S}\uparrow} + \underset{4}{4\text{K}_2\text{SO}_4} + \underset{6}{4\text{H}_2\text{O}}$$
100. $\text{KI}^{-1} + \text{K}_3[\text{Fe}^{+3}(\text{CN})_6] \longrightarrow \text{I}_2^0\downarrow + \text{K}_4[\text{Fe}^{+2}(\text{CN})_6]$

$$\begin{array}{l} 1 \quad 2\text{I}^{-1} - 2\bar{e} \longrightarrow \text{I}_2^0 \\ 2 \quad \text{Fe}^{+3} + 1\bar{e} \longrightarrow \text{Fe}^{+2} \end{array}$$

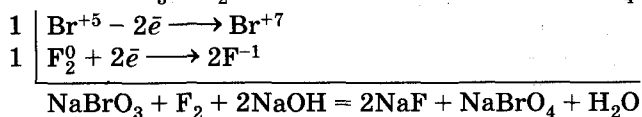
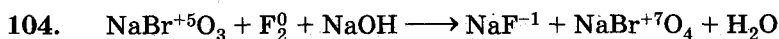
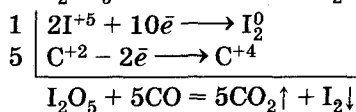
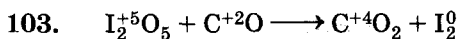
$$2\text{KI} + 2\text{K}_3[\text{Fe}(\text{CN})_6] = \text{I}_2^0\downarrow + 2\text{K}_4[\text{Fe}(\text{CN})_6]$$
101. $\text{KI}^{-1} + \text{O}_3^0 + \text{H}_2\text{O} \longrightarrow \text{I}_2^0\downarrow + \text{O}_2^0\uparrow + \text{KOH}$

$$\begin{array}{l} 1 \quad 2\text{I}^{-1} - 2\bar{e} \longrightarrow \text{I}_2^0\downarrow \\ 1 \quad \text{O}_3^0 + 2\bar{e} \longrightarrow \text{O}^{-2} + \text{O}_2^0 \end{array}$$

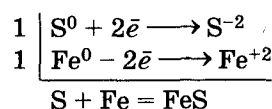
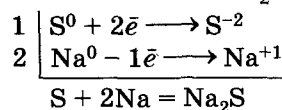
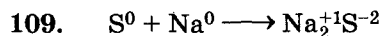
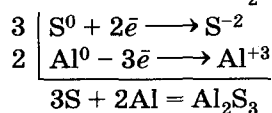
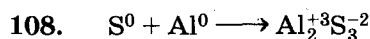
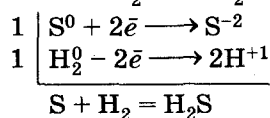
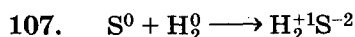
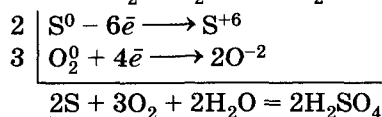
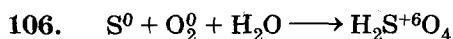
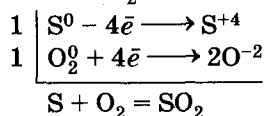
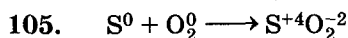
$$2\text{KI} + \text{O}_3 + \text{H}_2\text{O} = \text{I}_2\downarrow + \text{O}_2\uparrow + 2\text{KOH}$$
102. $\text{I}_2^{+5}\text{O}_5^{-2} \longrightarrow \text{I}_2^0 + \text{O}_2^0$

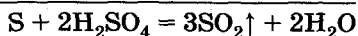
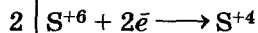
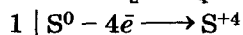
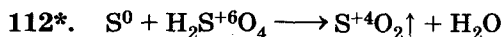
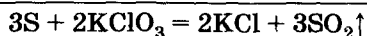
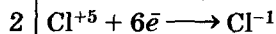
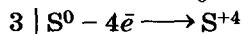
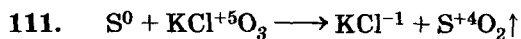
$$\begin{array}{l} 2 \quad 2\text{I}^{+5} + 10\bar{e} \longrightarrow \text{I}_2^0 \\ 5 \quad 2\text{O}^{-2} - 4\bar{e} \longrightarrow \text{O}_2^0 \end{array}$$

$$2\text{I}_2\text{O}_5 = 2\text{I}_2 + 5\text{O}_2$$

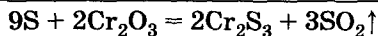
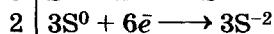
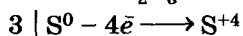
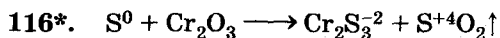
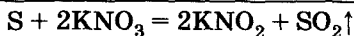
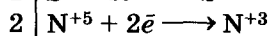
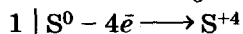
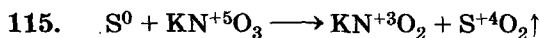
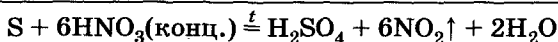
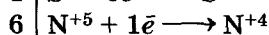
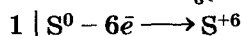
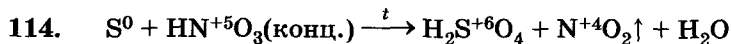
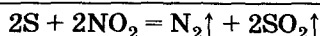
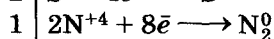
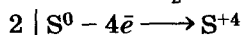
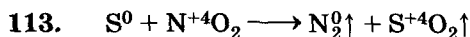


Сера и ее соединения

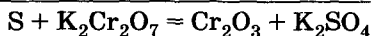
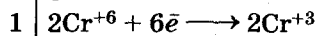
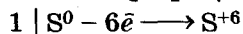
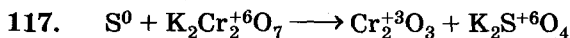


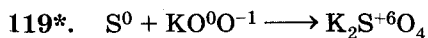
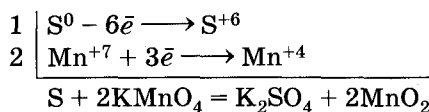
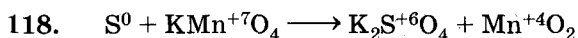


П о я с н е н и е. Коэффициент 3 перед SO_2 получается, если суммировать S^{+4} (из 1-й схемы ЭБ) и $2 \cdot 1\text{S}^{+4} = 2\text{S}^{+4}$ (из 2-й схемы ЭБ): $\text{S}^{+4} + 2\text{S}^{+4} = 3\text{S}^{+4}$.



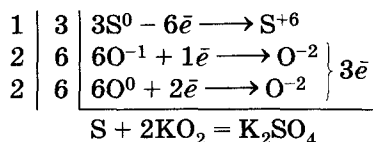
П о я с н е н и е. Часть атомов серы окисляется, а часть — восстанавливается. Суммарный коэффициент перед S равен 9: $3 \cdot 1 + 2 \cdot 3 = 9$.





надпероксид
калия

1-й вариант оформления ЭБ.



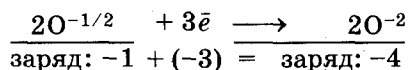
2-й вариант оформления 2-й схемы ЭБ:

1) схема: $2\text{O}^{-1/2} \longrightarrow \text{O}^{-2}$;

2) уравнивают кислород: $2\text{O}^{-1/2} \longrightarrow 2\text{O}^{-2}$;

3) уравнивают заряд.

Заряд в левой части $-1(2(-1/2))$, в правой -1 . Для уравнивания зарядов в левую часть добавляют $3\bar{e}$ (электроны отрицательно заряженные частицы).



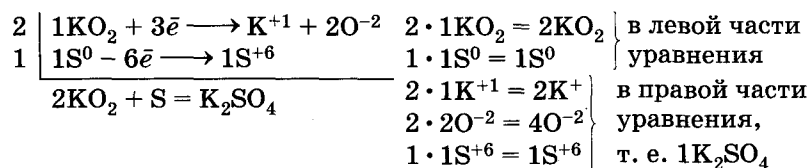
3-й вариант оформления ЭБ. Можно обойтись без определения степеней окисления элементов в KO_2 . В левой части ЭБ записывают «молекулярную» формулу KO_2 , в правой — эти же элементы, т. е. К и О, перешедшие в состав K_2SO_4 , где с. о. K^{+1} , O^{-2} :

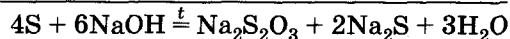
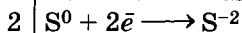
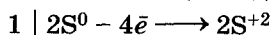
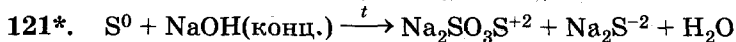
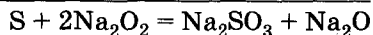
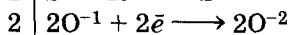
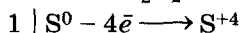
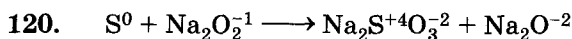
1) 1-я схема ЭБ: $\text{KO}_2 \longrightarrow \text{K}^{+1} + \text{O}^{-2}$;

2) уравнивается кислород: $\text{KO}_2 \longrightarrow \text{K}^{+1} + 2\text{O}^{-2}$;

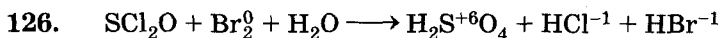
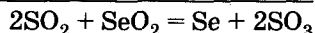
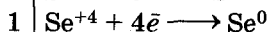
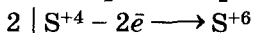
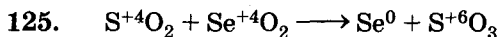
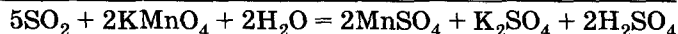
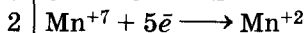
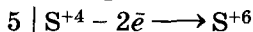
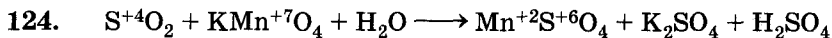
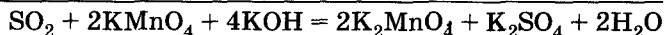
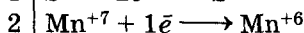
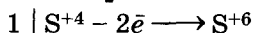
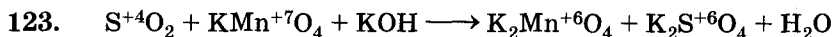
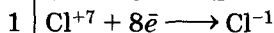
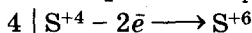
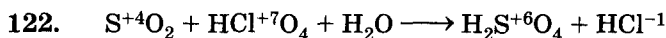
3) уравнивается заряд: $\text{KO}_2 + 3\bar{e} \longrightarrow \text{K}^{+1} + 2\text{O}^{-2}$;

4) полная схема ЭБ и расчет стехиометрических коэффициентов:



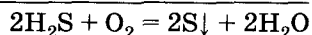
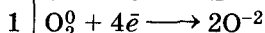
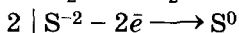
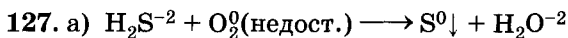
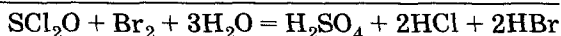
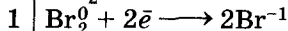
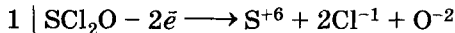


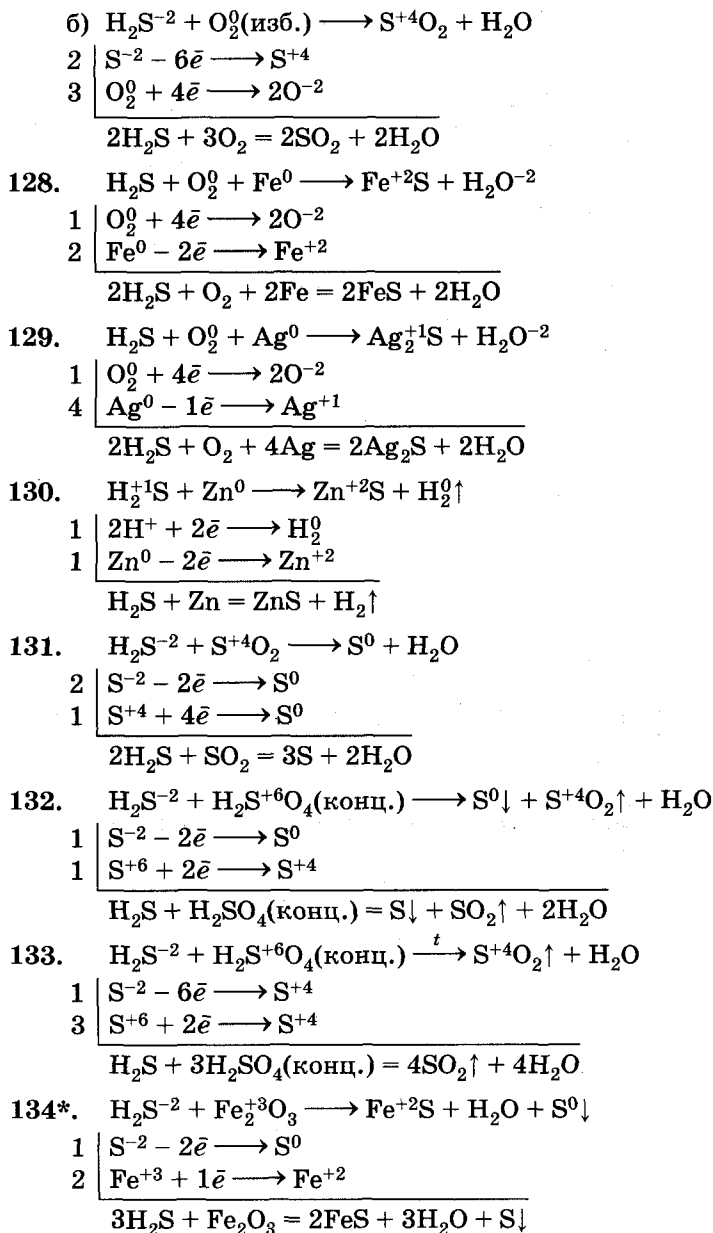
П о я с н е н и е. $\text{Na}_2\text{SO}_3\text{S}$, или $\text{Na}_2\text{S}_2\text{O}_3$, — тиосульфат натрия, соль тиосерной кислоты $\text{H}_2\text{S}_2\text{O}_3$. Строение $\text{S}_2\text{O}_3^{2-}$ см. на с. 10.



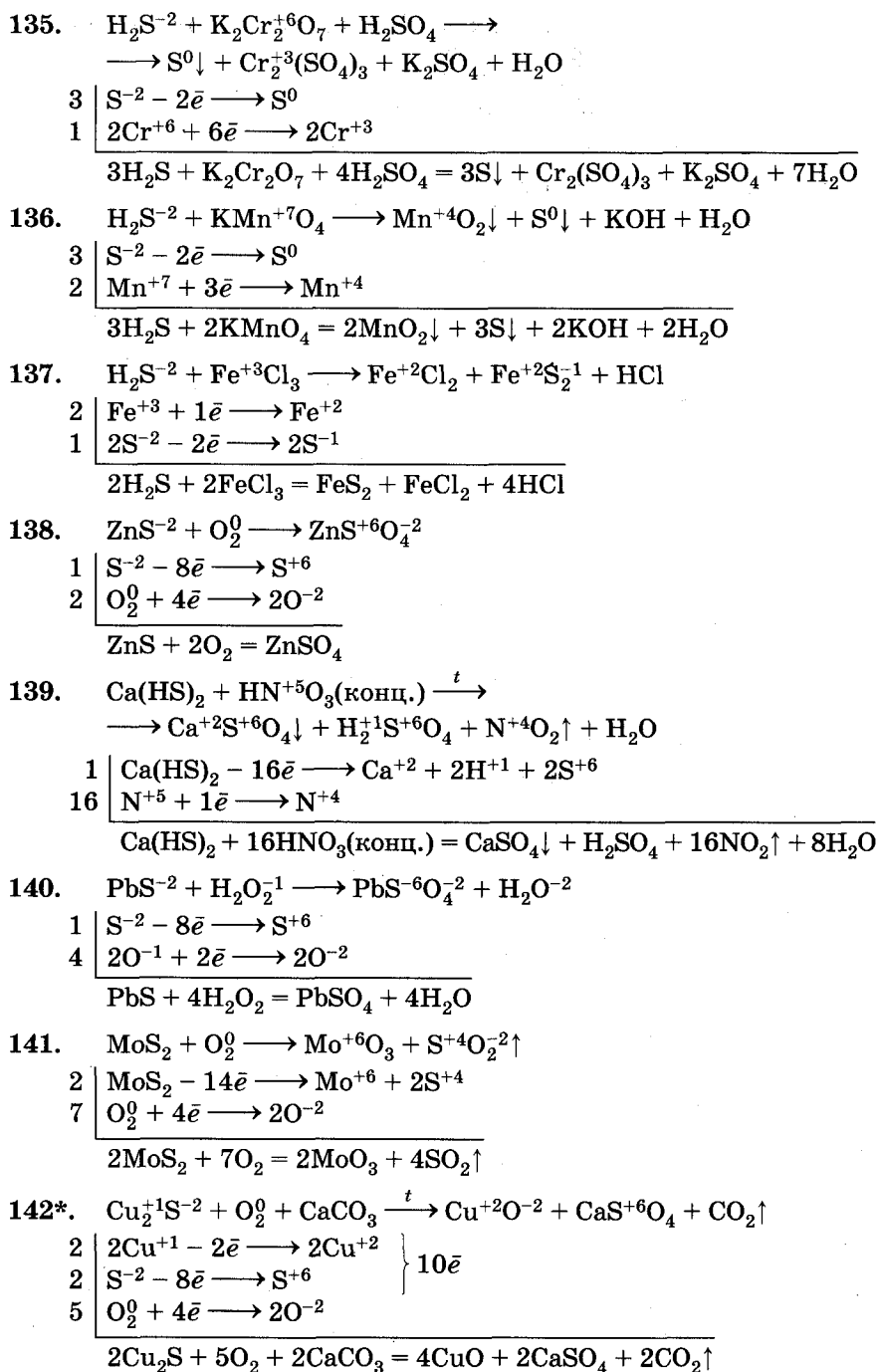
тионилхлорид бромная вода

(оксидихлорид серы)

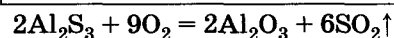
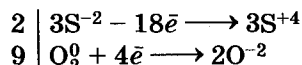
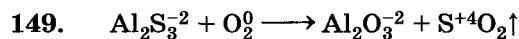
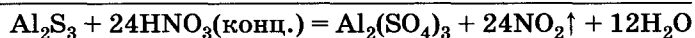
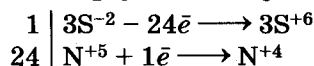
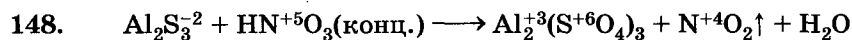
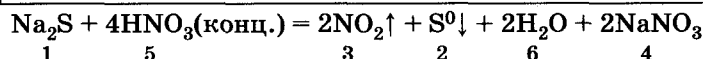
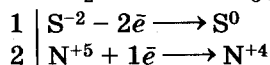
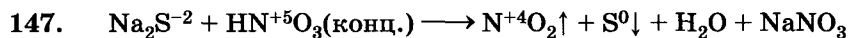
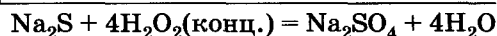
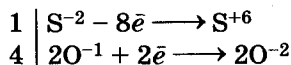
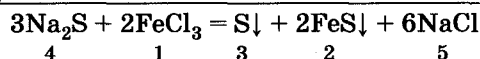
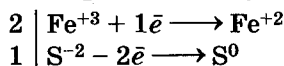
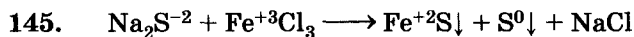
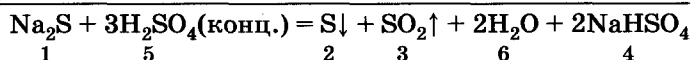
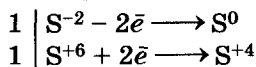
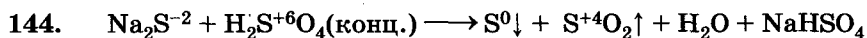
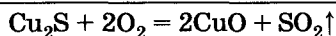
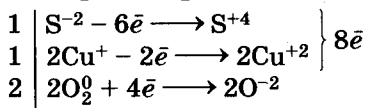
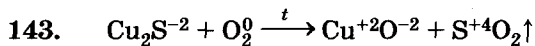
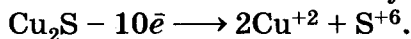




П о я с н е н и е. Часть серы (S^{-2}) из H_2S окисляется до S^0 , а другая часть переходит без изменения с. о. в состав FeS , поэтому коэффициент перед H_2S ставят после того, как уравнивают FeS и S .



П о я с н е н и е. 1-ю схему ЭБ можно оформить так:



156. $\text{H}_2\text{S}^{+6}\text{O}_4(\text{конц.}) + \text{Zn}^0 \longrightarrow \text{Zn}^{+2}\text{SO}_4 + \text{H}_2\text{S}^{-2}\uparrow + \text{H}_2\text{O}$

$$\begin{array}{l} 1 \quad \text{S}^{+6} + 8\bar{e} \longrightarrow \text{S}^{-2} \\ 4 \quad \text{Zn}^0 - 2\bar{e} \longrightarrow \text{Zn}^{+2} \end{array}$$

$$5\text{H}_2\text{SO}_4(\text{конц.}) + 4\text{Zn} = 4\text{ZnSO}_4 + \text{H}_2\text{S}\uparrow + 4\text{H}_2\text{O}$$

157. $\text{H}_2\text{S}^{+6}\text{O}_4(\text{конц.}) + \text{Al}^0 \longrightarrow \text{Al}_2^{+3}(\text{SO}_4)_3 + \text{H}_2\text{S}^{-2}\uparrow + \text{H}_2\text{O}$

$$\begin{array}{l} 3 \quad \text{S}^{+6} + 8\bar{e} \longrightarrow \text{S}^{-2} \\ 4 \quad 2\text{Al}^0 - 6\bar{e} \longrightarrow 2\text{Al}^{+3} \end{array}$$

$$15\text{H}_2\text{SO}_4(\text{конц.}) + 8\text{Al} = 4\text{Al}_2(\text{SO}_4)_3 + 3\text{H}_2\text{S}\uparrow + 12\text{H}_2\text{O}$$

158. $\text{H}_2\text{S}^{+6}\text{O}_4(\text{разб.}) + \text{K}^0 \longrightarrow \text{K}_2^{+1}\text{SO}_4 + \text{S}^0\downarrow + \text{S}^{+4}\text{O}_2\uparrow + \text{H}_2\text{O}$

$$\begin{array}{l} 1 \quad \text{S}^{+6} + 2\bar{e} \longrightarrow \text{S}^{+4} \\ 1 \quad \text{S}^{+6} + 6\bar{e} \longrightarrow \text{S}^0 \\ 8 \quad \text{K}^0 - 1\bar{e} \longrightarrow \text{K}^{+1} \end{array} \quad \left. \vphantom{\begin{array}{l} 1 \\ 1 \\ 8 \end{array}} \right\} 8\bar{e}$$

$$\begin{array}{ccccccc} 6\text{H}_2\text{SO}_4(\text{разб.}) & + & 8\text{K} & = & 4\text{K}_2\text{SO}_4 & + & \text{S}\downarrow + \text{SO}_2\uparrow + 6\text{H}_2\text{O} \\ 5 & & & & 1 & 2 & 3 & 4 & 6 \end{array}$$

159. $\text{H}_2\text{SO}_4 + \text{K}_2\text{Fe}^{+6}\text{O}_4^{-2} \longrightarrow \text{Fe}_2^{+3}(\text{SO}_4)_3 + \text{O}_2^0\uparrow + \text{H}_2\text{O} + \text{K}_2\text{SO}_4$

$$\begin{array}{l} 2 \quad 2\text{Fe}^{+6} + 6\bar{e} \longrightarrow 2\text{Fe}^{+3} \\ 3 \quad 2\text{O}^{-2} - 4\bar{e} \longrightarrow \text{O}_2^0 \end{array}$$

$$\begin{array}{ccccccc} 10\text{H}_2\text{SO}_4 & + & 4\text{K}_2\text{FeO}_4 & = & 2\text{Fe}_2(\text{SO}_4)_3 & + & 3\text{O}_2 + 10\text{H}_2\text{O} + 4\text{K}_2\text{SO}_4 \\ 5 & & 1 & & 2 & 3 & 6 & 4 \end{array}$$

160. $\text{H}_2\text{S}^{+6}\text{O}_4(\text{конц.}) + \text{Fe}^0 \xrightarrow{t} \text{Fe}_2^{+3}(\text{SO}_4)_3 + \text{S}^{+4}\text{O}_2\uparrow + \text{H}_2\text{O}$

$$\begin{array}{l} 3 \quad \text{S}^{+6} + 2\bar{e} \longrightarrow \text{S}^{+4} \\ 2 \quad \text{Fe}^0 - 3\bar{e} \longrightarrow \text{Fe}^{+3} \end{array}$$

$$6\text{H}_2\text{SO}_4(\text{конц.}) + 2\text{Fe} = \text{Fe}_2(\text{SO}_4)_3 + 3\text{SO}_2\uparrow + 6\text{H}_2\text{O}$$

$$\text{H}_2\text{SO}_4(\text{разб.}) + \text{Fe} \longrightarrow \text{FeSO}_4 + \text{H}_2\uparrow$$

$$\begin{array}{l} 1 \quad 2\text{H}^+ + 2\bar{e} \longrightarrow \text{H}_2^0 \\ 1 \quad \text{Fe}^0 - 2\bar{e} \longrightarrow \text{Fe}^{+2} \end{array}$$

$$\text{H}_2\text{SO}_4(\text{разб.}) + \text{Fe} = \text{FeSO}_4 + \text{H}_2\uparrow$$

161. $\text{H}_2\text{S}^{+6}\text{O}_4(\text{конц.}) + \text{C}^0 \longrightarrow \text{H}_2\text{O} + \text{C}^{+4}\text{O}_2\uparrow + \text{S}^{+4}\text{O}_2\uparrow$

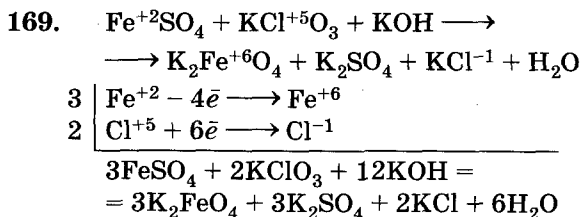
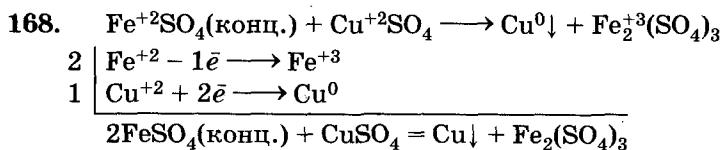
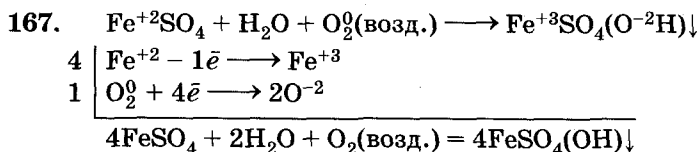
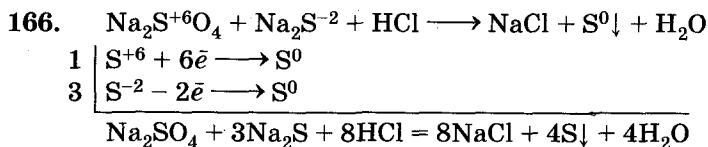
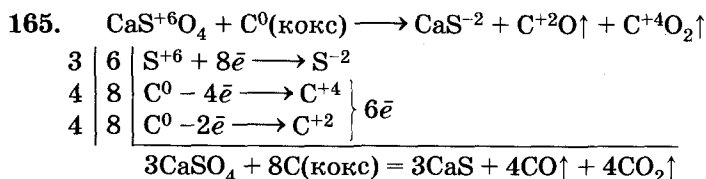
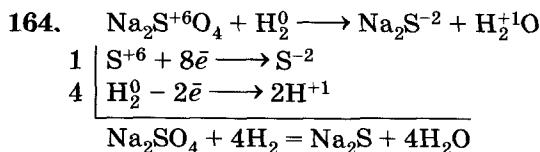
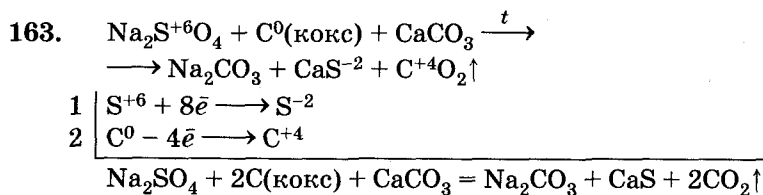
$$\begin{array}{l} 2 \quad \text{S}^{+6} + 2\bar{e} \longrightarrow \text{S}^{+4} \\ 1 \quad \text{C}^0 - 4\bar{e} \longrightarrow \text{C}^{+4} \end{array}$$

$$2\text{H}_2\text{SO}_4(\text{конц.}) + \text{C} = 2\text{H}_2\text{O} + \text{CO}_2\uparrow + 2\text{SO}_2\uparrow$$

162. $\text{Na}_2\text{S}^{+6}\text{O}_4 + \text{C}^0(\text{кокс}) \longrightarrow \text{Na}_2\text{S}^{-2} + \text{C}^{+2}\text{O}\uparrow$

$$\begin{array}{l} 1 \quad \text{S}^{+6} + 8\bar{e} \longrightarrow \text{S}^{-2} \\ 4 \quad \text{C}^0 - 2\bar{e} \longrightarrow \text{C}^{+2} \end{array}$$

$$\text{Na}_2\text{SO}_4 + 4\text{C}(\text{кокс}) = \text{Na}_2\text{S} + 4\text{CO}\uparrow$$



170. $\text{Fe}^{+2}\text{SO}_4 + \text{KMn}^{+7}\text{O}_4 + \text{H}_2\text{SO}_4 \longrightarrow$
 $\longrightarrow \text{Fe}_2^{+3}(\text{SO}_4)_3 + \text{Mn}^{+2}\text{SO}_4 + \text{H}_2\text{O} + \text{K}_2\text{SO}_4$

$$\begin{array}{l} 5 \quad 2\text{Fe}^{+2} - 2\bar{e} \longrightarrow 2\text{Fe}^{+3} \\ 2 \quad \text{Mn}^{+7} + 5\bar{e} \longrightarrow \text{Mn}^{+2} \end{array}$$

$$10\text{FeSO}_4 + 2\text{KMnO}_4 + 8\text{H}_2\text{SO}_4 =$$

$$= 5\text{Fe}_2(\text{SO}_4)_3 + 2\text{MnSO}_4 + 8\text{H}_2\text{O} + \text{K}_2\text{SO}_4$$
171. $\text{Fe}^{+2}\text{SO}_4 + \text{KCl}^{+5}\text{O}_3 + \text{H}_2\text{SO}_4 \longrightarrow \text{Fe}_2^{+3}(\text{SO}_4)_3 + \text{KCl}^{-1} + \text{H}_2\text{O}$

$$\begin{array}{l} 3 \quad 2\text{Fe}^{+2} - 2\bar{e} \longrightarrow 2\text{Fe}^{+3} \\ 1 \quad \text{Cl}^{+5} + 6\bar{e} \longrightarrow \text{Cl}^{-1} \end{array}$$

$$6\text{FeSO}_4 + \text{KClO}_3 + 3\text{H}_2\text{SO}_4 = 3\text{Fe}_2(\text{SO}_4)_3 + \text{KCl} + 3\text{H}_2\text{O}$$
172. $\text{Fe}^{+2}\text{SO}_4 + \text{K}_2\text{Cr}_2^{+6}\text{O}_7 + \text{H}_2\text{SO}_4 \longrightarrow$
 $\longrightarrow \text{Cr}_2^{+3}(\text{SO}_4)_3 + \text{Fe}_2^{+3}(\text{SO}_4)_3 + \text{K}_2\text{SO}_4 + \text{H}_2\text{O}$

$$\begin{array}{l} 3 \quad 2\text{Fe}^{+2} - 2\bar{e} \longrightarrow 2\text{Fe}^{+3} \\ 1 \quad 2\text{Cr}^{+6} + 6\bar{e} \longrightarrow 2\text{Cr}^{+3} \end{array}$$

$$6\text{FeSO}_4 + \text{K}_2\text{Cr}_2\text{O}_7 + 7\text{H}_2\text{SO}_4 =$$

$$= \text{Cr}_2(\text{SO}_4)_3 + 3\text{Fe}_2(\text{SO}_4)_3 + \text{K}_2\text{SO}_4 + 7\text{H}_2\text{O}$$
173. $\text{Fe}^{+2}\text{SO}_4 + \text{O}_2^0 + \text{H}_2\text{SO}_4 \longrightarrow \text{Fe}_2^{+3}(\text{SO}_4)_3 + \text{H}_2\text{O}^{-2}$

$$\begin{array}{l} 2 \quad 2\text{Fe}^{+2} - 2\bar{e} \longrightarrow 2\text{Fe}^{+3} \\ 1 \quad \text{O}_2^0 + 4\bar{e} \longrightarrow 2\text{O}^{-2} \end{array}$$

$$4\text{FeSO}_4 + \text{O}_2 + 2\text{H}_2\text{SO}_4 = 2\text{Fe}_2(\text{SO}_4)_3 + 2\text{H}_2\text{O}$$
174. $\text{CuS}^{+6}\text{O}_4^{-2} \longrightarrow \text{CuO} + \text{S}^{+4}\text{O}_2\uparrow + \text{O}_2^0\uparrow$

$$\begin{array}{l} 2 \quad \text{S}^{+6} + 2\bar{e} \longrightarrow \text{S}^{+4} \\ 1 \quad 2\text{O}^{-2} - 4\bar{e} \longrightarrow \text{O}_2^0 \end{array}$$

$$2\text{CuSO}_4 = 2\text{CuO} + 2\text{SO}_2\uparrow + \text{O}_2\uparrow$$
175. $\text{Al}_2(\text{S}^{+6}\text{O}_4^{-2})_3 \longrightarrow \text{Al}_2\text{O}_3 + \text{S}^{+4}\text{O}_2\uparrow + \text{O}_2^0\uparrow$

$$\begin{array}{l} 2 \quad 3\text{S}^{+6} + 6\bar{e} \longrightarrow 3\text{S}^{+4} \\ 3 \quad 2\text{O}^{-2} - 4\bar{e} \longrightarrow \text{O}_2^0 \end{array}$$

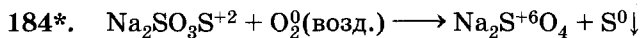
$$2\text{Al}_2(\text{SO}_4)_3 = 2\text{Al}_2\text{O}_3 + 6\text{SO}_2\uparrow + 3\text{O}_2\uparrow$$
176. $\text{Mn}^{+2}\text{SO}_4 + \text{NaBi}^{+5}\text{O}_3 + \text{HNO}_3 \longrightarrow$
 $\longrightarrow \text{HMn}^{+7}\text{O}_4 + \text{Bi}^{+3}(\text{NO}_3)_3 + \text{Na}_2\text{SO}_4 + \text{NaNO}_3 + \text{H}_2\text{O}$

$$\begin{array}{l} 2 \quad \text{Mn}^{+2} - 5\bar{e} \longrightarrow \text{Mn}^{+7} \\ 5 \quad \text{Bi}^{+5} + 2\bar{e} \longrightarrow \text{Bi}^{+3} \end{array}$$

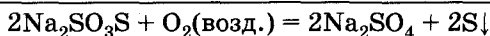
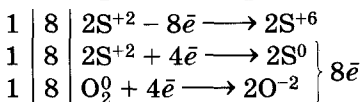
$$2\text{MnSO}_4 + 5\text{NaBiO}_3 + 16\text{HNO}_3 =$$

$$= 2\text{HMnO}_4 + 5\text{Bi}(\text{NO}_3)_3 + 2\text{Na}_2\text{SO}_4 + \text{NaNO}_3 + 7\text{H}_2\text{O}$$

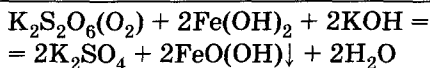
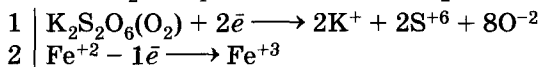
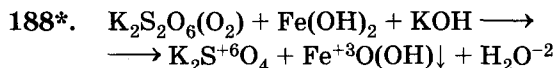
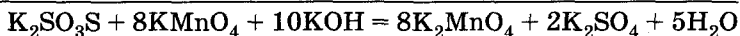
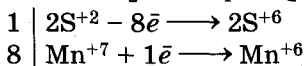
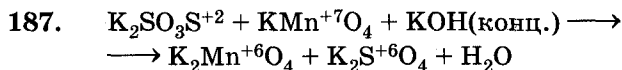
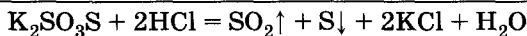
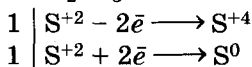
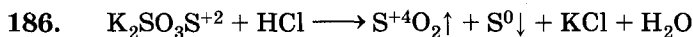
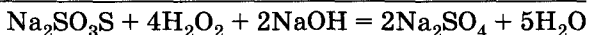
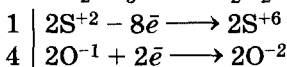
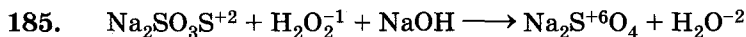
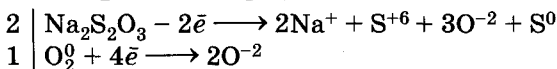
177. $\text{Cr}_2^{+3}(\text{SO}_4)_3 + \text{KMn}^{+7}\text{O}_4 + \text{KOH} \longrightarrow$
 $\longrightarrow \text{K}_2\text{Cr}^{+6}\text{O}_4 + \text{Mn}^{+4}\text{O}_2\downarrow + \text{K}_2\text{SO}_4 + \text{H}_2\text{O}$
 $\begin{array}{l} 1 \quad 2\text{Cr}^{+3} - 6\bar{e} \longrightarrow 2\text{Cr}^{+6} \\ 2 \quad \text{Mn}^{+7} + 3\bar{e} \longrightarrow \text{Mn}^{+4} \end{array}$
 $\text{Cr}_2(\text{SO}_4)_3 + 2\text{KMnO}_4 + 8\text{KOH} =$
 $= 2\text{K}_2\text{CrO}_4 + 2\text{MnO}_2\downarrow + 3\text{K}_2\text{SO}_4 + 4\text{H}_2\text{O}$
178. $\text{Ti}_2^{+3}(\text{SO}_4)_3 + \text{KCl}^{+5}\text{O}_3 + \text{H}_2\text{O} \longrightarrow \text{Ti}^{+4}\text{SO}_4\text{O} + \text{KCl}^{-1} + \text{H}_2\text{SO}_4$
 $\begin{array}{l} 3 \quad 2\text{Ti}^{+3} - 2\bar{e} \longrightarrow 2\text{Ti}^{+4} \\ 1 \quad \text{Cl}^{+5} + 6\bar{e} \longrightarrow \text{Cl}^{-1} \end{array}$
 $3\text{Ti}_2(\text{SO}_4)_3 + \text{KClO}_3 + 3\text{H}_2\text{O} = 6\text{TiSO}_4\text{O} + \text{KCl} + 3\text{H}_2\text{SO}_4$
179. $\text{Na}_2\text{S}^{+4}\text{O}_3 + \text{KI}^{+5}\text{O}_3 + \text{H}_2\text{SO}_4 \longrightarrow$
 $\longrightarrow \text{I}_2^0\downarrow + \text{Na}_2\text{S}^{+6}\text{O}_4 + \text{K}_2\text{S}^{+6}\text{O}_4 + \text{H}_2\text{O}$
 $\begin{array}{l} 5 \quad \text{S}^{+4} - 2\bar{e} \longrightarrow \text{S}^{+6} \\ 1 \quad 2\text{I}^{+5} + 10\bar{e} \longrightarrow \text{I}_2^0 \end{array}$
 $5\text{Na}_2\text{SO}_3 + 2\text{KIO}_3 + \text{H}_2\text{SO}_4 \longrightarrow \text{I}_2\downarrow + 5\text{Na}_2\text{SO}_4 + \text{K}_2\text{SO}_4 + \text{H}_2\text{O}$
180. $\text{Na}_2\text{S}^{+4}\text{O}_3 + \text{K}_2\text{Cr}_2^{+6}\text{O}_7 + \text{H}_2\text{O} \longrightarrow$
 $\longrightarrow \text{Na}_2\text{S}^{+6}\text{O}_4 + \text{KOH} + \text{Cr}^{+3}(\text{OH})_3\downarrow$
 $\begin{array}{l} 3 \quad \text{S}^{+4} - 2\bar{e} \longrightarrow \text{S}^{+6} \\ 1 \quad 2\text{Cr}^{+6} + 6\bar{e} \longrightarrow 2\text{Cr}^{+3} \end{array}$
 $3\text{Na}_2\text{SO}_3 + \text{K}_2\text{Cr}_2\text{O}_7 + 4\text{H}_2\text{O} = 3\text{Na}_2\text{SO}_4 + 2\text{KOH} + 2\text{Cr}(\text{OH})_3\downarrow$
181. $\text{Na}_2\text{S}^{+4}\text{O}_3 \xrightarrow{t} \text{Na}_2\text{S}^{-2} + \text{Na}_2\text{S}^{+6}\text{O}_4$
 $\begin{array}{l} 3 \quad \text{S}^{+4} - 2\bar{e} \longrightarrow \text{S}^{+6} \\ 1 \quad \text{S}^{+4} + 6\bar{e} \longrightarrow \text{S}^{-2} \end{array}$
 $\begin{array}{ccc} 4\text{Na}_2\text{SO}_3 & = & \text{Na}_2\text{S} + 3\text{Na}_2\text{SO}_4 \\ 3 & & 2 \quad 1 \end{array}$
182. $\text{K}_2\text{S}^{+4}\text{O}_3 + \text{KMn}^{+7}\text{O}_4 + \text{KOH}(\text{конц.}) \longrightarrow$
 $\longrightarrow \text{K}_2\text{Mn}^{+6}\text{O}_4 + \text{K}_2\text{S}^{+6}\text{O}_4 + \text{H}_2\text{O}$
 $\begin{array}{l} 1 \quad \text{S}^{+4} - 2\bar{e} \longrightarrow \text{S}^{+6} \\ 2 \quad \text{Mn}^{+7} + 1\bar{e} \longrightarrow \text{Mn}^{+6} \end{array}$
 $\text{K}_2\text{SO}_3 + 2\text{KMnO}_4 + 2\text{KOH} = 2\text{K}_2\text{MnO}_4 + \text{K}_2\text{SO}_4 + \text{H}_2\text{O}$
183. $\text{K}_2\text{S}^{+4}\text{O}_3 + \text{Ba}(\text{Cl}^{+1}\text{O})_2 \longrightarrow \text{BaS}^{+6}\text{O}_4 + \text{K}_2\text{SO}_4 + \text{KCl}^{-1}$
хлорит бария
 $\begin{array}{l} 2 \quad \text{S}^{+4} - 2\bar{e} \longrightarrow \text{S}^{+6} \\ 1 \quad 2\text{Cl}^{+1} + 4\bar{e} \longrightarrow 2\text{Cl}^{-1} \end{array}$
 $2\text{K}_2\text{SO}_3 + \text{Ba}(\text{ClO})_2 = \text{BaSO}_4\downarrow + \text{K}_2\text{SO}_4 + 2\text{KCl}$



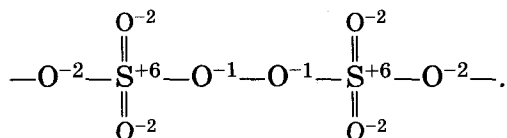
1-й вариант оформления ЭБ.



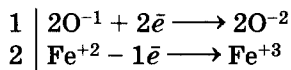
2-й вариант оформления ЭБ.



Пояснение. $\text{K}_2\text{S}_2\text{O}_8$, или $\text{K}_2\text{S}_2\text{O}_6(\text{O}_2)$, — пероксодисульфат калия, соль пероксодисерной (надсерной) кислоты — $\text{H}_2\text{S}_2\text{O}_8$. Структурная формула пероксодисульфат-иона:

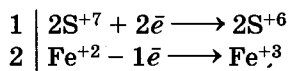


Исходя из указанных с. о. элементов схемы ЭБ можно составить следующим образом:

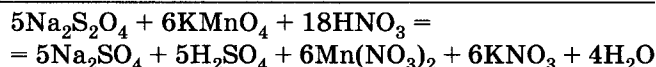
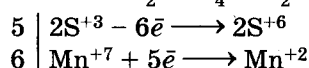
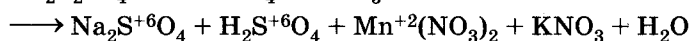
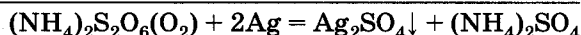
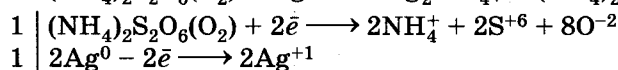
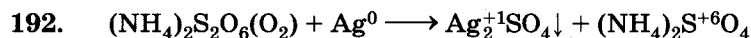
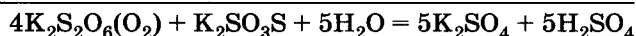
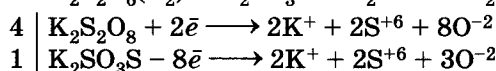
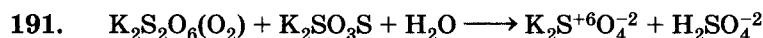
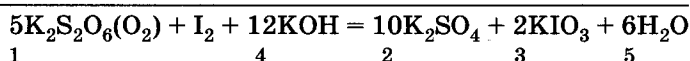
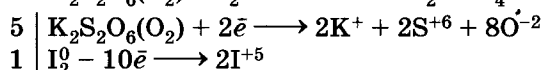
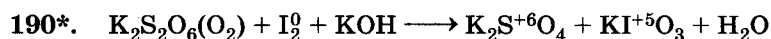
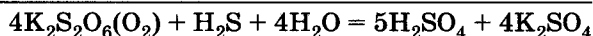
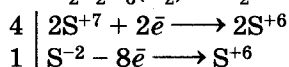
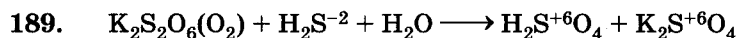


Степени окисления можно рассчитать и обычным способом: $\text{K}_2^{+1}\text{S}_2^x\text{O}_8^{-2}$; $2 \cdot (+1) + 2x + 8 \cdot (-2) = 0$; $2x = 14$; $x = +7$.

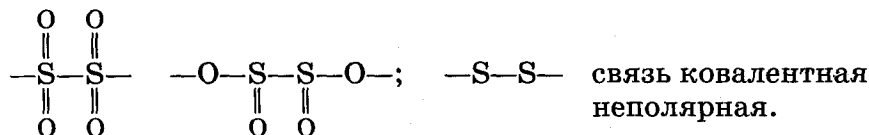
С учетом этих степеней окисления ЭБ можно оформить так:



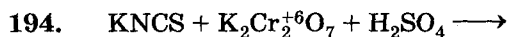
Для подбора стехиометрических коэффициентов в уравнении ОВР можно использовать оба варианта ЭБ.



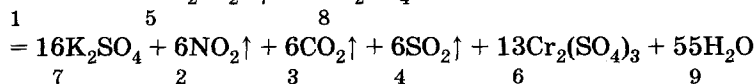
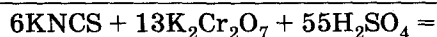
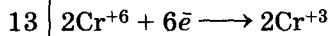
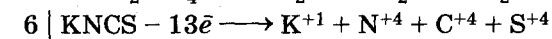
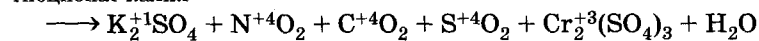
$\text{Na}_2\text{S}_2\text{O}_4$ — дитионит натрия. Возможные структурные формулы дитионит-иона:



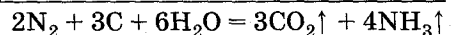
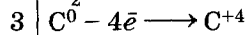
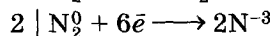
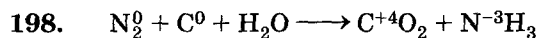
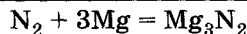
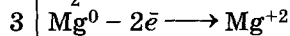
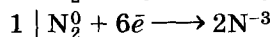
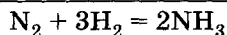
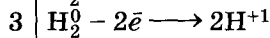
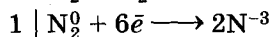
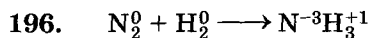
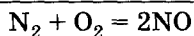
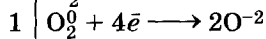
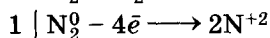
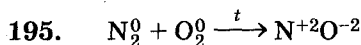
Степень окисления серы в $\text{Na}_2^{+1}\text{S}_2^x\text{O}_4^{-2} + 3(2 \cdot (+1) + 2x + 4 \cdot (-2) = 0; x = +3); \text{Na}_2^{+1}\text{S}_2^{+3}\text{O}_4^{-2}$.

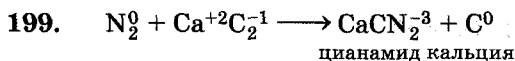


тиоционат калия

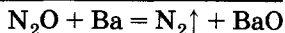
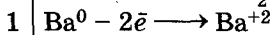
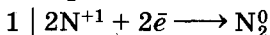
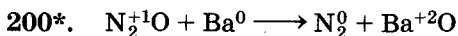
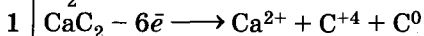
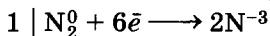


Азот и его соединения

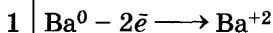
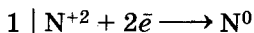




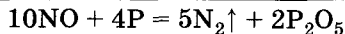
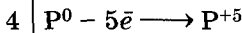
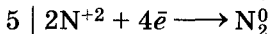
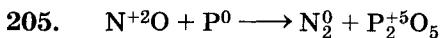
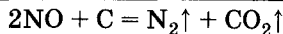
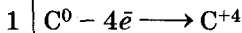
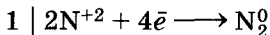
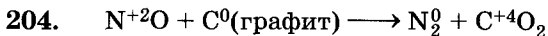
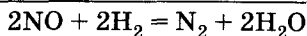
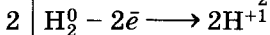
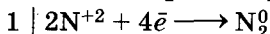
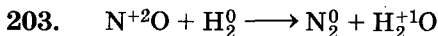
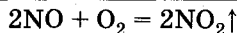
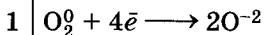
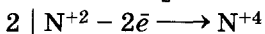
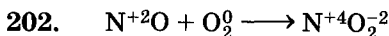
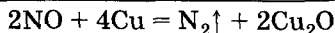
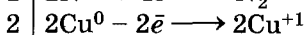
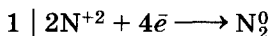
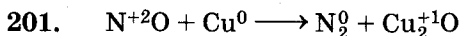
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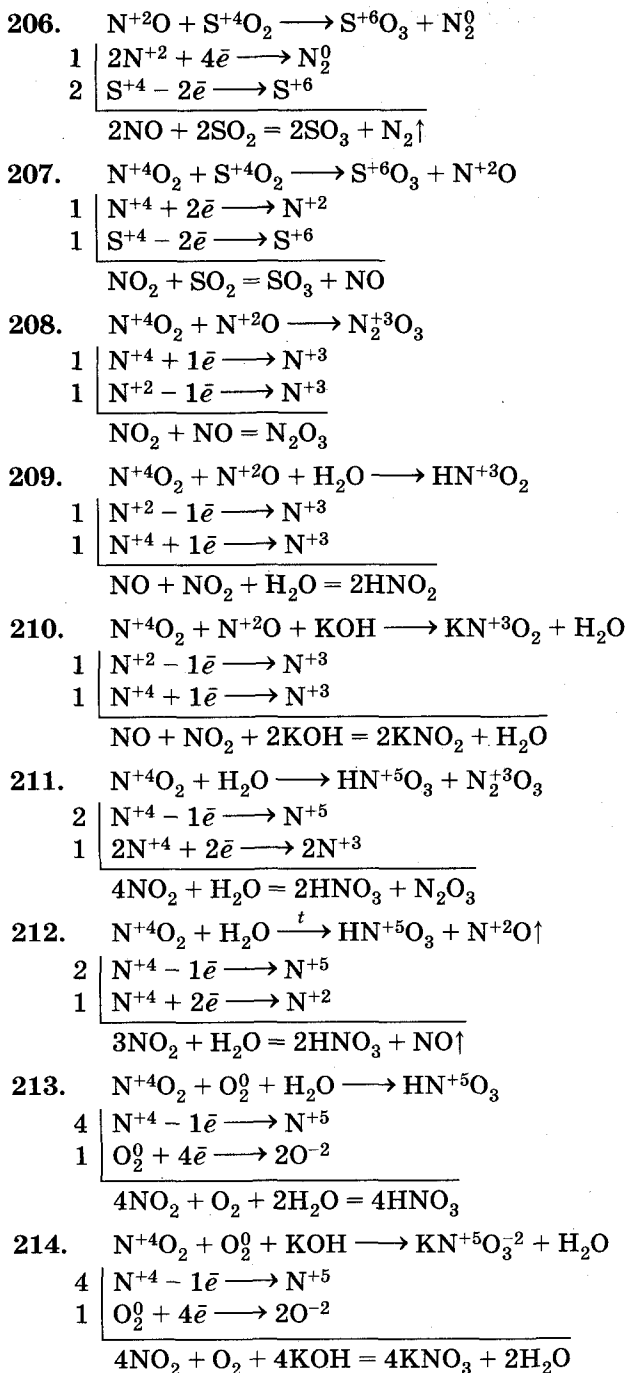


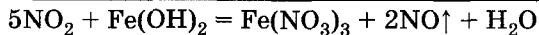
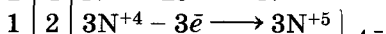
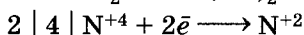
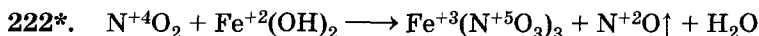
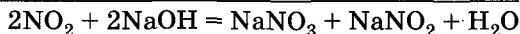
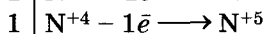
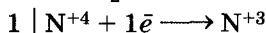
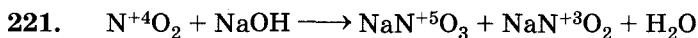
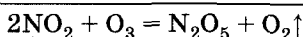
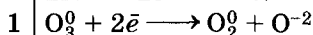
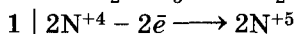
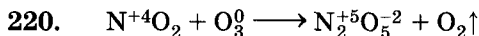
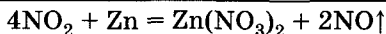
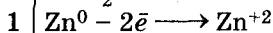
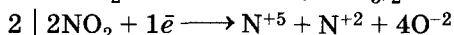
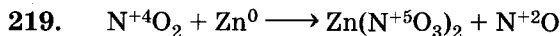
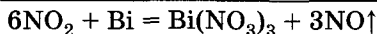
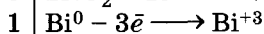
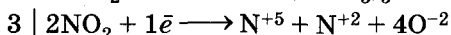
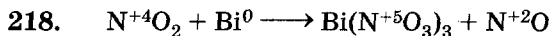
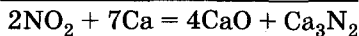
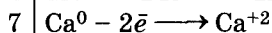
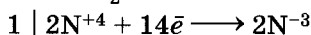
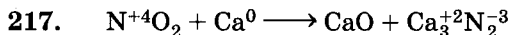
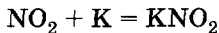
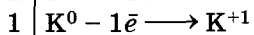
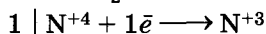
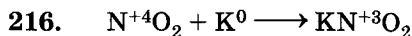
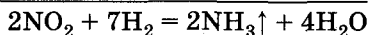
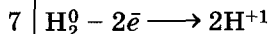
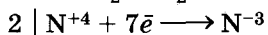
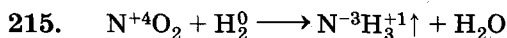
П о я с н е н и е. Структурная формула оксида азота (I)
 N_2O : $\text{N}^0 - \text{N}^{+2} = \text{O}^{-2}$. С учетом этих с. о. ЭБ оформляют так:



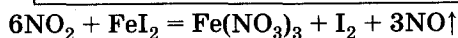
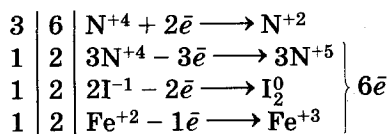
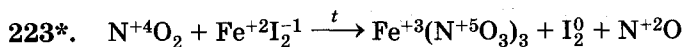
N^0 из N_2O не принимает участия в ОВР и без изменения с. о. переходит в N_2^0 .



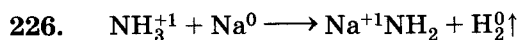
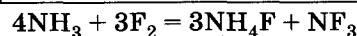
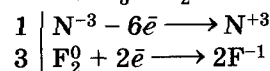
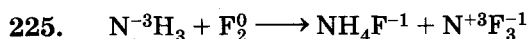
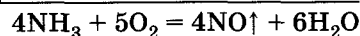
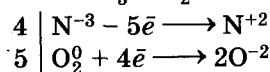
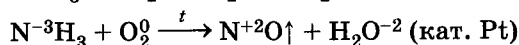
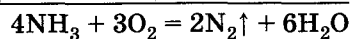
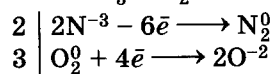
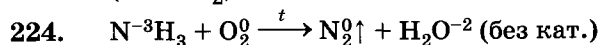




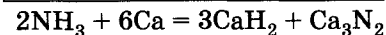
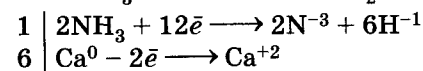
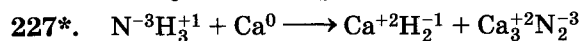
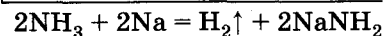
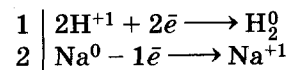
П о я с н е н и е. $2 \cdot (1\text{N}^{+4}) + 1 \cdot (3\text{N}^{+4}) = 5\text{N}^{+4}(5\text{N}^{+4}\text{O}_2)$.



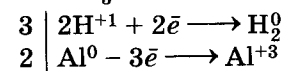
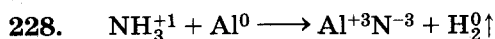
П о я с н е н и е. После сокращения множителей общее число N^{+4} рассчитывают так: $3 \cdot (1\text{N}^{+4}) + 1 \cdot (3\text{N}^{+4}) = 6\text{N}^{+4}(6\text{N}^{+4}\text{O}_2)$.

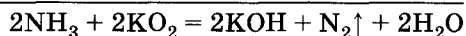
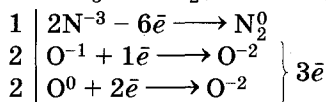
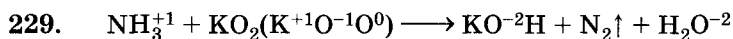


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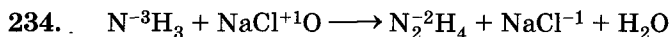
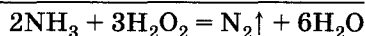
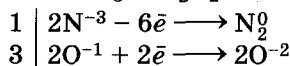
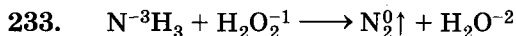
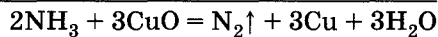
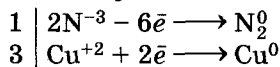
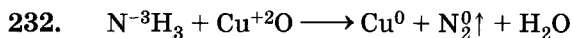
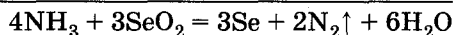
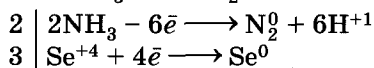
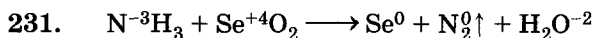
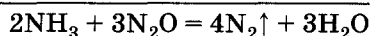
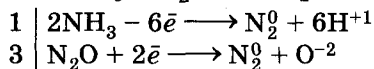
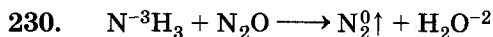


П о я с н е н и е. Формально NH_3 за счет атомов водорода выступает в роли окислителя.

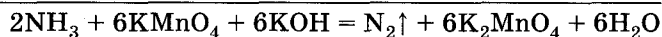
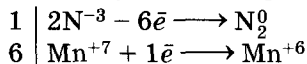
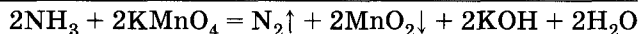
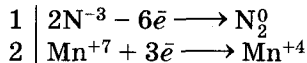
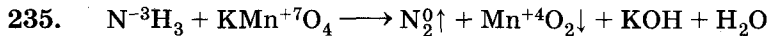
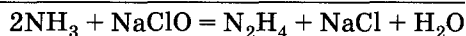
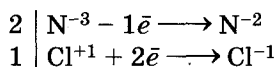


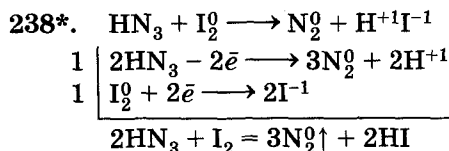
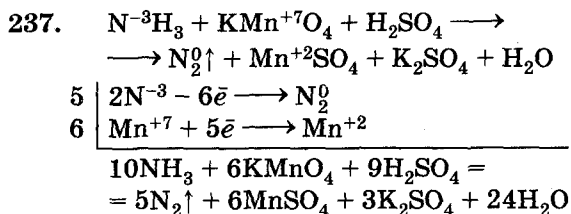


См. также задачу 119.

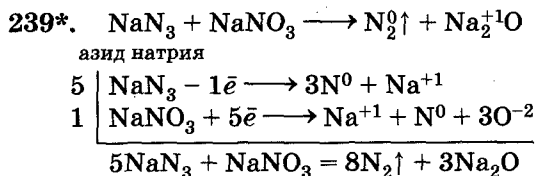


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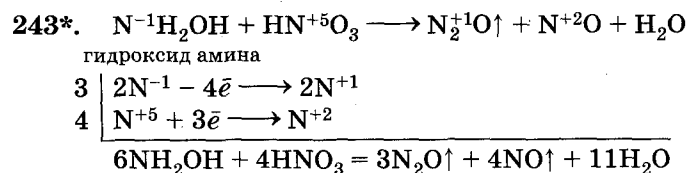
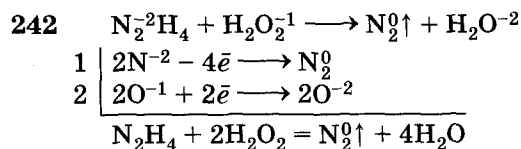
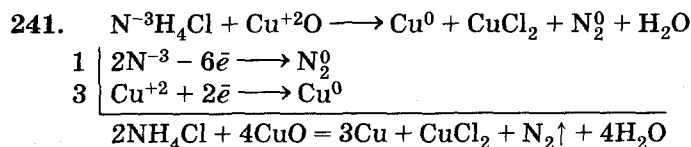
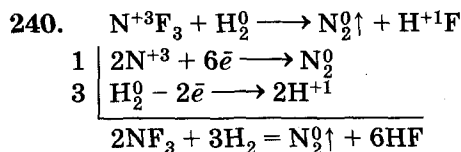


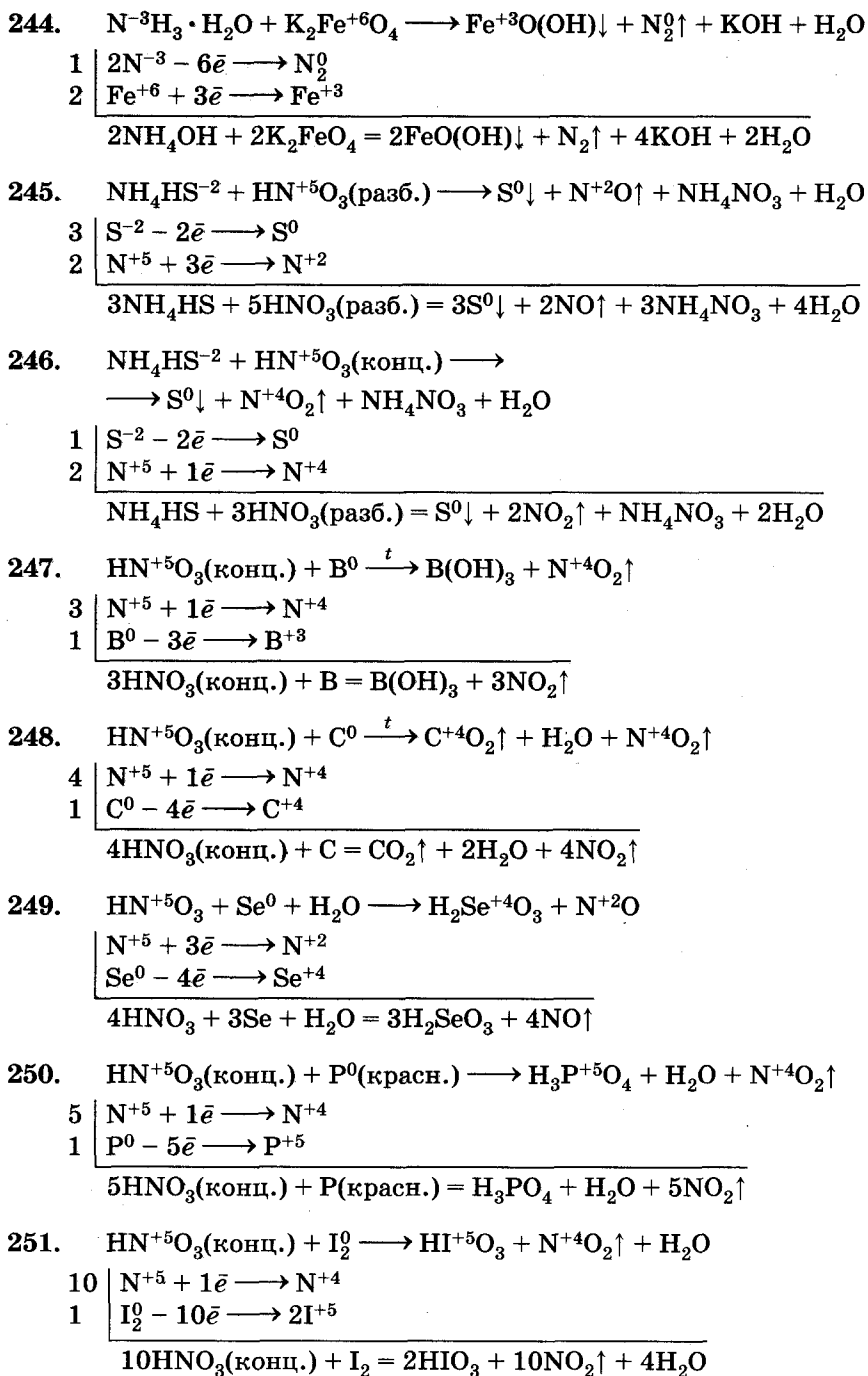


HN_3 — азотистоводородная кислота, соли этой кислоты называют азидами.



Пояснение. Коэффициент 8 перед N_2 рассчитывают так: $5 \cdot (3\text{N}^0) + 1 \cdot (1\text{N}^0) = 16\text{N}^0$, или 8N_2^0 .





252. $\text{HN}^{+5}\text{O}_3(\text{конц.}) + \text{Cu}^0 \longrightarrow \text{Cu}^{+2}(\text{NO}_3)_2 + \text{N}^{+4}\text{O}_2\uparrow + \text{H}_2\text{O}$

$$\begin{array}{l} 2 \left| \text{N}^{+5} + 1\bar{e} \longrightarrow \text{N}^{+4} \right. \\ 1 \left| \text{Cu}^0 - 2\bar{e} \longrightarrow \text{Cu}^{+2} \right. \end{array}$$

 $4\text{HNO}_3(\text{конц.}) + \text{Cu} = \text{Cu}(\text{NO}_3)_2 + 2\text{NO}_2\uparrow + 2\text{H}_2\text{O}$
253. $\text{HN}^{+5}\text{O}_3(\text{разб.}) + \text{Cu}^0 \longrightarrow \text{Cu}^{+2}(\text{NO}_3)_2 + \text{N}^{+2}\text{O}\uparrow + \text{H}_2\text{O}$

$$\begin{array}{l} 2 \left| \text{N}^{+5} + 3\bar{e} \longrightarrow \text{N}^{+2} \right. \\ 3 \left| \text{Cu}^0 - 2\bar{e} \longrightarrow \text{Cu}^{+2} \right. \end{array}$$

 $8\text{HNO}_3(\text{разб.}) + 3\text{Cu} = 3\text{Cu}(\text{NO}_3)_2 + 2\text{NO}\uparrow + 4\text{H}_2\text{O}$
254. $\text{HN}^{+5}\text{O}_3(\text{оч. разб.}) + \text{Al}^0 \longrightarrow \text{N}^{-3}\text{H}_4\text{NO}_3 + \text{Al}^{+3}(\text{NO}_3)_3 + \text{H}_2\text{O}$

$$\begin{array}{l} 3 \left| \text{N}^{+5} + 8\bar{e} \longrightarrow \text{N}^{-3} \right. \\ 8 \left| \text{Al}^0 - 3\bar{e} \longrightarrow \text{Al}^{+3} \right. \end{array}$$

 $30\text{HNO}_3(\text{оч. разб.}) + 8\text{Al} = 3\text{NH}_4\text{NO}_3 + 8\text{Al}(\text{NO}_3)_3 + 9\text{H}_2\text{O}$
255. $\text{HN}^{+5}\text{O}_3(\text{разб.}) + \text{Al}^0 \longrightarrow \text{Al}^{+3}(\text{NO}_3)_3 + \text{N}_2^{+1}\text{O}\uparrow + \text{H}_2\text{O}$

$$\begin{array}{l} 3 \left| 2\text{N}^{+5} + 8\bar{e} \longrightarrow 2\text{N}^{+1} \right. \\ 8 \left| \text{Al}^0 - 3\bar{e} \longrightarrow \text{Al}^{+3} \right. \end{array}$$

 $30\text{HNO}_3(\text{разб.}) + 8\text{Al} = 8\text{Al}(\text{NO}_3)_3 + 3\text{N}_2\text{O}\uparrow + 15\text{H}_2\text{O}$
256. $\text{HN}^{+5}\text{O}_3(\text{конц.}) + \text{Al}^0 \xrightarrow{t} \text{Al}^{+3}(\text{NO}_3)_3 + \text{N}^{+4}\text{O}_2\uparrow + \text{H}_2\text{O}$

$$\begin{array}{l} 3 \left| \text{N}^{+5} + 1\bar{e} \longrightarrow \text{N}^{+4} \right. \\ 1 \left| \text{Al}^0 - 3\bar{e} \longrightarrow \text{Al}^{+3} \right. \end{array}$$

 $6\text{HNO}_3(\text{конц.}) + \text{Al} = \text{Al}(\text{NO}_3)_3 + 3\text{NO}_2\uparrow + 3\text{H}_2\text{O}$
257. $\text{HN}^{+5}\text{O}_3(\text{разб.}) + \text{Fe}^0 \longrightarrow \text{Fe}^{+3}(\text{NO}_3)_3 + \text{N}^{+2}\text{O}\uparrow + \text{H}_2\text{O}$

$$\begin{array}{l} 1 \left| \text{N}^{+5} + 3\bar{e} \longrightarrow \text{N}^{+2} \right. \\ 1 \left| \text{Fe}^0 - 3\bar{e} \longrightarrow \text{Fe}^{+3} \right. \end{array}$$

 $4\text{HNO}_3(\text{разб.}) + \text{Fe} = \text{Fe}(\text{NO}_3)_3 + \text{NO}\uparrow + 2\text{H}_2\text{O}$
258. $\text{HN}^{+5}\text{O}_3(\text{разб.}) + \text{Ca}^0 \longrightarrow$
 $\text{N}_2^{+1}\text{O}\uparrow + \text{Ca}^{+2}(\text{NO}_3)_2 + \text{H}_2\text{O} (\text{или } \text{N}_2\uparrow)$

$$\begin{array}{l} 1 \left| 2\text{N}^{+5} + 8\bar{e} \longrightarrow 2\text{N}^{+1} \right. \\ 4 \left| \text{Ca}^0 - 2\bar{e} \longrightarrow \text{Ca}^{+2} \right. \end{array}$$

 $10\text{HNO}_3(\text{разб.}) + 4\text{Ca} = \text{N}_2\text{O} + 4\text{Ca}(\text{NO}_3)_2 + 5\text{H}_2\text{O}$
259. $\text{HN}^{+5}\text{O}_3(\text{оч. разб.}) + \text{Ca}^0 \longrightarrow \text{N}^{-3}\text{H}_4\text{NO}_3 + \text{Ca}^{+2}(\text{NO}_3)_2 + \text{H}_2\text{O}$

$$\begin{array}{l} 1 \left| \text{N}^{+5} + 8\bar{e} \longrightarrow \text{N}^{-3} \right. \\ 4 \left| \text{Ca}^0 - 2\bar{e} \longrightarrow \text{Ca}^{+2} \right. \end{array}$$

 $10\text{HNO}_3(\text{оч. разб.}) + 4\text{Ca} = \text{NH}_4\text{NO}_3 + 4\text{Ca}(\text{NO}_3)_2 + 3\text{H}_2\text{O}$

260. $\text{HN}^{+5}\text{O}_3(\text{конц.}) + \text{HCl}(\text{конц.}) + \text{Pt}^0 \longrightarrow$
 $\longrightarrow \text{N}^{+2}\text{O}\uparrow + \text{H}_2[\text{Pt}^{+4}\text{Cl}_6] + \text{H}_2\text{O}$

$$\begin{array}{l} 4 \left| \text{N}^{+5} + 3\bar{e} \longrightarrow \text{N}^{+2} \right. \\ 3 \left| \text{Pt}^0 - 4\bar{e} \longrightarrow \text{Pt}^{+4} \right. \end{array}$$

$$4\text{HNO}_3(\text{конц.}) + 18\text{HCl}(\text{конц.}) + 3\text{Pt} =$$

$$= 4\text{NO}\uparrow + 3\text{H}_2[\text{PtCl}_6] + 8\text{H}_2\text{O}$$
261. $\text{HN}^{+5}\text{O}_3(\text{конц.}) + \text{Te}^0 \longrightarrow \text{Te}^{+4}\text{O}_2 + \text{N}^{+4}\text{O}_2\uparrow + \text{H}_2\text{O}$

$$\begin{array}{l} 4 \left| \text{N}^{+5} + 1\bar{e} \longrightarrow \text{N}^{+4} \right. \\ 1 \left| \text{Te}^0 - 4\bar{e} \longrightarrow \text{Te}^{+4} \right. \end{array}$$

$$4\text{HNO}_3(\text{конц.}) + \text{Te} = \text{TeO}_2 + 4\text{NO}_2\uparrow + 2\text{H}_2\text{O}$$
262. $\text{HN}^{+5}\text{O}_3(\text{конц.}) + \text{As}^0 \longrightarrow \text{H}_3\text{As}^{+5}\text{O}_4 + \text{N}^{+4}\text{O}_2\uparrow + \text{H}_2\text{O}$

$$\begin{array}{l} 5 \left| \text{N}^{+5} + 1\bar{e} \longrightarrow \text{N}^{+4} \right. \\ 1 \left| \text{As}^0 - 5\bar{e} \longrightarrow \text{As}^{+5} \right. \end{array}$$

$$5\text{HNO}_3(\text{конц.}) + \text{As} = \text{H}_3\text{AsO}_4 + 5\text{NO}_2\uparrow + \text{H}_2\text{O}$$
263. $\text{HN}^{+5}\text{O}_3(\text{конц.}) + \text{Fe}^{+2}\text{O} \longrightarrow \text{Fe}^{+3}(\text{NO}_3)_3 + \text{N}^{+4}\text{O}_2\uparrow + \text{H}_2\text{O}$

$$\begin{array}{l} 1 \left| \text{N}^{+5} + 1\bar{e} \longrightarrow \text{N}^{+4} \right. \\ 1 \left| \text{Fe}^{+2} - 1\bar{e} \longrightarrow \text{Fe}^{+3} \right. \end{array}$$

$$4\text{HNO}_3(\text{конц.}) + \text{FeO} = \text{Fe}(\text{NO}_3)_3 + \text{NO}_2\uparrow + 2\text{H}_2\text{O}$$
264. $\text{HN}^{+5}\text{O}_3(\text{конц.}) + (\text{Fe}^{+2}\text{Fe}^{+3})\text{O}_4 \longrightarrow \text{Fe}^{+3}(\text{NO}_3)_3 + \text{N}^{+4}\text{O}_2\uparrow + \text{H}_2\text{O}$

$$\begin{array}{l} 1 \left| \text{N}^{+5} + 1\bar{e} \longrightarrow \text{N}^{+4} \right. \\ 1 \left| \text{Fe}^{+2} - 1\bar{e} \longrightarrow \text{Fe}^{+3} \right. \end{array}$$

$$10\text{HNO}_3(\text{конц.}) + (\text{FeFe}_2)\text{O}_4 = 3\text{Fe}(\text{NO}_3)_3 + \text{NO}_2\uparrow + 5\text{H}_2\text{O}$$
265. $\text{HN}^{+5}\text{O}_3(\text{конц.}) + \text{Fe}_3\text{C} \longrightarrow$
 $\longrightarrow \text{Fe}^{+3}(\text{NO}_3)_3 + \text{C}^{+4}\text{O}_2\uparrow + \text{N}^{+4}\text{O}_2\uparrow + \text{H}_2\text{O}$

$$\begin{array}{l} 13 \left| \text{N}^{+5} + 1\bar{e} \longrightarrow \text{N}^{+4} \right. \\ 1 \left| \text{Fe}_3\text{C} - 13\bar{e} \longrightarrow 3\text{Fe}^{+3} + \text{C}^{+4} \right. \end{array}$$

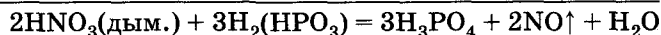
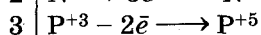
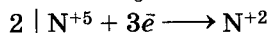
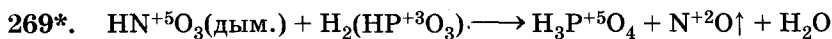
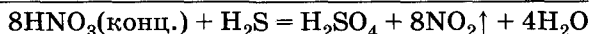
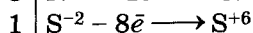
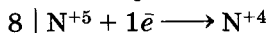
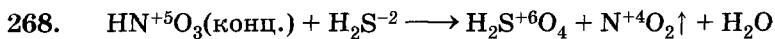
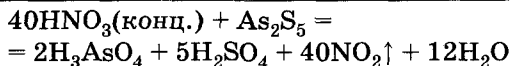
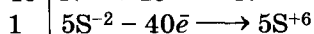
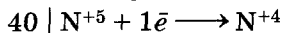
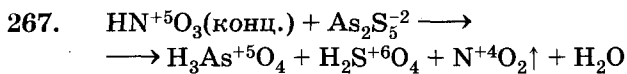
$$22\text{HNO}_3(\text{конц.}) + \text{Fe}_3\text{C} =$$

$$\begin{array}{ccccccc} 4 & & 1 & & & & \\ = & 3\text{Fe}(\text{NO}_3)_3 & + & \text{CO}_2\uparrow & + & 13\text{NO}_2\uparrow & + & 11\text{H}_2\text{O} \\ & 2 & & & & 3 & & 5 \end{array}$$
266. $\text{HN}^{+5}\text{O}_3(\text{конц.}) + \text{Cu}_2\text{S} \longrightarrow$
 $\longrightarrow \text{H}_2\text{S}^{+6}\text{O}_4 + \text{Cu}^{+2}(\text{NO}_3)_2 + \text{N}^{+4}\text{O}_2\uparrow + \text{H}_2\text{O}$

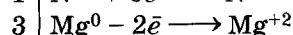
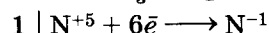
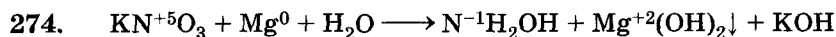
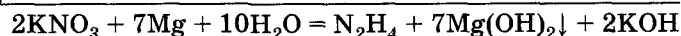
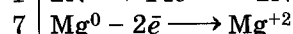
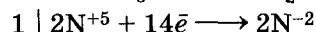
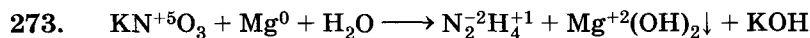
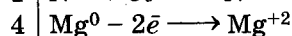
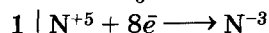
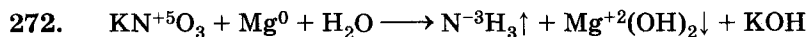
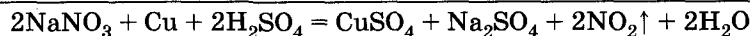
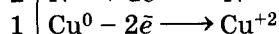
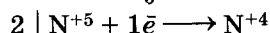
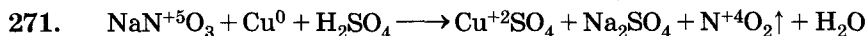
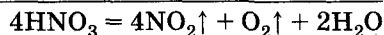
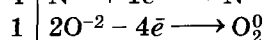
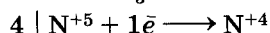
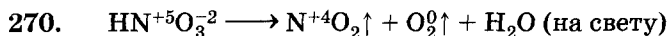
$$\begin{array}{l} 10 \left| \text{N}^{+5} + 1\bar{e} \longrightarrow \text{N}^{+4} \right. \\ 1 \left| \text{Cu}_2\text{S} - 10\bar{e} \longrightarrow 2\text{Cu}^{+2} + \text{S}^{+6} \right. \end{array}$$

$$14\text{HNO}_3(\text{конц.}) + \text{Cu}_2\text{S} =$$

$$= \text{H}_2\text{SO}_4 + 2\text{Cu}(\text{NO}_3)_2 + 10\text{NO}_2\uparrow + 6\text{H}_2\text{O}$$



$\text{H}_2(\text{PHO}_3)$, или H_3PO_3 , — фосфоновая кислота.



275. $\text{KN}^{+5}\text{O}_3 + \text{N}^{-3}\text{H}_4\text{Cl} \xrightarrow{t} \text{H}_2\text{O} + \text{KCl} + \text{N}_2^{+1}\text{O}\uparrow$

$$\begin{array}{l} 1 \quad \text{N}^{+5} + 4\bar{e} \longrightarrow \text{N}^{+1} \\ 1 \quad \text{N}^{-3} - 4\bar{e} \longrightarrow \text{N}^{+1} \end{array}$$

 $\text{KNO}_3 + \text{NH}_4\text{Cl} = 2\text{H}_2\text{O} + \text{KCl} + \text{N}_2\text{O}\uparrow$
276. $\text{KN}^{+5}\text{O}_3 + \text{Al}^0 + \text{KOH} + \text{H}_2\text{O} \longrightarrow \text{K}[\text{Al}^{+3}(\text{OH})_4] + \text{N}^{-3}\text{H}_3\uparrow$

$$\begin{array}{l} 3 \quad \text{N}^{+5} + 8\bar{e} \longrightarrow \text{N}^{-3} \\ 8 \quad \text{Al}^0 - 3\bar{e} \longrightarrow \text{Al}^{+3} \end{array}$$

 $3\text{KNO}_3 + 8\text{Al} + 5\text{KOH} + 18\text{H}_2\text{O} = 8\text{K}[\text{Al}(\text{OH})_4] + 3\text{NH}_3\uparrow$
277. $\text{KN}^{+5}\text{O}_3 + \text{Zn}^0 + \text{KOH} + \text{H}_2\text{O} \longrightarrow \text{K}_2[\text{Zn}^{+2}(\text{OH})_4] + \text{N}^{-3}\text{H}_3\uparrow$

$$\begin{array}{l} 1 \quad \text{N}^{+5} + 8\bar{e} \longrightarrow \text{N}^{-3} \\ 4 \quad \text{Zn}^0 - 2\bar{e} \longrightarrow \text{Zn}^{+2} \end{array}$$

 $\text{KNO}_3 + 4\text{Zn} + 7\text{KOH} + 6\text{H}_2\text{O} = 4\text{K}_2[\text{Zn}(\text{OH})_4] + \text{NH}_3\uparrow$
278. $\text{KN}^{+5}\text{O}_3 \xrightarrow{t} \text{O}_2^0\uparrow + \text{KN}^{+3}\text{O}_2$

$$\begin{array}{l} 2 \quad \text{N}^{+5} + 2\bar{e} \longrightarrow \text{N}^{+3} \\ 1 \quad 2\text{O}^{-2} - 4\bar{e} \longrightarrow \text{O}_2^0 \end{array}$$

 $2\text{KNO}_3 = \text{O}_2\uparrow + 2\text{KNO}_2$
279. $\text{Pb}^{+2}(\text{NO}_3)_2 + \text{H}_2\text{O}_2^{-1} + \text{NaOH} \longrightarrow \text{Pb}^{+4}\text{O}_2^{-2}\downarrow + \text{NaNO}_3 + \text{H}_2\text{O}$

$$\begin{array}{l} 1 \quad \text{Pb}^{+2} - 2\bar{e} \longrightarrow \text{Pb}^{+4} \\ 1 \quad 2\text{O}^{-1} + 2\bar{e} \longrightarrow 2\text{O}^{-2} \end{array}$$

 $\text{Pb}(\text{NO}_3)_2 + \text{H}_2\text{O}_2 + 2\text{NaOH} = \text{PbO}_2\downarrow + 2\text{NaNO}_3 + 2\text{H}_2\text{O}$
280. $\text{Pb}^{+2}(\text{NO}_3)_2 + \text{KCl}^{+1}\text{O} + \text{KOH} \xrightarrow{\text{сплав.}} \xrightarrow{\text{сплав.}} \text{K}_2\text{Pb}^{+4}\text{O}_3 + \text{KNO}_3 + \text{KCl}^{-1} + \text{H}_2\text{O}$

$$\begin{array}{l} 1 \quad \text{Pb}^{+2} - 2\bar{e} \longrightarrow \text{Pb}^{+4} \\ 1 \quad \text{Cl}^{+1} + 2\bar{e} \longrightarrow \text{Cl}^{-1} \end{array}$$

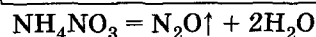
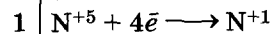
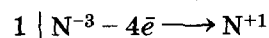
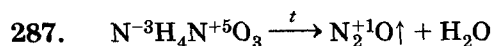
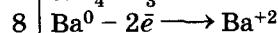
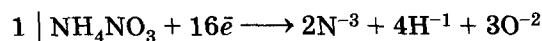
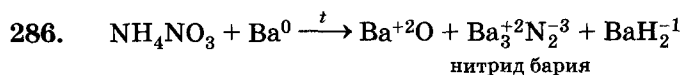
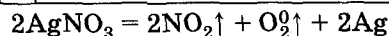
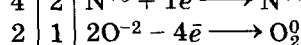
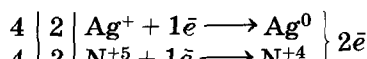
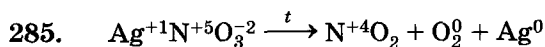
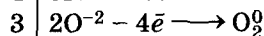
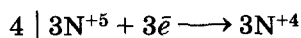
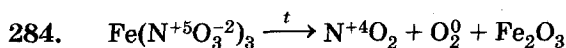
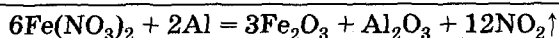
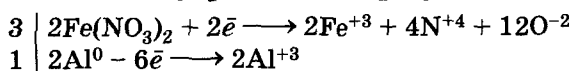
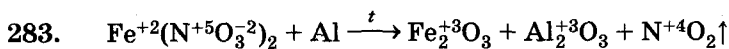
 $\text{Pb}(\text{NO}_3)_2 + \text{KClO} + 4\text{KOH} = \text{K}_2\text{PbO}_3 + 2\text{KNO}_3 + \text{KCl} + 2\text{H}_2\text{O}$
281. $\text{Mn}^{+2}(\text{NO}_3)_2 + \text{Pb}^0 + \text{HN}^{+5}\text{O}_3(\text{конц.}) \longrightarrow \longrightarrow \text{HMn}^{+7}\text{O}_4 + \text{Pb}^{+2}(\text{NO}_3)_2 + \text{H}_2\text{O} + \text{N}^{+4}\text{O}_2\uparrow$

$$\begin{array}{l} 7 \quad \text{N}^{+5} + 1\bar{e} \longrightarrow \text{N}^{+4} \\ 1 \quad \text{Mn}^{+2} - 5\bar{e} \longrightarrow \text{Mn}^{+7} \\ 1 \quad \text{Pb}^0 - 2\bar{e} \longrightarrow \text{Pb}^{+2} \end{array} \left. \vphantom{\begin{array}{l} 7 \\ 1 \\ 1 \end{array}} \right\} 7\bar{e}$$

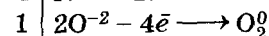
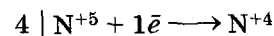
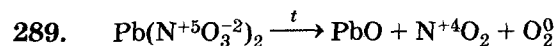
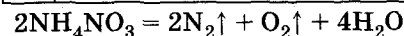
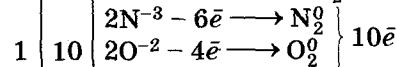
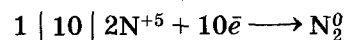
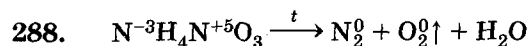
 $\text{Mn}(\text{NO}_3)_2 + \text{Pb}^0 + 7\text{HNO}_3(\text{конц.}) =$
 $= \text{HMnO}_4 + \text{Pb}(\text{NO}_3)_2 + 3\text{H}_2\text{O} + 7\text{NO}_2\uparrow$
282. $\text{Hg}^{+2}(\text{NO}_3)_2 + \text{H}_2\text{O}_2^{-1} + \text{NaOH} \longrightarrow \text{NaNO}_3 + \text{O}_2^0\uparrow + \text{H}_2\text{O} + \text{Hg}^0$

$$\begin{array}{l} 1 \quad \text{Hg}^{+2} + 2\bar{e} \longrightarrow \text{Hg}^0 \\ 1 \quad 2\text{O}^{-1} - 2\bar{e} \longrightarrow \text{O}_2^0 \end{array}$$

 $\text{Hg}(\text{NO}_3)_2 + \text{H}_2\text{O}_2 + 2\text{NaOH} = 2\text{NaNO}_3 + \text{O}_2\uparrow + 2\text{H}_2\text{O} + \text{Hg}$



оксид азота (I)



290. $\text{HN}^{+3}\text{O}_2 + \text{KMn}^{+7}\text{O}_4 + \text{H}_2\text{SO}_4 \longrightarrow$
 $\longrightarrow \text{HN}^{+5}\text{O}_3 + \text{Mn}^{+2}\text{SO}_4 + \text{K}_2\text{SO}_4 + \text{H}_2\text{O}$
 $\begin{array}{l} 5 \quad \text{N}^{+3} - 2\bar{e} \longrightarrow \text{N}^{+5} \\ 2 \quad \text{Mn}^{+7} + 5\bar{e} \longrightarrow \text{Mn}^{+2} \end{array}$

 $5\text{HNO}_2 + 2\text{KMnO}_4 + 3\text{H}_2\text{SO}_4 =$
 $= 5\text{HNO}_3 + 2\text{MnSO}_4 + \text{K}_2\text{SO}_4 + 3\text{H}_2\text{O}$
291. $\text{HN}^{+3}\text{O}_2 + \text{HMn}^{+7}\text{O}_4 \longrightarrow \text{Mn}^{+2}(\text{NO}_3)_2 + \text{HN}^{+5}\text{O}_3 + \text{H}_2\text{O}$
 $\begin{array}{l} 5 \quad \text{N}^{+3} - 2\bar{e} \longrightarrow \text{N}^{+5} \\ 2 \quad \text{Mn}^{+7} + 5\bar{e} \longrightarrow \text{Mn}^{+2} \end{array}$

 $5\text{HNO}_2 + 2\text{HMnO}_4 = 2\text{Mn}(\text{NO}_3)_2 + \text{HNO}_3 + 3\text{H}_2\text{O}$
292. $\text{HN}^{+3}\text{O}_2 \longrightarrow \text{N}^{+2}\text{O}\uparrow + \text{N}^{+4}\text{O}_2\uparrow + \text{H}_2\text{O}$
 $\begin{array}{l} 1 \quad \text{N}^{+3} + 1\bar{e} \longrightarrow \text{N}^{+2} \\ 1 \quad \text{N}^{+3} - 1\bar{e} \longrightarrow \text{N}^{+4} \end{array}$

 $2\text{HNO}_2 = \text{NO}\uparrow + \text{NO}_2\uparrow + \text{H}_2\text{O}$
293. $\text{HN}^{+3}\text{O}_2 + \text{O}_2^0(\text{возд.}) \longrightarrow \text{HN}^{+5}\text{O}_3^{-2}$
 $\begin{array}{l} 2 \quad \text{N}^{+3} - 2\bar{e} \longrightarrow \text{N}^{+5} \\ 1 \quad \text{O}_2^0 + 4\bar{e} \longrightarrow 2\text{O}^{-2} \end{array}$

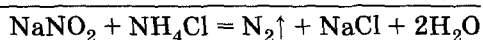
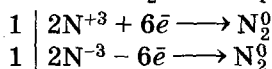
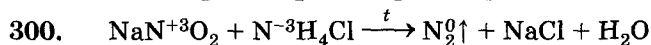
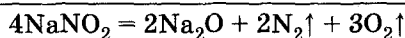
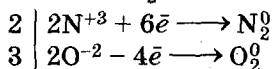
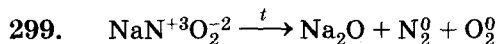
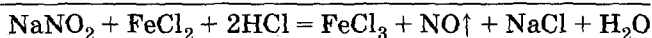
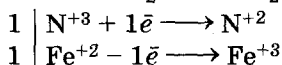
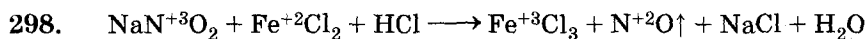
 $2\text{HNO}_2 + \text{O}_2(\text{возд.}) = 2\text{HNO}_3$
294. $\text{HN}^{+3}\text{O}_2 \longrightarrow \text{HN}^{+5}\text{O}_3 + \text{N}^{+2}\text{O} + \text{H}_2\text{O}$
 $\begin{array}{l} 1 \quad \text{N}^{+3} - 2\bar{e} \longrightarrow \text{N}^{+5} \\ 2 \quad \text{N}^{+3} + 1\bar{e} \longrightarrow \text{N}^{+2} \end{array}$

 $3\text{HNO}_2 = \text{HNO}_3 + 2\text{NO}\uparrow + \text{H}_2\text{O}$
295. $\text{HN}^{+3}\text{O}_2 + \text{Cl}_2^0 + \text{H}_2\text{O} \longrightarrow \text{HN}^{+5}\text{O}_3 + \text{HCl}^{-1}$
 $\begin{array}{l} 1 \quad \text{N}^{+3} - 2\bar{e} \longrightarrow \text{N}^{+5} \\ 1 \quad \text{Cl}_2^0 + 2\bar{e} \longrightarrow 2\text{Cl}^{-1} \end{array}$

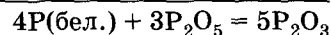
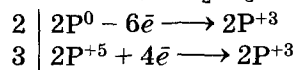
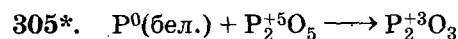
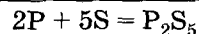
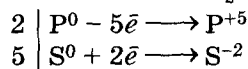
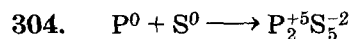
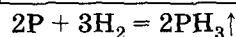
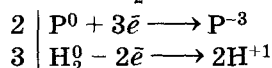
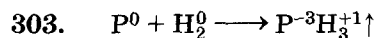
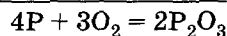
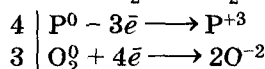
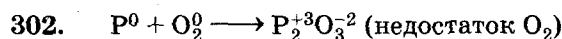
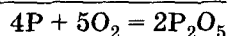
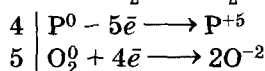
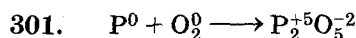
 $\text{HNO}_2 + \text{Cl}_2 + \text{H}_2\text{O} = \text{HNO}_3 + 2\text{HCl}$
296. $\text{HN}^{+3}\text{O}_2 + \text{HI}^{-1} \longrightarrow \text{I}_2^0\downarrow + \text{N}^{+2}\text{O}\uparrow + \text{H}_2\text{O}$
 $\begin{array}{l} 2 \quad \text{N}^{+3} + 1\bar{e} \longrightarrow \text{N}^{+2} \\ 1 \quad 2\text{I}^{-1} - 2\bar{e} \longrightarrow \text{I}_2^0 \end{array}$

 $2\text{HNO}_2 + 2\text{HI} = \text{I}_2\downarrow + 2\text{NO}\uparrow + 2\text{H}_2\text{O}$
297. $\text{NaN}^{+3}\text{O}_2 + \text{K}_2\text{Cr}_2^{+6}\text{O}_7 + \text{H}_2\text{SO}_4 \longrightarrow$
 $\longrightarrow \text{K}_2\text{SO}_4 + \text{NaN}^{+5}\text{O}_3 + \text{Cr}_2^{+3}(\text{SO}_4)_3 + \text{H}_2\text{O}$
 $\begin{array}{l} 3 \quad \text{N}^{+3} - 2\bar{e} \longrightarrow \text{N}^{+5} \\ 1 \quad 2\text{Cr}^{+6} + 6\bar{e} \longrightarrow 2\text{Cr}^{+3} \end{array}$

 $3\text{NaNO}_2 + \text{K}_2\text{Cr}_2\text{O}_7 + 4\text{H}_2\text{SO}_4 =$
 $= \text{K}_2\text{SO}_4 + 3\text{NaNO}_3 + \text{Cr}_2(\text{SO}_4)_3 + 4\text{H}_2\text{O}$

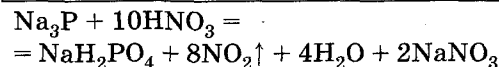
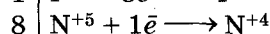
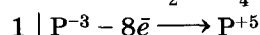
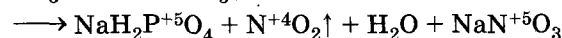
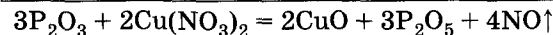
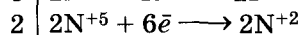
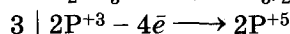
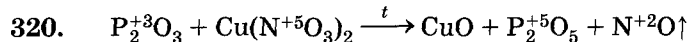
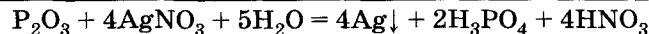
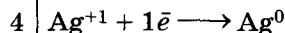
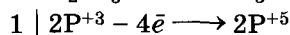
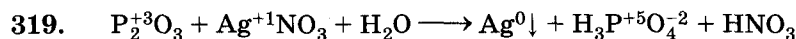
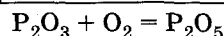
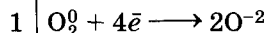
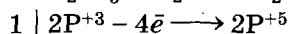
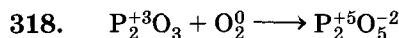
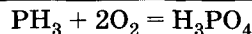
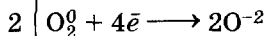
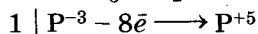
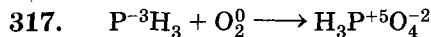
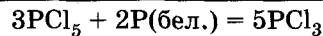
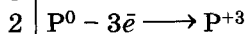
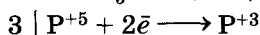
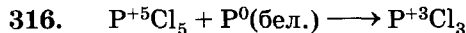
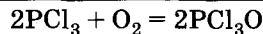
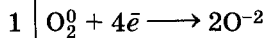
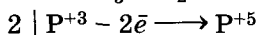
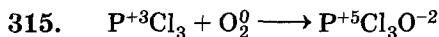
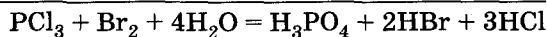
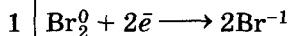
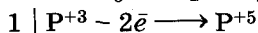
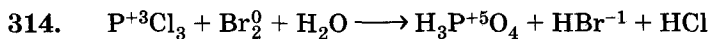


Фосфор и его соединения



Пояснение. $2 \cdot (2\text{P}^{+3}) + 3 \cdot (2\text{P}^{+3}) = 10\text{P}^{+3} (5\text{P}_2^{+3}\text{O}_3^{-2})$.

306. $P_4^0 + H_2^{+1}O \longrightarrow H_3P^{+5}O_4 + H_2^0\uparrow$
- | | |
|----|--|
| 1 | $4P^0 - 20\bar{e} \longrightarrow 4P^{+5}$ |
| 10 | $2H^{+1} + 2\bar{e} \longrightarrow H_2^0$ |
-
- $$P_4 + 16H_2O = 4H_3PO_4 + 10H_2^0\uparrow$$
307. $P^0(\text{бел.}) + KOH + H_2O \xrightarrow{t} P^{-3}H_3\uparrow + K(P^{+1}H_2O_2)$
- | | |
|---|---|
| 3 | $P^0 - 1\bar{e} \longrightarrow P^{+1}$ |
| 1 | $P^0 + 3\bar{e} \longrightarrow P^{-3}$ |
-
- $$4P(\text{бел.}) + 3KOH + 3H_2O = PH_3\uparrow + 3K(PH_2O_2)$$
308. $P^0 + Cl_2^0 \longrightarrow P^{+5}Cl_5^{-1}$
- | | |
|---|--|
| 2 | $P^0 - 5\bar{e} \longrightarrow P^{+5}$ |
| 5 | $Cl_2^0 + 2\bar{e} \longrightarrow 2Cl^{-1}$ |
-
- $$2P + 5Cl_2 = 2PCl_5$$
309. $P^0 + S^0 \longrightarrow P_2^{+3}S_3^{-2}$
- | | |
|---|---|
| 2 | $P^0 - 3\bar{e} \longrightarrow P^{+3}$ |
| 3 | $S^0 + 2\bar{e} \longrightarrow S^{-2}$ |
-
- $$2P + 3S = P_2S_3$$
310. $P^0 + Ca^0 \longrightarrow Ca_3^{+2}P_2^{-3}$
- | | |
|---|---|
| 1 | $2P^0 + 6\bar{e} \longrightarrow 2P^{-3}$ |
| 1 | $3Ca^0 - 6\bar{e} \longrightarrow 3Ca^{+2}$ |
-
- $$2P + 3Ca = Ca_3P_2$$
311. $P^0(\text{бел.}) + H_2S^{+6}O_4(\text{конц.}) \longrightarrow H_2(PHO_3) + S^{+4}O_2\uparrow$
- | | |
|---|--|
| 2 | $P^0 - 3\bar{e} \longrightarrow P^{+3}$ |
| 3 | $S^{+6} + 2\bar{e} \longrightarrow S^{+4}$ |
-
- $$2P(\text{бел.}) + 3H_2SO_4(\text{конц.}) = 2H_2(PHO_3) + 3SO_2\uparrow$$
312. $P^0 + Ba(OH)_2 + H_2O \longrightarrow P^{-3}H_3\uparrow + Ba(H_2P^{+1}O_2)_2$
фосфинат бария
- | | |
|---|---|
| 2 | $P^0 + 3\bar{e} \longrightarrow P^{-3}$ |
| 3 | $2P^0 - 2\bar{e} \longrightarrow 2P^{+1}$ |
-
- $$8P + 3Ba(OH)_2 + 6H_2O = 2PH_3\uparrow + 3Ba(H_2PO_2)_2$$
- 1 4 5 2 3
313. $P^0 + KMn^{+7}O_4 + H_2O(\text{гор.}) \longrightarrow$
 $\longrightarrow Mn^{+4}O_2 + K_2HP^{+5}O_4 + KH_2P^{+5}O_4 + KOH$
- | | |
|----|--|
| 3 | $2P^0 - 10\bar{e} \longrightarrow 2P^{+5}$ |
| 10 | $Mn^{+7} + 3\bar{e} \longrightarrow Mn^{+4}$ |
-
- $$6P + 10KMnO_4 + 5H_2O(\text{гор.}) =$$
- $$= 10MnO_2\downarrow + 3K_2HPO_4 + 3KH_2PO_4 + KOH$$



322. $\text{Ca}_3\text{P}_2^{-3} + \text{Ca}_3(\text{P}^{+5}\text{O}_4)_2 \longrightarrow \text{CaO} + \text{P}^0$

$$\begin{array}{l} 5 \quad 2\text{P}^{-3} - 6\bar{e} \longrightarrow 2\text{P}^0 \\ 3 \quad 2\text{P}^{+5} + 10\bar{e} \longrightarrow 2\text{P}^0 \end{array}$$

$$5\text{Ca}_3\text{P}_2 + 3\text{Ca}_3(\text{PO}_4)_2 = 24\text{CaO} + 16\text{P}$$
323. $\text{Ca}_3\text{P}_2^{-3} + \text{KMn}^{+7}\text{O}_4 + \text{H}_2\text{SO}_4 \longrightarrow$
 $\longrightarrow \text{Ca}_3(\text{P}^{+5}\text{O}_4)_2 + \text{Mn}^{+2}\text{SO}_4 + \text{K}_2\text{SO}_4 + \text{H}_2\text{O}$

$$\begin{array}{l} 5 \quad 2\text{P}^{-3} - 16\bar{e} \longrightarrow 2\text{P}^{+5} \\ 16 \quad \text{Mn}^{+7} + 5\bar{e} \longrightarrow \text{Mn}^{+2} \end{array}$$

$$5\text{Ca}_3\text{P}_2 + 16\text{KMnO}_4 + 24\text{H}_2\text{SO}_4 =$$

$$= 5\text{Ca}_3(\text{PO}_4)_2 + 16\text{MnSO}_4 + 8\text{K}_2\text{SO}_4 + 24\text{H}_2\text{O}$$
324. $\text{Ca}_3\text{P}_2^{-3} + \text{KMn}^{+7}\text{O}_4 + \text{H}_2\text{O} \longrightarrow \text{Ca}(\text{OH})_2 + \text{KOH} + \text{Mn}^{+4}\text{O}_2\downarrow + \text{P}^0$

$$\begin{array}{l} 1 \quad 2\text{P}^{-3} - 6\bar{e} \longrightarrow 2\text{P}^0 \\ 2 \quad \text{Mn}^{+7} + 3\bar{e} \longrightarrow \text{Mn}^{+4} \end{array}$$

$$\text{Ca}_3\text{P}_2 + 2\text{KMnO}_4 + 4\text{H}_2\text{O} = 3\text{Ca}(\text{OH})_2 + 2\text{KOH} + 2\text{MnO}_2\downarrow + 2\text{P}$$
325. $\text{Ca}_3\text{P}_2^{-3} + \text{Cl}_2^0 + \text{H}_2\text{O} \longrightarrow \text{Ca}_3(\text{P}^{+5}\text{O}_4)_2 + \text{HCl}^{-1}$

$$\begin{array}{l} 1 \quad 2\text{P}^{-3} - 16\bar{e} \longrightarrow 2\text{P}^{+5} \\ 8 \quad \text{Cl}_2^0 + 2\bar{e} \longrightarrow 2\text{Cl}^{-1} \end{array}$$

$$\text{Ca}_3\text{P}_2 + 8\text{Cl}_2 + 8\text{H}_2\text{O} = \text{Ca}_3(\text{PO}_4)_2 + 16\text{HCl}$$
- 326*. $\text{H}_3^{+1}\text{PO}_4(\text{разб.}) + \text{Fe}^0 \longrightarrow \text{Fe}^{+2}\text{HPO}_4\downarrow + \text{Fe}_3^{+2}(\text{PO}_4)_2\downarrow + \text{H}_2^0\uparrow$

$$\begin{array}{l} 1 \quad 4\text{Fe}^0 - 8\bar{e} \longrightarrow 4\text{Fe}^{+2} \\ 4 \quad 2\text{H}^{+1} + 2\bar{e} \longrightarrow \text{H}_2^0 \end{array}$$

$$3\text{H}_3\text{PO}_4(\text{разб.}) + 4\text{Fe} = \text{FeHPO}_4\downarrow + \text{Fe}_3(\text{PO}_4)_2\downarrow + 4\text{H}_2^0\uparrow$$

П о я с н е н и е. Из $3\text{H}_3\text{PO}_4$, где 9 атомов водорода, только 8 из них образуют 4H_2 , а оставшийся атом водорода без изменения с. о. переходит в состав FeHPO_4 .

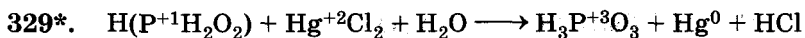
327. $\text{H}_3^{+1}\text{PO}_4(\text{разб.}) + \text{Mg}^0 \longrightarrow \text{Mg}_3^{+2}(\text{PO}_4)_2\downarrow + \text{H}_2^0\uparrow$

$$\begin{array}{l} 1 \quad 3\text{Mg}^0 - 6\bar{e} \longrightarrow 3\text{Mg}^{+2} \\ 3 \quad 2\text{H}^{+1} + 2\bar{e} \longrightarrow \text{H}_2^0 \end{array}$$

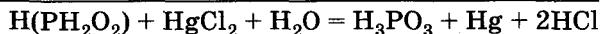
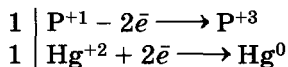
$$2\text{H}_3\text{PO}_4(\text{разб.}) + 3\text{Mg} = \text{Mg}_3(\text{PO}_4)_2\downarrow + 3\text{H}_2\uparrow$$
328. $\text{H}_2(\text{P}^{+3}\text{HO}_3) + \text{Ag}^{+1}\text{NO}_3 + \text{H}_2\text{O} \longrightarrow \text{H}_3\text{P}^{+5}\text{O}_4 + \text{Ag}^0\downarrow + \text{HNO}_3$

$$\begin{array}{l} 1 \quad \text{P}^{+3} - 2\bar{e} \longrightarrow \text{P}^{+5} \\ 2 \quad \text{Ag}^{+1} + 1\bar{e} \longrightarrow \text{Ag}^0 \end{array}$$

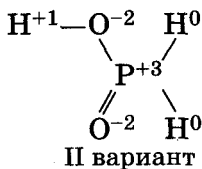
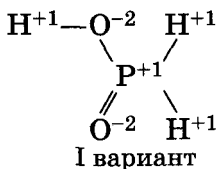
$$\text{H}_2(\text{PHO}_3) + 2\text{AgNO}_3 + \text{H}_2\text{O} = \text{H}_3\text{PO}_4 + 2\text{Ag}^0\downarrow + 2\text{HNO}_3$$



1-й вариант оформления ЭБ.

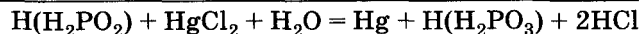
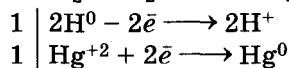
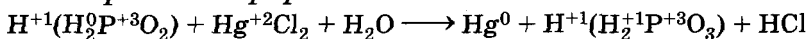


Пояснение. H_3PO_2 , или $\text{H}(\text{H}_2\text{PO}_2)$, — слабая фосфиновая кислота. Структурная формула и возможные степени окисления элементов:

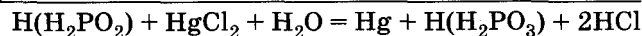
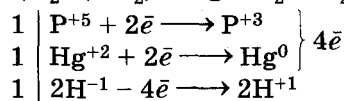
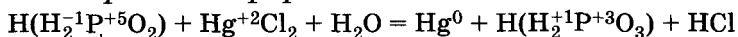


Степени окисления атома фосфора и двух атомов водорода по связи Р—Н разные, но везде электронейтральность молекулы соблюдается. Для нахождения стехиометрических коэффициентов можно использовать все три варианта степеней окисления фосфора и водорода.

2-й вариант оформления ЭБ.

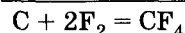
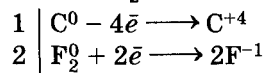
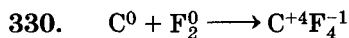


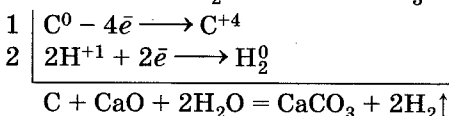
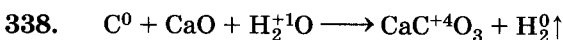
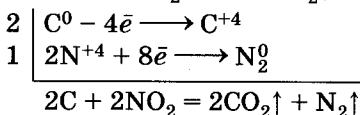
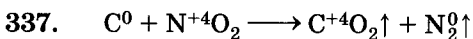
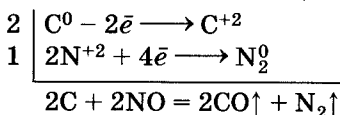
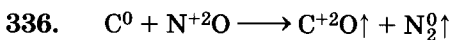
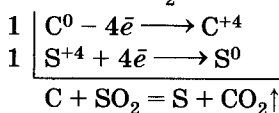
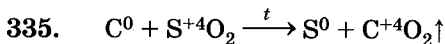
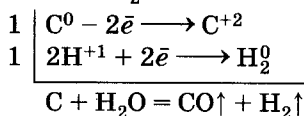
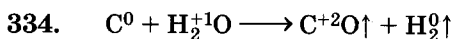
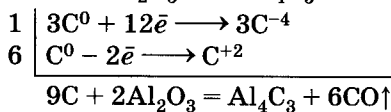
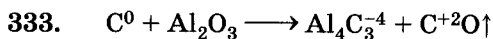
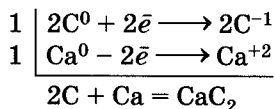
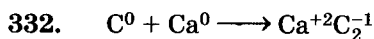
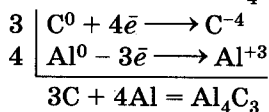
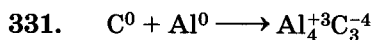
3-й вариант оформления ЭБ.

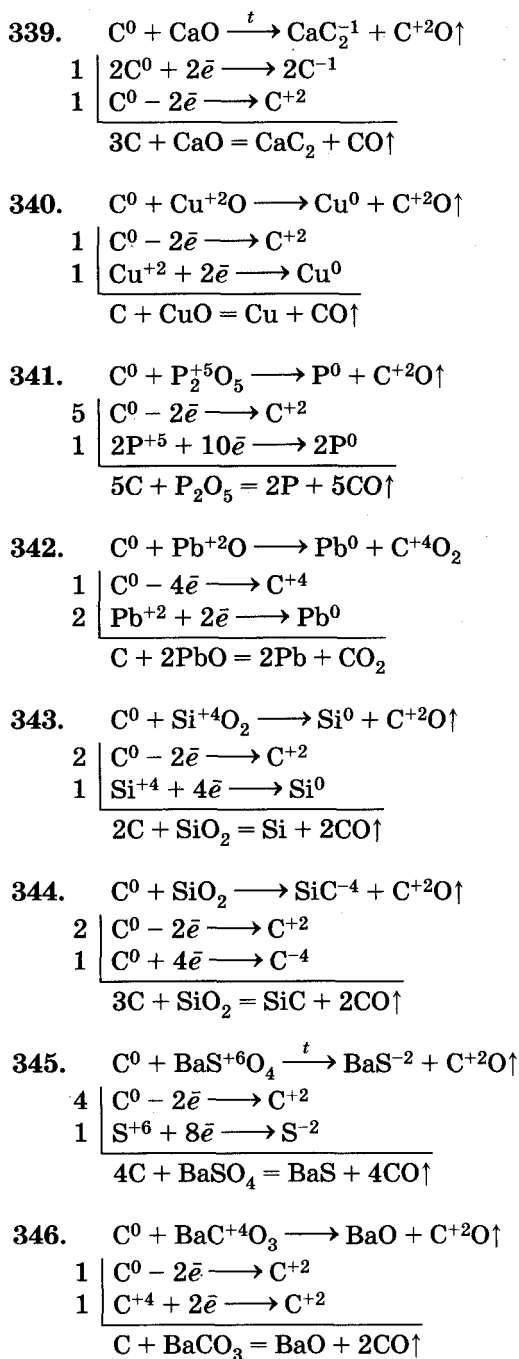


Пояснение. Приведенными примерами еще раз подчеркнем формальность понятия «степень окисления».

Углерод, кремний и их соединения

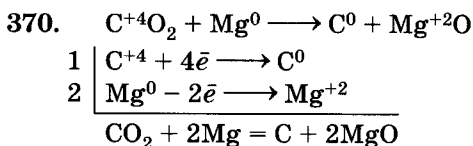
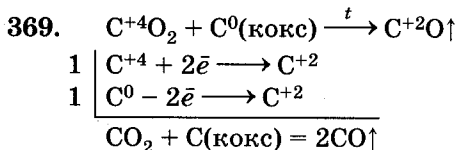
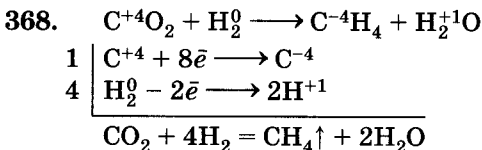
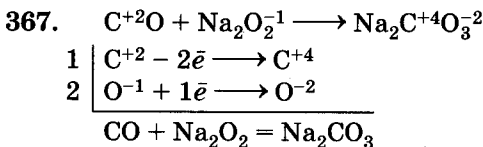
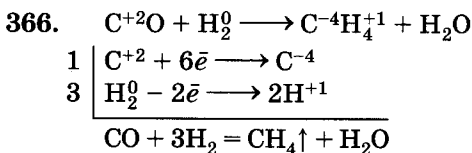
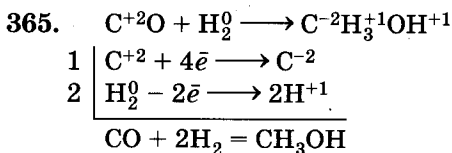
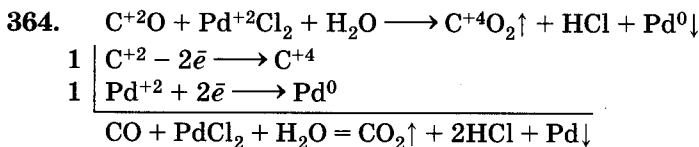
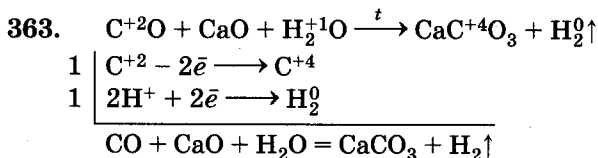


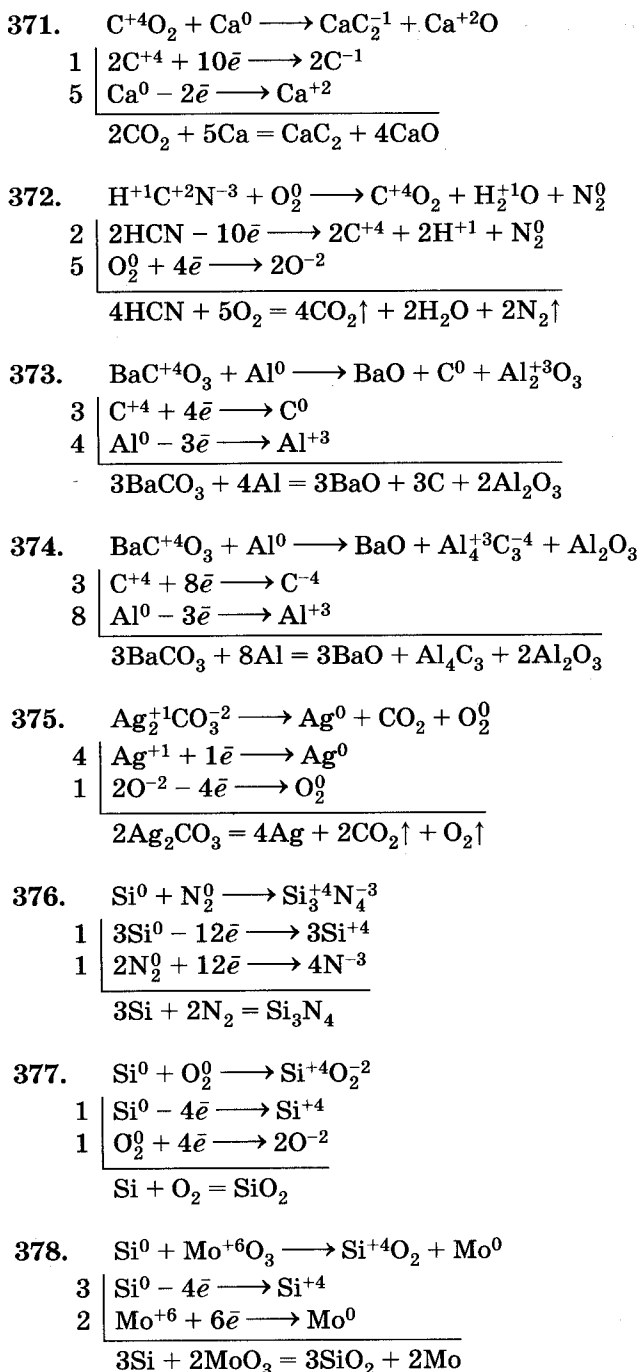


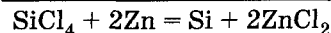
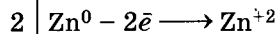
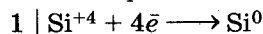
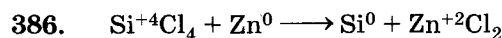
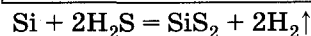
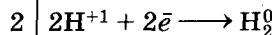
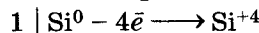
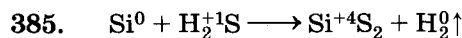
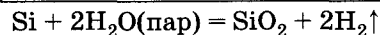
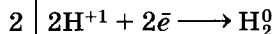
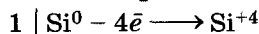
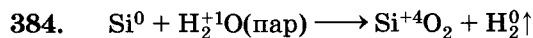
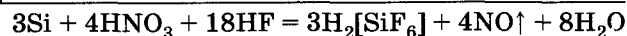
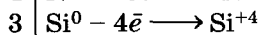
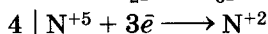
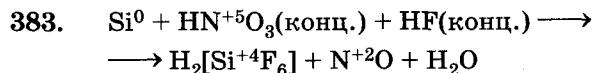
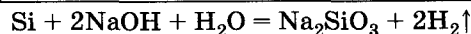
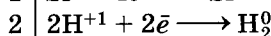
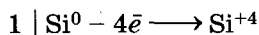
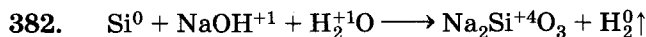
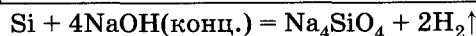
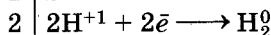
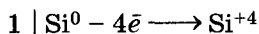
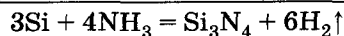
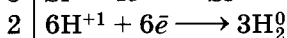
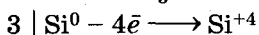
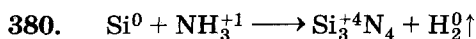
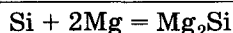
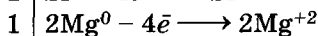
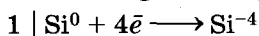
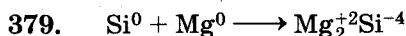


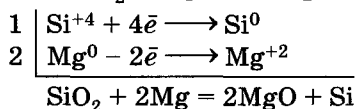
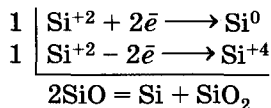
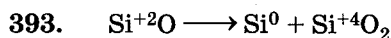
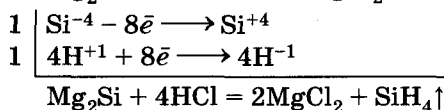
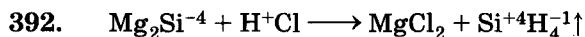
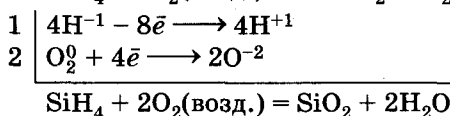
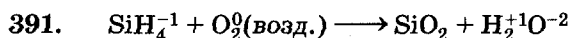
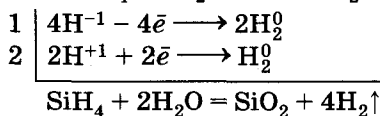
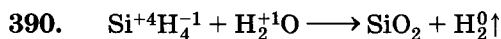
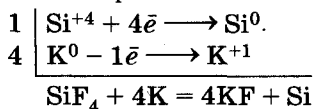
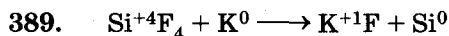
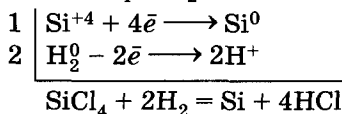
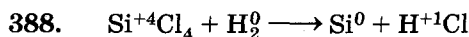
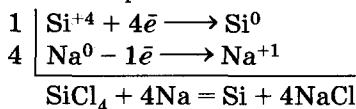
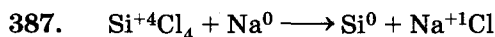
347. $C^0 + (Cr_2^{+3}Fe^{+2})O_4 \longrightarrow Fe^0 + Cr^0 + C^{+2}O\uparrow$
- | | | |
|---|---|------------|
| 4 | $C^0 - 2\bar{e} \longrightarrow C^{+2}$ | } 8\bar{e} |
| 1 | $2Cr^{+3} + 6\bar{e} \longrightarrow 2Cr^0$ | |
| 1 | $Fe^{+2} + 2\bar{e} \longrightarrow Fe^0$ | |
| $4C + (Cr_2Fe)O_4 = Fe + 2Cr + 4CO\uparrow$ | | |
348. $C^0 + K_2Cr_2^{+6}O_7 \longrightarrow Cr_2^{+3}O_3 + K_2C^{+4}O_3 + C^{+4}O_2\uparrow$
- | | |
|---|--|
| 3 | $C^0 - 4\bar{e} \longrightarrow C^{+4}$ |
| 2 | $2Cr^{+6} + 6\bar{e} \longrightarrow 2Cr^{+3}$ |
| $3C + 2K_2Cr_2O_7 = 2Cr_2O_3 + 2K_2CO_3 + CO_2\uparrow$ | |
349. $C^0 + CaCO_3 + Na_2S^{+6}O_4 \longrightarrow C^{+4}O_2\uparrow + Na_2CO_3 + CaS^{-2}$
- | | |
|---|--|
| 2 | $C^0 - 4\bar{e} \longrightarrow C^{+4}$ |
| 1 | $S^{+6} + 8\bar{e} \longrightarrow S^{-2}$ |
| $2C + CaCO_3 + Na_2SO_4 = 2CO_2\uparrow + Na_2CO_3 + CaS$ | |
350. $C^0 + KN^{+5}O_3 \longrightarrow K_2CO_3 + C^{+4}O_2\uparrow + N_2^0\uparrow$
- | | |
|---|---|
| 5 | $C^0 - 4\bar{e} \longrightarrow C^{+4}$ |
| 2 | $2N^{+5} + 10\bar{e} \longrightarrow N_2^0$ |
| $5C + 4KNO_3 = 2K_2CO_3 + 3CO_2\uparrow + 2N_2\uparrow$ | |
351. $C^0 + Ca_3(P^{+5}O_4)_2 + SiO_2 \longrightarrow CaSiO_3 + P^0 + C^{+2}O\uparrow$
- | | |
|--|---|
| 5 | $C^0 - 2\bar{e} \longrightarrow C^{+2}$ |
| 2 | $P^{+5} + 5\bar{e} \longrightarrow P^0$ |
| $5C + Ca_3(PO_4)_2 + 3SiO_2 = 3CaSiO_3 + 2P + 5CO\uparrow$ | |
352. $C^0 + Ca_3(P^{+5}O_4)_2 \longrightarrow C^{+2}O\uparrow + Ca_3P_2^{-3}$
- | | |
|---|---|
| 8 | $C^0 - 2\bar{e} \longrightarrow C^{+2}$ |
| 1 | $2P^{+5} + 16\bar{e} \longrightarrow 2P^{-3}$ |
| $8C + Ca_3(PO_4)_2 = 8CO\uparrow + Ca_3P_2$ | |
353. $C^0 + CaS^{+6}O_4 \longrightarrow S^{+4}O_2\uparrow + CaO + C^{+4}O_2\uparrow$
- | | |
|---|--|
| 1 | $C^0 - 4\bar{e} \longrightarrow C^{+4}$ |
| 2 | $S^{+6} + 2\bar{e} \longrightarrow S^{+4}$ |
| $C + 2CaSO_4 = 2SO_2\uparrow + 2CaO + CO_2\uparrow$ | |
354. $C^0 + Na_2S^{+6}O_4 \xrightarrow{t} Na_2S^{-2} + C^{+4}O_2\uparrow$
- | | |
|---|--|
| 2 | $C^0 - 4\bar{e} \longrightarrow C^{+4}$ |
| 1 | $S^{+6} + 8\bar{e} \longrightarrow S^{-2}$ |
| $2C + Na_2SO_4 = Na_2S + 2CO_2\uparrow$ | |

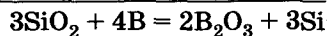
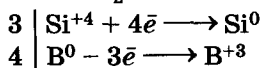
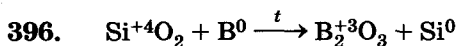
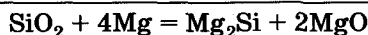
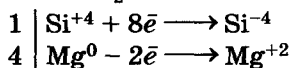
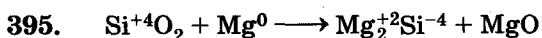
355. $C^0 + S^0 \longrightarrow C^{+4}S_2^{-2}$
- | | |
|---|---|
| 1 | $C^0 - 4\bar{e} \longrightarrow C^{+4}$ |
| 2 | $S^0 + 2\bar{e} \longrightarrow S^{-2}$ |
-
- $C + 2S = CS_2$
356. $C^0 + S^0 + KN^{+5}O_3 \longrightarrow N_2^0\uparrow + C^{+4}O_2\uparrow + K_2S^{-2}$
- | | | | |
|---|----|---|-------------|
| 3 | 12 | $C^0 - 4\bar{e} \longrightarrow C^{+4}$ | } 12\bar{e} |
| 1 | 4 | $S^0 + 2\bar{e} \longrightarrow S^{-2}$ | |
| 1 | 4 | $2N^{+5} + 10\bar{e} \longrightarrow N_2^0$ | |
-
- $3C + S + 2KNO_3 = N_2\uparrow + 3CO_2\uparrow + K_2S$
357. $C^0 + Cl_2^0 + Cr_2O_3 \longrightarrow CrCl_3^{-1} + C^{+2}O\uparrow$
- | | |
|---|---|
| 3 | $C^0 - 2\bar{e} \longrightarrow C^{+2}$ |
| 1 | $3Cl_2^0 + 6\bar{e} \longrightarrow 6Cl^{-1}$ |
-
- $3C + 3Cl_2 + Cr_2O_3 = 2CrCl_3 + 3CO\uparrow$
358. $C^0 + N_2^0 + BaC^{+4}O_3 \longrightarrow C^{+2}O\uparrow + Ba(C^{+2}N^{-3})_2$
- | | | |
|---|--|------------|
| 4 | $C^0 - 2\bar{e} \longrightarrow C^{+2}$ | } 8\bar{e} |
| 1 | $N_2^0 + 6\bar{e} \longrightarrow 2N^{-3}$ | |
| 1 | $C^{+4} + 2\bar{e} \longrightarrow C^{+2}$ | |
-
- $4C + N_2 + BaCO_3 = 3CO\uparrow + Ba(CN)_2$
359. $C^0 + K_2C^{+4}O_3 + NH_3 \longrightarrow KC^{+2}N + H_2O$
- | | |
|---|--|
| 1 | $C^0 - 2\bar{e} \longrightarrow C^{+2}$ |
| 1 | $C^{+4} + 2\bar{e} \longrightarrow C^{+2}$ |
-
- $C + K_2CO_3 + 2NH_3 = 2KCN + 3H_2O$
360. $C^{+2}O + (Fe^{+2}Fe_2^{+3})O_4 \longrightarrow Fe^0 + C^{+4}O_2\uparrow$
- | | | |
|---|---|------------|
| 4 | $C^{+2} - 2\bar{e} \longrightarrow C^{+4}$ | } 8\bar{e} |
| 1 | $2Fe^{+3} + 6\bar{e} \longrightarrow 2Fe^0$ | |
| 1 | $Fe^{+2} + 2\bar{e} \longrightarrow Fe^0$ | |
-
- $4CO + (FeFe_2)O_4 = 3Fe + 4CO_2\uparrow$
361. $C^{+2}O + O_2^0 \longrightarrow C^{+4}O_2\uparrow$
- | | |
|---|--|
| 2 | $C^{+2} - 2\bar{e} \longrightarrow C^{+4}$ |
| 1 | $O_2^0 + 4\bar{e} \longrightarrow 2O^{-2}$ |
-
- $2CO + O_2 = 2CO_2\uparrow$
362. $C^{+2}O + I_2^{+5}O_5 \longrightarrow C^{+4}O_2\uparrow + I_2^0$
- | | |
|---|---|
| 5 | $C^{+2} - 2\bar{e} \longrightarrow C^{+4}$ |
| 1 | $2I^{+5} + 10\bar{e} \longrightarrow I_2^0$ |
-
- $5CO + I_2O_5 = 5CO_2\uparrow + I_2$



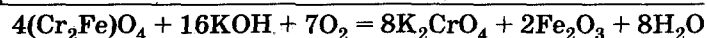
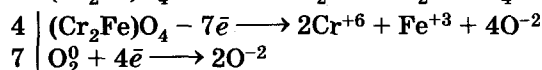
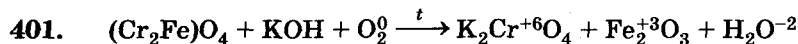
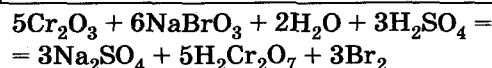
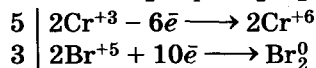
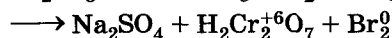
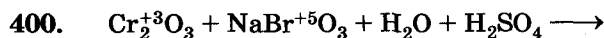
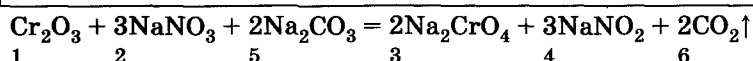
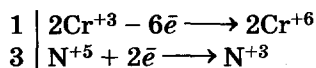
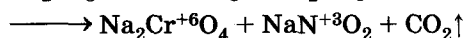
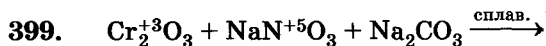
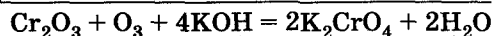
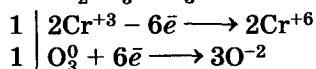
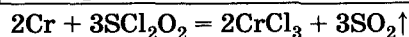
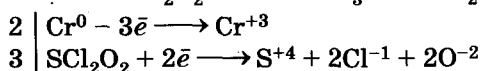
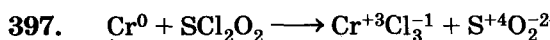


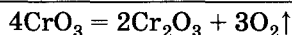
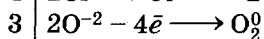
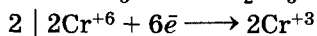
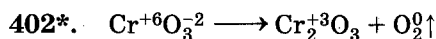




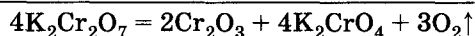
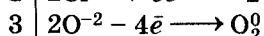
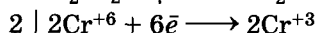
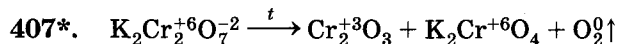
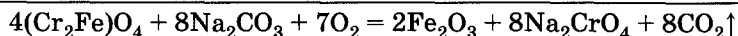
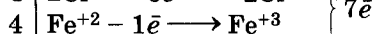
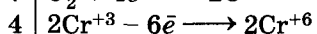
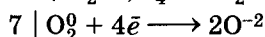
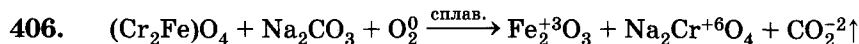
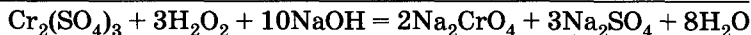
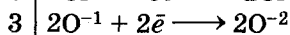
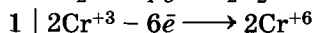
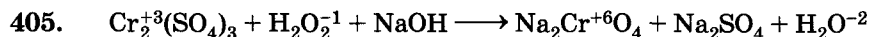
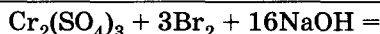
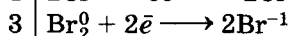
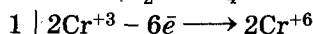
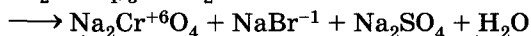
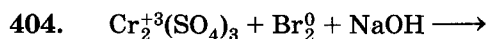
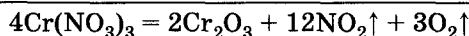
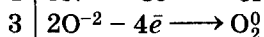
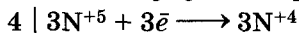
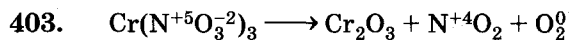


Хром. Соединения хрома и марганца

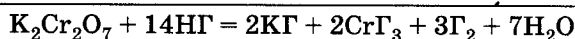
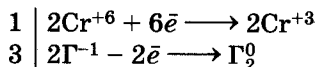
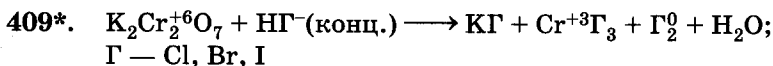
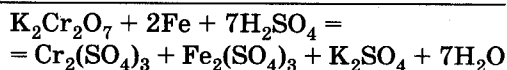
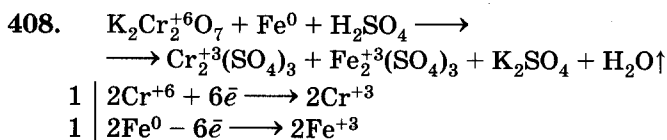




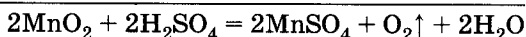
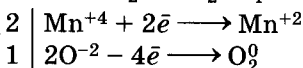
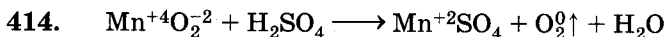
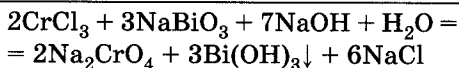
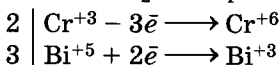
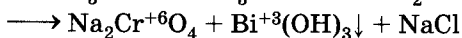
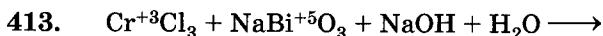
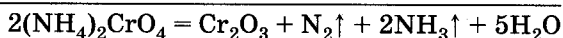
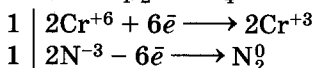
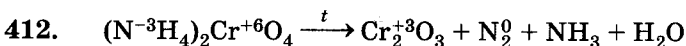
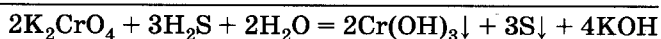
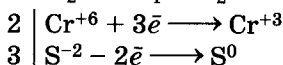
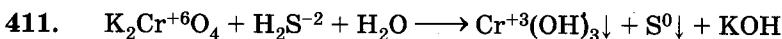
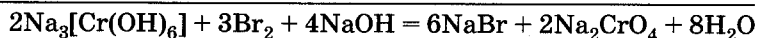
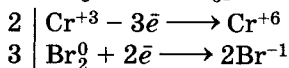
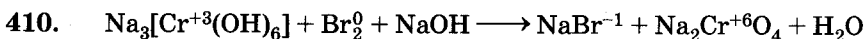
Пояснение. Из ЭБ следует, что из 12 атомов кислорода (4CrO_3) только 6O^{-2} ($3 \cdot 2\text{O}^{-2}$) превратятся в 3O_2^0 , остальные 6O^{-2} перейдут без изменения с. о. в $\text{Cr}_2\text{O}_3^{-2}$.

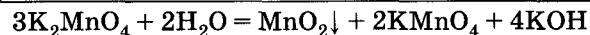
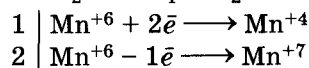
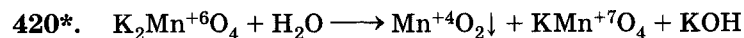
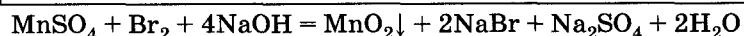
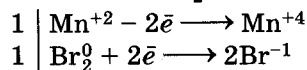
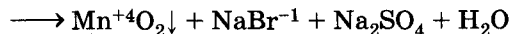
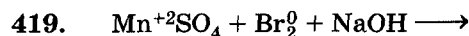
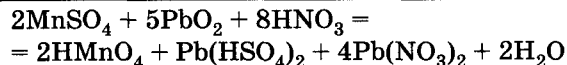
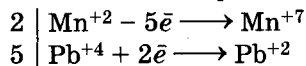
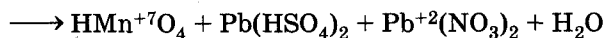
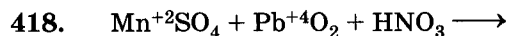
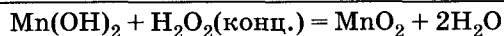
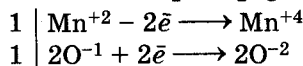
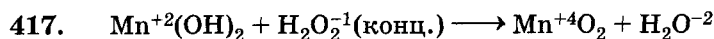
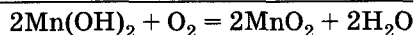
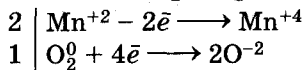
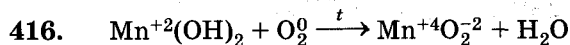
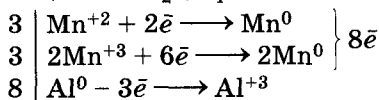
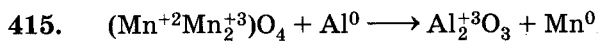


Пояснение. Расстановку коэффициентов рекомендуют начать с Cr_2O_3 , так как из $\text{K}_2\text{Cr}_2\text{O}_7$ часть хрома переходит без изменения степени окисления в состав K_2CrO_4 , другая — восстанавливается из Cr^{+6} до Cr^{+3} .

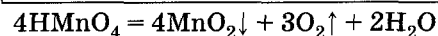
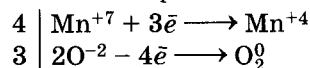
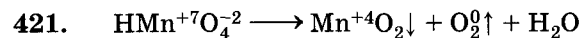


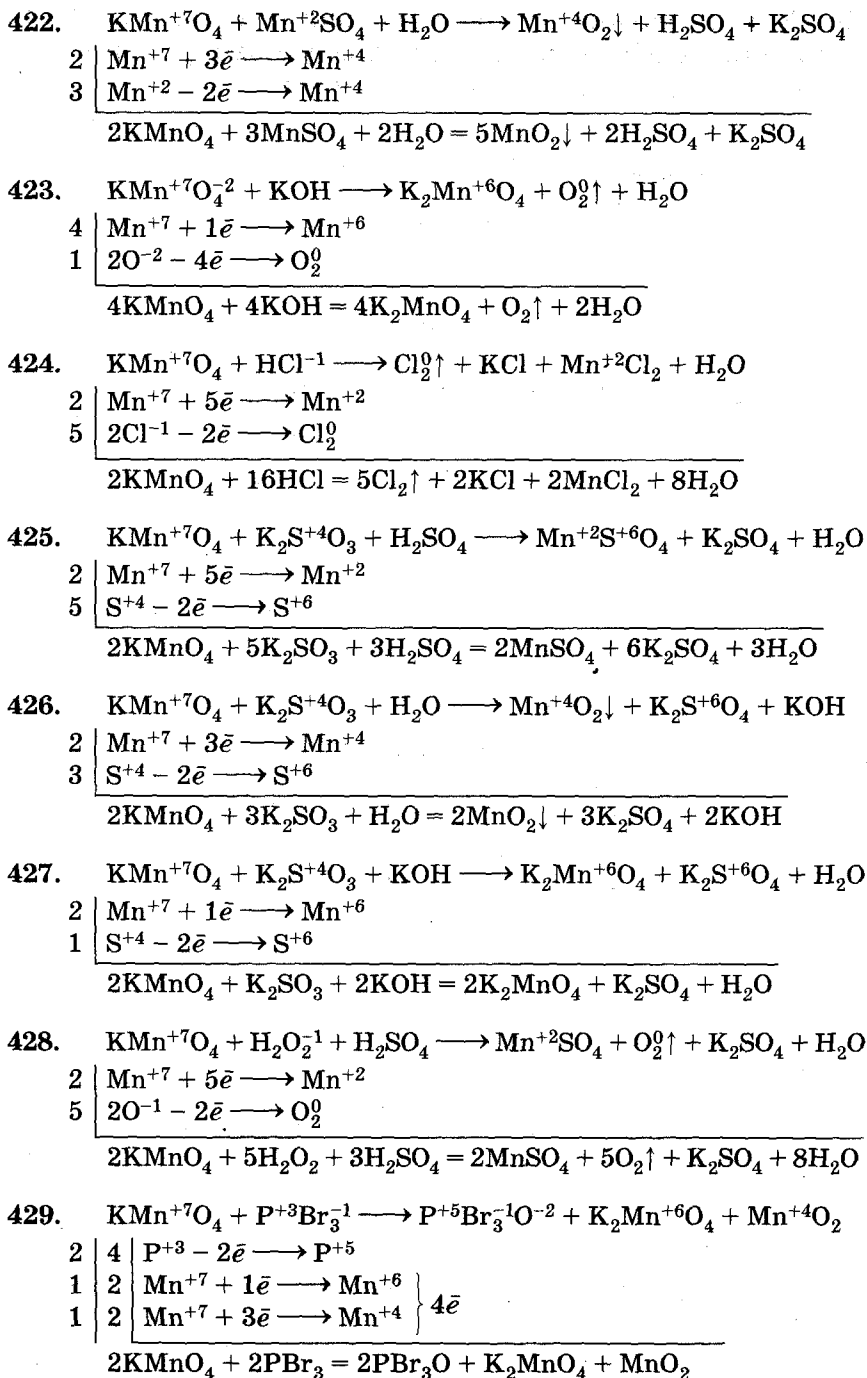
П о я с н е н и е. $\text{K}_2\text{Cr}_2\text{O}_7$ окисляет HI на холоде, а HBr и HCl при нагревании.



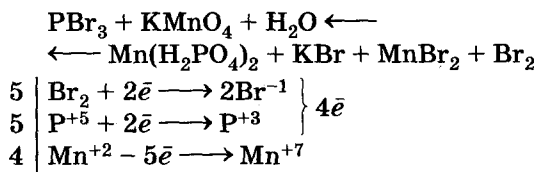


П о я с н е н и е. $1\text{Mn}^{+6} + 2\text{Mn}^{+6} = 3\text{Mn}^{+6}$ ($3\text{K}_2\text{Mn}^{+6}\text{O}_4$).





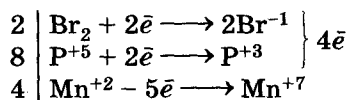
430*. В этом сложном уравнении, как и в примере 4 на с. 25, при подборе коэффициентов удобно рассмотреть процесс, идущий как бы справа налево:



Принято электронов: $5 \cdot 2\bar{e} + 5 \cdot 2\bar{e} = 20\bar{e}$,

отдано: $4 \cdot 5\bar{e} = 20\bar{e}$.

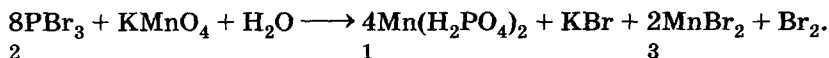
С учетом того, что в $\text{Mn}(\text{H}_2\text{PO}_4)_2$ отношение атомов Mn и P 1 : 2, подбираются коэффициенты: перед марганцем коэффициент 4, перед фосфором — 8. Перед бромом коэффициент 2 подбирался таким образом, чтобы общая сумма принятых электронов оставалась равной 20:



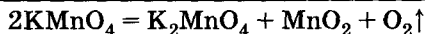
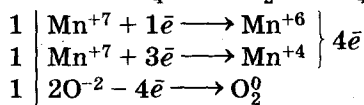
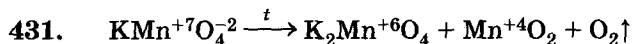
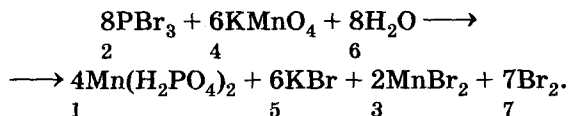
Принято электронов: $2 \cdot 2\bar{e} + 8 \cdot 2\bar{e} = 20\bar{e}$,

отдано: $4 \cdot 5\bar{e} = 20\bar{e}$.

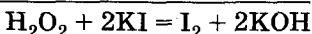
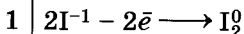
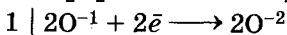
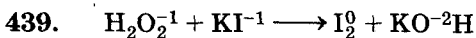
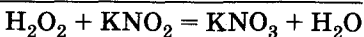
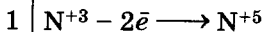
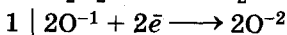
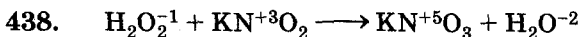
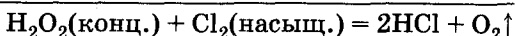
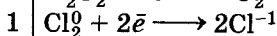
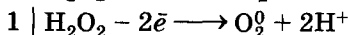
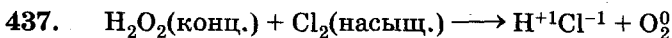
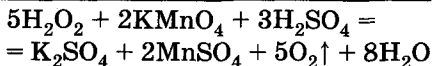
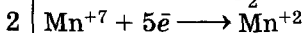
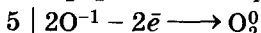
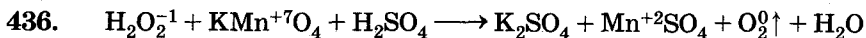
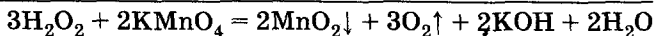
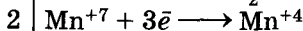
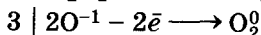
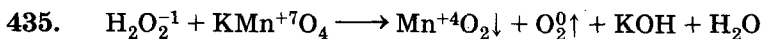
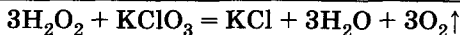
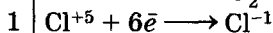
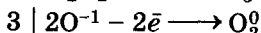
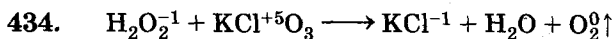
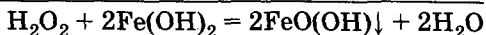
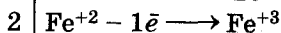
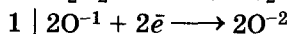
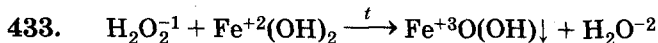
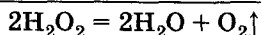
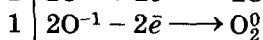
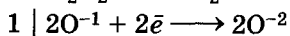
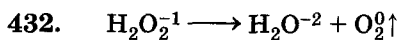
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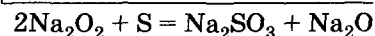
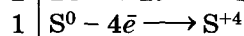
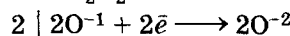
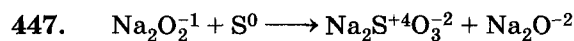
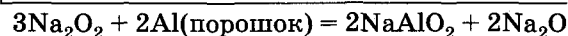
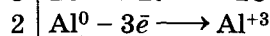
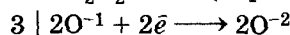
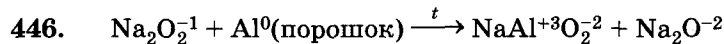
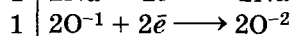
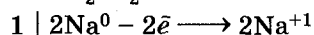
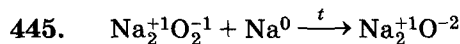
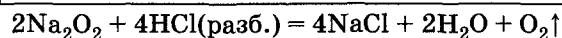
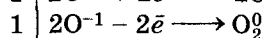
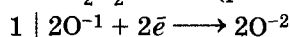
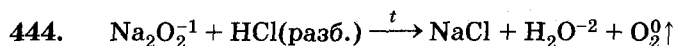
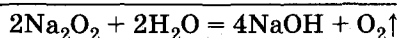
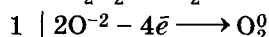
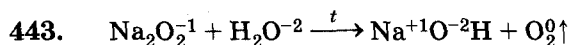
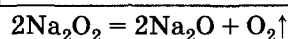
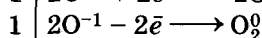
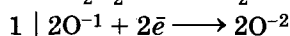
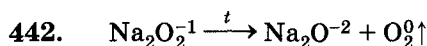
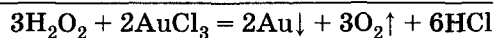
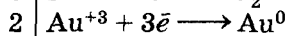
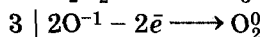
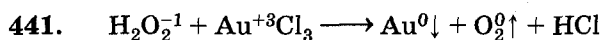
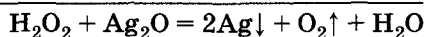
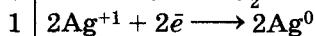
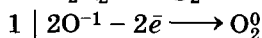
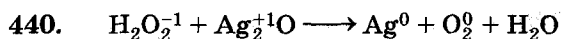


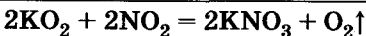
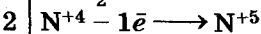
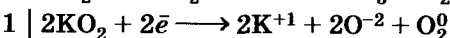
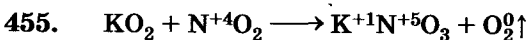
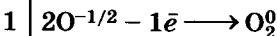
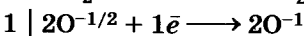
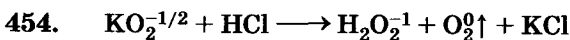
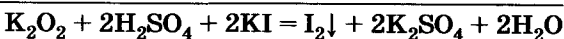
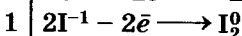
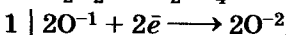
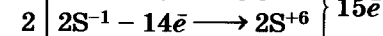
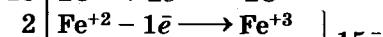
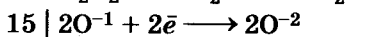
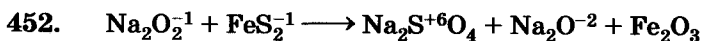
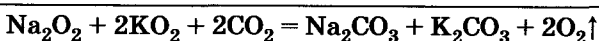
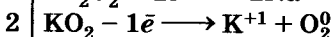
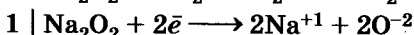
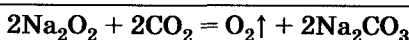
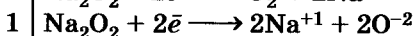
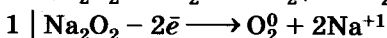
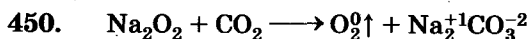
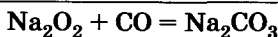
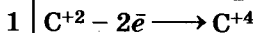
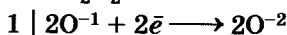
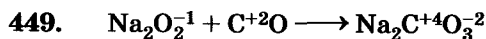
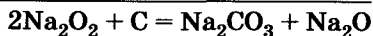
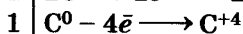
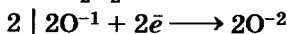
Уравнялся бром, который связан с марганцем. Далее уравниваются марганец и калий, потом водород и кислород. В последнюю очередь ставится коэффициент перед Br_2 , так как не весь бром из PBr_3 переходит в Br_2 :

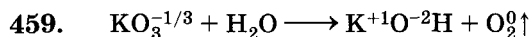
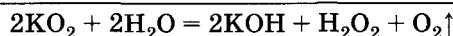
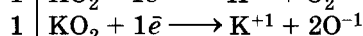
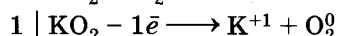
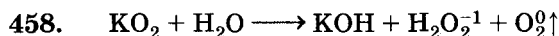
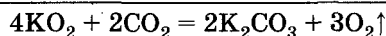
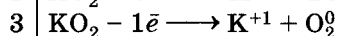
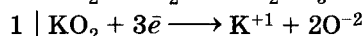
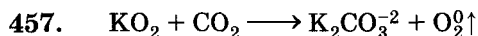
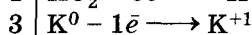
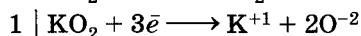
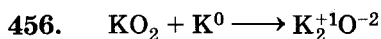


Пероксиды, надпероксиды и озониды

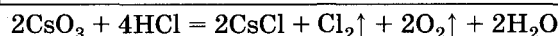
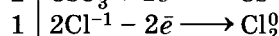
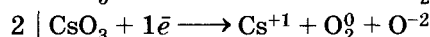
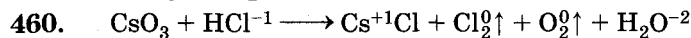
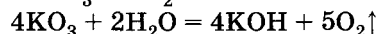
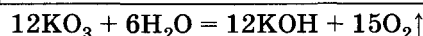
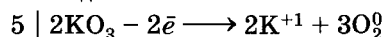




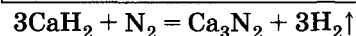
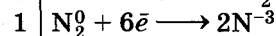
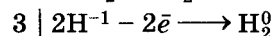
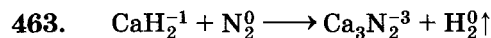
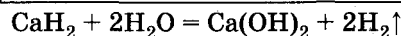
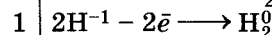
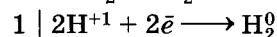
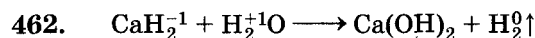
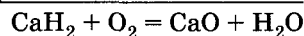
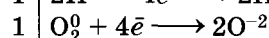
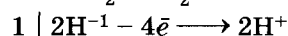
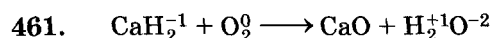


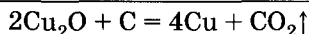
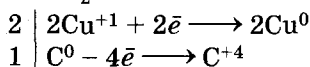
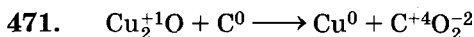
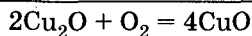
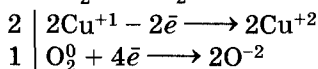
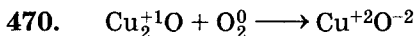
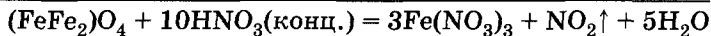
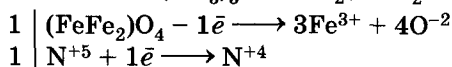
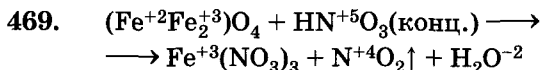
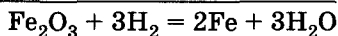
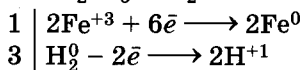
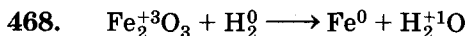
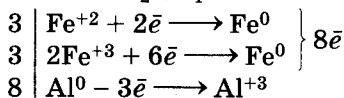
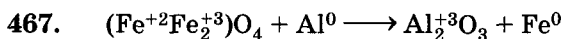
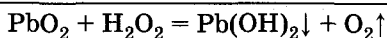
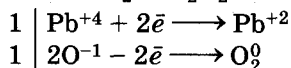
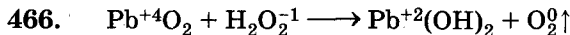
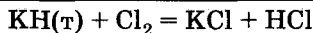
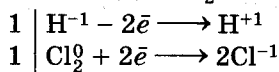
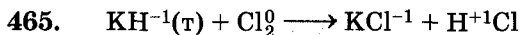
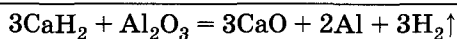
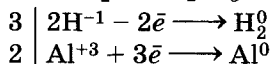
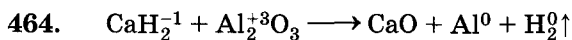


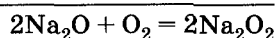
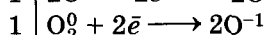
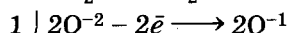
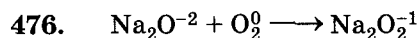
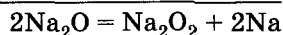
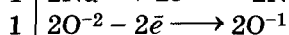
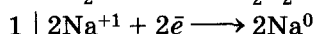
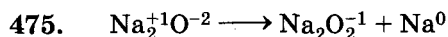
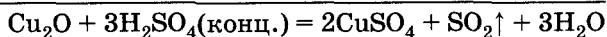
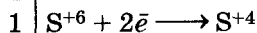
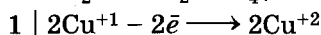
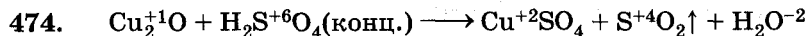
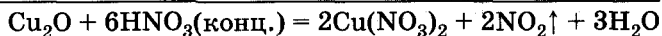
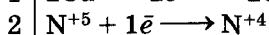
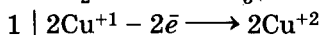
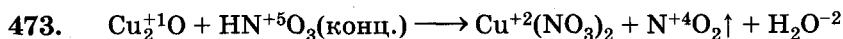
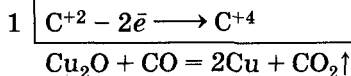
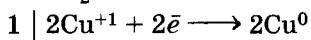
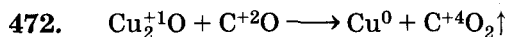
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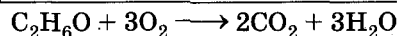
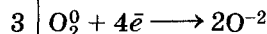
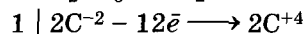
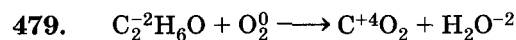
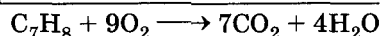
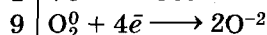
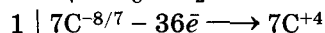
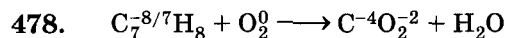
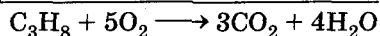
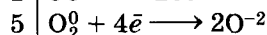
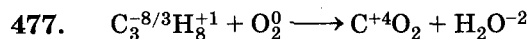
Гидриды и оксиды металлов

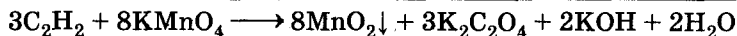
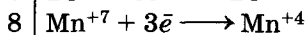
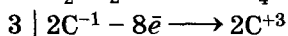
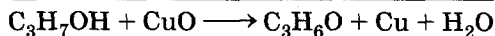
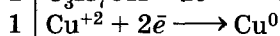
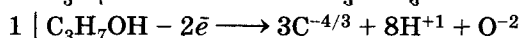
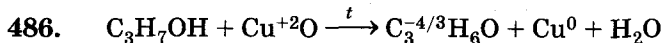
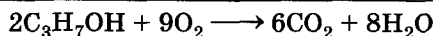
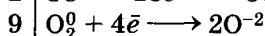
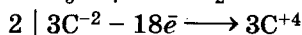
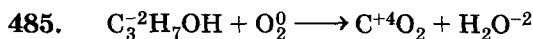
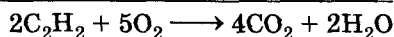
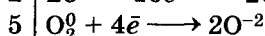
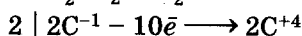
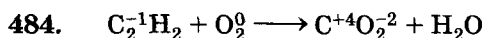
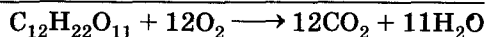
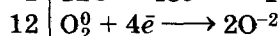
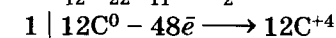
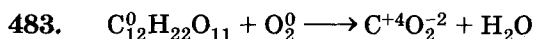
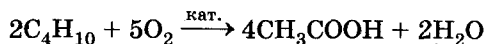
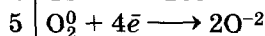
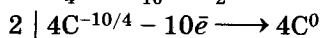
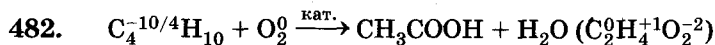
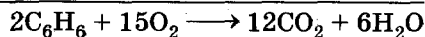
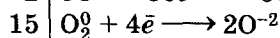
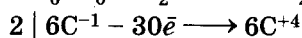
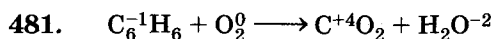
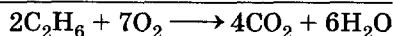
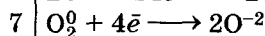
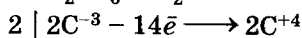
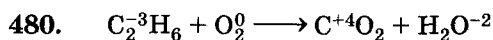


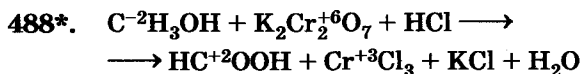




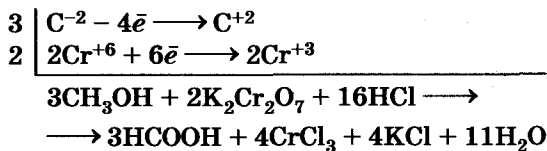
Органические соединения







1-й вариант оформления ЭБ.

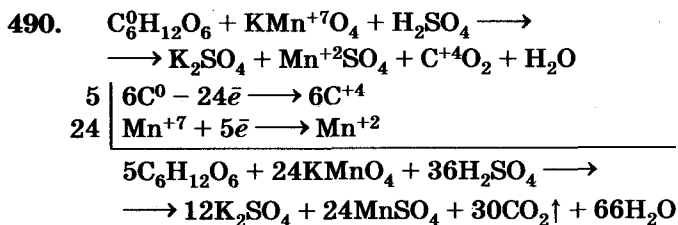
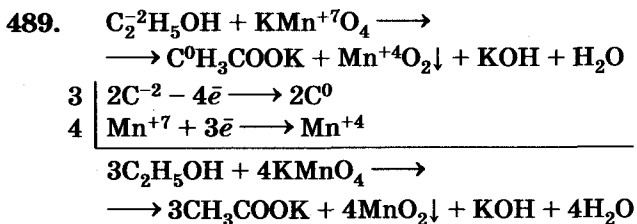


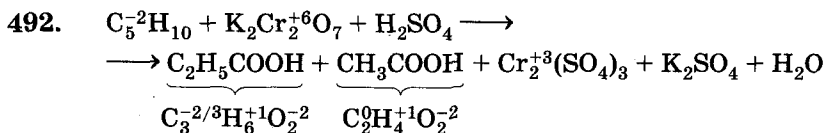
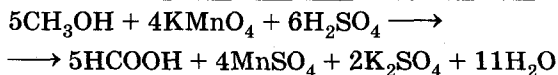
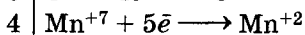
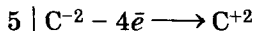
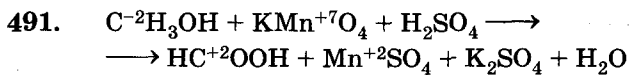
2-й вариант оформления ЭБ.

В уравнениях ОВР с участием органических соединений коэффициенты можно подобрать, не составляя электронного баланса.

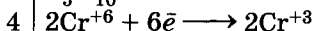
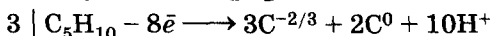
При анализе количественного состава $\text{CH}_3\text{OH} \longrightarrow \text{HCOOH}$ выясняется, что разница между ними в одном атоме кислорода (идет увеличение) и в двух атомах водорода (идет их уменьшение).

Атом кислорода, пошедший на окисление, соответствует переходу двух электронов. На связывание двух атомов водорода требуется еще один атом кислорода для образования H_2O . В сумме на окисление CH_3OH в HCOOH требуется два атома кислорода, что соответствует отдаче 4 электронов. На восстановление 2Cr^{+6} до 2Cr^{+3} требуется 6 электронов. Коэффициенты 4 и 6 сокращаются на два. Перед CH_3OH ставится коэффициент 3, перед $\text{K}_2\text{Cr}_2\text{O}_7$ — 2.

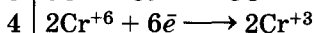
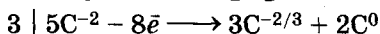




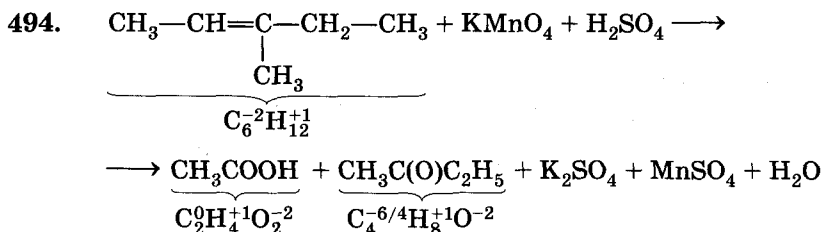
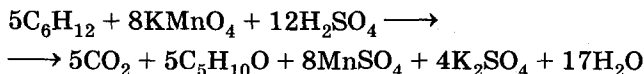
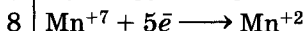
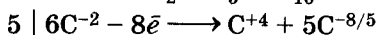
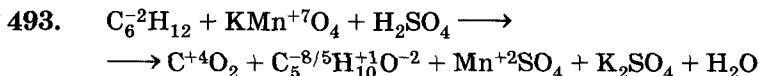
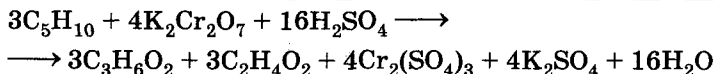
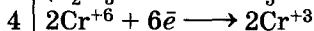
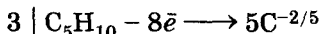
1-й вариант оформления ЭБ.

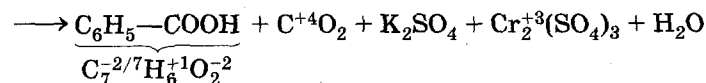
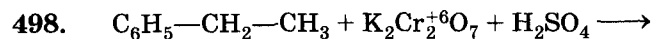
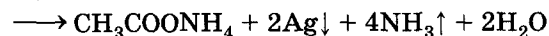
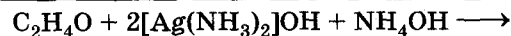
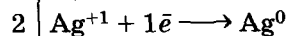
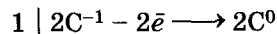
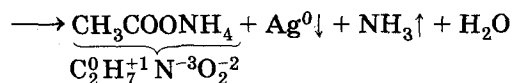
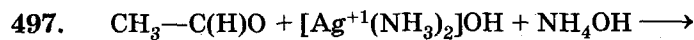
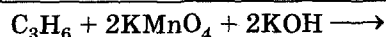
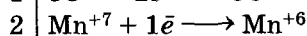
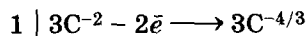
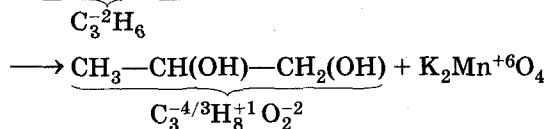
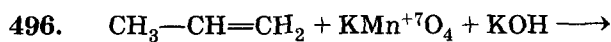
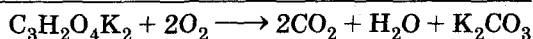
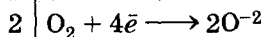
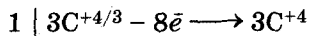
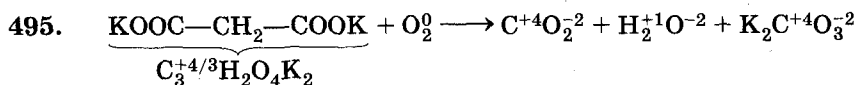
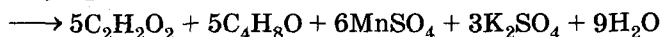
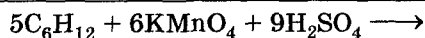
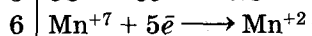
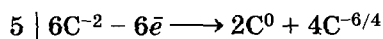


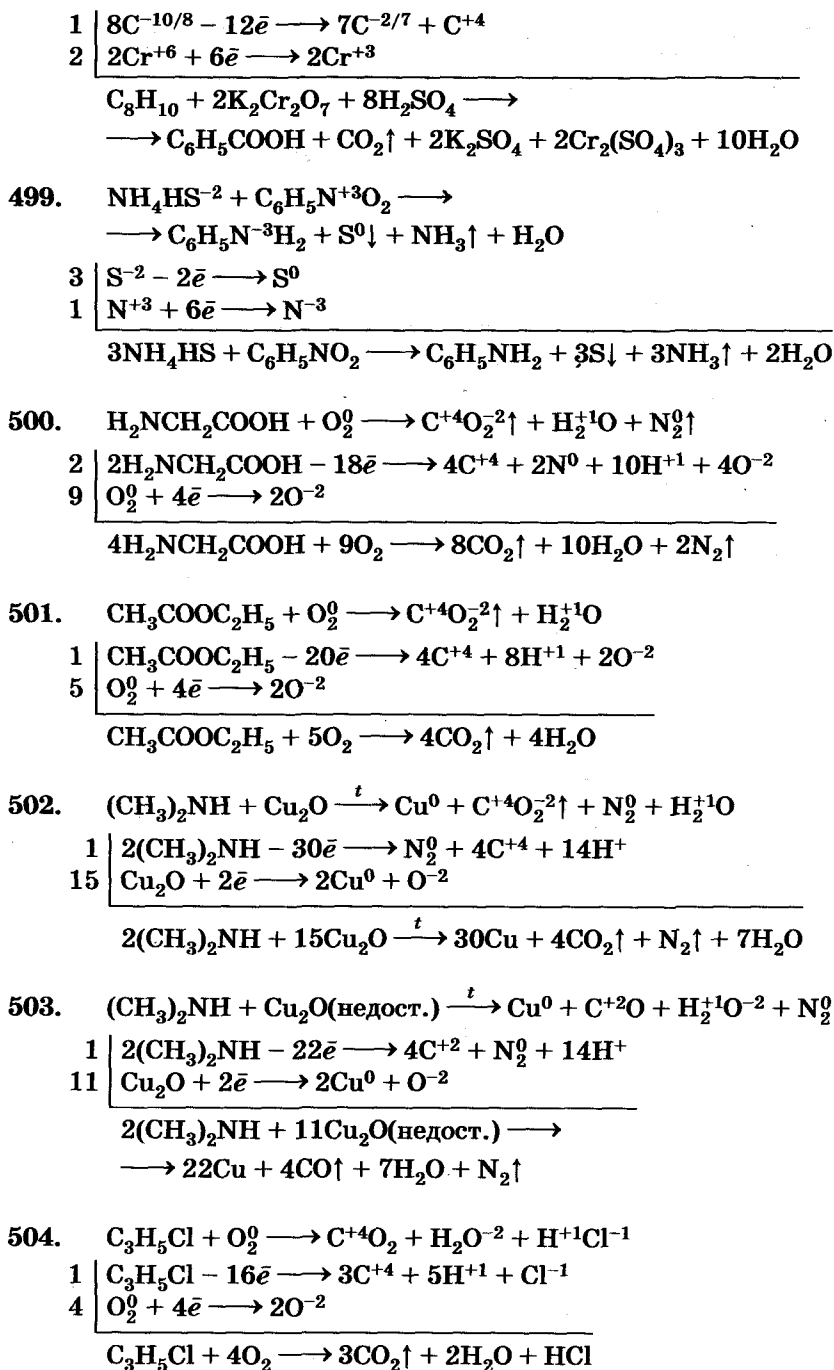
2-й вариант оформления ЭБ.

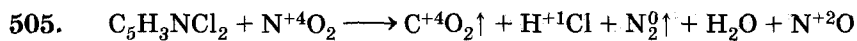


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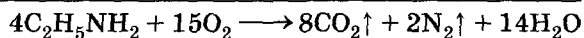
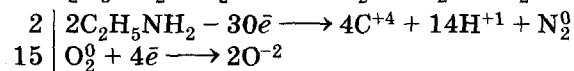
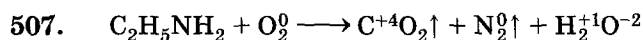
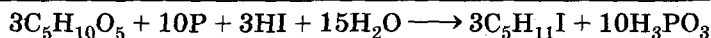
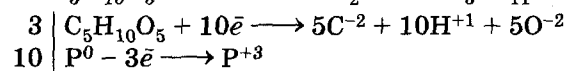
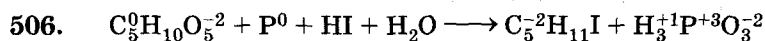
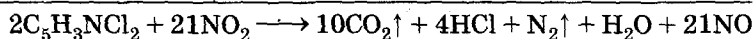
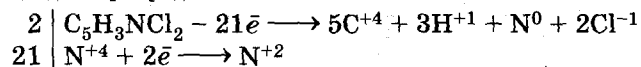








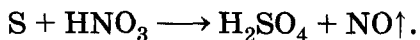
3,5-дихлорпиридин



Метод электронно-ионного баланса (метод полуреакций)

Правила оформления уравнений ОВР,
протекающих в кислотной среде

1. На первой строчке записывают схему реакции:



Определяют молекулы или ионы, которые участвуют в процессе окисления и восстановления.

В данном примере NO_3^- нитрат-ион — окислитель, а сера — восстановитель.

2. На второй и третьей строчках записывают в ионном виде полуреакции окисления и восстановления. Слабые электролиты, твердые и газообразные вещества записывают в молекулярном виде.

1-я полуреакция: $\text{S}^0 \longrightarrow \text{SO}_4^{2-}$ — процесс окисления;

2-я полуреакция: $\text{NO}_3^- \longrightarrow \text{NO}$ — процесс восстановления.

На основании закона сохранения массы и энергии при составлении уравнений полуреакций следует соблюдать баланс веществ и баланс зарядов.

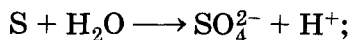
Для уравнивания числа атомов кислорода в 1-й полуреакции в ту часть, где он в избытке, добавляют столько катионов водорода H^+ , чтобы, связавшись с атомами кислорода, образовались молекулы воды.

1-я полуреакция. $\text{S}^0 \longrightarrow \text{SO}_4^{2-}$:

а) добавляют H^+ туда, где избыток кислорода:

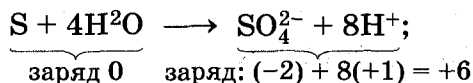


б) в противоположную часть добавляют H_2O :

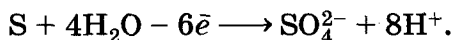


в) уравнивают кислород: $\text{S} + 4\text{H}_2\text{O} \longrightarrow \text{SO}_4^{2-} + \text{H}^+;$

г) уравнивают водород, подсчитывают заряды в полуреакциях:

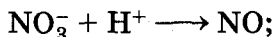


д) уравнивают заряд, для этого из левой части полуреакции отнимают 6 электронов:

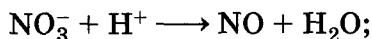


2-я полуреакция. $\text{NO}_3^- \longrightarrow \text{NO}$:

а) добавляют H^+ туда, где избыток кислорода:

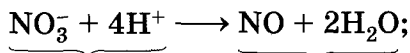


б) в противоположную часть добавляют H_2O :

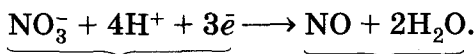


в) уравнивают кислород: $\text{NO}_3^- + \text{H}^+ \longrightarrow \text{NO} + 2\text{H}_2\text{O}$;

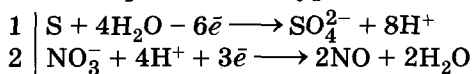
г) уравнивают водород, подсчитывают заряды в полуреакциях:



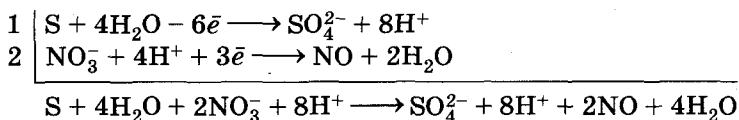
д) для уравнивания зарядов в левую часть полуреакции добавляют 3 электрона:



3. Балансируют (уравнивают) число отданных и принятых электронов в полуреакциях:



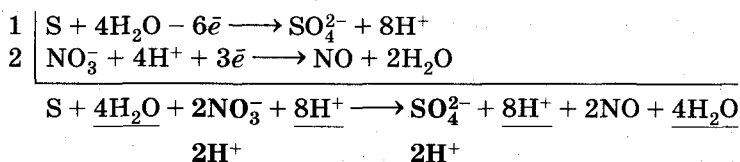
4. Суммируют сначала левые, а затем правые части полуреакций, не забывают предварительно у м н о ж и т ь множитель на коэффициент, если он стоит перед формулой. Результат — суммарное ионное уравнение:



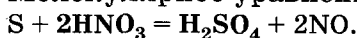
Подчеркивают и сокращают одинаковые ионы и молекулы. На следующем этапе добавляют недостающие ка-

тионы или анионы. Следует учесть, что количество добавляемых ионов в правую и левую части ионного уравнения должно быть одинаковым.

В рассматриваемом примере добавляют 2H^+ для 2NO_3^- и 2H^+ для SO_4^{2-} , из них образуются 2HNO_3 и H_2SO_4 :



Молекулярное уравнение:

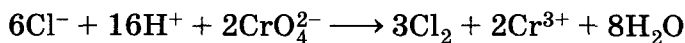


В полном виде оформление занимает шесть, иногда семь строчек.

Например:

- 1) схема уравнения: $\text{S} + \text{HNO}_3 \longrightarrow \text{H}_2\text{SO}_4 + \text{NO}$
- 2) 1-я полуреакция: $1 \quad \text{S} + 4\text{H}_2\text{O} - 6\bar{e} \longrightarrow \text{SO}_4^{2-} + 8\text{H}^+$
- 3) 2-я полуреакция: $2 \quad \text{NO}_3^- + 4\text{H}^+ + 3\bar{e} \longrightarrow \text{NO} + 2\text{H}_2\text{O}$
- 4) суммарное ион-
ное уравнение: $\text{S} + \underline{4\text{H}_2\text{O}} + \underline{2\text{NO}_3^-} + \underline{8\text{H}^+} \longrightarrow$
 $\longrightarrow \text{SO}_4^{2-} + \underline{8\text{H}^+} + \underline{2\text{NO}} + \underline{4\text{H}_2\text{O}}$
- 5) добавляемые
ионы: $2\text{H}^+ \longrightarrow 2\text{H}^+$
- 6) итоговое молеку-
лярное уравнение: $\text{S} + 2\text{HNO}_3 = \text{H}_2\text{SO}_4 + 2\text{NO}\uparrow$

Часто возникают трудности в определении количества добавляемых ионов. Например, в задаче 521 в ионном уравнении:



число добавляемых ионов рассчитывают следующим образом: к иону 2CrO_4^{2-} , где заряд -4 ($2 \cdot (-2)$), следует добавить столько катионов калия, чтобы в целом «молекула» стала нейтральной, т. е. 4K^+ , где заряд $+4$ ($4 \cdot (+1)$). Алгебраическая сумма положительных и отрицательных зарядов стала равной нулю. В итоге из 4K^+ и 2CrO_4^{2-} образуется $2\text{K}_2\text{CrO}_4$. 4K^+ записывают строго под 2CrO_4^{2-}

(см. пример 521). В правой части уравнения к 2Cr^{3+} , где заряд $+6$ ($2 \cdot (+3)$), следует добавить 6Cl^- , где заряд -6 ($6 \cdot (-1)$). Из 2Cr^{3+} и 6Cl^- образуется 2CrCl_3 . 6Cl^- также записывают строго под 2Cr^{3+} . Добавив 4K^+ в левую часть, столько же катионов калия следует добавить в правую часть уравнения для равенства. На эти 4K^+ потребуется 4Cl^- . В сумме из этих ионов образуется 4KCl . Следует обратить внимание на то, что ионы 4K^+ и 4Cl^- добавились в правую часть для равенства, поэтому их записывают как бы в стороне от ионного уравнения.

В правую часть в сумме добавлено 10Cl^- ($6\text{Cl}^- + 4\text{Cl}^-$), поэтому и в левой части уравнения добавленных хлорид-ионов также 10. Записывают их под одноименными ионами в ионном уравнении. Теперь в левой части в сумме стало 16Cl^- ($10\text{Cl}^- + 6\text{Cl}^-$). 16Cl^- свяжутся с имеющимися в ионном уравнении 16H^+ , из которых образуются 16HCl .

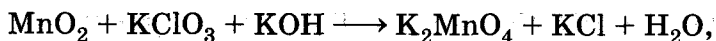
Формируется молекулярное уравнение. Для безошибочного воспроизведения его предлагается записывать формулы веществ, образующихся при суммировании ионов, под соответствующими ионами. Проверку проводят, как обычно, по числу атомов кислорода.

Из приведенных примеров видно, что при подборе коэффициентов методом ЭИБ отпадает необходимость нахождения степеней окисления элементов, легко определяются стехиометрические коэффициенты в молекулярном уравнении.

Правила оформления уравнений ОВР, протекающих в щелочной среде

Чтобы уравнивать число атомов водорода и кислорода в уравнениях для щелочной среды, добавляют воду в ту часть полуреакции, где избыток кислорода, а в противоположную часть — удвоенное число гидроксид-ионов. Перед H_2O ставят коэффициент, показывающий разницу в числе атомов кислорода в левой и в правой частях полуреакции, а перед OH^- — его удвоенный коэффициент. Получается так, что восстановитель присоединяет кислород из гидроксид-ионов.

ПРИМЕР. Расставьте коэффициенты в уравнении реакции, схема которой

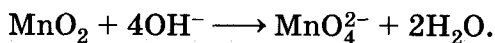


методом электронно-ионного баланса.

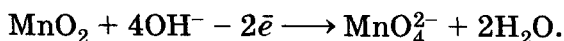
1-я полуреакция. $\text{MnO}_2 \longrightarrow \text{MnO}_4^{2-}$.

MnO_2 — восстановитель, он будет связывать гидроксид-ионы, в противоположную часть добавляют H_2O .

Так как в MnO_4^{2-} избыток кислорода, в эту часть полуреакции добавляют воду. Разница в числе атомов кислорода в полуреакции $\text{MnO}_2 \longrightarrow \text{MnO}_4^{2-}$ равна 2, поэтому перед H_2O ставят коэффициент 2, а перед OH^- — удвоенный коэффициент — 4:

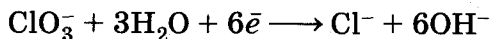


Уравнивают заряды. В левой части заряд -4 , справа -2 , поэтому из левой части полуреакции отнимают $2\bar{e}$:

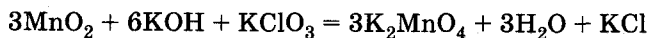
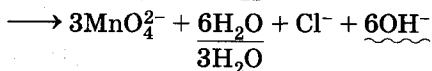
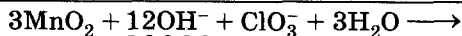
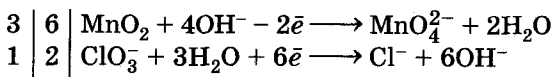


2-я полуреакция. $\text{ClO}_3^- \longrightarrow \text{Cl}^-$.

Во второй полуреакции Cl^- восстановитель, а ClO_3^- — окислитель. Избыток кислорода в окислителе. Разница в числе атомов кислорода равна 3, поэтому к иону ClO_3^- добавляют 3 молекулы воды, а к иону Cl^- — OH^- и перед ним ставят удвоенный коэффициент 6. Уравнивают заряды:

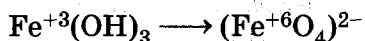


Далее:

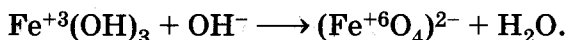


В некоторых уравнениях среди реагентов (примеры 815, 828 и др.) или среди продуктов реакции (пример 864) есть вещества, которые в своем составе уже содер-

жат гидроксид-группы или атомы водорода, что следует учитывать при уравнивании числа атомов кислорода и водорода в полуреакциях. В примере 815, в 1-й полуреакции:

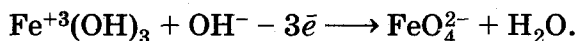


гидроксид-группы уже содержатся в составе $\text{Fe}(\text{OH})_3$, он же является восстановителем в этой реакции, поэтому будет связывать кислород из гидроксид-ионов:

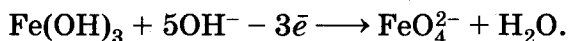


1-й способ оформления полуреакции.

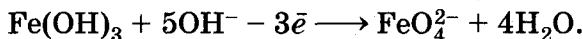
Для того чтобы железо из состояния +3 перешло в +6, оно должно отдать 3 электрона:



Уравниваются заряды в левой и правой частях полуреакции расстановкой коэффициентов:



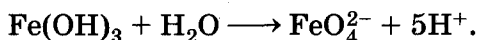
Уравнивают водород:



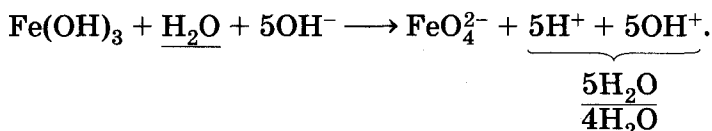
Кислород уравнивают одновременно с водородом.

2-й способ оформления полуреакции.

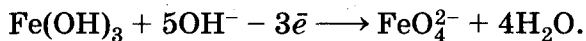
Полуреакцию составляют как бы для кислотной среды — добавляют H^+ в ту часть полуреакции, где больше кислорода, в противоположную часть — воду и уравнивают водород:



Чтобы нейтрализовать 5H^+ , добавляют столько же гидроксид-ионов, т. е. 5OH^- , в правую часть и такое же их количество в левую часть полуреакции:



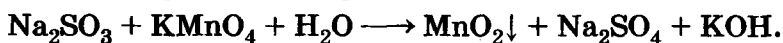
Уравнивают заряды:



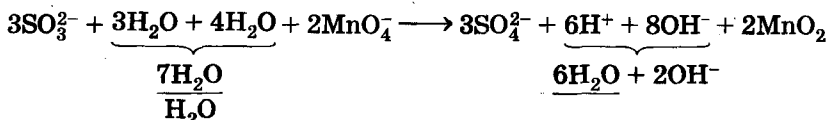
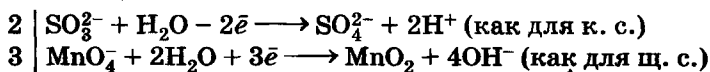
Правила оформления уравнений ОВР, протекающих в нейтральной среде

Среду нейтральной считают условно. На самом деле вследствие гидролиза соли среда может быть слабокислотной ($\text{pH} = 6-7$) или слабощелочной ($\text{pH} = 7-8$), поэтому полуреакции можно оформлять двумя способами.

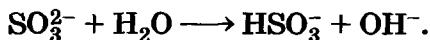
Способ 1 — без учета гидролиза соли. Так как среда нейтральная, в левые части полуреакций добавляют воду и тогда одну полуреакцию оформляют как для кислотной среды (добавляется $\text{H}_2\text{O} + \text{H}^+$), а другую — как для щелочной среды (добавляется $\text{H}_2\text{O} + \text{OH}^-$). В качестве примера рассмотрим схему уравнения реакции 984:



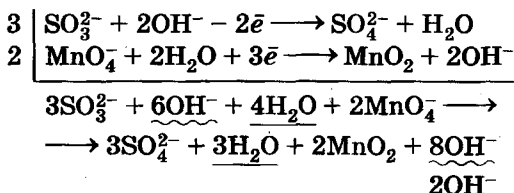
В левые части добавляют воду:



В правой части появляется дополнительная запись: $6\text{H}_2\text{O} + 2\text{OH}^-$, кстати, она же и показывает, что реально эта реакция протекает в щелочной среде. Щелочную среду создают освободившиеся вследствие гидролиза соли Na_2SO_3 гидроксид-ионы:

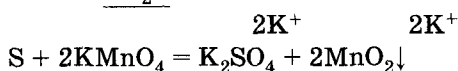
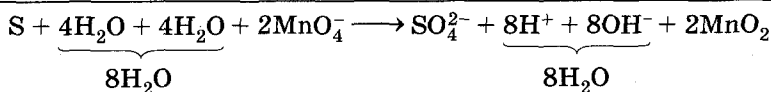
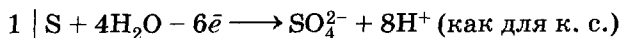
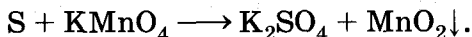


Способ 2. Если сразу учесть, что среда слабощелочная, то обе полуреакции удобнее и правильнее будет оформить как для щелочной среды:



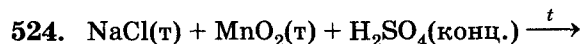
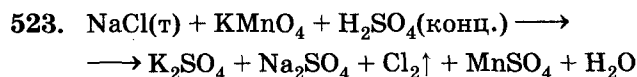
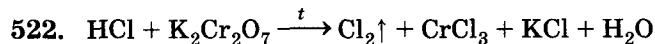
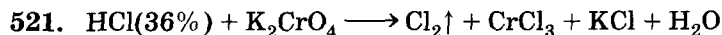
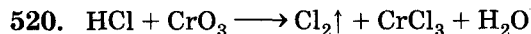
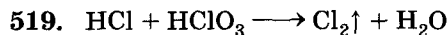
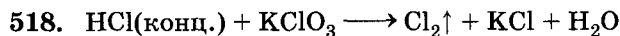
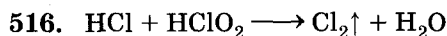
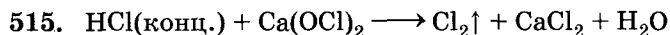
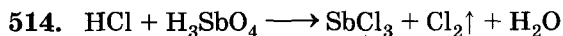
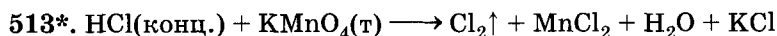
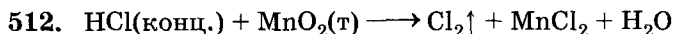
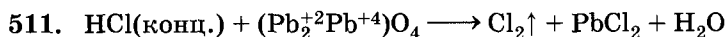
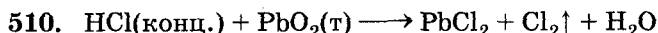
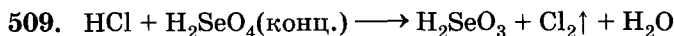
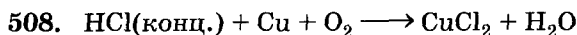
В этом случае отпадает необходимость в дополнительных записях. Если по схемам уравнений реакций можно определить среду, то полуреакции оформляют соответственно по способу 2 — или для кислотной, или для щелочной среды, в противном случае — по способу 1.

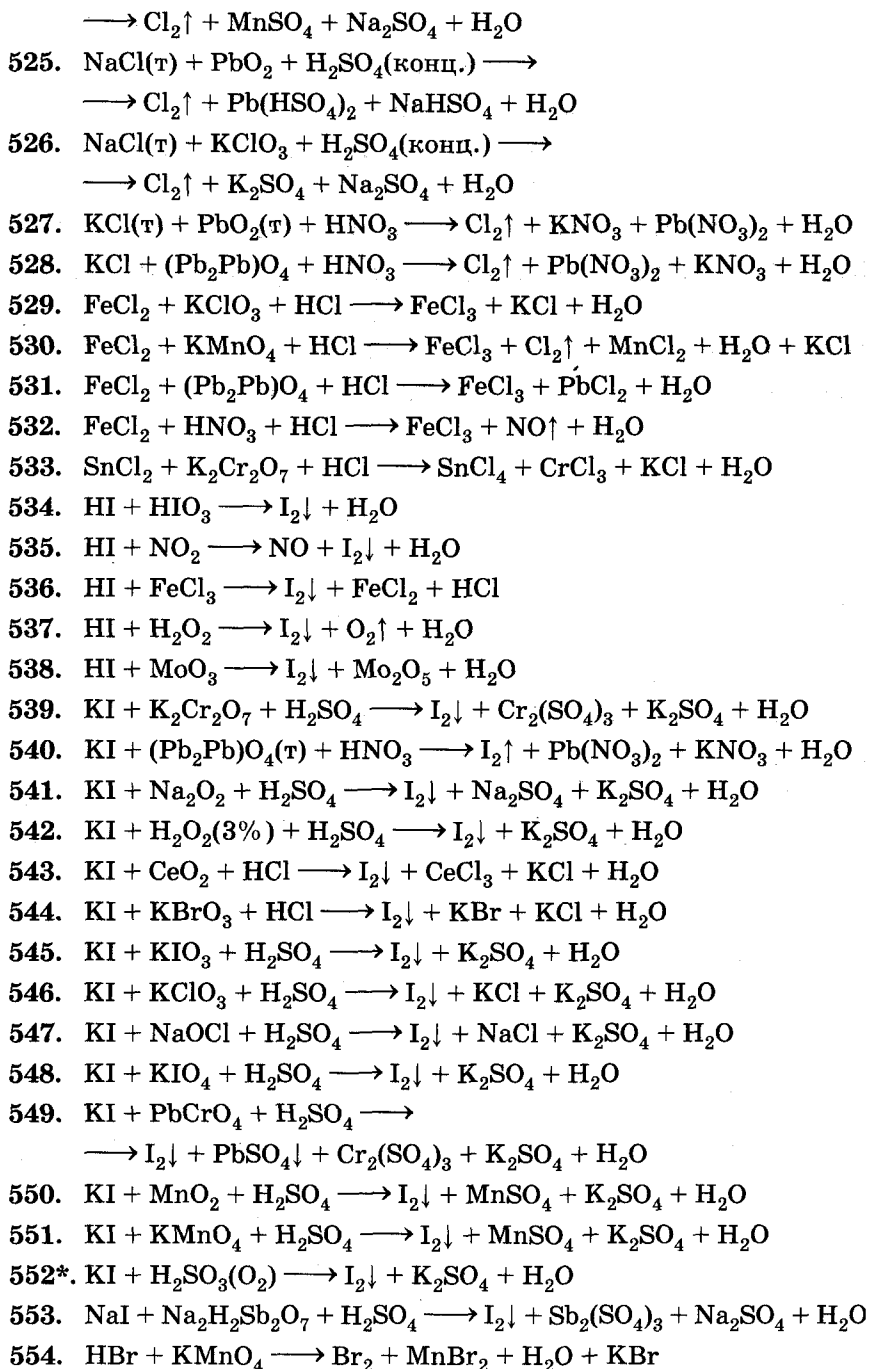
Приведем еще один пример уравнения реакции:



Схемы ОВР (условия задач)

Соединения галогенов (среда кислотная)





555. $\text{NaBr} + \text{NaBrO}_3 + \text{H}_2\text{SO}_4(\text{конц.}) \longrightarrow \text{Br}_2 + \text{Na}_2\text{SO}_4 + \text{H}_2\text{O}$
556. $\text{CuBr}(\text{т}) + \text{KMnO}_4 + \text{H}_2\text{SO}_4(\text{конц.}) \longrightarrow$
 $\longrightarrow \text{CuSO}_4 + \text{K}_2\text{SO}_4 + \text{Br}_2 + \text{MnSO}_4 + \text{H}_2\text{O}$
557. $\text{KBr} + \text{KBrO}_3(\text{конц.}) + \text{HCl} \longrightarrow \text{Br}_2 + \text{KCl} + \text{H}_2\text{O}$
558. $\text{KBr} + \text{KMnO}_4 + \text{H}_2\text{SO}_4 \longrightarrow$
 $\longrightarrow \text{K}_2\text{SO}_4 + \text{MnSO}_4 + \text{Br}_2 + \text{MnSO}_4 + \text{H}_2\text{O}$
559. $\text{Si}(\text{т}) + \text{HF}(\text{конц.}) + \text{HNO}_3(\text{конц.}) \longrightarrow \text{H}_2[\text{SiF}_6] + \text{NO}\uparrow + \text{H}_2\text{O}$
560. $\text{HClO} + \text{H}_2\text{O}_2 \longrightarrow \text{HCl} + \text{O}_2\uparrow + \text{H}_2\text{O}$
561. $\text{HClO} + \text{NO} + \text{H}_2\text{O} \longrightarrow \text{HCl} + \text{HNO}_3$
562. $\text{Ca}(\text{ClO})_2 + \text{Cu} + \text{H}_2\text{SO}_4 \longrightarrow \text{CaCl}_2 + \text{CuSO}_4 + \text{CaSO}_4\downarrow + \text{H}_2\text{O}$
- 563*. $\text{HClO}_2 \longrightarrow \text{ClO}_2\uparrow + \text{HClO}_3 + \text{HCl} + \text{H}_2\text{O}$
564. $\text{HClO}_3 + \text{P} + \text{H}_2\text{O} \longrightarrow \text{HCl} + \text{H}_3\text{PO}_4$
565. $\text{HClO}_3 + \text{Ag} \longrightarrow \text{AgCl}\downarrow + \text{AgClO}_3 + \text{H}_2\text{O}$
566. $\text{HClO}_3 + \text{S} + \text{H}_2\text{O} \longrightarrow \text{HCl} + \text{H}_2\text{SO}_4$
567. $\text{KClO}_3(\text{т}) + \text{H}_2\text{SO}_4(\text{конц.}) \longrightarrow \text{ClO}_2\uparrow + \text{KClO}_4 + \text{KHSO}_4 + \text{H}_2\text{O}$
568. $\text{NaClO}_3 + \text{H}_2\text{SO}_4 + \text{SO}_2 \longrightarrow \text{ClO}_2\uparrow + \text{NaHSO}_4$
569. $\text{HClO}_4 + \text{I}_2 \longrightarrow \text{HIO}_4 + \text{Cl}_2\uparrow$
570. $\text{KClO}_4 + \text{Al} + \text{H}_2\text{SO}_4(\text{разб.}) \longrightarrow \text{Al}_2(\text{SO}_4)_3 + \text{KCl} + \text{H}_2\text{O}$
571. $\text{HIO}_3 + \text{H}_2\text{O}_2 \longrightarrow \text{I}_2\downarrow + \text{O}_2\uparrow + \text{H}_2\text{O}$
572. $\text{HIO}_3 + \text{P} + \text{H}_2\text{O} \longrightarrow \text{HI}\downarrow + \text{H}_3\text{PO}_4$
573. $\text{NaIO}_3 + \text{CO} + \text{H}_2\text{SO}_4 \longrightarrow \text{I}_2 + \text{CO}_2\uparrow + \text{H}_2\text{O} + \text{Na}_2\text{SO}_4$

Сера, соединения серы и селена (среда кислотная)

- 574*. $\text{S} + \text{HNO}_3(\text{конц.}) \longrightarrow \text{H}_2\text{SO}_4 + \text{NO}_2\uparrow + \text{H}_2\text{O}$
575. $\text{S} + \text{H}_2\text{SO}_4(\text{конц.}) \longrightarrow \text{SO}_2\uparrow + \text{H}_2\text{O}$
576. $\text{SO}_2 + \text{K}_2\text{S}_2\text{O}_6(\text{O}_2) + \text{H}_2\text{O} \longrightarrow \text{K}_2\text{SO}_4 + \text{H}_2\text{SO}_4$
- 577*. $\text{SO}_2 + \text{KMnO}_4 + \text{H}_2\text{O} \longrightarrow \text{MnSO}_4 + \text{K}_2\text{SO}_4 + \text{H}_2\text{SO}_4$
578. $\text{SO}_2 + \text{K}_2\text{Cr}_2\text{O}_7 + \text{HCl} \longrightarrow \text{CrCl}_3 + \text{KCl} + \text{H}_2\text{SO}_4 + \text{H}_2\text{O}$
579. $\text{SO}_2 + \text{K}_2\text{Cr}_2\text{O}_7 + \text{H}_2\text{SO}_4 \longrightarrow \text{Cr}_2(\text{SO}_4)_3 + \text{K}_2\text{SO}_4 + \text{H}_2\text{O}$
580. $\text{SO}_2 + \text{I}_2 + \text{H}_2\text{O} \longrightarrow \text{HI} + \text{H}_2\text{SO}_4$
- 581*. $\text{SO}_2 + \text{NaIO}_3 + \text{H}_2\text{O} \longrightarrow \text{Na}_2\text{SO}_4 + \text{H}_2\text{SO}_4 + \text{I}_2\downarrow$
- 582*. $\text{SO}_2 + \text{NO}_2 + \text{H}_2\text{O} \longrightarrow \text{NO}\uparrow + \text{H}_2\text{SO}_4$
583. $\text{SO}_2 + \text{HNO}_3(\text{конц.}) \longrightarrow \text{H}_2\text{SO}_4 + \text{NO}_2\uparrow$
584. $\text{SO}_2 + \text{HClO}_4 + \text{H}_2\text{O} \longrightarrow \text{H}_2\text{SO}_4 + \text{HCl}$

585. $\text{SO}_2 + \text{H}_2\text{SeO}_3 + \text{H}_2\text{O} \longrightarrow \text{Se}\downarrow + \text{H}_2\text{SO}_4$
586. $\text{SO}_2 + \text{H}_2\text{TeO}_3 + \text{H}_2\text{O} \longrightarrow \text{Te}\downarrow + \text{H}_2\text{SO}_4$
587. $\text{H}_2\text{S} + \text{HIO}_3 \longrightarrow \text{S}\downarrow + \text{H}_2\text{O} + \text{I}_2\downarrow$
588. $\text{H}_2\text{S} + \text{HClO}_4 \longrightarrow \text{H}_2\text{SO}_4 + \text{HCl}$
589. $\text{H}_2\text{S} + \text{HClO}_3 \longrightarrow \text{H}_2\text{SO}_4 + \text{HCl}$
590. $\text{H}_2\text{S} + \text{Cl}_2 + \text{H}_2\text{O} \longrightarrow \text{H}_2\text{SO}_4 + \text{HCl}$
591. $\text{H}_2\text{S} + \text{FeBr}_3 \longrightarrow \text{HBr} + \text{FeBr}_2 + \text{S}\downarrow$
- 592*. $\text{H}_2\text{S} + \text{H}[\text{AuCl}_4] + \text{H}_2\text{O} \longrightarrow \text{Au}\downarrow + \text{S}\downarrow + \text{H}_2\text{SO}_4 + \text{HCl}$
593. $\text{H}_2\text{S} + \text{FeCl}_3 \longrightarrow \text{FeS} + \text{S}\downarrow + \text{HCl}$ (может образоваться FeS_2 вместо FeS)
594. $\text{H}_2\text{S} + \text{NaVO}_3 + \text{HCl} \longrightarrow \text{S}\downarrow + (\text{VO})\text{Cl}_2 + \text{H}_2\text{O} + \text{NaCl}$
595. $\text{H}_2\text{S} + \text{SO}_2 \longrightarrow \text{S}\downarrow + \text{H}_2\text{O}$
596. $\text{H}_2\text{S} + \text{H}_2\text{SO}_4(\text{конц.}) \longrightarrow \text{S}\downarrow + \text{SO}_2\uparrow + \text{H}_2\text{O}$
597. $\text{H}_2\text{S} + \text{H}_2\text{SO}_4(\text{конц.}) \longrightarrow \text{SO}_2\uparrow + \text{H}_2\text{O}$
598. $\text{H}_2\text{S}(\text{насыщ.}) + \text{HNO}_3(\text{конц.}) \longrightarrow \text{S}\downarrow + \text{NO}_2\uparrow + \text{H}_2\text{O}$
599. $\text{H}_2\text{S} + \text{HNO}_3 \longrightarrow \text{H}_2\text{SO}_4 + \text{H}_2\text{O} + \text{NO}\uparrow$
600. $\text{H}_2\text{S} + \text{K}_2\text{Cr}_2\text{O}_7 + \text{H}_2\text{SO}_4 \longrightarrow \text{Cr}_2(\text{SO}_4)_3 + \text{S}\downarrow + \text{H}_2\text{O} + \text{K}_2\text{SO}_4$
601. $\text{H}_2\text{S} + \text{KMnO}_4 + \text{H}_2\text{SO}_4(\text{разб.}) \longrightarrow \text{S}\downarrow + \text{MnSO}_4 + \text{K}_2\text{SO}_4 + \text{H}_2\text{O}$
602. $\text{H}_2\text{S} + \text{H}_2\text{O}_2 \longrightarrow \text{H}_2\text{SO}_4 + \text{H}_2\text{O}$
603. $\text{Na}_2\text{S} + \text{Na}_3\text{AsO}_4 + \text{HCl} \longrightarrow \text{As}_2\text{S}_3\downarrow + \text{S}\downarrow + \text{NaCl} + \text{H}_2\text{O}$
604. $\text{Na}_2\text{S} + \text{K}_2\text{Cr}_2\text{O}_7 + \text{H}_2\text{SO}_4 \longrightarrow$
 $\longrightarrow \text{S}\downarrow + \text{Cr}_2(\text{SO}_4)_3 + \text{K}_2\text{SO}_4 + \text{Na}_2\text{SO}_4 + \text{H}_2\text{O}$
605. $\text{Na}_2\text{S} + \text{K}_2\text{Cr}_2\text{O}_7(\text{конц.}) + \text{H}_2\text{SO}_4(\text{конц.}) \longrightarrow$
 $\longrightarrow \text{Cr}_2(\text{SO}_4)_3 + \text{Na}_2\text{SO}_4 + \text{K}_2\text{SO}_4 + \text{H}_2\text{O}$
606. $\text{Na}_2\text{S} + \text{H}_2\text{SO}_4(\text{конц.}) \longrightarrow \text{S}\downarrow + \text{SO}_2\uparrow + \text{NaHSO}_4 + \text{H}_2\text{O}$
607. $\text{Na}_2\text{S} + \text{HNO}_3(\text{конц.}) \longrightarrow \text{S}\downarrow + \text{NO}_2\uparrow + \text{H}_2\text{O} + \text{NaNO}_3$
608. $\text{Na}_2\text{S} + \text{Na}_2\text{SO}_3 + \text{HCl} \longrightarrow \text{S}\downarrow + \text{NaCl} + \text{H}_2\text{O}$
609. $\text{Na}_2\text{S} + \text{H}_2\text{O}_2(\text{конц.}) \longrightarrow \text{Na}_2\text{SO}_4 + \text{H}_2\text{O}$
610. $\text{Ba}(\text{HS})_2 + \text{HNO}_3 \longrightarrow \text{S}\downarrow + \text{NO}\uparrow + \text{H}_2\text{O} + \text{Ba}(\text{NO}_3)_2$
611. $\text{MnS}(\text{т}) + \text{HNO}_3(\text{конц.}) \longrightarrow \text{MnSO}_4 + \text{NO}_2\uparrow + \text{H}_2\text{O}$
612. $\text{HgS}(\text{т}) + \text{HNO}_3(\text{конц.}) + \text{HCl}(\text{конц.}) \longrightarrow$
 $\longrightarrow \text{HgCl}_2 + \text{NO}\uparrow + \text{H}_2\text{O} + \text{H}_2\text{SO}_4$
613. $\text{PbS}(\text{т}) + \text{HNO}_3(\text{разб.}) \longrightarrow \text{PbSO}_4\downarrow + \text{H}_2\text{O} + \text{NO}\uparrow$
614. $\text{PbS}(\text{т}) + \text{HNO}_3(\text{конц.}) \longrightarrow \text{PbSO}_4\downarrow + \text{NO}_2\uparrow + \text{H}_2\text{O}$

615. $\text{PbS(т)} + \text{H}_2\text{O}_2(30\%) \longrightarrow \text{PbSO}_4\downarrow + \text{H}_2\text{O}$
616. $\text{NH}_4\text{HS} + \text{HNO}_3(\text{конц.}) \longrightarrow \text{S}\downarrow + \text{H}_2\text{O} + \text{NO}_2\uparrow + \text{NH}_4\text{NO}_3$
- 617*. $\text{FeS}_2 + \text{Br}_2 + \text{HNO}_3 + \text{H}_2\text{O} \longrightarrow \text{Fe(NO}_3)_3 + \text{H}_2\text{SO}_4 + \text{HBr}$
618. $\text{FeS}_2(\text{т}) + \text{HNO}_3(\text{конц.}) \longrightarrow \text{Fe(NO}_3)_3 + \text{H}_2\text{SO}_4 + \text{NO}_2\uparrow + \text{H}_2\text{O}$
619. $\text{FeS}_2(\text{т}) + \text{H}_2\text{SO}_4(\text{конц.}) \longrightarrow \text{Fe}_2(\text{SO}_4)_3 + \text{SO}_2\uparrow + \text{H}_2\text{O}$
- 620*. $\text{FeS(т)} + \text{H}_2\text{SO}_4(\text{конц.}) + \text{HNO}_3(\text{конц.}) \longrightarrow$
 $\longrightarrow \text{Fe}_2(\text{SO}_4)_3 + \text{NO}_2\uparrow + \text{H}_2\text{O}$
621. $\text{CuFeS(т)} + \text{HNO}_3 \longrightarrow \text{Cu(NO}_3)_2 + \text{Fe(NO}_3)_3 + \text{NO}_2\uparrow + \text{SO}_2\uparrow + \text{H}_2\text{O}$
622. $\text{CuS(т)} + \text{HNO}_3(\text{конц.}) \longrightarrow \text{CuSO}_4 + \text{NO}_2\uparrow + \text{H}_2\text{O}$
623. $\text{Cu}_2\text{S(т)} + \text{HNO}_3 \longrightarrow \text{Cu(NO}_3)_2 + \text{CuSO}_4 + \text{H}_2\text{O} + \text{NO}_2\uparrow$
- 624*. $\text{Sb}_2\text{S}_3(\text{т}) + \text{HNO}_3(\text{конц.}) \longrightarrow \text{Sb}_2\text{O}_5 + \text{H}_2\text{SO}_4 + \text{NO}_2\uparrow + \text{H}_2\text{O}$
625. $\text{Al}_2\text{S}_3(\text{т}) + \text{HNO}_3(\text{конц.}) \longrightarrow \text{Al}_2(\text{SO}_4)_3 + \text{H}_2\text{O} + \text{NO}_2\uparrow$
626. $\text{NiS} + \text{H}_2\text{O}_2 + \text{H}_2\text{SO}_4 \longrightarrow \text{S}\downarrow + \text{NiSO}_4 + \text{H}_2\text{O}$
627. $(\text{U}_2\text{U})\text{O}_8 + \text{H}_2\text{SO}_4 \longrightarrow \text{UO}_2\text{SO}_4 + \text{U}(\text{SO}_4)_2 + \text{H}_2\text{O}$
628. $\text{H}_2\text{SO}_4(\text{конц.}) + \text{Zn} \longrightarrow \text{H}_2\text{S}\uparrow + \text{H}_2\text{O} + \text{ZnSO}_4$
629. $\text{H}_2\text{SO}_4(\text{конц.}) + \text{Na} \longrightarrow \text{H}_2\text{S}\uparrow + \text{H}_2\text{O} + \text{Na}_2\text{SO}_4$
630. $\text{H}_2\text{SO}_4(\text{конц.}) + \text{C} \longrightarrow \text{SO}_2\uparrow + \text{H}_2\text{O} + \text{CO}_2$
631. $\text{H}_2\text{SO}_4(\text{конц.}) + \text{Cu} \longrightarrow \text{CuSO}_4 + \text{SO}_2\uparrow + \text{H}_2\text{O}$
632. $\text{H}_2\text{SO}_4(\text{конц.}) + \text{K} \longrightarrow$
 $\longrightarrow \text{K}_2\text{SO}_4 + \text{SO}_2\uparrow + \text{S}\downarrow + \text{H}_2\text{O} (+ \text{примесь } \text{H}_2\text{S})$
633. $\text{H}_2\text{SO}_4(\text{конц.}) + \text{Al} \longrightarrow \text{Al}_2(\text{SO}_4)_3 + \text{H}_2\text{S}\uparrow + \text{H}_2\text{O}$
634. $\text{H}_2\text{SO}_4(\text{конц.}) + \text{Al} \longrightarrow \text{Al}_2(\text{SO}_4)_3 + \text{S}\downarrow + \text{H}_2\text{O}$
635. $\text{H}_2\text{SO}_4(\text{конц.}) + \text{Ca}_3\text{P}_2(\text{т}) \longrightarrow \text{Ca}_3(\text{PO}_4)_2\downarrow + \text{SO}_2\uparrow + \text{H}_2\text{O}$
636. $\text{H}_2\text{SO}_4(\text{конц.}) + \text{P(бел)} \longrightarrow \text{H}_2(\text{PHO}_3) + \text{SO}_2$
637. $\text{H}_2\text{SO}_4(\text{конц.}) + \text{PH}_3 \longrightarrow \text{SO}_2\uparrow + \text{H}_2(\text{PHO}_3) + \text{H}_2\text{O}$
638. $\text{H}_2\text{SO}_4 + \text{BaO}_2 \longrightarrow \text{BaSO}_4 + \text{O}_3\uparrow + \text{H}_2\text{O}$
639. $\text{H}_2\text{SO}_4(\text{конц.}) + \text{CuBr} \longrightarrow \text{CuBr}_2 + \text{CuSO}_4 + \text{SO}_2\uparrow + \text{H}_2\text{O}$
640. $\text{H}_2\text{SO}_4(\text{конц.}) + \text{KI(т)} \longrightarrow \text{I}_2\downarrow + \text{H}_2\text{S}\uparrow + \text{KHSO}_4 + \text{H}_2\text{O}$
641. $\text{H}_2\text{SO}_4 + \text{NaBr} \longrightarrow \text{Br}_2 + \text{SO}_2\uparrow + \text{Na}_2\text{SO}_4 + \text{H}_2\text{O}$
642. $\text{H}_2\text{SO}_4 + \text{NaI} \longrightarrow \text{I}_2\downarrow + \text{H}_2\text{S}\uparrow + \text{Na}_2\text{SO}_4 + \text{H}_2\text{O}$
643. $\text{H}_2\text{SO}_4(\text{конц.}) + \text{HI} \longrightarrow \text{I}_2\downarrow + \text{S}\downarrow + \text{H}_2\text{O}$
644. $\text{H}_2\text{SO}_4(\text{конц.}) + \text{HBr} \longrightarrow \text{Br}_2 + \text{SO}_2\uparrow + \text{H}_2\text{O} (\text{но не с } \text{HCl})$
645. $\text{H}_2\text{SO}_4(\text{конц.}) + \text{KBr(т)} \xrightarrow{t} \text{Br}_2 + \text{SO}_2\uparrow + \text{H}_2\text{O} + \text{KHSO}_4$
646. $\text{H}_2\text{SO}_4(60\%) + \text{KMnO}_4(\text{т}) \longrightarrow \text{MnSO}_4 + \text{O}_2\uparrow + \text{K}_2\text{SO}_4 + \text{H}_2\text{O}$

647. $\text{H}_2\text{SO}_4 + \text{K}_2\text{FeO}_4 \longrightarrow \text{Fe}_2(\text{SO}_4)_3 + \text{O}_2\uparrow + \text{K}_2\text{SO}_4 + \text{H}_2\text{O}$
648. $\text{H}_2\text{SO}_4(\text{конц.}) + \text{KClO}_3 \longrightarrow \text{HClO}_4 + \text{ClO}_2\uparrow + \text{KHSO}_4 + \text{H}_2\text{O}$
649. $\text{FeSO}_4 + \text{KNO}_2 + \text{H}_2\text{SO}_4 \longrightarrow \text{Fe}_2(\text{SO}_4)_3 + \text{NO}\uparrow + \text{K}_2\text{SO}_4 + \text{H}_2\text{O}$
650. $\text{FeSO}_4 + \text{O}_2 + \text{H}_2\text{SO}_4 \longrightarrow \text{Fe}_2(\text{SO}_4)_3 + \text{H}_2\text{O}$
651. $\text{FeSO}_4 + \text{KClO}_3 + \text{H}_2\text{SO}_4 \longrightarrow \text{Fe}_2(\text{SO}_4)_3 + \text{KCl} + \text{H}_2\text{O}$
652. $\text{FeSO}_4 + \text{HNO}_3(\text{конц.}) + \text{H}_2\text{SO}_4 \longrightarrow \text{Fe}_2(\text{SO}_4)_3 + \text{NO}_2\uparrow + \text{H}_2\text{O}$
653. $\text{FeSO}_4 + \text{HNO}_3(\text{конц.}) \longrightarrow \text{Fe}(\text{NO}_3)_3 + \text{NO}_2\uparrow + \text{H}_2\text{SO}_4 + \text{H}_2\text{O}$
654. $\text{FeSO}_4 + \text{K}_2\text{Cr}_2\text{O}_7 + \text{H}_2\text{SO}_4 \longrightarrow$
 $\longrightarrow \text{Fe}_2(\text{SO}_4)_3 + \text{Cr}_2(\text{SO}_4)_3 + \text{K}_2\text{SO}_4 + \text{H}_2\text{O}$
655. $\text{FeSO}_4 + \text{KMnO}_4 + \text{H}_2\text{SO}_4 \longrightarrow \text{Fe}_2(\text{SO}_4)_3 + \text{MnSO}_4 + \text{K}_2\text{SO}_4 + \text{H}_2\text{O}$
656. $\text{FeSO}_4 + \text{H}_2\text{O}_2 + \text{H}_2\text{SO}_4 \longrightarrow \text{Fe}_2(\text{SO}_4)_3 + \text{H}_2\text{O}$
657. $\text{FeSO}_4 + \text{BaO}_2(\text{т}) + \text{H}_2\text{SO}_4 \longrightarrow \text{Fe}_2(\text{SO}_4)_3 + \text{BaSO}_4\downarrow + \text{H}_2\text{O}$
658. $\text{FeSO}_4 + \text{Br}_2 + \text{H}_2\text{SO}_4 \longrightarrow \text{HBr} + \text{Fe}_2(\text{SO}_4)_3$
659. $\text{FeSO}_4 + \text{HIO}_3 + \text{H}_2\text{SO}_4 \longrightarrow \text{I}_2\downarrow + \text{Fe}_2(\text{SO}_4)_3 + \text{H}_2\text{O}$
660. $\text{FeSO}_4 + \text{CrO}_3 + \text{H}_2\text{SO}_4 \longrightarrow \text{Fe}_2(\text{SO}_4)_3 + \text{Cr}_2(\text{SO}_4)_3 + \text{H}_2\text{O}$
661. $\text{SnSO}_4 + \text{KIO}_3 + \text{H}_2\text{SO}_4 \longrightarrow \text{Sn}(\text{SO}_4)_2 + \text{I}_2\downarrow + \text{K}_2\text{SO}_4 + \text{H}_2\text{O}$
- 662*. $\text{MnSO}_4 + \text{PbO}_2 + \text{H}_2\text{SO}_4 \longrightarrow \text{HMnO}_4 + \text{PbSO}_4\downarrow + \text{H}_2\text{O}$
663. $\text{MnSO}_4 + \text{PbO}_2 + \text{HNO}_3(\text{конц.}) \longrightarrow$
 $\longrightarrow \text{HMnO}_4 + \text{Pb}(\text{HSO}_4)_2 + \text{Pb}(\text{NO}_3)_2 + \text{H}_2\text{O}$
- 664*. $\text{MnSO}_4 + \text{NaBiO}_3(\text{т}) + \text{HNO}_3 \longrightarrow$
 $\longrightarrow \text{HMnO}_4 + \text{Bi}(\text{NO}_3)_3\downarrow + \text{Na}_2\text{SO}_4 + \text{NaNO}_3 + \text{H}_2\text{O}$
665. $\text{SO}_2 + \text{HBrO} \longrightarrow \text{H}_2\text{SO}_4 + \text{HBr}$
666. $\text{SO}_2 + \text{H}_2\text{O}_2 \longrightarrow \text{H}_2\text{SO}_4$
667. $\text{Na}_2\text{SO}_3 + \text{KMnO}_4 + \text{H}_2\text{SO}_4 \longrightarrow$
 $\longrightarrow \text{MnSO}_4 + \text{Na}_2\text{SO}_4 + \text{K}_2\text{SO}_4 + \text{H}_2\text{O}$
- 668*. $\text{Na}_2\text{SO}_3 + \text{SnCl}_2 + \text{HCl} \longrightarrow \text{SnS}_2 + \text{H}_2[\text{SnCl}_6] + \text{NaCl} + \text{H}_2\text{O}$
669. $\text{Na}_2\text{SO}_3 + \text{Zn} + \text{HCl} \longrightarrow \text{ZnCl}_2 + \text{NaCl} + \text{H}_2\text{S}\uparrow + \text{H}_2\text{O}$
670. $\text{Na}_2\text{SO}_3 + \text{Mg} + \text{CH}_3\text{COOH} \longrightarrow$
 $\longrightarrow \text{Mg}(\text{CH}_3\text{COO})_2 + \text{CH}_3\text{COONa} + \text{H}_2\text{S}\uparrow + \text{H}_2\text{O}$
671. $\text{BaSO}_3 + \text{HNO}_3 + \text{KMnO}_4 \longrightarrow \text{BaSO}_4 + \text{Mn}(\text{NO}_3)_2 + \text{KNO}_3 + \text{H}_2\text{O}$
672. $\text{Na}_2\text{SO}_3 + \text{I}_2 + \text{H}_2\text{O} \longrightarrow \text{Na}_2\text{SO}_4 + \text{HI}$
673. $\text{Na}_2\text{SO}_3 + \text{CrO}_3 + \text{H}_2\text{SO}_4 \longrightarrow \text{Na}_2\text{SO}_4 + \text{Cr}_2(\text{SO}_4)_3 + \text{H}_2\text{O}$
674. $\text{Na}_2\text{SO}_3 + \text{KClO}_3 + \text{H}_2\text{SO}_4 \longrightarrow \text{Cl}_2\uparrow + \text{Na}_2\text{SO}_4 + \text{K}_2\text{SO}_4 + \text{H}_2\text{O}$
675. $\text{K}_2\text{SO}_3 + \text{K}_2\text{Cr}_2\text{O}_7 + \text{H}_2\text{SO}_4 \longrightarrow \text{K}_2\text{SO}_4 + \text{Cr}_2(\text{SO}_4)_3 + \text{H}_2\text{O}$

- 676*. $\text{Na}_2\text{SO}_3\text{S} + \text{KMnO}_4 + \text{H}_2\text{SO}_4 \longrightarrow$
 $\longrightarrow \text{MnSO}_4 + \text{Na}_2\text{SO}_4 + \text{K}_2\text{SO}_4 + \text{H}_2\text{O}$
677. $\text{Na}_2\text{SO}_3\text{S} + \text{HCl} \longrightarrow \text{S} + \text{SO}_2\uparrow + \text{NaCl} + \text{H}_2\text{O}$
678. $\text{Na}_2\text{SO}_3\text{S} + \text{HCl}(\text{конц.}) + \text{H}_2\text{O} \longrightarrow \text{H}_2\text{SO}_4 + \text{H}_2\text{S}\uparrow + \text{NaCl}$
- 679*. $\text{Na}_2\text{SO}_3\text{S} + \text{HOCl} + \text{H}_2\text{O} \longrightarrow \text{H}_2\text{SO}_4 + \text{HCl} + \text{NaCl}$
- 680*. $\text{Na}_2\text{SO}_3\text{S} + \text{KMnO}_4 + \text{HNO}_3 \longrightarrow$
 $\longrightarrow \text{Na}_2\text{SO}_4 + \text{H}_2\text{SO}_4 + \text{Mn}(\text{NO}_3)_2 + \text{KNO}_3 + \text{H}_2\text{O}$
681. $\text{Na}_2\text{SO}_3\text{S} + \text{HNO}_3(\text{конц.}) \longrightarrow \text{Na}_2\text{SO}_4 + \text{S}\downarrow + \text{NO}_2\uparrow + \text{H}_2\text{O}$
- 682*. $\text{Na}_2\text{SO}_3\text{S} + \text{Cl}_2(\text{насыщ.}) + \text{H}_2\text{O} \longrightarrow \text{H}_2\text{SO}_4 + \text{HCl} + \text{NaCl}$
- 683*. $\text{Na}_2\text{SO}_3\text{S} + \text{I}_2(\text{иодная вода}) \longrightarrow \text{Na}_2\text{S}_4\text{O}_6 + \text{NaI}$
- 684*. $\text{Na}_2\text{S}_2\text{O}_4 + \text{KMnO}_4 + \text{HNO}_3 \longrightarrow$
 $\longrightarrow \text{Na}_2\text{SO}_4 + \text{H}_2\text{SO}_4 + \text{Mn}(\text{NO}_3)_2 + \text{KNO}_3 + \text{H}_2\text{O}$
685. $\text{SeO}_2 + \text{SO}_2 + \text{H}_2\text{O} \longrightarrow \text{H}_2\text{SO}_4 + \text{Se}\downarrow$
686. $\text{H}_2\text{SeO}_4(\text{конц.}) + \text{Au} \longrightarrow \text{Au}_2(\text{SeO}_4)_3 + \text{SeO}_2 + \text{H}_2\text{O}$
687. $\text{H}_2\text{SeO}_3 + \text{H}_2\text{O}_2 \longrightarrow \text{H}_2\text{SeO}_4 + \text{H}_2\text{O}$
688. $\text{H}_2\text{SeO}_3 + \text{HClO}_3 \longrightarrow \text{H}_2\text{SeO}_4 + \text{HCl}$
689. $\text{Na}_2\text{SeO}_3 + \text{Na}_2\text{SO}_3 + \text{HCl} \longrightarrow \text{Na}_2\text{SO}_4 + \text{Se}\downarrow + \text{H}_2\text{O} + \text{NaCl}$

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690. $\text{NO}_2 + \text{HI}(\text{разб.}) \longrightarrow \text{NO}\uparrow + \text{I}_2\downarrow + \text{H}_2\text{O}$
691. $\text{N}_2\text{O} + \text{KMnO}_4 + \text{H}_2\text{SO}_4 \longrightarrow \text{NO}_2\uparrow + \text{MnSO}_4 + \text{K}_2\text{SO}_4 + \text{H}_2\text{O}$
692. $\text{HNO}_3(\text{оч. разб.}) + \text{Ca} \longrightarrow \text{Ca}(\text{NO}_3)_2 + \text{NH}_4\text{NO}_3 + \text{H}_2\text{O}$
693. $\text{HNO}_3(\text{разб.}) + \text{Ca} \longrightarrow \text{Ca}(\text{NO}_3)_2 + \text{N}_2\text{O}\uparrow + \text{H}_2\text{O}$
694. $\text{HNO}_3(\text{оч. разб.}) + \text{Co} \longrightarrow \text{Co}(\text{NO}_3)_2 + \text{N}_2\uparrow + \text{H}_2\text{O}$
- 695*. $\text{HNO}_3(\text{разб.}) + \text{Fe} \longrightarrow \text{Fe}(\text{NO}_3)_3 + \text{NO}\uparrow + \text{H}_2\text{O}$
696. $\text{HNO}_3(35\%) + \text{Cu} \longrightarrow \text{Cu}(\text{NO}_3)_2 + \text{NO}\uparrow + \text{H}_2\text{O}$
697. $\text{HNO}_3(\text{конц.}) + \text{Cu} \longrightarrow \text{Cu}(\text{NO}_3)_2 + \text{NO}_2\uparrow + \text{H}_2\text{O}$
698. $\text{HNO}_3(\text{разб.}) + \text{Zn} \longrightarrow \text{Zn}(\text{NO}_3)_2 + \text{N}_2\uparrow + \text{H}_2\text{O}$
699. $\text{HNO}_3(\text{разб.}) + \text{Zn} \longrightarrow \text{Zn}(\text{NO}_3)_2 + \text{N}_2\text{O}\uparrow + \text{H}_2\text{O}$
700. $\text{HNO}_3(\text{разб.}) + \text{Bi} \longrightarrow \text{Bi}(\text{NO}_3)_3 + \text{NO}\uparrow + \text{H}_2\text{O}$
701. $\text{HNO}_3(\text{оч. разб.}) + \text{Mg} \longrightarrow \text{Mg}(\text{NO}_3)_2 + \text{NH}_4\text{NO}_3 + \text{H}_2\text{O}$
702. $\text{HNO}_3(\text{разб.}) + \text{Mg} \longrightarrow \text{Mg}(\text{NO}_3)_2 + \text{N}_2\text{O}\uparrow + \text{H}_2\text{O}$
703. $\text{HNO}_3(\text{разб.}) + \text{Sn} \longrightarrow \text{Sn}(\text{NO}_3)_2 + \text{N}_2\uparrow + \text{H}_2\text{O}$
704. $\text{HNO}_3(\text{конц.}) + \text{Sn} \longrightarrow \text{SnO}_2\downarrow + \text{NO}_2\uparrow + \text{H}_2\text{O}$

705. $\text{HNO}_3 + \text{Sb} \longrightarrow \text{Sb}_2\text{O}_5\downarrow + \text{NO}_2\uparrow + \text{H}_2\text{O}$
706. $\text{HNO}_3(\text{разб.}) + \text{Se} + \text{H}_2\text{O} \longrightarrow \text{H}_2\text{SeO}_3 + \text{NO}\uparrow$
707. $\text{HNO}_3(\text{разб.}) + \text{Al} \longrightarrow \text{Al}(\text{NO}_3)_3 + \text{N}_2\text{O}\uparrow + \text{H}_2\text{O}$
708. $\text{HNO}_3(\text{оч. разб.}) + \text{Al} \longrightarrow \text{Al}(\text{NO}_3)_3 + \text{NH}_4\text{NO}_3 + \text{H}_2\text{O}$
709. $\text{HNO}_3 + \text{Cd} \longrightarrow \text{Cd}(\text{NO}_3)_2 + \text{NO}_2\uparrow + \text{H}_2\text{O}$
710. $\text{HNO}_3 + \text{C} \longrightarrow \text{CO}_2\uparrow + \text{NO}_2\uparrow + \text{H}_2\text{O}$
711. $\text{HNO}_3 + \text{I}_2 \longrightarrow \text{HIO}_3 + \text{NO}\uparrow + \text{H}_2\text{O}$
712. $\text{HNO}_3(\text{конц.}) + \text{I}_2 \longrightarrow \text{HIO}_3 + \text{NO}_2\uparrow + \text{H}_2\text{O}$
713. $\text{HNO}_3 + \text{B} \longrightarrow \text{B}(\text{OH})_3 + \text{NO}_2\uparrow$
714. $\text{HNO}_3(30\%) + \text{P} + \text{H}_2\text{O} \longrightarrow \text{H}_3\text{PO}_4 + \text{NO}\uparrow$
715. $\text{HNO}_3(\text{конц.}) + \text{P}(\text{красн.}) \longrightarrow \text{H}_3\text{PO}_4 + \text{NO}_2\uparrow + \text{H}_2\text{O}$
716. $\text{HNO}_3 + \text{Ca}_3\text{P}_2 \longrightarrow \text{Ca}(\text{NO}_3)_2 + \text{H}_3\text{PO}_4 + \text{NO}_2\uparrow + \text{H}_2\text{O}$
717. $\text{HNO}_3(\text{конц.}) + \text{PCl}_3 \longrightarrow \text{H}_3\text{PO}_4 + \text{NO}_2\uparrow + \text{HCl}$
718. $\text{HNO}_3(\text{разб.}) + \text{H}(\text{PH}_2\text{O}_2) \longrightarrow \text{H}_2(\text{PHO}_3) + \text{NO}\uparrow + \text{H}_2\text{O}$
719. $\text{HNO}_3(\text{дым.}) + \text{H}_2(\text{PHO}_3) \longrightarrow \text{H}_3\text{PO}_4 + \text{NO}\uparrow + \text{H}_2\text{O}$
720. $\text{HNO}_3(\text{конц.}) + \text{Na}_3\text{P}(\text{т}) \longrightarrow \text{NaH}_2\text{PO}_4 + \text{NaNO}_3 + \text{NO}_2\uparrow + \text{H}_2\text{O}$
721. $\text{HNO}_3(\text{конц.}) + \text{PH}_3 \longrightarrow \text{H}_3\text{PO}_4 + \text{NO}_2\uparrow + \text{H}_2\text{O}$
722. $\text{HNO}_3(\text{конц.}) + \text{Na}_2\text{SO}_3 \longrightarrow \text{Na}_2\text{SO}_4 + \text{NO}_2\uparrow + \text{H}_2\text{O}$
723. $\text{HNO}_3(\text{конц.}) + \text{Cu}_2\text{O}(\text{т}) \longrightarrow \text{Cu}(\text{NO}_3)_2 + \text{NO}_2\uparrow + \text{H}_2\text{O}$
724. $\text{HNO}_3(\text{конц.}) + \text{Fe}(\text{OH})_2 \longrightarrow \text{Fe}(\text{NO}_3)_3 + \text{NO}_2\uparrow + \text{H}_2\text{O}$
725. $\text{HNO}_3(\text{конц.}) + \text{FeO}(\text{т}) \longrightarrow \text{Fe}(\text{NO}_3)_3 + \text{NO}_2\uparrow + \text{H}_2\text{O}$
726. $\text{HNO}_3(\text{конц.}) + (\text{FeFe}_2)\text{O}_4(\text{т}) \longrightarrow \text{Fe}(\text{NO}_3)_3 + \text{NO}_2\uparrow + \text{H}_2\text{O}$
727. $\text{HNO}_3(\text{разб.}) + (\text{FeFe}_2)\text{O}_4(\text{т}) \longrightarrow \text{Fe}(\text{NO}_3)_3 + \text{NO}\uparrow + \text{H}_2\text{O}$
728. $\text{HNO}_3 + \text{CuI} \longrightarrow \text{Cu}(\text{NO}_3)_2 + \text{I}_2\downarrow + \text{NO}_2\uparrow + \text{H}_2\text{O}$
729. $\text{HNO}_3 + \text{CaI}_2 \longrightarrow \text{Ca}(\text{NO}_3)_2 + \text{I}_2\downarrow + \text{NO}\uparrow + \text{H}_2\text{O}$
730. $\text{HNO}_3(\text{конц.}) + \text{FeCl}_2 \longrightarrow \text{Fe}(\text{NO}_3)_3 + \text{NO}_2\uparrow + \text{HCl}\uparrow + \text{H}_2\text{O}$
731. $\text{HNO}_3(\text{конц.}) + \text{CuBr}(\text{т}) \longrightarrow \text{Cu}(\text{NO}_3)_2 + \text{HBr} + \text{NO}_2\uparrow + \text{H}_2\text{O}$
732. $\text{HNO}_3(\text{конц.}) + \text{HCl} \longrightarrow \text{Cl}_2 + \text{NO} + \text{H}_2\text{O}$
733. $\text{HNO}_3(\text{конц.}) + \text{HCl}(\text{конц.}) + \text{Pt} \longrightarrow \text{H}_2[\text{PtCl}_6] + \text{NO}\uparrow + \text{H}_2\text{O}$
734. $\text{HNO}_3(\text{конц.}) + \text{HCl}(\text{конц.}) + \text{Au} \longrightarrow \text{H}[\text{AuCl}_4] + \text{NO}\uparrow + \text{H}_2\text{O}$
735. $\text{HNO}_3(60\%) + \text{HI} \longrightarrow \text{HIO}_3 + \text{NO}_2\uparrow + \text{H}_2\text{O}$
736. $\text{HNO}_3(\text{конц.}) + \text{Fe}_3\text{C}(\text{т}) \longrightarrow \text{Fe}(\text{NO}_3)_3 + \text{CO}_2\uparrow + \text{NO}_2\uparrow + \text{H}_2\text{O}$
737. $\text{HNO}_3(\text{конц.}) + \text{Fe}(\text{As})\text{S} \longrightarrow \text{FeAsO}_4\downarrow + \text{NO}_2\uparrow + \text{H}_2\text{SO}_4 + \text{H}_2\text{O}$
738. $\text{NaNO}_3 + \text{Hg} + \text{H}_2\text{SO}_4 \longrightarrow \text{Hg}_2\text{SO}_4 + \text{NO}\uparrow + \text{H}_2\text{O} + \text{Na}_2\text{SO}_4$

739. $\text{NaNO}_3 + \text{Zn} + \text{HCl} \longrightarrow \text{ZnCl}_2 + \text{NaNO}_2 + \text{H}_2\text{O}$
740. $\text{HNO}_2 + \text{Br}_2 + \text{H}_2\text{O} \longrightarrow \text{HNO}_3 + \text{HBr}$
741. $\text{HNO}_2 + \text{HMnO}_4 \longrightarrow \text{Mn}(\text{NO}_3)_2 + \text{HNO}_3 + \text{H}_2\text{O}$
742. $\text{HNO}_2 + \text{HI} \longrightarrow \text{I}_2\downarrow + \text{NO}\uparrow + \text{H}_2\text{O}$
743. $\text{HNO}_2 + \text{SnCl}_2 + \text{HCl} \longrightarrow \text{H}_2[\text{SnCl}_6] + \text{NO}\uparrow + \text{H}_2\text{O}$
744. $\text{HNO}_2 + \text{H}_2\text{O}_2(\text{конц.}) \longrightarrow \text{HNO}_3 + \text{H}_2\text{O}$
745. $\text{HNO}_2 + \text{HCl} + \text{KMnO}_4 \longrightarrow \text{KNO}_3 + \text{Mn}(\text{NO}_3)_2 + \text{H}_2\text{O} + \text{KCl}$
746. $\text{HNO}_2 + \text{KI} + \text{H}_2\text{SO}_4 \longrightarrow \text{NO}\uparrow + \text{I}_2\downarrow + \text{K}_2\text{SO}_4 + \text{H}_2\text{O}$
- 747*. $\text{HNO}_2 + \text{FeCl}_2(\text{конц.}) + \text{H}_2\text{SO}_4 \longrightarrow$
 $\longrightarrow \text{FeCl}_3 + \text{N}_2\uparrow + \text{Fe}_2(\text{SO}_4)_3 + \text{H}_2\text{O}$
748. $\text{KNO}_2 + \text{KI} + \text{H}_2\text{SO}_4 \longrightarrow \text{NO}\uparrow + \text{H}_2\text{O} + \text{I}_2\downarrow + \text{K}_2\text{SO}_4$
749. $\text{KNO}_2 + \text{KI} + \text{CH}_3\text{COOH} \longrightarrow \text{NO}\uparrow + \text{H}_2\text{O} + \text{I}_2 + \text{CH}_3\text{COOK}$
750. $\text{KNO}_2 + \text{KMnO}_4 + \text{H}_2\text{SO}_4 \longrightarrow \text{MnSO}_4 + \text{KNO}_3 + \text{K}_2\text{SO}_4 + \text{H}_2\text{O}$
751. $\text{NaNO}_2 + \text{FeCl}_2 + \text{HCl} \longrightarrow \text{FeCl}_3 + \text{NO}\uparrow + \text{NaCl} + \text{H}_2\text{O}$
752. $\text{NaNO}_2 + \text{K}_2\text{Cr}_2\text{O}_7 + \text{H}_2\text{SO}_4 \longrightarrow$
 $\longrightarrow \text{NaNO}_3 + \text{Cr}_2(\text{SO}_4)_3 + \text{K}_2\text{SO}_4 + \text{H}_2\text{O}$
753. $\text{NaNO}_2 + \text{PbO}_2 + \text{HNO}_3 \longrightarrow \text{NaNO}_3 + \text{Pb}(\text{NO}_3)_2 + \text{H}_2\text{O}$
754. $\text{NaNO}_2(\text{т}) + \text{HNO}_3(\text{конц.}) \longrightarrow \text{NO}_2\uparrow + \text{NO}\uparrow + \text{NaNO}_3 + \text{H}_2\text{O}$
755. $\text{H}_2(\text{PHO}_3) + \text{Pb}(\text{NO}_3)_2 + \text{H}_2\text{O} \longrightarrow \text{Pb}\downarrow + \text{HNO}_3 + \text{H}_3\text{PO}_4$
756. $\text{H}_2(\text{PHO}_3) + \text{HgCl}_2 + \text{H}_2\text{O} \longrightarrow \text{H}_3\text{PO}_4 + \text{HCl} + \text{Hg}_2\text{Cl}_2\downarrow$
757. $\text{H}_2(\text{PHO}_3) + \text{AgNO}_3 + \text{H}_2\text{O} \longrightarrow \text{H}_3\text{PO}_4 + \text{HNO}_3 + \text{Ag}\downarrow$
758. $\text{H}_2(\text{PHO}_3) + \text{Cl}_2 + \text{H}_2\text{O} \longrightarrow \text{H}_3\text{PO}_4 + \text{HCl}$
759. $\text{PBr}_3 + \text{HNO}_3 \longrightarrow \text{H}_3\text{PO}_4 + \text{Br}_2 + \text{NO}_2\uparrow + \text{H}_2\text{O}$
760. $\text{Mg}_3\text{P}_2 + \text{HNO}_3(\text{конц.}) \longrightarrow \text{H}_3\text{PO}_4 + \text{Mg}(\text{NO}_3)_2 + \text{NO}_2\uparrow + \text{H}_2\text{O}$
761. $\text{Ca}_3\text{P}_2 + \text{K}_2\text{Cr}_2\text{O}_7 + \text{HCl} \longrightarrow \text{Ca}_3(\text{PO}_4)_2 + \text{CrCl}_3 + \text{KCl} + \text{H}_2\text{O}$
762. $\text{Ca}_3\text{P}_2 + \text{KMnO}_4 + \text{H}_2\text{SO}_4 \longrightarrow \text{Ca}_3(\text{PO}_4)_2 + \text{MnSO}_4 + \text{K}_2\text{SO}_4 + \text{H}_2\text{O}$
763. $\text{P}_2\text{O}_3 + \text{H}_2\text{SO}_4(\text{конц.}) + \text{H}_2\text{O} \longrightarrow \text{H}_3\text{PO}_4 + \text{SO}_2\uparrow$
764. $\text{H}(\text{PH}_2\text{O}_2) + \text{HgCl}_2 + \text{H}_2\text{O} \xrightarrow{t} \text{H}_2(\text{PHO}_3) + \text{HCl} + \text{Hg}\downarrow$
765. $\text{H}(\text{PH}_2\text{O}_2) + \text{I}_2 + \text{H}_2\text{O} \longrightarrow \text{H}_2(\text{PHO}_3) + \text{HI}$
766. $\text{PH}_3 + \text{KMnO}_4 + \text{H}_2\text{SO}_4 \longrightarrow \text{H}_3\text{PO}_4 + \text{MnSO}_4 + \text{K}_2\text{SO}_4 + \text{H}_2\text{O}$
767. $\text{AsH}_3 + \text{KMnO}_4 + \text{H}_2\text{SO}_4 \longrightarrow \text{H}_3\text{AsO}_4 + \text{MnSO}_4 + \text{K}_2\text{SO}_4 + \text{H}_2\text{O}$
768. $\text{As}_2\text{O}_3 + \text{Zn} + \text{H}_2\text{SO}_4 \longrightarrow \text{AsH}_3\uparrow + \text{ZnSO}_4 + \text{H}_2\text{O}$
769. $\text{As}_2\text{O}_3 + \text{HClO} + \text{H}_2\text{O} \longrightarrow \text{H}_3\text{AsO}_4 + \text{HCl}$
770. $\text{HNO}_3(50\%) + \text{As}_2\text{O}_3 \longrightarrow \text{HAsO}_3 + \text{NO}_2\uparrow + \text{N}_2\text{O}$

771. $\text{As}_2\text{O}_3 + \text{H}(\text{PH}_2\text{O}_2) \longrightarrow \text{As}\downarrow + \text{H}_2(\text{PHO}_3)$
 772. $\text{As}_2\text{O}_3 + \text{KBrO}_3 + \text{HCl} \longrightarrow \text{As}_2\text{O}_5 + \text{HBr}_2 + \text{KCl}$
 773. $\text{H}_3\text{AsO}_3 + \text{Zn} + \text{H}_2\text{SO}_4 \longrightarrow \text{AsH}_3\uparrow + \text{ZnSO}_4 + \text{H}_2\text{O}$
 774. $\text{Na}_3\text{AsO}_4 + \text{KI} + \text{HCl} \longrightarrow \text{As}_2\text{O}_3 + \text{I}_2\downarrow + \text{KCl} + \text{H}_2\text{O} + \text{NaCl}$
 775*. $\text{Na}_3\text{AsO}_4 + \text{Zn} + \text{H}_2\text{SO}_4 \longrightarrow \text{AsH}_3\uparrow + \text{ZnSO}_4 + \text{Na}_2\text{SO}_4 + \text{H}_2\text{O}$
 776. $\text{As}_2\text{S}_5 + \text{HNO}_3(\text{конц.}) \longrightarrow \text{H}_3\text{AsO}_4 + \text{H}_2\text{SO}_4 + \text{NO}_2\uparrow + \text{H}_2\text{O}$
 777. $\text{As}_2\text{S}_3 + \text{HNO}_3 + \text{H}_2\text{O} \longrightarrow \text{H}_3\text{AsO}_4 + \text{H}_2\text{SO}_4 + \text{NO}\uparrow$

Соединения марганца (среда кислотная)

778. $\text{HMnO}_4 + \text{MnSO}_4 + \text{H}_2\text{O} \longrightarrow \text{MnO}_2\downarrow + \text{H}_2\text{SO}_4$
 779. $\text{KMnO}_4 + \text{FeCO}_3 + \text{H}_2\text{SO}_4 \longrightarrow$
 $\longrightarrow \text{Fe}_2(\text{SO}_4)_3 + \text{CO}_2\uparrow + \text{MnSO}_4 + \text{K}_2\text{SO}_4 + \text{H}_2\text{O}$
 780. $\text{KMnO}_4 + \text{Al} + \text{H}_2\text{SO}_4 \longrightarrow \text{Al}_2(\text{SO}_4)_3 + \text{MnSO}_4 + \text{KCl} + \text{H}_2\text{O}$
 781*. $\text{KMnO}_4 + \text{SmCl}_2 + \text{H}_2\text{SO}_4 \longrightarrow$
 $\longrightarrow \text{Sm}_2(\text{SO}_4)_3 + \text{SmCl}_3 + \text{MnCl}_2 + \text{KCl} + \text{H}_2\text{O} + \text{HCl}$
 782*. $\text{KMnO}_4 + \text{FeCl}_2 + \text{H}_2\text{SO}_4 \longrightarrow$
 $\longrightarrow \text{Fe}_2(\text{SO}_4)_3 + \text{FeCl}_3 + \text{MnSO}_4 + \text{KCl} + \text{H}_2\text{O}$
 783*. $\text{KMnO}_4 + (\text{NH}_4)_2\text{Fe}(\text{SO}_4)_2 + \text{H}_2\text{SO}_4 \longrightarrow$
 $\longrightarrow \text{Fe}_2(\text{SO}_4)_3 + (\text{NH}_4)_2\text{SO}_4 + \text{MnSO}_4 + \text{K}_2\text{SO}_4 + \text{H}_2\text{O}$
 784. $\text{KMnO}_4 + \text{Ag} + \text{H}_2\text{SO}_4 \longrightarrow \text{Ag}_2\text{SO}_4\downarrow + \text{MnSO}_4 + \text{K}_2\text{SO}_4 + \text{H}_2\text{O}$
 785. $\text{KMnO}_4 + \text{Na}_2\text{O}_2 + \text{H}_2\text{SO}_4 \longrightarrow$
 $\longrightarrow \text{MnSO}_4 + \text{O}_2\uparrow + \text{Na}_2\text{SO}_4 + \text{H}_2\text{O} + \text{K}_2\text{SO}_4$
 786. $\text{Mn}(\text{NO}_3)_2 + \text{PbO}_2(\tau) + \text{HNO}_3 \longrightarrow \text{Pb}(\text{NO}_3)_2 + \text{HMnO}_4 + \text{H}_2\text{O}$
 787. $\text{Mn}(\text{NO}_3)_2 + \text{NaBiO}_3(\tau) + \text{HNO}_3 \longrightarrow$
 $\longrightarrow \text{Bi}(\text{NO}_3)_3 + \text{HMnO}_4 + \text{NaNO}_3 + \text{H}_2\text{O}$
 788. $\text{Mn}(\text{NO}_3)_2 + \text{NaBiO}_3(\tau) + \text{H}_2\text{O} + \text{HNO}_3 \longrightarrow$
 $\longrightarrow \text{HMnO}_4 + \text{BiNO}_3(\text{OH})_2\downarrow + \text{NaNO}_3$
 789. $\text{Mn}(\text{NO}_3)_2 + (\text{Pb}_2\text{Pb})\text{O}_4(\tau) + \text{HNO}_3 \longrightarrow \text{Pb}(\text{NO}_3)_2 + \text{HMnO}_4 + \text{H}_2\text{O}$
 790. $\text{MnO}_2 + \text{Fe}(\text{NO}_3)_2 + \text{HNO}_3 \longrightarrow \text{Fe}(\text{NO}_3)_3 + \text{Mn}(\text{NO}_3)_2 + \text{H}_2\text{O}$
 791. $\text{MnO}_2 + \text{H}_2\text{SO}_4 \longrightarrow \text{MnSO}_4 + \text{H}_2\text{O} + \text{O}_2\uparrow$
 792. $\text{MnO}_2 + \text{FeCl}_2 + \text{HCl} \longrightarrow \text{FeCl}_3 + \text{MnCl}_2 + \text{H}_2\text{O}$
 793. $\text{MnO}_2 + \text{H}_2\text{O}_2 + \text{H}_2\text{SO}_4 \longrightarrow \text{MnSO}_4 + \text{O}_2\uparrow + \text{H}_2\text{O}$
 794. $\text{MnO}_2 + \text{NaBiO}_3(\tau) + \text{HNO}_3 + \text{H}_2\text{O} \longrightarrow$
 $\longrightarrow \text{BiNO}_3(\text{OH})_2\downarrow + \text{HMnO}_4 + \text{NaNO}_3$
 795. $\text{K}_2\text{MnO}_4 + \text{HCl}(\text{конц.}) \longrightarrow \text{KCl} + \text{MnCl}_2 + \text{Cl}_2\uparrow + \text{H}_2\text{O}$

Соединения хрома (среда кислотная)

796. $\text{K}_2\text{Cr}_2\text{O}_7 + \text{KNCS} + \text{H}_2\text{SO}_4 \longrightarrow$
 $\longrightarrow \text{CO}_2\uparrow + \text{Cr}_2(\text{SO}_4)_3 + \text{SO}_2\uparrow + \text{NO}_2\uparrow + \text{K}_2\text{SO}_4 + \text{H}_2\text{O}$
797. $\text{K}_2\text{Cr}_2\text{O}_7 + \text{Zn} + \text{H}_2\text{SO}_4 \longrightarrow \text{ZnSO}_4 + \text{CrSO}_4 + \text{K}_2\text{SO}_4 + \text{H}_2\text{O}$
798. $\text{K}_2\text{Cr}_2\text{O}_7 + \text{H}_2\text{O}_2 + \text{H}_2\text{SO}_4 \longrightarrow \text{O}_2\uparrow + \text{Cr}_2(\text{SO}_4)_3 + \text{K}_2\text{SO}_4 + \text{H}_2\text{O}$
- 799*. $\text{K}_2\text{Cr}_2\text{O}_7 + \text{H}_2\text{O}_2 + \text{H}_2\text{SO}_4 \longrightarrow \text{CrO}(\text{O}_2)_2 + \text{K}_2\text{SO}_4 + \text{H}_2\text{O}$
800. $\text{H}_2\text{Cr}_2\text{O}_7 + \text{SO}_2 \longrightarrow \text{Cr}_2(\text{SO}_4)_3 + \text{H}_2\text{O}$
801. $\text{K}_2\text{Cr}_2\text{O}_7 + \text{Fe} + \text{H}_2\text{SO}_4 \longrightarrow \text{Fe}_2(\text{SO}_4)_3 + \text{CrSO}_4 + \text{K}_2\text{SO}_4 + \text{H}_2\text{O}$
802. $\text{K}_2\text{Cr}_2\text{O}_7 + \text{HClO}_4 + \text{HI} \longrightarrow \text{Cr}(\text{ClO}_4)_3 + \text{KClO}_4 + \text{I}_2\downarrow + \text{H}_2\text{O}$
803. $\text{Cr}(\text{NO}_3)_3 + \text{NaBiO}_3 + \text{HNO}_3 \longrightarrow$
 $\longrightarrow \text{Bi}(\text{NO}_3)_3 + \text{NaNO}_3 + \text{H}_2\text{Cr}_2\text{O}_7 + \text{H}_2\text{O}$

Пероксид водорода (среда кислотная)

804. $\text{H}_2\text{O}_2 + \text{PbO}_2 + \text{HNO}_3 \longrightarrow \text{Pb}(\text{NO}_3)_2 + \text{O}_2 + \text{H}_2\text{O}$
805. $\text{H}_2\text{O}_2 + \text{NiO}(\text{OH}) + \text{HCl} \longrightarrow \text{NiCl}_2 + \text{O}_2\uparrow + \text{H}_2\text{O}$

Галогены и их соединения (среда щелочная)

806. $\text{Cl}_2 + \text{FeO}(\text{OH}) + \text{KOH} \longrightarrow \text{K}_2\text{FeO}_4 + \text{KCl} + \text{H}_2\text{O}$
807. $\text{Cl}_2 + \text{NaOH}(\text{гор.}, \text{конц.}) \longrightarrow \text{NaClO}_3 + \text{NaCl} + \text{H}_2\text{O}$
808. $\text{Cl}_2 + \text{NaOH}(\text{хол.}, \text{конц.}) \longrightarrow \text{NaClO} + \text{NaCl} + \text{H}_2\text{O}$
809. $\text{Cl}_2 + \text{Ca}(\text{OH})_2 \longrightarrow \text{Ca}(\text{ClO})_2 + \text{CaCl}_2 + \text{H}_2\text{O}$
810. $\text{Cl}_2 + \text{Ba}(\text{OH})_2 \longrightarrow \text{BaCl}_2 + \text{Ba}(\text{ClO}_3)_2 + \text{H}_2\text{O}$
- 811*. $\text{Cl}_2 + \text{NaIO}_3 + \text{NaOH} \longrightarrow \text{Na}_3\text{H}_2\text{IO}_6\downarrow + \text{NaCl} + \text{H}_2\text{O}$
812. $\text{Cl}_2 + \text{Na}_2\text{SeO}_3 + \text{NaOH} \longrightarrow \text{Na}_2\text{SeO}_4 + \text{NaCl} + \text{H}_2\text{O}$
813. $\text{Cl}_2 + \text{Br}_2 + \text{KOH} \longrightarrow \text{KCl} + \text{KBrO}_3 + \text{H}_2\text{O}$
814. $\text{ClO}_2 + \text{NaOH} \longrightarrow \text{NaClO}_3 + \text{NaClO}_2 + \text{H}_2\text{O}$
815. $\text{FeCl}_2 + \text{H}_2\text{O}_2 + \text{KOH} \longrightarrow \text{FeO}(\text{OH})\downarrow + \text{KCl} + \text{H}_2\text{O}$
816. $\text{FeCl}_2 + \text{O}_2 + \text{KOH} \longrightarrow \text{FeO}(\text{OH})\downarrow + \text{KCl} + \text{H}_2\text{O}$
817. $\text{AuCl}_3 + \text{H}_2\text{O}_2 + \text{KOH} \longrightarrow \text{Au}\downarrow + \text{O}_2\uparrow + \text{KCl} + \text{H}_2\text{O}$
818. $\text{BiCl}_3 + \text{Sn}(\text{OH})_2 + \text{NaOH} \longrightarrow \text{Bi}\downarrow + \text{Na}_2[\text{Sn}(\text{OH})_6] + \text{NaCl}$
819. $\text{BiCl}_3 + \text{SnCl}_2 + \text{KOH} \longrightarrow \text{Bi}\downarrow + \text{K}_2[\text{Sn}(\text{OH})_6] + \text{KCl}$
820. $\text{CuCl}_2 + \text{NH}_2\text{OH} + \text{NaOH} \longrightarrow \text{N}_2\uparrow + \text{CuCl} + \text{NaCl} + \text{H}_2\text{O}$
821. $\text{Br}_2 + \text{Ni}(\text{OH})_2 + \text{NaOH} \longrightarrow \text{NiO}(\text{OH}) + \text{NaBr} + \text{H}_2\text{O}$
822. $\text{Br}_2 + \text{ZnS} + \text{NaOH} \longrightarrow \text{Na}_2[\text{Zn}(\text{OH})_4] + \text{S}\downarrow + \text{NaBr}$

823. $\text{Br}_2 + \text{MnSO}_4 + \text{NaOH} \longrightarrow \text{MnO}_2\downarrow + \text{NaBr} + \text{Na}_2\text{SO}_4 + \text{H}_2\text{O}$
 824. $\text{Br}_2 + \text{Sb}_2\text{O}_3 + \text{KOH} + \text{H}_2\text{O} \longrightarrow \text{K}_2[\text{Sb}(\text{OH})_6] + \text{KBr}$
 825. $\text{Br}_2 + \text{Bi}(\text{OH})_3 + \text{KOH} \longrightarrow \text{KBiO}_3 + \text{KBr} + \text{H}_2\text{O}$
 826. $\text{Br}_2 + \text{FeO}(\text{OH}) + \text{KOH}(\text{конц.}) \longrightarrow \text{K}_2\text{FeO}_4 + \text{KBr} + \text{H}_2\text{O}$
 827. $\text{KBrO} + \text{Fe}(\text{OH})_2 + \text{KOH} \longrightarrow \text{K}_2\text{FeO}_4 + \text{KBr} + \text{H}_2\text{O}$
 828. $\text{F}_2 + \text{NaOH} \longrightarrow \text{NaF} + \text{OF}_2 + \text{H}_2\text{O}$
 829. $\text{F}_2 + \text{NaBrO}_3 + \text{NaOH} \longrightarrow \text{NaBrO}_4 + \text{NaF} + \text{H}_2\text{O}$
 830. $\text{Cl}_2 + \text{NaAsO}_2 + \text{NaOH} \longrightarrow \text{Na}_3\text{AsO}_4 + \text{NaCl} + \text{H}_2\text{O}$
 831. $\text{Cl}_2 + \text{K}_3\text{AsO}_3 + \text{KOH} \longrightarrow \text{K}_3\text{AsO}_4 + \text{KCl} + \text{H}_2\text{O}$
 832. $\text{I}_2 + \text{N}_2\text{H}_4 + \text{NaOH} \longrightarrow \text{NaI} + \text{N}_2\uparrow + \text{H}_2\text{O}$

Сера и ее соединения (среда щелочная)

833. $\text{S} + \text{NaOH}(\text{конц.}) \longrightarrow \text{Na}_2\text{SO}_3\text{S} + \text{Na}_2\text{S} + \text{H}_2\text{O}$
 834. $\text{S} + \text{KOH} \longrightarrow \text{K}_2\text{S} + \text{K}_2\text{SO}_3 + \text{H}_2\text{O}$
 835. $\text{S} + \text{F}_2 + \text{NaOH} \longrightarrow \text{Na}_2\text{SO}_4 + \text{NaF} + \text{H}_2\text{O}$
 836. $\text{SO}_2 + \text{H}_2\text{S} + \text{NaOH} \longrightarrow \text{Na}_2\text{SO}_3\text{S} + \text{H}_2\text{O}$
 837. $\text{Na}_2\text{SO}_3\text{S} + \text{H}_2\text{O}_2 + \text{NaOH} \longrightarrow \text{Na}_2\text{SO}_4 + \text{H}_2\text{O}$
 838. $\text{K}_2\text{S}_2\text{O}_6(\text{O}_2) + \text{AgNO}_3 + \text{KOH} \longrightarrow$
 $\longrightarrow (\text{Ag}^{+1}\text{Ag}^{+3})\text{O}_2 + \text{K}_2\text{SO}_4 + \text{KNO}_3 + \text{H}_2\text{O}$
 839. $\text{K}_2\text{S}_2\text{O}_6(\text{O}_2) + \text{I}_2(\text{т}) + \text{KOH} \longrightarrow \text{K}_2\text{SO}_4 + \text{KIO}_3 + \text{H}_2\text{O}$
 840. $\text{K}_2\text{S}_2\text{O}_6(\text{O}_2) + \text{Fe}(\text{OH})_2 + \text{KOH}(\text{конц.}) \longrightarrow$
 $\longrightarrow \text{K}_2\text{SO}_4 + \text{FeO}(\text{OH})\downarrow + \text{H}_2\text{O}$
 841. $\text{K}_2\text{S}_2\text{O}_6(\text{O}_2) + \text{NaOH}(\text{конц.}) + \text{Bi}(\text{OH})_3 \longrightarrow$
 $\longrightarrow \text{K}_2\text{SO}_4 + \text{Na}_2\text{SO}_4 + \text{NaBiO}_3\downarrow + \text{H}_2\text{O}$
 842. $\text{Na}_2\text{SO}_3 + \text{Cl}_2 + \text{NaOH} \longrightarrow \text{NaCl} + \text{Na}_2\text{SO}_4 + \text{H}_2\text{O}$
 843. $\text{Na}_2\text{SO}_3 + \text{AgNO}_3 + \text{NaOH} \longrightarrow \text{Na}_2\text{SO}_4 + \text{Ag}\downarrow + \text{H}_2\text{O} + \text{NaNO}_3$
 844. $\text{FeSO}_4 + \text{H}_2\text{O}_2 + \text{NaOH} \longrightarrow \text{FeO}(\text{OH})\downarrow + \text{Na}_2\text{SO}_4 + \text{H}_2\text{O}$
 845. $\text{CoS} + \text{H}_2\text{O}_2 + \text{NaOH} \longrightarrow \text{CoO}(\text{OH})\downarrow + \text{Na}_2\text{SO}_4 + \text{H}_2\text{O}$

Соединения азота и фосфора (среда щелочная)

- 846*. $\text{NH}_3 + \text{KMnO}_4 + \text{KOH} \longrightarrow \text{KNO}_3 + \text{K}_2\text{MnO}_4 + \text{H}_2\text{O}$
 847. $\text{NO} + \text{KOH} \longrightarrow \text{N}_2\uparrow + \text{KNO}_2 + \text{H}_2\text{O}$
 или $\text{NO} + \text{KOH} \longrightarrow \text{N}_2\text{O}\uparrow + \text{KNO}_2 + \text{H}_2\text{O}$
 848. $\text{NO}_2 + \text{O}_2 + \text{KOH} \longrightarrow \text{KNO}_3 + \text{H}_2\text{O}$
 849. $\text{NO}_2 + \text{KOH}(\text{разб.}) \longrightarrow \text{KNO}_2 + \text{KNO}_3 + \text{H}_2\text{O}$

850. $\text{NO}_2 + \text{Ba}(\text{OH})_2 \longrightarrow \text{Ba}(\text{NO}_3)_2 + \text{Ba}(\text{NO}_2)_2 + \text{H}_2\text{O}$
 851. $\text{NO}_2 + \text{NO} + \text{KOH}(\text{разб.}) \longrightarrow \text{KNO}_2 + \text{H}_2\text{O}$
 852. $\text{Cr}(\text{NO}_3)_3 + \text{H}_2\text{O}_2 + \text{NaOH} \longrightarrow \text{Na}_2\text{CrO}_4 + \text{NaNO}_3 + \text{H}_2\text{O}$
 853. $\text{KNO}_3 + \text{Os} + \text{KOH} + \text{H}_2\text{O} \longrightarrow \text{KNO}_2 + \text{K}_2[\text{OsO}_2(\text{OH})_4]$
 854. $\text{KNO}_3 + \text{Fe} + \text{KOH} \longrightarrow \text{KNO}_2 + \text{K}_2\text{FeO}_4 + \text{H}_2\text{O}$
 855*. $\text{NaNO}_3 + \text{Al} + \text{NaOH} + \text{H}_2\text{O} \longrightarrow \text{Na}[\text{Al}(\text{OH})_4] + \text{NH}_3\uparrow$
 856. $\text{Pb}(\text{NO}_3)_2 + \text{H}_2\text{O}_2 + \text{NaOH} \longrightarrow \text{PbO}_2\downarrow + \text{NaNO}_3 + \text{H}_2\text{O}$
 857. $\text{AgNO}_3 + \text{N}_2\text{H}_4(\text{конц.}) + \text{KOH} \longrightarrow \text{Ag}\downarrow + \text{N}_2\uparrow + \text{KNO}_3 + \text{H}_2\text{O}$
 858. $\text{AgNO}_3 + \text{H}_2\text{O}_2 + \text{KOH} \longrightarrow \text{Ag}\downarrow + \text{O}_2\uparrow + \text{KNO}_3 + \text{H}_2\text{O}$
 859. $\text{KNO}_3 + \text{Zn} + \text{KOH} + \text{H}_2\text{O} \longrightarrow \text{K}_2[\text{Zn}(\text{OH})_4] + \text{NH}_3\uparrow$
 860. $\text{NaNO}_2 + \text{Al} + \text{NaOH} + \text{H}_2\text{O} \longrightarrow \text{Na}[\text{Al}(\text{OH})_4] + \text{NH}_3\uparrow$
 861. $\text{NH}_3 \cdot \text{H}_2\text{O}(\text{конц.}) + \text{Br}_2(\text{разб.}) \longrightarrow \text{N}_2\uparrow + \text{NH}_4\text{Br} + \text{H}_2\text{O}$
 862. $\text{NH}_3 \cdot \text{H}_2\text{O}(\text{конц.}) + \text{I}_2(\tau) \longrightarrow \text{I}_3\text{N}\downarrow + \text{NH}_4\text{I} + \text{H}_2\text{O}$
 863. $\text{NH}_3 \cdot \text{H}_2\text{O} + \text{K}_2\text{FeO}_4 \longrightarrow \text{FeO}(\text{OH})\downarrow + \text{N}_2\uparrow + \text{KOH} + \text{H}_2\text{O}$
 864. $\text{P} + \text{H}_2\text{O} + \text{KOH} \longrightarrow \text{PH}_3\uparrow + \text{K}(\text{PH}_2\text{O}_2)$
 865. $\text{P} + \text{Ba}(\text{OH})_2 + \text{H}_2\text{O} \longrightarrow \text{PH}_3\uparrow + \text{Ba}(\text{PH}_2\text{O}_2)_2$
 866. $\text{PH}_3 + \text{KClO} + \text{KOH} \longrightarrow \text{KCl} + \text{K}(\text{PH}_2\text{O}_2) + \text{H}_2\text{O}$

Соединения германия, кремния и теллура (среда щелочная)

867. $\text{Ge} + \text{NaOH} + \text{H}_2\text{O}_2 \longrightarrow \text{Na}_2\text{GeO}_3 + \text{H}_2\text{O}$
 868. $\text{Si} + \text{NaOH} + \text{H}_2\text{O} \longrightarrow \text{Na}_2\text{SiO}_3 + \text{H}_2\uparrow$
 869. $\text{Te} + \text{KOH} \longrightarrow \text{K}_2\text{Te} + \text{K}_2\text{TeO}_3 + \text{H}_2\text{O}$

Соединения марганца (среда щелочная)

870. $\text{Mn}(\text{OH})_2 + \text{Cl}_2 + \text{KOH} \longrightarrow \text{MnO}_2\downarrow + \text{KCl} + \text{H}_2\text{O}$
 871. $\text{MnCl}_2 + \text{KBrO} + \text{KOH} \longrightarrow \text{MnO}(\text{OH})\downarrow + \text{KBr} + \text{KCl} + \text{H}_2\text{O}$
 872. $\text{KMnO}_4 + \text{KOH}(15\%) \longrightarrow \text{K}_2\text{MnO}_4 + \text{H}_2\text{O} + \text{O}_2\uparrow$
 873. $\text{KMnO}_4 + \text{K}_2\text{SO}_3(\text{недост.}) + \text{NaOH} \longrightarrow$
 $\longrightarrow \text{K}_2\text{MnO}_4 + \text{Na}_2\text{SO}_4 + \text{H}_2\text{O}$
 874. $\text{KMnO}_4 + \text{K}_2\text{SO}_3(\text{изб.}) + \text{KOH}(\text{конц.}) \longrightarrow$
 $\longrightarrow \text{K}_3\text{MnO}_4 + \text{K}_2\text{SO}_4 + \text{H}_2\text{O}$
 875. $\text{KMnO}_4 + \text{SO}_2 + \text{KOH} \longrightarrow \text{MnO}_2\downarrow + \text{K}_2\text{SO}_4 + \text{H}_2\text{O}$
 876. $\text{KMnO}_4 + \text{SO}_2 + \text{KOH} \longrightarrow \text{K}_2\text{MnO}_4 + \text{K}_2\text{SO}_4 + \text{H}_2\text{O}$
 877. $\text{KMnO}_4 + \text{K}_2\text{SO}_3\text{S} + \text{KOH} \longrightarrow \text{K}_2\text{MnO}_4 + \text{K}_2\text{SO}_4 + \text{H}_2\text{O}$

878. $\text{KMnO}_4(\text{конц.}) + \text{K}(\text{PH}_2\text{O}_2) + \text{KOH}(\text{конц.}) \longrightarrow$
 $\longrightarrow \text{K}_3\text{PO}_4 + \text{K}_2\text{MnO}_4 + \text{H}_2\text{O}$
879. $\text{KMnO}_4 + \text{K}_2(\text{PHO}_3) + \text{KOH}(\text{конц.}) \longrightarrow \text{K}_3\text{PO}_4 + \text{K}_2\text{MnO}_4 + \text{H}_2\text{O}$
880. $\text{KMnO}_4 + \text{SnCl}_2 + \text{KOH} \longrightarrow \text{K}_2\text{MnO}_4 + \text{K}_2[\text{Sn}(\text{OH})_6] + \text{KCl}$
881. $\text{KMnO}_4 + \text{Al} + \text{KOH} \longrightarrow \text{K}_2\text{MnO}_4 + \text{K}[\text{Al}(\text{OH})_4]$
882. $\text{MnSO}_4 + \text{H}_2\text{O}_2 + \text{NaOH} \longrightarrow \text{MnO}_2 + \text{Na}_2\text{SO}_4 + \text{H}_2\text{O}$
883. $\text{Mn}(\text{NO}_3)_2 + \text{H}_2\text{O}_2 + \text{NaOH} \longrightarrow \text{MnO}_2\downarrow + \text{NaNO}_3 + \text{H}_2\text{O}$

Соединения хрома (среда щелочная)

884. $\text{Cr}_2\text{O}_3 + \text{Br}_2 + \text{NaOH} \longrightarrow \text{Na}_2\text{CrO}_4 + \text{NaBr} + \text{H}_2\text{O}$
885. $\text{Cr}(\text{OH})_3 + \text{Cl}_2 + \text{NaOH} \longrightarrow \text{Na}_2\text{CrO}_4 + \text{NaCl} + \text{H}_2\text{O}$
886. $\text{Cr}(\text{OH})_3 + \text{PbO}_2 + \text{KOH} \longrightarrow \text{K}_2\text{CrO}_4 + \text{K}[\text{Pb}(\text{OH})_3] + \text{H}_2\text{O}$
887. $\text{CrCl}_2 + \text{O}_2 + \text{KOH} + \text{H}_2\text{O} \longrightarrow \text{K}_3[\text{Cr}(\text{OH})_6] + \text{KCl}$
888. $\text{CrCl}_3 + \text{Br}_2 + \text{KOH} \longrightarrow \text{K}_2\text{CrO}_4 + \text{KBr} + \text{KCl} + \text{H}_2\text{O}$
889. $\text{CrCl}_3 + \text{H}_2\text{O}_2 + \text{NaOH} \longrightarrow \text{Na}_2\text{CrO}_4 + \text{NaCl} + \text{H}_2\text{O}$
890. $\text{CrCl}_3 + \text{NaClO} + \text{NaOH} \longrightarrow \text{Na}_2\text{CrO}_4 + \text{NaCl} + \text{H}_2\text{O}$
891. $\text{CrCl}_3 + \text{NaBiO}_3 + \text{NaOH} + \text{H}_2\text{O} \longrightarrow$
 $\longrightarrow \text{Na}_2\text{CrO}_4 + \text{Bi}(\text{OH})_3\downarrow + \text{NaCl}$
892. $\text{Cr}_2(\text{SO}_4)_3 + \text{Br}_2 + \text{NaOH} \longrightarrow \text{Na}_2\text{CrO}_4 + \text{NaBr} + \text{Na}_2\text{SO}_4 + \text{H}_2\text{O}$
893. $\text{Cr}_2(\text{SO}_4)_3 + \text{H}_2\text{O}_2 + \text{NaOH} \longrightarrow \text{Na}_2\text{CrO}_4 + \text{Na}_2\text{SO}_4 + \text{H}_2\text{O}$
894. $\text{Cr}_2(\text{SO}_4)_3 + \text{KMnO}_4 + \text{KOH} \longrightarrow$
 $\longrightarrow \text{K}_2\text{CrO}_4 + \text{MnO}_2\downarrow + \text{K}_2\text{SO}_4 + \text{H}_2\text{O}$
895. $\text{KCr}(\text{SO}_4)_2 + \text{H}_2\text{O}_2(\text{конц.}) + \text{KOH}(\text{конц.}) \longrightarrow$
 $\longrightarrow \text{K}_2\text{CrO}_4 + \text{K}_2\text{SO}_4 + \text{H}_2\text{O}$
896. $\text{Na}_3[\text{Cr}(\text{OH})_6] + \text{Br}_2 + \text{NaOH} \longrightarrow \text{Na}_2\text{CrO}_4 + \text{NaBr} + \text{H}_2\text{O}$
897. $\text{K}_3[\text{Cr}(\text{OH})_6] + \text{PbO}_2 + \text{KOH} \longrightarrow \text{K}_2\text{CrO}_4 + \text{K}[\text{Pb}(\text{OH})_3] + \text{H}_2\text{O}$
898. $\text{Na}_3[\text{Cr}(\text{OH})_6] + \text{Br}_2 + \text{NaOH} \longrightarrow \text{Na}_2\text{CrO}_4 + \text{NaBr} + \text{H}_2\text{O}$
899. $\text{Na}_3[\text{Cr}(\text{OH})_6] + \text{PbO}_2 + \text{NaOH} \longrightarrow \text{Na}_2\text{CrO}_4 + \text{Na}_2\text{PbO}_2 + \text{H}_2\text{O}$
900. $\text{K}_3[\text{Cr}(\text{OH})_6] + \text{H}_2\text{O}_2 \longrightarrow \text{K}_2\text{CrO}_4 + \text{H}_2\text{O} + \text{KOH}$
901. $\text{Na}_2\text{CrO}_4 + \text{H}_2\text{O}_2(30\%) + \text{NaOH}(\text{изб.}) \longrightarrow \text{Na}_3\text{CrO}_8 + \text{H}_2\text{O}$

Галогены и их соединения (среда нейтральная)

902. $\text{Cl}_2 + \text{Br}_2 + \text{H}_2\text{O} \longrightarrow \text{HCl} + \text{HBrO}_3$
903. $\text{Cl}_2(\text{насыщ.}) + \text{K}_2\text{MnO}_4 \longrightarrow \text{KCl} + \text{KMnO}_4$
904. $\text{Cl}_2 + \text{FeSO}_4 + \text{Na}_2\text{SO}_4 \longrightarrow \text{NaCl} + \text{Fe}_2(\text{SO}_4)_3$

905. $\text{Cl}_2 + \text{I}_2 + \text{H}_2\text{O} \longrightarrow \text{HCl} + \text{HIO}_3$
906. $\text{Cl}_2 + \text{NH}_4\text{HS} + \text{NH}_3 \cdot \text{H}_2\text{O} \longrightarrow \text{S}\downarrow + \text{NH}_4\text{Cl} + \text{H}_2\text{O}$
907. $\text{Cl}_2 + \text{PH}_3 + \text{H}_2\text{O} \longrightarrow \text{H}_3\text{PO}_4 + \text{HCl}$
908. $\text{Cl}_2 + \text{H}_2\text{O} \longrightarrow \text{HClO} + \text{HCl}$
909. $\text{Cl}_2 + \text{As} + \text{H}_2\text{O} \longrightarrow \text{H}_3\text{AsO}_4 + \text{HCl}$
910. $\text{Cl}_2 + \text{Ca}_3\text{P}_2 + \text{H}_2\text{O} \longrightarrow \text{Ca}_3(\text{PO}_4)_2\downarrow + \text{HCl}$
911. $\text{Br}_2 + \text{PCl}_3 + \text{H}_2\text{O} \longrightarrow \text{H}_3\text{PO}_4 + \text{HCl} + \text{HBr}$
912. $\text{Br}_2 + \text{SO}_2 + \text{H}_2\text{O} \longrightarrow \text{HBr} + \text{H}_2\text{SO}_4$
913. $\text{I}_2 + \text{H}_2\text{S} \longrightarrow \text{HI} + \text{S}\downarrow$
914. $\text{I}_2 + \text{SO}_2 + \text{H}_2\text{O} \longrightarrow \text{HI} + \text{H}_2\text{SO}_4$
915. $\text{I}_2 + \text{Na}_3\text{AsO}_3 + \text{H}_2\text{O} \longrightarrow \text{Na}_3\text{AsO}_4 + \text{HI}$
916. $\text{I}_2 + \text{H}_2\text{O}_2(30\%) \longrightarrow \text{HIO}_3 + \text{H}_2\text{O}$
917. $\text{I}_2 + \text{PH}_3 + \text{H}_2\text{O} \longrightarrow \text{HI} + \text{H}(\text{PH}_2\text{O}_2)$
918. $\text{KI} + \text{FeCl}_3 \longrightarrow \text{FeCl}_2 + \text{KCl} + \text{I}_2\downarrow$
919. $\text{KI} + \text{H}_2\text{O}_2 \longrightarrow \text{I}_2\downarrow + \text{KOH}$
920. $\text{FeI}_2 + \text{NO}_2 + \text{H}_2\text{O} \longrightarrow \text{I}_2\downarrow + \text{FeO}(\text{OH})\downarrow + \text{NO}\uparrow$
921. $\text{KI} + \text{O}_3 + \text{H}_2\text{O} \longrightarrow \text{I}_2\downarrow + \text{O}_2 + \text{KOH}$
922. $\text{KI} + \text{Cl}_2(\text{p-p, изб.}) + \text{H}_2\text{O} \longrightarrow \text{KIO}_3 + \text{HCl}$
923. $\text{CuI} + \text{NO}_2 + \text{H}_2\text{O} \longrightarrow \text{I}_2\downarrow + \text{Cu}(\text{OH})_2\downarrow + \text{NO}\uparrow$
924. $\text{KI} + \text{CuCl}_2 \longrightarrow \text{CuCl}\downarrow + \text{I}_2\downarrow + \text{KCl}$
925. $\text{NaI} + \text{K}_2\text{CrO}_4 + \text{H}_2\text{O} \longrightarrow \text{Cr}(\text{OH})_3\downarrow + \text{I}_2\downarrow + \text{KOH} + \text{NaOH}$
926. $\text{SnCl}_2 + \text{AuCl}_3 + \text{H}_2\text{O} \longrightarrow \text{SnO}_2\downarrow + \text{HCl} + \text{AuCl}_2$
927. $\text{AuCl}_3 + \text{Se} + \text{H}_2\text{O} \longrightarrow \text{H}_2\text{SeO}_3 + \text{HCl} + \text{Au}\downarrow$
928. $\text{HgCl}_2 + \text{SO}_2 + \text{H}_2\text{O} \longrightarrow \text{H}_2\text{SO}_4 + \text{Hg}_2\text{Cl}_2\downarrow + \text{HCl}$
929. $\text{KBr} + \text{KMnO}_4 + \text{H}_2\text{O} \longrightarrow \text{Br}_2 + \text{MnO}_2\downarrow + \text{KOH}$
930. $\text{ICl} + \text{H}_2\text{O} \longrightarrow \text{I}_2\downarrow + \text{HCl} + \text{HIO}_3$
931. $\text{KClO} + \text{Na}_2\text{SO}_3\text{S} \longrightarrow \text{Na}_2\text{SO}_4 + \text{S}\downarrow + \text{KCl}$
932. $\text{NaClO} + \text{NaCN} + \text{H}_2\text{O} \longrightarrow \text{NaHCO}_3 + \text{NaCl} + \text{N}_2\uparrow$
933. $\text{NaClO} + \text{Ni}(\text{OH})_2 \longrightarrow \text{NaCl} + \text{NiO}(\text{OH}) + \text{H}_2\text{O}$
934. $\text{Ca}(\text{ClO})_2 + \text{H}_2\text{O}_2(\text{конц.}) \longrightarrow \text{O}_2 + \text{H}_2\text{O} + \text{CaCl}_2$
935. $\text{Ca}(\text{ClO})_2 + \text{NaBr} + \text{H}_2\text{O} \longrightarrow \text{CaCl}_2 + \text{Br}_2 + \text{NaOH}$
936. $\text{Ca}(\text{ClO})_2 + \text{SO}_2 + \text{H}_2\text{O} \longrightarrow \text{CaSO}_4 + \text{HCl} + \text{H}_2\text{SO}_4$
937. $\text{NaBrO}_3 + \text{NH}_3 \longrightarrow \text{N}_2\uparrow + \text{NaBr} + \text{H}_2\text{O}$
938. $\text{NaBrO} + \text{Fe}(\text{OH})_2 \longrightarrow \text{NaBr} + \text{FeO}(\text{OH}) + \text{H}_2\text{O}$

**Соединения азота и фосфора
(среда нейтральная)**

939. $\text{NO} + \text{NO}_2 + \text{H}_2\text{O} \longrightarrow \text{HNO}_2$
940. $\text{Ca}(\text{NO}_2)_2 + (\text{NH}_4)_2\text{SO}_4 \longrightarrow \text{N}_2\uparrow + \text{CaSO}_4\downarrow + \text{H}_2\text{O}$
941. $\text{KNO}_2 + \text{Br}_2 + \text{H}_2\text{O} \longrightarrow \text{KNO}_3 + \text{HBr}$
942. $\text{KNO}_3 + \text{Mg} + \text{H}_2\text{O} \longrightarrow \text{NH}_2\text{OH} + \text{KOH} + \text{Mg}(\text{OH})_2\downarrow$
943. $\text{KNO}_3 + \text{Mg} + \text{H}_2\text{O} \longrightarrow \text{N}_2\text{H}_4 + \text{KOH} + \text{Mg}(\text{OH})_2\downarrow$
944. $\text{KNO}_3 + \text{Mg} + \text{H}_2\text{O} \longrightarrow \text{NH}_3\uparrow + \text{KOH} + \text{Mg}(\text{OH})_2\downarrow$
945. $\text{Cu}(\text{NO}_3)_2 + \text{KI} \longrightarrow \text{CuI}\downarrow + \text{I}_2\downarrow + \text{KNO}_3$
946. $\text{AgNO}_3 + \text{AsH}_3 + \text{H}_2\text{O} \longrightarrow \text{Ag}\downarrow + \text{As}_2\text{O}_3\downarrow + \text{HNO}_3$
947. $\text{NH}_3 \cdot \text{H}_2\text{O}(\text{насыщ.}) + \text{KMnO}_4 \longrightarrow \text{MnO}_2\downarrow + \text{N}_2\uparrow + \text{KOH} + \text{H}_2\text{O}$
948. $\text{PH}_3 + \text{AgNO}_3 + \text{H}_2\text{O} \longrightarrow \text{Ag}\downarrow + \text{H}_3\text{PO}_4 + \text{HNO}_3$
949. $\text{PBr}_3 + \text{Cl}_2 + \text{H}_2\text{O} \longrightarrow \text{H}_3\text{PO}_4 + \text{Br}_2 + \text{HCl}$
950. $\text{P}_2\text{O}_3 + \text{AgNO}_3 + \text{H}_2\text{O} \longrightarrow \text{H}_3\text{PO}_4 + \text{HNO}_3 + \text{Ag}\downarrow$

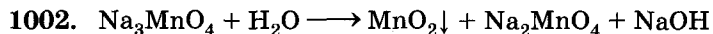
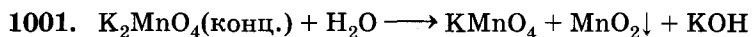
Соединения серы (среда нейтральная)

951. $\text{SO}_2 + \text{SeO}_2 + \text{H}_2\text{O} \longrightarrow \text{Se}\downarrow + \text{H}_2\text{SO}_4$
952. $\text{SO}_2 + \text{Cl}_2 + \text{H}_2\text{O} \longrightarrow \text{H}_2\text{SO}_4 + \text{HCl}$
953. $\text{SO}_2 + \text{KMnO}_4 + \text{H}_2\text{O} \longrightarrow \text{MnO}_2\downarrow + \text{H}_2\text{SO}_4 + \text{K}_2\text{SO}_4$
954. $\text{SO}_2 + \text{KMnO}_4 + \text{H}_2\text{O} \longrightarrow \text{MnSO}_4 + \text{H}_2\text{SO}_4 + \text{K}_2\text{SO}_4$
955. $\text{H}_2\text{S} + \text{K}_2\text{Cr}_2\text{O}_7 + \text{H}_2\text{O} \longrightarrow \text{S}\downarrow + \text{KOH} + \text{Cr}(\text{OH})_3\downarrow$
956. $\text{H}_2\text{S}(\text{насыщ.}) + \text{KMnO}_4 \longrightarrow \text{MnO}_2\downarrow + \text{S}\downarrow + \text{H}_2\text{O} + \text{KOH}$
957. $\text{H}_2\text{S} + \text{H}_2\text{O} + \text{K}_2\text{S}_2\text{O}_6(\text{O}_2) \longrightarrow \text{H}_2\text{SO}_4 + \text{K}_2\text{SO}_4$
958. $\text{H}_2\text{S} + \text{K}_2\text{FeO}_4 \longrightarrow \text{S}\downarrow + \text{KOH} + \text{FeO}(\text{OH})$
959. $\text{H}_2\text{S} + \text{K}_2\text{CrO}_4 + \text{H}_2\text{O} \longrightarrow \text{S}\downarrow + \text{KOH} + \text{Cr}(\text{OH})_3$
960. $\text{H}_2\text{S} + \text{Cl}_2 + \text{H}_2\text{O} \longrightarrow \text{H}_2\text{SO}_4 + \text{HCl}$
961. $\text{Na}_2\text{S}(\text{разб.}) + \text{FeCl}_3(\text{разб.}) \longrightarrow \text{S}\downarrow + \text{FeCl}_2 + \text{FeS}\downarrow + \text{NaCl}$
962. $\text{Na}_2\text{S} + \text{H}_2\text{O}_2(\text{конц.}) \longrightarrow \text{Na}_2\text{SO}_4 + \text{H}_2\text{O}$
963. $\text{Na}_2\text{S} + \text{FeCl}_3 + \text{H}_2\text{O} \longrightarrow \text{Na}_2\text{SO}_4 + \text{FeCl}_2 + \text{HCl}$
964. $\text{Na}_2\text{S} + \text{Ca}(\text{ClO})_2 + \text{H}_2\text{O} \longrightarrow \text{S}\downarrow + \text{CaCl}_2 + \text{NaOH}$
965. $\text{Na}_2\text{S} + \text{Na}_2\text{CrO}_4 + \text{H}_2\text{O} \longrightarrow \text{Cr}(\text{OH})_3\downarrow + \text{NaOH} + \text{S}\downarrow$
966. $\text{Na}_2\text{S} + \text{Na}_2\text{CrO}_4 + \text{H}_2\text{O} \longrightarrow \text{Na}_3[\text{Cr}(\text{OH})_6] + \text{NaOH} + \text{S}\downarrow$
967. $\text{Na}_2\text{S} + \text{O}_2 + \text{H}_2\text{O} \longrightarrow \text{Na}_2\text{SO}_3\text{S} + \text{NaOH}$
968. $\text{NaHS} + \text{NO}_2 + \text{H}_2\text{O} \longrightarrow \text{HNO}_2 + \text{S}\downarrow + \text{NaNO}_3$

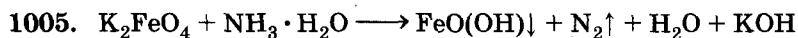
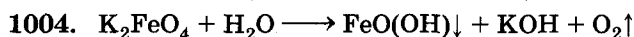
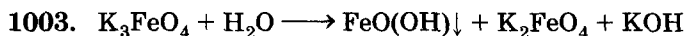
969. $\text{NaHS} + \text{NO}_2 + \text{H}_2\text{O} \longrightarrow \text{NaHSO}_4 + \text{HNO}_3 + \text{HNO}_2$
970. $\text{K}_2\text{S} + \text{KMnO}_4 + \text{H}_2\text{O} \longrightarrow \text{S} \downarrow + \text{MnO}_2 \downarrow + \text{KOH}$
971. $\text{PbS}(\tau) + \text{H}_2\text{O}_2 \longrightarrow \text{PbSO}_4 \downarrow + \text{H}_2\text{O}$
972. $\text{NH}_4\text{HS} + \text{K}_2\text{CrO}_4 + \text{H}_2\text{O} \longrightarrow \text{S} \downarrow + \text{Cr}(\text{OH})_3 + \text{NH}_3 \cdot \text{H}_2\text{O} + \text{KOH}$
973. $\text{K}_2\text{SO}_3 + \text{I}_2 + \text{H}_2\text{O} \longrightarrow \text{K}_2\text{SO}_4 + \text{HI}$
974. $\text{Na}_2\text{SO}_3 + \text{K}_2\text{Cr}_2\text{O}_7 + \text{H}_2\text{O} \longrightarrow \text{Na}_2\text{SO}_4 + \text{Cr}(\text{OH})_3 \downarrow + \text{KOH}$
975. $\text{Na}_2\text{SO}_3 + \text{KMnO}_4 + \text{H}_2\text{O} \longrightarrow \text{Na}_2\text{SO}_4 + \text{MnO}_2 \downarrow + \text{KOH}$
976. $\text{Na}_2\text{SO}_3 + \text{Cl}_2 + \text{H}_2\text{O} \longrightarrow \text{Na}_2\text{SO}_4 + \text{HCl}$
977. $\text{Na}_2\text{SO}_3 + \text{KMnO}_4 + \text{H}_2\text{SO}_4 \longrightarrow \text{Na}_2\text{SO}_4 + \text{MnO}_2 \downarrow + \text{K}_2\text{SO}_4 + \text{H}_2\text{O}$
978. $\text{K}_2\text{SO}_3 + \text{K}_2\text{CrO}_4 + \text{H}_2\text{O} \longrightarrow \text{K}_2\text{SO}_4 + \text{Cr}(\text{OH})_3 \downarrow + \text{KOH}$
979. $\text{NaHSO}_3 + \text{Cl}_2 + \text{H}_2\text{O} \longrightarrow \text{Na}_2\text{SO}_4 + \text{HCl} + \text{H}_2\text{SO}_4$
980. $\text{FeSO}_4(\text{конц.}) + \text{CuSO}_4 \longrightarrow \text{Fe}_2(\text{SO}_4)_3 + \text{Cu} \downarrow$
981. $\text{FeSO}_4 + \text{O}_2 + \text{H}_2\text{O} \longrightarrow (\text{FeOH})\text{SO}_4 \downarrow$
982. $\text{MnSO}_4 + \text{KMnO}_4 + \text{H}_2\text{O} \longrightarrow \text{MnO}_2 \downarrow + \text{H}_2\text{SO}_4 + \text{K}_2\text{SO}_4$
983. $\text{CuSO}_4 + \text{KI} \longrightarrow \text{CuI} + \text{I}_2 \downarrow + \text{K}_2\text{SO}_4$
984. $\text{Ce}(\text{SO}_4)_2 + \text{H}_2\text{O}_2 \longrightarrow \text{Ce}_2(\text{SO}_4)_3 + \text{O}_2 \uparrow + \text{H}_2\text{SO}_4$
985. $\text{U}(\text{SO}_4)_2 + \text{O}_2 + \text{H}_2\text{O} \longrightarrow (\text{UO}_2)\text{SO}_4 + \text{H}_2\text{SO}_4$
986. $\text{Na}_2\text{SO}_3\text{S} + \text{SeO}_2 + \text{H}_2\text{O} \longrightarrow \text{Se} + \text{Na}_2\text{S}_4\text{O}_6 + \text{NaOH}$
987. $\text{Na}_2\text{SO}_3\text{S} + \text{I}_2 \longrightarrow \text{Na}_2\text{S}_4\text{O}_6 + \text{NaI}$
988. $\text{Na}_2\text{SO}_3\text{S} + \text{Cl}_2 + \text{H}_2\text{O} \longrightarrow \text{H}_2\text{SO}_4 + \text{NaCl} + \text{HCl}$
989. $\text{K}_2\text{SO}_3\text{S} + \text{K}_3[\text{Fe}(\text{CN})_6] \longrightarrow \text{K}_2\text{S}_4\text{O}_6 + \text{K}_4[\text{Fe}(\text{CN})_6]$
990. $\text{H}_2\text{S}_2\text{O}_6(\text{O}_2) + \text{H}_2\text{O} \longrightarrow \text{H}_2\text{SO}_4 + \text{H}_2\text{O}_2$
991. $\text{K}_2\text{S}_2\text{O}_6(\text{O}_2) + \text{Cr}_2(\text{SO}_4)_3 + \text{H}_2\text{O} \longrightarrow \text{K}_2\text{Cr}_2\text{O}_7 + \text{K}_2\text{SO}_4 + \text{H}_2\text{SO}_4$
992. $\text{K}_2\text{S}_2\text{O}_6(\text{O}_2) + \text{H}_2\text{O} \xrightarrow{\text{MnO}_2, t} \text{K}_2\text{SO}_4 + \text{H}_2\text{SO}_4 + \text{O}_2 \uparrow$
- 993*. $\text{K}_2\text{S}_2\text{O}_6(\text{O}_2) + \text{K}_2\text{SO}_3\text{S} + \text{H}_2\text{O} \longrightarrow \text{K}_2\text{SO}_4 + \text{H}_2\text{SO}_4$
994. $\text{K}_2\text{S}_2\text{O}_6(\text{O}_2) + \text{K}_2\text{SO}_3 + \text{H}_2\text{O} \longrightarrow \text{K}_2\text{SO}_4 + \text{H}_2\text{SO}_4$
995. $\text{K}_2\text{S}_2\text{O}_6(\text{O}_2)(\text{конц.}) + \text{K}_2\text{MnO}_4 \longrightarrow \text{KMnO}_4 + \text{K}_2\text{SO}_4$
996. $\text{K}_2\text{S}_2\text{O}_6(\text{O}_2) + \text{Pb}(\text{NO}_3)_2 + \text{H}_2\text{O} \longrightarrow \text{PbO}_2 \downarrow + \text{KNO}_3 + \text{H}_2\text{SO}_4$
997. $(\text{NH}_4)_2\text{S}_2\text{O}_8 + \text{MnSO}_4 + \text{H}_2\text{O} \longrightarrow \text{MnO}_2 \downarrow + (\text{NH}_4)_2\text{SO}_4 + \text{H}_2\text{SO}_4$

Соединения марганца (среда нейтральная)

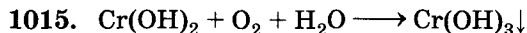
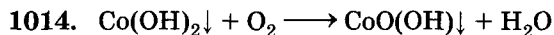
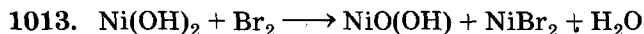
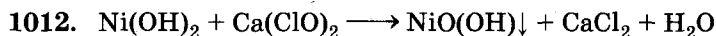
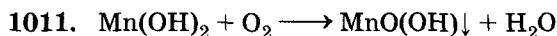
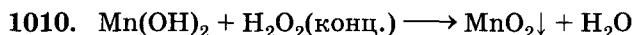
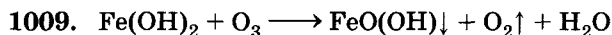
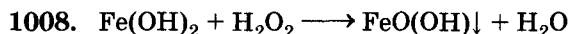
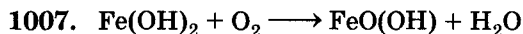
998. $\text{KMnO}_4 + \text{H}_2\text{O}_2 \longrightarrow \text{O}_2 \uparrow + \text{H}_2\text{O} + \text{KOH} + \text{MnO}_2 \downarrow$
999. $\text{KMnO}_4 + \text{KHSO}_3 \longrightarrow \text{K}_2\text{SO}_4 + \text{KHSO}_4 + \text{MnO}_2 \downarrow + \text{H}_2\text{O}$
1000. $\text{K}_2\text{MnO}_4 + \text{CO}_2 + \text{H}_2\text{O} \longrightarrow \text{MnO}_2 \downarrow + \text{KMnO}_4 + \text{KHCO}_3$



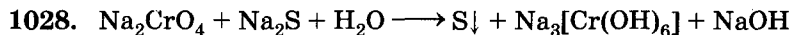
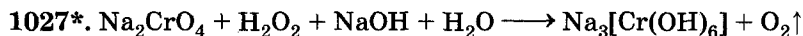
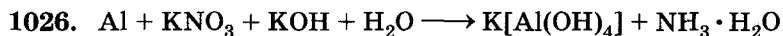
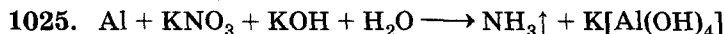
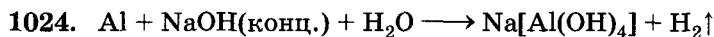
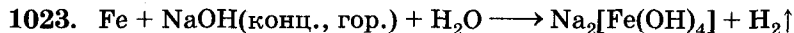
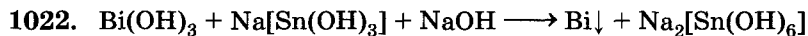
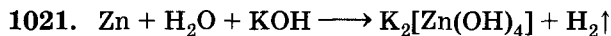
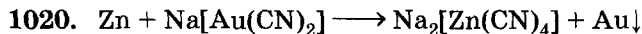
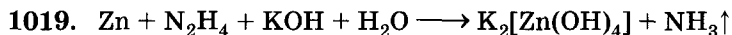
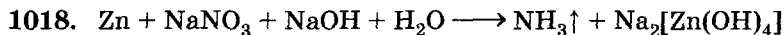
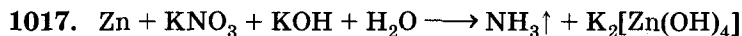
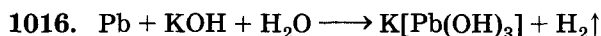
Соединения железа (среда нейтральная)



Гидроксиды железа, марганца, никеля, кобальта и хрома (среда нейтральная)



Комплексные соединения



1029. $\text{K}_2\text{CrO}_4 + \text{NH}_4\text{HS} + \text{H}_2\text{O} + \text{KOH} \longrightarrow$
 $\longrightarrow \text{S}\downarrow + \text{K}_3[\text{Cr}(\text{OH})_6] + \text{NH}_3 \cdot \text{H}_2\text{O}$
1030. $\text{Na}_2\text{CrO}_4 + \text{SO}_2 + \text{NaOH} + \text{H}_2\text{O} \longrightarrow \text{Na}_3[\text{Cr}(\text{OH})_6] + \text{Na}_2\text{SO}_4$
1031. $\text{K}_2\text{Cr}_2\text{O}_7 + \text{SnCl}_2 + \text{KOH} + \text{H}_2\text{O} \longrightarrow$
 $\longrightarrow \text{K}_3[\text{Cr}(\text{OH})_6] + \text{KCl} + \text{K}_2[\text{Sn}(\text{OH})_6]$
1032. $\text{Na}_3[\text{Cr}(\text{OH})_6] + \text{Cl}_2 + \text{NaOH} \longrightarrow \text{Na}_2\text{CrO}_4 + \text{NaCl} + \text{H}_2\text{O}$
1033. $\text{Na}_3[\text{Cr}(\text{OH})_6] + \text{H}_2\text{O}_2 \longrightarrow \text{Na}_2\text{CrO}_4 + \text{H}_2\text{O} + \text{NaOH}$
1034. $\text{Na}_3[\text{Cr}(\text{OH})_6] + \text{PbO}_2 + \text{NaOH} \longrightarrow$
 $\longrightarrow \text{Na}_2\text{CrO}_4 + \text{Na}[\text{Pb}(\text{OH})_3] + \text{H}_2\text{O}$
1035. $\text{Na}_3[\text{Cr}(\text{OH})_6] + \text{Na}_2\text{O}_2 \longrightarrow \text{Na}_2\text{CrO}_4 + \text{NaOH} + \text{H}_2\text{O}$
1036. $\text{Na}_3[\text{Cr}(\text{OH})_6] + \text{Br}_2 + \text{NaOH} \longrightarrow \text{Na}_2\text{CrO}_4 + \text{NaBr} + \text{H}_2\text{O}$
1037. $\text{Na}_3[\text{Cr}(\text{OH})_6] + \text{NaBiO}_3(\tau) + \text{H}_2\text{O} \longrightarrow$
 $\longrightarrow \text{Na}_2\text{CrO}_4 + \text{Bi}(\text{OH})_3\downarrow + \text{NaOH}$
1038. $\text{Na}_3[\text{Cr}(\text{OH})_6] + \text{NaBiO}_3(\tau) \longrightarrow$
 $\longrightarrow \text{Na}_2\text{CrO}_4 + \text{BiO}(\text{OH})\downarrow + \text{NaOH} + \text{H}_2\text{O}$
1039. $[\text{Ag}(\text{NH}_3)_2]\text{OH} + \text{HCHO} \longrightarrow \text{CO}_2\uparrow + \text{Ag} + \text{NH}_3\uparrow + \text{H}_2\text{O}$
1040. $\text{Ta} + \text{HF} + \text{HNO}_3 \longrightarrow \text{H}_2[\text{TaF}_7] + \text{NO}\uparrow + \text{H}_2\text{O}$
1041. $\text{Au} + \text{NaCN} + \text{O}_2 + \text{H}_2\text{O} \longrightarrow \text{Na}[\text{Au}(\text{CN})_2] + \text{NaOH}$
1042. $\text{K}_4[\text{Fe}(\text{CN})_6] + \text{Br}_2 \longrightarrow \text{K}_3[\text{Fe}(\text{CN})_6] + \text{KBr}$
1043. $\text{K}_4[\text{Fe}(\text{CN})_6] + \text{KMnO}_4 + \text{H}_2\text{SO}_4 \longrightarrow$
 $\longrightarrow \text{K}_3[\text{Fe}(\text{CN})_6] + \text{K}_2\text{SO}_4 + \text{MnSO}_4 + \text{H}_2\text{O}$
1044. $\text{K}_4[\text{Fe}(\text{CN})_6] + \text{H}_2\text{O}_2 + \text{H}_2\text{SO}_4 \longrightarrow \text{K}_3[\text{Fe}(\text{CN})_6] + \text{K}_2\text{SO}_4 + \text{H}_2\text{O}$
1045. $\text{K}_4[\text{Fe}(\text{CN})_6] + \text{Cl}_2 \longrightarrow \text{K}_3[\text{Fe}(\text{CN})_6] + \text{KCl}$
1046. $\text{K}_3[\text{Fe}(\text{CN})_6] + \text{KOH}(\text{конц.}) \longrightarrow \text{K}_4[\text{Fe}(\text{CN})_6] + \text{O}_2\uparrow + \text{H}_2\text{O}$
1047. $\text{K}_3[\text{Fe}(\text{CN})_6] + \text{H}_2\text{O}_2 + \text{KOH} \longrightarrow \text{K}_4[\text{Fe}(\text{CN})_6] + \text{O}_2\uparrow + \text{H}_2\text{O}$
1048. $\text{K}_3[\text{Fe}(\text{CN})_6] + \text{KI} \longrightarrow \text{K}_4[\text{Fe}(\text{CN})_6] + \text{I}_2\downarrow$
1049. $\text{Hg}_2(\text{NO}_3)_2 + \text{KCN} \longrightarrow \text{Hg}\downarrow + \text{K}_2[\text{Hg}(\text{CN})_4] + \text{KNO}_3$
1050. $\text{Hg}_2\text{I}_2 + \text{KI} \longrightarrow \text{K}_2[\text{HgI}_4] + \text{Hg}\downarrow$
1051. $\text{C}_6\text{H}_5\text{NO}_2 + \text{Al} + \text{KOH} + \text{H}_2\text{O} \longrightarrow \text{C}_6\text{H}_5\text{NH}_2 + \text{K}[\text{Al}(\text{OH})_4]$

Органические соединения

1052. $\text{CH}_3-\text{CH}_2-\text{CH}=\text{CH}-\text{CH}_3 + \text{K}_2\text{Cr}_2\text{O}_7 + \text{H}_2\text{SO}_4 \longrightarrow$
 $\longrightarrow \text{CH}_3\text{COOH} + \text{CH}_3-\text{CH}_2-\text{COOH} + \text{Cr}_2(\text{SO}_4)_3 + \text{K}_2\text{SO}_4 + \text{H}_2\text{O}$
1053. $\text{C}_2\text{H}_4 + \text{KMnO}_4 + \text{H}_2\text{O} \longrightarrow \text{CH}_2\text{OH}-\text{CH}_2\text{OH} + \text{MnO}_2\downarrow + \text{KOH}$

1054. $\text{C}_2\text{H}_2 + \text{KMnO}_4 + \text{H}_2\text{SO}_4 \longrightarrow \text{H}_2\text{C}_2\text{O}_4 + \text{MnSO}_4 + \text{K}_2\text{SO}_4 + \text{H}_2\text{O}$
1055. $\text{C}_2\text{H}_2 + \text{KMnO}_4 \longrightarrow \text{MnO}_2\downarrow + \text{K}_2\text{C}_2\text{O}_4 + \text{KOH} + \text{H}_2\text{O}$
- 1056*. $\text{C}_2\text{H}_4 + \text{KMnO}_4 + \text{H}_2\text{SO}_4 \xrightarrow{t} \text{CO}_2\uparrow + \text{MnSO}_4 + \text{K}_2\text{SO}_4 + \text{H}_2\text{O}$
1057. $\text{CH}_2=\text{C}(\text{CH}_3)-\text{CH}_2-\text{CH}_2-\text{CH}_3 + \text{KMnO}_4 + \text{H}_2\text{SO}_4 \longrightarrow$
 $\longrightarrow \text{CO}_2\uparrow + \text{CH}_3-\text{C}(\text{O})-\text{CH}_2-\text{CH}_2-\text{CH}_3 + \text{K}_2\text{SO}_4 +$
 $+ \text{MnSO}_4 + \text{H}_2\text{O}$
1058. $\text{CH}_3-\text{CH}=\text{C}(\text{CH}_3)\text{CH}_2-\text{CH}_3 + \text{KMnO}_4 + \text{H}_2\text{SO}_4 \longrightarrow$
 $\longrightarrow \text{CH}_3\underset{\text{O}}{\underset{\parallel}{\text{C}}}\text{OH} + \text{CH}_3-\underset{\text{O}}{\underset{\parallel}{\text{C}}}-\text{CH}_2-\text{CH}_3 + \text{K}_2\text{SO}_4 + \text{MnSO}_4 + \text{H}_2\text{O}$
1059. $\text{CH}_2=\text{CH}-\text{CH}_2-\text{NH}_2 + \text{KMnO}_4 + \text{H}_2\text{O} \longrightarrow$
 $\longrightarrow \text{CH}_2(\text{OH})-\text{CHOH}-\text{CH}_2-\text{NH}_2 + \text{MnO}_2\downarrow + \text{KOH}$
1060. $\text{CH}_3-\text{C}\equiv\text{CH} + \text{KMnO}_4 + \text{H}_2\text{SO}_4 \xrightarrow{t}$
 $\longrightarrow \text{CH}_3\text{COOH} + \text{CO}_2\uparrow + \text{K}_2\text{SO}_4 + \text{MnSO}_4 + \text{H}_2\text{O}$
1061. $\text{CH}_3-\text{CH}_2-\text{CH}_2-\text{C}\equiv\text{CH} + \text{K}_2\text{Cr}_2\text{O}_7 + \text{H}_2\text{SO}_4 \longrightarrow$
 $\longrightarrow \text{CH}_3\text{COOH} + \text{CO}_2\uparrow + \text{Cr}_2(\text{SO}_4)_3 + \text{K}_2\text{SO}_4 + \text{H}_2\text{O}$
1062. $\text{C}_6\text{H}_5\text{CH}_3 + \text{MnO}_2 + \text{H}_2\text{SO}_4 \longrightarrow \text{C}_6\text{H}_5\text{C}(\text{H})\text{O} + \text{MnSO}_4 + \text{H}_2\text{O}$
1063. $\text{C}_6\text{H}_5\text{CH}_3 + \text{KMnO}_4 + \text{H}_2\text{SO}_4 \longrightarrow$
 $\longrightarrow \text{C}_6\text{H}_5\text{COOH} + \text{MnSO}_4 + \text{K}_2\text{SO}_4 + \text{H}_2\text{O}$
1064. $\text{C}_6\text{H}_5\text{CH}_3 + \text{KMnO}_4 \longrightarrow \text{C}_6\text{H}_5\text{COOK} + \text{MnO}_2\downarrow + \text{KOH} + \text{H}_2\text{O}$
1065. $\text{C}_6\text{H}_5\text{C}_2\text{H}_5 + \text{KMnO}_4 \longrightarrow$
 $\longrightarrow \text{C}_6\text{H}_5\text{COOK} + \text{K}_2\text{CO}_3 + \text{MnO}_2\downarrow + \text{KOH} + \text{H}_2\text{O}$
1066. $\text{C}_6\text{H}_5\text{C}_2\text{H}_5 + \text{K}_2\text{Cr}_2\text{O}_7 + \text{H}_2\text{SO}_4 \longrightarrow$
 $\longrightarrow \text{C}_6\text{H}_5\text{COOH} + \text{CO}_2\uparrow + \text{K}_2\text{SO}_4 + \text{Cr}_2(\text{SO}_4)_3 + \text{H}_2\text{O}$
1067. $\text{C}_6\text{H}_5\text{C}_2\text{H}_5 + \text{KMnO}_4 + \text{H}_2\text{SO}_4 \longrightarrow$
 $\longrightarrow \text{C}_6\text{H}_5\text{COOH} + \text{CO}_2\uparrow + \text{K}_2\text{SO}_4 + \text{MnSO}_4 + \text{H}_2\text{O}$
1068. $\text{C}_6\text{H}_5\text{C}_3\text{H}_7 + \text{K}_2\text{Cr}_2\text{O}_7 + \text{H}_2\text{SO}_4 \longrightarrow$
 $\longrightarrow \text{C}_6\text{H}_5\text{COOH} + \text{CO}_2\uparrow + \text{K}_2\text{SO}_4 + \text{Cr}_2(\text{SO}_4)_3 + \text{H}_2\text{O}$
1069. $\text{C}_6\text{H}_5\text{CH}(\text{CH}_3)_2 + \text{KMnO}_4 + \text{H}_2\text{SO}_4 \longrightarrow$
 $\longrightarrow \text{C}_6\text{H}_5\text{COOH} + \text{CO}_2\uparrow + \text{MnSO}_4 + \text{K}_2\text{SO}_4 + \text{H}_2\text{O}$
1070. $\text{C}_6\text{H}_4(\text{CH}_3)_2 + \text{KMnO}_4 \longrightarrow \text{C}_6\text{H}_4(\text{COOK})_2 + \text{MnO}_2\downarrow + \text{KOH} + \text{H}_2\text{O}$
1071. $\text{C}_6\text{H}_5\text{NO}_2 + \text{Fe} + \text{H}_2\text{O} \longrightarrow \text{C}_6\text{H}_5\text{NH}_2 + (\text{Fe}^\text{II}\text{Fe}^\text{III})\text{O}_4\downarrow$
1072. $\text{C}_6\text{H}_5\text{NO}_2 + \text{Zn} + \text{H}_2\text{SO}_4 \longrightarrow \text{C}_6\text{H}_5\text{NH}_2 + \text{ZnSO}_4 + \text{H}_2\text{O}$
1073. $\text{C}_6\text{H}_4\text{NO}_2\text{C}_4\text{H}_9 + \text{KMnO}_4 + \text{H}_2\text{SO}_4 \xrightarrow{t}$
 $\longrightarrow \text{C}_6\text{H}_4\text{NO}_2\text{COOH} + \text{CO}_2\uparrow + \text{K}_2\text{SO}_4 + \text{MnSO}_4 + \text{H}_2\text{O}$

1074. $\text{C}_6\text{H}_4\text{NO}_2\text{C}_4\text{H}_9 + \text{KMnO}_4 + \text{H}_2\text{SO}_4 \longrightarrow$
 $\longrightarrow \text{C}_6\text{H}_4\text{NH}_2\text{COOH} + \text{CO}_2\uparrow + \text{K}_2\text{SO}_4 + \text{MnSO}_4 + \text{H}_2\text{O}$
1075. $(\text{CH}_3)_2\text{C}_6\text{H}_3\text{C}(\text{H})\text{O} + [\text{Ag}(\text{NH}_3)_2]\text{OH} \longrightarrow$
 $\longrightarrow (\text{CH}_3)_2\text{C}_6\text{H}_3\text{COONH}_4 + \text{Ag}\downarrow + \text{NH}_3\uparrow + \text{H}_2\text{O}$
1076. $(\text{CH}_3)_2\text{C}_6\text{H}_3\text{C}(\text{H})\text{O} + \text{KMnO}_4 + \text{H}_2\text{SO}_4 \longrightarrow$
 $\longrightarrow (\text{HOOC})_2\text{C}_6\text{H}_3\text{COOH} + \text{MnSO}_4 + \text{K}_2\text{SO}_4 + \text{H}_2\text{O}$
1077. $\text{C}_6\text{H}_5\text{CH}=\text{CH}_2 + \text{KMnO}_4 + \text{H}_2\text{O} \longrightarrow$
 $\longrightarrow \text{C}_6\text{H}_5\text{CH}(\text{OH})\text{CH}_2(\text{OH}) + \text{MnO}_2\downarrow + \text{KOH}$
1078. $\text{C}_6\text{H}_5\text{CH}=\text{CH}_2 + \text{KMnO}_4 + \text{H}_2\text{SO}_4 \longrightarrow$
 $\longrightarrow \text{C}_6\text{H}_5\text{COOH} + \text{CO}_2\uparrow + \text{K}_2\text{SO}_4 + \text{MnSO}_4 + \text{H}_2\text{O}$
1079. $\text{C}_6\text{H}_5\text{CH}=\text{CH}_2 + \text{KMnO}_4 \longrightarrow$
 $\longrightarrow \text{C}_6\text{H}_5\text{COOK} + \text{K}_2\text{CO}_3 + \text{MnO}_2\downarrow + \text{KOH} + \text{H}_2\text{O}$
1080. $\text{C}_6\text{H}_5\text{CH}=\text{CH}-\text{CH}_2-\text{CH}_3 + \text{KMnO}_4 + \text{H}_2\text{SO}_4 \longrightarrow$
 $\longrightarrow \text{C}_6\text{H}_5\text{COOH} + \text{C}_2\text{H}_5\text{COOH} + \text{MnSO}_4 + \text{K}_2\text{SO}_4 + \text{H}_2\text{O}$
1081. $\text{C}_6\text{H}_5\text{C}\equiv\text{CH} + \text{KMnO}_4 + \text{H}_2\text{SO}_4 \xrightarrow{t}$
 $\longrightarrow \text{C}_6\text{H}_5\text{COOH} + \text{CO}_2\uparrow + \text{K}_2\text{SO}_4 + \text{MnSO}_4 + \text{H}_2\text{O}$
1082. $\text{R}-\text{NH}_2 + \text{HNO}_2 \longrightarrow \text{R}-\text{OH} + \text{N}_2\uparrow + \text{H}_2\text{O}$
1083. $\text{CH}_3\text{NH}_2 + \text{KNO}_2 + \text{HCl} \longrightarrow \text{CH}_3\text{OH} + \text{N}_2\uparrow + \text{H}_2\text{O} + \text{KCl}$
- 1084*. $\text{H}_2\text{C}-\text{CH}_2 + \text{KMnO}_4 \longrightarrow \text{K}_2\text{C}_2\text{O}_4 + \text{MnO}_2\downarrow + \text{KOH} + \text{H}_2\text{O}$
 $\quad \quad \quad \text{O}$
1085. $\text{CH}_3\text{OH} + \text{KMnO}_4 \xrightarrow{t} \text{MnO}_2\downarrow + \text{H}_2\text{O} + \text{K}_2\text{CO}_3$
1086. $\text{CH}_3\text{OH} + \text{K}_2\text{Cr}_2\text{O}_7 + \text{HCl} \longrightarrow \text{HCOOH} + \text{CrCl}_3 + \text{KCl} + \text{H}_2\text{O}$
1087. $\text{CH}_3\text{OH} + \text{KMnO}_4 + \text{H}_2\text{SO}_4 \longrightarrow$
 $\longrightarrow \text{HCOOH} + \text{MnSO}_4 + \text{K}_2\text{SO}_4 + \text{H}_2\text{O}$
1088. $\text{C}_2\text{H}_5\text{OH} + \text{KMnO}_4 \longrightarrow \text{CH}_3\text{COOK} + \text{MnO}_2\downarrow + \text{KOH} + \text{H}_2\text{O}$
1089. $\text{C}_2\text{H}_5\text{OH} + \text{KMnO}_4 + \text{H}_2\text{SO}_4 \longrightarrow$
 $\longrightarrow \text{CH}_3\text{COOH} + \text{K}_2\text{SO}_4 + \text{MnSO}_4 + \text{H}_2\text{O}$
1090. $\text{C}_2\text{H}_5\text{OH} + \text{KMnO}_4 \longrightarrow \text{MnO}_2\downarrow + \text{CH}_3\text{C}(\text{H})\text{O} + \text{KOH} + \text{H}_2\text{O}$
1091. $\text{C}_2\text{H}_5\text{OH} + \text{K}_2\text{FeO}_4 \longrightarrow \text{Fe}_2\text{O}_3\downarrow + \text{CH}_3\text{C}(\text{H})\text{O} + \text{KOH} + \text{H}_2\text{O}$
1092. $\text{C}_2\text{H}_5\text{OH} + \text{K}_2\text{MnO}_4 \longrightarrow \text{CH}_3\text{C}(\text{H})\text{O} + \text{MnO}_2\downarrow + \text{KOH}$
1093. $\text{C}_2\text{H}_5\text{OH} + \text{CrO}_3 + \text{H}_2\text{SO}_4 \longrightarrow \text{CH}_3\text{COOH} + \text{Cr}_2(\text{SO}_4)_3 + \text{H}_2\text{O}$
1094. $\text{C}_2\text{H}_5\text{OH} + \text{CrO}_3 \longrightarrow \text{CO}_2\uparrow + \text{Cr}_2\text{O}_3\downarrow + \text{H}_2\text{O}$
1095. $\text{C}_2\text{H}_5\text{OH} + \text{Cl}_2 + \text{NaOH} \longrightarrow \text{CH}_3\text{COONa} + \text{NaCl} + \text{H}_2\text{O}$
1096. $\text{C}_2\text{H}_5\text{OH} + \text{K}_2\text{Cr}_2\text{O}_7 + \text{H}_2\text{SO}_4 \longrightarrow$
 $\longrightarrow \text{CH}_3\text{C}(\text{H})\text{O} + \text{Cr}_2(\text{SO}_4)_3 + \text{K}_2\text{SO}_4 + \text{H}_2\text{O}$

1097. $\text{CH}_3\text{CH}(\text{OH})\text{CH}_2-\text{CH}_3 + \text{KMnO}_4 \longrightarrow$
 $\longrightarrow \text{CH}_3\text{C}(\text{O})-\text{CH}_2\text{CH}_3 + \text{MnO}_2\downarrow + \text{KOH} + \text{H}_2\text{O}$
1098. $\text{HOOC}-\text{CH}(\text{CH}_3)\text{CH}_2\text{OH} + \text{KMnO}_4 + \text{H}_2\text{SO}_4 \longrightarrow$
 $\longrightarrow \text{HOOC}-\text{CH}(\text{CH}_3)\text{COOH} + \text{K}_2\text{SO}_4 + \text{MnSO}_4 + \text{H}_2\text{O}$
1099. $\text{HOCH}_2\text{CH}_2\text{OH} + \text{KMnO}_4 \longrightarrow \text{K}_2\text{C}_2\text{O}_4 + \text{MnO}_2\downarrow + \text{KOH} + \text{H}_2\text{O}$
1100. $\text{HCOOH} + \text{Cu}(\text{OH})_2 \xrightarrow{t} \text{Cu}_2\text{O}\downarrow + \text{CO}_2\uparrow + \text{H}_2\text{O}$
1101. $\text{HC}(\text{H})\text{O} + \text{KMnO}_4 + \text{H}_2\text{SO}_4 \longrightarrow$
 $\longrightarrow \text{HCOOH} + \text{MnSO}_4 + \text{K}_2\text{SO}_4 + \text{H}_2\text{O}$
1102. $\text{CH}_3\text{C}(\text{H})\text{O} + \text{Cu}(\text{OH})_2 + \text{NaOH} \xrightarrow{t} \text{CH}_3\text{COONa} + \text{Cu}_2\text{O}\downarrow + \text{H}_2\text{O}$
1103. $\text{HOCH}_2(\text{CHOH})_4\text{C}(\text{H})\text{O} + \text{Cu}(\text{OH})_2 + \text{NaOH} \xrightarrow{t}$
 $\longrightarrow \text{HOCH}_2(\text{CHOH})_4\text{COONa} + \text{Cu}_2\text{O}\downarrow + \text{H}_2\text{O}$
1104. $\text{CH}_3\text{C}(\text{H})\text{O} + \text{KMnO}_4 + \text{H}_2\text{SO}_4 \longrightarrow$
 $\longrightarrow \text{CH}_3\text{COOH} + \text{K}_2\text{SO}_4 + \text{MnSO}_4 + \text{H}_2\text{O}$
1105. $\text{CH}_3\text{CH}_2\text{C}(\text{H})\text{O} + \text{Cu}(\text{OH})_2 \xrightarrow{t}$
 $\longrightarrow \text{CH}_3-\text{CH}_2-\text{COOH} + \text{CuOH}\downarrow + \text{H}_2\text{O}$
1106. $\text{C}_2\text{H}_5\text{C}(\text{H})\text{O} + \text{KMnO}_4 + \text{H}_2\text{SO}_4 \longrightarrow$
 $\longrightarrow \text{C}_2\text{H}_5\text{COOH} + \text{K}_2\text{SO}_4 + \text{MnSO}_4 + \text{H}_2\text{O}$
1107. $\text{CH}_3\text{C}(\text{H})\text{O} + \text{NH}_3 \cdot \text{H}_2\text{O} + [\text{Ag}(\text{NH}_3)_2]\text{OH} \xrightarrow{t}$
 $\longrightarrow \text{CH}_3\text{COONH}_4 + \text{Ag}\downarrow + \text{NH}_3\uparrow + \text{H}_2\text{O}$
1108. $\text{R}-\text{CH}(\text{OH})\text{CH}_3 + \text{KOH} + \text{I}_2 \longrightarrow \text{RCOOK} + \text{CH}_3\text{I} + \text{KI} + \text{H}_2\text{O}$
1109. $\text{CH}_2\text{O} + \text{K}_2\text{Cr}_2\text{O}_7 + \text{H}_2\text{SO}_4 \longrightarrow \text{CO}_2\uparrow + \text{Cr}_2(\text{SO}_4)_3 + \text{K}_2\text{SO}_4 + \text{H}_2\text{O}$
1110. $\text{RCOCH}_3 + \text{I}_2 + \text{KOH} \longrightarrow \text{RCOOK} + \text{CH}_3\text{I} + \text{KI} + \text{H}_2\text{O}$
1111. $\text{HC}(\text{H})\text{O} + [\text{Ag}(\text{NH}_3)_2]\text{OH} \xrightarrow{t} \text{CO}_2\uparrow + \text{Ag}\downarrow + \text{NH}_3\uparrow + \text{H}_2\text{O}$
1112. $\text{HC}(\text{H})\text{O} + \text{Cu}(\text{OH})_2 \xrightarrow{t} \text{Cu}_2\text{O}\downarrow + \text{HCOOH} + \text{H}_2\text{O}$
1113. $\text{HCOOH} + [\text{Ag}(\text{NH}_3)_2]\text{OH} + \text{NH}_3 \cdot \text{H}_2\text{O} \xrightarrow{t}$
 $\longrightarrow \text{Ag}\downarrow + (\text{NH}_4)_2\text{CO}_3 + \text{NH}_3\uparrow + \text{H}_2\text{O}$
1114. $\text{HCOOH} + \text{HgCl}_2 \longrightarrow \text{Hg}\downarrow + \text{CO}_2\uparrow + \text{HCl}$
- 1115*. $\text{CH}_3\text{COOH} + \text{KMnO}_4 \longrightarrow \text{K}_2\text{CO}_3 + \text{MnO}_2\downarrow + \text{H}_2\text{O} + \text{CO}_2\uparrow$
1116. $\text{H}_2\text{C}_2\text{O}_4 + \text{K}_2\text{Cr}_2\text{O}_7 + \text{H}_2\text{SO}_4 \longrightarrow$
 $\longrightarrow \text{Cr}_2(\text{SO}_4)_3 + \text{CO}_2\uparrow + \text{K}_2\text{SO}_4 + \text{H}_2\text{O}$
1117. $\text{H}_2\text{C}_2\text{O}_4 + \text{KMnO}_4 + \text{H}_2\text{SO}_4 \longrightarrow \text{CO}_2\uparrow + \text{K}_2\text{SO}_4 + \text{MnSO}_4 + \text{H}_2\text{O}$
1118. $(\text{NH}_4)_2\text{C}_2\text{O}_4 + \text{KMnO}_4 + \text{H}_2\text{SO}_4 \longrightarrow$
 $\longrightarrow \text{MnSO}_4 + \text{CO}_2\uparrow + (\text{NH}_4)_2\text{SO}_4 + \text{K}_2\text{SO}_4 + \text{H}_2\text{O}$

1119. $\text{Na}_2\text{C}_2\text{O}_4 + \text{KBrO}_3 + \text{H}_2\text{O} \longrightarrow \text{CO}_2\uparrow + \text{KBr} + \text{NaOH}$
 1120. $\text{H}_2\text{C}_2\text{O}_4 + \text{KMnO}_4 \longrightarrow \text{CO}_2\uparrow + \text{K}_2\text{MnO}_4 + \text{MnO}_2\downarrow + \text{H}_2\text{O}$
 1121. $\text{C}_6\text{H}_{12}\text{O}_6 + \text{KMnO}_4 + \text{H}_2\text{SO}_4 \longrightarrow \text{CO}_2\uparrow + \text{MnSO}_4 + \text{K}_2\text{SO}_4 + \text{H}_2\text{O}$
 1122. $(\text{C}_5\text{H}_{11}\text{O}_5)\text{C}(\text{H})\text{O} + [\text{Ag}(\text{NH}_3)_2]\text{OH} + \text{NH}_3 \cdot \text{H}_2\text{O} \xrightarrow{t}$
 $\longrightarrow (\text{C}_5\text{H}_{11}\text{O}_5)\text{COONH}_4 + \text{NH}_3\uparrow + \text{H}_2\text{O} + \text{Ag}\downarrow$
 1123. $(\text{C}_5\text{H}_{11}\text{O}_5)\text{C}(\text{H})\text{O} + \text{Br}_2 + \text{H}_2\text{O} \longrightarrow (\text{C}_5\text{H}_{11}\text{O}_5)\text{COOH} + \text{HBr}$
 1124. $\text{C}_{12}\text{H}_{22}\text{O}_{11} + \text{KMnO}_4 + \text{H}_2\text{SO}_4 \longrightarrow$
 $\longrightarrow \text{CO}_2\uparrow + \text{K}_2\text{SO}_4 + \text{MnSO}_4 + \text{H}_2\text{O}$

Решения задач

Соединения галогенов (среда кислотная)

508. $\text{HCl}(\text{конц.}) + \text{Cu} + \text{O}_2 \longrightarrow \text{CuCl}_2 + \text{H}_2\text{O}$

$$\begin{array}{l} 2 \quad \text{Cu} - 2\bar{e} \longrightarrow \text{Cu}^{2+} \\ 1 \quad \text{O}_2 + 4\text{H}^+ + 4\bar{e} \longrightarrow 2\text{H}_2\text{O} \end{array}$$

$$\begin{array}{ccc} 2\text{Cu} + \text{O}_2 + 4\text{H}^+ & \longrightarrow & 2\text{Cu}^{2+} + 2\text{H}_2\text{O} \\ & & 4\text{Cl}^- \quad 4\text{Cl}^- \end{array}$$

$$2\text{Cu} + \text{O}_2 + 4\text{HCl}(\text{конц.}) = 2\text{CuCl}_2 + 2\text{H}_2\text{O}$$

509. $\text{HCl} + \text{H}_2\text{SeO}_4(\text{конц.}) \longrightarrow \text{H}_2\text{SeO}_3 + \text{Cl}_2\uparrow + \text{H}_2\text{O}$

$$\begin{array}{l} 1 \quad 2\text{Cl}^- - 2\bar{e} \longrightarrow \text{Cl}_2 \\ 1 \quad \text{SeO}_4^{2-} + 2\text{H}^+ + 2\bar{e} \longrightarrow \text{SeO}_3^{2-} + \text{H}_2\text{O} \end{array}$$

$$\begin{array}{ccc} 2\text{Cl}^- + 2\text{H}^+ + \text{SeO}_4^{2-} & \longrightarrow & \text{Cl}_2 + \text{SeO}_3^{2-} + \text{H}_2\text{O} \\ 2\text{H}^+ & & 2\text{H}^+ \end{array}$$

$$2\text{HCl} + \text{H}_2\text{SeO}_4 = \text{Cl}_2 + \text{H}_2\text{SeO}_3 + \text{H}_2\text{O}$$

510. $\text{HCl}(\text{конц.}) + \text{PbO}_2(\text{т}) \longrightarrow \text{PbCl}_2 + \text{Cl}_2\uparrow + \text{H}_2\text{O}$

$$\begin{array}{l} 1 \quad 2\text{Cl}^- - 2\bar{e} \longrightarrow \text{Cl}_2 \\ 1 \quad \text{PbO}_2 + 4\text{H}^+ + 2\bar{e} \longrightarrow \text{Pb}^{2+} + 2\text{H}_2\text{O} \end{array}$$

$$\begin{array}{ccc} 2\text{Cl}^- + 4\text{H}^+ + \text{PbO}_2 & \longrightarrow & \text{Cl}_2 + \text{Pb}^{2+} + 2\text{H}_2\text{O} \\ 2\text{Cl}^- & & 2\text{Cl}^- \end{array}$$

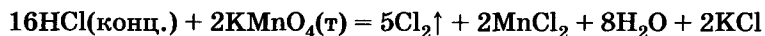
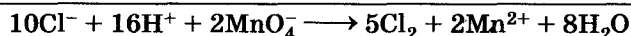
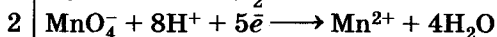
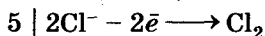
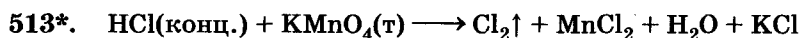
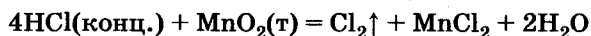
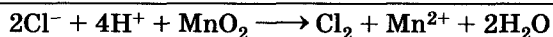
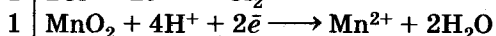
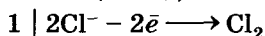
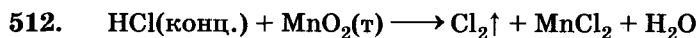
$$4\text{HCl}(\text{конц.}) + \text{PbO}_2(\text{т}) = \text{Cl}_2\uparrow + \text{PbCl}_2 + 2\text{H}_2\text{O}$$

511. $\text{HCl}(\text{конц.}) + (\text{Pb}_2\text{Pb})\text{O}_4 \longrightarrow \text{Cl}_2\uparrow + \text{PbCl}_2 + \text{H}_2\text{O}$

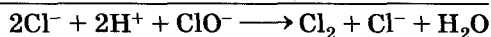
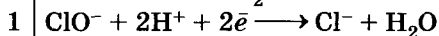
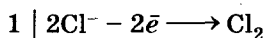
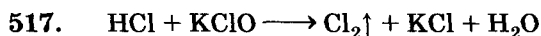
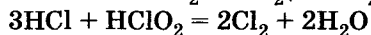
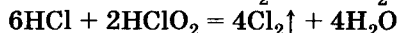
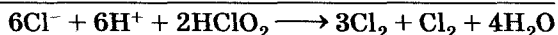
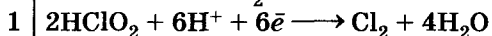
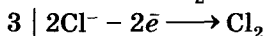
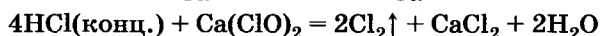
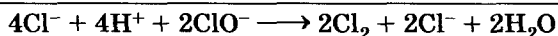
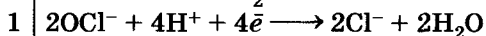
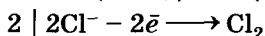
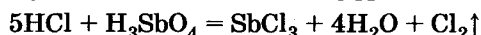
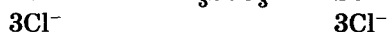
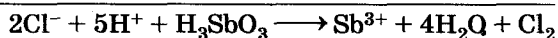
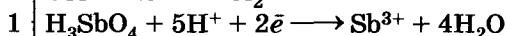
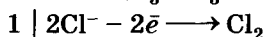
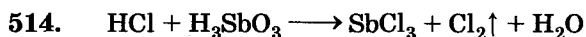
$$\begin{array}{l} 1 \quad 2\text{Cl}^- - 2\bar{e} \longrightarrow \text{Cl}_2 \\ 1 \quad \text{Pb}_3\text{O}_4 + 8\text{H}^+ + 2\bar{e} \longrightarrow 3\text{Pb}^{2+} + 4\text{H}_2\text{O} \end{array}$$

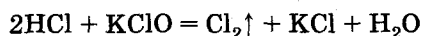
$$\begin{array}{ccc} 2\text{Cl}^- + 8\text{H}^+ + \text{Pb}_3\text{O}_4 & \longrightarrow & \text{Cl}_2 + 3\text{Pb}^{2+} + 4\text{H}_2\text{O} \\ 6\text{Cl}^- & & 6\text{Cl}^- \end{array}$$

$$8\text{HCl}(\text{конц.}) + \text{Pb}_3\text{O}_4 = \text{Cl}_2\uparrow + 3\text{PbCl}_2 + 4\text{H}_2\text{O}$$

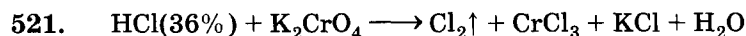
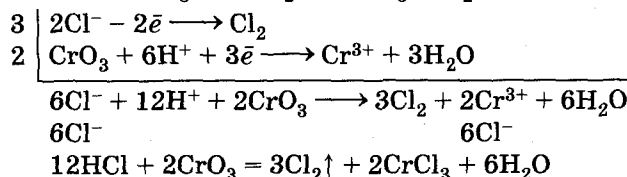
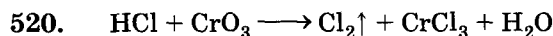
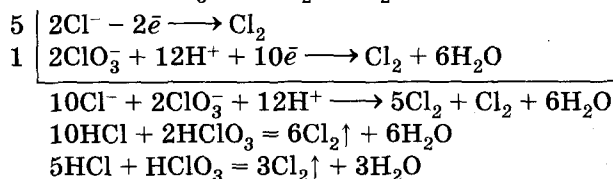
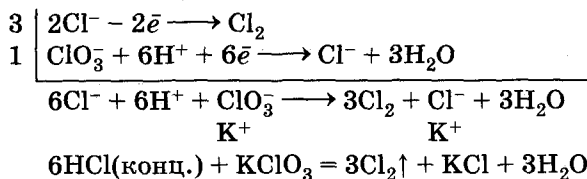


На 16H^+ требуется 16Cl^- . В ионном уравнении 10Cl^- уже есть, поэтому следует добавить еще 6Cl^- .

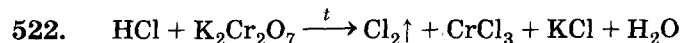
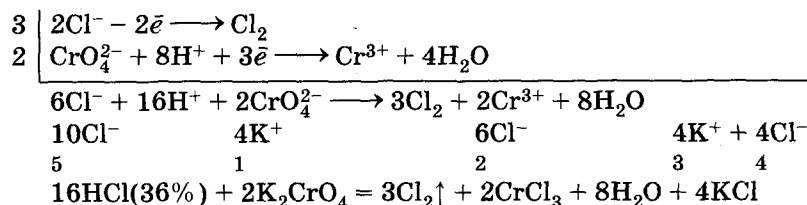




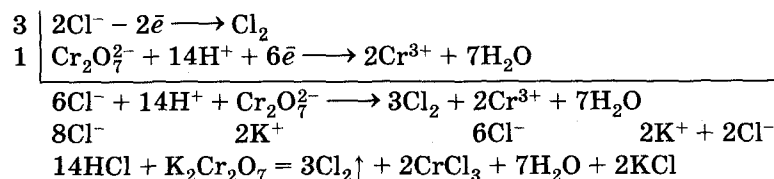
бертолетова соль
(хлорат калия)

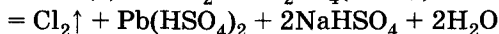
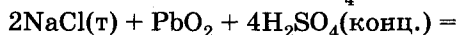
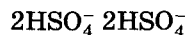
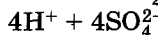
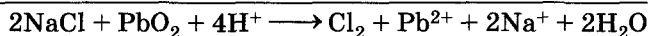
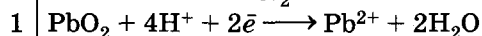
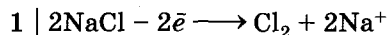
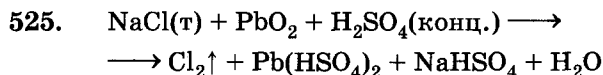
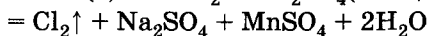
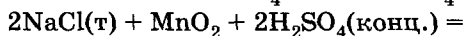
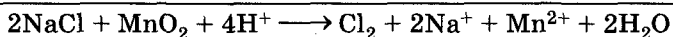
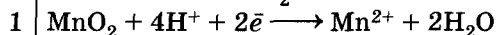
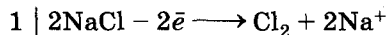
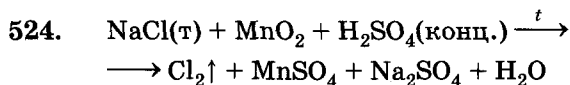
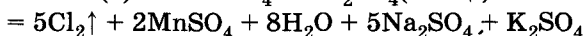
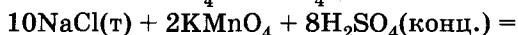
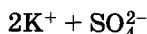
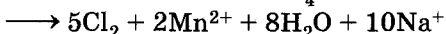
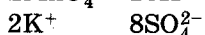
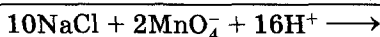
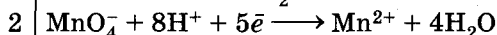
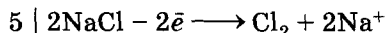
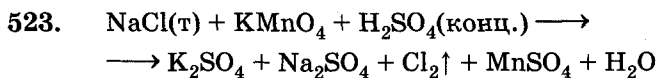


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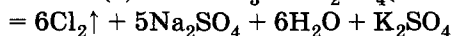
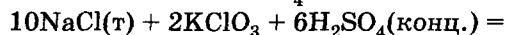
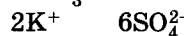
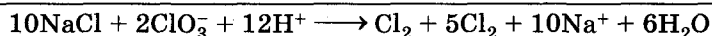
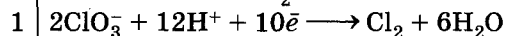
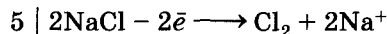


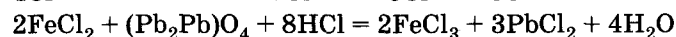
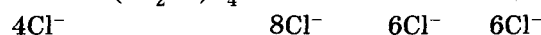
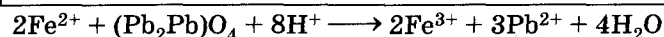
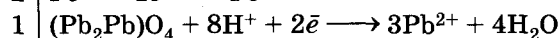
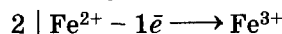
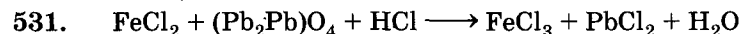
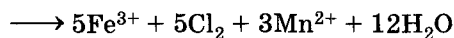
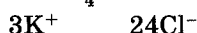
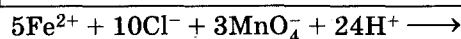
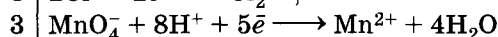
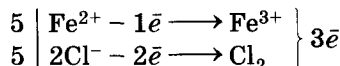
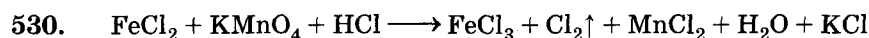
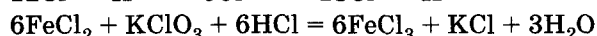
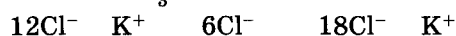
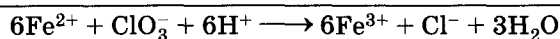
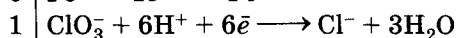
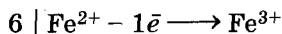
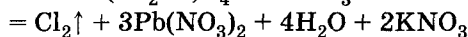
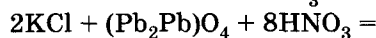
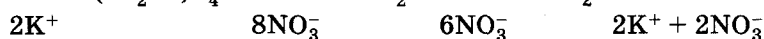
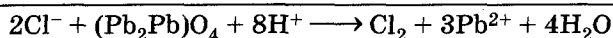
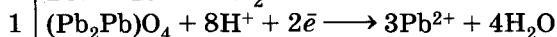
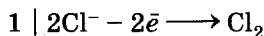
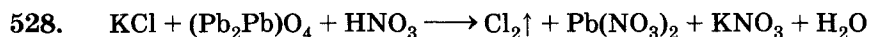
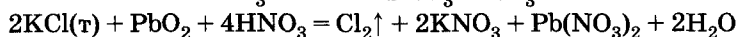
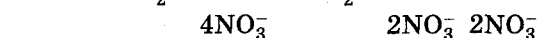
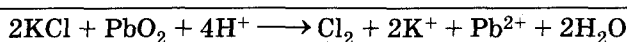
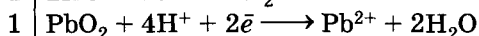
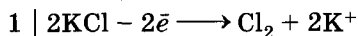
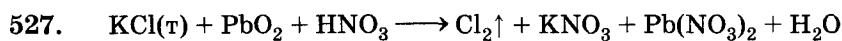
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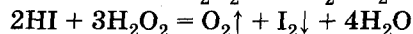
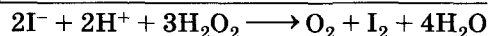
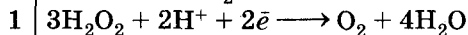
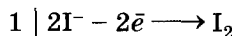
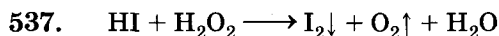
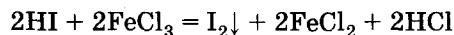
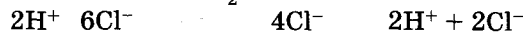
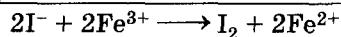
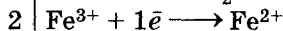
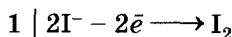
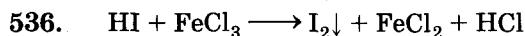
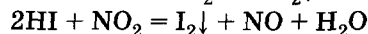
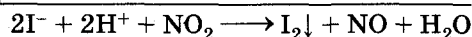
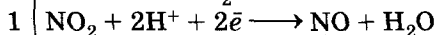
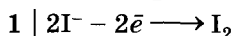
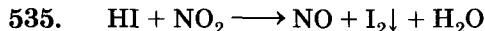
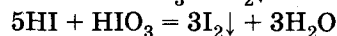
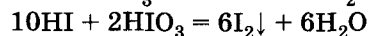
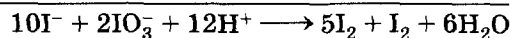
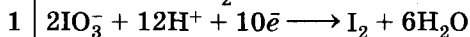
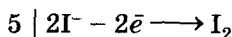
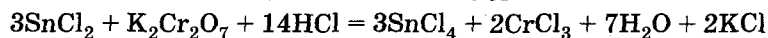
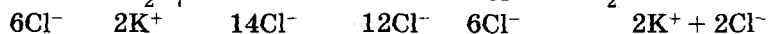
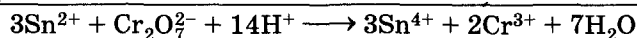
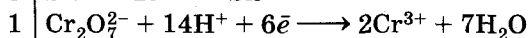
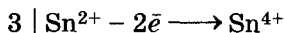
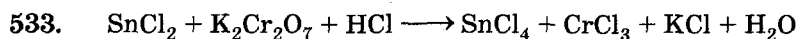
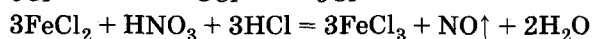
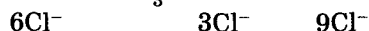
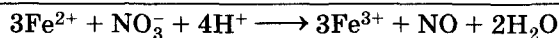
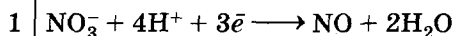
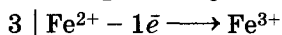


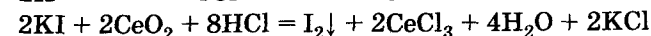
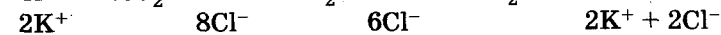
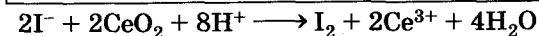
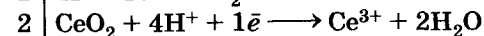
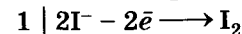
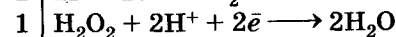
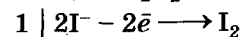
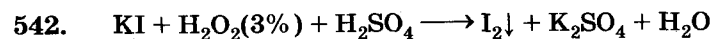
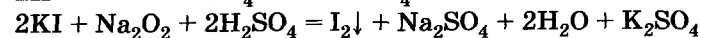
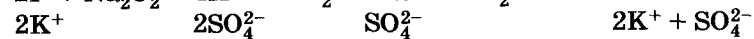
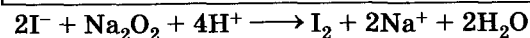
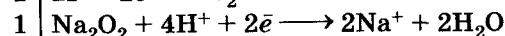
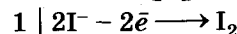
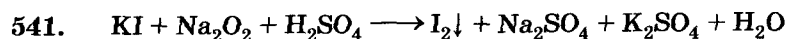
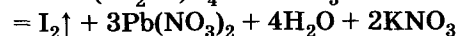
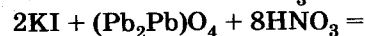
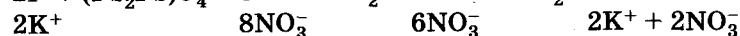
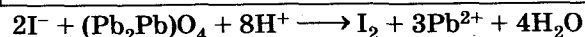
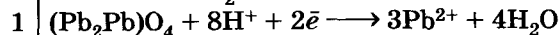
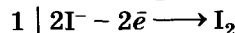
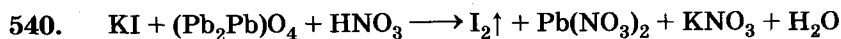
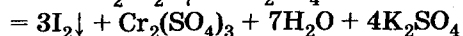
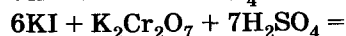
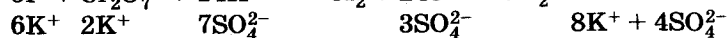
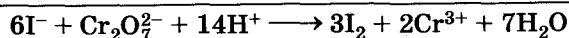
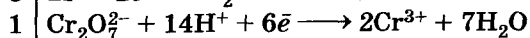
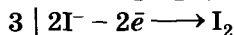
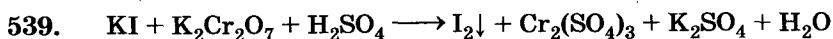
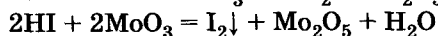
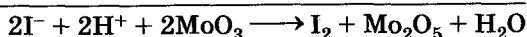
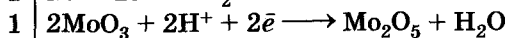
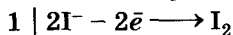
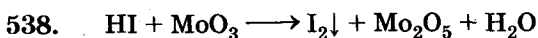


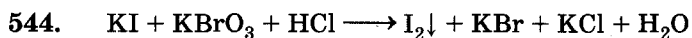
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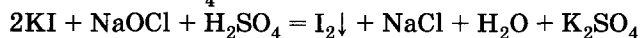
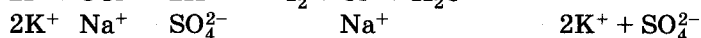
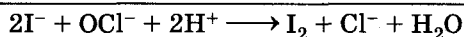
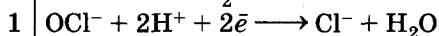
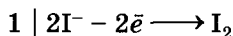
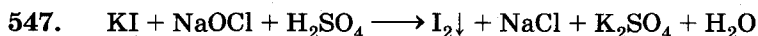
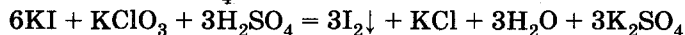
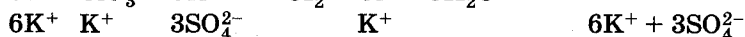
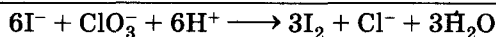
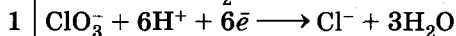
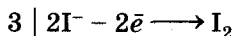
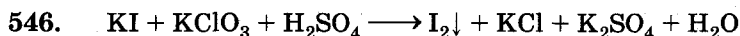
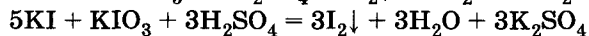
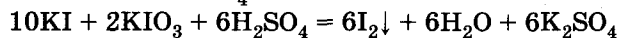
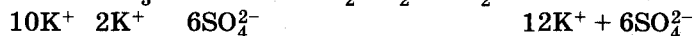
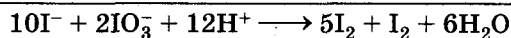
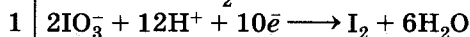
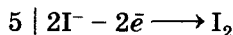
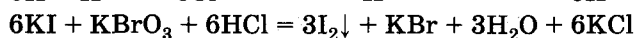
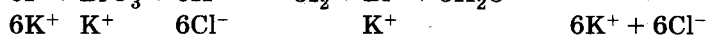
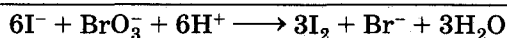
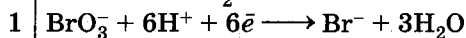
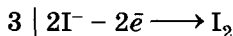






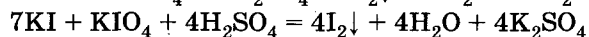
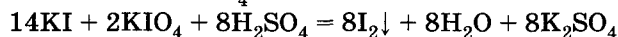
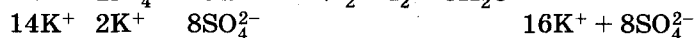
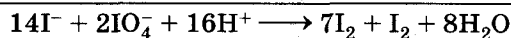
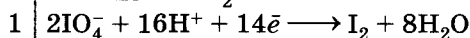
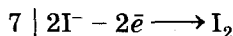


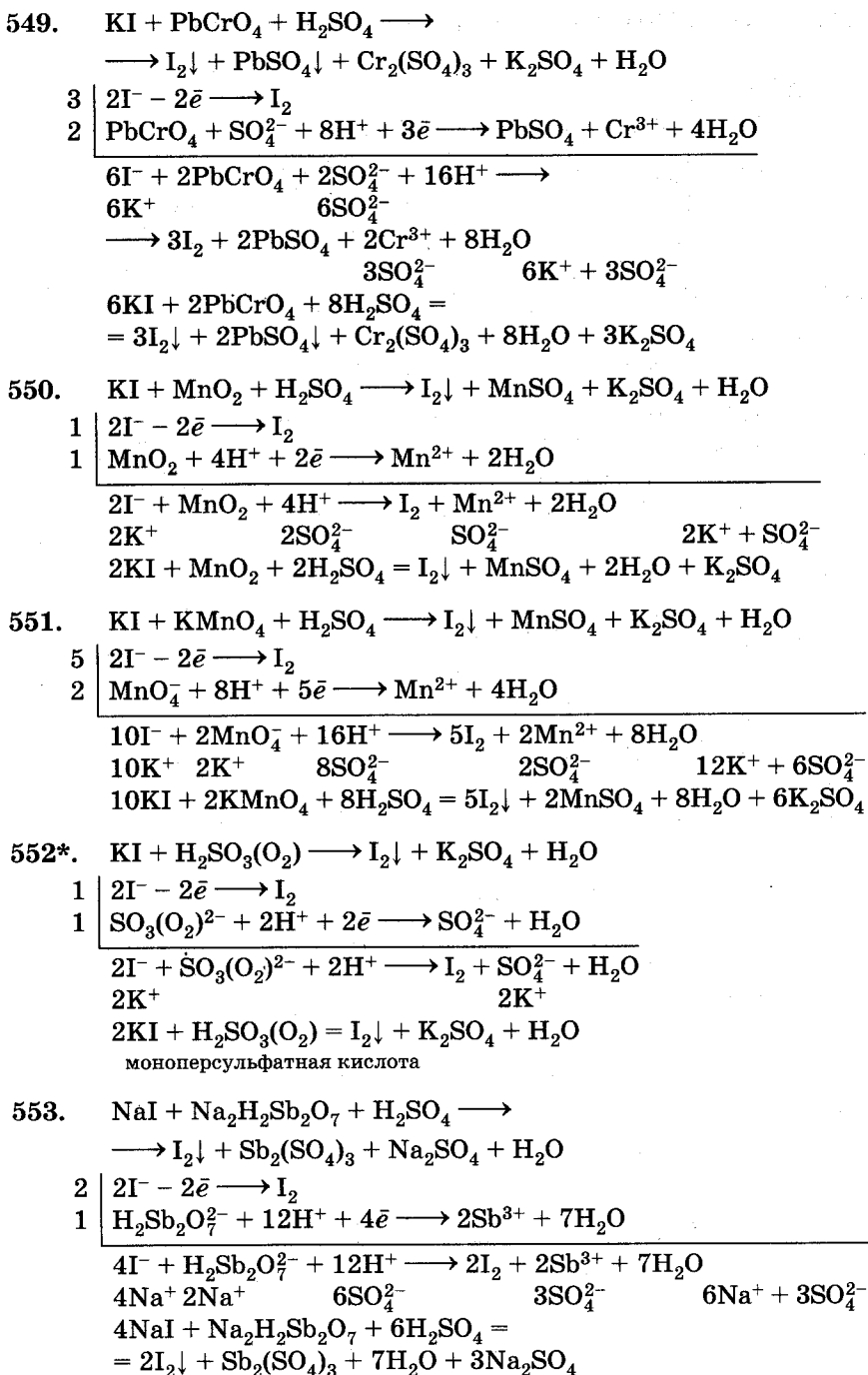
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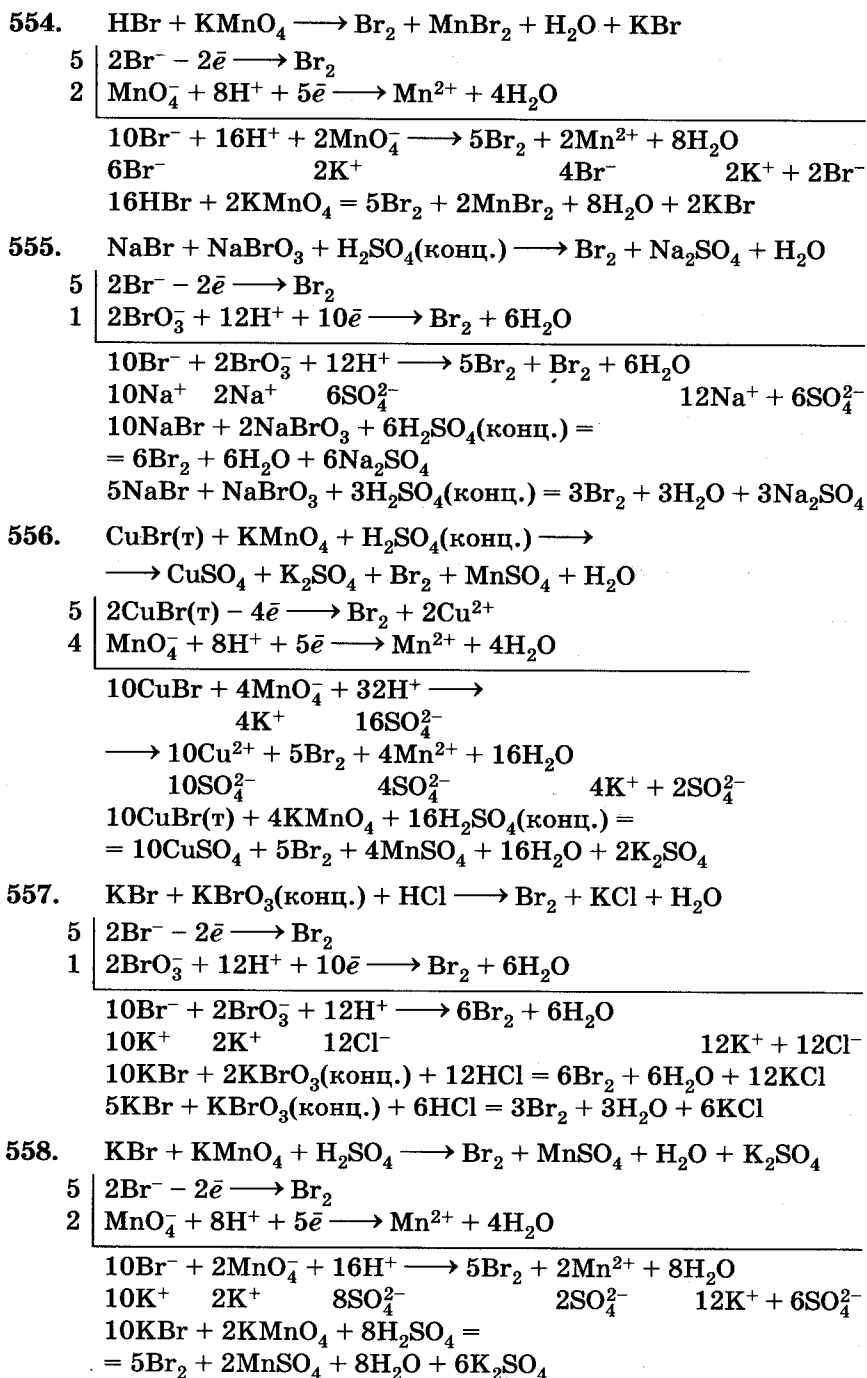


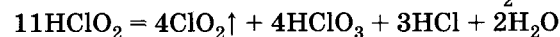
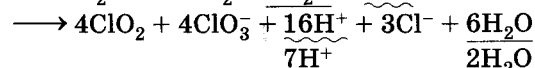
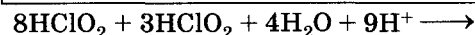
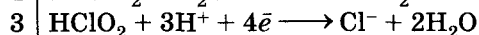
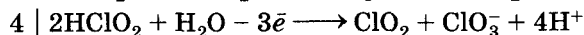
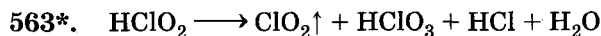
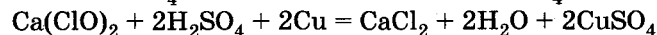
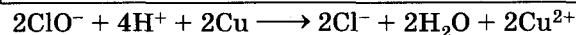
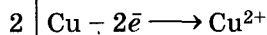
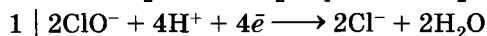
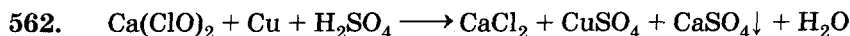
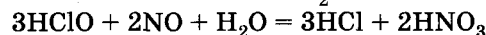
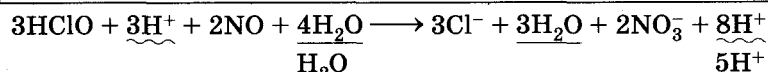
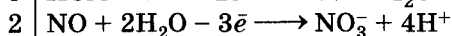
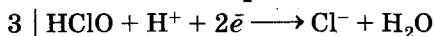
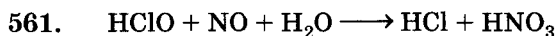
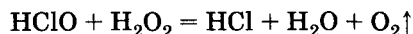
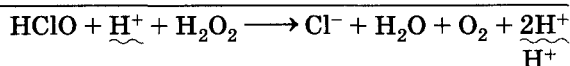
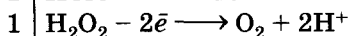
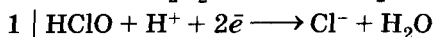
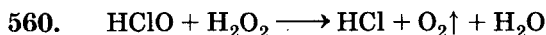
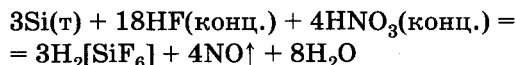
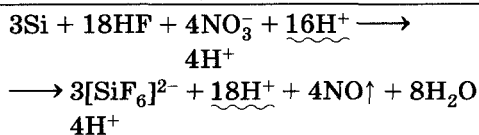
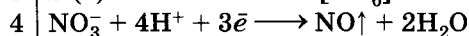
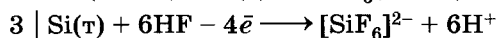
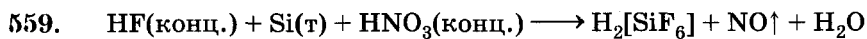
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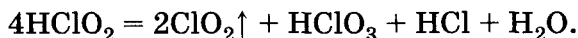


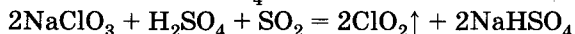
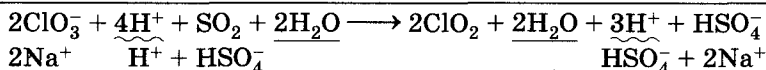
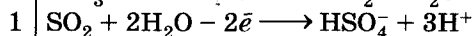
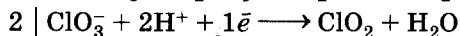
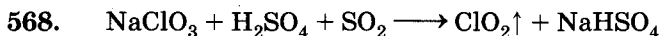
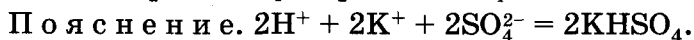
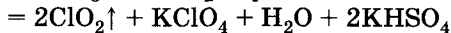
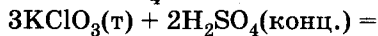
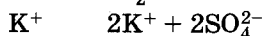
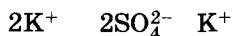
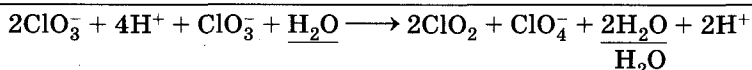
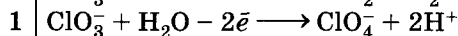
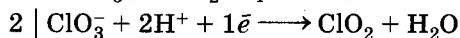
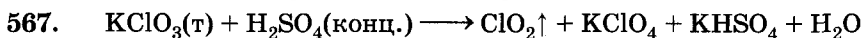
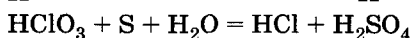
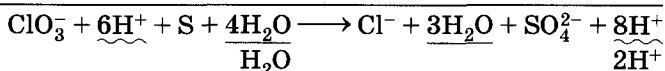
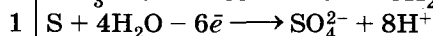
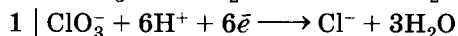
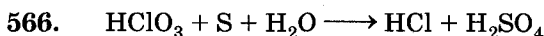
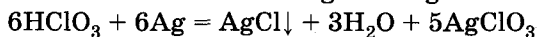
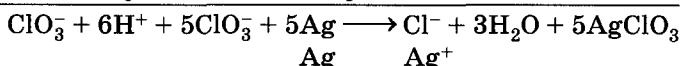
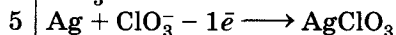
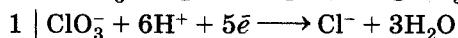
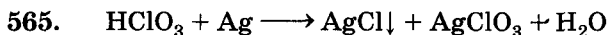
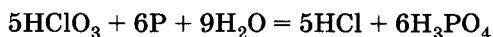
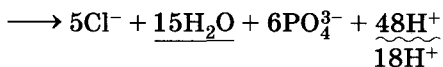
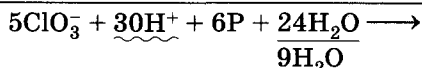
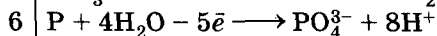
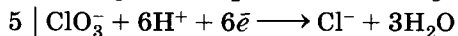
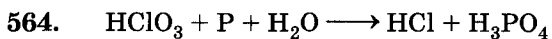


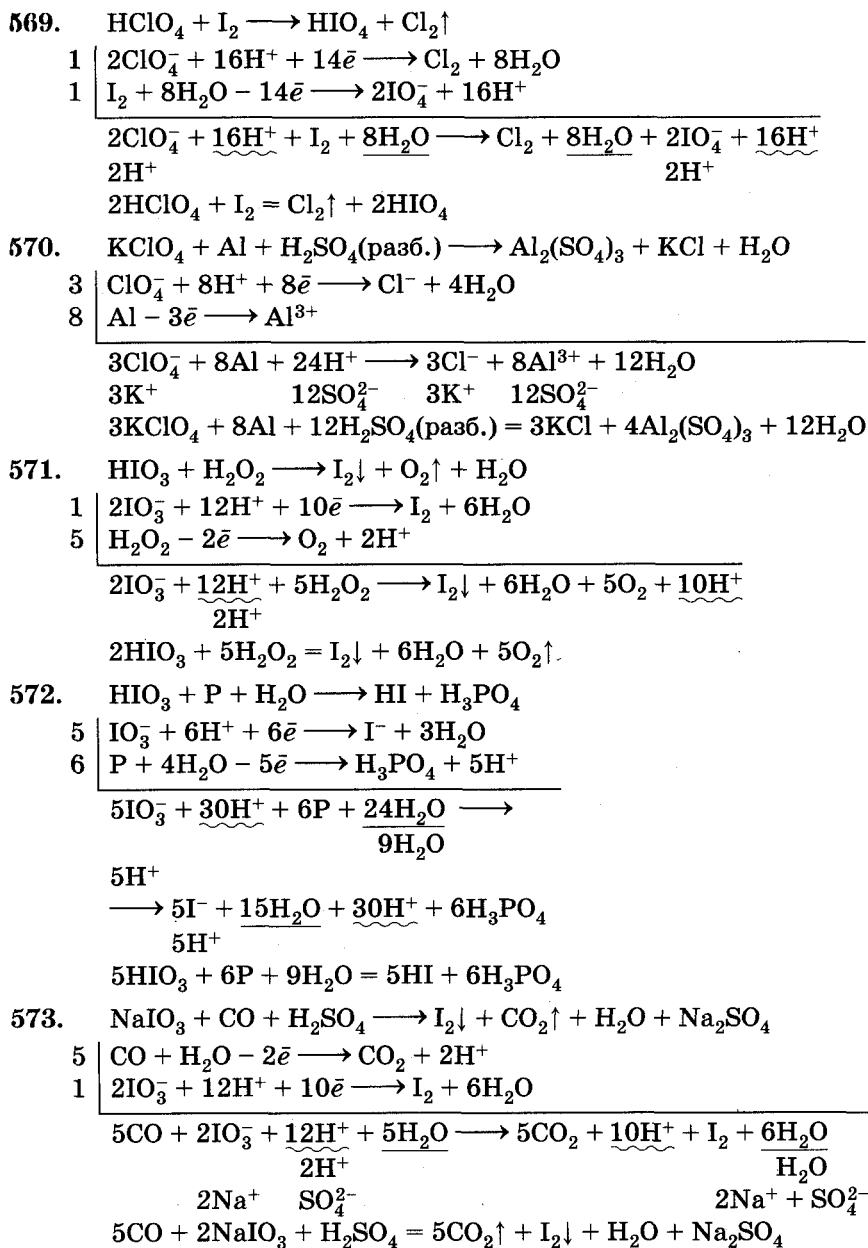




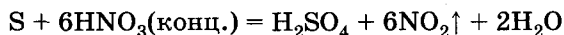
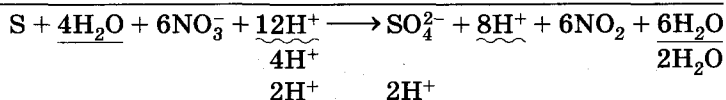
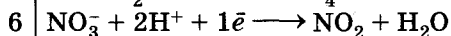
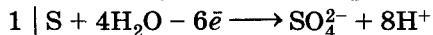
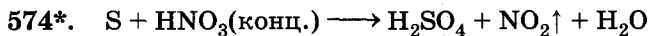
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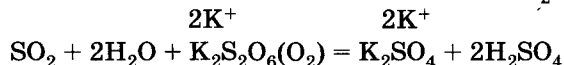
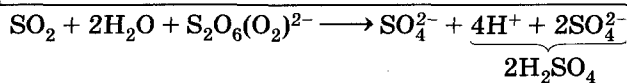
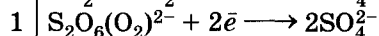
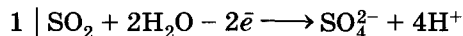
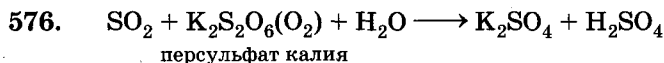
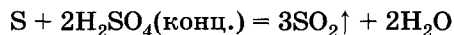
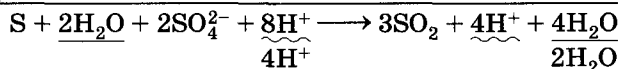
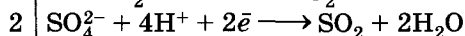
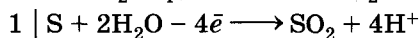
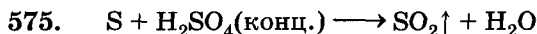
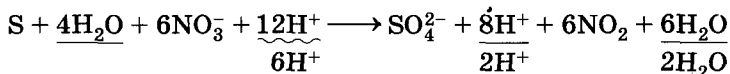


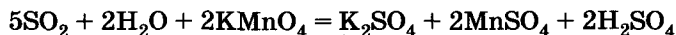
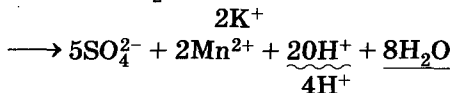
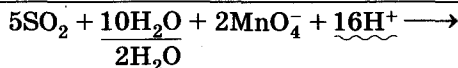
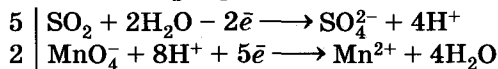
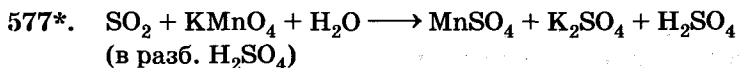


**Сера, соединения серы
и селена (среда кислотная)**

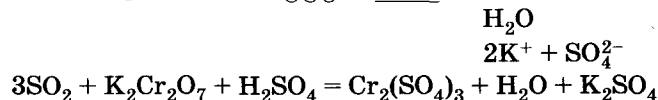
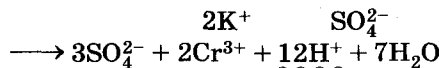
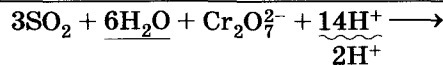
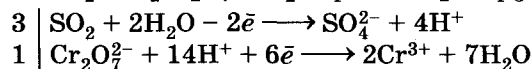
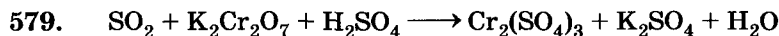
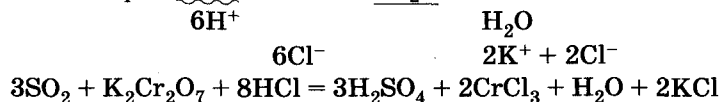
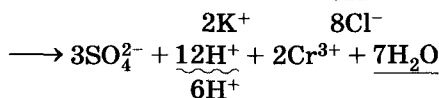
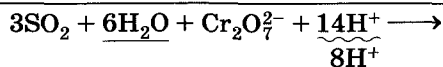
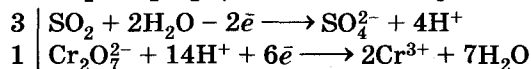
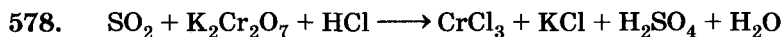
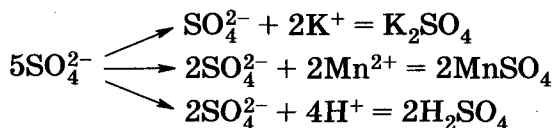


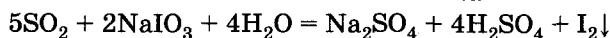
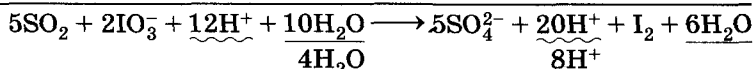
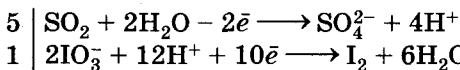
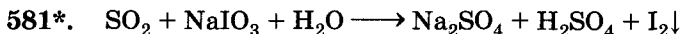
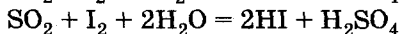
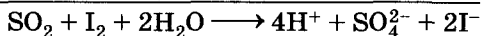
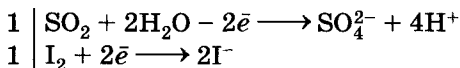
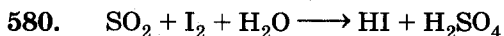
П о я с н е н и е. Традиционно в ионном уравнении производят максимальное сокращение катионов водорода, что не всегда приемлемо. При определенном навыке можно определить, какое количество H^+ можно сократить. В данном примере сокращение следует произвести не на все 8H^+ , а на 6H^+ , так как из ионного уравнения видно, что на 6NO_3^- требуется 6H^+ (в левой части), а на $\text{SO}_4^{2-} - 2\text{H}^+$. Оформление полуреакций упрощается:



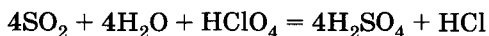
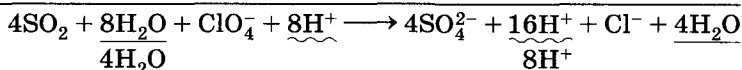
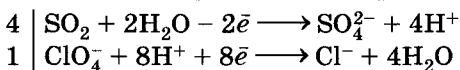
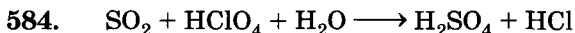
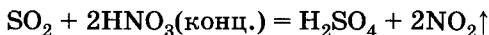
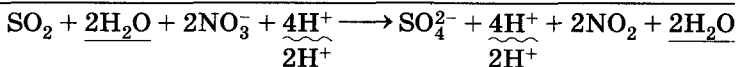
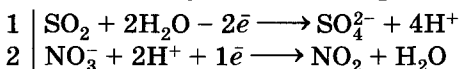
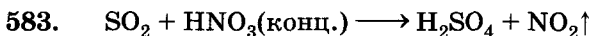
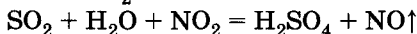
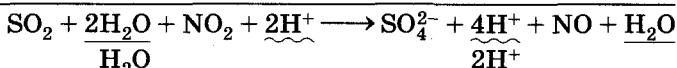
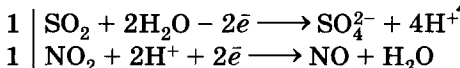
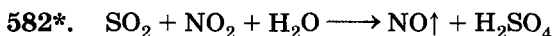


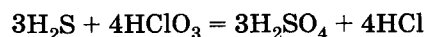
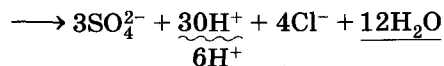
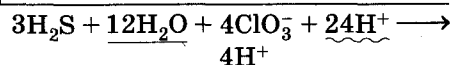
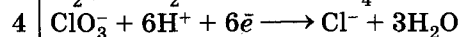
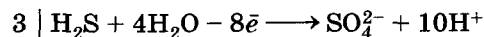
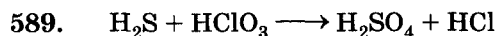
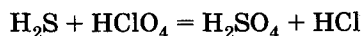
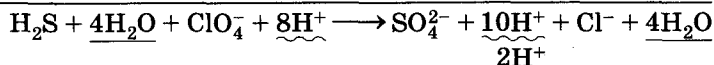
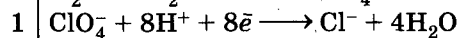
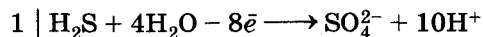
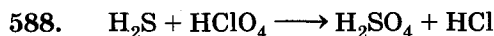
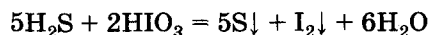
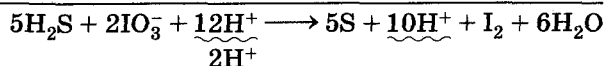
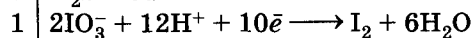
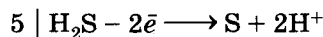
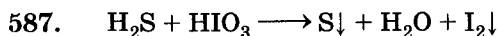
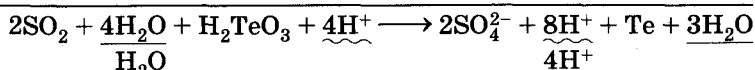
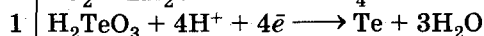
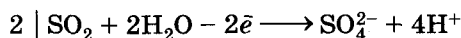
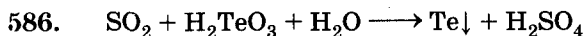
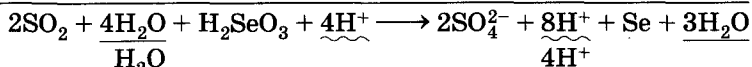
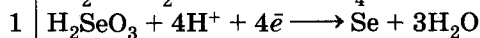
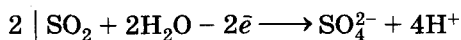
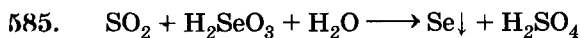
Пояснение. Катионы (K^+ , Mn^{2+} и H^+) в соответствии со своими зарядами связываются с необходимым количеством сульфат-ионов (из имеющихся 5SO_4^{2-}) так:

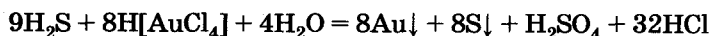
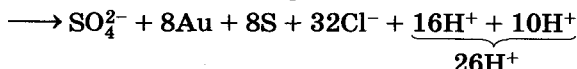
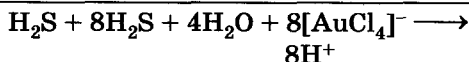
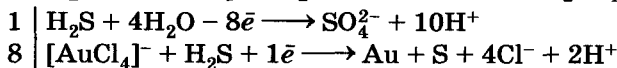
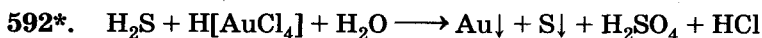
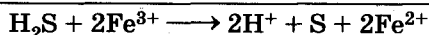
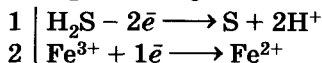
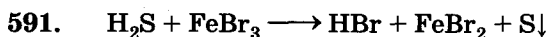
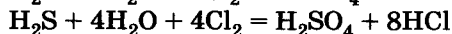
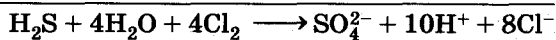
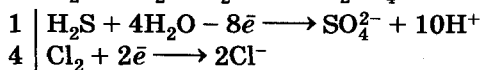
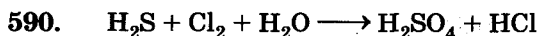




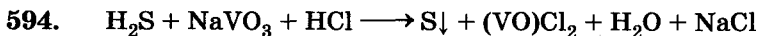
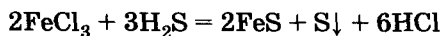
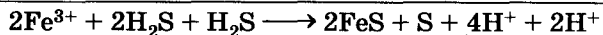
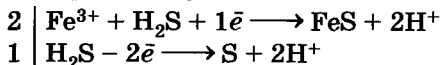
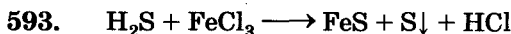
Пояснение. 8H^+ свяжут 4SO_4^{2-} из 5SO_4^{2-} , оставшийся SO_4^{2-} свяжется с 2Na^+ , которые добавляют в правую часть ионного уравнения.



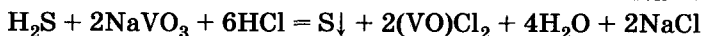
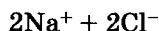
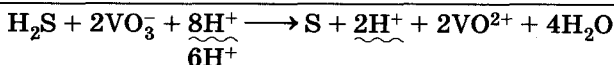
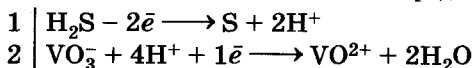


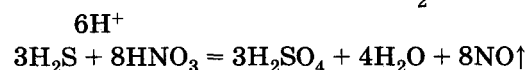
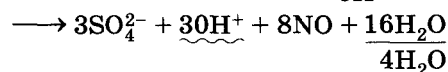
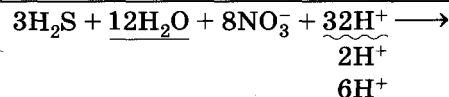
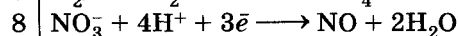
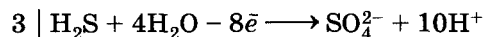
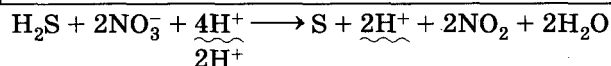
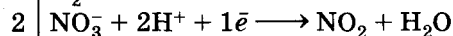
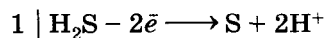
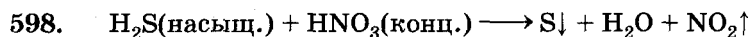
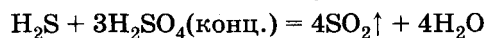
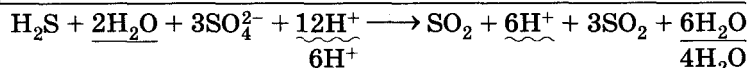
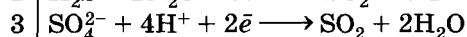
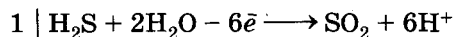
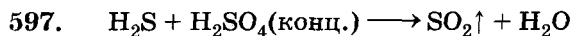
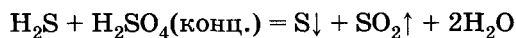
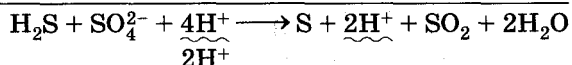
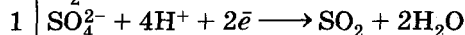
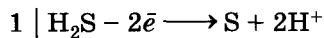
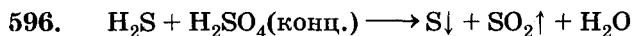
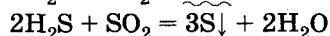
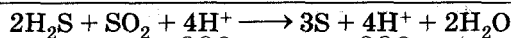
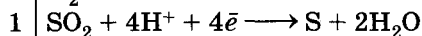
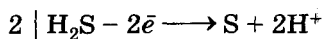
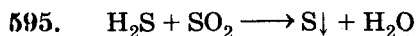


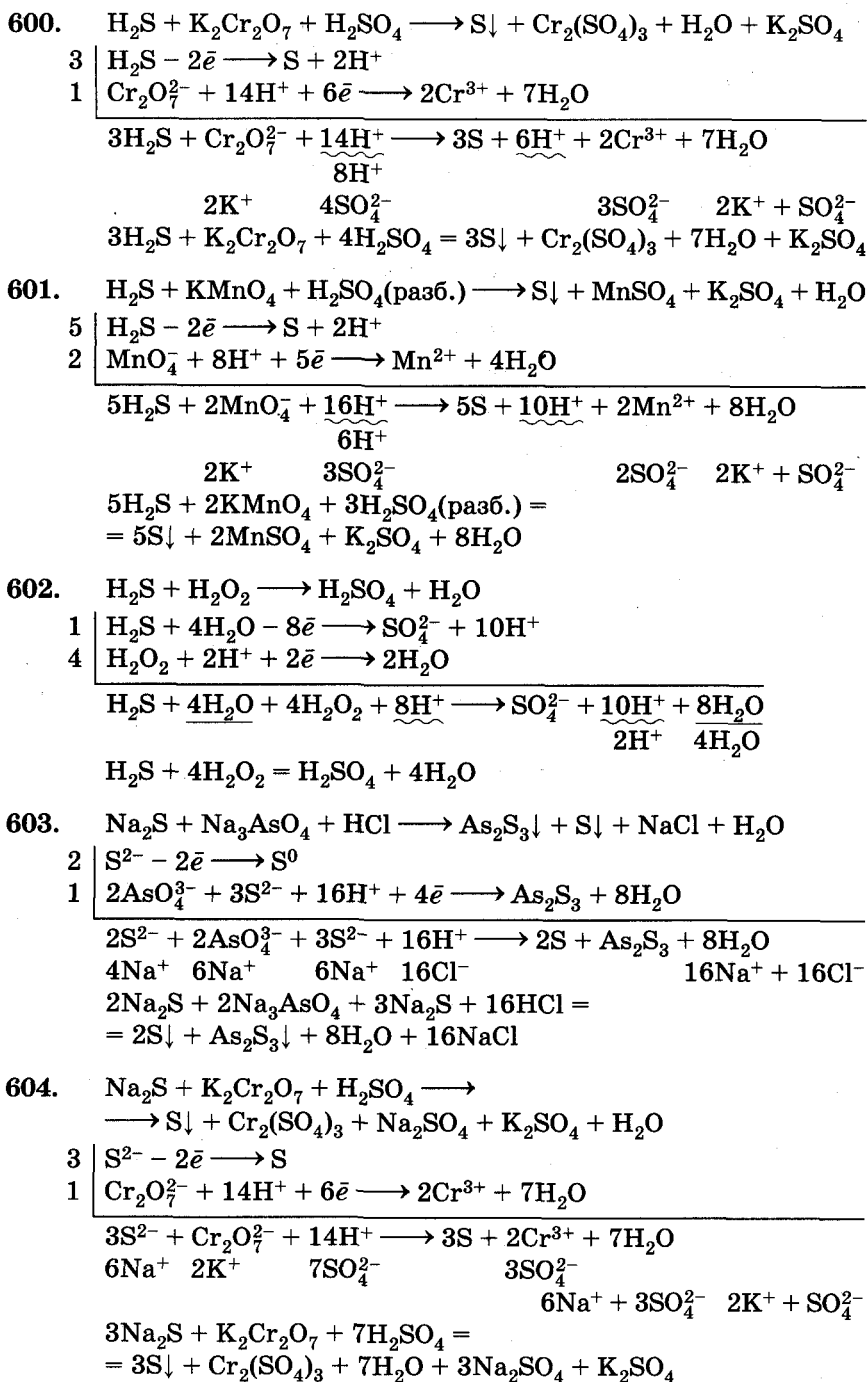
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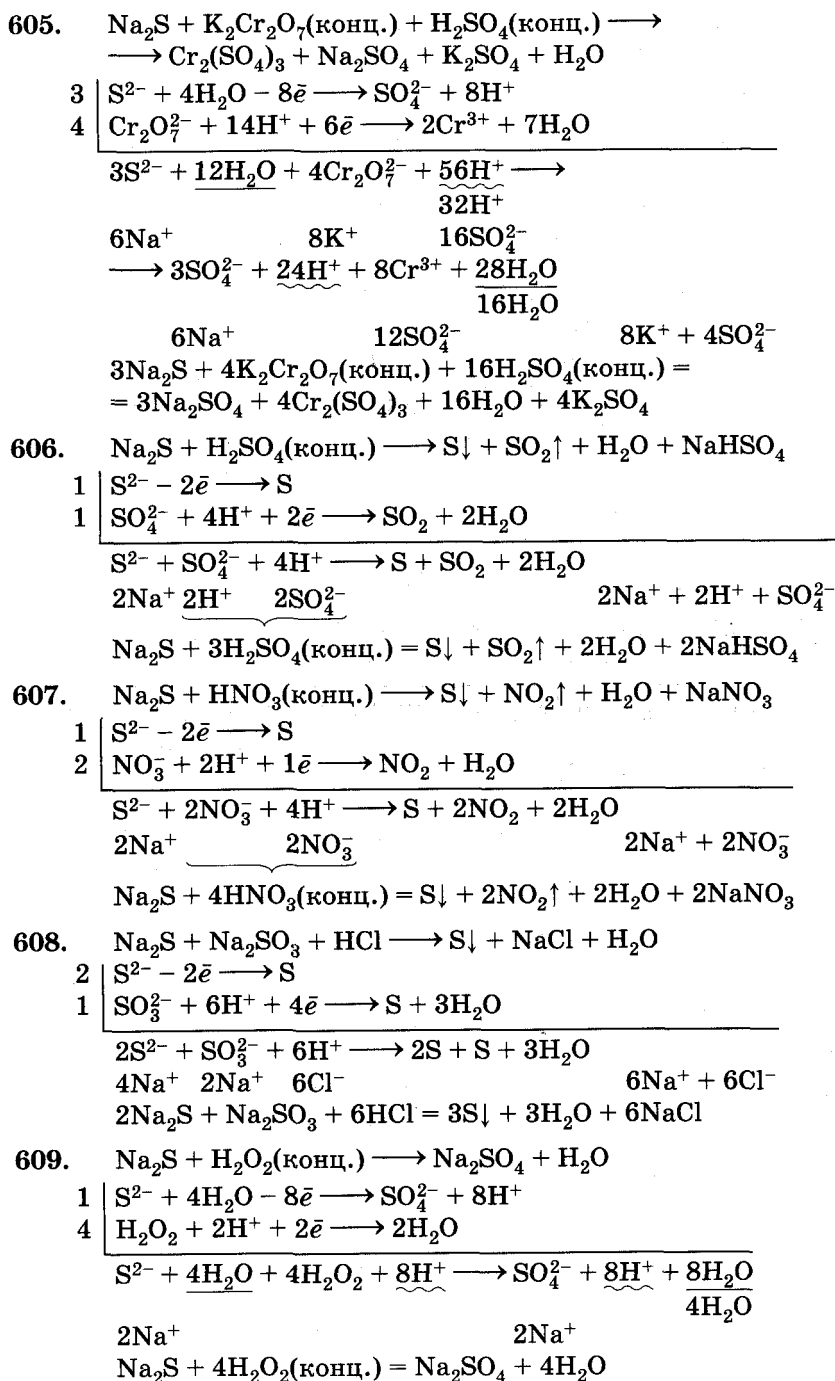


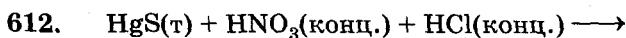
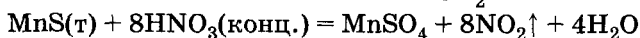
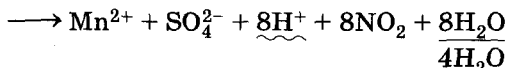
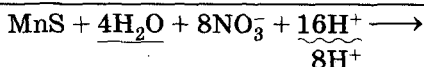
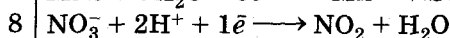
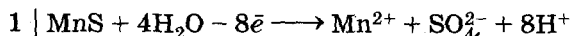
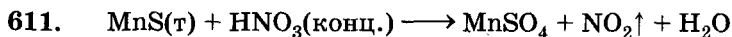
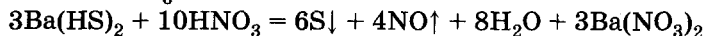
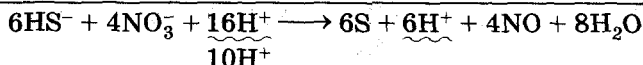
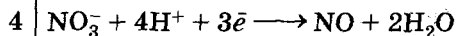
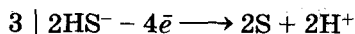
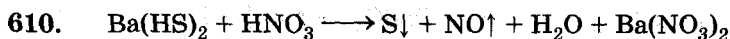
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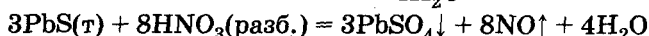
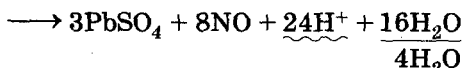
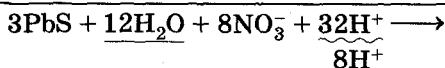
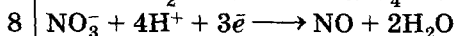
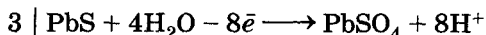
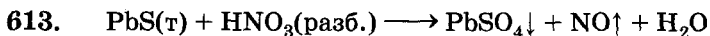
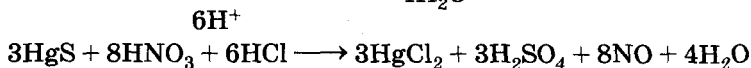
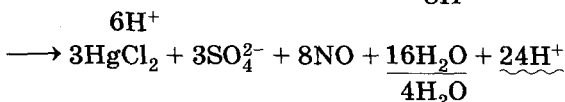
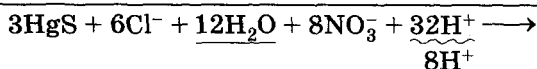
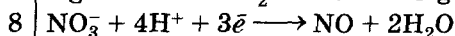
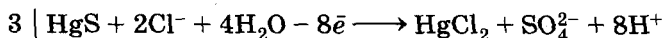
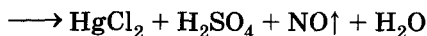


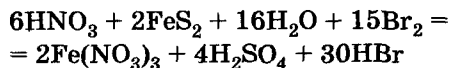
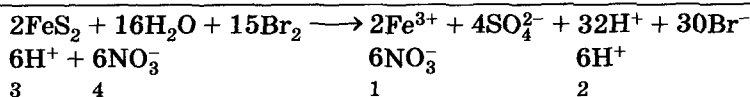
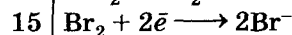
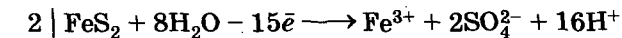
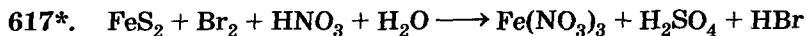
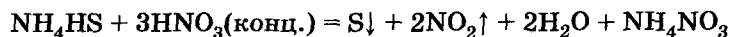
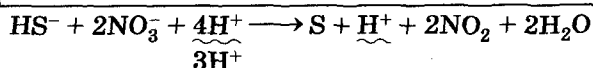
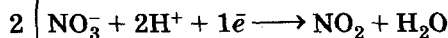
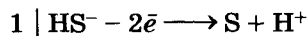
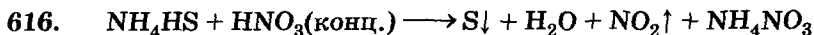
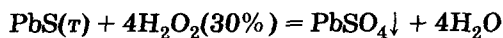
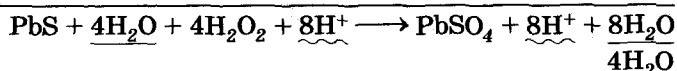
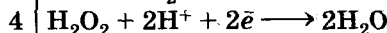
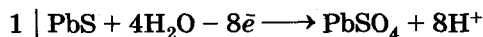
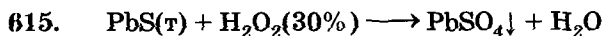
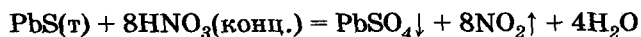
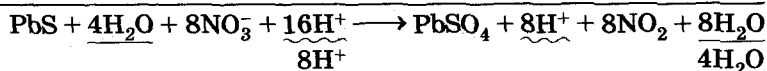
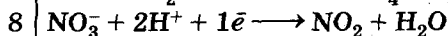
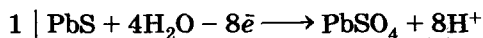
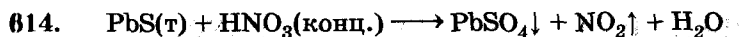






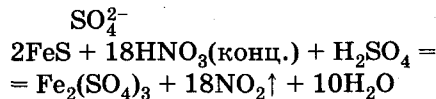
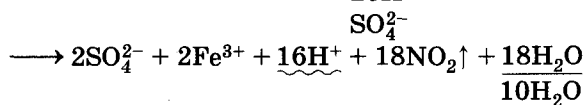
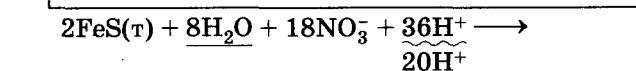
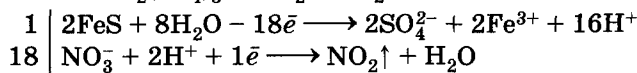
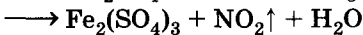
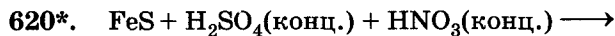
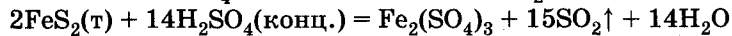
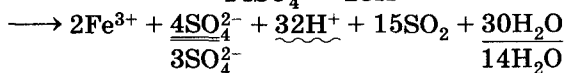
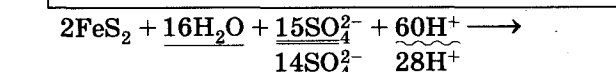
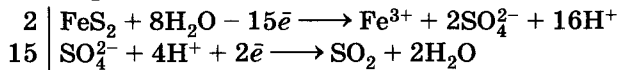
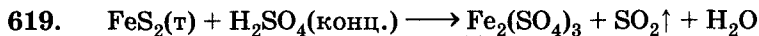
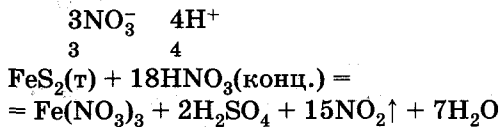
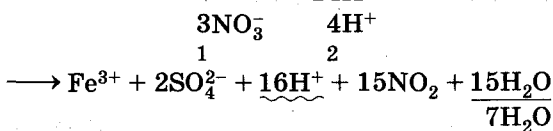
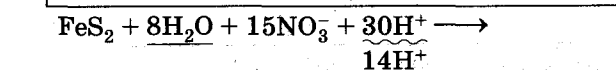
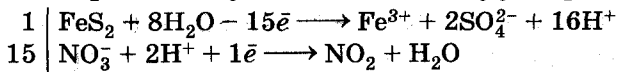
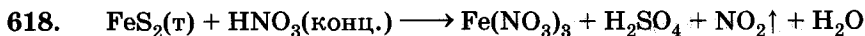
«царская водка»



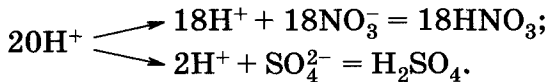


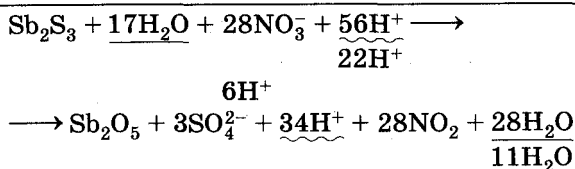
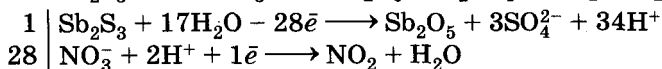
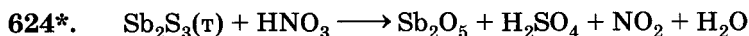
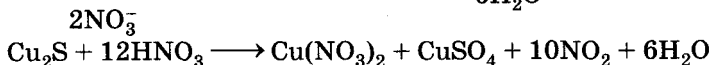
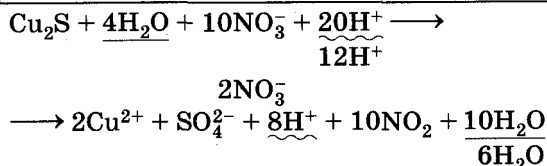
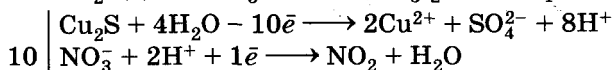
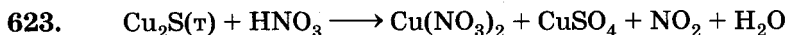
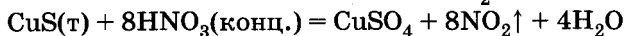
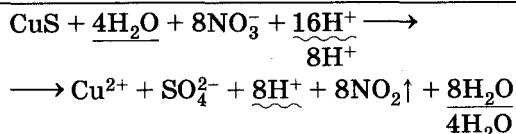
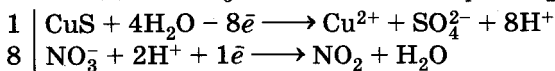
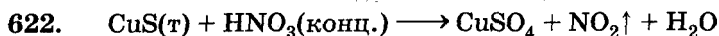
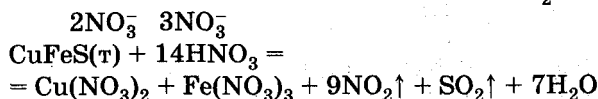
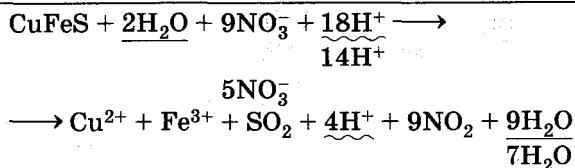
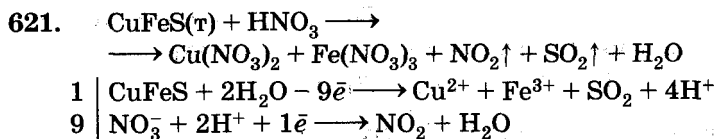
П о я с н е н и е. На 4SO_4^{2-} и 30Br^- требуется всего 38H^+ , в ионном уравнении уже есть 32H^+ , поэтому к 32H^+ добавляются 6H^+ .

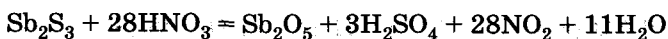
Обратите внимание, в этом примере недостающие ионы 6H^+ и 6NO_3^- добавляются не в правую часть, а в левую.



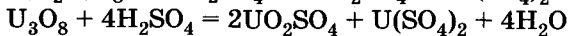
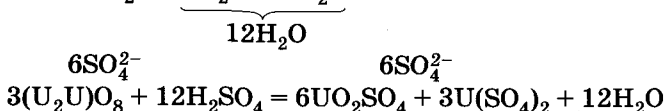
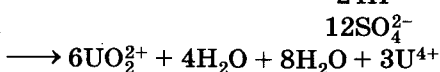
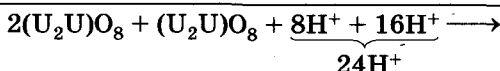
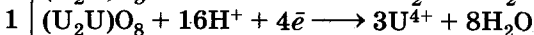
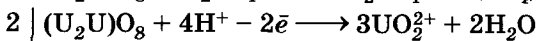
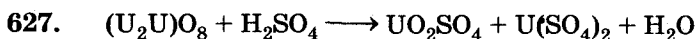
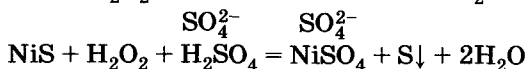
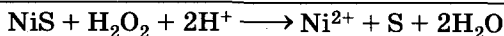
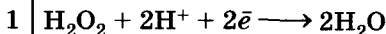
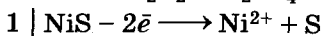
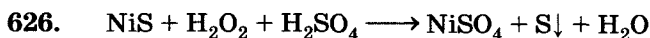
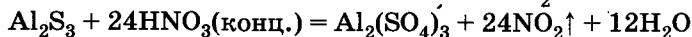
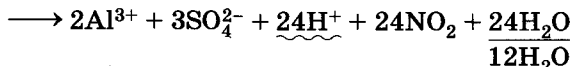
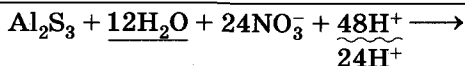
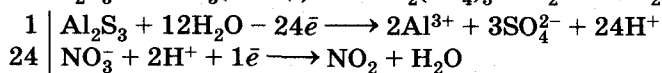
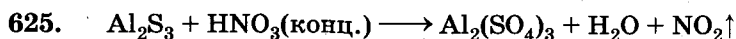
Пояснение. Ионны 2SO_4^{2-} и добавленный ион SO_4^{2-} (в сумме 3SO_4^{2-}) связываются с 2Fe^{3+} с образованием $\text{Fe}_2(\text{SO}_4)_3$. Катионы водорода (20H^+) распределяются так:



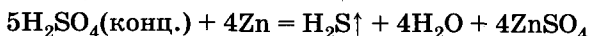
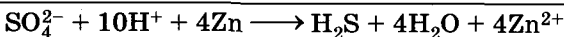
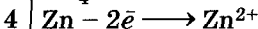
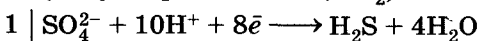
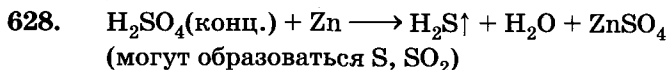


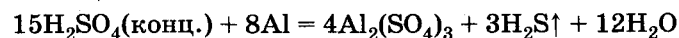
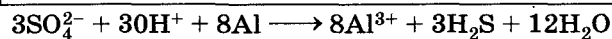
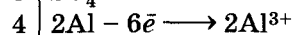
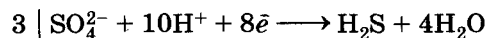
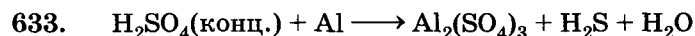
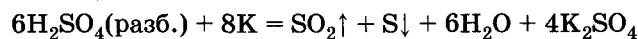
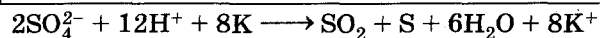
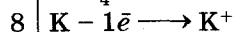
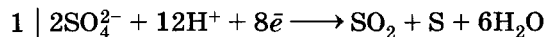
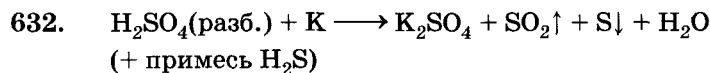
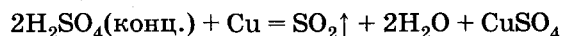
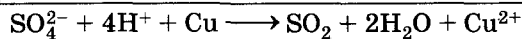
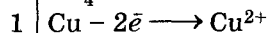
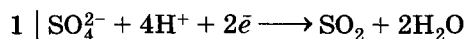
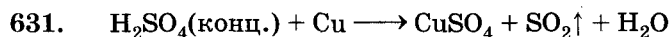
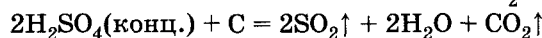
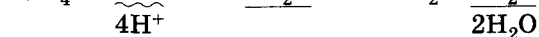
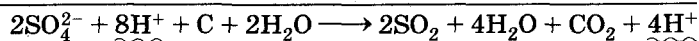
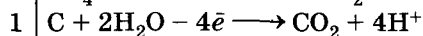
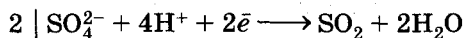
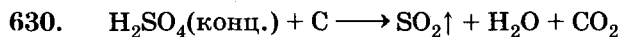
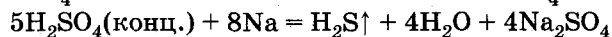
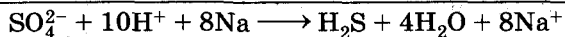
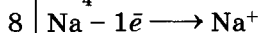
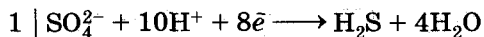
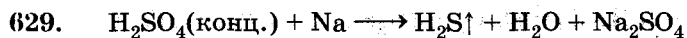


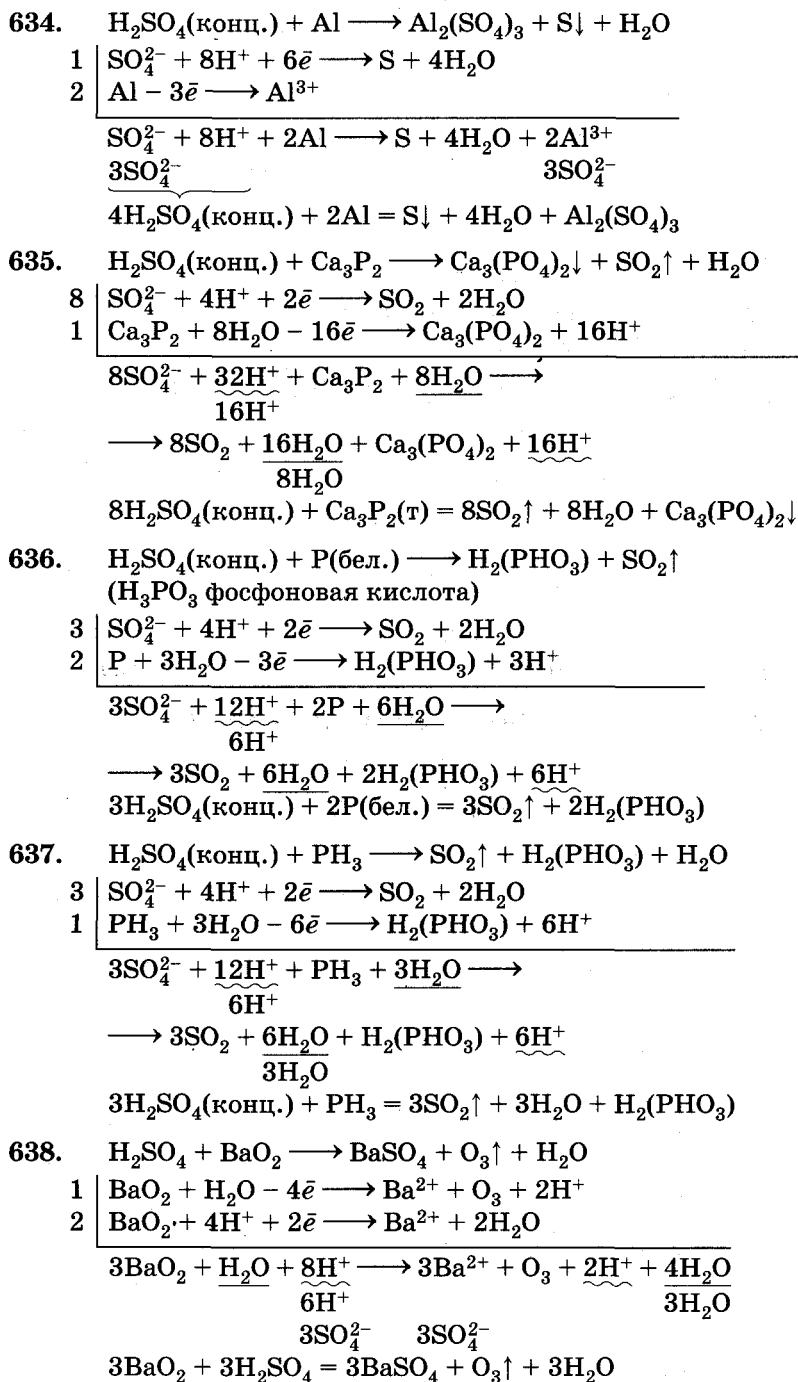
Пояснение. $28\text{NO}_3^- + 22\text{H}^+ + 6\text{H}^+ = 28\text{HNO}_3$.

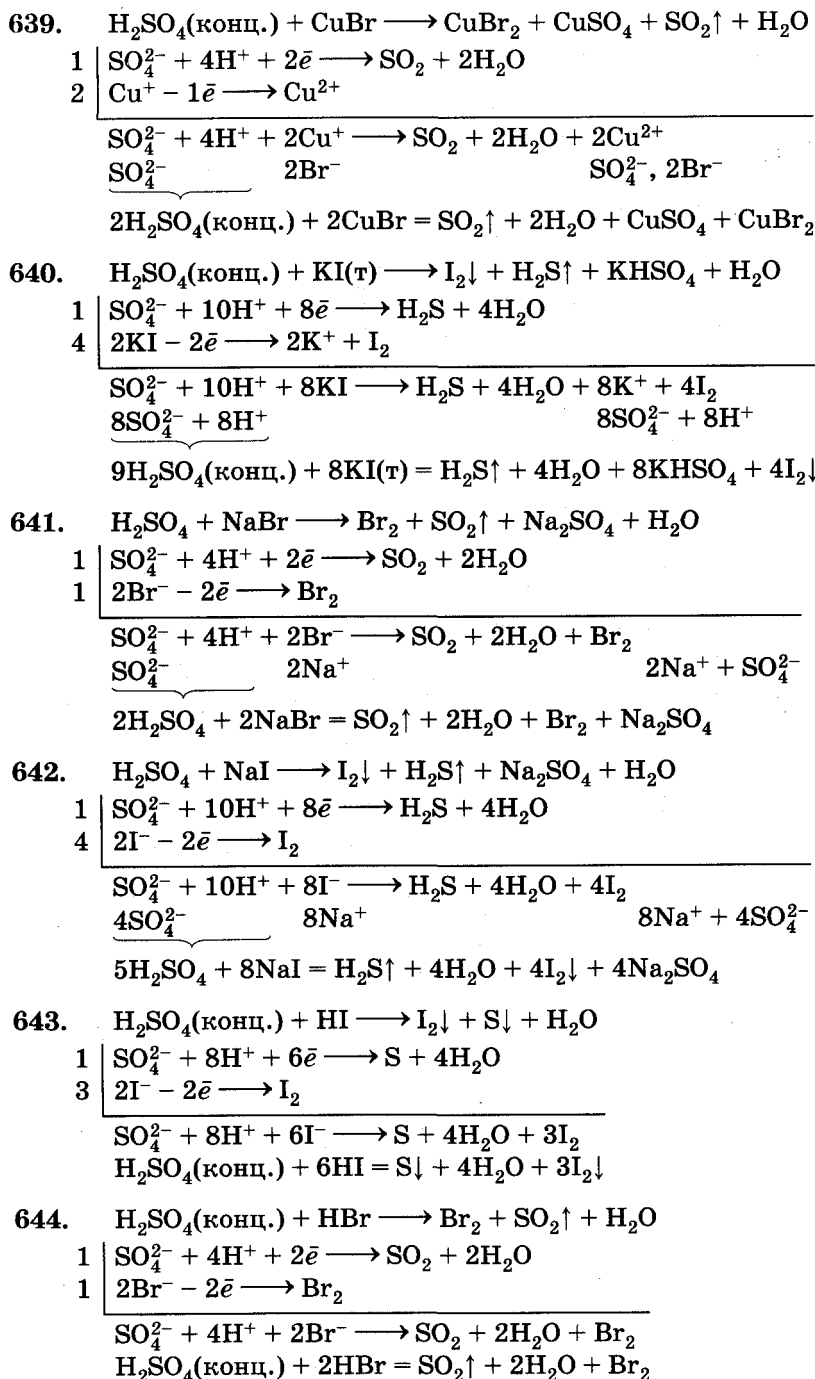


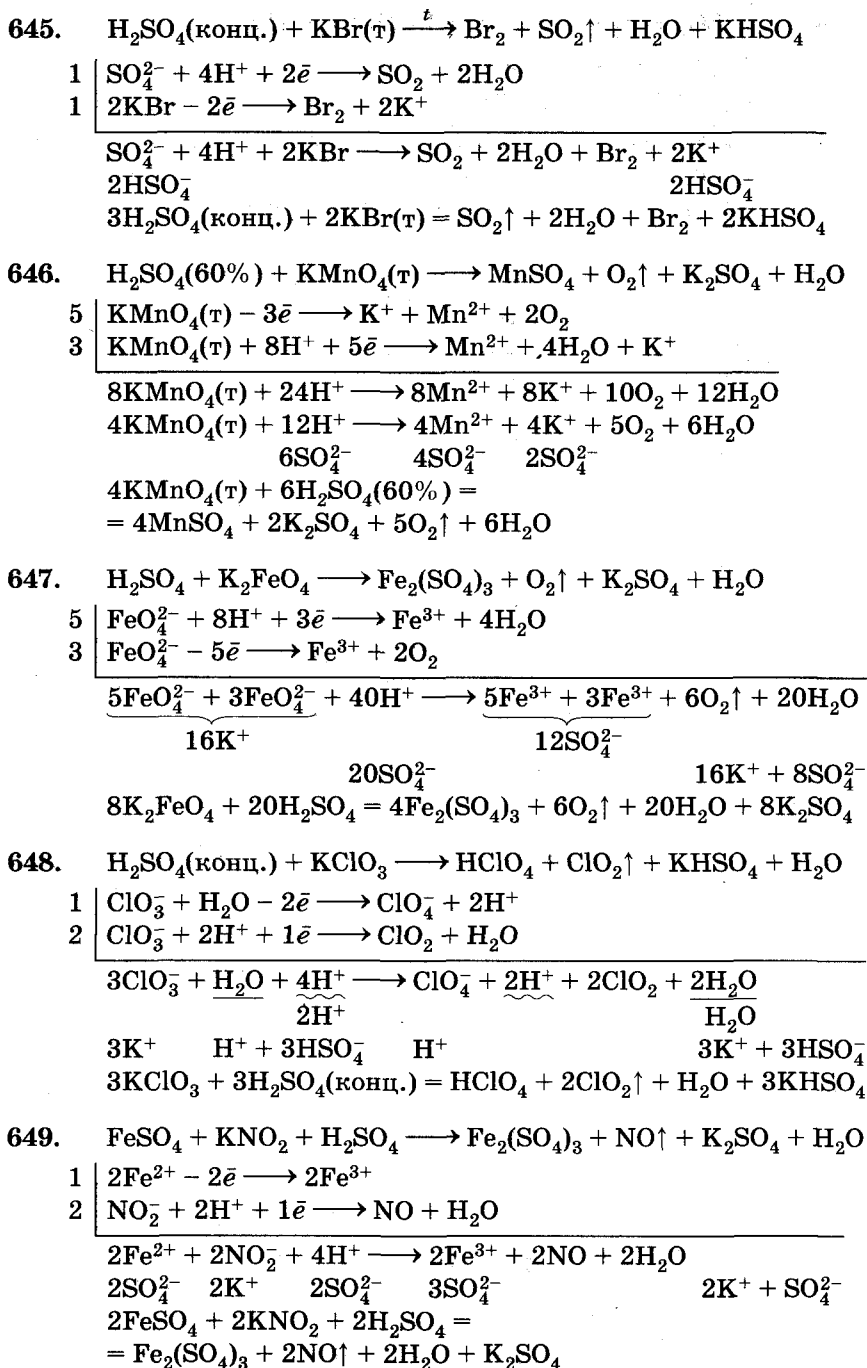
сульфат уранила

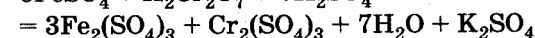
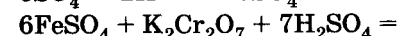
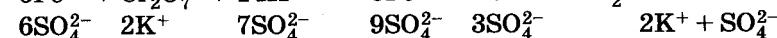
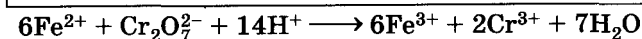
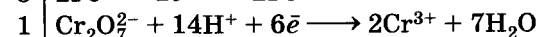
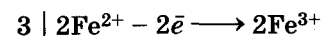
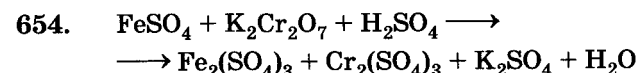
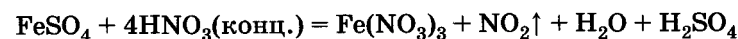
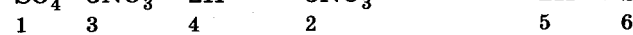
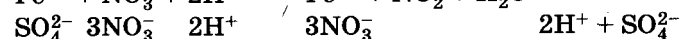
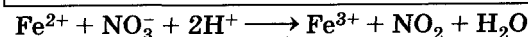
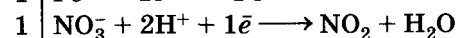
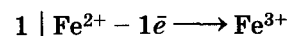
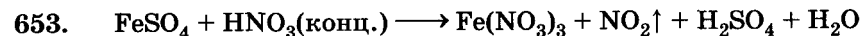
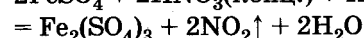
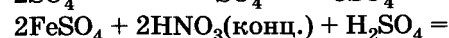
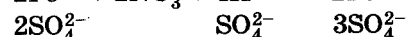
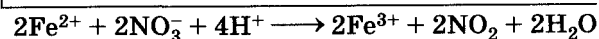
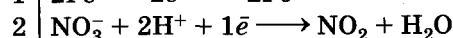
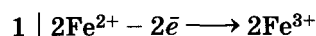
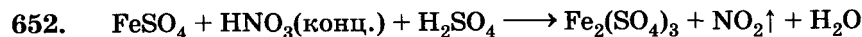
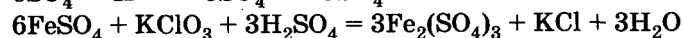
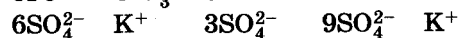
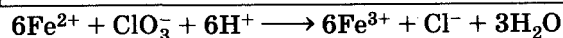
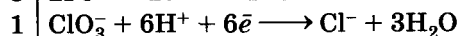
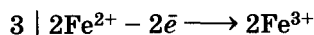
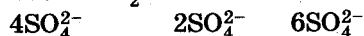
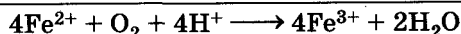
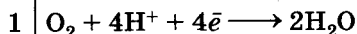
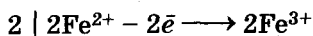


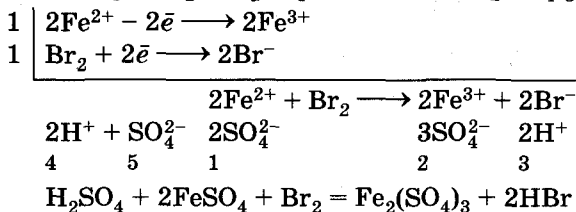
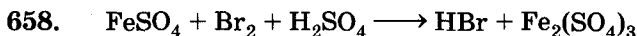
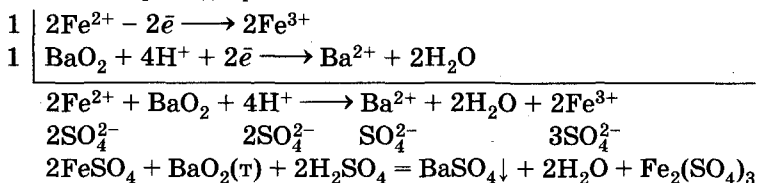
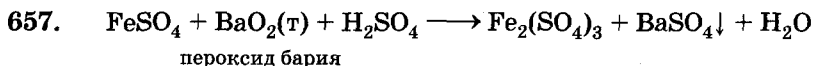
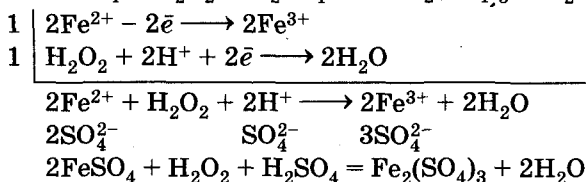
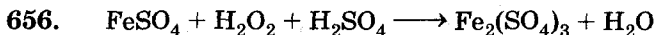
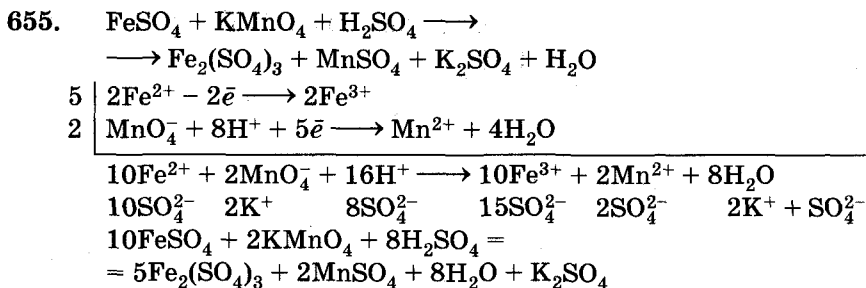




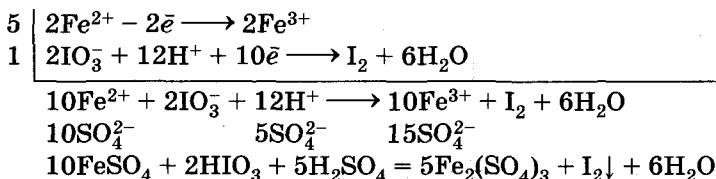




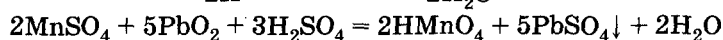
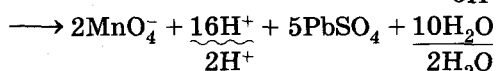
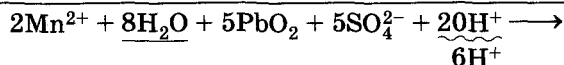
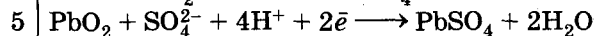
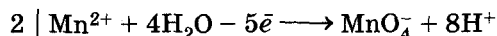
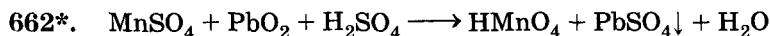
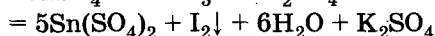
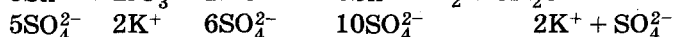
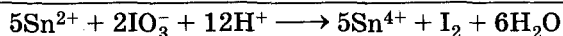
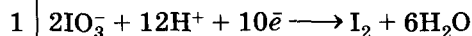
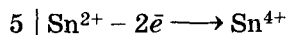
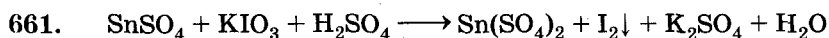
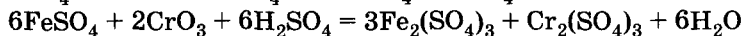
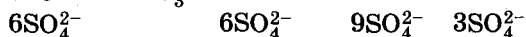
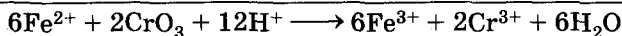
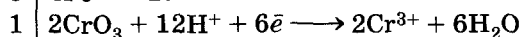
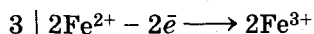
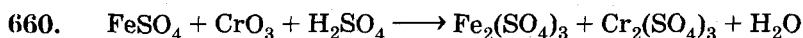




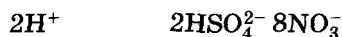
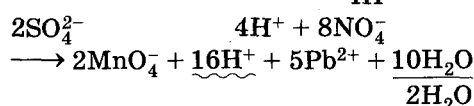
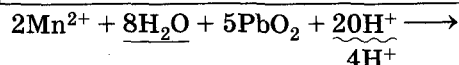
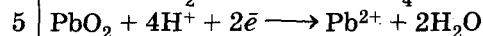
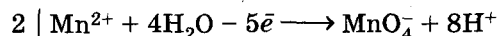
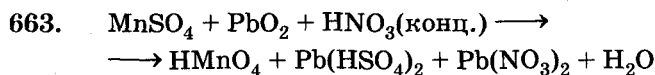
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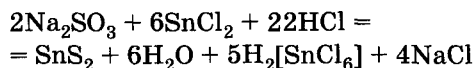
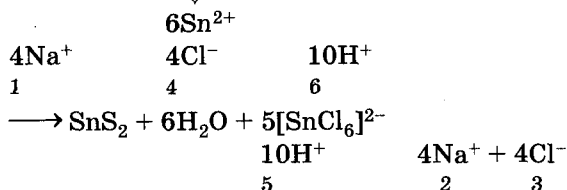
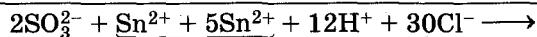
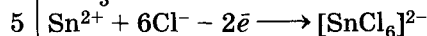
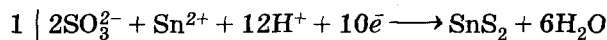
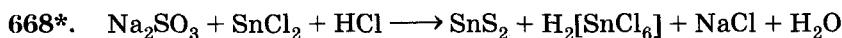
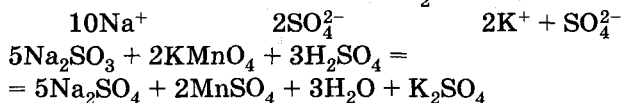
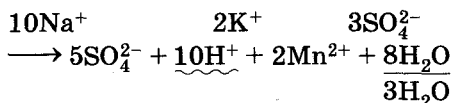
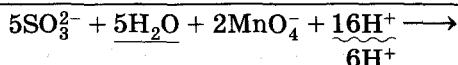
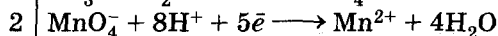
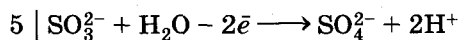
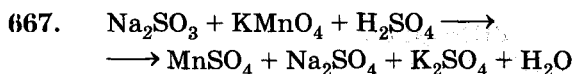


Пояснение. Распределение $12\text{H}^+ : 10\text{H}^+ + 5\text{SO}_4^{2-} = 5\text{H}_2\text{SO}_4, 2\text{H}^+ + 2\text{IO}_3^- = 2\text{HIO}_3$.



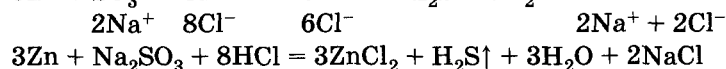
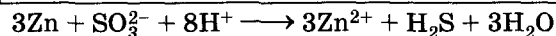
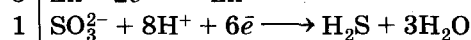
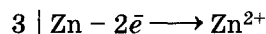
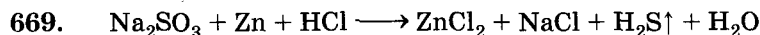
Пояснение. Распределение $5\text{SO}_4^{2-} : 3\text{SO}_4^{2-} + 6\text{H}^+ = 3\text{H}_2\text{SO}_4, 2\text{SO}_4^{2-} + 2\text{Mn}^{2+} = 2\text{MnSO}_4$.

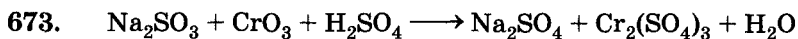
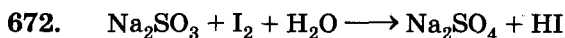
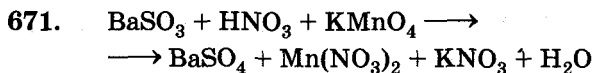
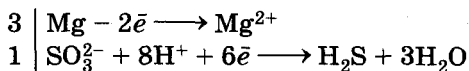


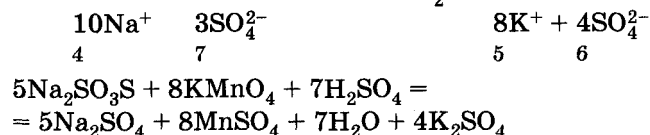
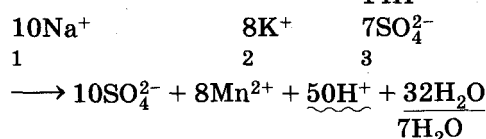
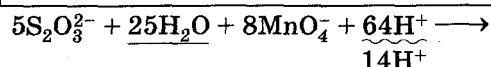
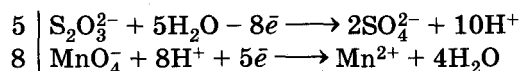
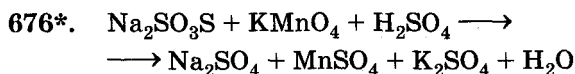
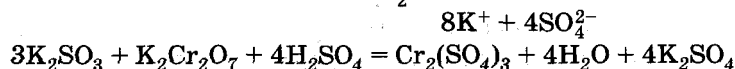
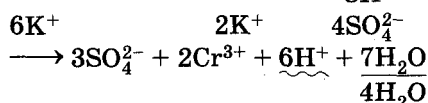
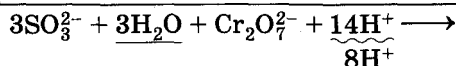
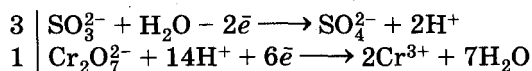
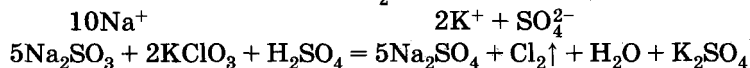
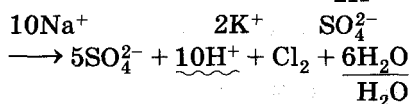
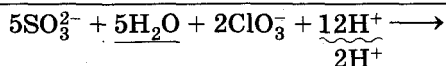
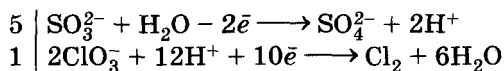
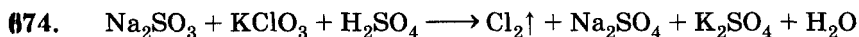


П о я с н е н и е. В левой части ионного уравнения всего 34Cl^- ($30\text{Cl}^- + 4\text{Cl}^-$); катионов водорода 22H^+ ($12\text{H}^+ + 10\text{H}^+$).

Распределение 34Cl^- : $12\text{Cl}^- + 6\text{Sn}^{2+} = 6\text{SnCl}_2$;
 $22\text{Cl}^- + 22\text{H}^+ = 12\text{HCl}$.





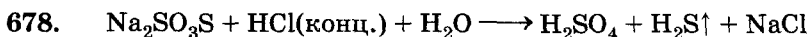
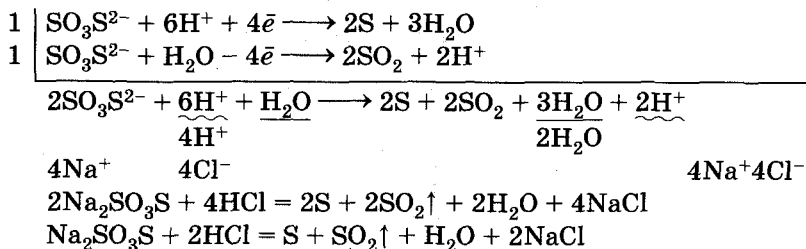


Пояснение. Необходимо обратить внимание на последовательность добавления ионов. В правой части ионного уравнения всего 13SO_4^{2-} ($10\text{SO}_4^{2-} + 3\text{SO}_4^{2-}$), кото-

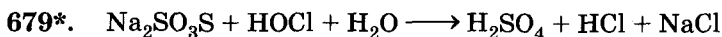
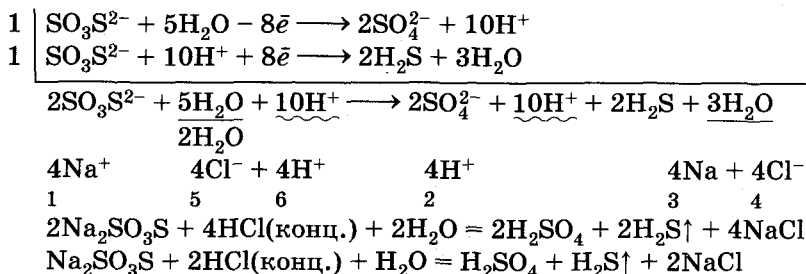
рые распределяются так: $5\text{SO}_4^{2-} + 10\text{Na}^+ = 5\text{Na}_2\text{SO}_4$;
 $8\text{SO}_4^{2-} + 8\text{Mn}^{2+} = 8\text{MnSO}_4$.



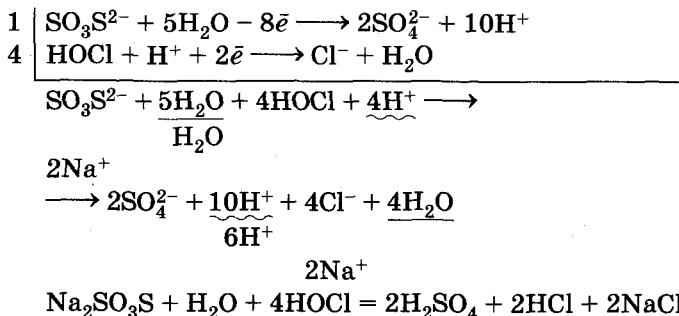
тиосульфат натрия



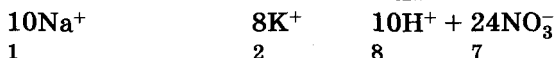
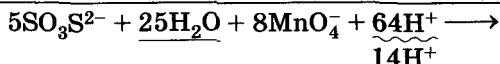
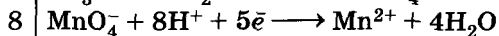
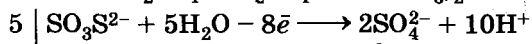
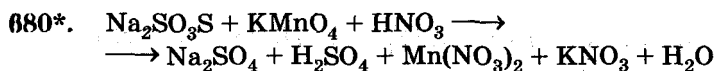
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тиосульфат натрия



П о я с н е н и е. Распределение 6H^+ : $4\text{H}^+ + 2\text{SO}_4^{2-} = 2\text{H}_2\text{SO}_4$, $2\text{H}^+ + 2\text{Cl}^- = 2\text{HCl}$; оставшиеся 2Cl^- из 4Cl^- связываются с 2Na^+ , и образуется 2NaCl .

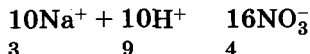
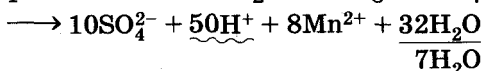


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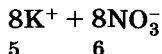
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4



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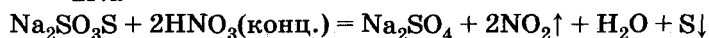
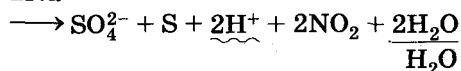
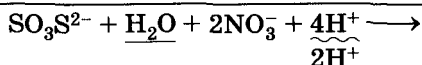
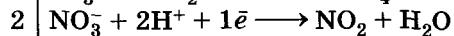
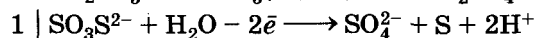
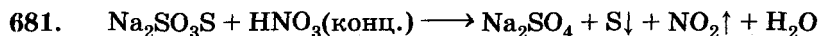
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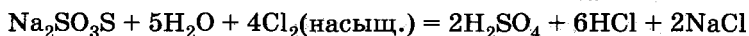
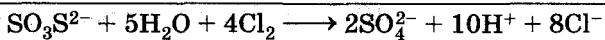
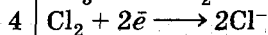
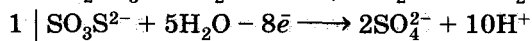
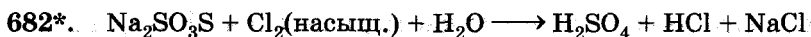


Порядок добавления ионов.

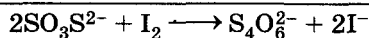
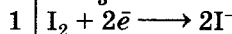
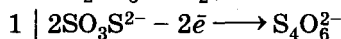
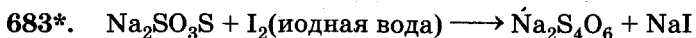
Способ 1. Исходя из ионного уравнения, легко рассчитать число и порядок добавления ионов по порядку: 1, 2, 3, 4, 5, 6. Далее в правой части нитрат-ионов в сумме стало $16\text{NO}_3^- + 8\text{NO}_3^- = 24\text{NO}_3^-$, следовательно, в левую часть их добавляется столько же (порядок 7). На 24NO_3^- потребуется 24H^+ , из которых 14H^+ уже есть в ионном уравнении, поэтому в левую часть добавляются 10H^+ (порядок 8) и столько же в правую часть (порядок 9).

Способ 2. Добавленные 10Na^+ в правой части (порядок 3) свяжут из 10SO_4^{2-} , имеющих в ионном уравнении, только 5SO_4^{2-} , из которых образуется $5\text{Na}_2\text{SO}_4$. К оставшимся 5SO_4^{2-} добавляется 10H^+ . Теперь в левой части в сумме $10\text{H}^+ + 14\text{H}^+ = 24\text{H}^+$, на которые следует добавить 24NO_3^- .

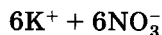
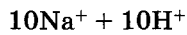
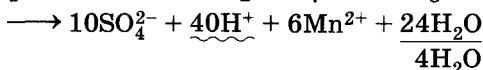
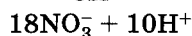
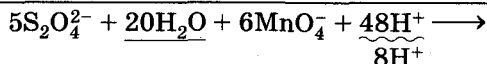
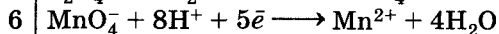
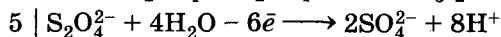
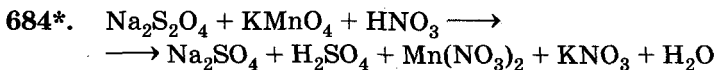




Пояснение. Распределение 10H^+ : $4\text{H}^+ + 2\text{SO}_4^{2-} = 2\text{H}_2\text{SO}_4$; $6\text{H}^+ + 6\text{Cl}^- = 6\text{HCl}$. Оставшиеся 2Cl^- ($8\text{Cl}^- - 6\text{Cl}^-$) соединяются с 2Na^+ с образованием 2NaCl .

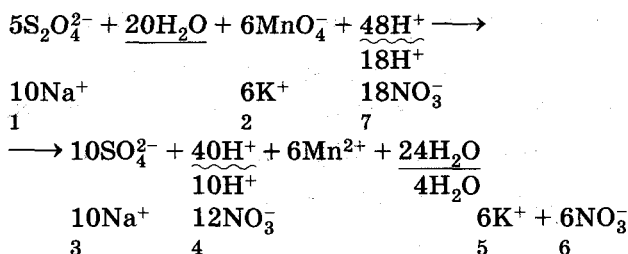


Пояснение. Как видно из примеров 682 и 683, тиосульфат-ион SO_3S^{2-} превращается в сульфат-ион SO_4^{2-} , если взаимодействует с сильным окислителем, а со слабым окислителем (иодной водой) — в ион $\text{S}_4\text{O}_6^{2-}$.



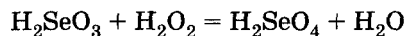
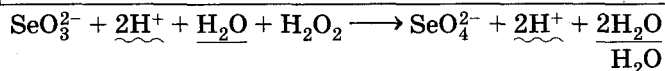
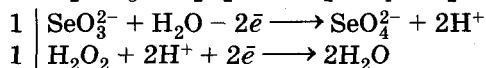
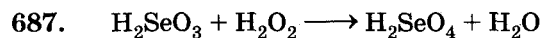
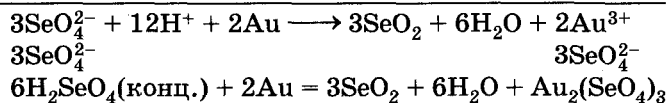
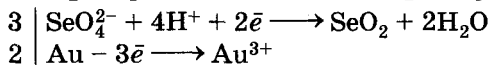
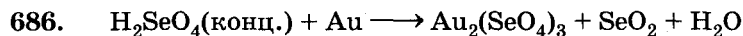
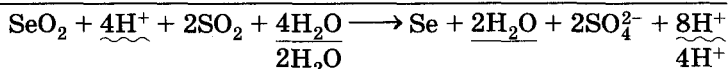
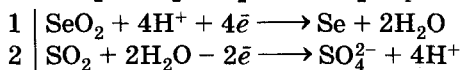
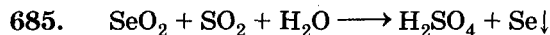
Пояснение. Добавив основные ионы согласно ионному уравнению (1, 2, 3, 4), для равенства в правую часть добавляется 6K^+ , на которые требуется добавить 6NO_3^- . В сумме нитрат-ионов становится 18 ($12\text{NO}_3^- + 6\text{NO}_3^-$),

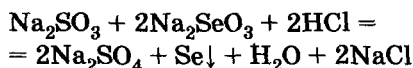
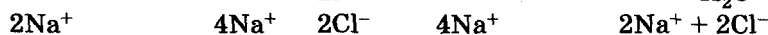
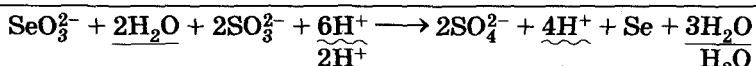
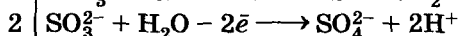
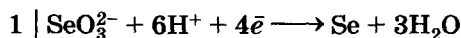
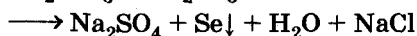
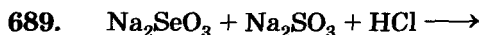
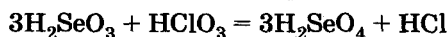
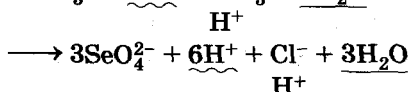
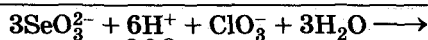
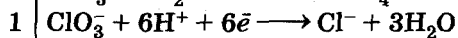
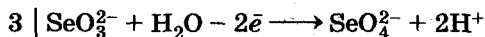
следовательно, и в левую часть добавятся 18NO_3^- , на которые необходимы 18H^+ . С учетом изложенного имеющиеся в ионном уравнении катионы водорода $48\text{H}^+ \longrightarrow 40\text{H}^+$ логичней сократить не на 40, а на 30, что упростит оформление:



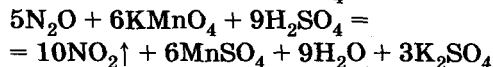
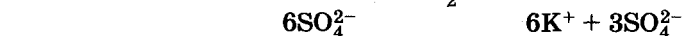
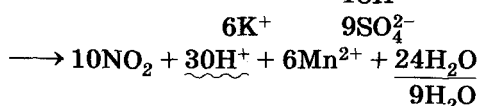
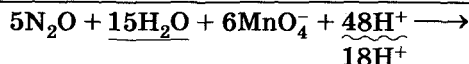
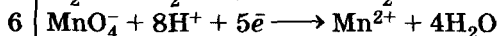
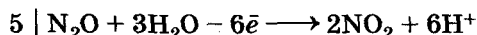
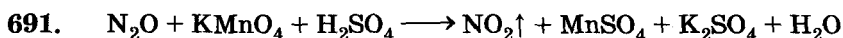
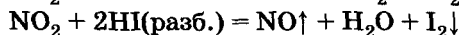
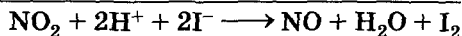
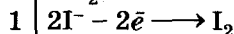
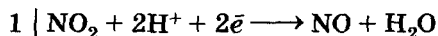
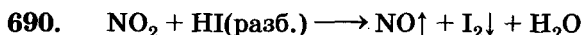
Распределение сульфат-ионов.

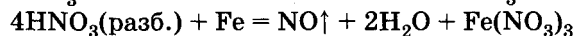
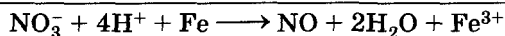
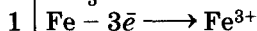
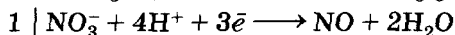
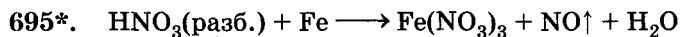
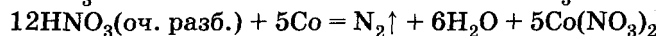
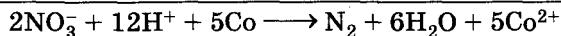
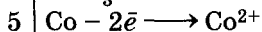
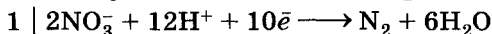
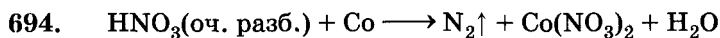
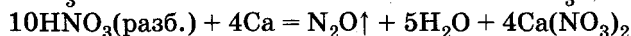
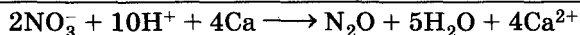
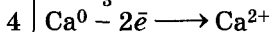
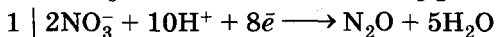
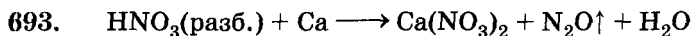
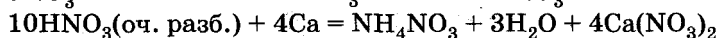
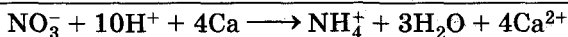
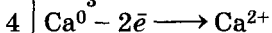
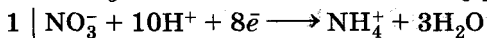
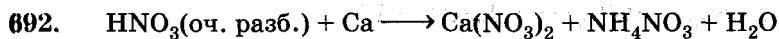
Помня о равенстве количества добавляемых ионов, понятно, что, добавив в левую часть 10Na^+ , столько же их следует добавить в правую часть, хотя сульфат-ионов в ионном уравнении больше: $10\text{Na}^+ + 5\text{SO}_4^{2-} = 5\text{Na}_2\text{SO}_4$. Оставшиеся 5SO_4^{2-} свяжутся с имеющимися в ионном уравнении 10H^+ , из которых образуется $5\text{H}_2\text{SO}_4$.



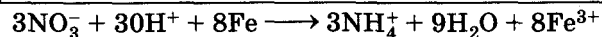
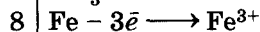
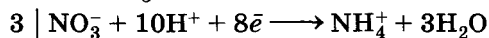
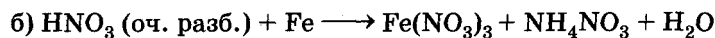
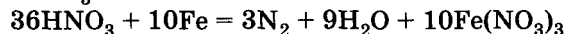
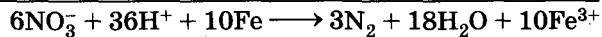
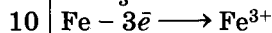
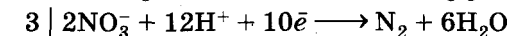
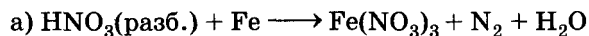


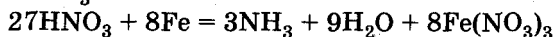
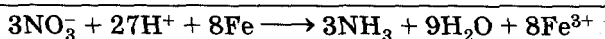
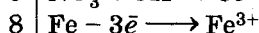
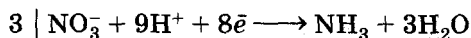
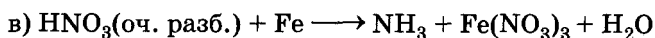
Соединения азота, фосфора и мышьяка (среда кислотная)



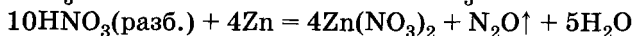
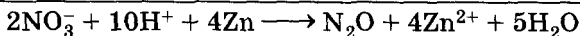
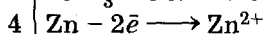
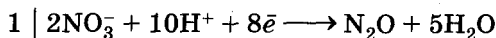
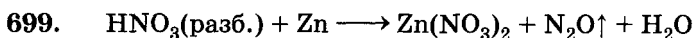
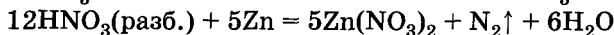
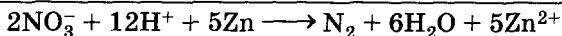
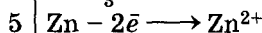
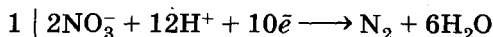
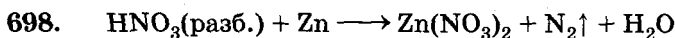
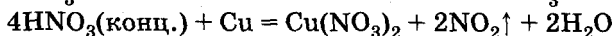
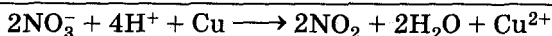
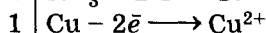
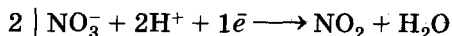
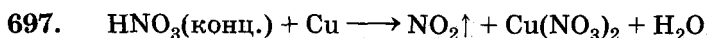
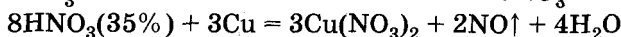
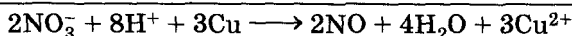
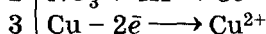
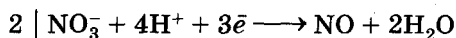
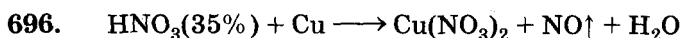


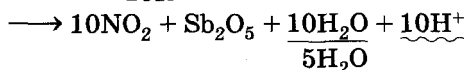
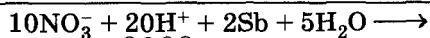
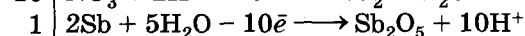
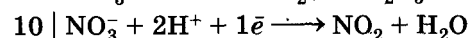
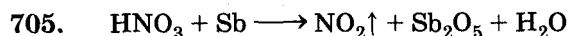
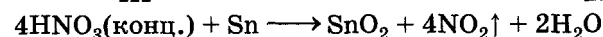
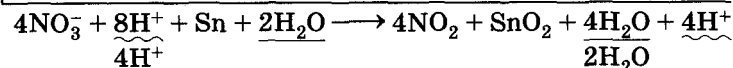
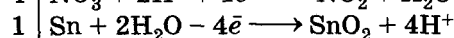
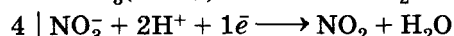
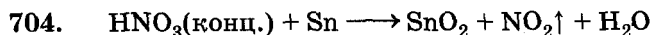
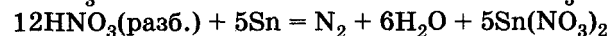
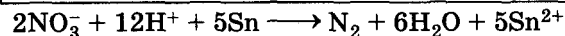
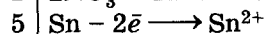
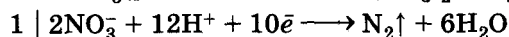
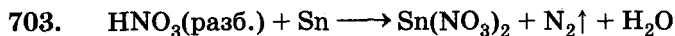
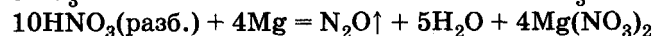
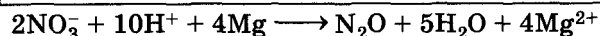
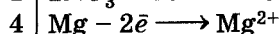
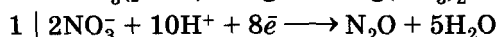
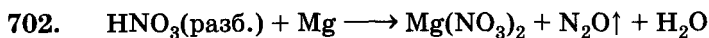
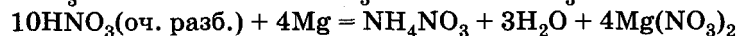
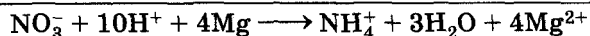
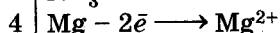
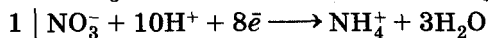
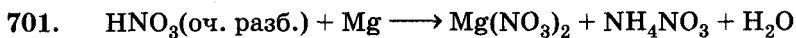
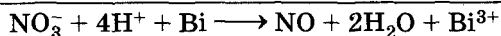
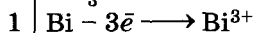
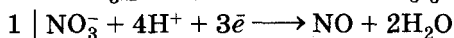
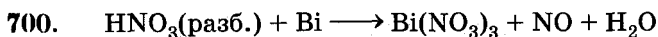
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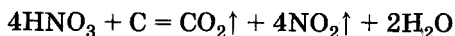
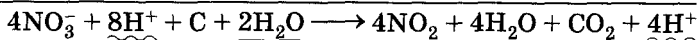
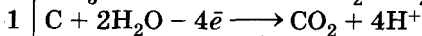
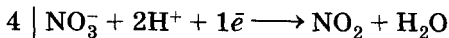
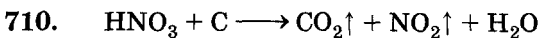
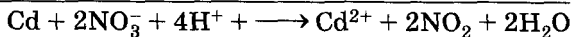
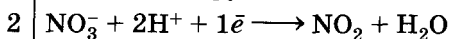
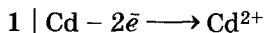
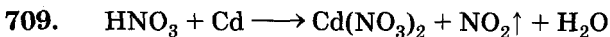
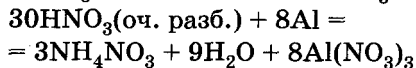
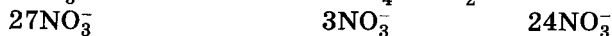
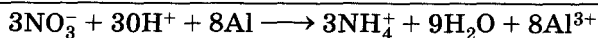
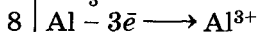
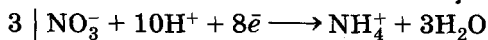
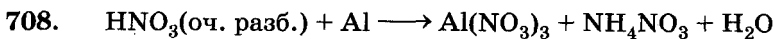
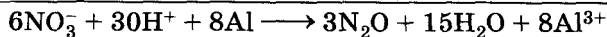
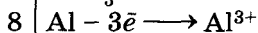
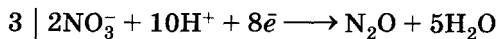
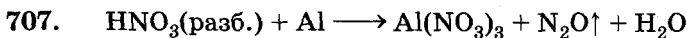
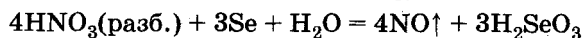
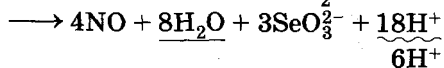
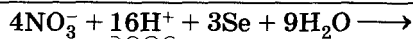
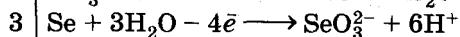
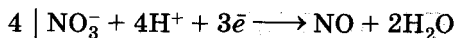
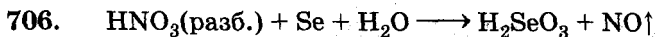


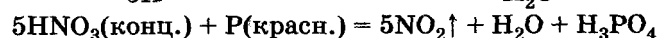
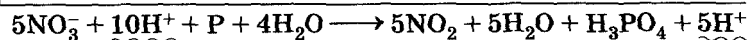
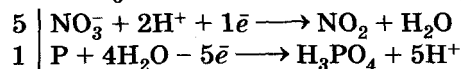
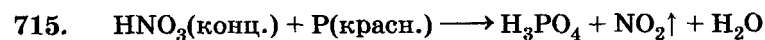
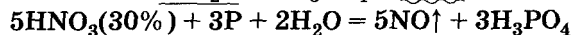
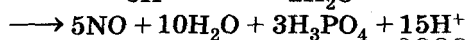
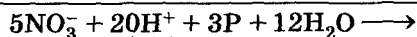
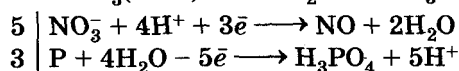
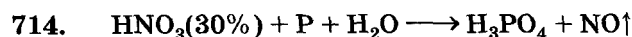
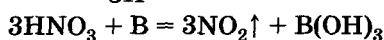
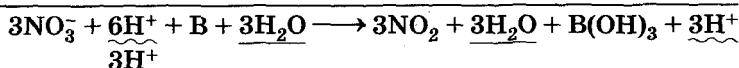
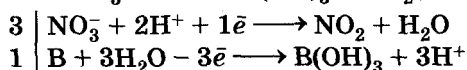
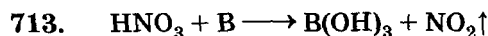
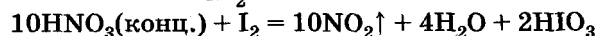
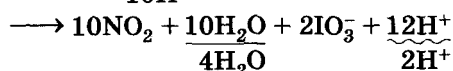
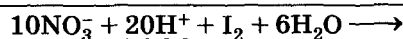
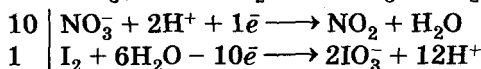
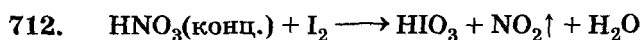
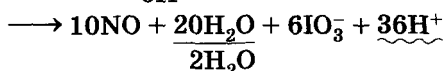
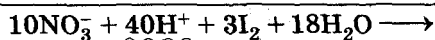
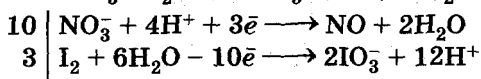
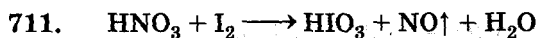


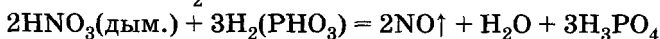
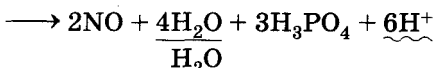
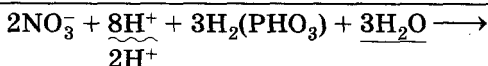
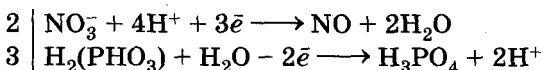
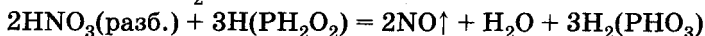
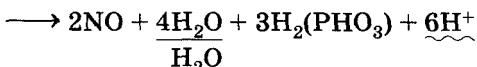
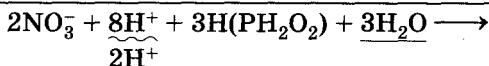
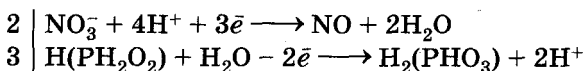
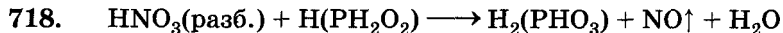
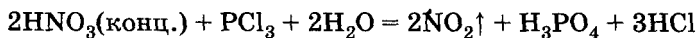
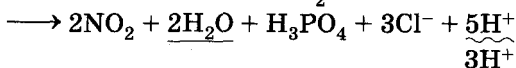
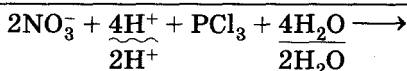
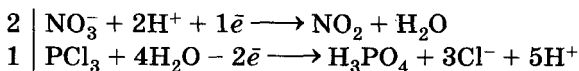
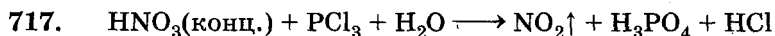
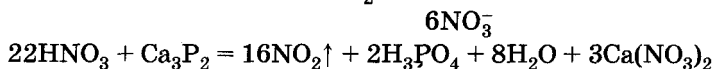
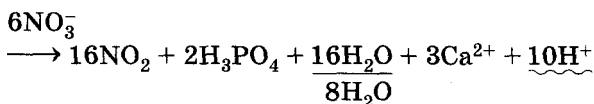
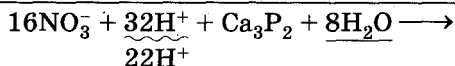
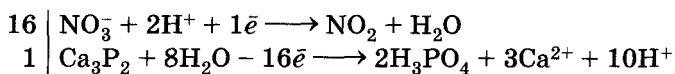
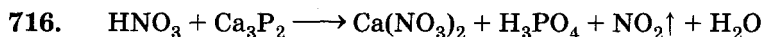
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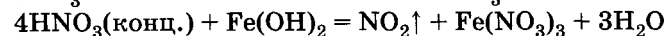
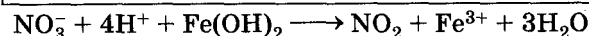
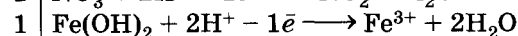
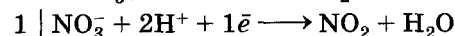
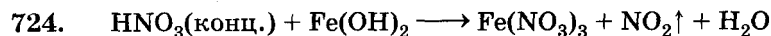
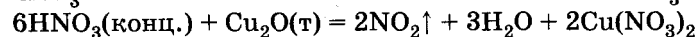
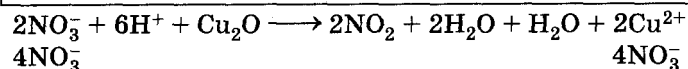
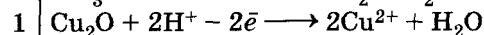
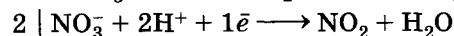
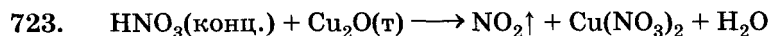
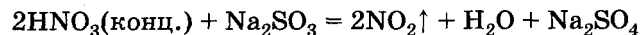
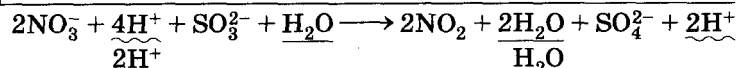
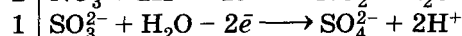
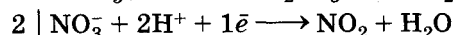
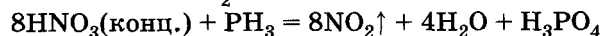
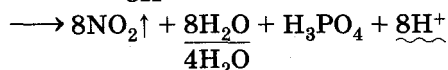
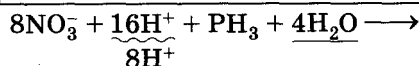
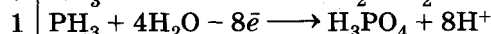
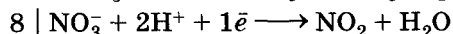
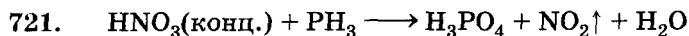
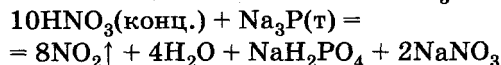
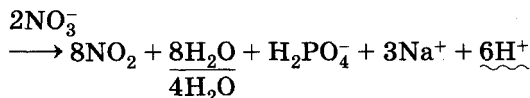
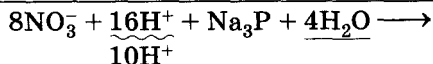
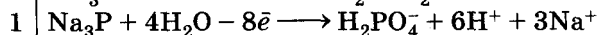
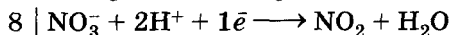
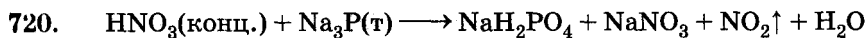


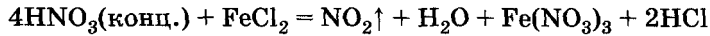
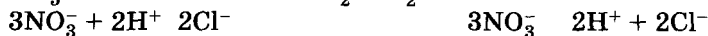
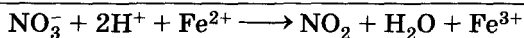
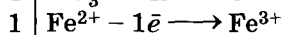
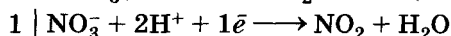
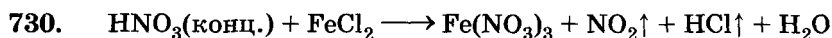
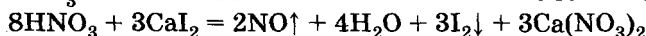
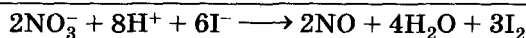
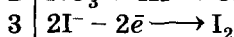
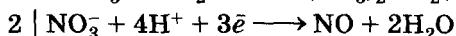
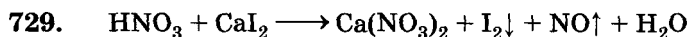
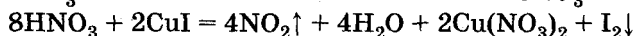
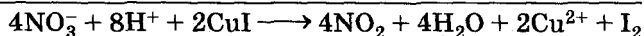
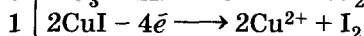
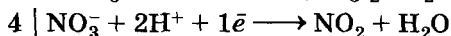
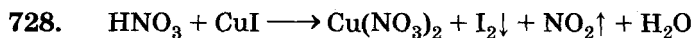
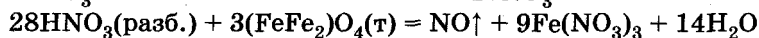
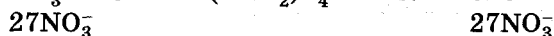
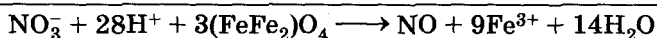
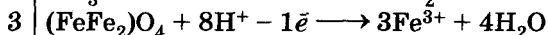
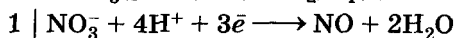
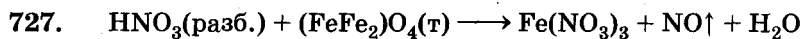
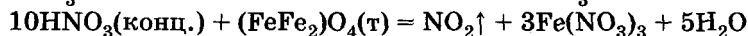
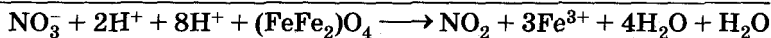
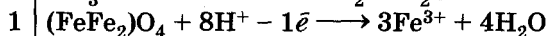
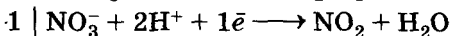
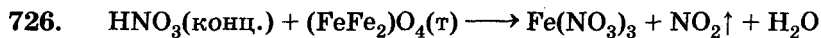
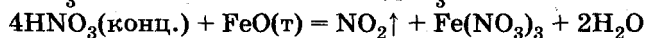
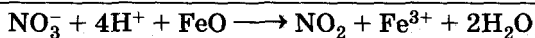
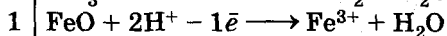
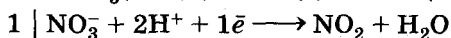
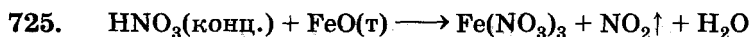


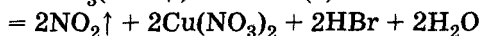
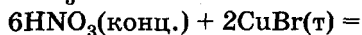
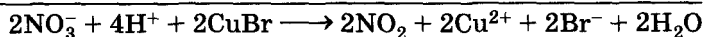
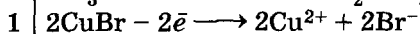
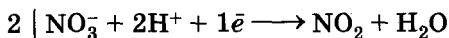
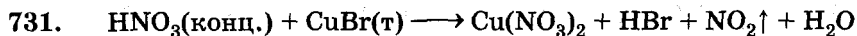




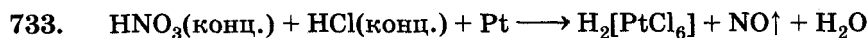
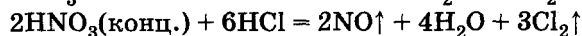
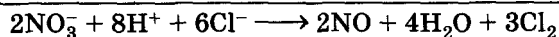
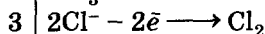
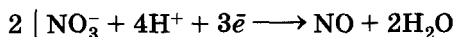




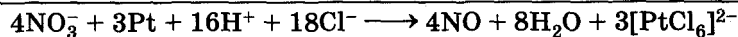
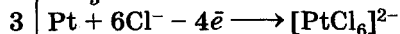
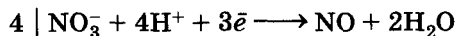




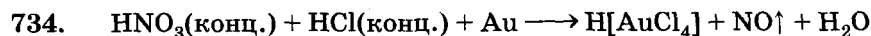
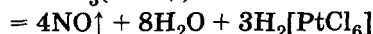
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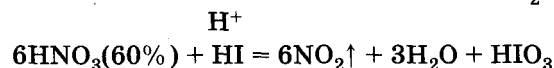
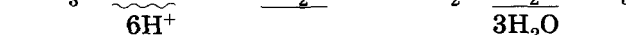
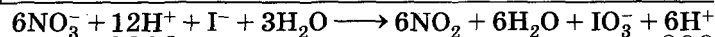
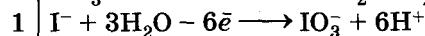
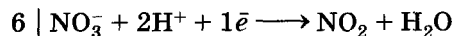
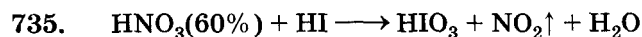
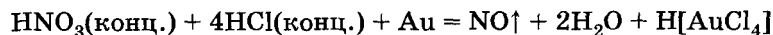
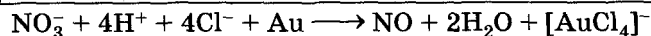
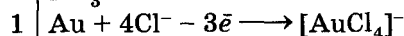
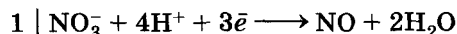


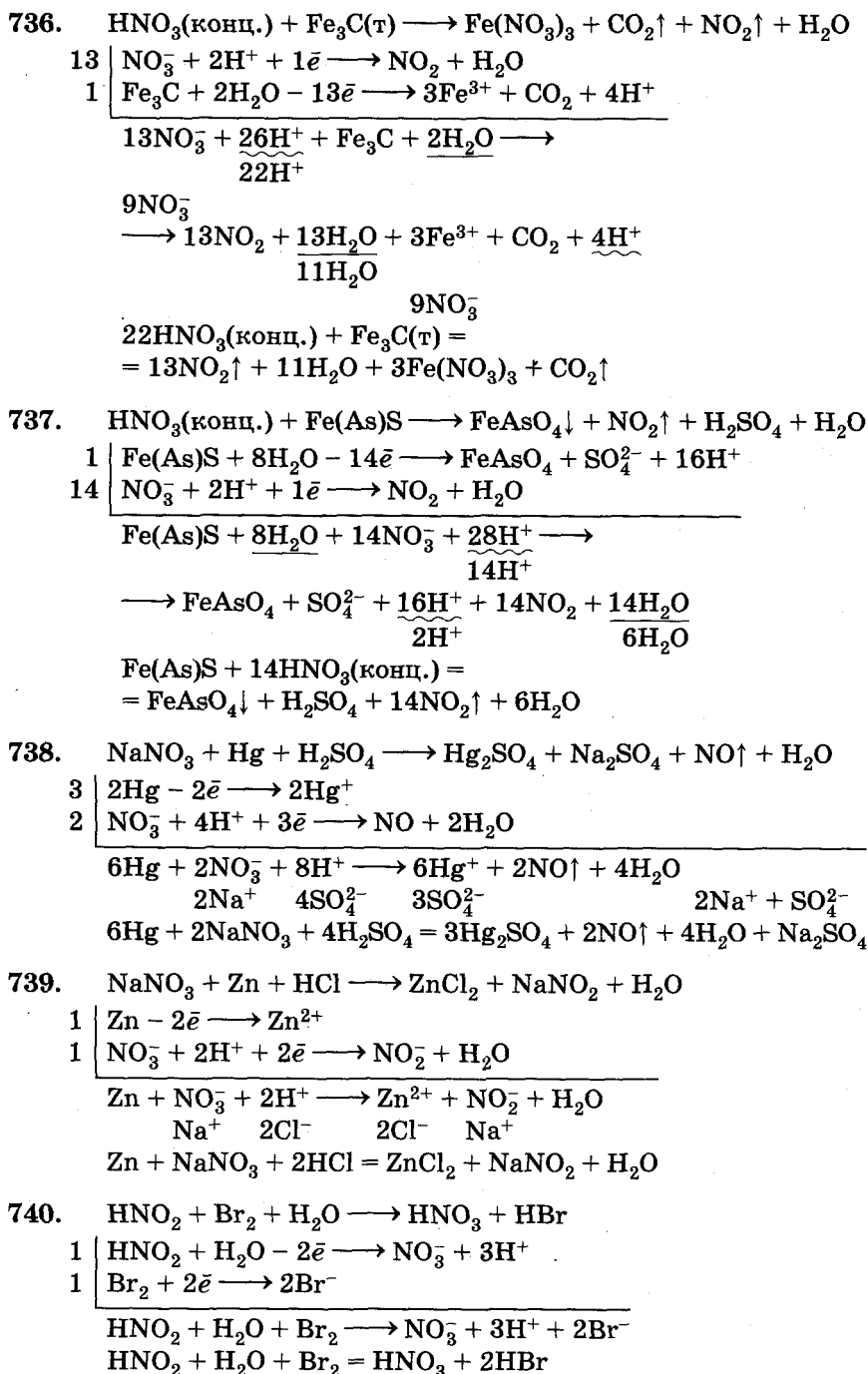
гексахлороплатинат (IV)

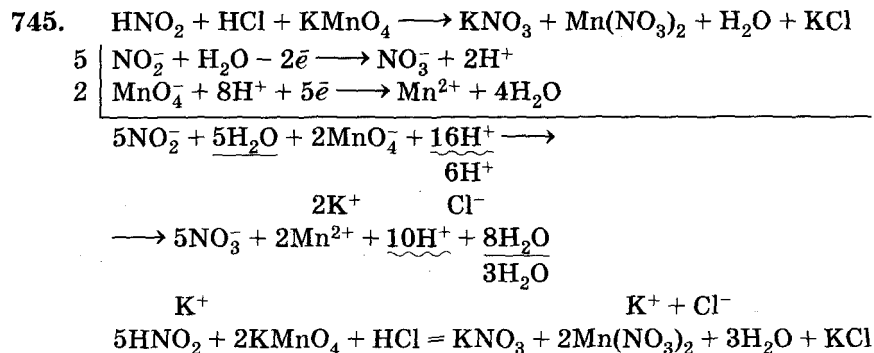
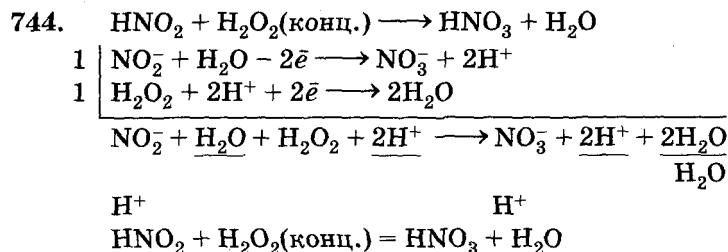
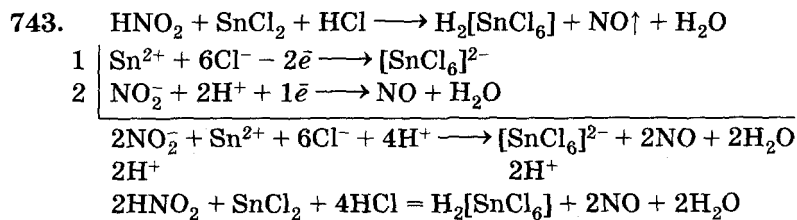
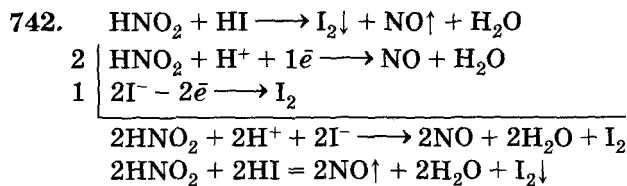
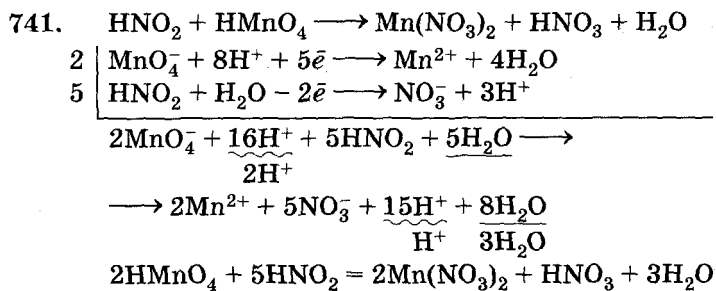


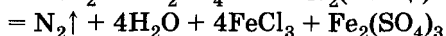
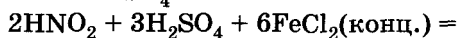
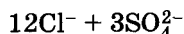
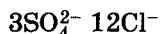
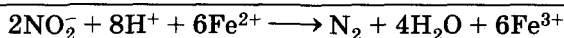
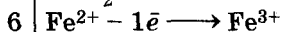
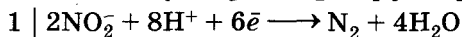
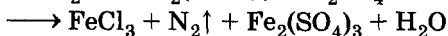
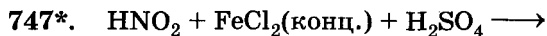
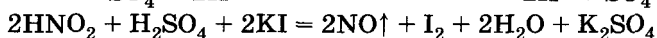
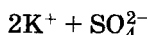
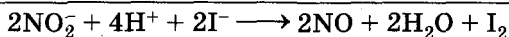
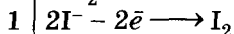
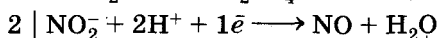
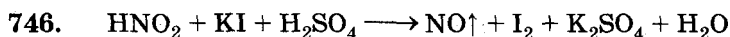
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тетрахлороаурат (III)

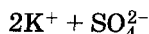
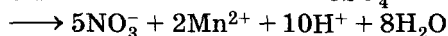
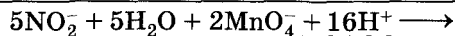
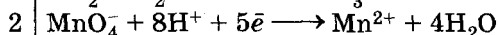
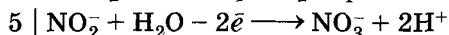
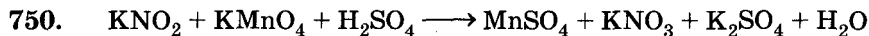
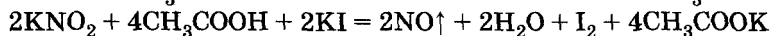
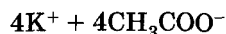
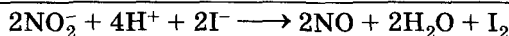
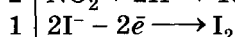
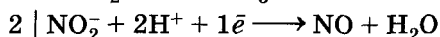
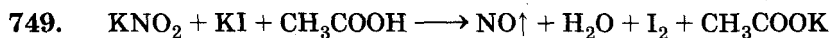
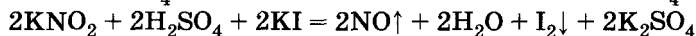
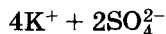
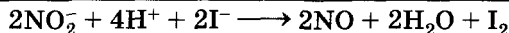
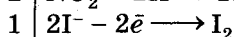
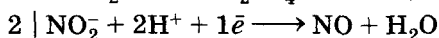
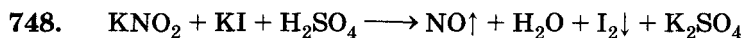


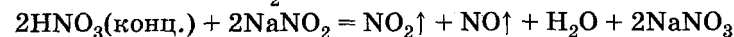
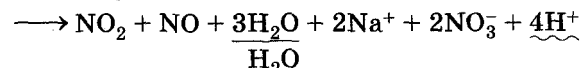
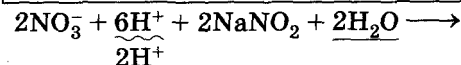
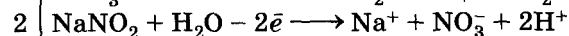
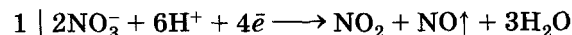
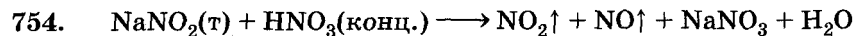
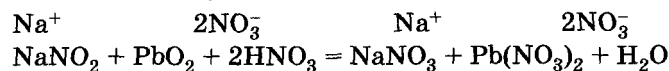
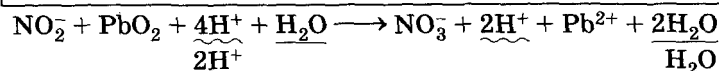
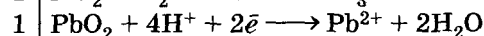
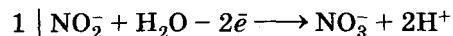
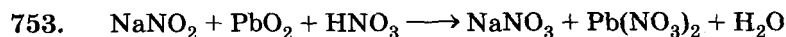
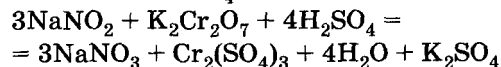
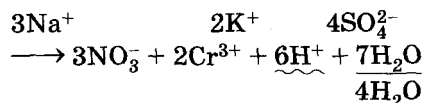
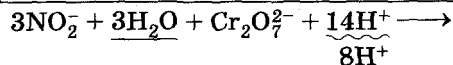
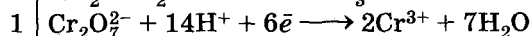
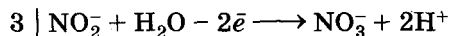
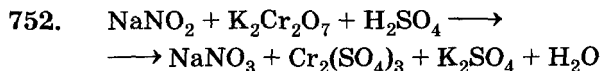
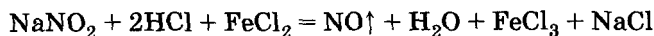
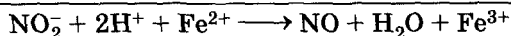
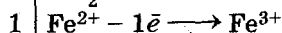
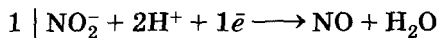
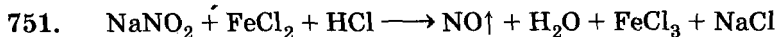
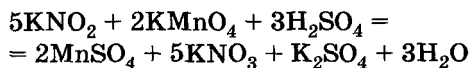


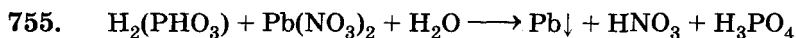




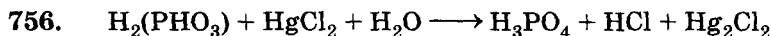
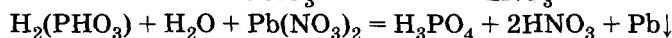
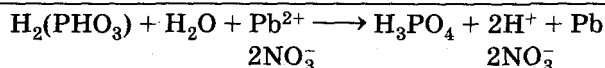
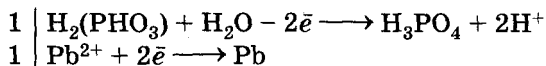
П о я с н е н и е. Распределение 6Fe^{3+} : $4\text{Fe}^{3+} + 12\text{Cl}^- = 4\text{FeCl}_3$, $2\text{Fe}^{3+} + 3\text{SO}_4^{2-} + \text{Fe}_2(\text{SO}_4)_3$.



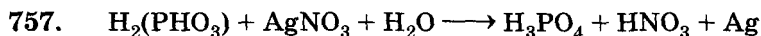
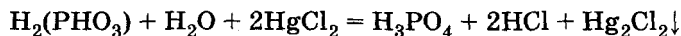
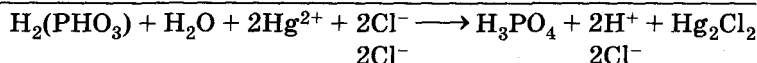
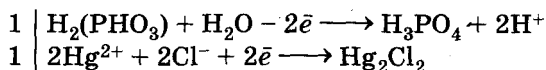




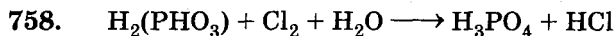
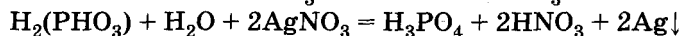
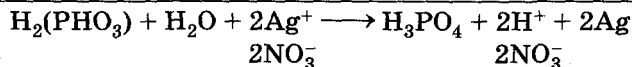
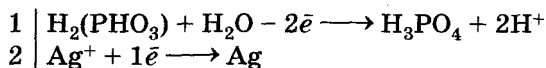
фосфористая кислота



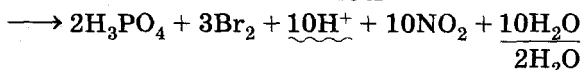
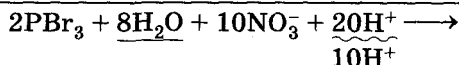
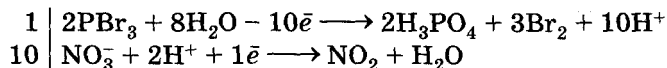
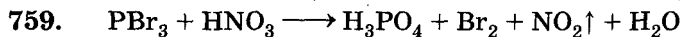
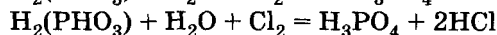
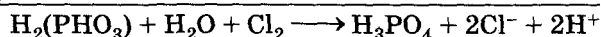
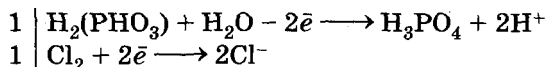
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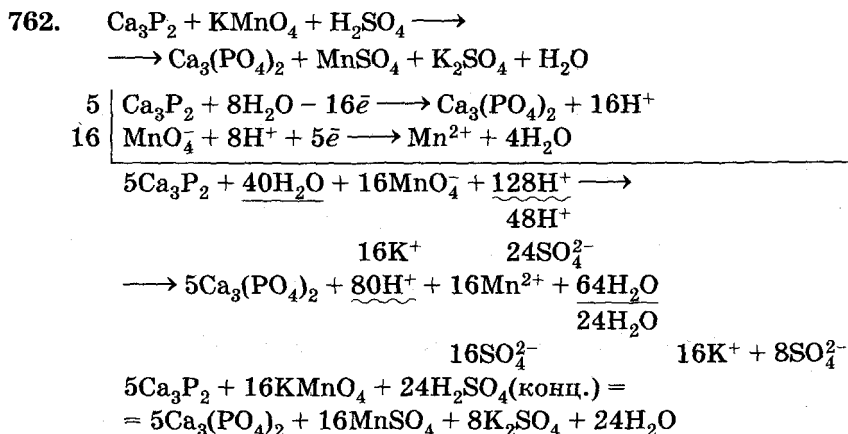
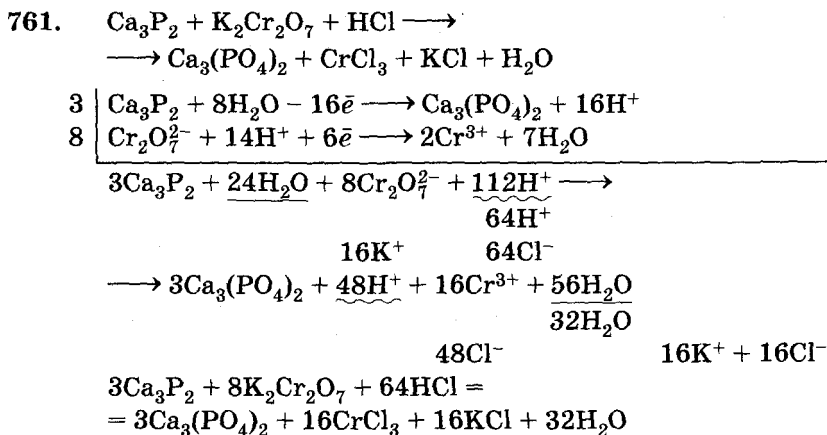
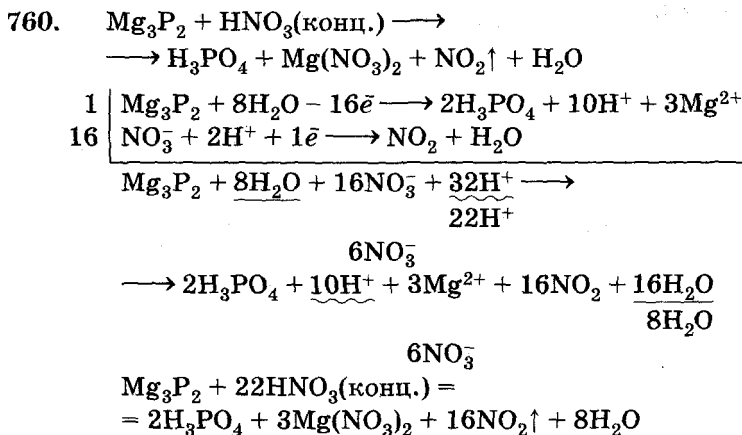


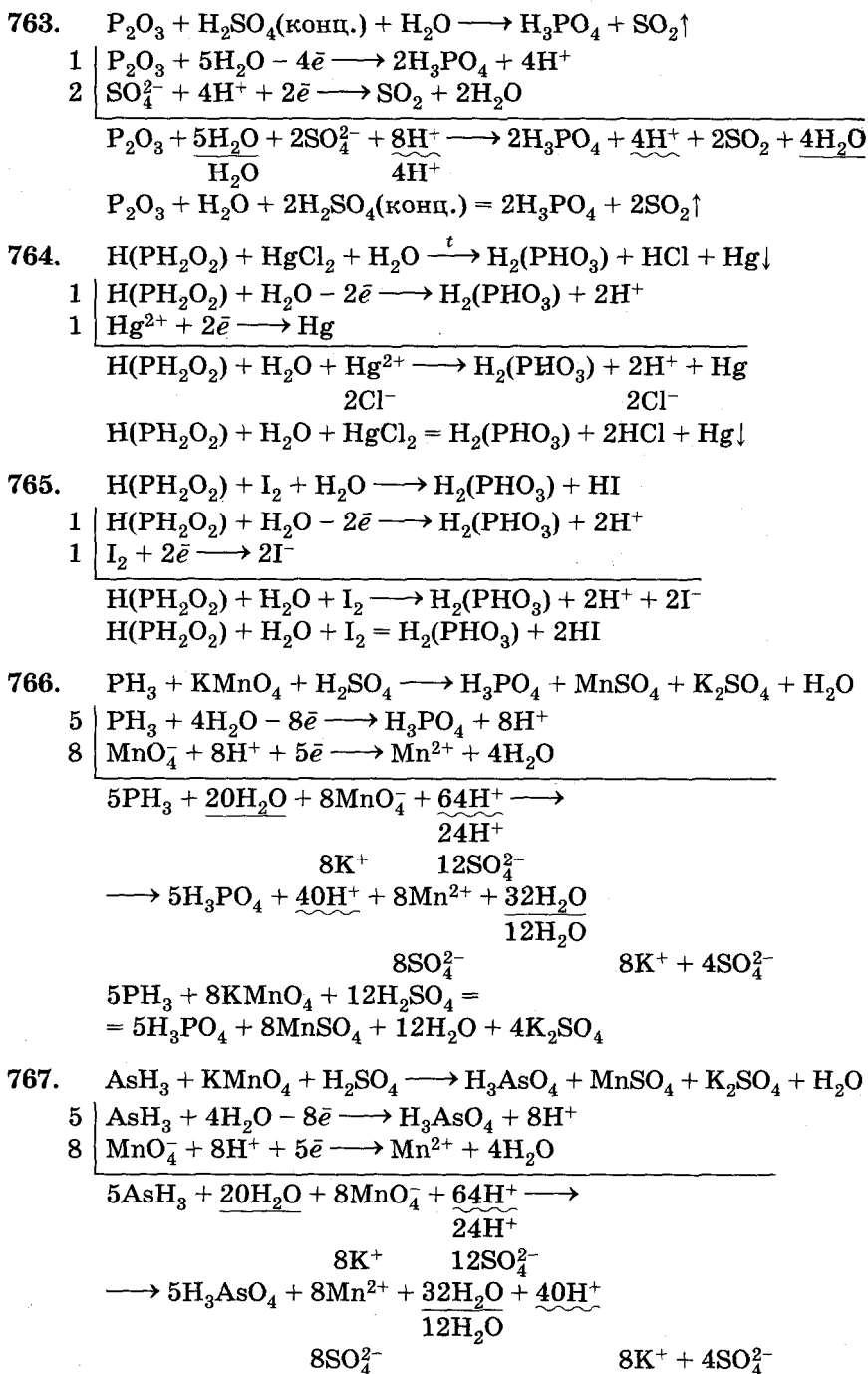
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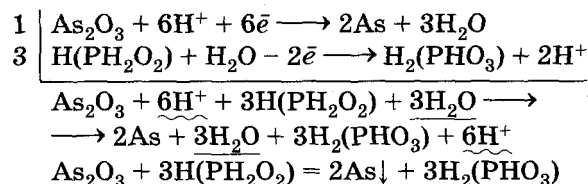
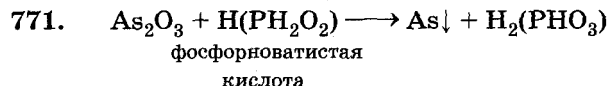
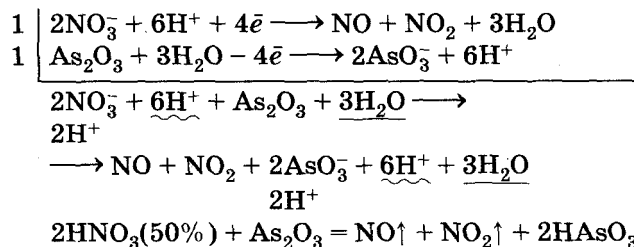
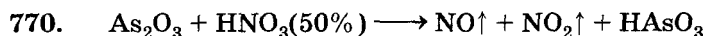
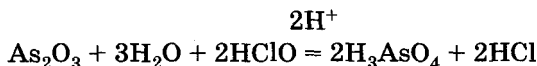
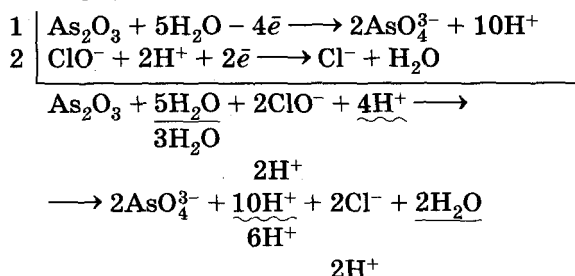
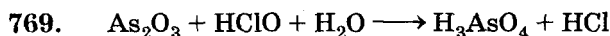
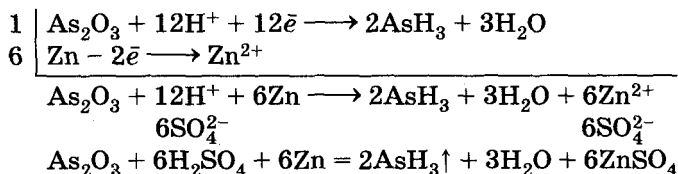
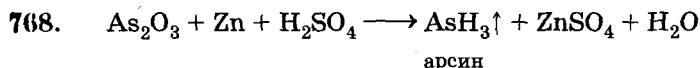
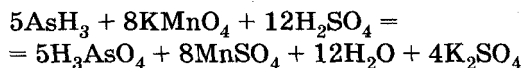


фосфористая кислота



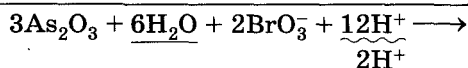
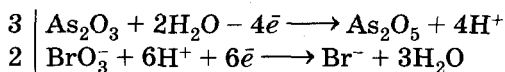








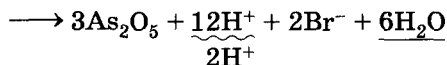
бромат калия



2H^+

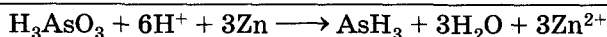
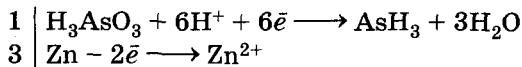
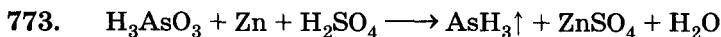
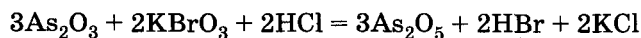
2K^+

2Cl^-



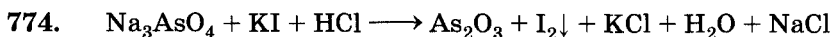
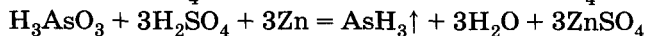
2H^+

$2\text{K}^+ + 6\text{Cl}^-$

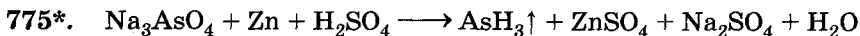
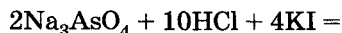
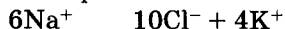
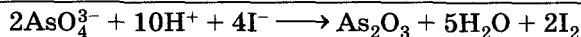
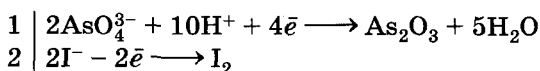


3SO_4^{2-}

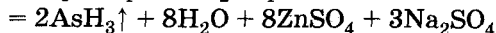
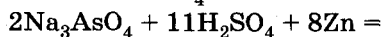
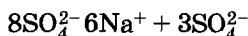
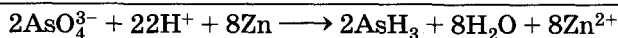
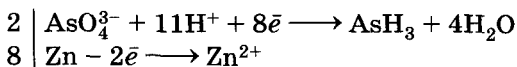
3SO_4^{2-}



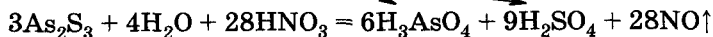
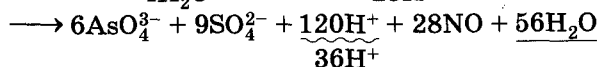
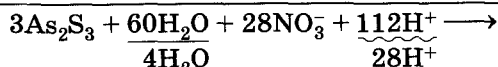
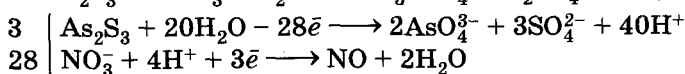
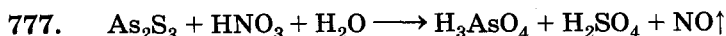
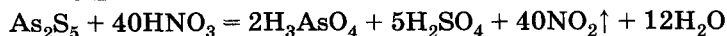
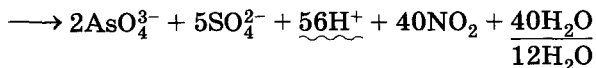
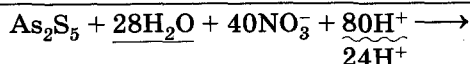
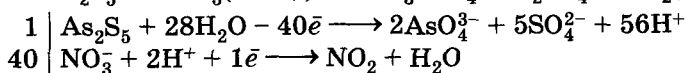
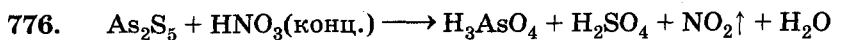
арсенат натрия



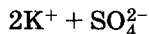
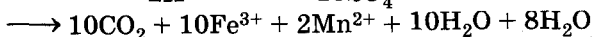
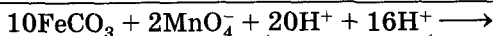
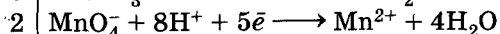
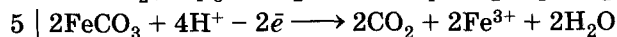
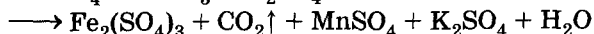
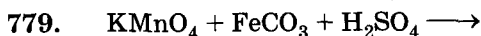
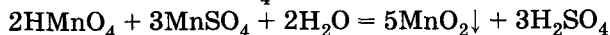
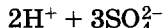
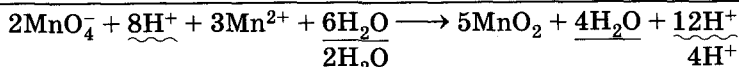
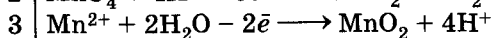
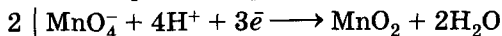
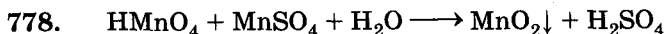
арсенат натрия

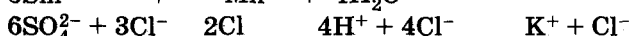
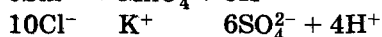
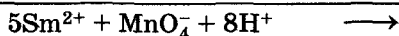
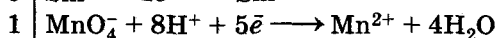
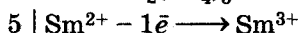
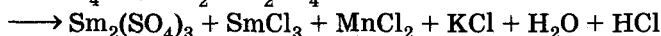
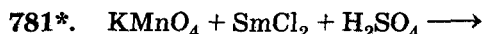
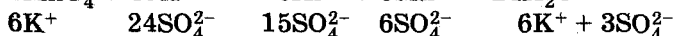
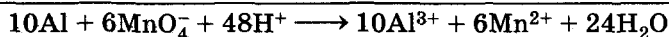
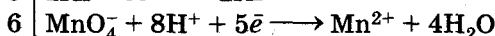
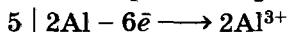
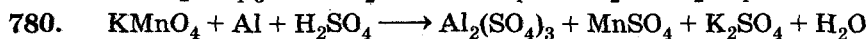
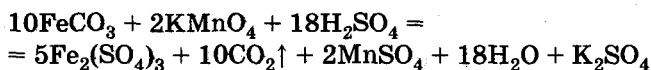


П о я с н е н и е. Если сократить множители 2 и 8, то перед H_2SO_4 окажется дробный коэффициент.

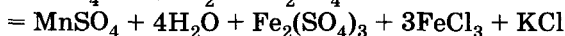
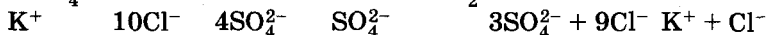
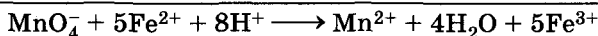
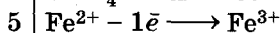
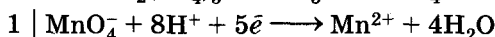
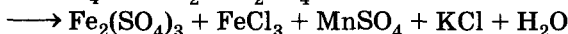
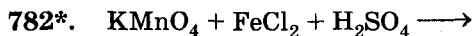
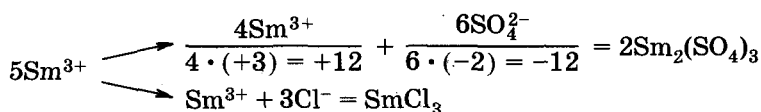


Соединения марганца (среда кислотная)

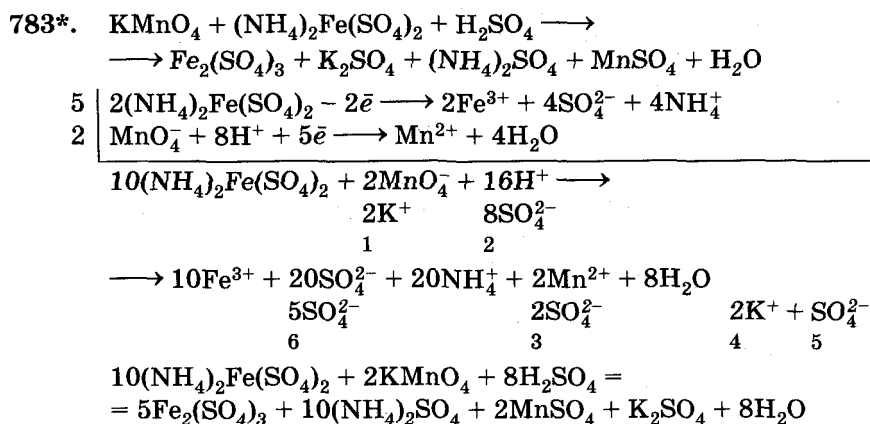




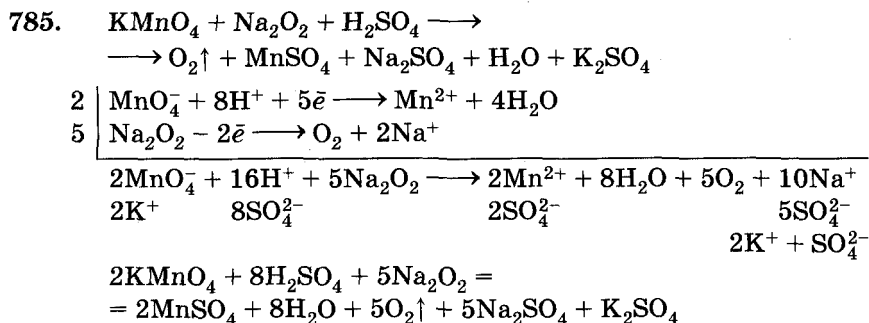
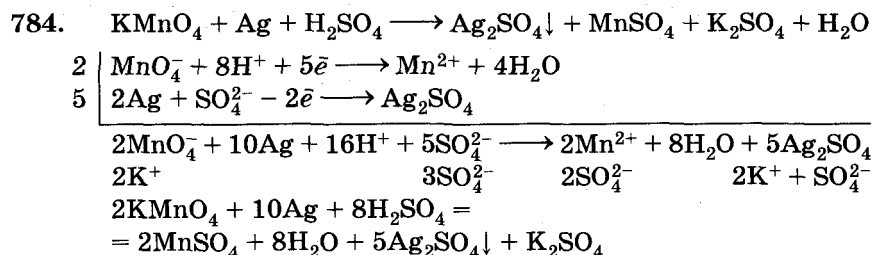
Пояснение. В молекулярном уравнении в продуктах реакции образуются $\text{Sm}_2(\text{SO}_4)_3$ и SmCl_3 . В ионном уравнении после стрелки 5Sm^{3+} , которые логично будет распределить как 4 : 1 (4Sm^{3+} и Sm^{3+}), т. е. $2\text{Sm}_2(\text{SO}_4)_3$ и SmCl_3 , другого варианта нет.

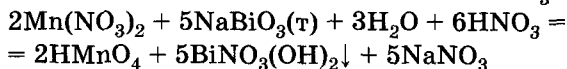
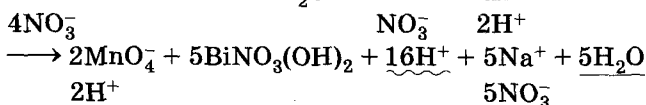
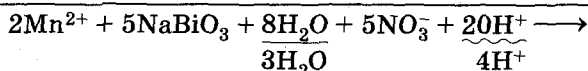
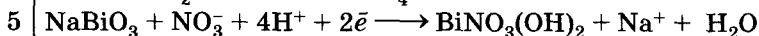
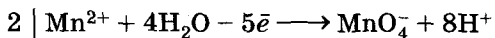
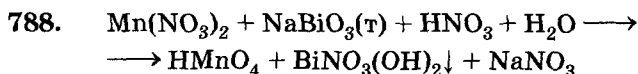
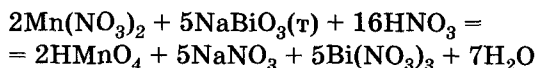
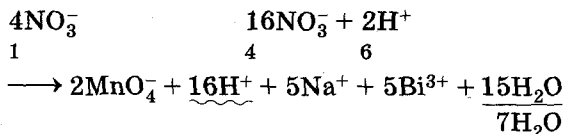
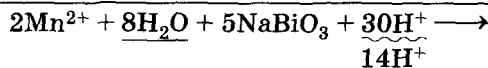
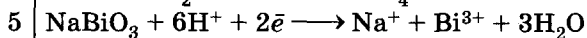
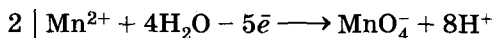
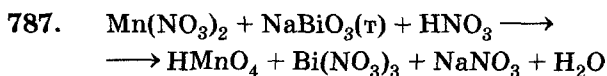
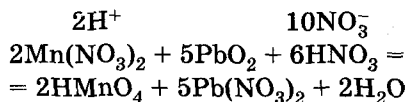
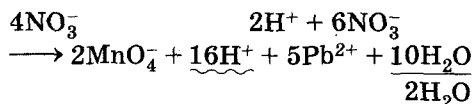
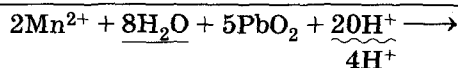
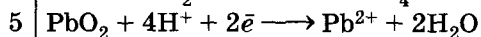
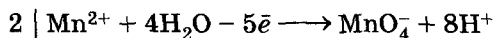
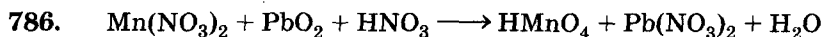


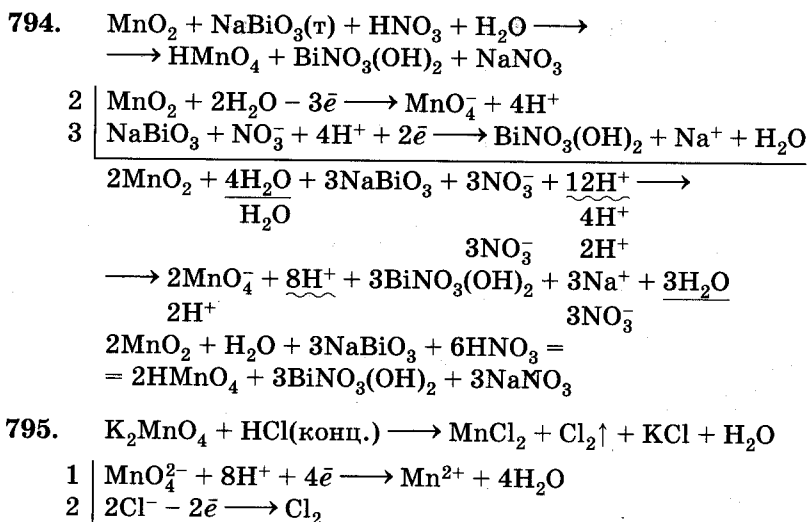
Пояснение. Распределяя 5Fe^{3+} между 3SO_4^{2-} и 9Cl^- , обратите внимание на заряды анионов. На 9Cl^- (заряд -9) потребуется 3Fe^{3+} (заряд $3 \cdot (+3) = +9$), а на 3SO_4^{2-} (заряд $3 \cdot (-2) = -6$) потребуется 2Fe^{3+} (заряд $2 \cdot (+3) = +6$).



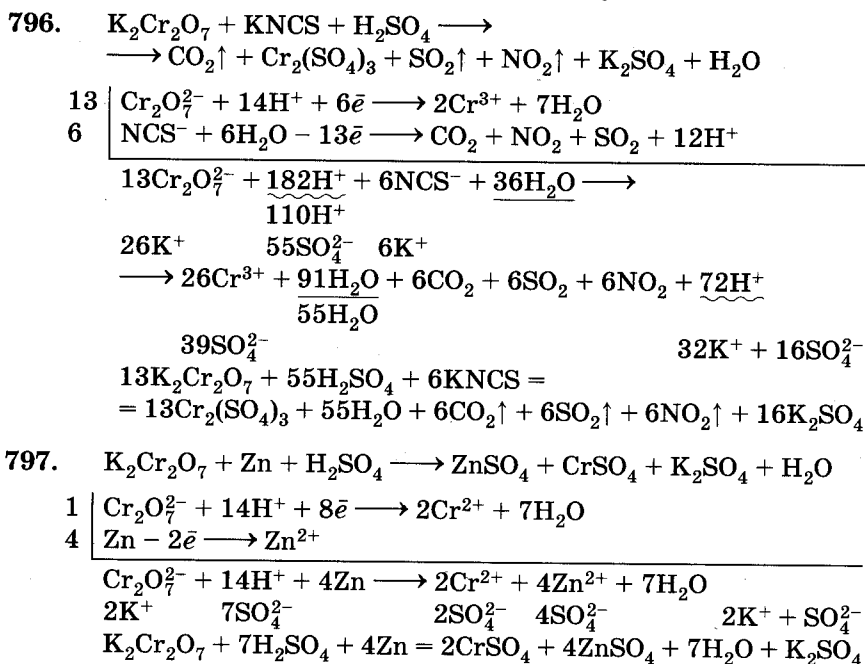
Пояснение. Распределение 25SO_4^{2-}
 $(20\text{SO}_4^{2-} + 5\text{SO}_4^{2-}): 15\text{SO}_4^{2-} + 10\text{Fe}^{3+} + 5\text{Fe}_2(\text{SO}_4)_3;$
 $10\text{SO}_4^{2-} + 20\text{NH}_4^+ = 10(\text{NH}_4)_2\text{SO}_4.$

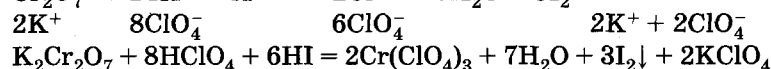
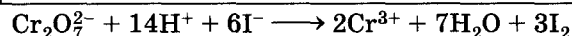
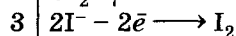
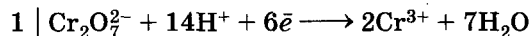
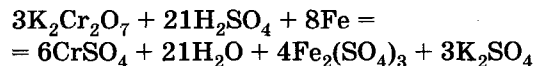
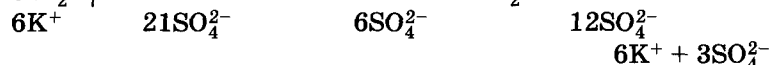
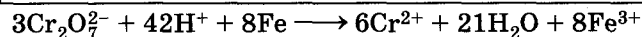
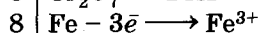
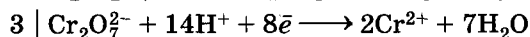
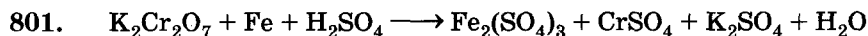
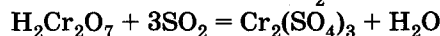
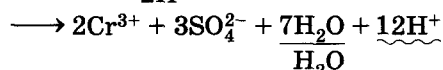
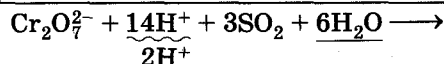
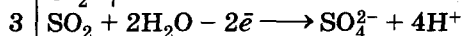
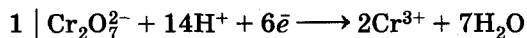
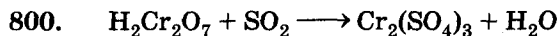
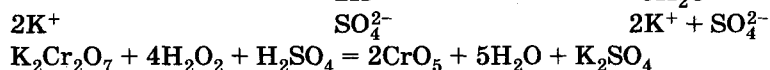
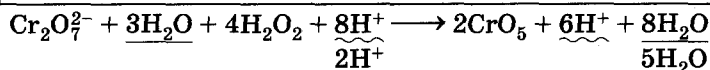
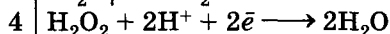
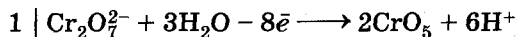
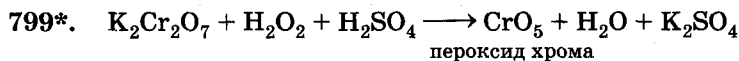
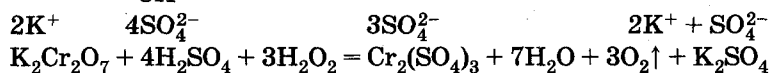
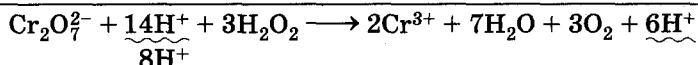
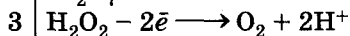
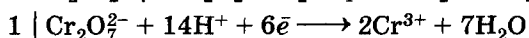
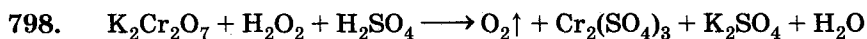


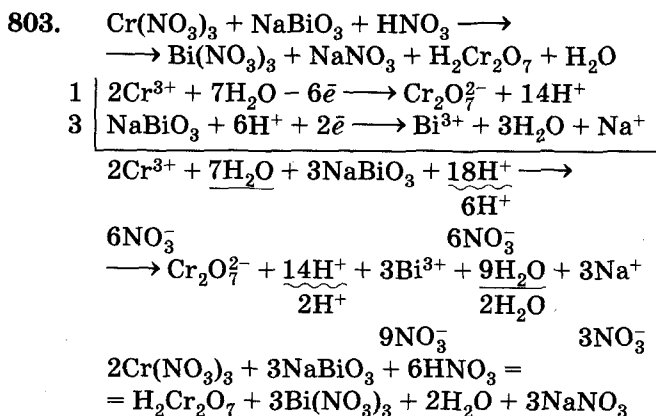




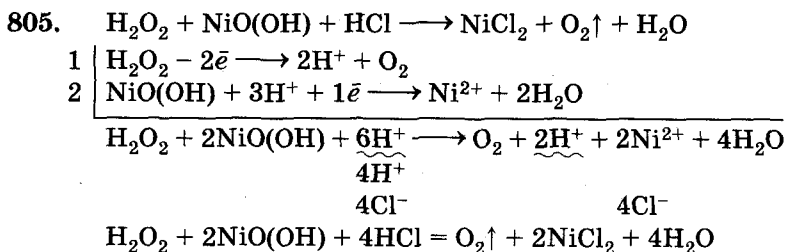
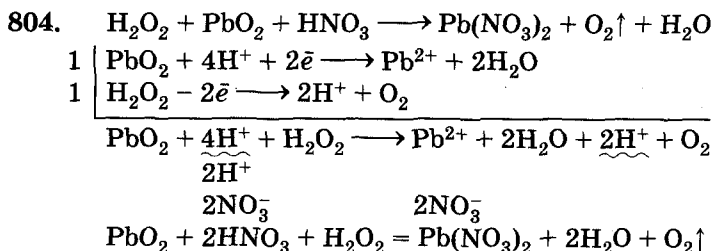
Соединения хрома (среда кислотная)



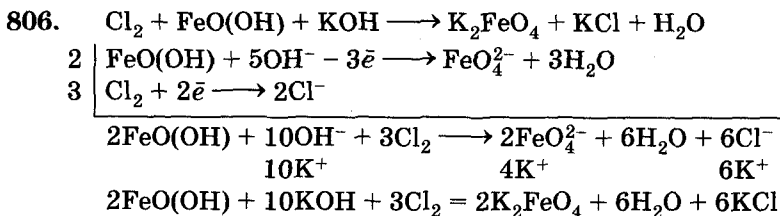




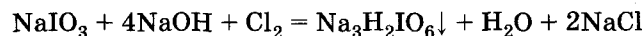
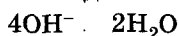
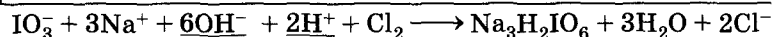
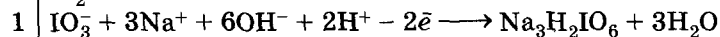
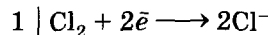
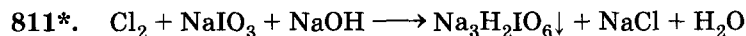
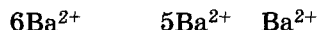
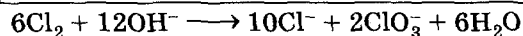
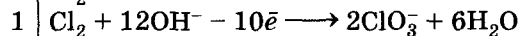
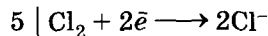
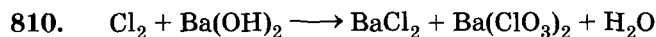
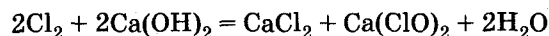
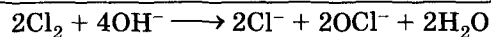
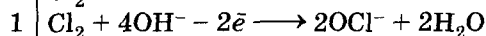
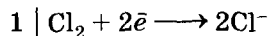
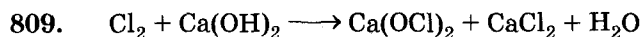
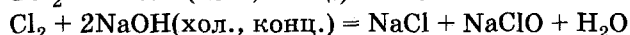
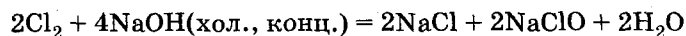
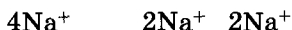
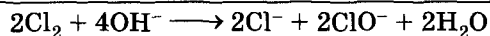
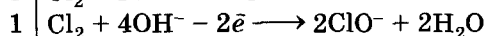
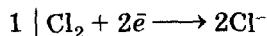
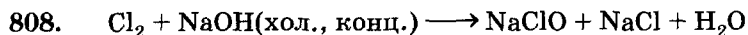
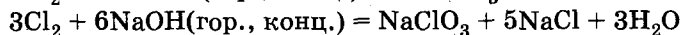
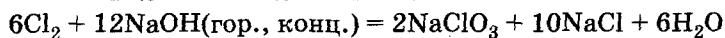
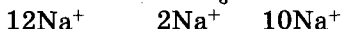
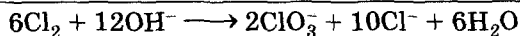
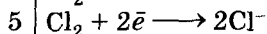
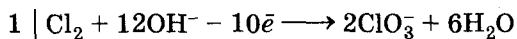
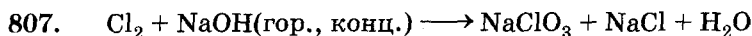
Пероксид водорода (среда кислотная)¹



Галогены и их соединения (среда щелочная)



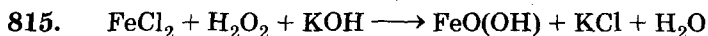
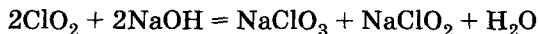
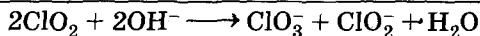
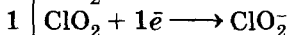
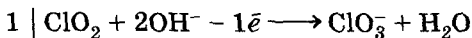
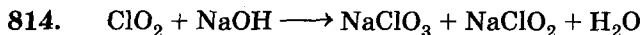
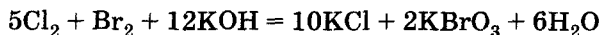
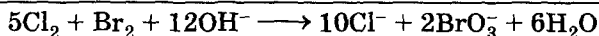
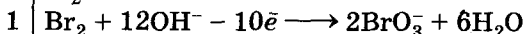
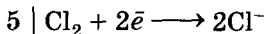
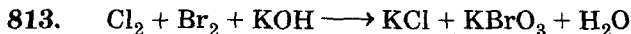
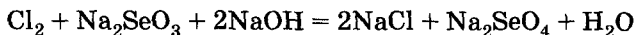
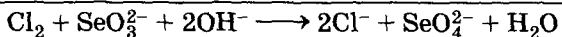
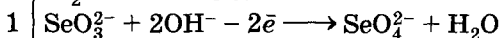
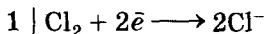
¹ См. также примеры 537, 541, 542, 560, 571, 602, 609, 615, 626, 666, 744, 785, 793, 798, 799.



П о я с н е н и е. $\text{Na}_3\text{H}_2\text{IO}_6$ — соль $\text{HIO}_4 \cdot 2\text{H}_2\text{O}(\text{H}_5\text{IO}_6)$.

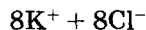
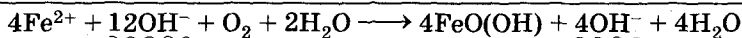
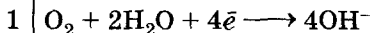
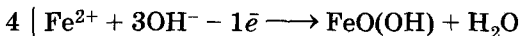
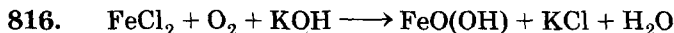
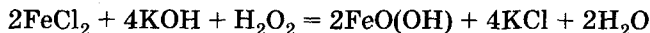
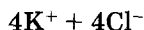
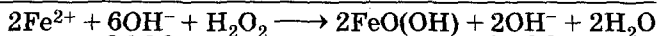
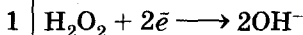
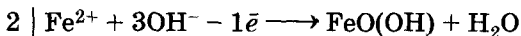


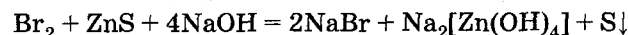
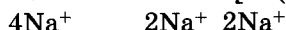
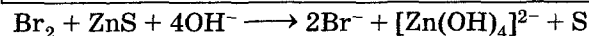
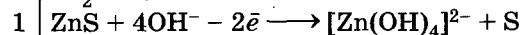
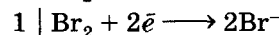
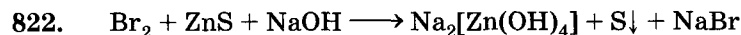
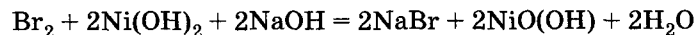
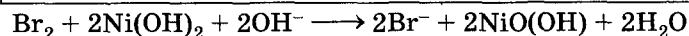
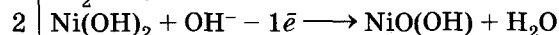
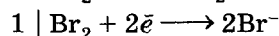
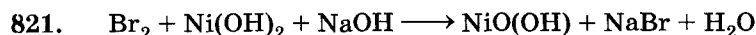
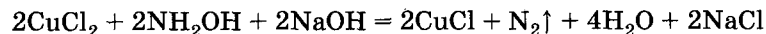
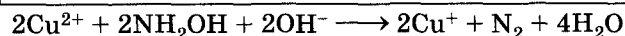
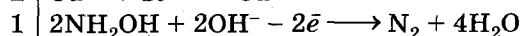
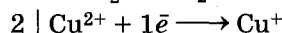
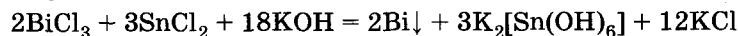
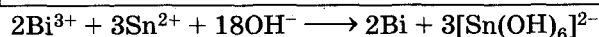
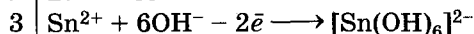
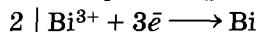
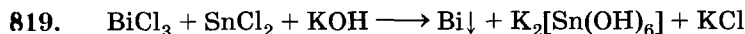
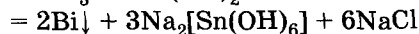
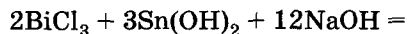
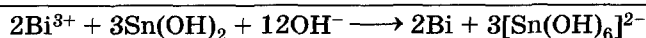
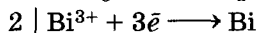
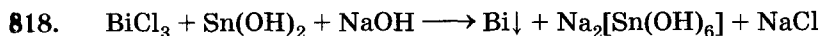
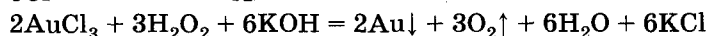
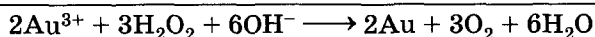
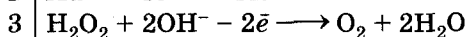
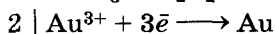
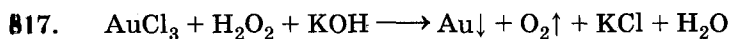
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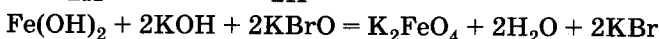
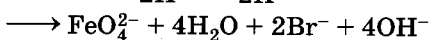
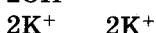
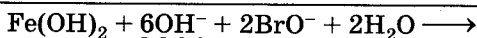
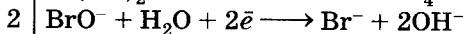
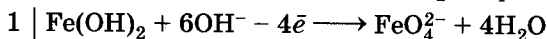
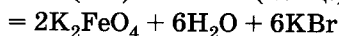
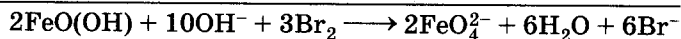
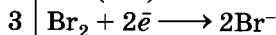
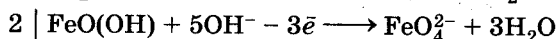
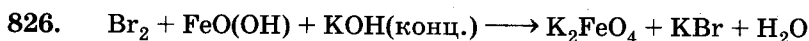
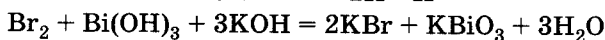
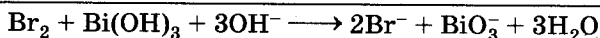
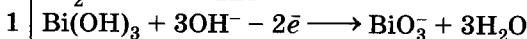
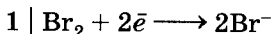
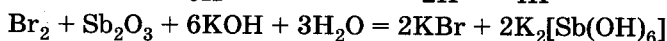
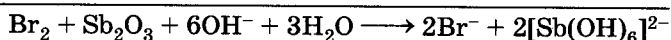
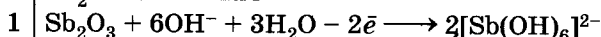
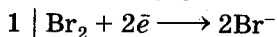
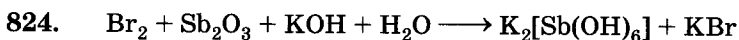
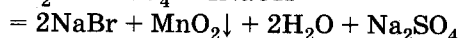
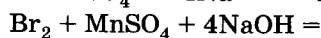
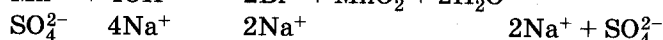
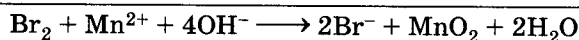
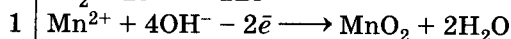
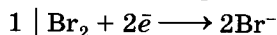
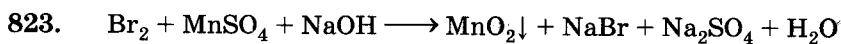


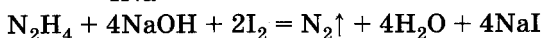
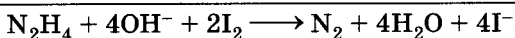
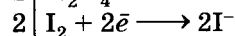
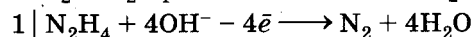
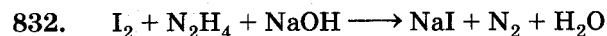
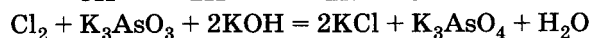
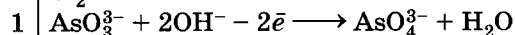
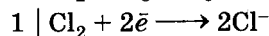
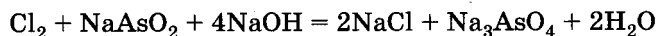
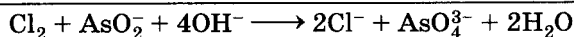
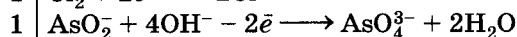
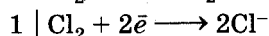
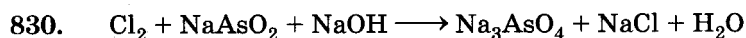
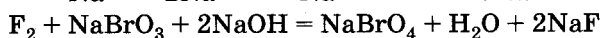
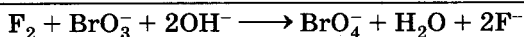
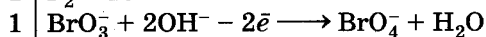
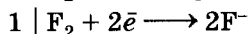
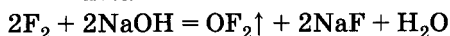
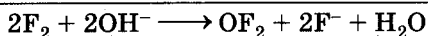
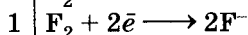
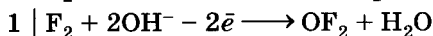
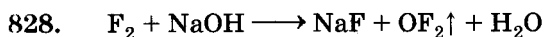
метагидроксид

железа (III)

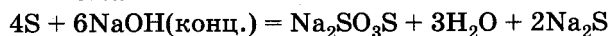
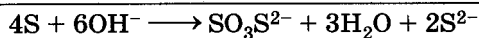
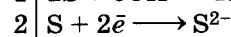
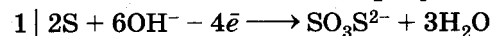
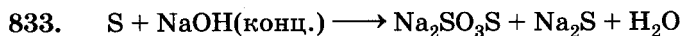


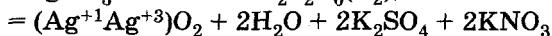
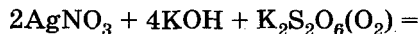
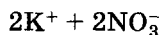
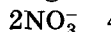
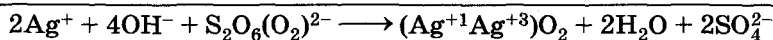
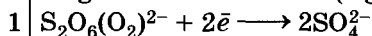
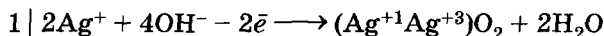
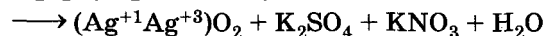
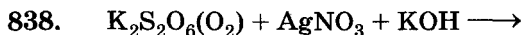
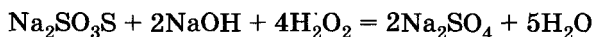
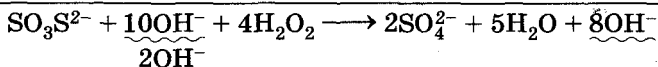
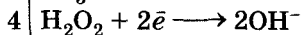
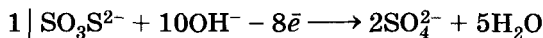
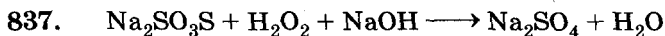
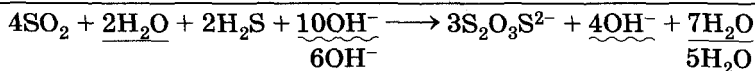
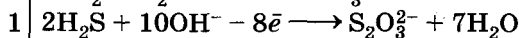
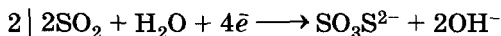
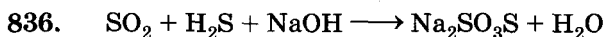
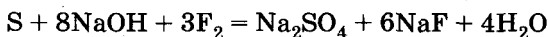
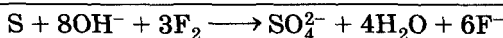
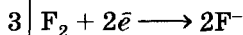
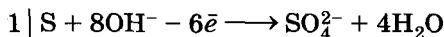
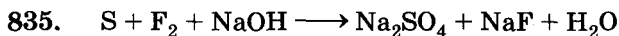
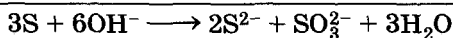
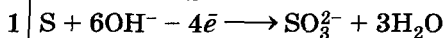
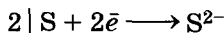
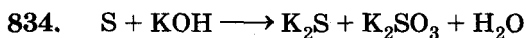


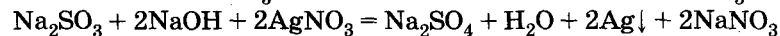
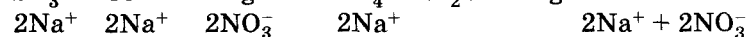
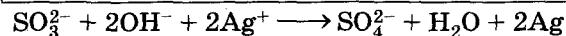
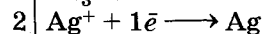
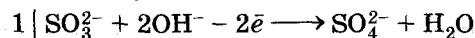
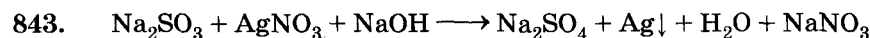
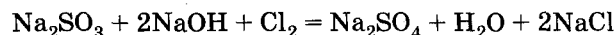
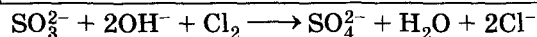
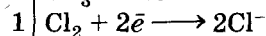
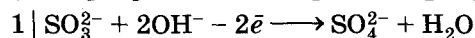
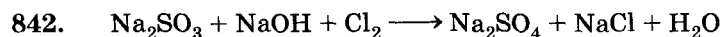
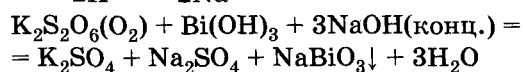
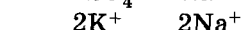
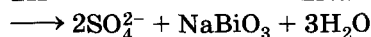
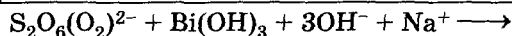
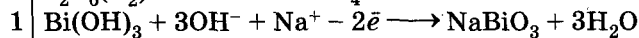
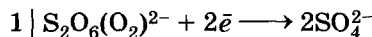
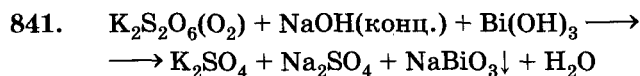
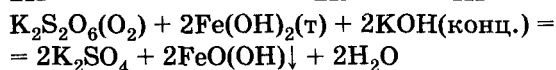
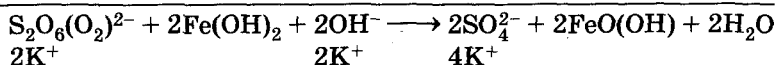
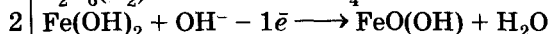
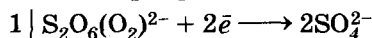
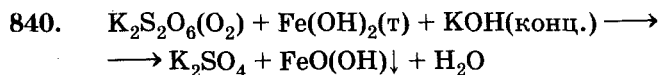
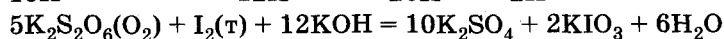
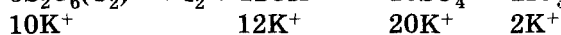
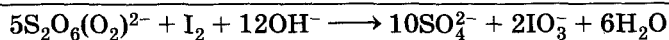
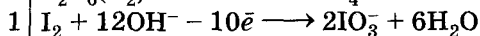
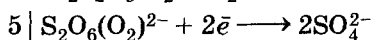


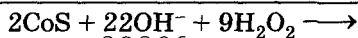
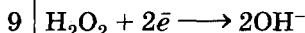
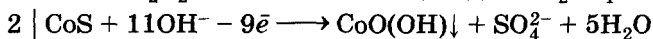
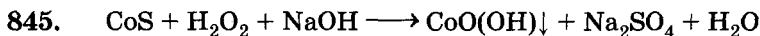
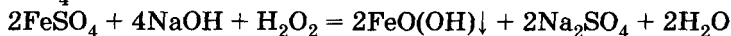
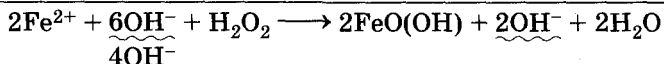
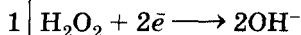
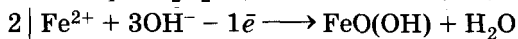
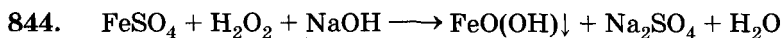


Серa и ее соединения (среда щелочная)

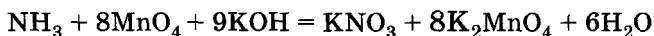
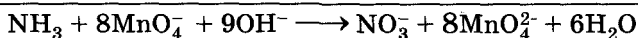
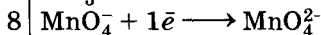
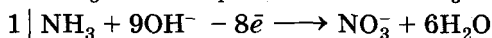
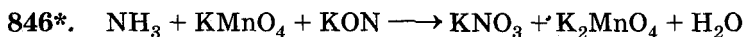




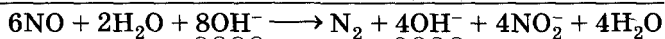
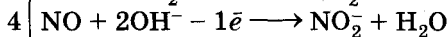
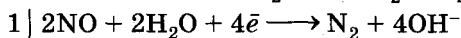
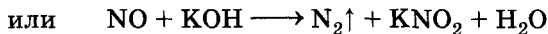
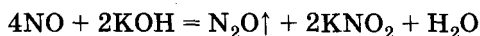
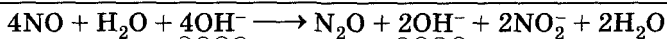
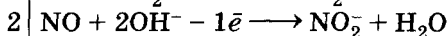
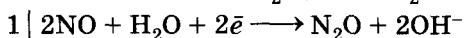
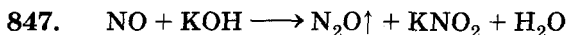


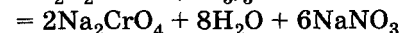
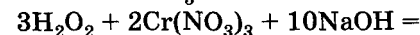
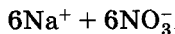
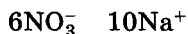
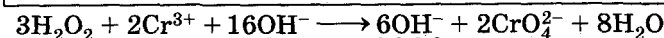
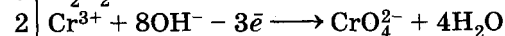
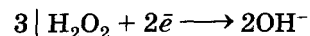
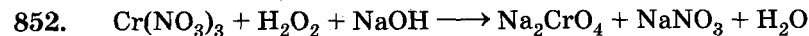
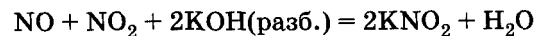
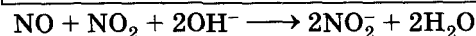
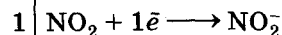
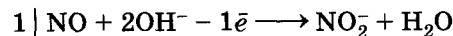
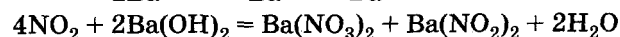
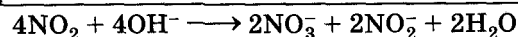
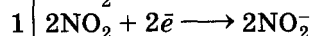
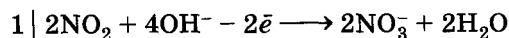
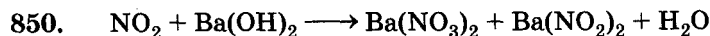
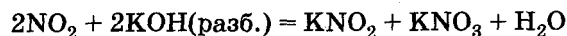
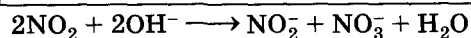
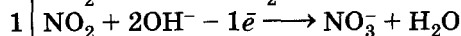
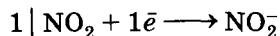
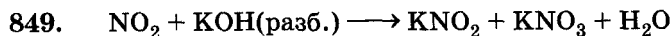
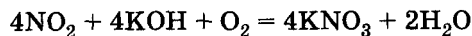
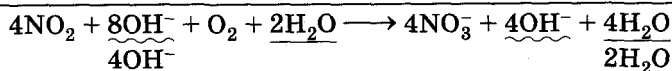
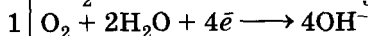
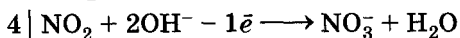
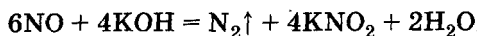


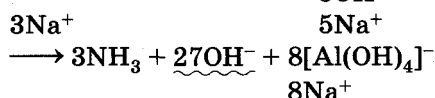
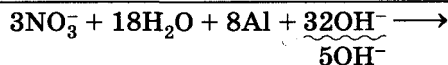
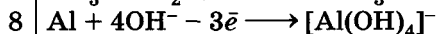
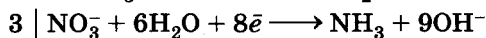
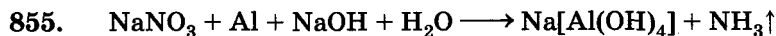
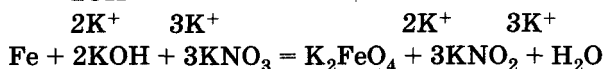
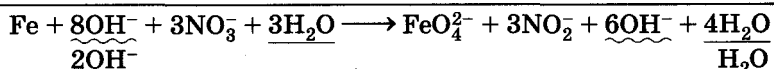
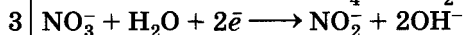
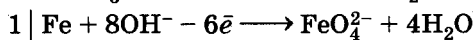
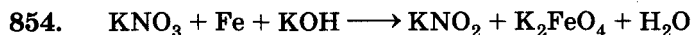
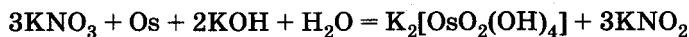
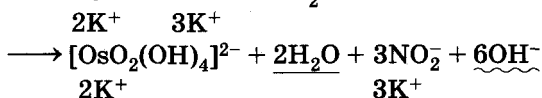
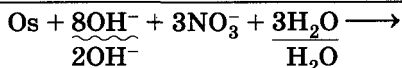
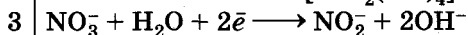
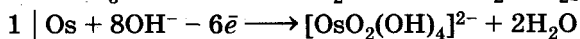
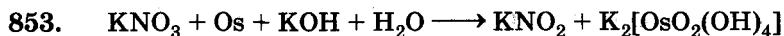
Соединения азота и фосфора (среда щелочная)



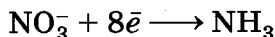
П о я с н е н и е. См. решение задачи 855.







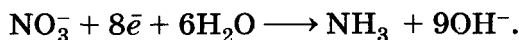
П о я с н е н и е. Проблема «лишнего» водорода в 1-й полуреакции: $\text{N}^{+5}\text{O}_3 \longrightarrow \text{N}^{-3}\text{H}_3$. По одному способу число молекул воды и гидроксид-ионов можно рассчитать по изменению степени окисления азота. Для перехода N^{+5} в N^{-3} следует добавить $8\bar{e}$ в левую часть схемы:



Заряд (-9)

Заряд 0

Уравниваются заряды добавлением пары $\text{H}_2\text{O}-\text{OH}^-$:

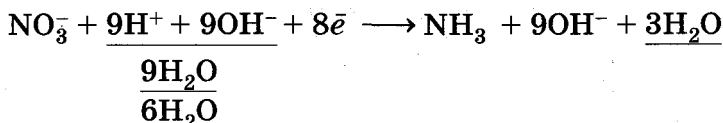


Заряд (-9)

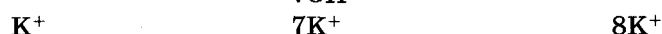
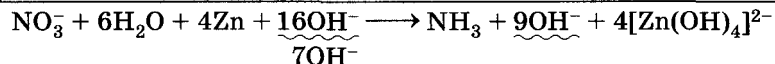
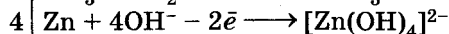
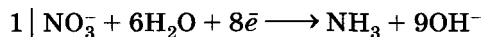
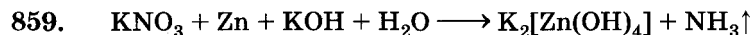
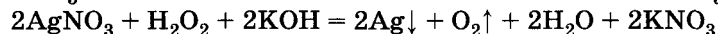
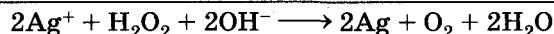
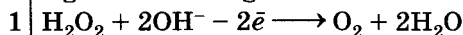
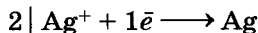
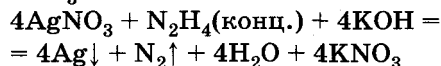
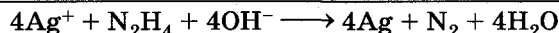
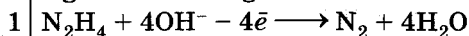
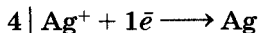
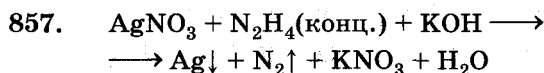
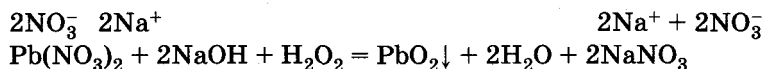
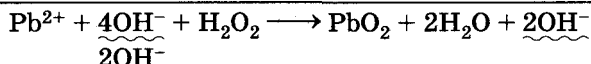
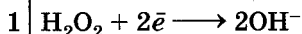
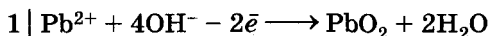
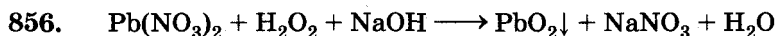
Заряд (-9)

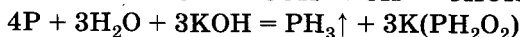
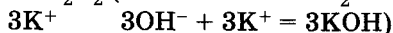
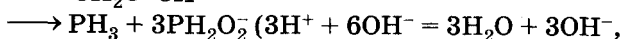
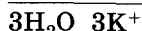
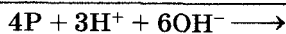
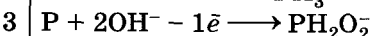
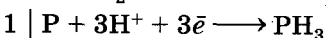
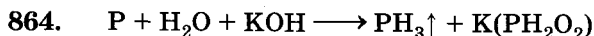
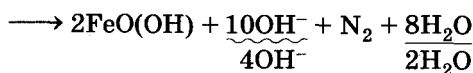
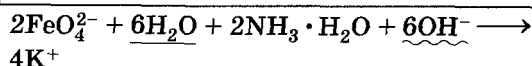
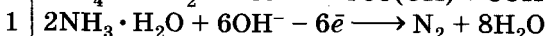
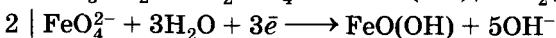
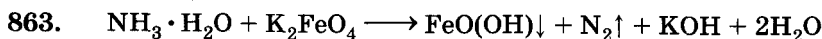
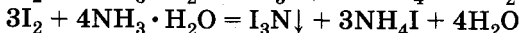
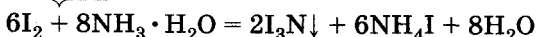
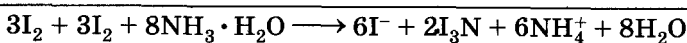
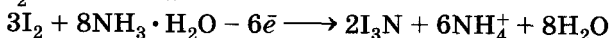
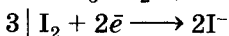
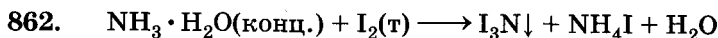
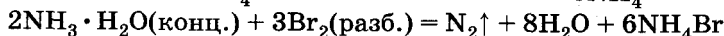
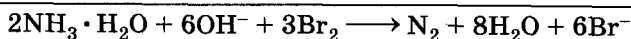
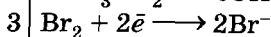
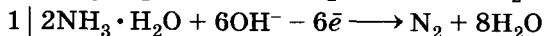
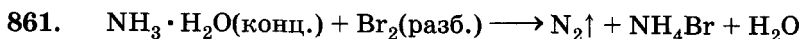
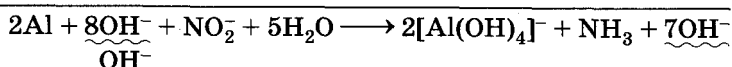
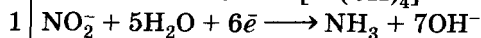
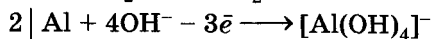
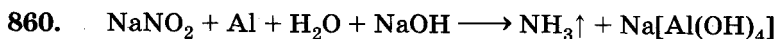
По другому способу 1-ю полуреакцию составляют как бы для кислотной среды: $\text{NO}_3^- + 9\text{H}^+ + 8\bar{e} \longrightarrow \text{NH}_3 + 3\text{H}_2\text{O}$.

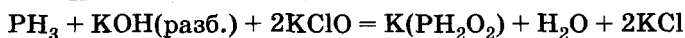
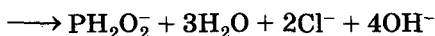
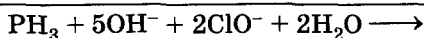
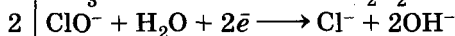
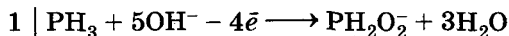
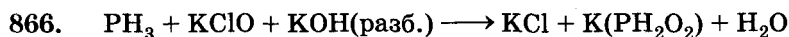
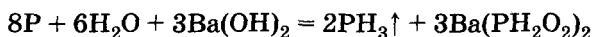
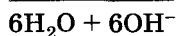
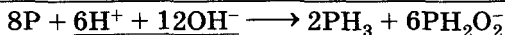
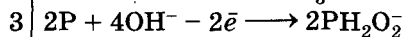
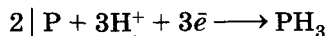
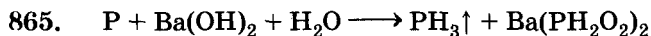
Теперь, чтобы нейтрализовать 9H^+ , добавим 9OH^- в левую часть и столько же в правую, для равенства зарядов.



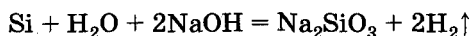
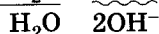
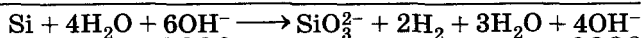
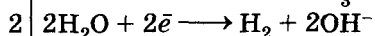
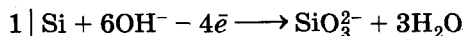
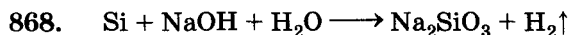
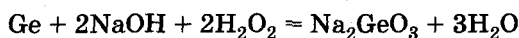
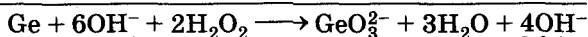
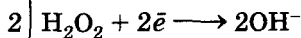
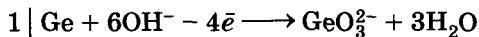
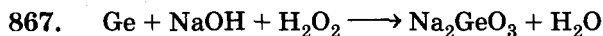
После сокращений: $\text{NO}_3^- + 6\text{H}_2\text{O} + 8\bar{e} \longrightarrow \text{NH}_3 + 9\text{OH}^-$.

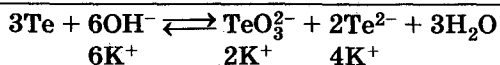
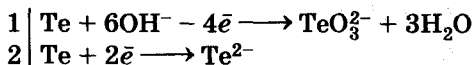




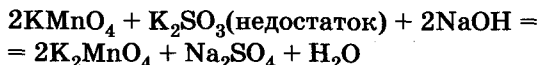
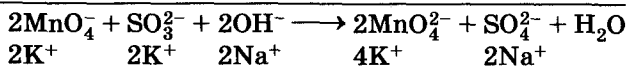
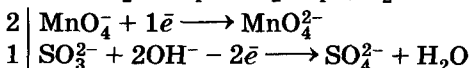
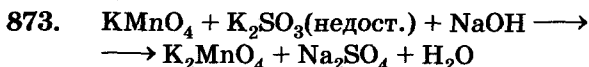
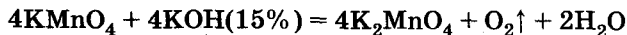
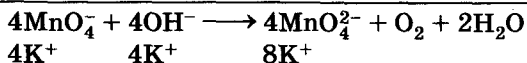
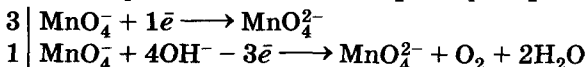
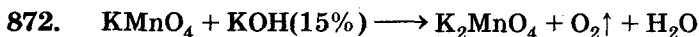
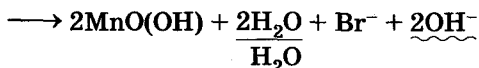
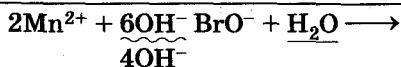
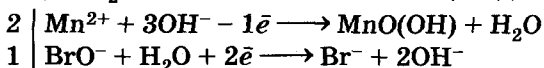
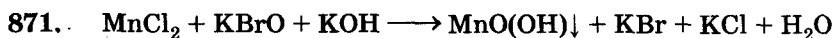
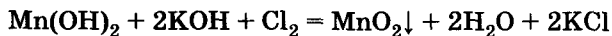
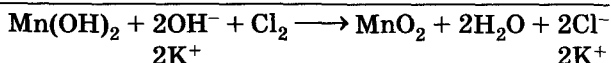
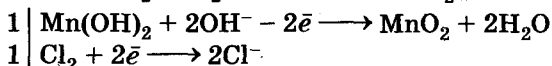


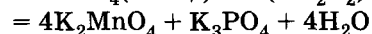
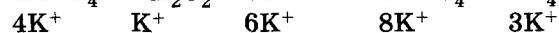
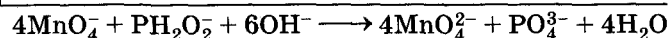
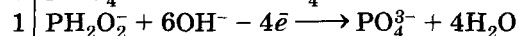
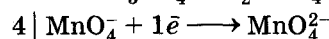
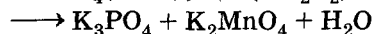
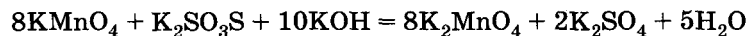
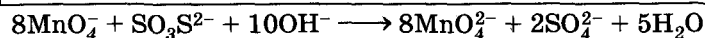
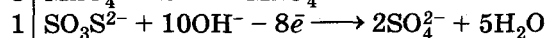
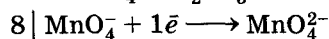
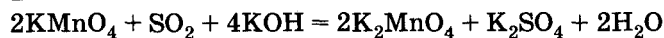
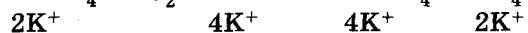
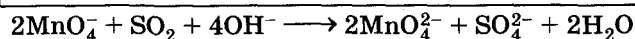
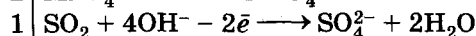
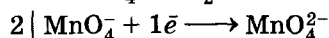
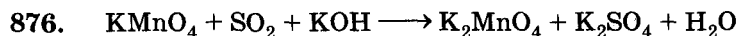
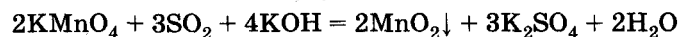
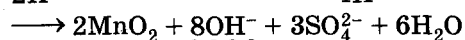
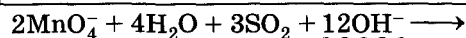
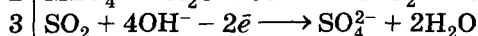
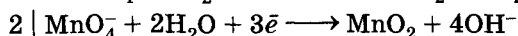
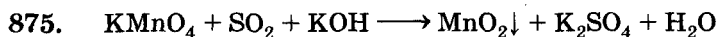
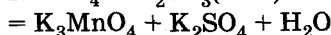
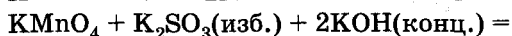
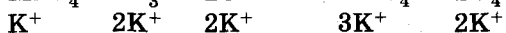
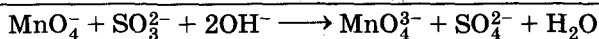
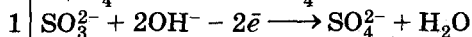
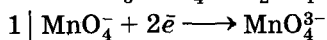
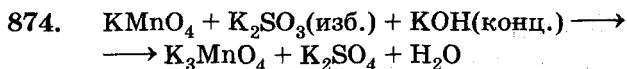
Соединения германия, кремния и теллура (среда щелочная)

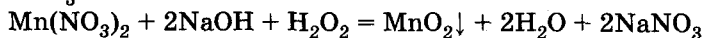
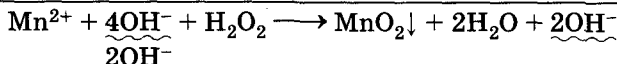
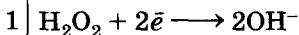
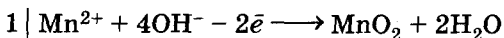
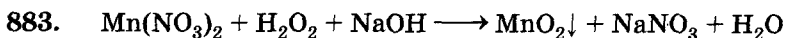
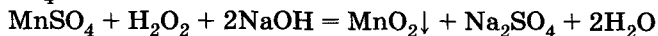
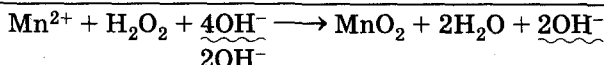
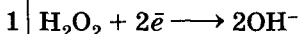
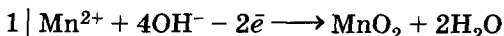
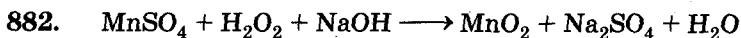
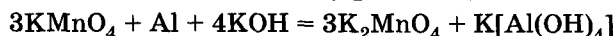
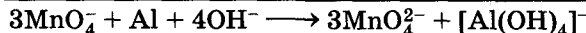
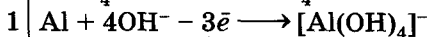
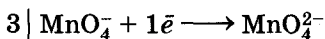
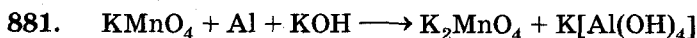
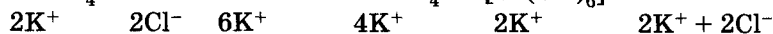
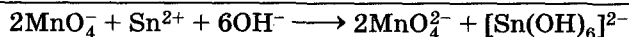
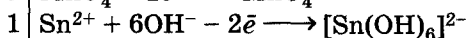
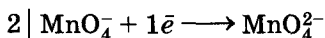
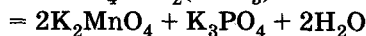
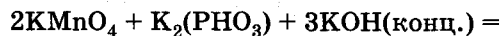
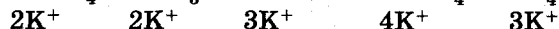
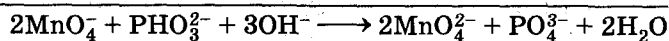
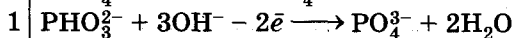
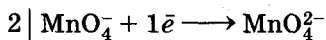
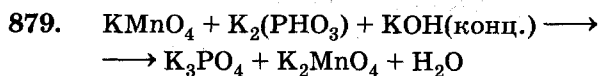




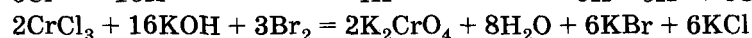
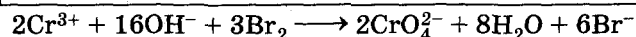
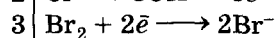
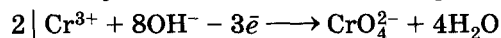
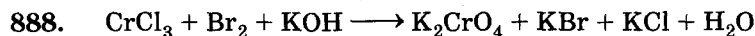
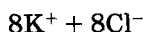
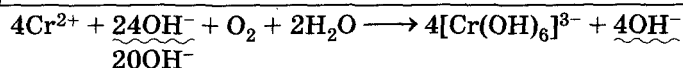
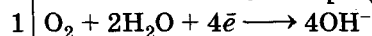
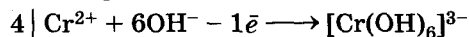
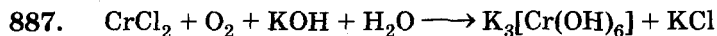
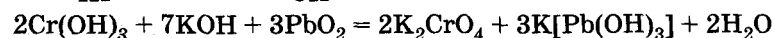
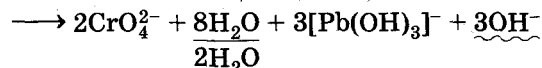
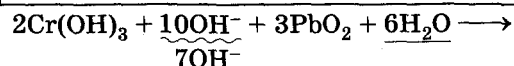
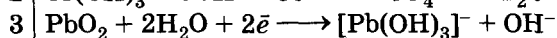
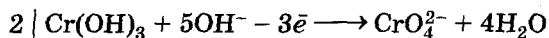
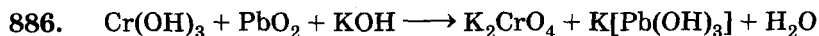
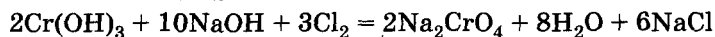
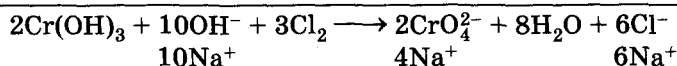
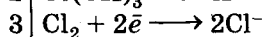
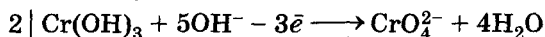
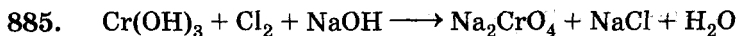
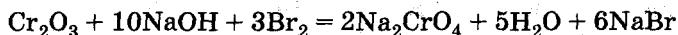
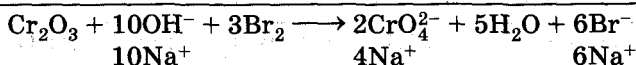
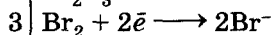
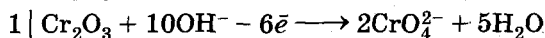
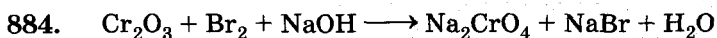
Соединения марганца (среда щелочная)

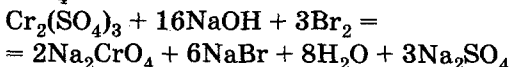
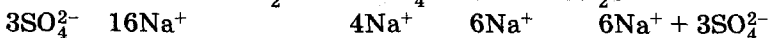
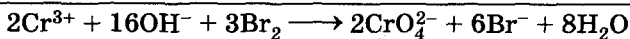
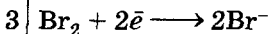
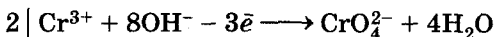
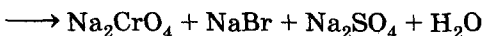
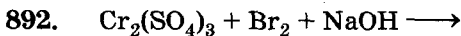
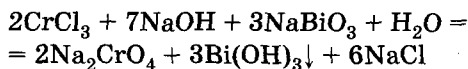
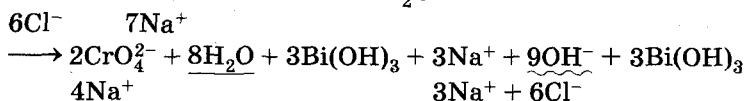
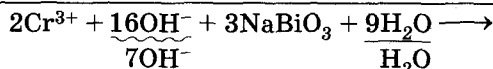
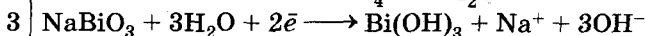
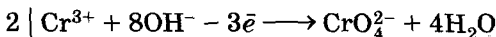
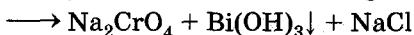
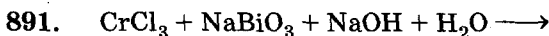
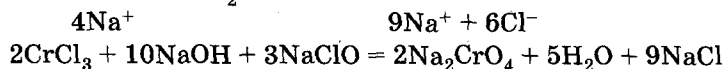
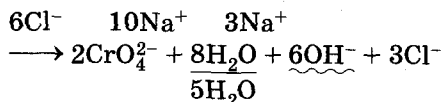
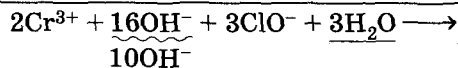
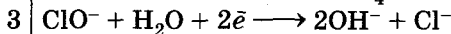
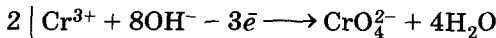
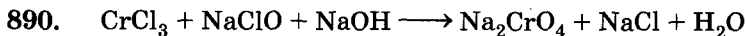
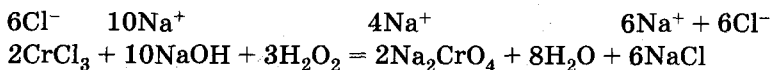
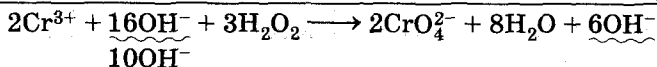
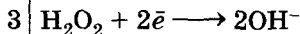
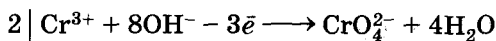
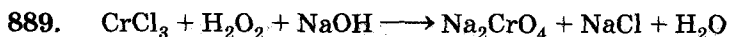


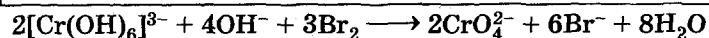
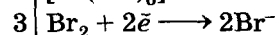
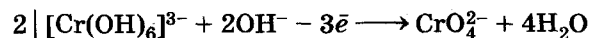
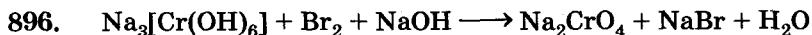
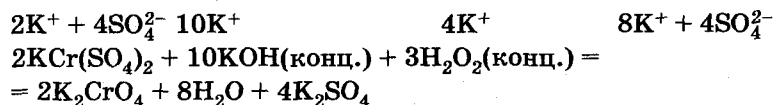
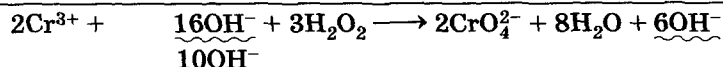
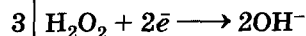
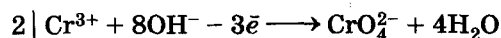
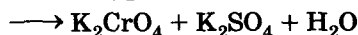
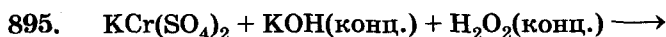
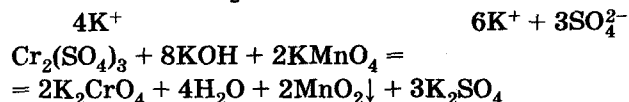
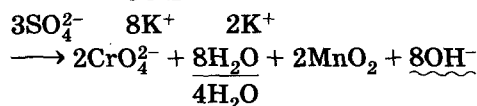
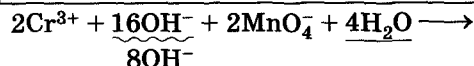
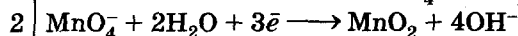
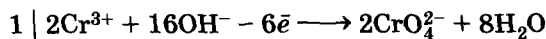
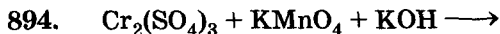
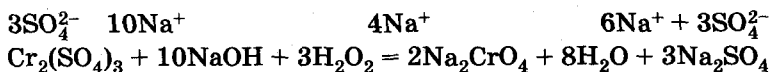
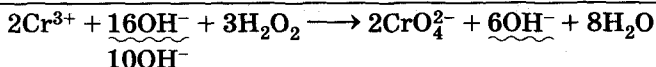
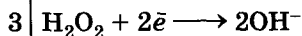
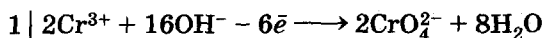
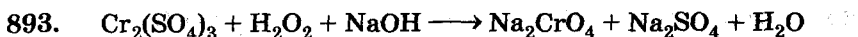


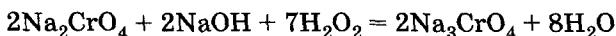
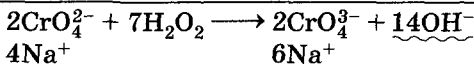
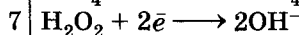
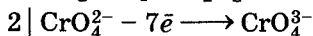
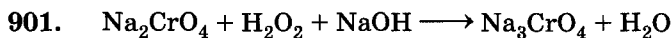
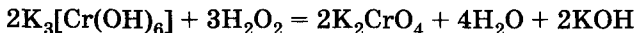
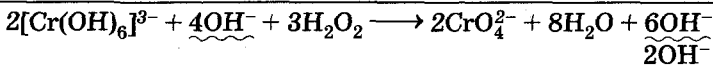
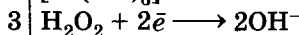
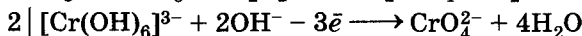
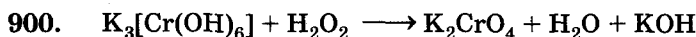
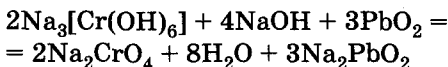
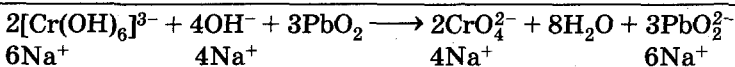
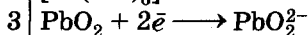
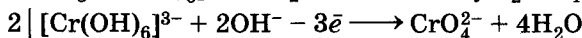
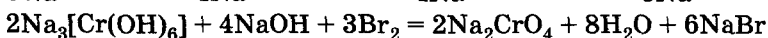
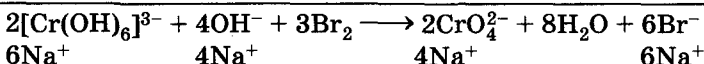
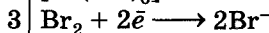
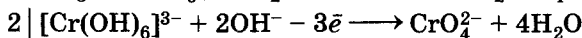
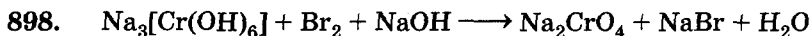
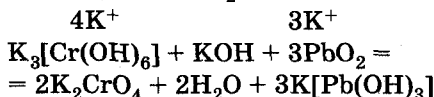
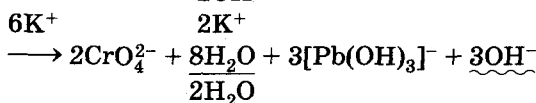
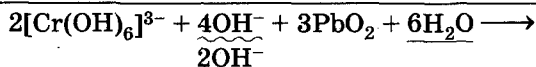
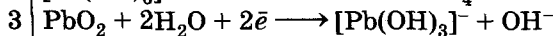
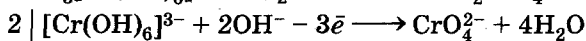
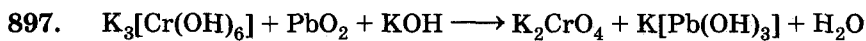


Соединения хрома (среда щелочная)

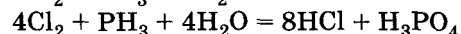
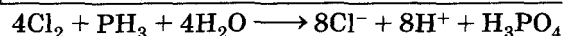
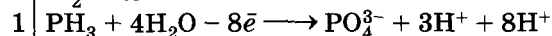
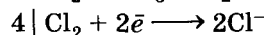
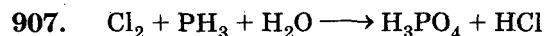
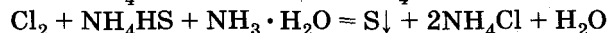
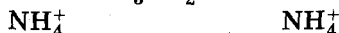
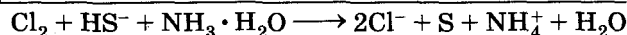
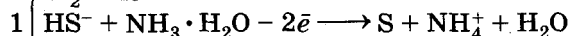
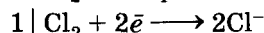
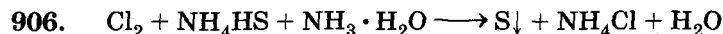
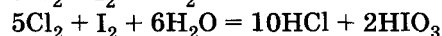
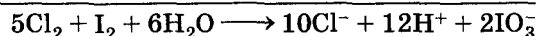
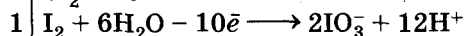
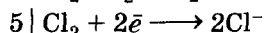
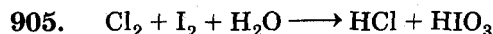
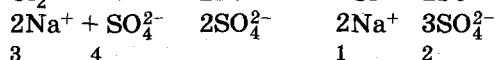
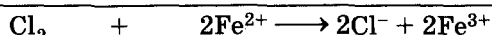
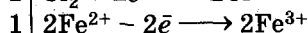
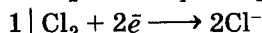
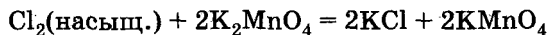
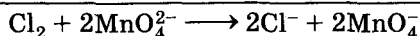
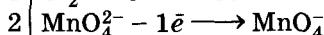
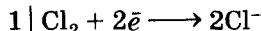
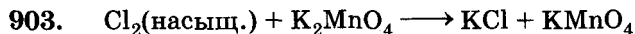
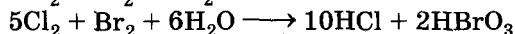
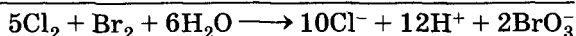
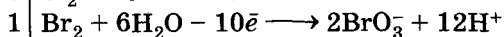
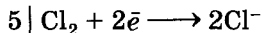
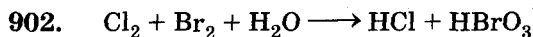


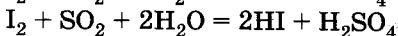
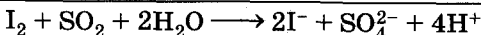
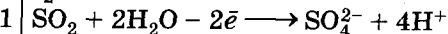
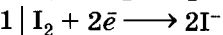
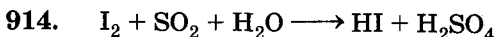
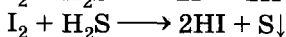
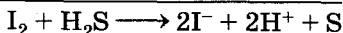
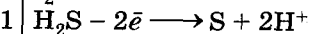
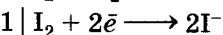
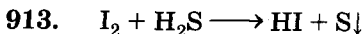
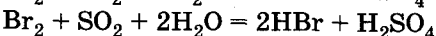
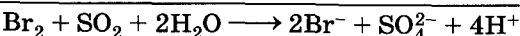
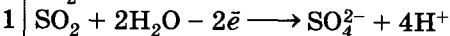
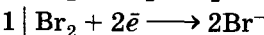
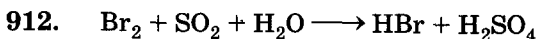
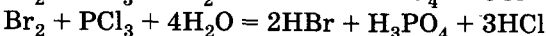
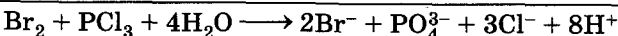
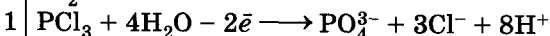
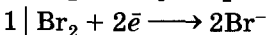
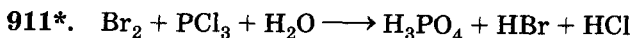
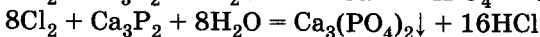
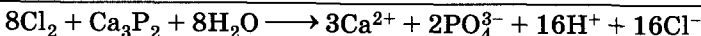
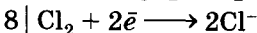
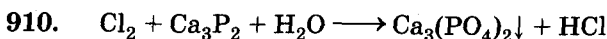
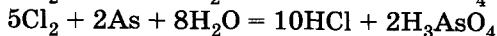
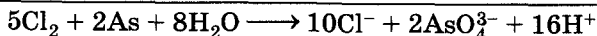
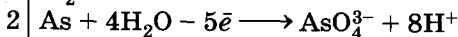
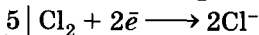
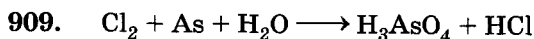
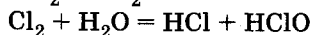
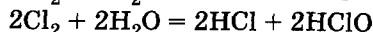
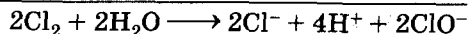
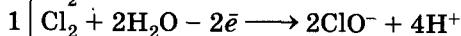
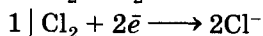
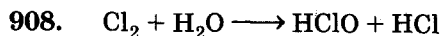


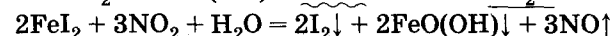
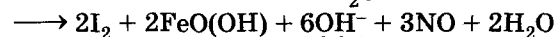
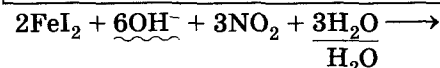
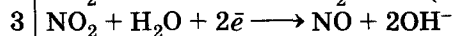
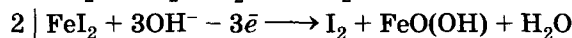
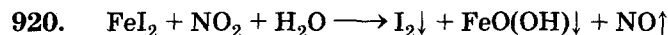
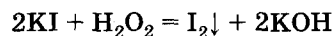
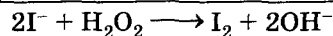
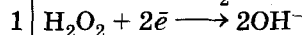
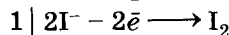
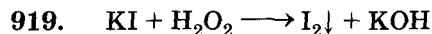
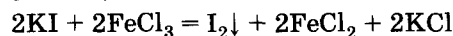
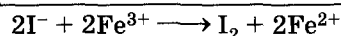
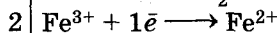
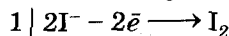
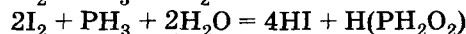
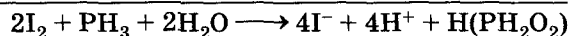
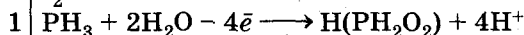
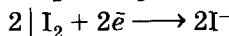
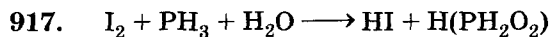
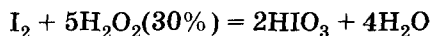
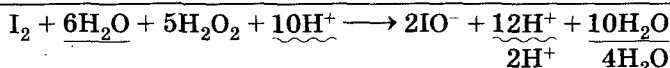
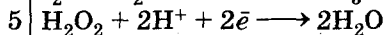
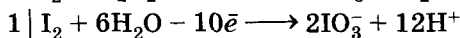
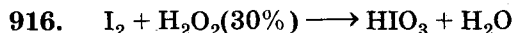
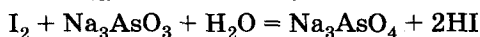
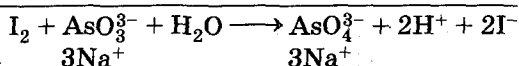
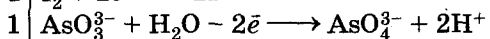
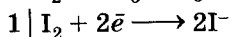
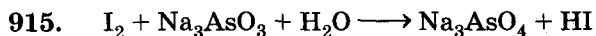


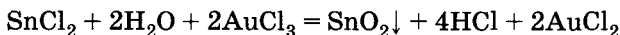
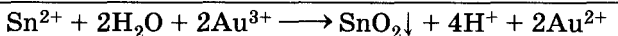
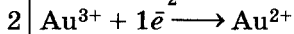
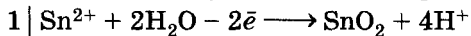
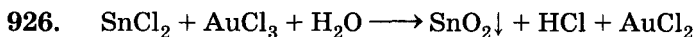
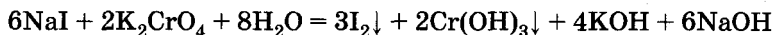
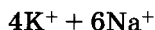
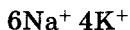
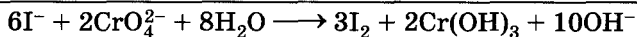
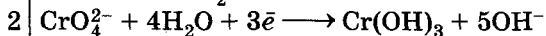
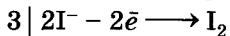
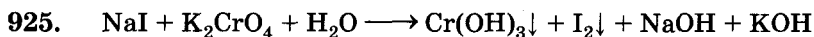
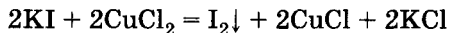
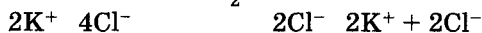
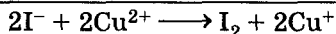
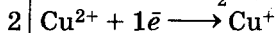
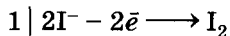
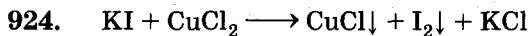
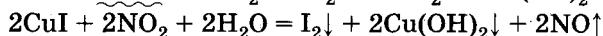
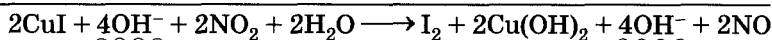
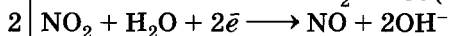
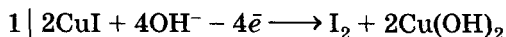
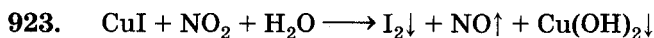
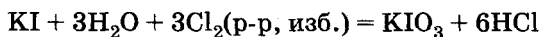
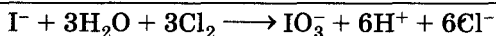
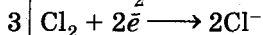
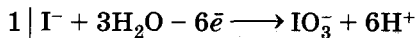
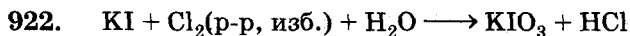
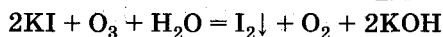
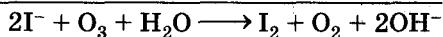
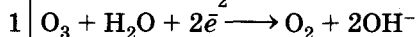
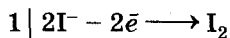
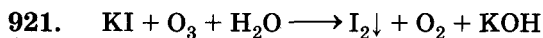


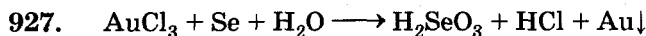
Галогены и их соединения (среда нейтральная)



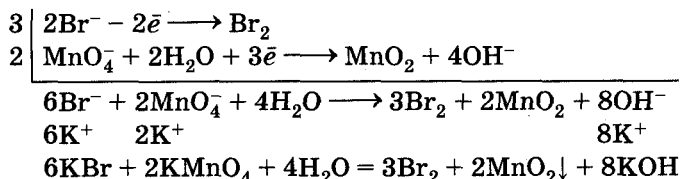
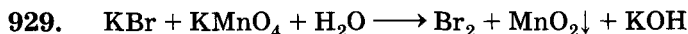
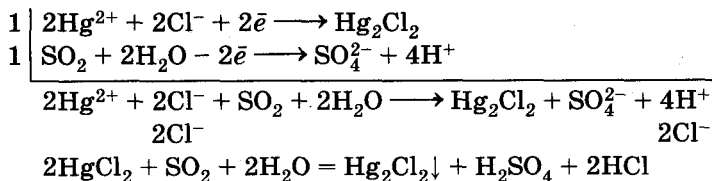
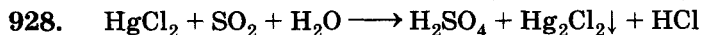
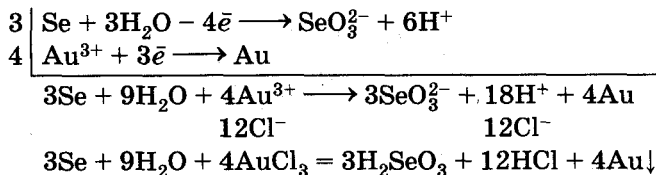




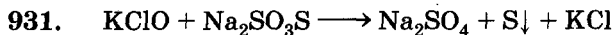
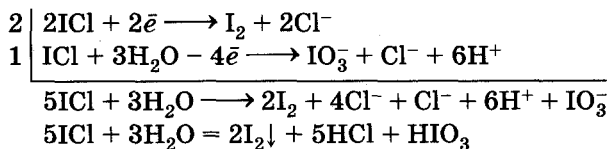




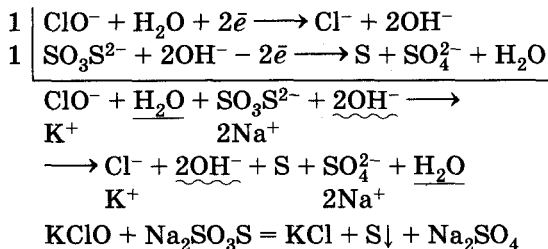
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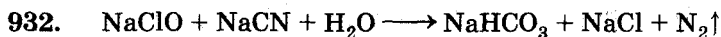


хлорид
иода (I)

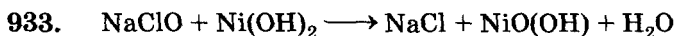
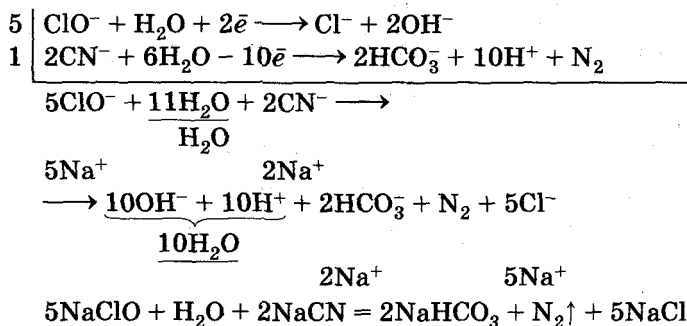


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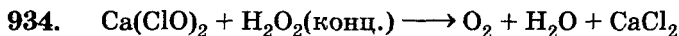
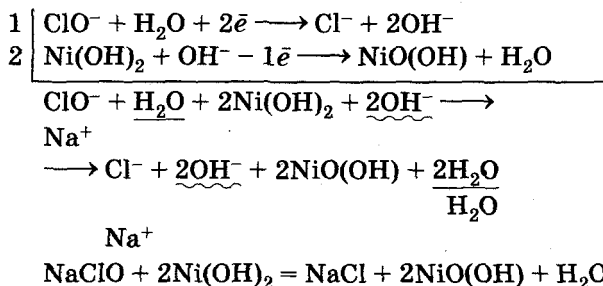




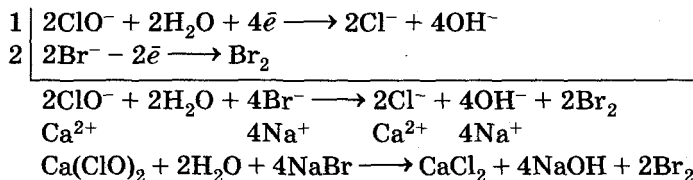
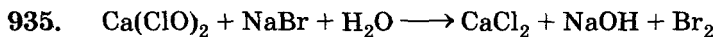
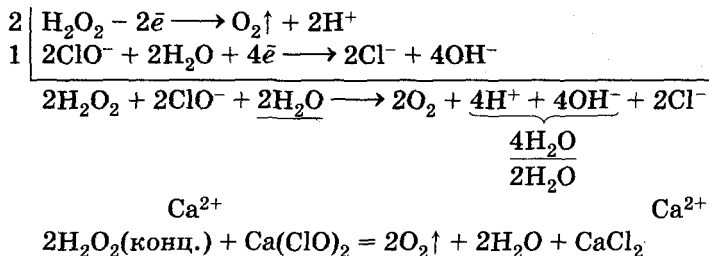
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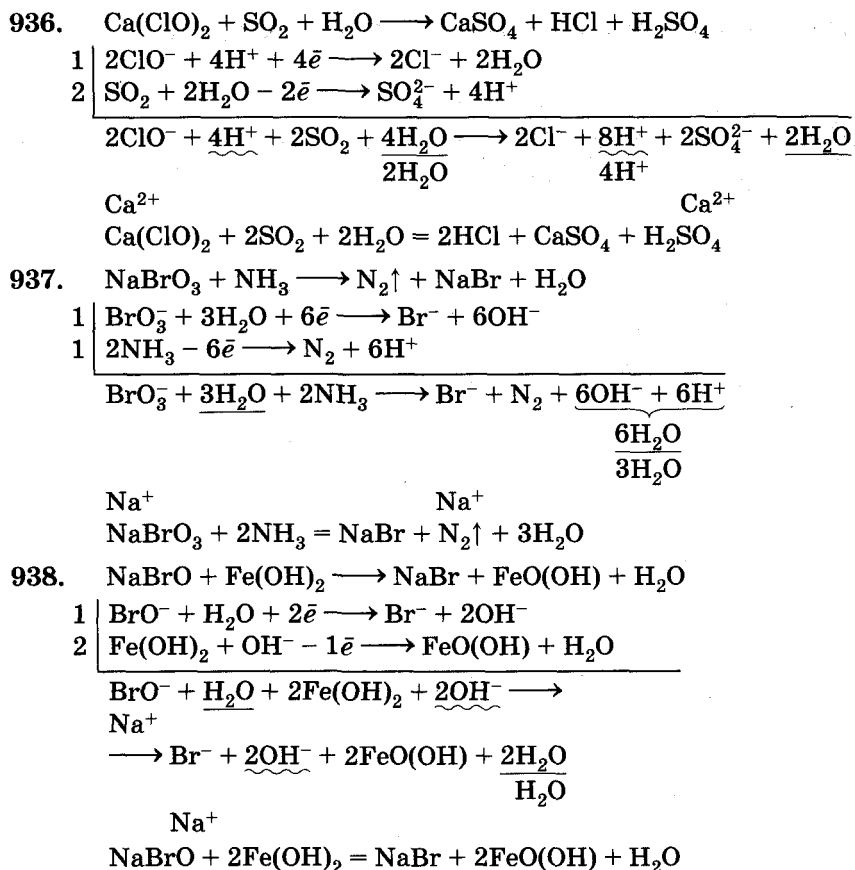


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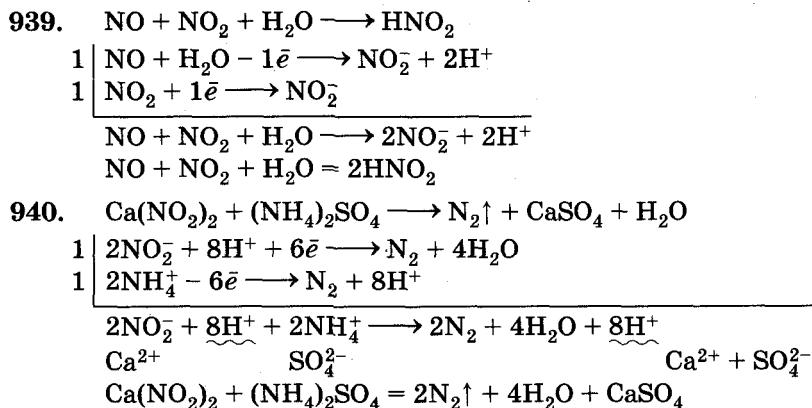


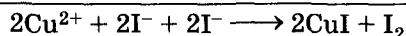
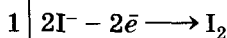
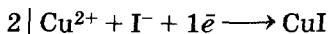
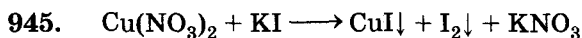
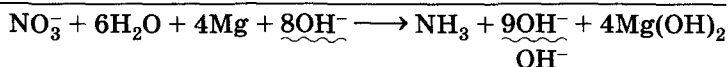
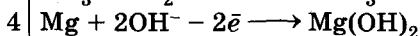
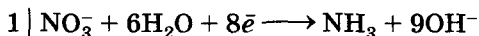
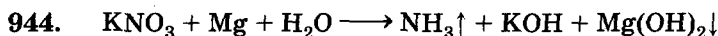
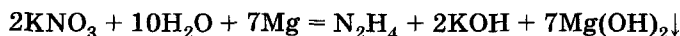
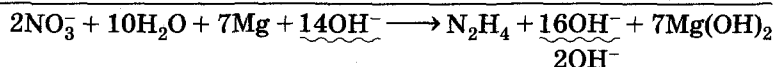
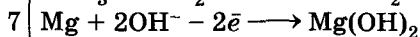
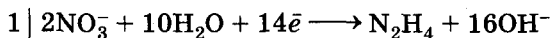
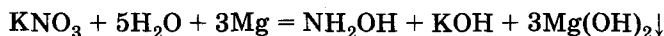
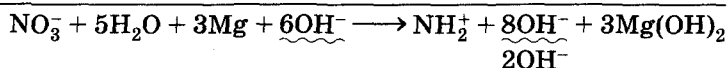
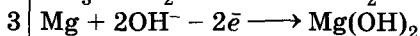
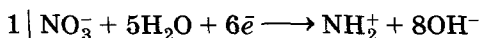
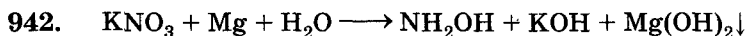
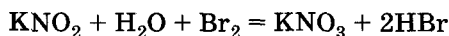
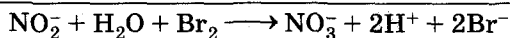
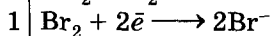
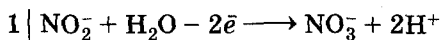
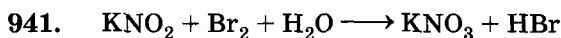
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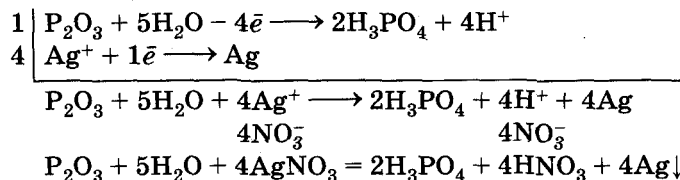
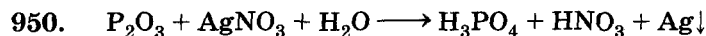
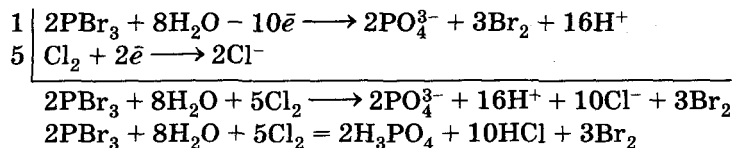
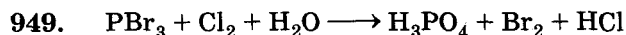
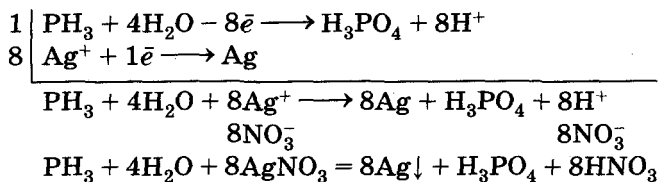
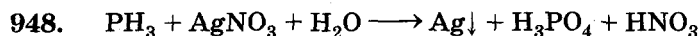
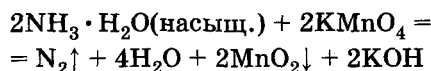
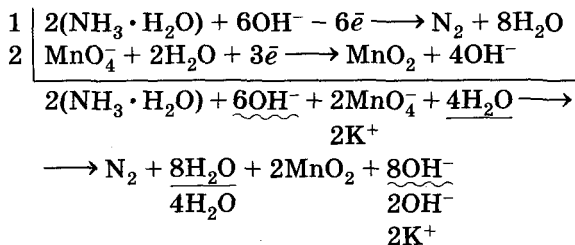
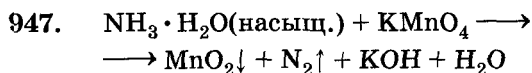
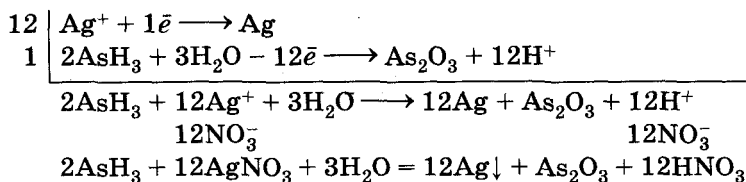
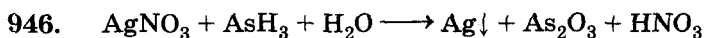




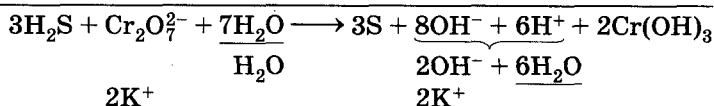
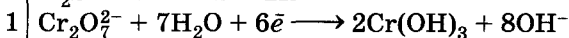
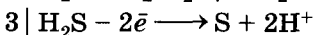
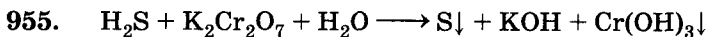
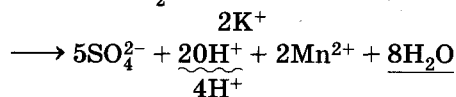
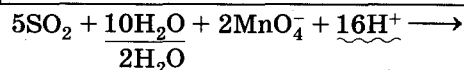
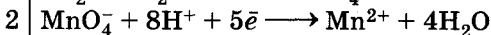
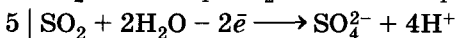
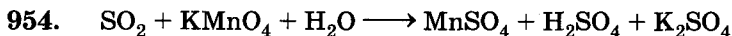
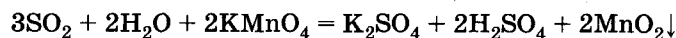
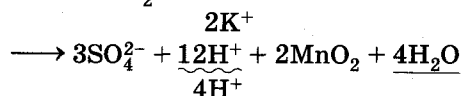
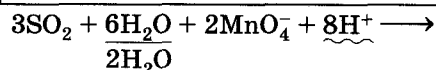
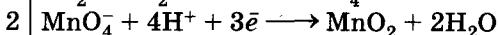
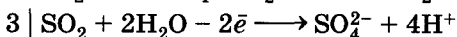
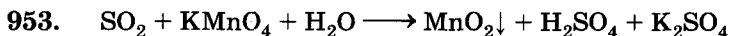
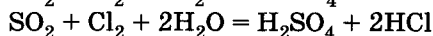
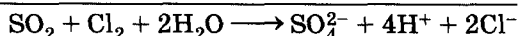
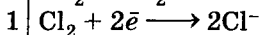
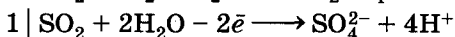
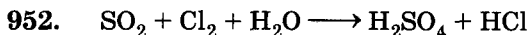
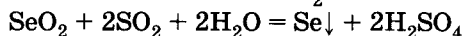
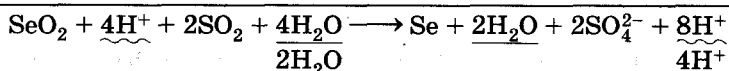
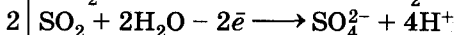
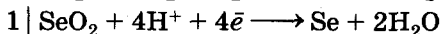
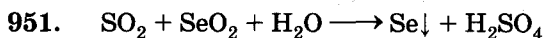
Соединения азота и фосфора (среда нейтральная)

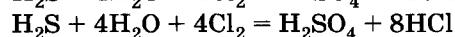
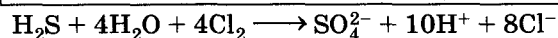
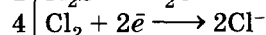
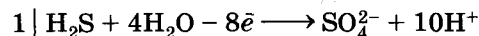
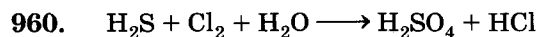
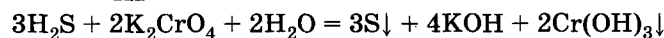
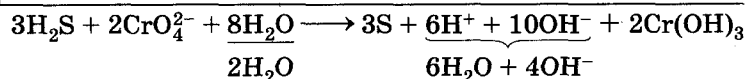
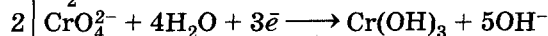
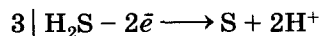
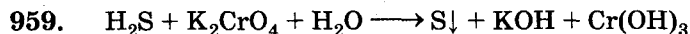
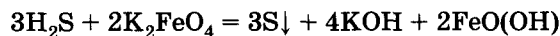
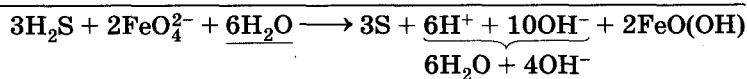
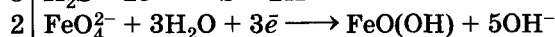
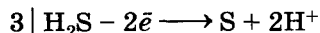
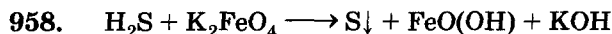
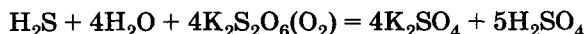
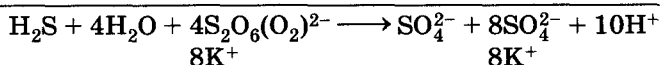
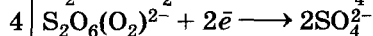
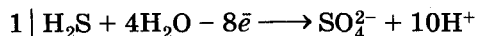
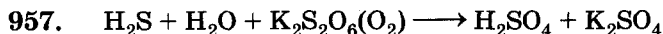
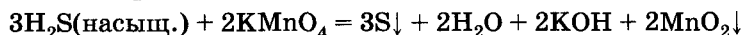
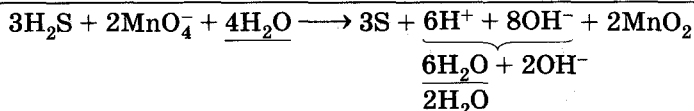
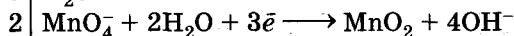
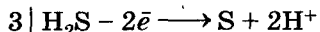
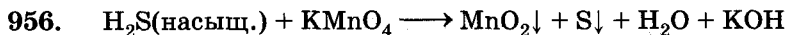
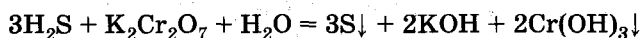


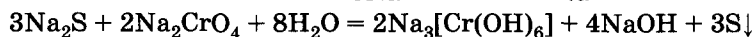
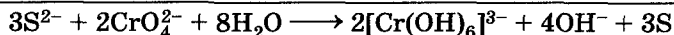
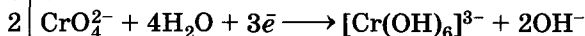
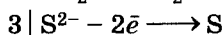
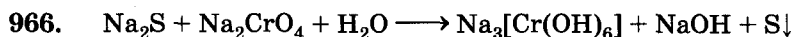
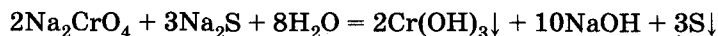
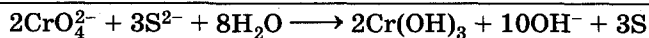
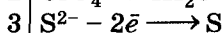
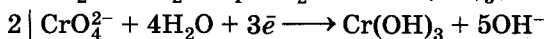
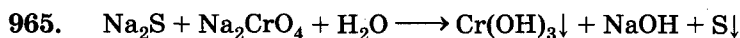
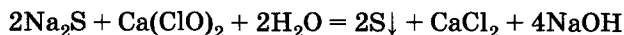
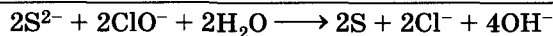
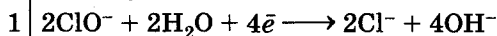
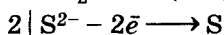
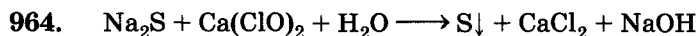
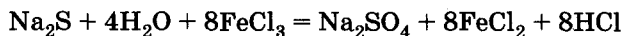
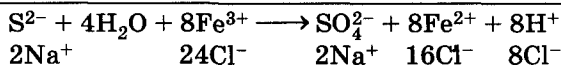
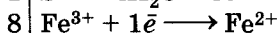
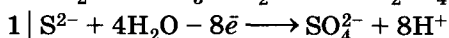
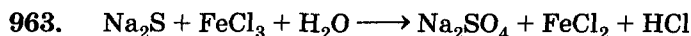
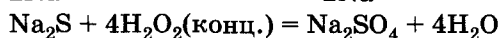
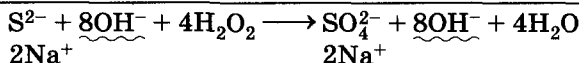
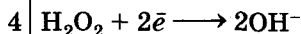
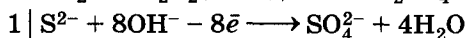
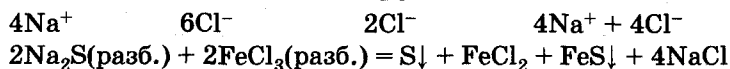
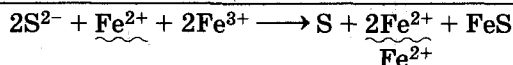
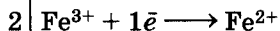
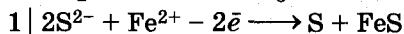
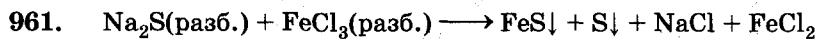


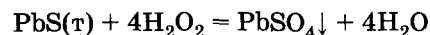
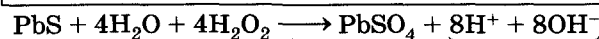
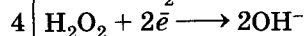
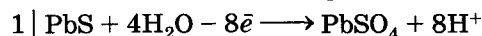
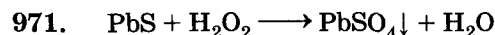
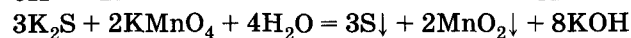
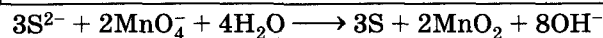
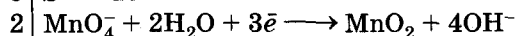
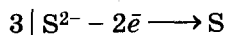
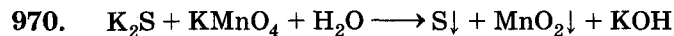
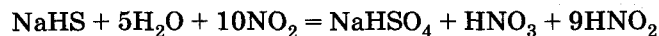
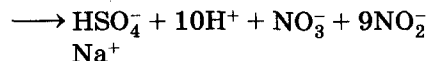
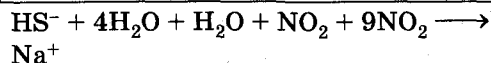
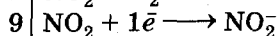
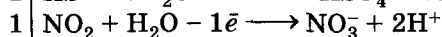
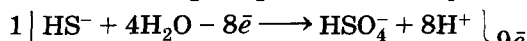
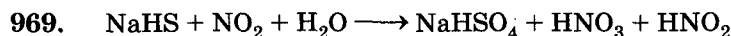
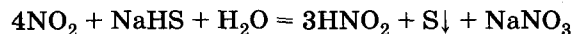
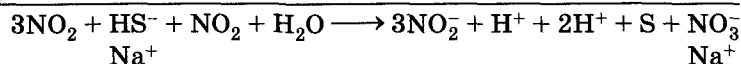
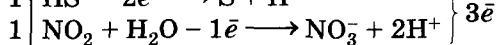
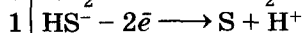
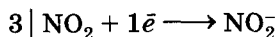
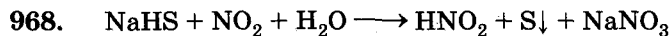
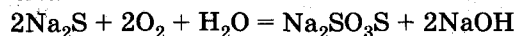
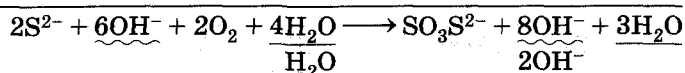
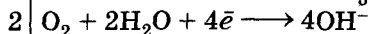
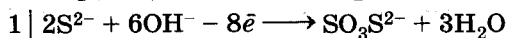
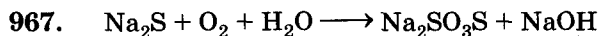


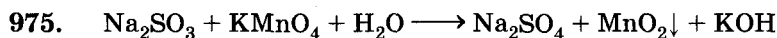
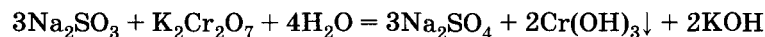
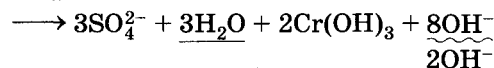
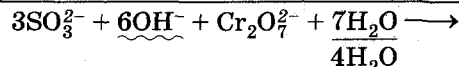
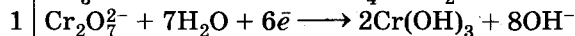
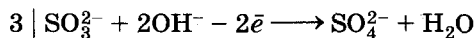
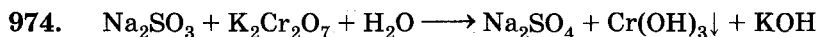
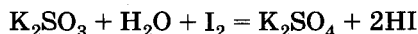
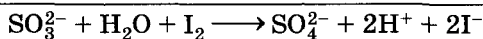
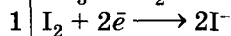
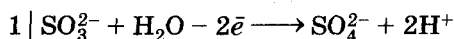
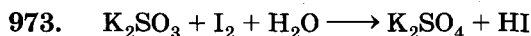
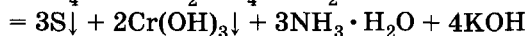
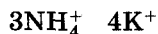
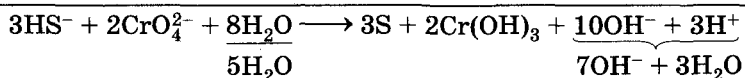
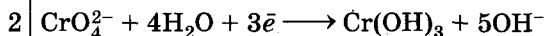
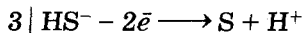
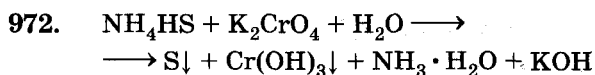
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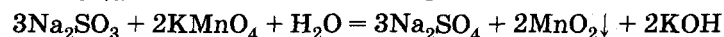
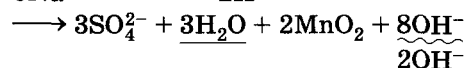
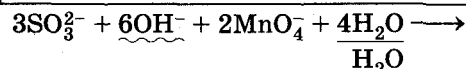
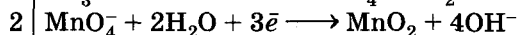
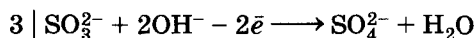




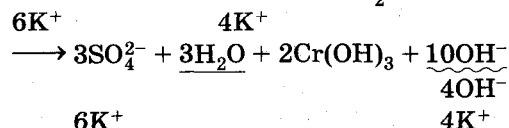
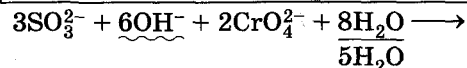
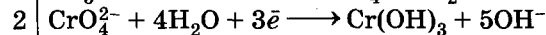
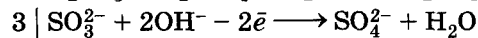
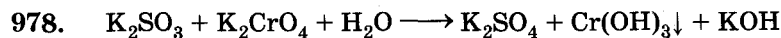
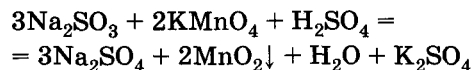
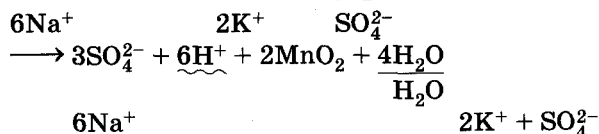
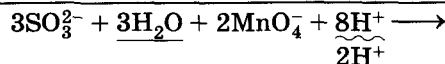
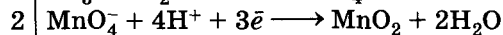
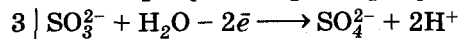
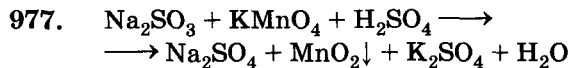
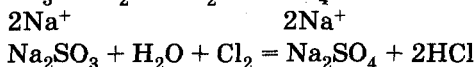
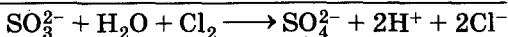
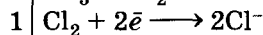
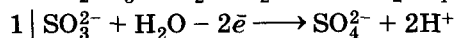
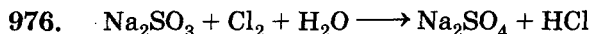
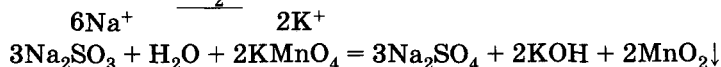
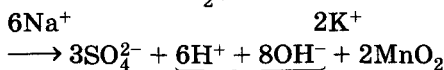
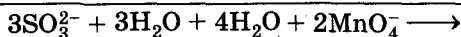
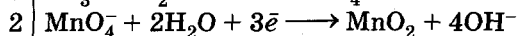
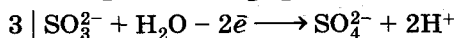


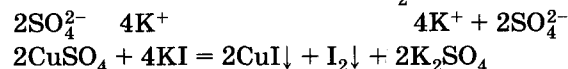
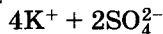
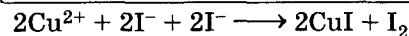
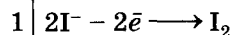
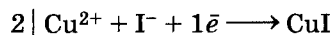
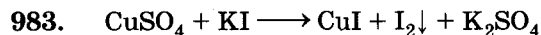
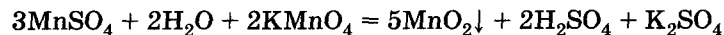
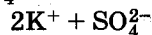
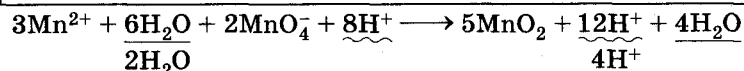
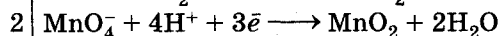
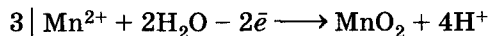
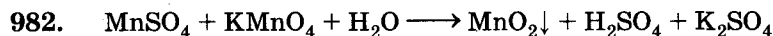
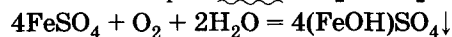
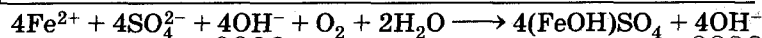
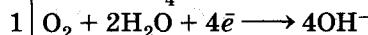
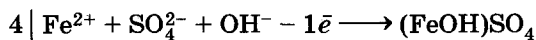
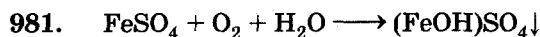
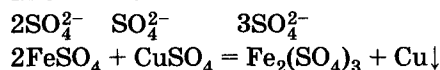
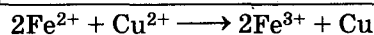
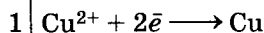
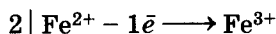
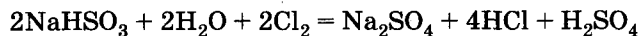
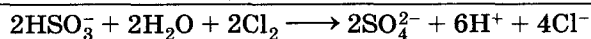
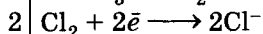
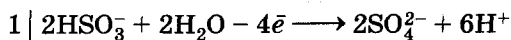
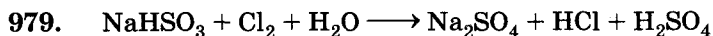
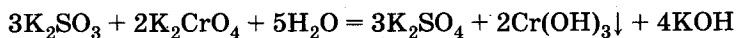


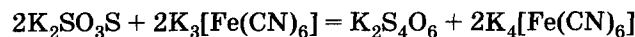
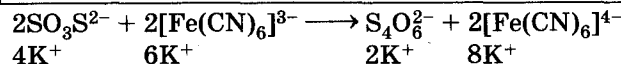
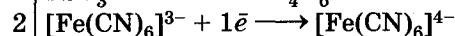
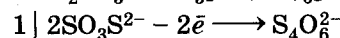
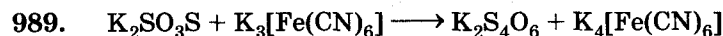
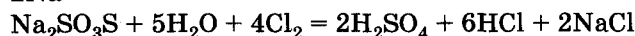
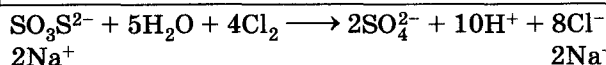
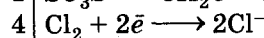
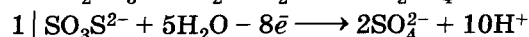
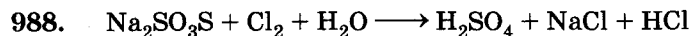
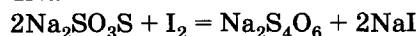
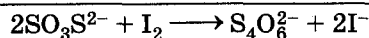
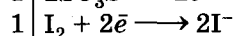
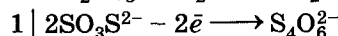
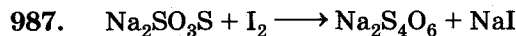
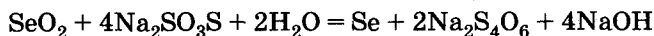
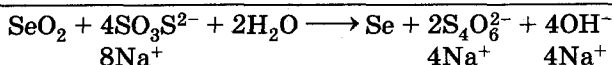
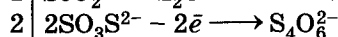
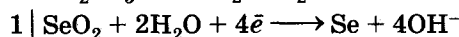
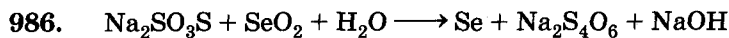
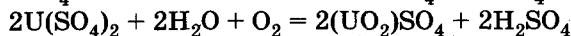
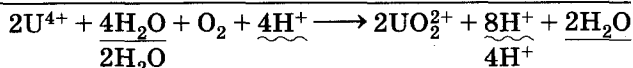
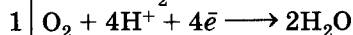
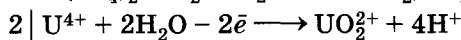
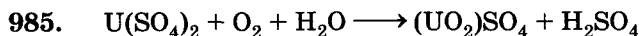
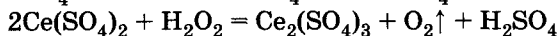
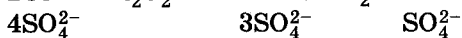
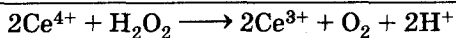
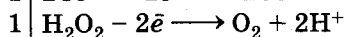
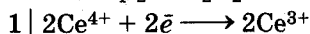
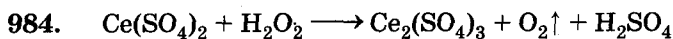
1-й вариант оформления ЭИБ.

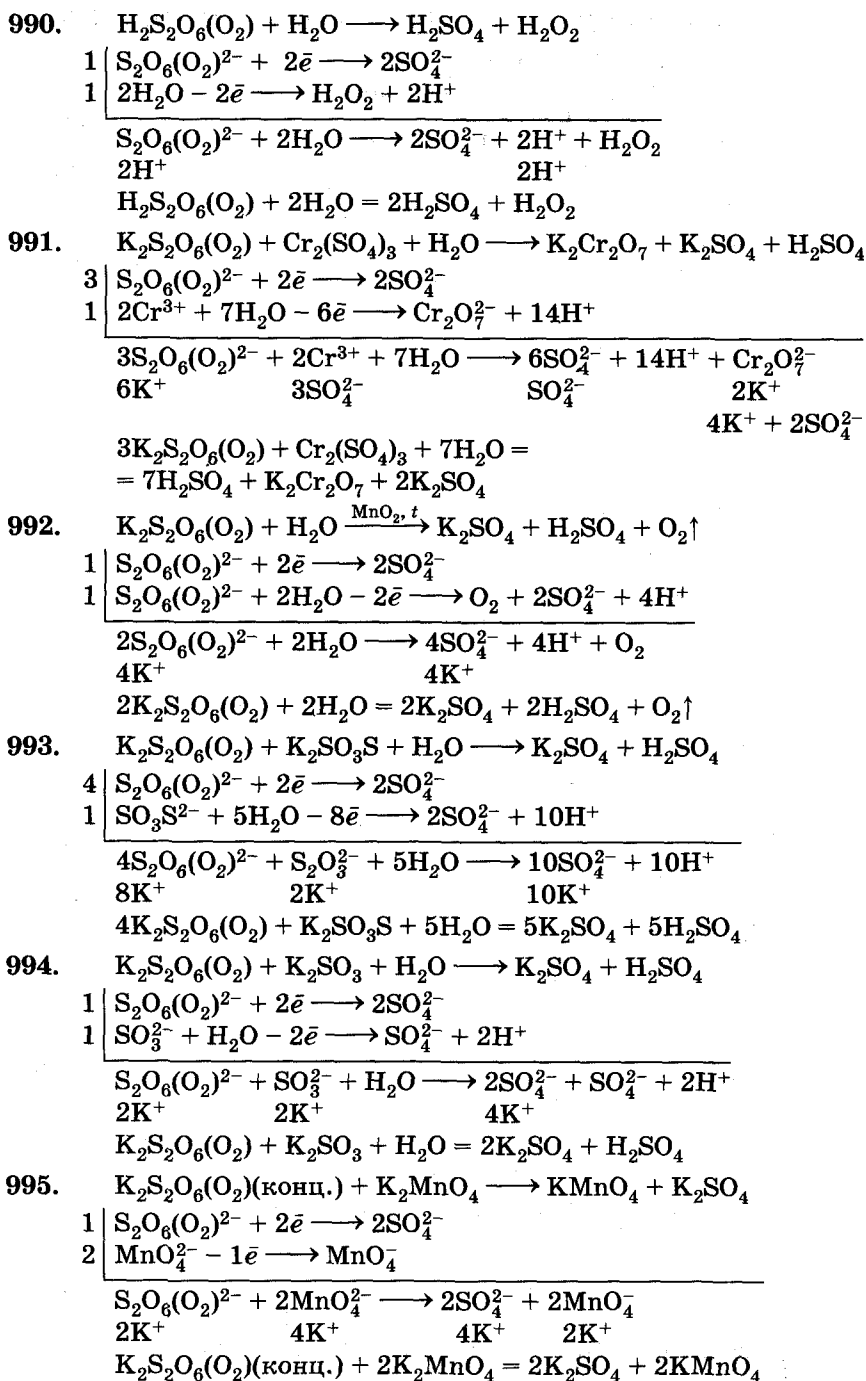


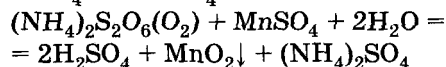
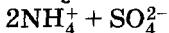
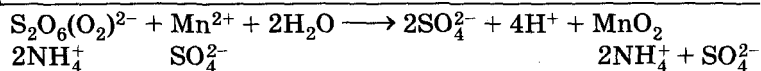
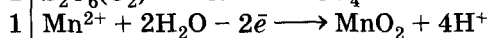
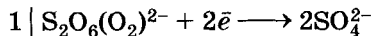
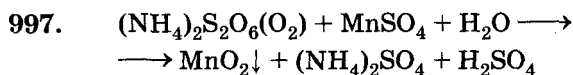
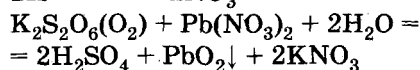
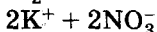
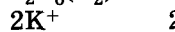
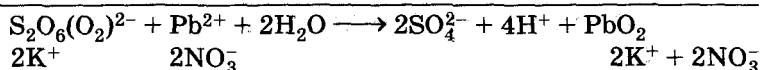
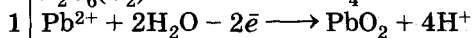
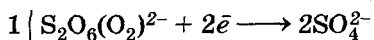
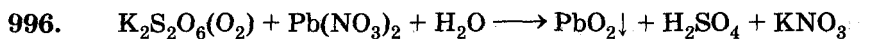
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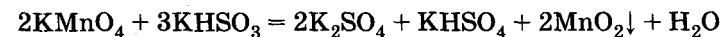
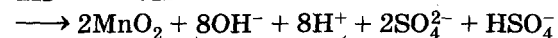
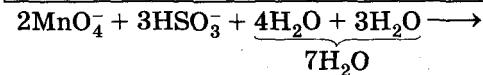
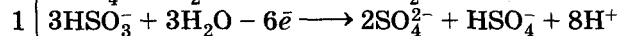
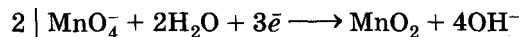
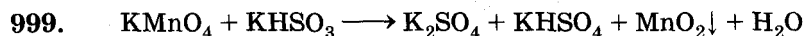
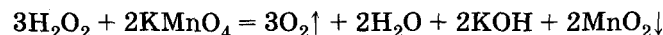
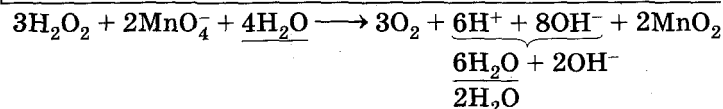
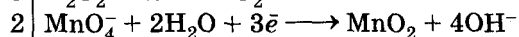
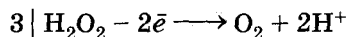
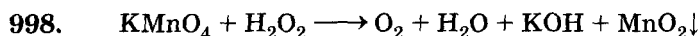


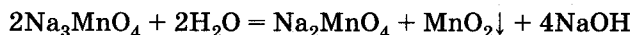
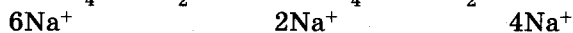
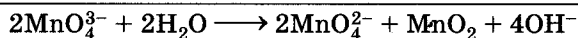
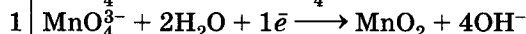
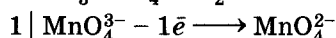
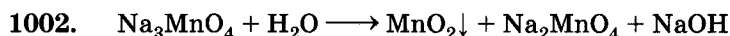
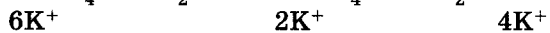
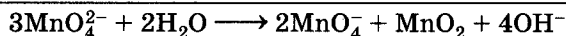
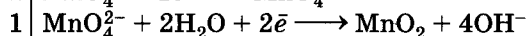
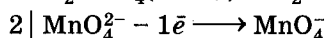
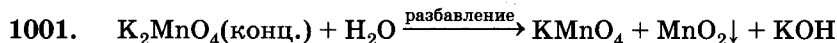
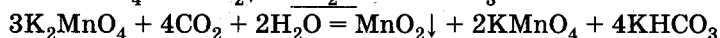
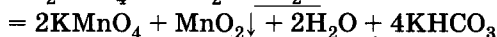
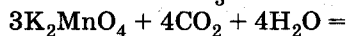
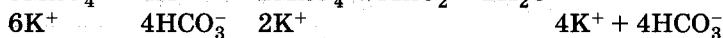
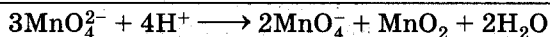
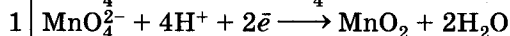
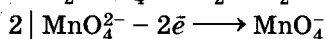
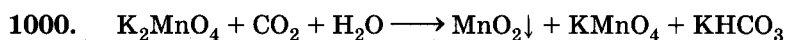




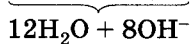
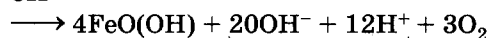
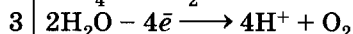
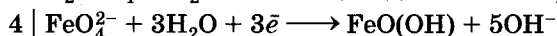
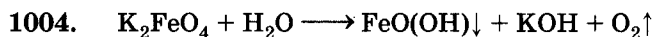
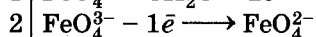
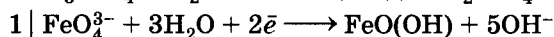
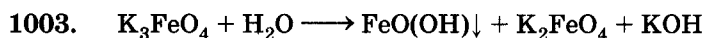


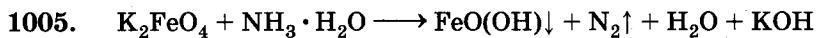
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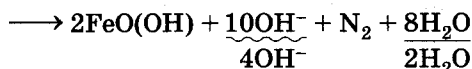
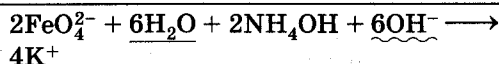
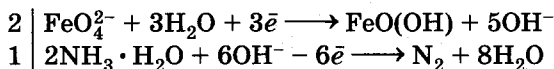


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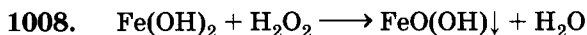
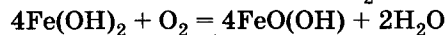
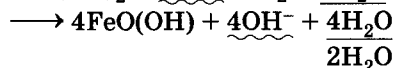
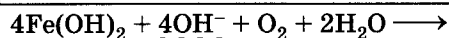
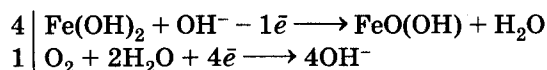
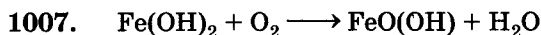
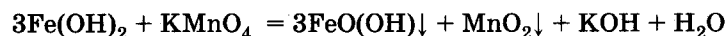
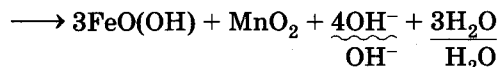
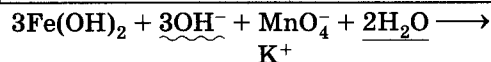
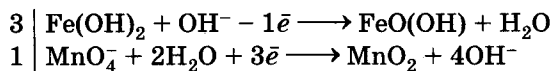




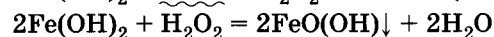
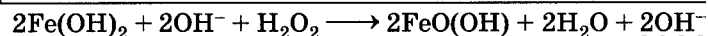
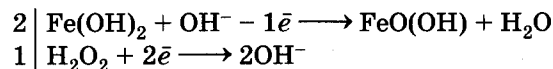
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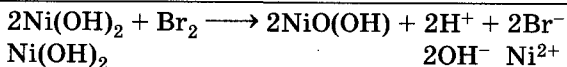
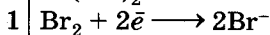
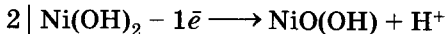
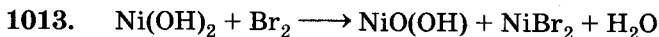
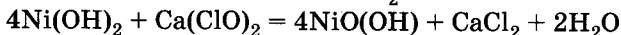
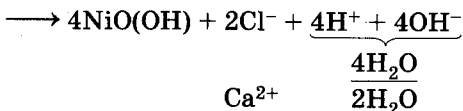
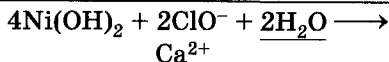
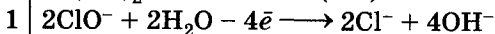
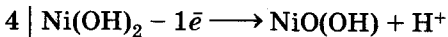
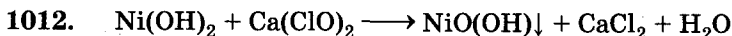
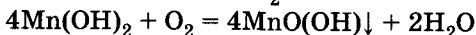
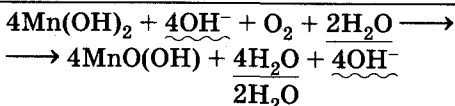
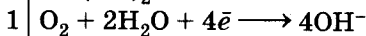
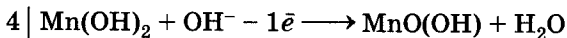
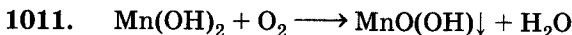
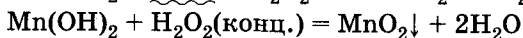
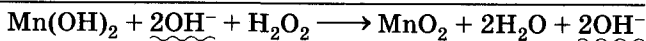
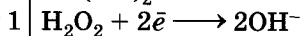
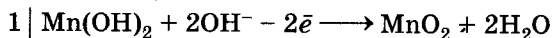
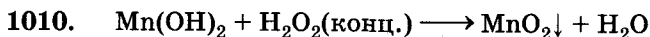
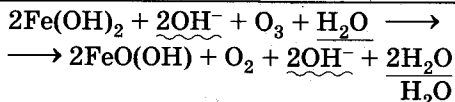
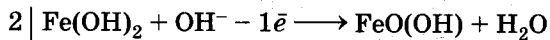
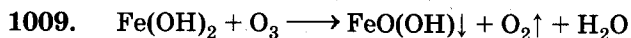


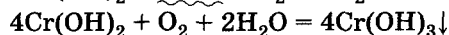
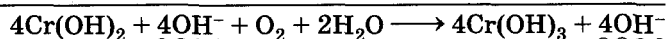
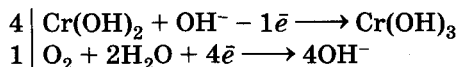
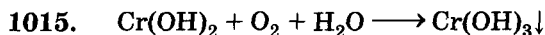
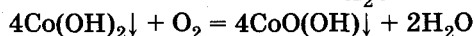
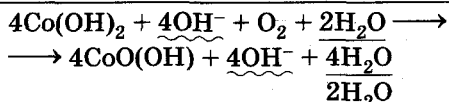
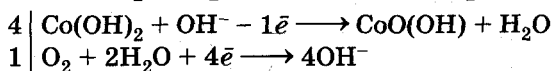
**Гидроксиды железа, марганца, никеля,
кобальта и хрома (среда нейтральная)**



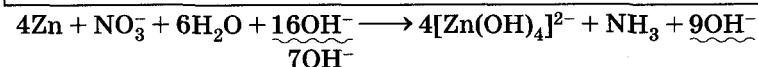
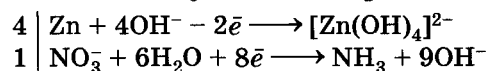
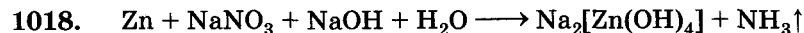
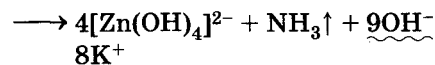
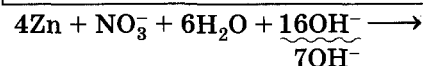
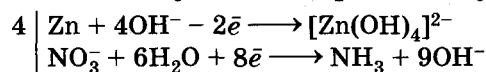
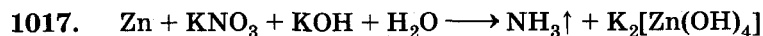
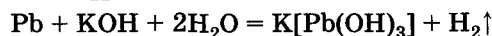
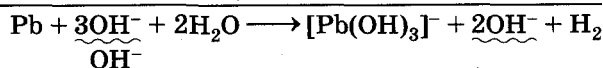
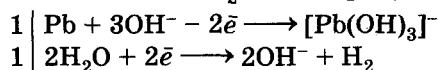
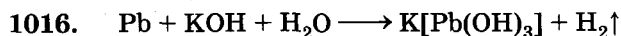
метагидроксид
железа (III)

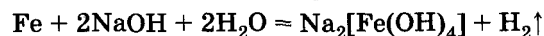
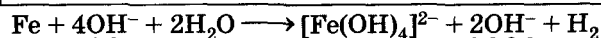
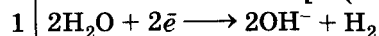
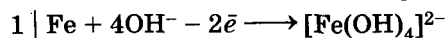
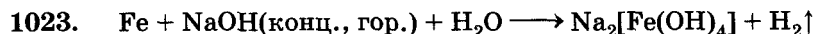
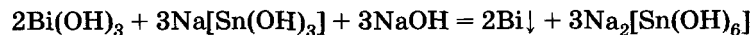
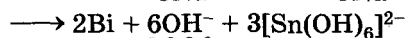
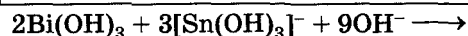
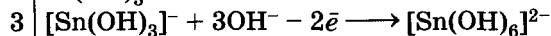
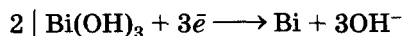
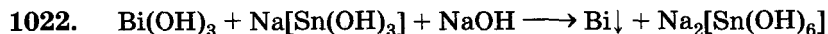
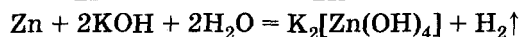
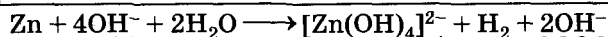
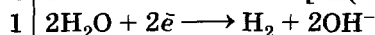
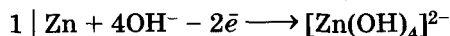
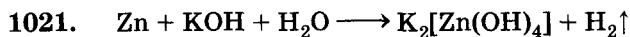
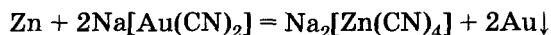
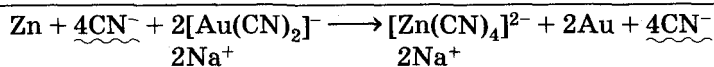
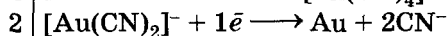
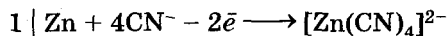
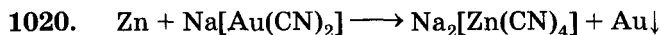
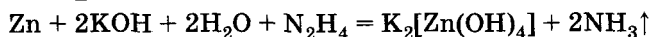
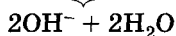
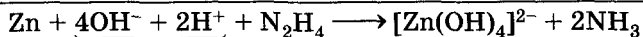
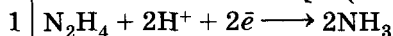
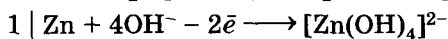
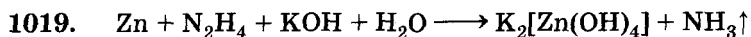


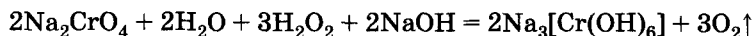
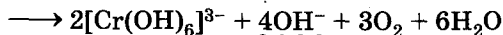
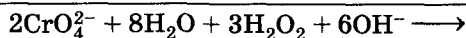
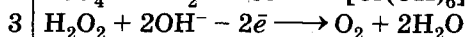
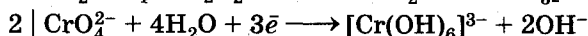
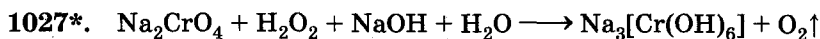
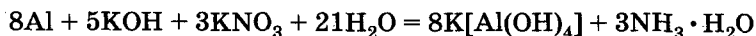
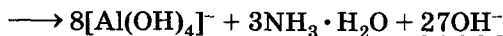
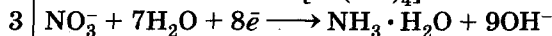
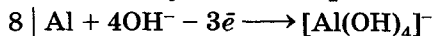
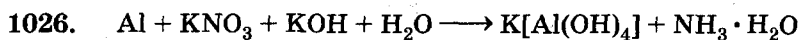
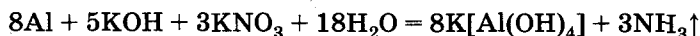
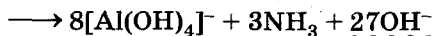
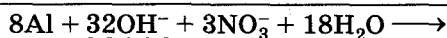
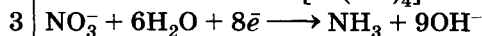
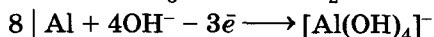
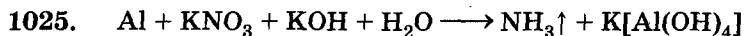
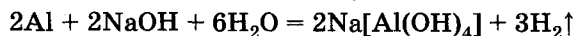
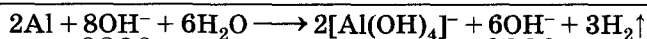
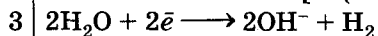
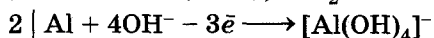
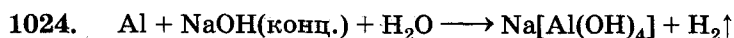




Комплексные соединения

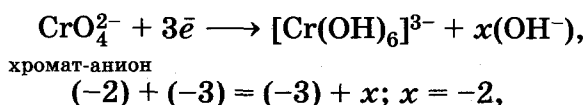




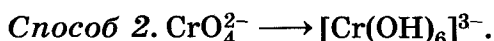


Пояснение. В правой части 1-й полуреакции в составе $[\text{Cr}(\text{OH})_6]^{3-}$ содержится уже 6 атомов водорода, поэтому трудно рассчитать число добавляемых молекул воды и гидроксид-ионов.

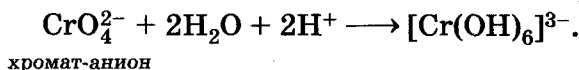
Способ 1 оформления полуреакции. Если вспомнить, что с. о. хрома в $\text{CrO}_4^{2-} + 6$, а в $[\text{Cr}(\text{OH})_6]^{3-} + 3$, то для перехода Cr^{+6} в Cr^{+3} в левую часть полуреакции следует добавить три электрона: $\text{CrO}_4^{2-} + 3\bar{e} \longrightarrow [\text{Cr}(\text{OH})_6]^{3-}$. Для уравнивания зарядов в правую часть следует добавить 2OH^- , в левую — $4\text{H}_2\text{O}$. Далее уравниваются кислород и водород: $\text{CrO}_4^{2-} + 3\bar{e} + 4\text{H}_2\text{O} \longrightarrow [\text{Cr}(\text{OH})_6]^{3-} + 2\text{OH}^-$. Число добавляемых гидроксид-ионов можно было рассчитать и по уравнению:



т. е. следует добавить 2OH^- в правую часть.



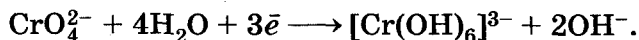
Поскольку в правой части избыток кислорода и водорода, в левую часть добавим H_2O и H^+ . Уравняем сначала кислород, потом водород:



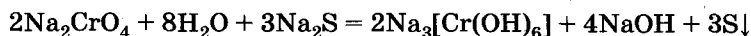
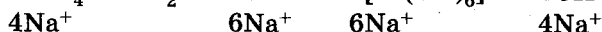
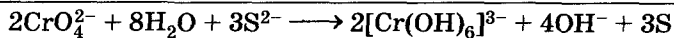
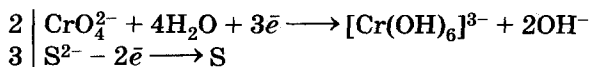
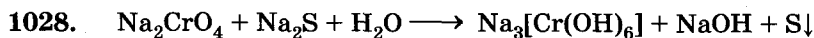
Для нейтрализации 2H^+ добавим два гидроксид-иона в левую часть и для равенства столько же гидроксид-ионов в правую часть:



2H^+ и 2OH^- образуют $2\text{H}_2\text{O}$, а в сумме $4\text{H}_2\text{O}$ в левой части. Уравниваем заряды, в левую часть добавляем три электрона:

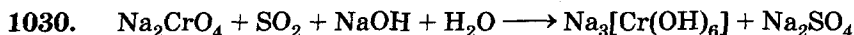
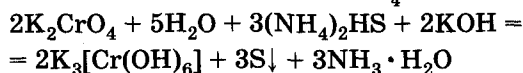
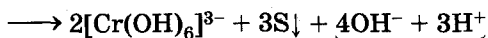
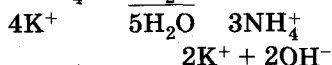
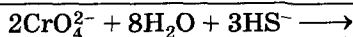
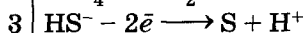
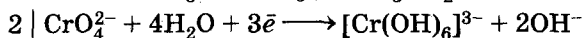
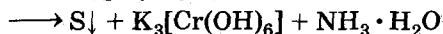


хромат-анион

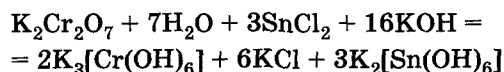
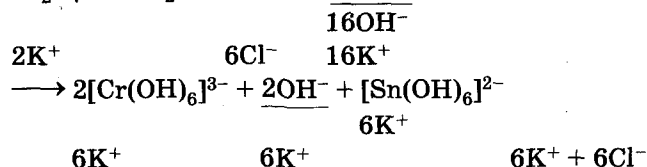
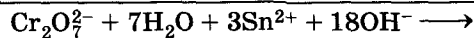
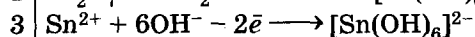
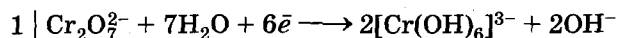
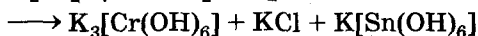
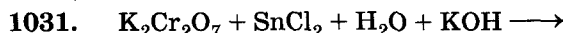
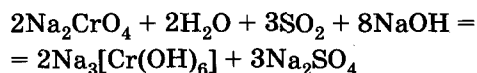
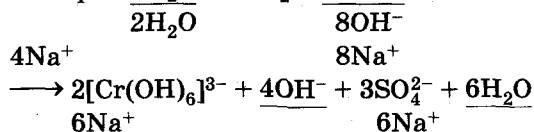
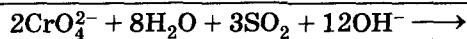
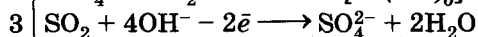
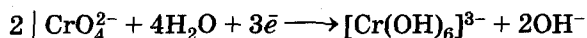


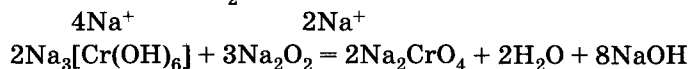
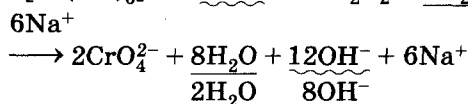
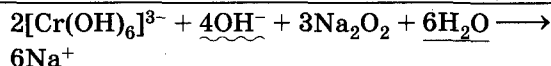
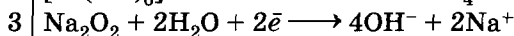
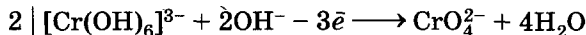
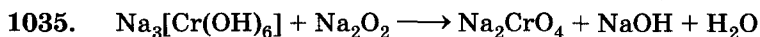
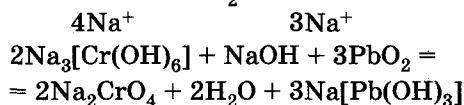
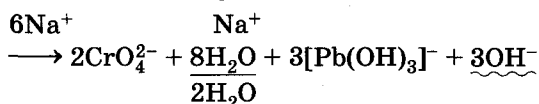
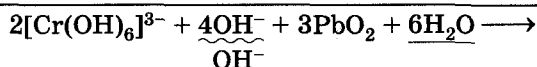
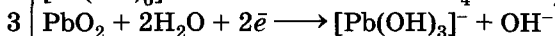
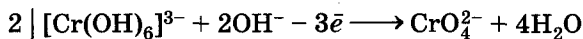
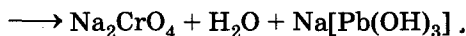
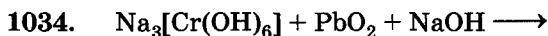
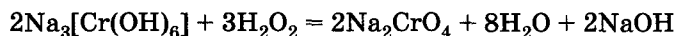
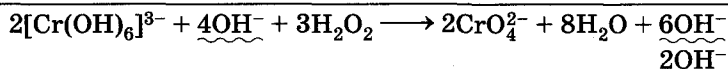
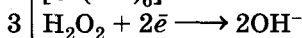
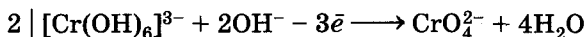
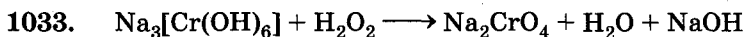
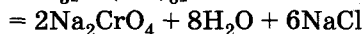
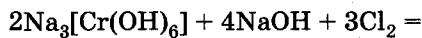
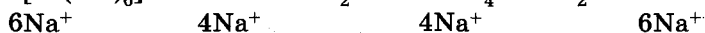
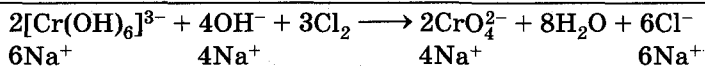
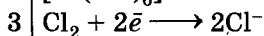
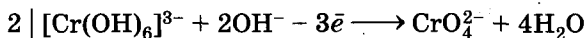
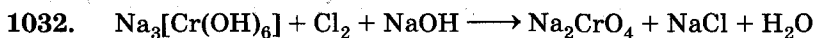


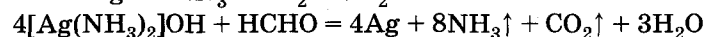
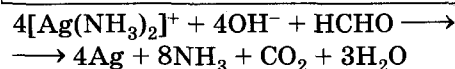
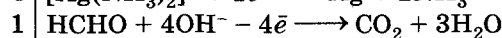
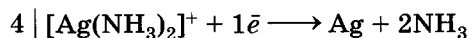
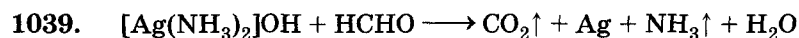
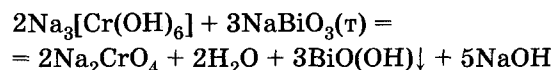
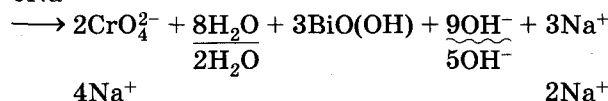
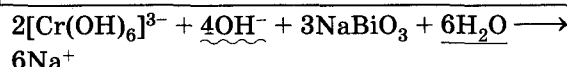
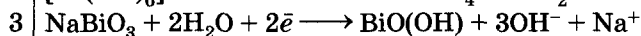
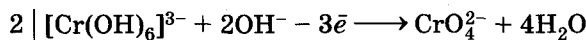
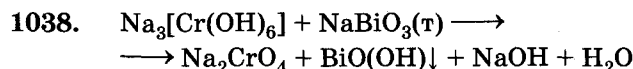
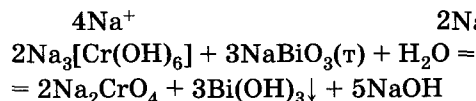
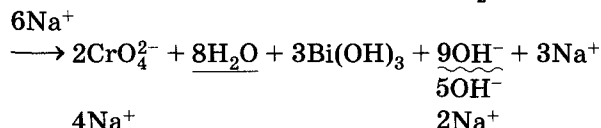
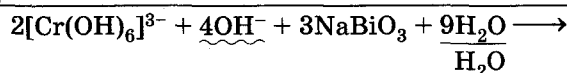
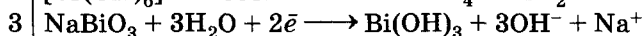
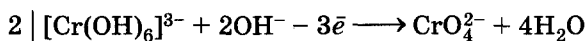
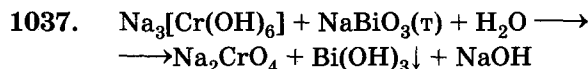
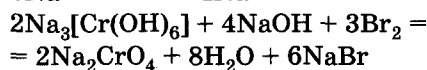
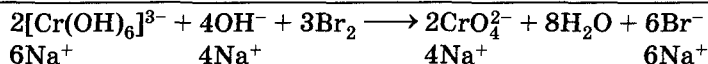
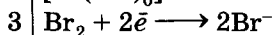
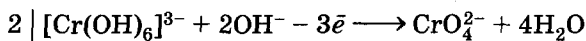
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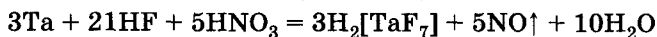
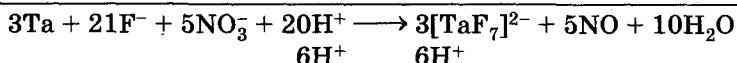
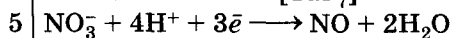
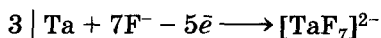
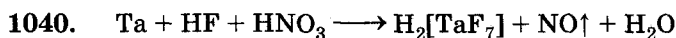


гексагидроксохромат (III) калия

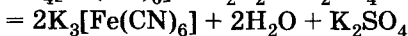
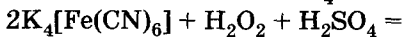
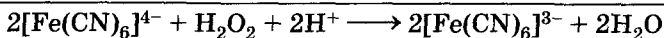
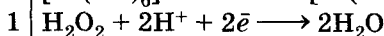
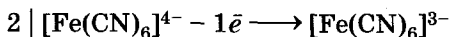
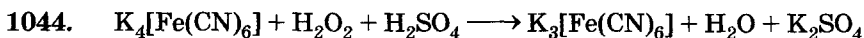
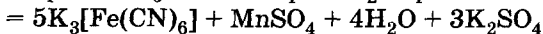
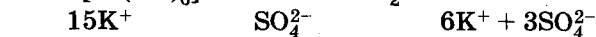
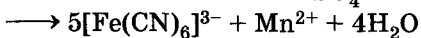
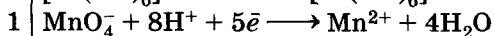
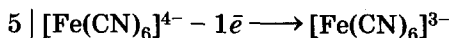
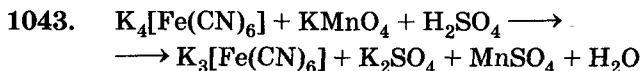
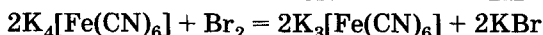
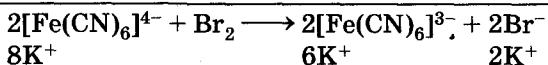
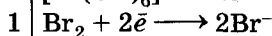
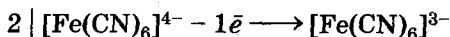
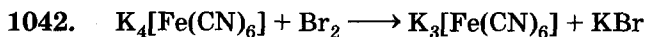
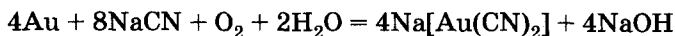
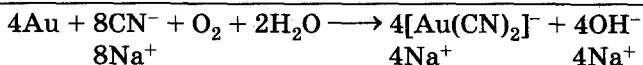
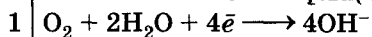
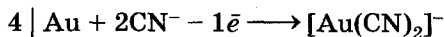
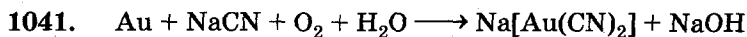


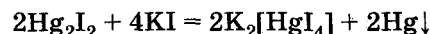
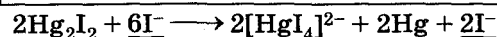
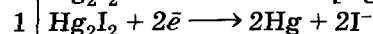
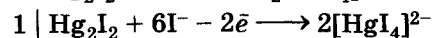
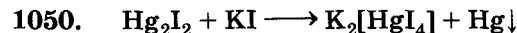
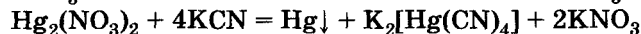
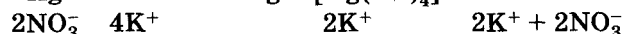
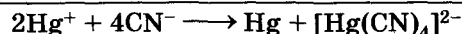
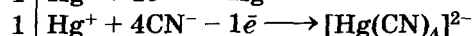
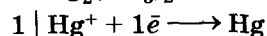
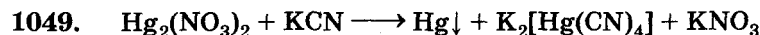
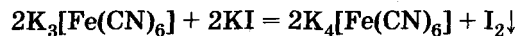
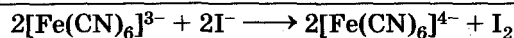
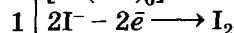
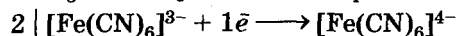
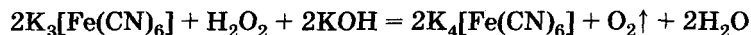
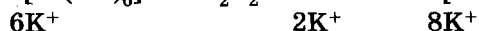
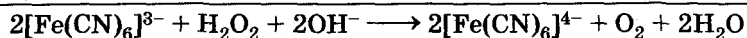
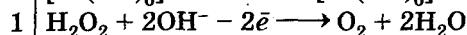
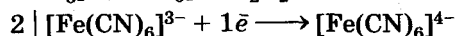
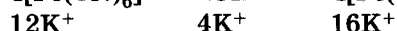
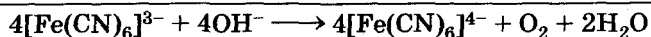
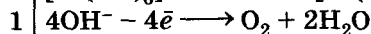
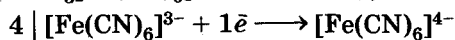
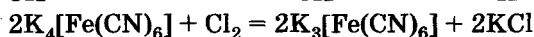
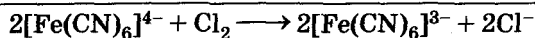
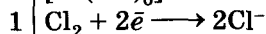
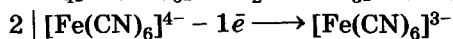
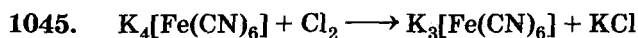


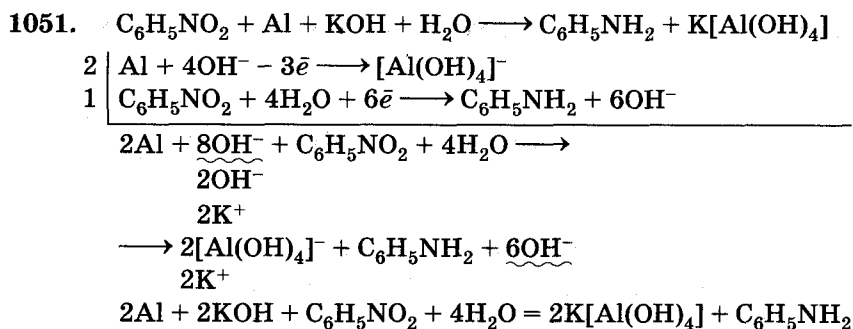




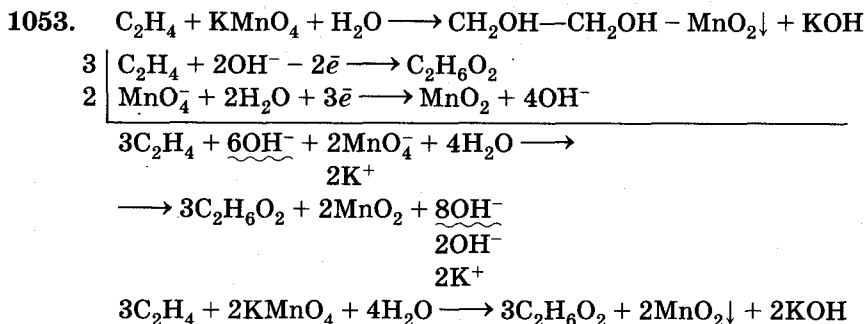
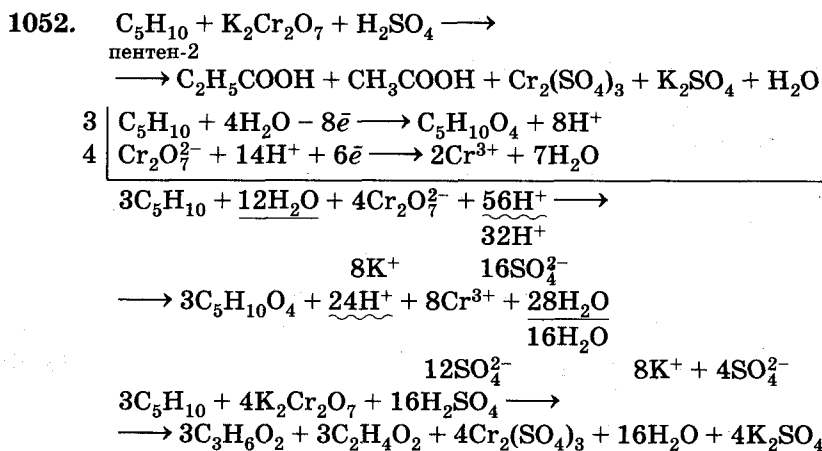
П о я с н е н и е. Распределение 26H^+ ($21\text{H}^+ + 5\text{H}^+$).



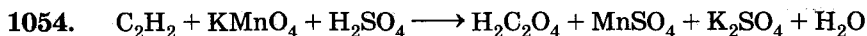




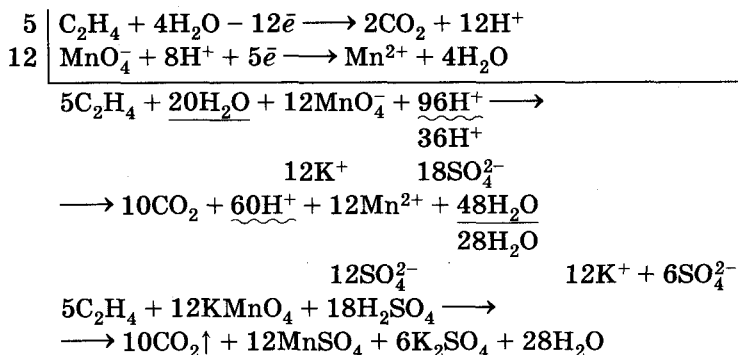
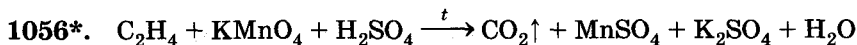
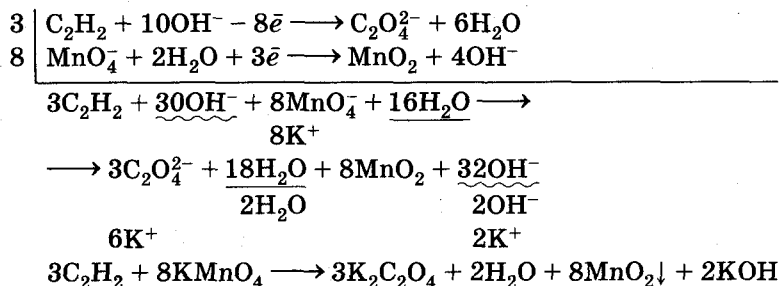
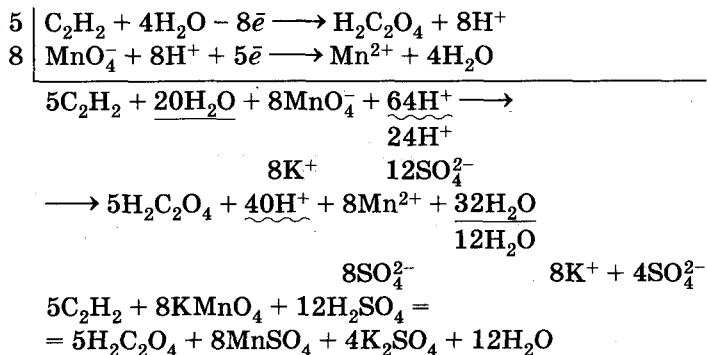
Органические соединения¹



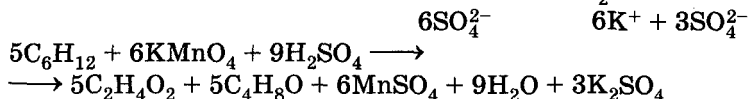
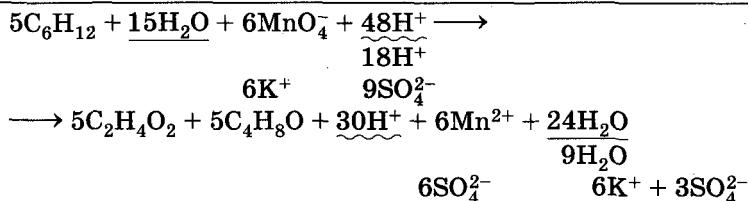
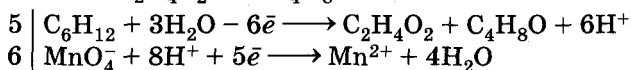
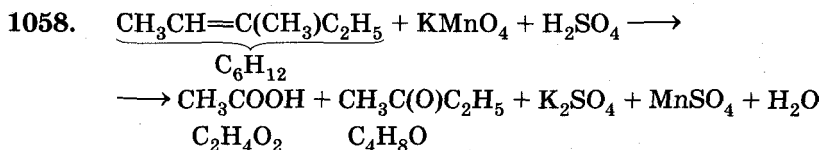
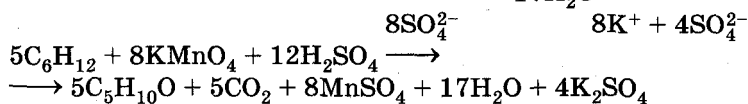
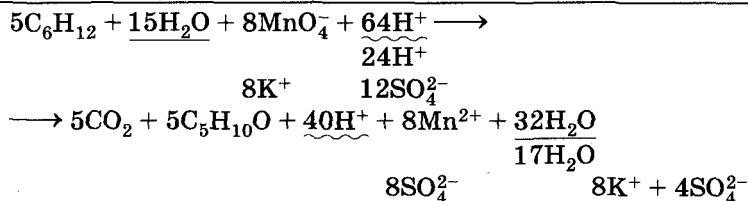
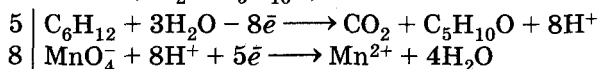
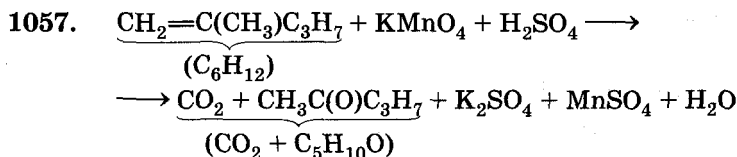
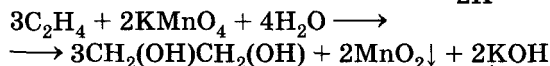
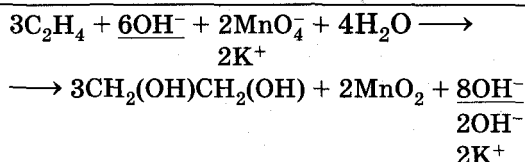
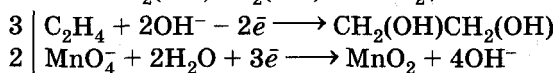
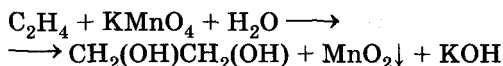
¹ Для облегчения процесса расстановки коэффициентов в уравнениях реакций с участием органических соединений в решениях задач вместо структурных формул часто приведены молекулярные формулы.

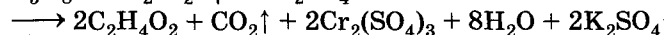
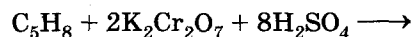
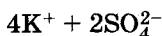
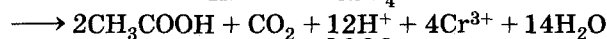
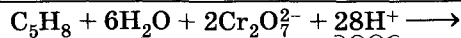
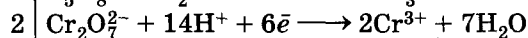
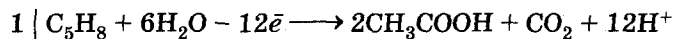
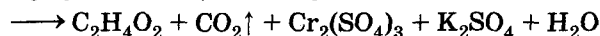
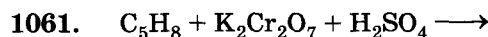
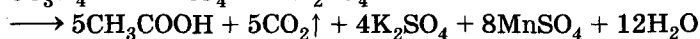
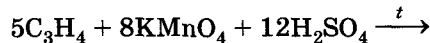
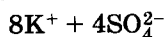
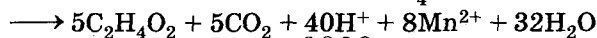
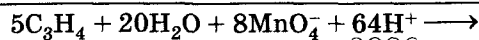
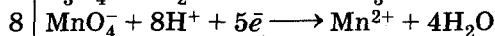
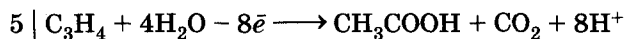
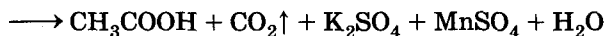
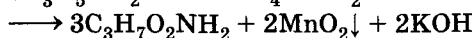
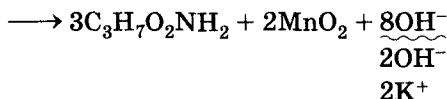
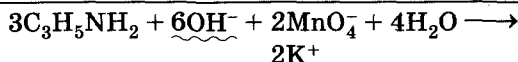
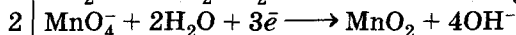
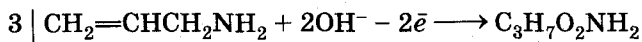
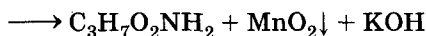
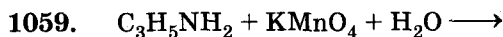


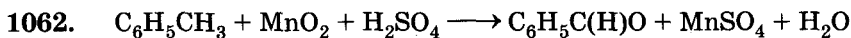
щавелевая кислота



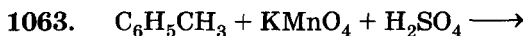
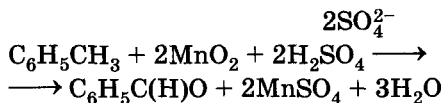
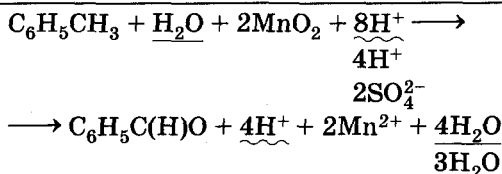
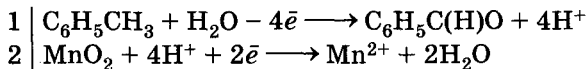
Пояснение. В кислотной среде образующиеся вначале гликоли окисляются до углекислого газа — CO_2 . В нейтральной среде при комнатной температуре алкены образуют алкандиолы:



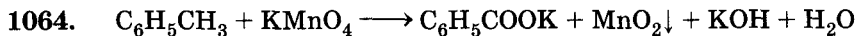
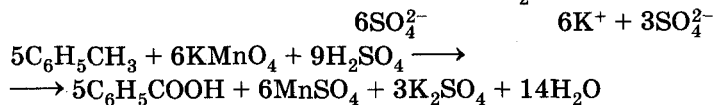
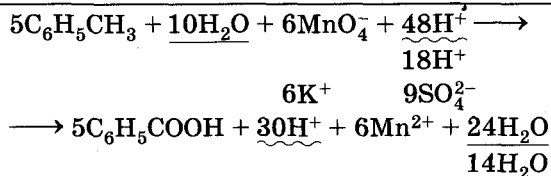
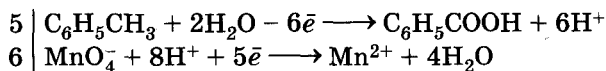




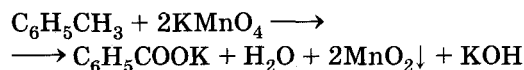
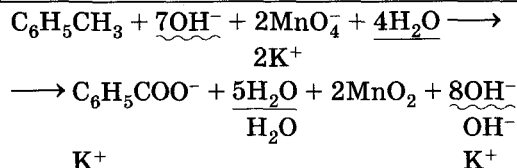
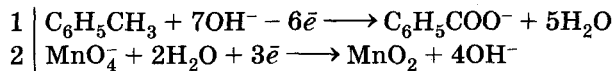
бензальдегид

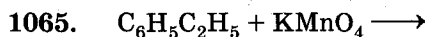


бензойная кислота

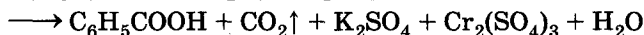
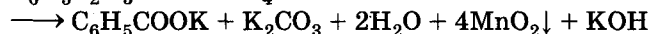
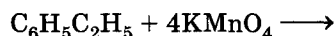
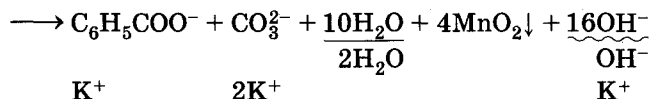
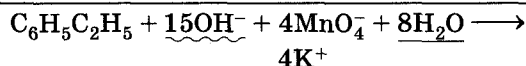
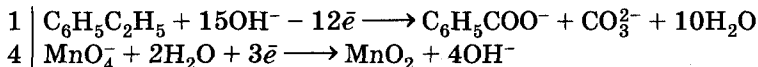
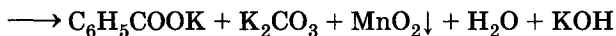


толуол

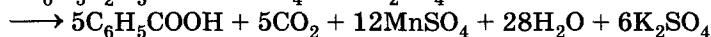
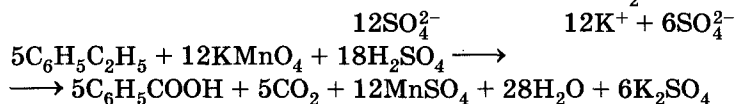
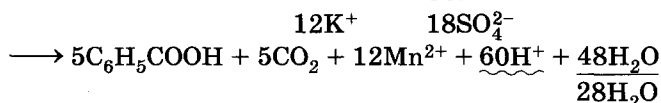
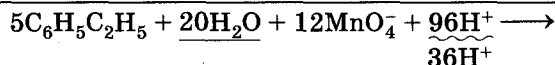
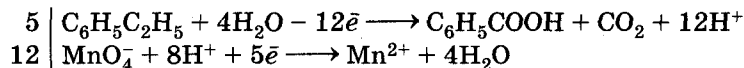
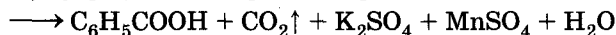
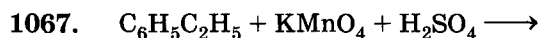
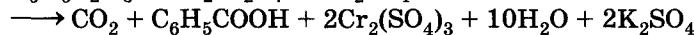
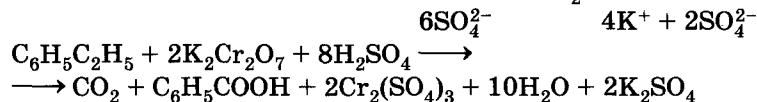
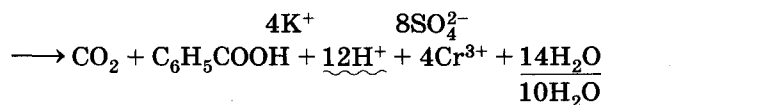
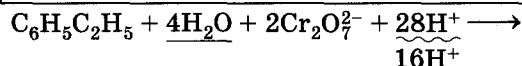
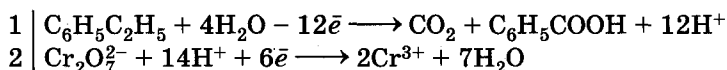


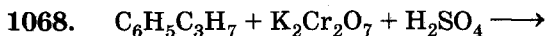


этилбензол



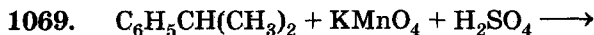
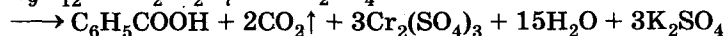
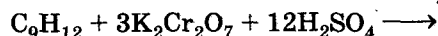
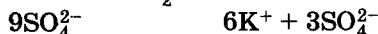
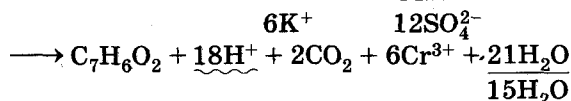
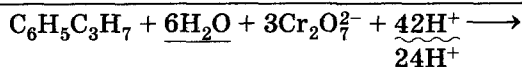
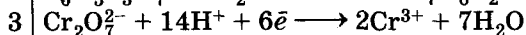
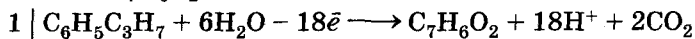
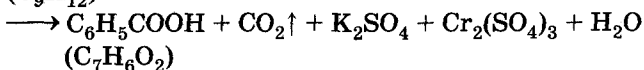
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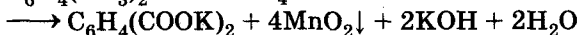
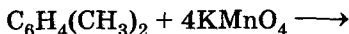
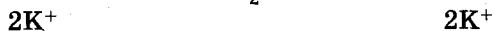
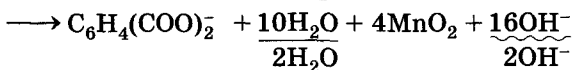
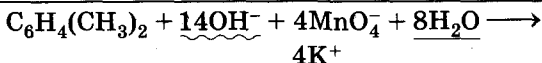
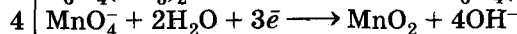
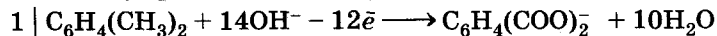
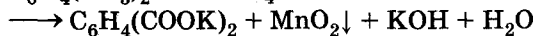
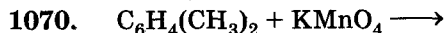
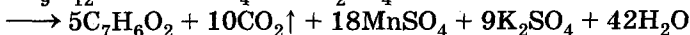
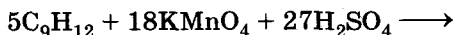
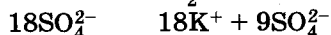
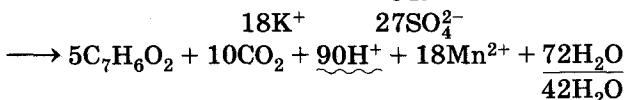
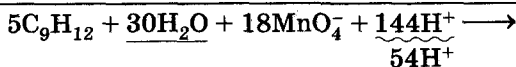
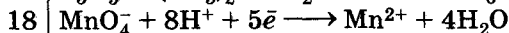
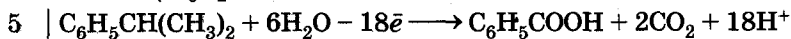
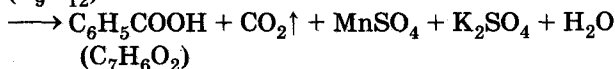


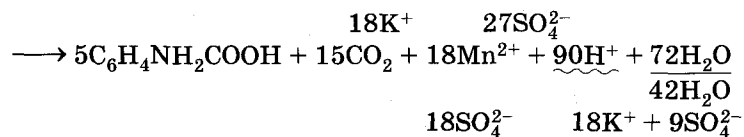
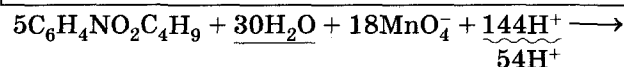
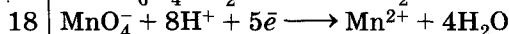
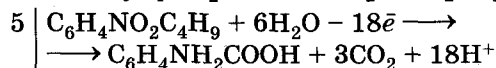
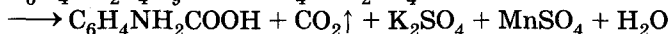
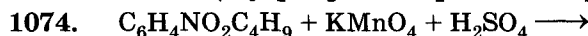
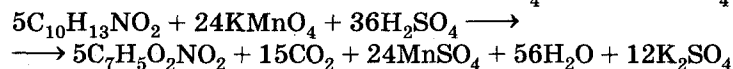
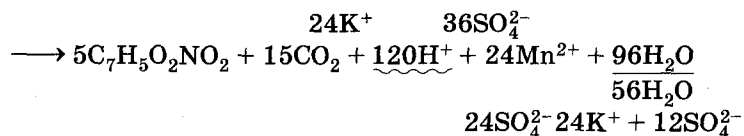
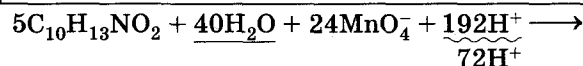
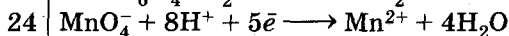
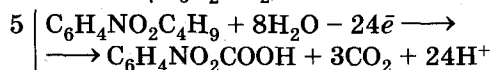
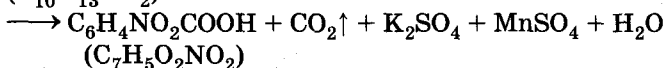
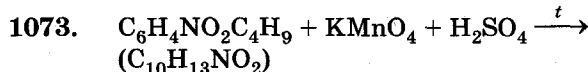
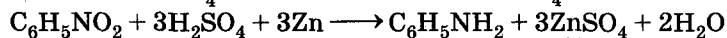
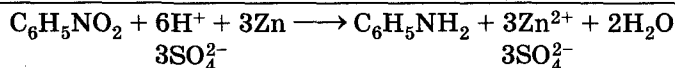
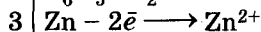
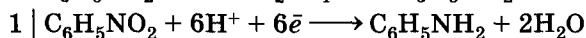
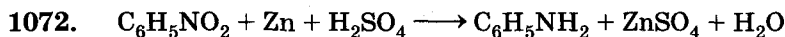
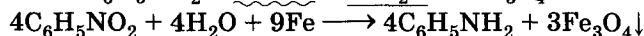
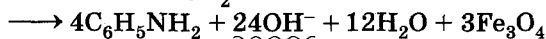
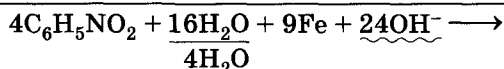
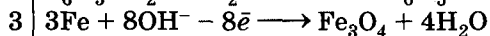
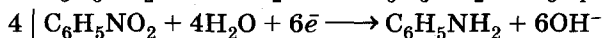
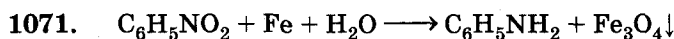
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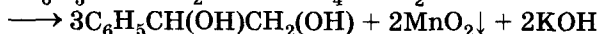
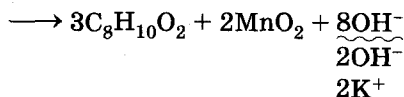
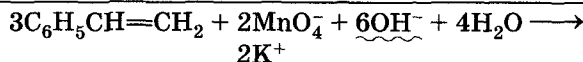
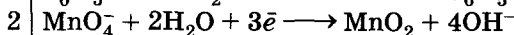
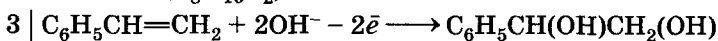
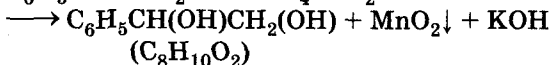
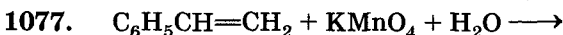
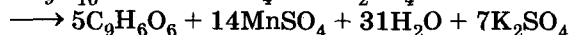
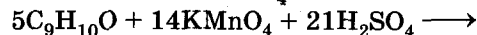
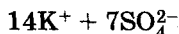
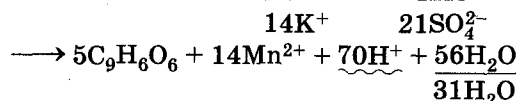
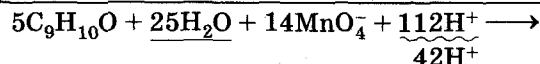
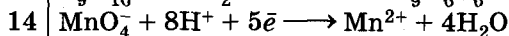
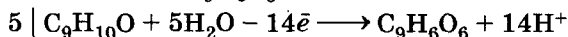
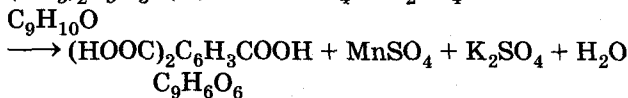
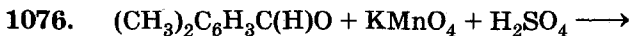
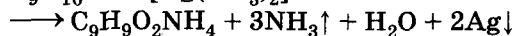
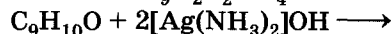
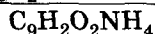
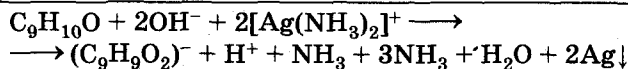
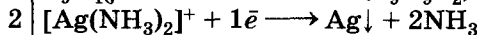
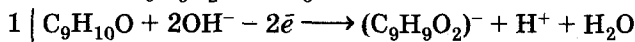
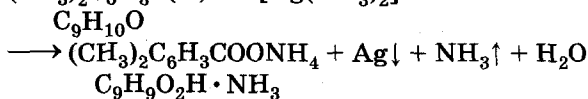
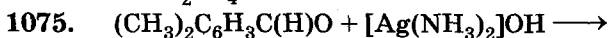
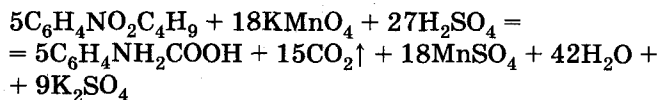
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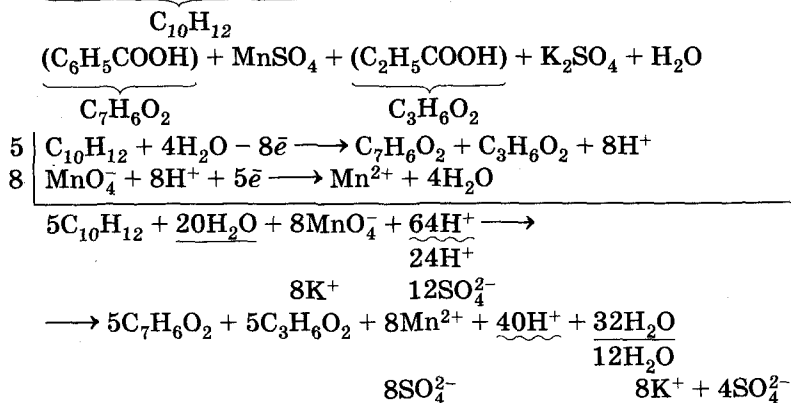
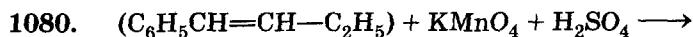
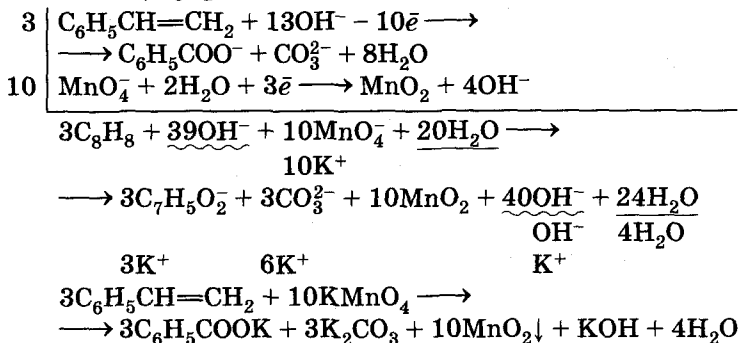
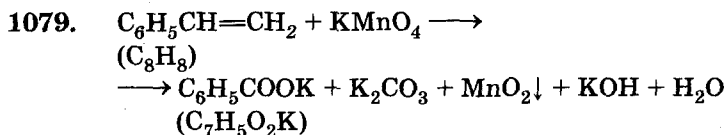
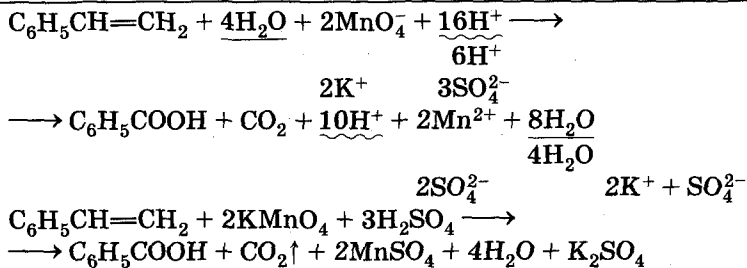
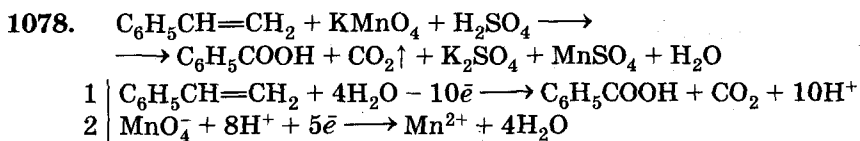


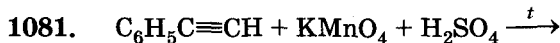
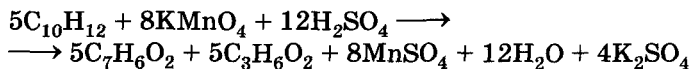
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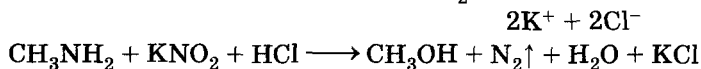
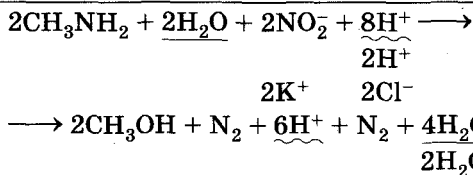
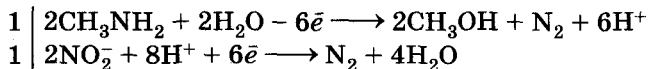
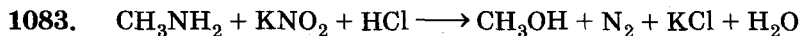
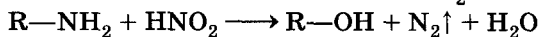
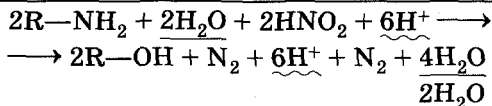
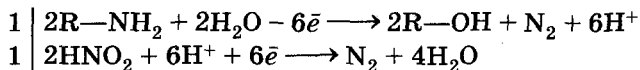
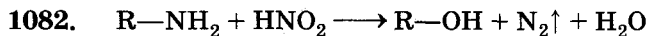
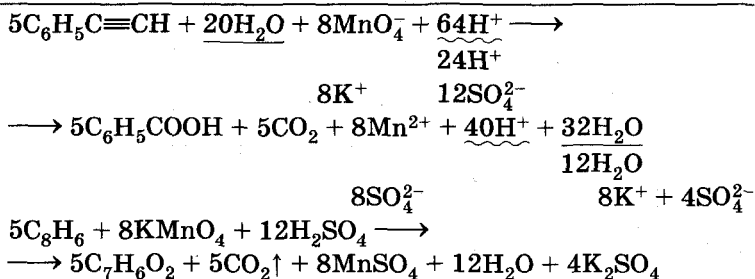
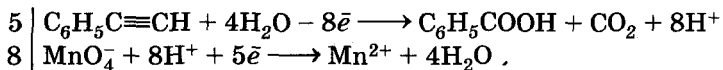
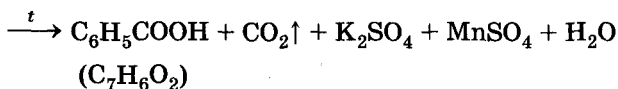


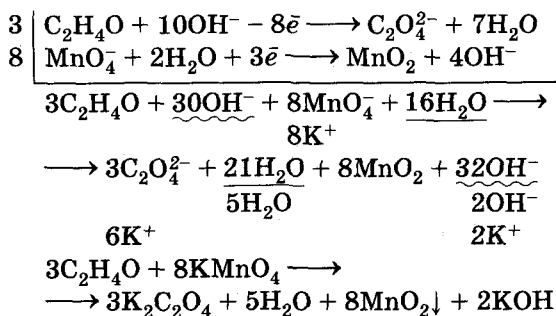
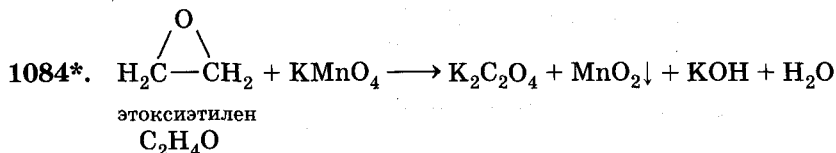




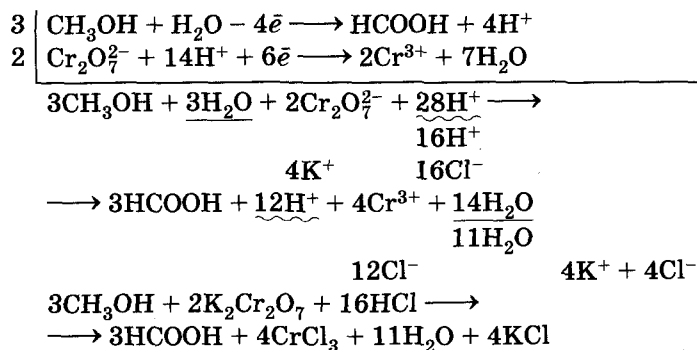
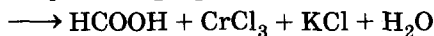
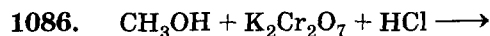
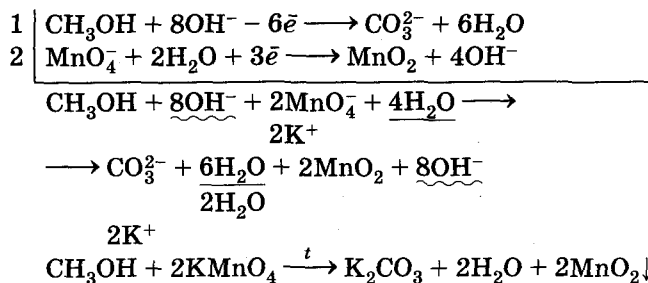
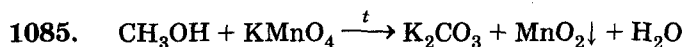


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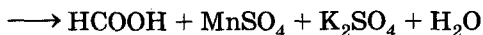


П о я с н е н и е. Степень окисления углерода в исходном соединении равна -1.

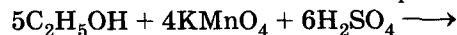
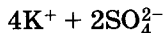
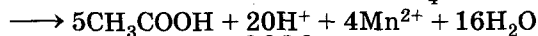
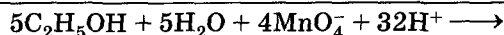
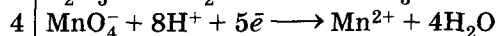
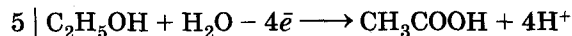
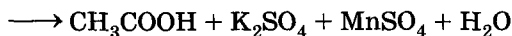
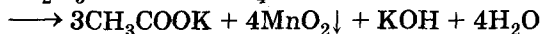
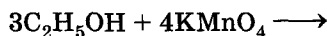
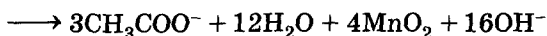
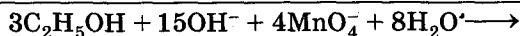
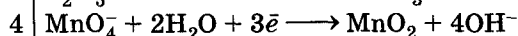
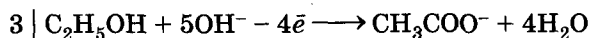
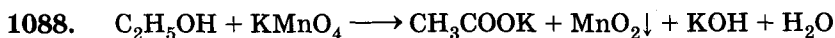
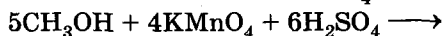
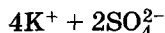
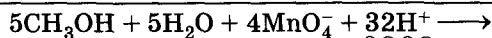
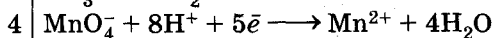
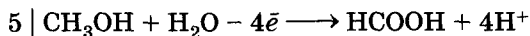


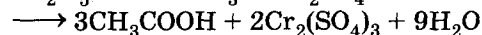
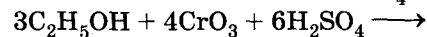
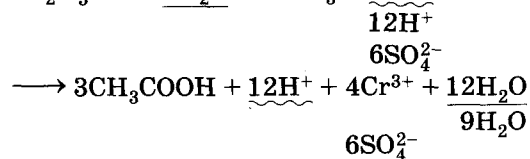
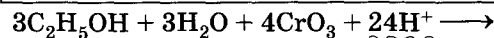
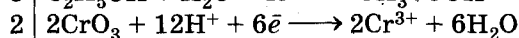
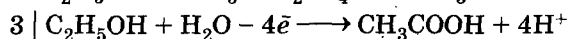
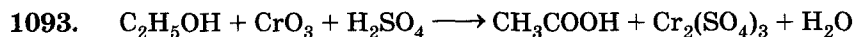
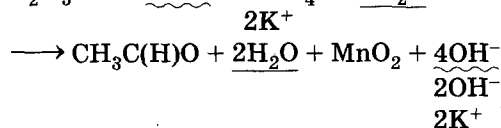
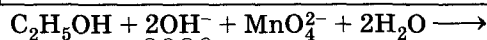
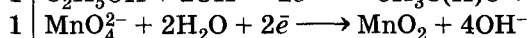
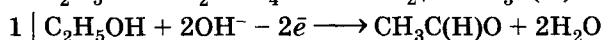
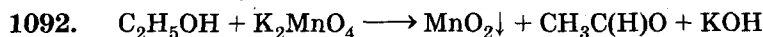
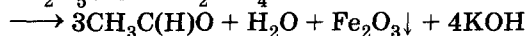
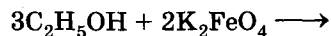
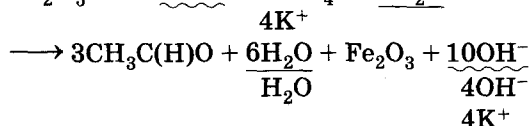
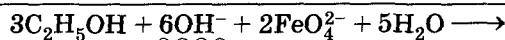
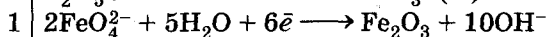
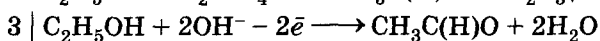
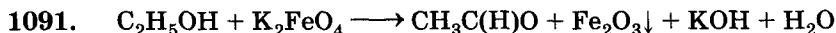
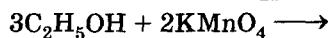
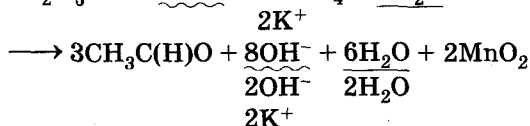
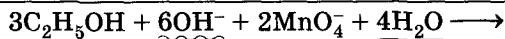
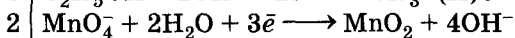
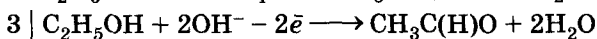
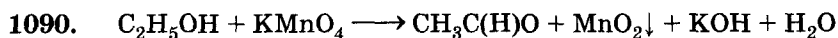


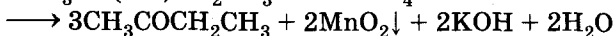
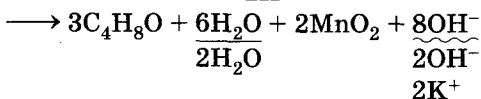
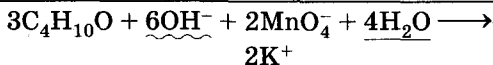
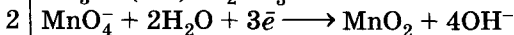
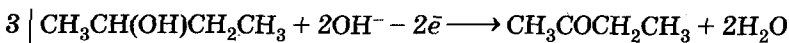
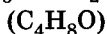
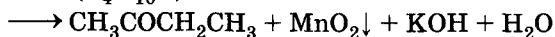
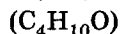
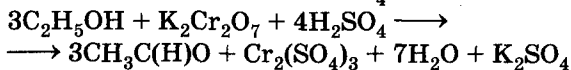
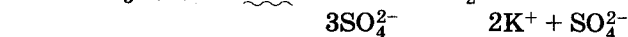
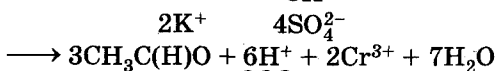
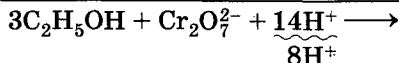
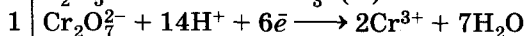
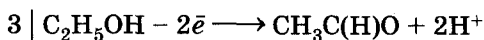
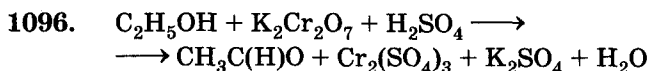
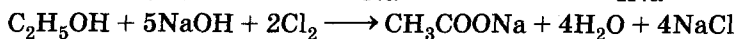
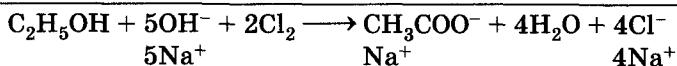
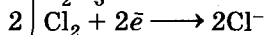
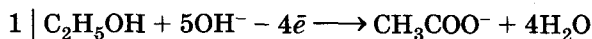
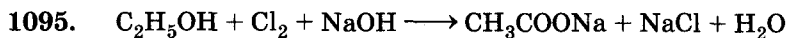
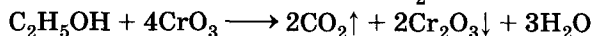
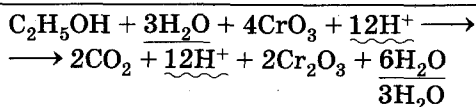
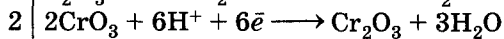
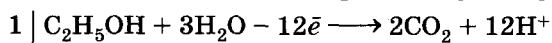
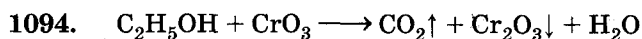
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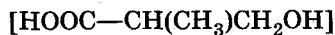


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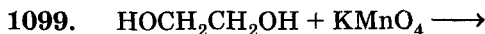
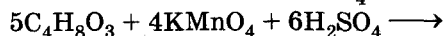
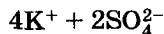
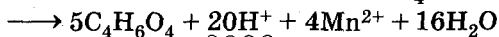
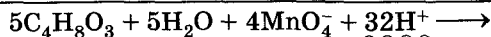
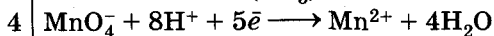
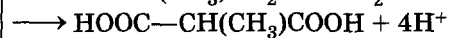




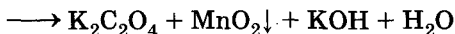
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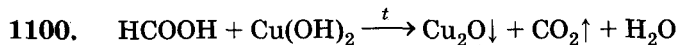
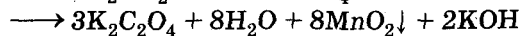
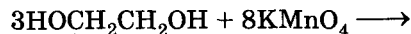
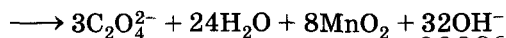
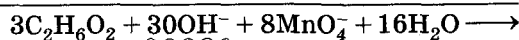
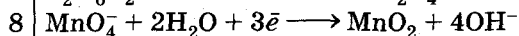
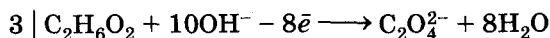
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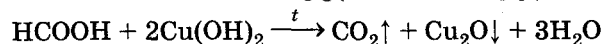
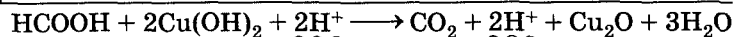
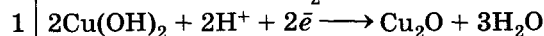
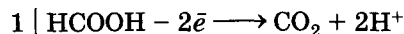
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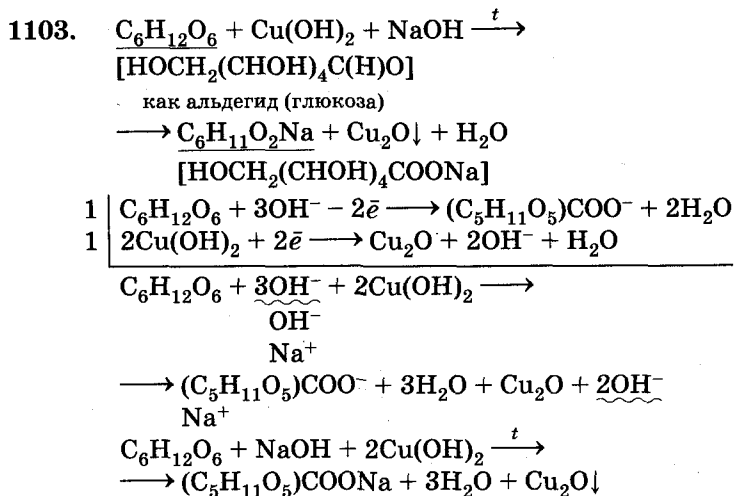
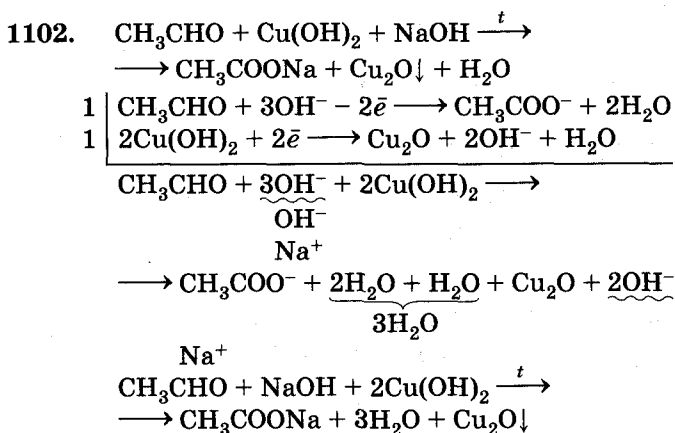
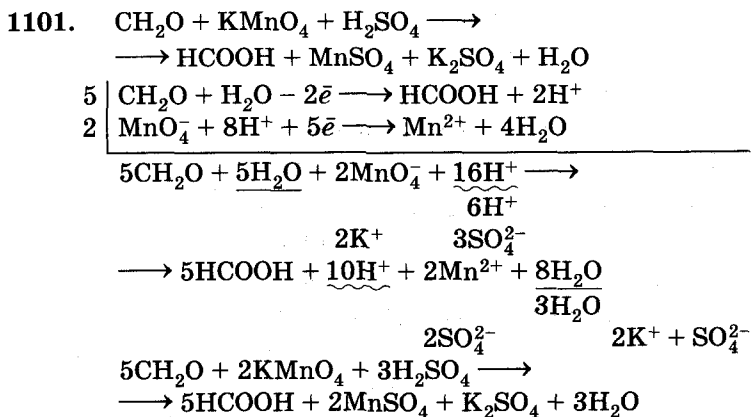


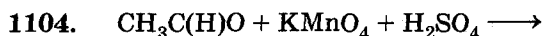
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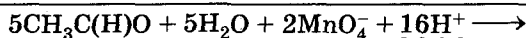
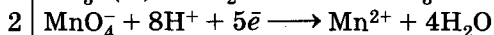
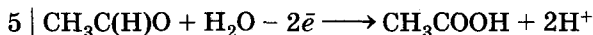
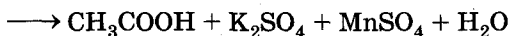
как альдегид







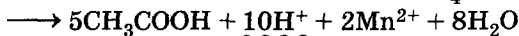
этаналь



6H^+

2K^+

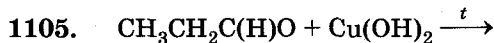
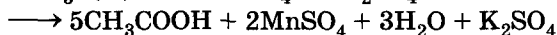
3SO_4^{2-}



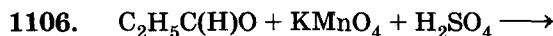
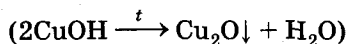
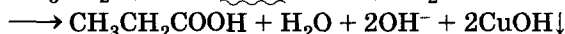
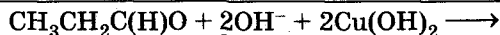
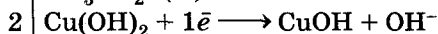
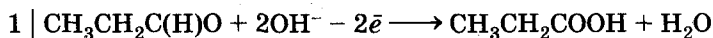
$3\text{H}_2\text{O}$

2SO_4^{2-}

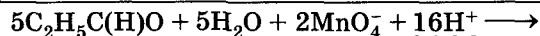
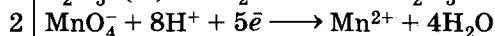
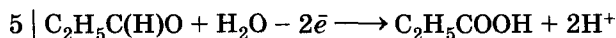
$2\text{K}^+ + \text{SO}_4^{2-}$



пропаналь



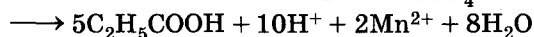
пропаналь



6H^+

2K^+

3SO_4^{2-}

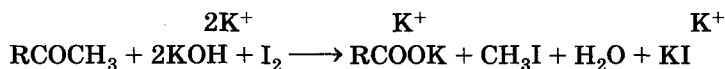
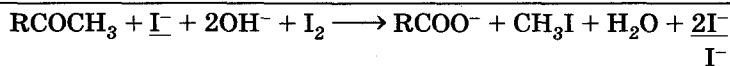
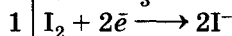
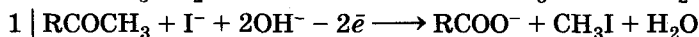
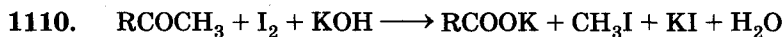
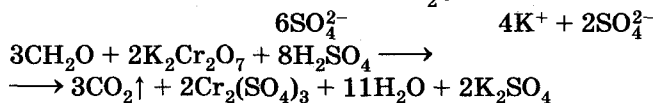
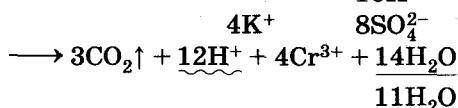
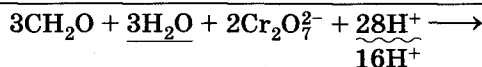
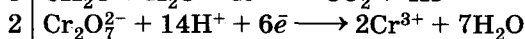
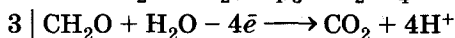
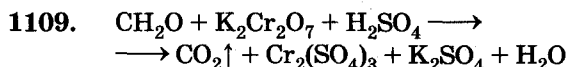
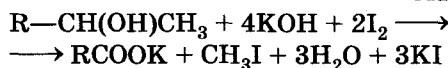
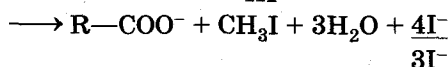
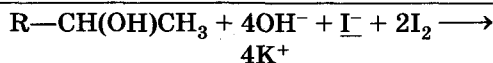
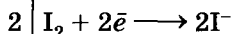
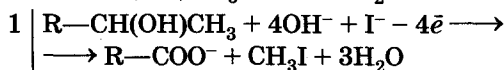
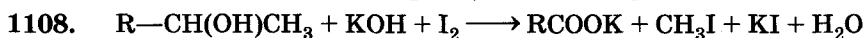
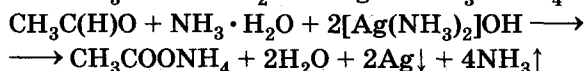
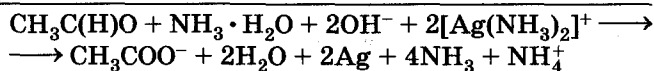
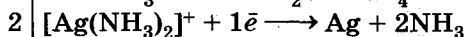
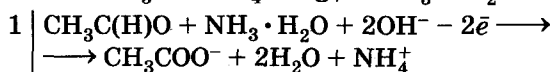
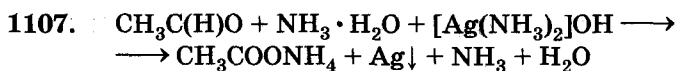


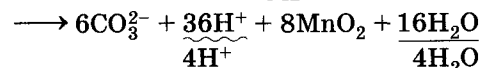
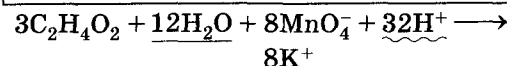
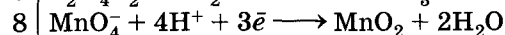
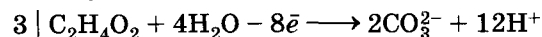
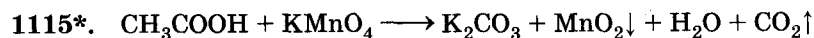
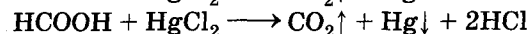
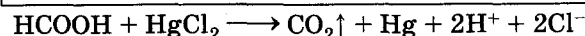
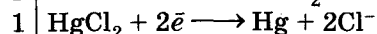
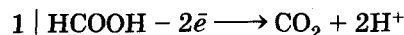
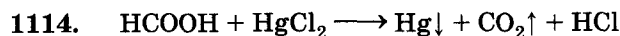
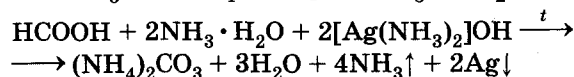
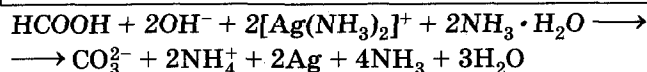
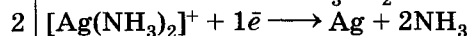
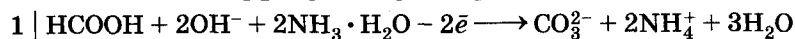
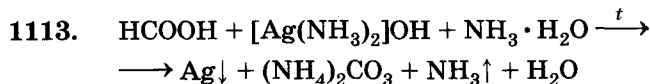
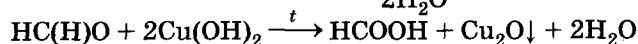
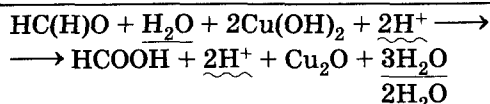
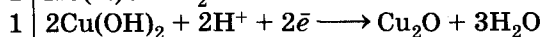
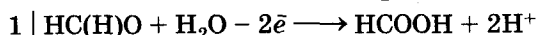
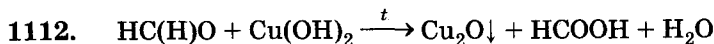
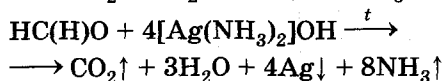
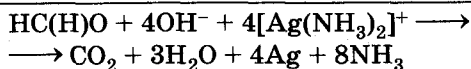
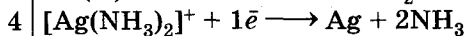
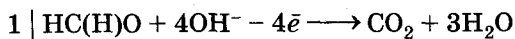
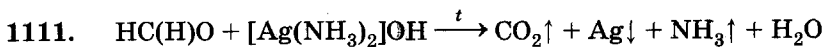
$3\text{H}_2\text{O}$

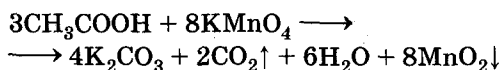
2SO_4^{2-}

$2\text{K}^+ + \text{SO}_4^{2-}$





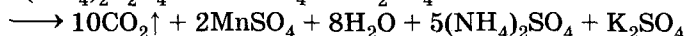
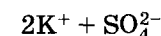
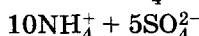
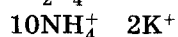
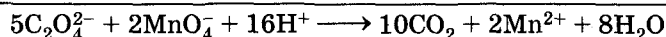
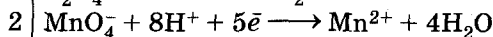
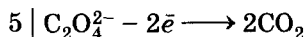
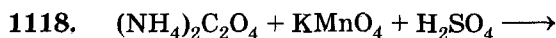
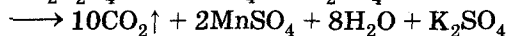
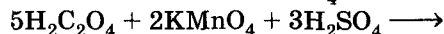
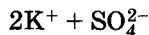
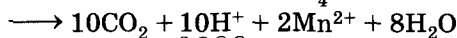
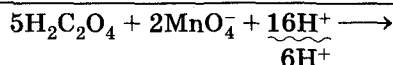
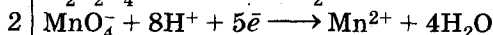
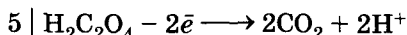
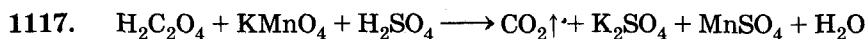
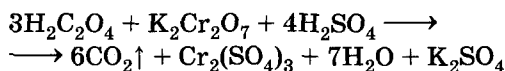
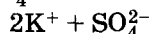
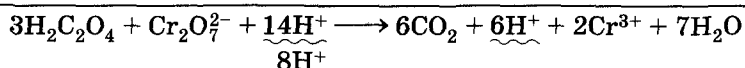
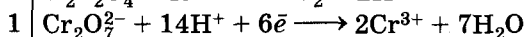
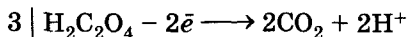
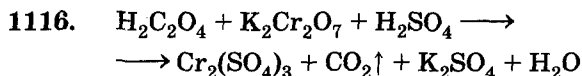


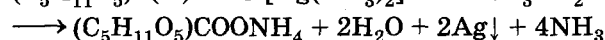
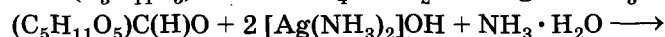
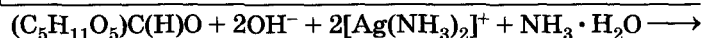
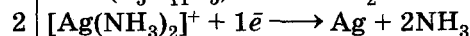
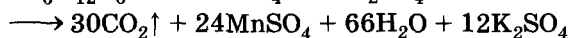
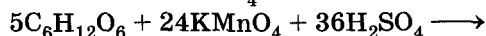
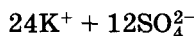
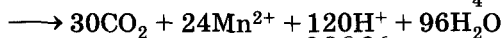
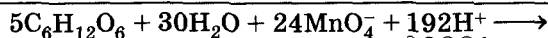
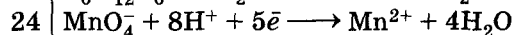
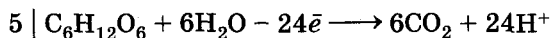
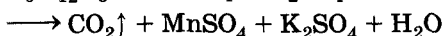
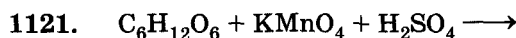
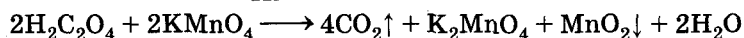
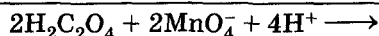
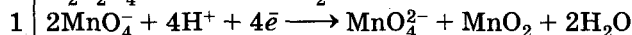
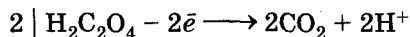
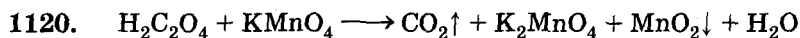
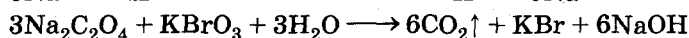
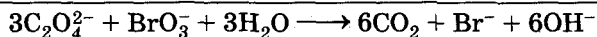
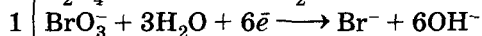
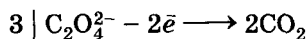
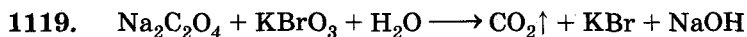


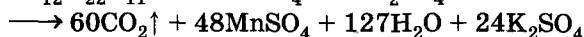
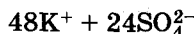
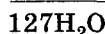
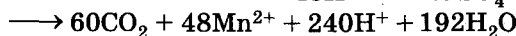
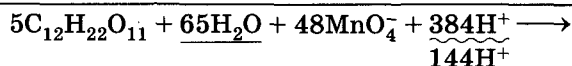
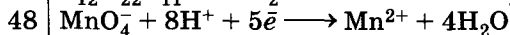
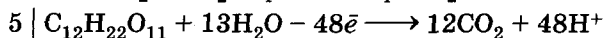
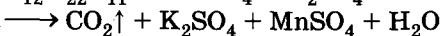
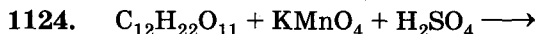
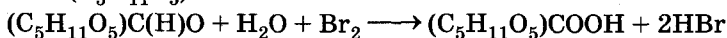
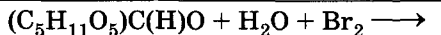
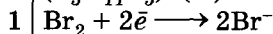
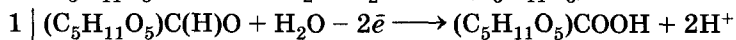
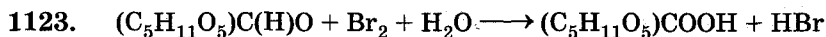
П о я с н е н и е. Распределение карбонат-ионов:



В сумме в правой части уравнения образуются $6\text{H}_2\text{O}$ ($4\text{H}_2\text{O} + 2\text{H}_2\text{O}$).







НЕОРГАНИЧЕСКИЕ ВЕЩЕСТВА

Ag 129, 192, 285, 319, 328, 375, 440, 497, 565, 757, 784, 843, 946,
948, 950, 1039, 1075, 1107, 1111, 1113, 1122

AgCl 6, 15, 565

AgClO₃ 6, 565

Ag₂CO₃ 375

AgNO₃ 15, 285, 328, 757, 838, 843, 857, 858, 946, 948, 950

[Ag(NH₃)₂]OH 497, 1039

(Ag⁺¹Ag⁺³)O₂ 838

Ag₂O 440

Ag₂S 129

Ag₂SO₄ 192, 784

Al 52, 70, 75, 108, 157, 228, 254, 276, 281, 331, 373, 415, 464, 467,
570, 633, 634, 707, 708, 780, 855, 881, 1024, 1025, 1026, 1051

Al₄C₃ 20, 331, 333

AlCl₃ 20, 70

AlI₃ 52

AlN 28

Al(NO₃)₃ 254, 319, 707, 708

Al₂O₃ 75, 149, 175, 283, 333, 373, 415, 464, 467

Al₂S₃ 108, 149, 625

Al₂(SO₄)₃ 148, 157, 175, 570, 626, 633, 634, 780

As 17, 909

AsH₃ 767, 768, 773, 775, 946

As₂O₃ 768—772, 946

As₂O₅ 772

As₂S₃ 603, 777

As₂S₅ 267, 776

Au 441, 502, 686, 734, 817, 927, 1020, 1041

AuCl₃ 441, 817, 926, 927

AuCl₂ 926

Au₂(SeO₄)₃ 686, 695

¹ Формулы веществ расположены в алфавитном порядке с указанием номеров уравнений ОВР, в которых они встречаются.

В 247, 396, 713
 B_2O_3 396
 $\text{B}(\text{OH})_3$, или H_3BO_3 , 247, 713
 Ва 200, 286
 BaCO_3 346, 358, 373
 $\text{Ba}(\text{CN})_2$ 358
 BaCl_2 810
 $\text{Ba}(\text{ClO})_2$ 183
 $\text{Ba}(\text{ClO}_3)_2$ 810
 BaH_2 286
 $\text{Ba}(\text{HS})_2$ 610
 $\text{Ba}(\text{H}_2\text{PO}_2)_2$ 312, 865
 Ba_3N_2 286
 $\text{Ba}(\text{NO}_2)_2$ 850
 $\text{Ba}(\text{NO}_3)_2$ 610, 850
 BaO 200, 286, 346, 373, 374
 BaO_2 638, 657
 $\text{Ba}(\text{OH})_2$ 312, 810, 850, 865
 BaS 345
 BaSO_3 671
 BaSO_4 183, 345, 638, 657, 671
 Би 218, 700, 818, 819, 1022
 BiCl_3 818, 819
 $\text{Bi}(\text{NO}_3)_3$ 176, 218, 664, 700, 787, 788, 803
 $\text{BiNO}_3(\text{OH})_2$ 788, 794
 $\text{Bi}(\text{OH})_3$ 413, 841, 891, 1022, 1037
 $\text{BiO}(\text{OH})$ 1038
 Br_2 4, 32, 34—47, 126, 314, 400, 404, 405, 410, 419, 430, 554—
 558, 617, 641, 644, 645, 658, 740, 759, 779, 813, 821—826,
 861, 884, 888, 892, 896, 898, 902, 911, 912, 929, 935, 941, 949,
 1013, 1036, 1042, 1123
 Br_2O 36
 С 21, 26, 161—163, 165, 198, 199, 204, 248, 330—359, 369, 370,
 373, 448, 471, 630, 710
 CO 18, 103, 162, 163, 165, 333, 334, 336, 339, 340, 341, 343, 347,
 352, 353, 357, 358, 360—367, 369, 449, 472, 503, 573
 CO_2 26, 49, 59, 60, 103, 142, 161, 163, 165, 194, 198, 204, 248,
 265, 335, 337, 342, 348, 349, 350, 351, 354, 356, 360—362,
 364, 368, 371, 372, 375, 399, 406, 450, 457, 471, 472, 477—
 481, 484, 485, 490, 494, 498, 500, 501, 502, 504, 505, 507, 573,
 630, 710, 736, 779, 796, 1000, 1039, 1056, 1060, 1061, 1066—

1069, 1073, 1074, 1078, 1081, 1100, 1109, 1111, 1115, 1117,
 1119—1121, 1124
 CCl_2O 18
 CCl_4 10, 20
 CS_2 355
 Ca 63, 217, 227, 258, 259, 310, 332, 371, 692, 693
 CaCN_2 199
 CaC_2 19, 199, 332, 339, 371
 CaCO_3 142, 163, 338, 349, 363
 CaCl_2 19, 24, 63, 64, 69, 84, 514, 515, 562, 809, 934, 935, 964, 1012
 $\text{Ca}(\text{ClO})_2$ 24, 84, 514, 515, 562, 809, 934—936, 964, 1021
 $\text{Ca}(\text{OH})_2$ 34, 324, 462, 809
 CaH_2 64, 69, 80, 227, 461—464
 $\text{Ca}(\text{HS})_2$ 139
 CaI_2 729
 Ca_3N_2 217, 227, 463
 $\text{Ca}(\text{NO}_3)_2$ 258, 259, 692, 693, 716, 729
 CaO 80, 217, 322, 338, 339, 353, 363, 461, 464
 Ca_3P_2 310, 322—325, 352, 635, 716, 761, 762, 910
 $\text{Ca}_3(\text{PO}_4)_2$ 322, 323, 325, 351, 352, 635, 761, 762, 910
 CaS 163, 349
 CaSO_4 139, 142, 165, 353, 562, 936, 940
 CaSiO_3 351
 CF_4 330
 Cd 709
 $\text{Cd}(\text{NO}_3)_2$ 709
 CeCl_2 19
 CeCl_3 543
 CeO_2 543
 $\text{Ce}(\text{SO}_4)_2$ 984
 $\text{Ce}_2(\text{SO}_4)_3$ 498, 984
 Cl_2 1—32, 34, 38—40, 59, 60, 62, 66, 67, 76, 86, 87, 92, 295, 308,
 325, 357, 424, 437, 460, 465, 509—528, 530, 569, 674, 682,
 732, 758, 795, 806—813, 830, 831, 842, 870, 885, 902—910,
 922, 949, 952, 960, 976, 979, 988, 1032, 1045, 1095, 1116
 ClO_2 6, 77, 81, 86—90, 563, 567, 568, 648, 814
 ClO_3 1
 Cl_2O 86
 Cl_2O_6 89, 91
 ClF 3
 Cl_3N 12, 13
 Co 18, 694

CoO(OH) 845, 1014
 $\text{Co(NO}_3)_2$ 694
 CoS 845
 Cr 61, 347, 397
 CrCl_2 887
 CrCl_3 61, 357, 397, 413, 488, 520, 521, 522, 533, 578, 761, 888—891, 1086
 $\text{Cr(ClO}_4)_3$ 802
 $(\text{Cr}_2\text{Fe})\text{O}_4$ 347, 401, 406
 $\text{Cr(NO}_3)_3$ 403, 803, 852
 Cr(OH)_2 1015
 Cr(OH)_3 180, 411, 885, 886, 925, 955, 959, 965, 972, 974, 978, 1015
 CrO_3 402, 520, 660, 673, 1093, 1094
 Cr_2O_3 116, 117, 348, 357, 398—400, 402, 403, 407, 412, 884, 1094
 CrO_5 ($\text{CrO(O}_2)_2$) 790
 Cr_2S_3 116
 $\text{Cr}_2(\text{SO}_4)_3$ 135, 172, 177, 194, 297, 404, 405, 408, 492, 539, 549, 579, 600, 604, 605, 654, 660, 673, 675, 752, 796, 798, 800, 892—894, 991, 1052, 1061, 1066, 1068, 1093, 1096, 1109, 1116
 CrSO_4 797, 801
 CsCl 460
 CsO_3 460
 Cu 70, 155, 168, 201, 232, 241, 252, 253, 271, 340, 471, 472, 486, 502, 503, 508, 562, 631, 696, 697, 980
 CuBr 556, 639, 731
 CuBr_2 639
 CuCl 820, 924
 CuCl_2 70, 241, 508, 820, 924
 CuFeS 621
 CuI 728, 923, 945, 983
 $\text{Cu(NO}_3)_2$ 252, 253, 266, 320, 473, 623, 625, 696, 697, 723, 728, 731, 945
 CuO 142, 143, 174, 232, 241, 320, 340, 470, 486
 Cu_2O 201, 407—474, 502, 503, 723, 1110, 1102, 1103, 1112
 CuOH 1105
 Cu(OH)_2 923, 1100, 1102, 1103, 1105, 1112
 CuS 622
 Cu_2S 142, 143, 266, 623
 CuSO_4 155, 168, 174, 271, 474, 556, 562, 622, 623, 631, 639, 980, 983
 F_2 3, 53—58, 104, 330, 828, 829, 835

Fe 31, 65, 110, 128, 160, 257, 326, 347, 360, 408, 467, 468, 695,
 801, 854, 1023, 1071
 FeAsS 737
 FeAsO₄ 737
 FeBr₂ 591
 FeBr₃ 591
 FeHPO₄ 326
 Fe₃C 265, 736
 FeCO₃ 779
 FeCl₂ 65, 66, 68, 98, 137, 298, 529—532, 536, 730, 742, 782, 792,
 815, 816, 918, 961, 963
 FeCl₃ 31, 67, 68, 98, 137, 145, 298, 529, 530—532, 536, 593, 742,
 782, 793, 918, 961, 963
 FeI₂ 223, 920
 Fe(NO₃)₃ 152, 222, 223, 257, 263—265, 284, 617, 618, 621, 653,
 695, 724—727, 730, 736, 790
 Fe(NO₃)₂ 283, 790
 FeO 264, 725
 Fe₂O₃ 67, 134, 153, 154, 283, 284, 406, 452, 468, 1091
 Fe₃O₄ 167, 264, 360, 469, 727, 1071
 Fe(OH)₂ 188, 222, 433, 724, 827, 840, 938, 1006—1009
 FeO(OH) 66, 188, 244, 433, 806, 815, 816, 826, 840, 844, 863, 920,
 938, 958, 1003—1009
 Fe₃(PO₄)₂ 23, 26
 FeS 107, 128, 134, 145, 150, 154, 593, 620, 961
 FeS₂ 137, 151—153, 452, 617—619
 Fe₂(SO₄)₃ 150, 151, 159, 160, 167, 168, 170—173, 408, 619, 620,
 650—652, 654—660, 742, 779, 782, 783, 801, 904, 980
 FeSO₄ 160, 168—173, 650—660, 844, 904, 980, 981
 (FeSO₄)OH 167, 981
 HMnO₄ 176, 281, 291, 418, 663, 664, 741, 778, 786—789, 794
 Ge 867
 H₂ 2, 63, 64, 65, 68, 69, 72, 107, 130, 160, 195, 215, 226, 228, 240,
 303, 306, 326, 327, 334, 338, 363, 365, 380—382, 384, 385,
 388, 390, 462, 464, 468, 1016, 1021, 1023, 1024
 HAsO₃ 770
 H₃AsO₄ 17, 267, 767, 769, 776, 777, 909
 H[AuCl₄] 592, 734
 HBr 33, 126, 314, 554, 591, 644, 658, 665, 731, 740, 911, 912, 941,
 1123
 HBrO 655

HBrO₃ 32, 902

HCN 372

HCl 2, 5, 8, 10, 12, 16, 17, 25—28, 30, 32, 38, 59—69, 76, 85, 98, 122, 126, 137, 166, 186, 260, 295, 297, 314, 325, 329, 364, 388, 424, 437, 441, 444, 454, 460, 465, 488, 504, 505, 508—522, 529—533, 543, 544, 557, 560, 561, 563, 564, 566, 578, 584, 588—590, 592—594, 603, 608, 612, 668, 669, 677—679, 682, 683, 689, 717, 730—734, 739, 743, 745, 751, 756, 761, 764, 769, 772, 774, 781, 792, 795, 805, 902, 905, 907—911, 922, 926—928, 930, 936, 949, 952, 960, 963, 967, 979, 988, 1083, 1086, 1114

HClO 37, 85, 560, 561, 679, 769, 908

HClO₂ 90, 516, 563

HClO₃ 39, 519, 563, 564—566, 588, 688

HClO₄ 40, 81, 122, 569, 584, 588, 802

H₂Cr₂O₇ 400, 803

HF 58, 240, 383, 559, 1040

HI 33, 42, 43, 47, 48, 85, 93—95, 97, 98, 238, 296, 506, 534—548, 572, 580, 643, 670, 672, 735, 742, 765, 802, 913—915, 917, 973

HIO₃ 38, 39, 45, 50, 97, 251, 534, 571, 572, 587, 659, 711, 712, 735, 905, 916, 930

HIO₄ 40, 569

HNO₃ 45, 97, 114, 139, 147, 148, 150, 176, 193, 211—213, 243, 245—270, 281, 290, 291, 293—295, 319, 321, 328, 383, 418, 469, 473, 527, 528, 532, 540, 559, 561, 574, 583, 598, 599, 610—614, 652, 653, 663, 664, 671, 680, 681, 684, 692—737, 740, 741, 744, 753—755, 759, 770, 776, 777, 787—790, 794, 803, 804, 946, 948, 950, 969, 1040

HNO₂ 209, 290—296, 740—747, 939, 968, 969, 1082

H₂O 5, 8, 10, 17, 26—28, 32, 38, 43, 47—49, 58, 66, 84, 101, 106, 124, 126, 136, 178, 180, 212, 213, 306, 307, 313, 325, 328, 334, 363, 364, 382, 384, 390, 400, 411, 413, 420, 426, 443, 458, 459, 506, 561, 564, 566, 572, 576, 577, 580—582, 584, 585, 590, 672, 685, 706, 714, 740, 757, 758, 863, 902, 905, 907, 909, 910, 911, 914, 920—932, 934—936, 939, 941—946, 948—955, 957, 959—970, 972, 973, 975, 976, 978, 979, 981, 982, 985, 986, 988, 990—994, 996, 997, 1000—1004, 1015—1019, 1021, 1023—1030, 1037, 1041, 1051, 1053, 1059, 1071, 1077

H₂O₂ 25, 50, 90, 140, 146, 185, 233, 282, 405, 417, 432—441, 454, 466, 537, 542, 560, 571, 602, 609, 615, 626, 656, 666, 697, 747, 793, 798, 799, 804, 805, 837, 844, 845, 852, 856, 850, 867, 882,

883, 889, 893, 895, 900, 901, 916, 919, 934, 962, 971, 984, 990,
 998, 1008, 1010, 1027, 1033, 1044, 1047
 H_3PO_3 47, 506
 $\text{H}(\text{PH}_2\text{O}_3)$ 311
 $\text{H}_2(\text{HPO}_3)$ 269, 328, 636, 637, 718, 719, 755, 758, 764, 765, 771
 H_3PO_4 250, 306, 314, 319, 326, 327, 328, 564, 572, 714—717, 719,
 721, 755—760, 763, 766, 907, 948, 949, 950
 H_3PO_2 ($\text{H}(\text{PH}_2\text{O}_2)$) 48, 329, 718, 764, 771, 917
 H_3SbO_4 514
 $\text{H}_2[\text{PtCl}_6]$ 260, 733
 H_2S 8, 42, 99, 104, 127—137, 156, 157, 189, 268, 285, 587, 602,
 628, 629, 633, 642, 669, 678, 836, 913, 955—960
 $\text{H}_2\text{SO}_3(\text{O}_2)$ 552
 H_2SO_4 8, 10, 43, 77, 94, 96, 99, 106, 112, 114, 122, 124, 126, 132,
 133, 135, 139, 144, 150, 152, 155—161, 170—173, 178, 189,
 191, 194, 237, 267, 268, 290, 297, 311, 323, 400, 408, 411, 414,
 422, 425, 436, 453, 474, 492—494, 498, 523—526, 539, 541,
 542, 545—551, 553, 555, 556, 558, 562, 566—586, 588—590,
 592, 596—602, 604—606, 612, 617—620, 624, 626—648, 650—
 662, 665—667, 670, 673—676, 678—680, 682, 684, 691, 738,
 746—748, 750, 752, 762, 763, 766—768, 773, 775—789, 782—
 785, 791, 796, 798, 801, 912, 914, 928, 936, 951—954, 957,
 960, 977, 979, 982, 984, 985, 988, 990—994, 996, 997, 1043,
 1044, 1052, 1054, 1056—1058, 1060—1063, 1066—1069,
 1072—1074, 1076, 1078, 1080, 1081, 1087, 1089, 1093, 1096,
 1098, 1101, 1104, 1106, 1109, 1116—1118, 1121, 1124
 H_2SeO_4 509, 687, 688
 H_2SeO_3 249, 509, 585, 927
 $\text{H}_2\text{S}_2\text{O}_8$ 990
 $\text{H}_2[\text{SnCl}_6]$ 668, 743
 $\text{H}_2[\text{SiF}_6]$ 383, 559
 $\text{H}_2[\text{TaF}_7]$ 1040
 H_2TeO_3 586
 Hg 282, 329, 738, 764, 1049, 1050
 HgBr_2 36
 HgCl_2 329, 612, 756, 764, 928, 1114
 Hg_2Cl_2 756, 928
 $\text{Hg}(\text{NO}_3)_2$ 282
 $\text{Hg}_2(\text{NO}_3)_2$ 1049
 HgO 36
 Hg_2I_2 1050
 HgS 612

Hg₂SO₄ 738

I₂ 33, 38—52, 85, 93—96, 98—103, 190, 223, 238, 250, 296, 362, 439, 453, 534—553, 569, 571, 573, 580, 581, 587, 640, 642, 643, 659, 661, 672, 683, 690, 711, 712, 728, 729, 742, 746, 749, 765, 774, 802, 839, 862, 905, 913—921, 923, 925, 930, 945, 973, 983, 987, 1048, 1108, 1110

ICI 930

I₂O₅ 102, 103, 362

I₃N 862

K 158, 216, 389, 456, 632

K₃AsO₃ 831

K₃AsO₄ 17, 831

K[Al(OH)₄] 276, 1025, 1026

KBr 4, 430, 544, 554, 557, 558, 645, 824—826, 871, 888, 929, 1042, 1119

KBrO 871

KBrO₃ 34, 41, 544, 557, 772, 813, 1119

KBiO₃ 825

K₂CO₃ 348, 350, 359, 451, 457, 495, 1065, 1079, 1085, 1115

KCN 359, 1049

KCl 4, 22, 23, 29, 30, 60, 62, 71, 74—76, 78—80, 82, 111, 169, 171, 178, 183, 186, 275, 280, 424, 434, 454, 465, 488, 513, 517, 518, 521, 522, 527—529, 530, 533, 543, 544, 546, 557, 570, 578, 651, 745, 761, 772, 774, 781, 782, 795, 806, 813, 815—817, 819, 831, 866, 870, 871, 887, 903, 918, 924, 931, 1031, 1045, 1083, 1086

KClO 22, 280, 517, 866, 931

KClO₂ 88

KClO₃ 23, 34, 74—80, 88, 91, 111, 169—171, 178, 434, 518, 526, 529, 546, 567, 648, 651, 674

KClO₄ 77, 82, 91, 567, 570, 802

K₂CrO₄ 177, 393, 407, 411, 886, 888, 895, 897, 900, 925, 959, 972, 978, 1029

K₂Cr₂O₇ 117, 135, 172, 180, 194, 297, 348, 407—409, 488, 492, 498, 522, 533, 539, 578, 579, 600, 604, 605, 654, 675, 752, 761, 796—799, 801, 802, 955, 974, 991, 1031, 1052, 1061, 1066, 1068, 1086, 1096, 1109, 1116

K₃[Cr(OH)₆] 887, 897, 900, 1029, 1031

KCr(SO₄)₂ 895

KF 389

K_2FeO_4 159, 169, 244, 647, 806, 826, 827, 854, 863, 958, 1003—1005, 1091
 K_3FeO_4 1003
 $K_3[Fe(CN)_6]$ 30, 100, 989, 1042—1048
 $K_4[Fe(CN)_6]$ 30, 989
 KH 465
 $KHCO_3$ 1000
 $K_2[HgI_4]$ 1050
 $K_2(PO_3)$ 879
 KH_2PO_2 307, 878
 $KHSO_4$ 77, 567, 640, 645, 648, 999
 $KHSO_3$ 999
 KI 5, 44, 99—101, 439, 453, 539—552, 640, 746, 748, 749, 774, 918, 919, 921, 922, 924, 1048, 1050, 1108, 1110
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 KIO_4 548
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 KNCS 194, 794
 KNO_2 115, 210, 216, 278, 438, 649, 748—750, 846—849, 851, 853, 854, 941, 1083
 KNO_3 71, 115, 193, 214, 272—278, 280, 350, 353, 432, 455, 527, 528, 540, 671, 680, 684, 745, 750, 838, 848, 849, 853, 854, 857—859, 941—945, 996, 1017, 1025, 1026, 1049
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834, 839—840, 847—849, 851, 853, 854, 857—859, 864, 866,
869, 870, 872, 874—881, 888, 894, 895, 900, 919, 921, 925,
929, 942—944, 947, 955, 956, 958, 959, 970, 972, 974, 975,
978, 998, 1001, 1003—1006, 1016, 1017, 1019, 1021, 1025,
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1065, 1070, 1077, 1079, 1084, 1088, 1090—1092, 1097, 1108,
1110

$K_2[OsO_2(OH)_4]$ 853

K_3PO_4 878, 879

K_2PHO_3 313

KPH_2O_2 307, 313, 864, 866

K_2PbO_3 280

$K[Pb(OH)_3]$ 886, 897, 1016

K_2S 356, 834, 970

K_2SO_3 44, 182, 183, 425—427, 834, 873, 874, 973, 978, 944

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750, 752, 762, 766, 767, 779, 783, 785, 796—779, 801, 838—
841, 874—878, 894, 895, 954, 957, 963, 973, 977, 978, 982,
983, 991—995, 999, 1043, 1044, 1052, 1054, 1056—1058,
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1081, 1087, 1089, 1096, 1098, 1101, 1104, 1106, 1109, 1116—
1118, 1121, 1124

$K_2S_2O_3$ 186, 187, 191, 877, 989, 993

$K_2S_2O_8$ 188—191, 576, 838—841, 957, 991—996

$K_2S_4O_6$ 989

$K_2[Sb(OH)_3]$ 824

$K_2[Sn(OH)_6]$ 819, 880, 1031

K_2Te 869

K_2TeO_3 869

$K_2[Zn(OH)_4]$ 277, 859, 1017, 1019, 1021

Mg 72, 197, 272—274, 327, 370, 379, 394, 395, 670, 701, 702,
942—944

$Mg(CH_3COO)_2$ 670

$MgCl_2$ 72, 392, 424

Mg_3N_2 197

$Mg(NO_3)_2$ 701, 702

MgO 370, 395

Mg_3P_2 760
 $\text{Mg}_3(\text{PO}_4)_2$ 327
 Mg_2Si 379, 392, 395
 Mn 415
 MnBr_2 430, 554
 MnCl_2 60, 62, 512, 513, 520, 781, 792, 795, 871
 $\text{Mn}(\text{H}_2\text{PO}_4)_2$ 430
 $\text{Mn}(\text{NO}_3)_2$ 193, 281, 291, 671, 680, 684, 741, 745, 760, 786—789, 750, 883
 MnO_2 118, 136, 177, 235, 313, 324, 414, 417, 419, 421, 422, 426, 429, 431, 487, 489, 512, 524, 550, 778, 790—794, 823, 870, 875, 882, 883, 894, 929, 947, 953, 956, 970, 975, 977, 982, 997—1002, 1006, 1010, 1053, 1055, 1059, 1062, 1064, 1065, 1070, 1077, 1079, 1084, 1085, 1088, 1090, 1092, 1097, 1099, 1115—1120
 Mn_3O_4 415
 $\text{MnO}(\text{OH})$ 871, 1011
 $\text{Mn}(\text{OH})_2$ 416, 417, 870, 1010, 1011
 MnS 611
 MnSO_4 170, 176, 237, 290, 323, 413, 418, 419, 422, 425, 428, 490, 491, 493, 494, 523, 524, 550, 551, 556, 558, 577, 601, 611, 640, 655, 664, 667, 676, 691, 750, 762, 766, 767, 778, 779, 780, 782, 783, 785, 791, 793, 823, 882, 954, 982, 997, 1054, 1056—1058, 1060, 1062, 1063, 1069, 1073, 1074, 1076, 1078, 1080, 1081, 1087, 1089, 1098, 1101, 1104, 1106, 1117, 1118, 1121, 1124
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 NH_4Br 861
 $(\text{NH}_4)_2\text{CO}_3$ 1113
 NH_4Cl 12, 13, 14, 71, 72, 240, 275, 300, 906
 $(\text{NH}_4)_2\text{CrO}_4$ 412

NH_4F 225
 $(\text{NH}_4)_2\text{Fe}(\text{SO}_4)_2$ 783
 NH_2OH 243, 274, 820
 NH_4OH 244, 497, 861—863, 906, 942, 947, 972, 1005, 1026, 1038,
 1107, 1113, 1122
 NH_4HS 245, 246, 499, 616, 906, 972, 1029
 $(\text{NH}_4)_2\text{S}_2\text{O}_8$ 192, 998
 $(\text{NH}_4)_2\text{SO}_4$ 192, 783, 940, 997, 1118
 NH_4I 862
 NH_4NO_3 245, 246, 254, 259, 286—288, 616, 692, 701, 708
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 N_2O_3 208, 211
 N_2O_5 15, 220
 NOCl 11, 547
 Na 109, 226, 387, 445, 475, 629
 NaAsO_2 49, 830
 Na_3AsO_3 915
 Na_3AsO_4 603, 774, 775, 830, 915
 NaAlO_2 446
 $\text{Na}[\text{Al}(\text{OH})_4]$ 855, 860, 1024
 $\text{Na}[\text{Au}(\text{CN})_2]$ 1020
 NaBr 37, 404, 410, 419, 555, 641, 821—823, 884, 892, 896, 898,
 935, 937, 938
 NaBrO 37, 938
 NaBrO_3 104, 400, 555, 829, 937
 NaBrO_4 104, 829
 NaBiO_3 176, 413, 664, 788, 794, 803, 841, 891, 1037, 1038
 NaCN 932, 1041
 Na_2CO_3 49, 163, 349, 367, 399—406, 448—452

NaCl 73, 83, 145, 166, 234, 298, 300, 387, 413, 444, 523—526, 547, 594, 603, 608, 668, 669, 677—679, 682, 689, 751, 774, 807, 808, 811, 812, 820, 830, 842, 885, 889—893, 904, 932, 933, 961, 988, 1032, 1095
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 NaClO_2 83, 814
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 Na_2CrO_4 404, 405, 406, 410, 413, 852, 884, 885, 889, 890, 896, 898, 899, 901, 965, 966, 1027, 1028, 1030, 1032—1038
 Na_3CrO_8 901
 $\text{Na}_3[\text{Cr}(\text{OH})_6]$ 410, 896, 898, 899, 966, 1027, 1028, 1030, 1032—1038
 NaF 57, 104, 828, 829, 835
 $\text{Na}_2[\text{Fe}(\text{OH})_4]$ 1023
 Na_2GeO_3 867
 NaHCO_3 932
 NaH_2AsO_4 49
 NaH_2PO_4 321, 720
 $\text{Na}_2\text{H}_2\text{Sb}_2\text{O}_7$ 553
 NaHS 968, 969
 NaHSO_3 979
 NaHSO_4 144, 525, 568, 606, 969
 $\text{Na}_3\text{H}_2\text{IO}_6$ 811
 NaI 49, 51, 96, 553, 642, 684, 832, 987
 NaIO_3 51, 96, 581, 573, 811
 Na_2MnO_4 1002
 Na_3MnO_4 1002
 NaN_3 239
 NaNH_2 226
 NaNO_2 221, 297—300, 393, 739, 750—754
 NaNO_3 147, 176, 221, 239, 271, 279, 282, 297, 321, 399, 607, 664, 720, 752—754, 788, 794, 803, 843, 852, 855, 856, 860, 883, 968, 1018
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Na_3P 720
 Na_2PbO_2 899
 NaPb(OH)_3 1034
 Na_2S 109, 121, 144—147, 162, 166, 181, 354, 603—609, 833, 961—967, 1028
 $\text{Na}_2\text{S}_2\text{O}_3$ 121, 184, 185, 676—683, 833, 836, 837, 931, 967, 986—988
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 Na_2SeO_3 812
 Na_2SeO_4 812
 Na_2SiO_3 382, 868
 Na_4SiO_4 381
 $\text{Na}_2[\text{Sn(OH)}_6]$ 818, 1022
 $\text{Na}_2[\text{Zn(OH)}_4]$ 822, 1018
 $\text{Na}_2[\text{Zn(CN)}_4]$ 1020
 NaVO_3 594
 NiBr_2 1073
 NiCl_2 805
 Ni(OH)_2 821, 933, 1012, 1013
 NiO(OH) 805, 821, 933, 1012, 1013
 NiS 626
 NiSO_4 626
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 PCl_5 316
 PCl_3O 315
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 PI_3 46
 POBr_3 429
 P_2O_3 302, 303, 318—320, 763, 950
 P_2O_5 79, 205, 301, 305, 318, 320
 PbS 613—615
 P_2S_3 309
 P_2S_5 304
 Pb 281, 342, 755, 1016
 PbCl_2 510, 511, 531
 PbCrO_4 549
 $\text{Pb}(\text{HSO}_4)_2$ 418, 525, 663
 $\text{Pb}(\text{NO}_3)_2$ 279, 280, 281, 289, 418, 527, 528, 540, 663, 753, 755,
 786, 789, 804, 856, 996
 PbO 289, 342
 PbO_2 279, 418, 466, 510, 525, 527, 662, 663, 753, 786, 804, 856,
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 PbSO_4 140, 549, 613—615, 662, 971
 Pd 364
 PdCl_2 364
 Pt 290, 733

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 606—608, 610, 616, 626, 632, 634, 643, 677, 681, 822, 833—
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SO_3 125, 206, 207

Sb 705

Sb_2O_3 824

Sb_2O_5 624, 705

Sb_2S_3 624

SbCl_3 514

$\text{Sb}_2(\text{SO}_4)_3$ 553

Se 125, 231, 585, 685, 689, 706, 927, 951, 986

SeO_2 125, 231, 685, 686, 951, 986

Si 55, 343, 376—389, 393, 394, 396, 559

SiC 344

SiCl_4 21, 386—388

SiF_4 55, 56, 389

SiH_4 390, 391, 392

Si_3N_4 376, 380

SiO 393

SiO_2 343, 344, 351, 377, 378, 384, 390, 391, 393—396

SiS_2 385

SmCl_2 781

SmCl_3 781

$\text{Sm}_2(\text{SO}_4)_3$ 781

Sn 703, 704

SnCl_2 533, 668, 743, 819, 880, 926, 1031

SnCl_4 533

$\text{Sn}(\text{NO}_3)_2$ 703

$\text{Sn}(\text{OH})_2$ 818

SnO_2 704, 927

SnSO_4 661

$\text{Sn}(\text{SO}_4)_2$ 661

SnS_2 668

Ta 1040

Te 261, 869

TeO_2 261

TiCl_3 17

TiOSO_4 178

$\text{Ti}_2(\text{SO}_4)_3$ 178

U_3O_8 627

UO_2SO_4 627, 985

$\text{U}(\text{SO}_4)_2$ 627, 985

VOCl_2 594

Zn 130, 156, 219, 276, 386, 628, 669, 698, 699, 739, 768, 773, 775,
797, 859, 1018—1021, 1072

ZnCl_2 384, 669, 739

$\text{Zn}(\text{NO}_3)_2$ 219, 698, 690

ZnS 130, 138, 822

ZnSO_4 138, 156, 628, 768, 773, 775, 797, 1072

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C_2H_4 1053, 1056

C_3H_6 496

C_5H_{10} 492, 1052

C_6H_{12} $(\text{CH}_3\text{CH}=\text{C}(\text{CH}_3)\text{CH}_2\text{CH}_3)$ 493, 1058

C_6H_{12} $(\text{CH}_2=\text{C}(\text{CH}_3)\text{CH}_2\text{CH}_2\text{CH}_3)$ 494, 1057

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Алкины

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C_3H_4 $(\text{CH}_3-\text{C}\equiv\text{CH})$ 1060

C_5H_8 $(\text{CH}\equiv\text{CCH}_2\text{CH}_2\text{CH}_3)$ 1061

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 $C_6H_5CH(CH_3)_2$ 1069
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 $C_6H_5C_3H_7$ 1068
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 $C_6H_5CH=CH-CH_2-CH_3$ 1080
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 $C_6H_5NO_2$ 499, 1051, 1071, 1072
 $NO_2C_6H_4C_4H_9$ 1073, 1074
 $NH_2C_6H_5$ 499, 1071, 1072
 $C_5H_3NCl_2$ 505

Спирты

$RCH(OH)CH_3$ 1108
 $R-OH$ 1082
 CH_3OH 365, 488, 1083, 1085—1087
 C_2H_5OH 489, 1088—1096
 C_3H_7OH 486
 C_4H_9OH ($CH_3CH(OH)CH_2CH_3$) 1097
 $C_2H_4(OH)_2$ (CH_2OH-CH_2OH) 1053, 1099
 $C_3H_6(OH)_2$ ($CH_3CHONCH_2OH$) 496
 $C_3H_5(OH)_2NH_2$ ($CH_2ONCHONCH_2NH_2$) 1059
 $C_6H_5-C_2H_3(OH)_2$ ($C_6H_5CHONCH_2OH$) 1077

Альдегиды

CH_2O 1101, 1109, 1111, 1112
 C_2H_4O 497, 1090—1092, 1096, 1102, 1104
 C_3H_6O 486, 1105—1107
 $C_5H_{10}O$ 493
 C_6H_5CHO 1062
 $C_9H_{10}O$ ($(CH_3)_2C_6H_3CHO$) 1075, 1076

Кетоны

$R-C(O)CH_3$ 1110
 C_4H_8O ($CH_3C(O)CH_2CH_3$) 1058, 1097
 $C_5H_{10}O$ ($CH_3C(O)C_3H_7$) 1057

Органические кислоты

НСООН 488, 491, 1039, 1086, 1087, 1100, 1101, 1112—1114
 CH_3COOH 482, 491, 749, 1052, 1058, 1060, 1061, 1089, 1093, 1104, 1115
 $\text{C}_2\text{H}_5\text{COOH}$ 492, 1052, 1080, 1105, 1106
 $\text{H}_2\text{C}_2\text{O}_4$ 1054, 1116, 1117, 1120
 $\text{C}_5\text{H}_{11}\text{O}_5\text{COOH}$ 1123
 $\text{C}_3\text{H}_6(\text{OH})\text{COOH}$ ($\text{HOOCCH}(\text{CH}_3)\text{CH}_2\text{OH}$) 1098
 $\text{C}_2\text{H}_4(\text{COOH})_2$ ($\text{HOOCCH}(\text{CH}_3)\text{COOH}$) 1098
 $\text{NH}_2\text{CH}_2\text{COOH}$ 500
 $\text{C}_6\text{H}_5\text{COOH}$ 498, 1063, 1066—1069, 1078, 1080, 1081
 $\text{C}_6\text{H}_3(\text{COOH})_3$ 1076
 $\text{NO}_2\text{C}_6\text{H}_4\text{COOH}$ 1073
 $\text{NH}_2\text{C}_6\text{H}_4\text{COOH}$ 1074

Соли органических кислот

RCOOK 1108, 1110
 $\text{CH}_3\text{COONH}_4$ 497, 1107
 CH_3COOK 489, 749, 1065, 1088
 CH_3COONa 670, 1095, 1102
 $\text{C}_6\text{H}_5\text{COOK}$ 1064, 1079
 $(\text{C}_5\text{H}_{11}\text{O}_5)\text{COONH}_4$ 1122
 $(\text{C}_5\text{H}_{11}\text{O}_5)\text{COONa}$ ($\text{HOCH}_2(\text{CHOH})_4\text{COONa}$) 1003
 $(\text{CH}_3)_2\text{C}_6\text{H}_3\text{COONH}_4$ 1075
 $\text{C}_3\text{H}_2\text{K}_2\text{O}_4$ ($\text{KOOCCH}_2\text{COOK}$) 495
 $(\text{NH}_4)_2\text{C}_2\text{O}_4$ 1118
 $\text{K}_2\text{C}_2\text{O}_4$ 487, 1055, 1084, 1099
 $\text{Na}_2\text{C}_2\text{O}_4$ 1119
 $\text{C}_6\text{H}_4(\text{COOK})_2$ 1070

Эфиры


 $\text{H}_2\text{C}-\text{CH}_2$ 1084
 $\text{CH}_3\text{COOC}_2\text{H}_5$ 501

Углеводы

$\text{C}_5\text{H}_{10}\text{O}_5$ 506
 $\text{C}_6\text{H}_{12}\text{O}_6$ 490, 1103, 1121—1123
 $\text{C}_{12}\text{H}_{22}\text{O}_{11}$ 483, 1124

КОМПЛЕКСНЫЕ СОЕДИНЕНИЯ

$[\text{Ag}(\text{NH}_3)_2]\text{OH}$ 497, 1039, 1075, 1107, 1111, 1113, 1122

$\text{H}[\text{AuCl}_4]$ 592, 734

$\text{H}_2[\text{PtCl}_6]$ 260, 733

$\text{H}_2[\text{SiF}_6]$ 383, 559

$\text{H}_2[\text{SnCl}_6]$ 668, 743

$\text{H}_2[\text{TaF}_6]$ 1040

$\text{K}[\text{Al}(\text{OH})_4]$ 276, 881, 1026, 1051, 1052

$\text{K}_3[\text{Cr}(\text{OH})_6]$ 887, 897, 900, 1029, 1031

$\text{K}_2[\text{HgI}_4]$ 1050

$\text{K}_2[\text{Hg}(\text{CN})_4]$ 1049

$\text{K}_2[\text{OsO}_2(\text{OH})_4]$ 853

$\text{K}_3[\text{Fe}(\text{CN})_6]$ 30, 100, 989, 1042—1048

$\text{K}[\text{Pb}(\text{OH})_3]$ 886, 897, 1016

$\text{K}_2[\text{Sn}(\text{OH})_6]$ 819, 880, 1031

$\text{K}_2[\text{Sb}(\text{OH})_6]$ 824

$\text{K}_2[\text{Zn}(\text{OH})_4]$ 277, 859, 1017, 1019, 1021

$\text{Na}[\text{Au}(\text{CN})_2]$ 1020, 1041

$\text{Na}[\text{Al}(\text{OH})_4]$ 855, 860, 1024

$\text{Na}_3[\text{Cr}(\text{OH})_6]$ 410, 896, 898, 899, 966, 1027, 1028, 1030, 1032—1038

$\text{Na}_2[\text{Pb}(\text{OH})_3]$ 1034

$\text{Na}_2[\text{Fe}(\text{OH})_4]$ 1023

$\text{Na}[\text{Sn}(\text{OH})_3]$ 1022

$\text{Na}_2[\text{Sn}(\text{OH})_6]$ 818, 1022

$\text{Na}_2[\text{Zn}(\text{OH})_4]$ 822, 1018

$\text{Na}_2[\text{Zn}(\text{CN})_4]$ 1020

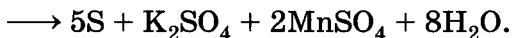
Приложения

1. Относительные электроотрицательности элементов (по Полингу)

[illegible]

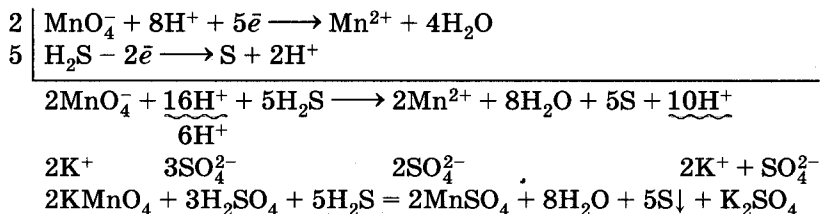
2. Примеры на восстановление левой части уравнений ОВР

ПРИМЕР 1. Дана правая часть уравнения реакции:

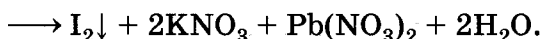


Восстановите левую часть уравнения.

Уравнение составлено для кислотной среды, так как среди продуктов восстановленный ион марганца Mn^{2+} . Перед серой S стоит коэффициент 5, следовательно, окислитель MnO_4^- взял $5\bar{e}$. Восстановитель — S^{2-} , так как, во-первых, образовалась сера S^0 , простое вещество, во-вторых, перед MnSO_4 стоит коэффициент 2, следовательно, восстановитель отдал $2\bar{e}$. Возможные полуреакции:



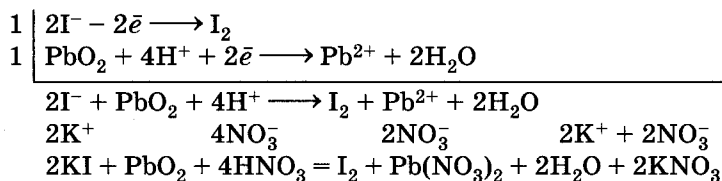
ПРИМЕР 2. Дана правая часть уравнения реакции:



Восстановите левую часть уравнения.

Уравнение составлено для кислотной среды. Окислитель взял $2\bar{e}$, так как образовался Pb^{2+} : $\text{Pb}^{+4} + 2\bar{e} \longrightarrow \text{Pb}^{2+}$, а ионы иода были восстановителями.

Возможные полуреакции:

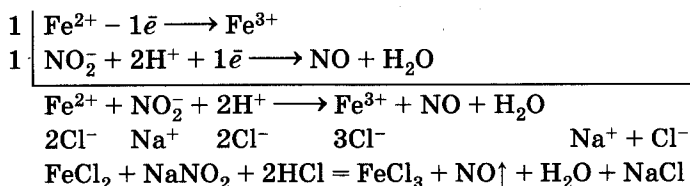


ПРИМЕР 3. Даны формулы продуктов реакции NO , FeCl_3 , NaCl , H_2O .

Восстановите уравнение реакции.

Видно, что азот восстановился, а ионы железа окислились, среда кислотная. Перед Fe^{3+} — коэффициент 1, следовательно, окислитель взял $1\bar{e}$.

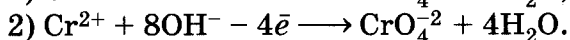
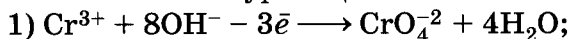
Возможные полуреакции:



ПРИМЕР 4. Восстановите левую часть уравнения, если даны продукты реакции с коэффициентами $2\text{K}_2\text{CrO}_4$, 7KCl , $5\text{H}_2\text{O}$.

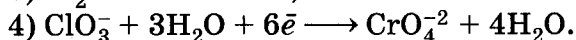
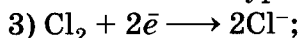
Уравнение составлено для щелочной среды, так как K_2CrO_4 образуется в щелочной среде. Перед хроматионом стоит коэффициент 2, следовательно, окислитель принял $2\bar{e}$ или кратное двум число электронов.

Возможные полуреакции:

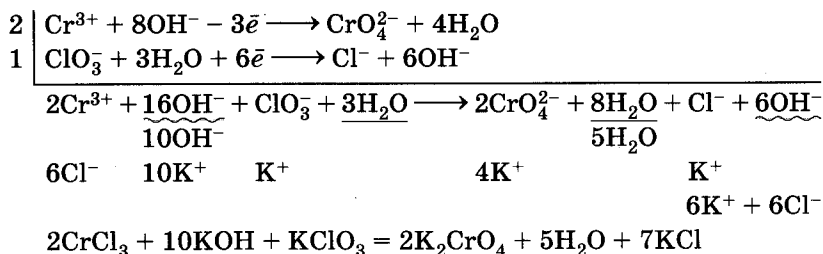


В качестве окислителя выступает хлор.

Возможные полуреакции:

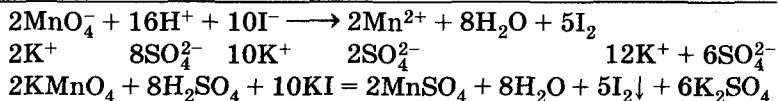
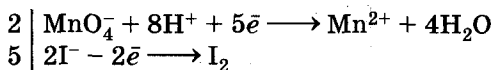


Подойдут варианты 1 и 4, так как число отданных и взятых электронов соответственно 3 и 6 после сокращения 1 и 2, т. е. перед CrO_4^{2-} появится коэффициент 2.



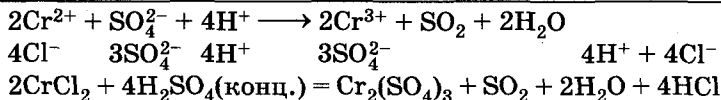
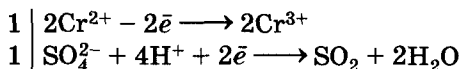
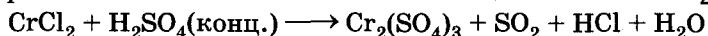
ПРИМЕР 5. Восстановите левую часть уравнения, если образовались продукты MnSO_4 , I_2 , K_2SO_4 , H_2O .

Среда кислотная, так как восстанавливается Mn^{2+} . Кислотная среда создавалась за счет серной кислоты. Исходные вещества: H_2SO_4 , KMnO_4 (окислитель), KI (восстановитель).



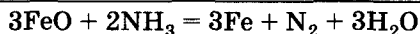
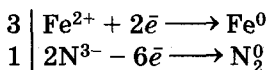
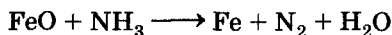
ПРИМЕР 6. Какие два вещества вступили в реакцию, если образовались $\text{Cr}_2(\text{SO}_4)_3$, SO_2 , HCl , H_2O ?

Определяем окислитель и восстановитель. Так как образовался SO_2 , можно предположить, что окислитель — H_2SO_4 , восстановитель — Cr^{+2} , а конкретнее — CrCl_2 .



ПРИМЕР 7. Какие два вещества вступили в реакцию, если образовались Fe , H_2O , N_2 ?

Если восстановилось железо, значит, до реакции оно было Fe^{2+} — окислитель, тогда восстановитель вещество, которое содержало N^{3-} , — это NH_3 .



3. Растворимость, осаждение и гидролиз солей при комнатной температуре

Ион	Br ⁻	CH ₃ COO ⁻	CO ₃ ²⁻	Cl ⁻	F ⁻	I ⁻	NO ₃ ⁻	PO ₄ ³⁻	S ²⁻	SO ₄ ²⁻
Ag ⁺	H	M	H	H	P*	H	P	H	H	M
Al ³⁺	*P	+	—	*P	M	*P	*P	H	+	*P
Ba ²⁺	P	P*	H	P	M	P	P	H	P*	H
Be ²⁺	*P	+	[H]	*P	*P	*P	*P	H	+	*P
Ca ²⁺	P	P*	H	P	H	P	P	H	P*	M
Cd ²⁺	*P	*P*	[H]	*P	*P*	*P	*P	H	H	*P
Co ²⁺	*P	*P*	[H]	*P	*P*	*P	*P	H	H	*P
Cr ³⁺	*P	+	—	*P	M	H	*P	H	[H]	*P
Cu ²⁺	*P	*P*	[H]	*P	*P*	—	*P	H	H	*P
Fe ²⁺	*P	*P*	[H]	*P	M	*P	*P	H	H	*P
Fe ³⁺	*P	—	—	*P	H	—	*P	H	—	*P
Hg ²⁺	M	*P*	—	*P	+	H	+	H	H	+
Hg ₂ ²⁺	H	M	H	H	M	H	+	H	—	H
K ⁺	P	P*	P*	P	P*	P	P	P*	P*	P
Li ⁺	P	P*	P*	P	H	P	P	M	P*	P

Ион	Br^-	CH_3COO^-	CO_3^{2-}	Cl^-	F^-	I^-	NO_3^-	PO_4^{3-}	S^{2-}	SO_4^{2-}
Mg^{2+}	*Р	*Р*	М	*Р	М	*Р	*Р	Н	Н	*Р
Mn^{2+}	*Р	*Р*	[Н]	*Р	*Р*	*Р	*Р	Н	Н	*Р
NH_4^+	*Р	*Р*	*Р*	*Р	*Р*	*Р	*Р	—	+	*Р
Na^+	Р	Р*	Р*	Р	Р*	Р	Р	Р*	Р*	Р
Ni^{2+}	*Р	*Р*	[Н]	*Р	*Р*	*Р	*Р	Н	Н	*Р
Pb^{2+}	М	*Р*	[Н]	М	М	М	*Р	Н	Н	Н
Sn^{2+}	+	+	—	+	*Р*	М	+	Н	Н	*Р
Sr^{2+}	Р	Р*	Н	Р	Н	Р	Р	Н	Р*	Р
Tl^+	М	Р*	М	М	Р*	Н	Р	Н	Н	М
Zn^{2+}	*Р	*Р*	[Н]	Р	М	*Р	*Р	Н	Н	*Р

Обозначения:

Р — хорошо растворимая соль ($> 0,1$ моль/л);М — малорастворимая соль ($0,1—0,001$ моль/л);Н — практически нерастворимая соль ($< 0,001$ моль/л);[Н] — не осаждается из раствора вследствие необратимого гидролиза (выпадает осадок (основная соль) и выделяется углекислый газ, а для Cr_2S_3 — осадок $\text{Cr}(\text{OH})_3$ и газ H_2S);

+ — полностью реагирует с водой;

— — не существует (соль не получена);

*Р — гидролизуется по катиону;

Р* — гидролизуется по аниону.

4. Характерные степени окисления элементов

Символ	Название	Степень окисления	Символ	Название	Степень окисления
${}_{89}\text{Ac}$	Актиний	0, +3	${}_{87}\text{Fr}$	Франций	0, +1
${}_{47}\text{Ag}$	Серебро	0, +1	${}_{31}\text{Ga}$	Галлий	0, +3
${}_{13}\text{Al}$	Алюминий	0, +3	${}_{64}\text{Gd}$	Гадолиний	0, +3
${}_{95}\text{Am}$	Америций	0, +2, +3, +4	${}_{32}\text{Ge}$	Германий	0, +2, +4
${}_{18}\text{Ar}$	Аргон	0	${}_{1}\text{H}$	Водород	0, +3
${}_{33}\text{As}$	Мышьяк	-3, 0, +3, +5	${}_{2}\text{He}$	Гелий	0
${}_{85}\text{At}$	Астат	-1, 0, +1, +5	${}_{72}\text{Hf}$	Гафний	0, +4
${}_{79}\text{Au}$	Золото	0, +1, +3	${}_{80}\text{Hg}$	Ртуть	0, +1, +2
${}_{5}\text{B}$	Бор	-3, 0, +3	${}_{67}\text{Ho}$	Гольмий	0, +3
${}_{56}\text{Ba}$	Барий	0, +2	${}_{53}\text{I}$	Иод	-1, 0, +1, +5, +7
${}_{1}\text{Be}$	Бериллий	0, +2	${}_{49}\text{In}$	Индий	0, +3
${}_{83}\text{Bi}$	Висмут	0, +3, +5	${}_{77}\text{Ir}$	Иридий	0, +3, +4
${}_{97}\text{Bk}$	Берклий	0, +3, +4	${}_{19}\text{K}$	Калий	0, +1

Символ	Название	Степень окисления	Символ	Название	Степень окисления
$_{35}\text{Br}$	Бром	-1, 0, +1, +5, +7	$_{36}\text{Kr}$	Криптон	0, +2
$_6\text{C}$	Углерод	-4, -1, 0, +2, +4	$_{104}\text{Ku}$	Курчатовий	0, +4
$_{20}\text{Ca}$	Кальций	0, +2	$_{57}\text{La}$	Лантан	0, +3
$_{48}\text{Cd}$	Кадмий	0, +2	$_3\text{Li}$	Литий	0, +1
$_{58}\text{Ce}$	Церий	0, +3, +4	$_{103}\text{Lr}$	Лоуренций	0, +3
$_{98}\text{Cf}$	Калифорний	0, +3, +4	$_{71}\text{Lu}$	Лютеций	0, +3
$_{17}\text{Cl}$	Хлор	-1, 0, +1, +3, +4, +5, +6, +7	$_{101}\text{Md}$	Менделевий	0, +2, +3
$_{96}\text{Cm}$	Кюрий	0, +3, +4	$_{12}\text{Mg}$	Магний	0, +2
$_{27}\text{Co}$	Кобальт	0, +2, +3	$_{25}\text{Mn}$	Марганец	0, +2, +4, +6, +7
$_{24}\text{Cr}$	Хром	0, +2, +3, +4	$_{42}\text{Mo}$	Молибден	0, +4, +6
$_{55}\text{Cs}$	Цезий	0, +1	$_7\text{N}$	Азот	-3, 0, +1, +2, +3, +4, +5
$_{29}\text{Cu}$	Медь	0, +1, +2	$_{11}\text{Na}$	Натрий	0, +1

${}_{66}\text{Dy}$	Диспрозий	0, +3	${}_{41}\text{Nb}$	Ниобий	0, +4, +5
${}_{68}\text{Er}$	Эрбий	0, +3	${}_{60}\text{Nd}$	Неодим	0, +3
${}_{99}\text{Es}$	Эйнштейний	0, +2, +3	${}_{10}\text{Ne}$	Неон	0
${}_{63}\text{Eu}$	Европий	0, +2, +3	${}_{28}\text{Ni}$	Никель	0, +2, +3
${}_{9}\text{F}$	Фтор	-1, 0	${}_{103}\text{No}$	Нобелий	0, +2, +3
${}_{26}\text{Fe}$	Железо	0, +2, +3, +6	${}_{93}\text{Np}$	Нептуний	0, +3, +4, +6, +7
${}_{100}\text{Fm}$	Фермий	0, +2, +3	${}_{21}\text{Sc}$	Скандий	0, +3
${}_{105}\text{Ns}$	Нильсборий	0, +5	${}_{34}\text{Se}$	Селен	-2, 0, +4, +6
${}_{8}\text{O}$	Кислород	-2, -1, 0, +2	${}_{14}\text{Si}$	Кремний	-4, 0, +2, +4
${}_{76}\text{Os}$	Осмий	0, +4, +6, +8	${}_{62}\text{Sm}$	Самарий	0, +2, +3
${}_{15}\text{P}$	Фосфор	-3, 0, +1, +3, +5	${}_{50}\text{Sn}$	Олово	0, +2, +4
${}_{91}\text{Pa}$	Протактиний	0, +4, +5	${}_{38}\text{Sr}$	Стронций	0, +2
${}_{82}\text{Pb}$	Свинец	0, +2, +4	${}_{73}\text{Ta}$	Тантал	0, +4, +5
${}_{46}\text{Pd}$	Палладий	0, +2, 4	${}_{65}\text{Tb}$	Тербий	0, +3, +4
${}_{61}\text{Pm}$	Прометий	0, +3	${}_{43}\text{Tc}$	Технеций	0, +4, +7
${}_{84}\text{Po}$	Полоний	0, +2, +4	${}_{53}\text{Te}$	Теллур	-2, 0, +4, +6

Символ	Название	Степень окисления	Символ	Название	Степень окисления
${}_{59}\text{Pr}$	Празеодим	0, +3, +4	${}_{90}\text{Th}$	Торий	0, +4
${}_{78}\text{Pt}$	Платина	0, +2, +4	${}_{22}\text{Ti}$	Титан	0, +2, +3, +4
${}_{94}\text{Pu}$	Плутоний	0, +3, +4, +5, +6	${}_{81}\text{Tl}$	Таллий	0, +1, +3
${}_{88}\text{Ra}$	Радий	0, +2	${}_{69}\text{Tm}$	Тулий	0, +3
${}_{37}\text{Rb}$	Рубидий	0, +1	${}_{92}\text{U}$	Уран	0, +3, +4, +6
${}_{75}\text{Re}$	Рений	0, +4, +7	${}_{23}\text{V}$	Ванадий	0, +2, +3, +4, +5
${}_{45}\text{Rh}$	Родий	0, +3, +4	${}_{74}\text{W}$	Вольфрам	0, +4, +6
${}_{86}\text{Rn}$	Радон	0, +2, +4, +6, +8	${}_{54}\text{Xe}$	Ксенон	0, +2, +4, +6, +8
${}_{44}\text{Ru}$	Рутений	0, +2, +4, +6, +8	${}_{39}\text{Y}$	Иттрий	0, +3
${}_{16}\text{S}$	Сера	-2, 0, +4, +6	${}_{70}\text{Yb}$	Иттербий	0, +2, +3
${}_{51}\text{Sb}$	Сурьма	0, +3, +5	${}_{30}\text{Zn}$	Цинк	0, +2
			${}_{40}\text{Zr}$	Цирконий	0, +4

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Учебное издание

**Дзудцова Донара Дмитриевна
Бестаева Лана Борисовна**

ОКИСЛИТЕЛЬНО-ВОССТАНОВИТЕЛЬНЫЕ РЕАКЦИИ

Ответственный редактор *А. В. Яшукова*
Оформление *Э. К. Реоли*
Художественный редактор *Э. К. Реоли*
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Компьютерная верстка *Т. В. Рыбина*
Корректор *Г. И. Мосякина*

Санитарно-эпидемиологическое заключение
№ 77.99.15.953.Д.005481.08.04 от 25.08.2004.

Подписано к печати 27.06.05. Формат 60х90^{1/16}.

Бумага типографская. Гарнитура «Школьная». Печать офсетная.
Усл. печ. л. 20,0. Тираж 3000 экз. Заказ № 4510081.

ООО «Дрофа». 127018, Москва, Суцевский вал, 49.

Предложения и замечания по содержанию и оформлению книги
просим направлять в учебную редакцию издательства «Дрофа»:
127018, Москва, а/я 79. Тел.: (095) 795-05-41. E-mail: chief@drofa.ru

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издательства «Дрофа» обращаться по адресу:
127018, Москва, Суцевский вал, 49.

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Торговый дом «Школьник».

109172, Москва, Малые Каменщики, д. 6, стр. 1А.

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- правила определения степеней окисления элементов;
- правила подбора коэффициентов в уравнениях ОВР методами электронного и электронно-ионного баланса;
- подробную информацию об окислительно-восстановительных свойствах элементов в неорганических и органических соединениях;
- более 1000 примеров уравнений ОВР.



ISBN 5-7107-8550-4



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